IFA WORLD FERTILIZER USE MANUAL

IFA World Fertilizer Use Manual. 1992.

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The IFA World Fertilizer Use Manual has not been updated since 1992 and this pdf version contains the original texts.
IFA does not intent to update this version.

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The manual is a 600-page handbook, containing detailed information on current fertilizer use recommendations and practices, for a wide range of world crops.

Following an introduction on fertilizers and their efficient use, there are chapters on individual crops or groups of crops, each written by an author who is a recognized expert in the crop(s) in question. There is information on more than 100 crops grown for food and renewable raw materials.

Each chapter contains:

- information on the biology of the crop,
- plant and soil analysis data,
- nutrient uptake and removal figures,
- recommendations for fertilizer use,
- current fertilizer practice in different countries,
- further reading.

80 crop specialists contributed to the manual. **Dr. W. Wichmann** of BASF co-ordinated and edited the contributions of the different authors and it is he who made the manual possible. **Mr. D.J. Halliday, Dr. G. Kemmler and Dr. M.E. Trenkel** also reviewed the texts. **Professor A. Finck** prepared the introduction.

Crop index - Type of crops

Cereal crops

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Rubber

Rubber

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Temperate Grassland: Permanent Grass and Sown Grass or Leys
Temperate Grassland: Grass/Legume Swards
Tropical Grasses
Tropical and Sub-Tropical Grass/Legume-Associations
Fodder Legumes
Fodderbeet

Mulberry

Mulberry

Other, Miscellaneous Crops

Oils and Fibres
Drugs and Spices
Dates and Figs
Bush Fruits

Fertilizers and their efficient use

Introductory chapter of the IFA World Fertilizer Use Manual

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Fertilizers and their efficient use

Summary

1. Fertilizers are substances that supply plant nutrients or amend soil fertility. They are the most effective means of increasing crop production and of improving the quality of food and fodder. With them, food for more people can be produced than this planet would otherwise support.

2. Fertilizer use is most effective (for obtaining high crop yields) on soils with high natural or improved fertility, but even on low fertility soils crop growth can be substantially improved.

3. Fertilizers are used in order to supplement the natural nutrient supply in the soil, especially to correct the (yield-limiting) minimum factor.

4. Some mineral and organic substances can be used directly as fertilizers, but most must be chemically processed in order to adapt them to plant needs (manufactured fertilizers).

5. The most suitable type of macro- or micronutrient fertilizer for a particular purpose depends on the speed of nutrient uptake desired (leaf spray or quick soil supply in water-soluble form, or slow-acting for a continuous supply), on the combination of nutrients wanted and on growth-enhancing side-effects such as mobilization of other soil nutrients).

6. The amount of fertilizer applied should be based on diagnostic methods, e.g. graded according to the content of available nutrients in the soil; plant analysis can also reveal hidden minimum factors (which may be eliminated by additonal fertilizer application). The upper limit of fertilizer use from the production point of view is determined by the limit of economic return.

7. Fertilizers should be distributed in such a way that all the crop plants in the field obtain an adequate amount; this is generally achieved by broadcasting granular fertilizers, and to some extent by spraying fertilizers in solution. The granules are either worked into the top layer of the soil or left for slow penetration by rain water.

8. Food quality is definitely improved by adequate use of fertilizers in general, and of mineral fertilizers in particular, provided they are applied in accordance with the latest concepts and knowledge. Quality in this context is understood to include not only the presence of quality components but also the absence of unwanted surplus nutrients and of toxic substances in plant products. Food production using adequate mineral and organic fertilizers prevents, and increases resistance to, many diseases. Agriculture which is based on the most up-to-date fertilizer practices may rightly be described as "health-promoting".

9. There is a risk that fertilizer use may adversely affect the environment (soil, water or air) beyond the low and harmless level which is unavoidable in primary production ; but, in practice, any pollution which may occur beyond this level is mainly the result of faulty manurial practices, often of excessive use of organic manures.

10. Fertilizer use has been shown to be an effective means of enhancing crop production for more than a century. It has contributed largely to the major increases in yields which have been achieved worldwide and to a substantial improvement of human and animal health. The degree of pollution, which is to some extent unavoidable, can be kept to an insignificant level typical of primary production.

In brief, fertilizers applied with the latest 'know-how', are an important source of wealth for the farmer, and of food and health for the population.

Why use fertilizers ?

Use of fertilizers is needed for all types of long-term crop production in order to achieve yield levels which make the effort of cropping worthwhile.

Modern fertilizer practices, first introduced more than a century ago and based on the chemical concept of plant nutrition, have contributed very widely to the immense increase in agricultural production and have resulted in better quality food and fodder. As a beneficial side-effect, the fertility of soils has been improved resulting in more stable yield levels, as well as in a better (nutrition-induced) resistance to some diseases and climatic stress. Furthermore, the farmer's economic returns have increased due to more effective production.

The purpose of fertilizer use, especially for higher yields, is identical in temperate and tropical climates:
- to supplement the natural soil nutrient supply in order to satisfy the demand of crops with a high yield potential
- to compensate for the nutrients lost by the removal of plant products or by leaching, etc.
- to improve unfavourable or to maintain good soil conditions for cropping.

Nutrients required by plants

Plants contain practically all (92) natural elements but need only 16 for good growth. Thirteen of these are essential mineral nutrient elements, commonly abbreviated, though with less precision, to "nutrients" (Box 1). They must be provided either by the soil or by animal manure or mineral fertilizer. Some other mineral nutrient elements, e.g. Na, Si, Co, have a beneficial effect on some plants but are not essential.

Box 1: Essential and beneficial mineral nutrients for plants.

Essential mineral nutrients (13) required for growth (of equal importance physiologically):

Macronutrients (6) of which the critical contents in plants are 2-30 g/kg of dry matter:

Major nutrients (3), applied in fertilizers for almost all crops on most soils:
N = nitrogen (taken up as NO_3^- or NH_4^+)
P = phosphorus (taken up as $H_2PO_4^-$ etc.)
K = potassium (taken up as K^+)

Secondary nutrients (3), applied in fertilizers mainly for certain crops on some soils:
S = sulphur (taken up as SO_4^{2-})
Ca = calcium (taken up as Ca^{2+})
Mg = magnesium (taken up as Mg^{2+})

Micronutrients (7) of which the critical contents in plants are 0.3-50 mg/kg of dry matter:

Heavy metals (5):
Fe = Iron
Mn = manganese (Fe, Mn, Zn, Cu taken up as divalent cation or chelate)
Zn = zinc
Cu = copper
Mo = molybdenum (taken up as molybdate MoO_4^{2-})

Non-metals (2):
Cl = chlorine (taken up as Cl^-)
B = boron (taken up as $H_2BO_3^-$, etc.)

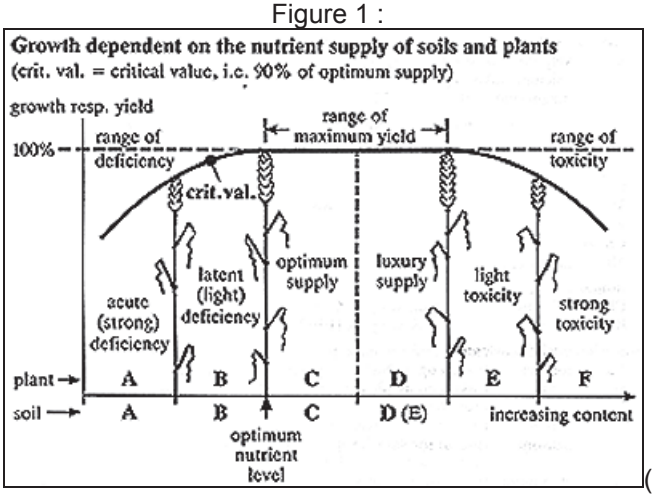
Some beneficial nutrients useful for some plants:
Na = sodium (taken up as Na+; can partly replace K for some crops)
Si = silicon (taken up as silicate, etc., e.g. for strengthening cereal stems to resist lodging)
Co = cobalt (mainly for N-fixation of legumes)
Cl = chlorine (useful for some crops in greater than essential amounts, for osmotic regulation and improved resistance to some fungi)
Al = aluminium (perhaps beneficial for some plants, e.g. tea ?).

Fertilizer use must also take into account the nutritional requirements of animals and human beings consuming the crops. It may, for example, be necessary or advisable to supply, for the benefit of grazing animals, increased amounts of elements which are not essential to the plants, e.g. of Na, or of essential elements like selenium and cobalt as a precaution against nutritional disorders caused by deficiencies.

Since hardly any soil can supply all the nutrients needed in sufficient amounts to meet the demands of high-yielding crops, the deficit must be made good by adding fertilizers and/or manures.

As a general rule, nutrient uptake should always be ahead of the production of plant dry matter during the vegetation period. The total amount needed at a given stage can be estimated from the critical nutrient level required in plant tissue for the optimum functioning of biochemical processes. In fact the nutrient contents of crops are usually somewhat higher than the critical levels, i.e. they are in the "normal" or optimum nutrient range (Fig. 1). On the other hand, a large surplus or luxury supply is unwanted, not only because of the money wasted on excess fertilizer but also because it may upset the nutrient balance.

Figure 1 :



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Plants, in general, contain maximum amounts of nutrients in the later stages of growth shortly before maturity (usually more than is actually needed), but nutrient balance calculations are for the most part based on the somewhat smaller amounts that are, or could be, removed from the field at the time of harvest. Relevant nutrient removal data (corrected, or not, for possible luxury uptake) should be established for all crops and farming systems.

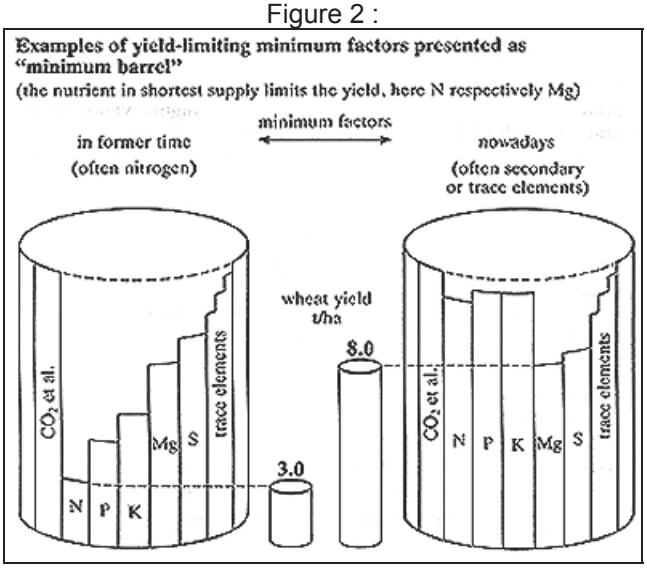
The amounts of nutrients which need to be added in fertilizers and manures depend on:

- the nutrient requirement of a crop for the desired yield level;
- the nutrient supply in the soil, which can be estimated by diagnostic methods.

No fertilizer or manure is needed if the uptake of a nutrient from the soil does not lead, even in the longer term, to any significant depletion of the soil reserves. This is often the case with micronutrients.

Fertilizer use should concentrate especially on the so-called "minimum factor" which, according to the Law of the Minimum, critically limits plant growth (Fig. 2).

Figure 2 :



Soil fertility and its improvement

The best results from fertilizer use are obtained on the basis of a high fertility level. Soils differ widely in their ability to meet plant requirements; most have only a moderate natural soil fertility but can be considerably improved by soil amendments. For successful farming the natural fertility of the soil is often less important than its potential productivity after the removal of its inherent limiting factors.
The term "soil fertility" comprises a complex of properties which should be optimized as far as possible (Box 2).

Box 2. Components of soil fertility.

Soil depth (determining the volume of soil accessible to the root system). Most arable crops prefer about one metre without any obstructing layer.

Soil structure (based on size distribution and aggregation of particles). This determines the distribution of pore sizes which is decisive for the supply of air and water to the roots.

Soil reaction (an indicator and regulator of chemical processes and equilibria).

Content of nutrients in different degrees of availability.

Storage capacity for soluble nutrients from the soil and fertilizers.

Humus content and quality (including proportion in mineralizable form).

Quantity and activity of soil organisms as agents of transformation processes.

Content of detrimental or toxic substances, either naturally occurring (e.g. salts in saline soils, aluminium in extremely acid soils) or man-made (e.g. from pollution).

A highly productive soil with high fertility, natural or improved,

- mobilizes soil nutrients from the reserves;
- transforms fertilizer nutrients into easily available forms;
- stores water-soluble nutrients in easily available forms, thus preventing leaching;
- offers the plants a balanced nutrient supply, due to its self-regulating system;
- stores and supplies sufficient water;
- maintains good soil aeration for the oxygen requirements of roots;
- does not 'fix' nutrients, i.e. convert them into unavailable form.

Soils with a high natural fertility can produce substantial crop yields even without added fertilizer, but can produce even higher yields with an additional supply of the critical nutrients. Good soil fertility provides the basis for all other measures for successful farming and should not be neglected.

There are alternative ways of making use of soil fertility in farming:

- exploitation i.e. farming without any added fertilizer (e.g. in shifting cultivation);
- utilization of as many components of soil fertility as possible without compensation and yet without negative yield effects (e.g. by applying only moderate amounts of fertilizer N and P);
- maintenance and improvement of soil fertility to assure consistent high yields (e.g. by compensating for losses due to removal and by soil amendments to improve fertility).

The large differences in fertility between different soil types and sub-types must be taken into account.
Some indications are given below for regional particularities.

Soils of the humid tropics
- are partly very acid (liming is required, generally to pH 5.5-6.5);
- are often low in available P or liable to P-fixation (use of phosphate fertilizer is therefore often essential, combined if necessary with liming);
- in very humid areas are often low in available K, Mg and S (therefore there are high fertilizer requirements for these nutrients);
- often have a low sorption or storage capacity for nutrients (so fertilizer application should be split between several dressings);
- are often low in available N, although the decomposable organic matter is rapidly mineralized.

Soils of the sub-tropics are characterized by:
- water shortage (so, without irrigation, fertilizer use must be suitably adapted to efficient water use);
- N being the main critical nutrient, on account of the low humus content;
- widespread P deficiency, especially in sandy soils;
- neutral soil reaction (therefore often a shortage of available Fe and Zn);
- a generally good supply of S, Mn and B;
- risk of salinity.

Soils of humid temperate zones are marked by:
- widespread soil acidity which requires liming;
- partly obstacles to root growth (e.g. hard layers in subsoil);
- often insufficient aeration (poor natural drainage of heavy soils);
- generally shortage of available N and often of P, K, Mg;
- low nutrient reserves in sandy soils, also only little storage and therefore considerable leaching with water surplus;
- partly fixation of P and Mo (due to natural soil acidity) and Cu (in organic soils);

- climatic cold stress retarding nutrient uptake, etc..

Soil reaction and liming

For soils to be highly productive, they must first be in the optimum pH range (the values mentioned below refer to measurements in water suspension).

Values under pH 4.5-5.0 can be very damaging to plants ("soil acidity syndrome") by causing nutrient deficiencies (of P, Mg, etc.) and toxicities (of Al, Fe, Mn). Liming should be employed to raise the pH to at least about 5.5. A pH range of 5.5-6.5 seems to be satisfactory for moderate yields of most crops. Optimum pH values, resp. ranges, for high yields have been established for different soils and rotations.

All kinds of liming materials may be used to increase the pH. In many acid soils, lime containing magnesium carbonate provides a double benefit by adjusting pH and by supplying Mg. However, liming experiments, especially in the tropics, often give disappointing results, not because liming is ineffective but because it is not complemented by appropriate fertilizer use to correct other critical factors.

In neutral to slightly alkaline soils under high yield conditions, use of acidifying N fertilizers can be advantageous through resulting in a better supply of micronutrients such as Mn or Zn (which tend to be somewhat immobilized).

Gypsum is a useful soil amendment on saline/sodic soils (for removal of Na and supply of structure improving Ca).

Organic fertilizers inclusive manures

A number of organic materials can be useful soil amendments and suppliers of nutrients (Box 3). Since many are waste products, they can sometimes be cheaply available, especially if used near to where they are produced. Some farm wastes are used because recycling is the only, and moreover beneficial, means of disposing of them. If waste material of any kind has to be bought in by the farmer, it must be comparatively cheap and have no detrimental or toxic effect, and it must be profitable.

Box 3. Types of organic fertilizers inclusive manures.

Naturally occurring material, e.g. peat.

Farm wastes:
- crop residues (straw, leaves, etc.)
- animal manures (farmyard manure, liquid manure, slurry)
- compost (mixture of decomposed plant residues etc.)
- green manures (leguminous or other crops incorporated into the soil).

Residues from processing of plant products, e.g.:
- fibres (from paper industry) and pressed cakes (from oilseeds)
- wood materials (bark, sawdust; lignin from paper industry)
- molasses (from sugar industry).

Residues from processing of animal products, e.g.:
- blood-, horn- and bone-meal
- leather dust, etc.

Town wastes:
- composted household refuse
- sewage sludge.

Soil inoculants (e.g. living micro-organisms.).

Processed organic wastes, especially if they are to be sold, generally require mechanical and chemical preparation, i.e. they must be dried, ground, mixed, granulated, neutralized, complemented by the addition of particular nutrients, and free of pathogenic germs.

Important criteria for organic fertilizers are:
- dry matter content;
- total and easily mineralizable humus;
- total and quick-acting N:
- C/N ratio;
- contents of substances detrimental to plant growth or product quality (heavy metals in particular should be below established critical limits).

The rates at which they are applied should take into account both the expected nutrient supply available to the crop (for N in farmyard manure, about 20-30 % of the amount applied is available to the crop during the first year) and the need to minimize nutrient losses or detrimental effects.

Organic matter, applied as a fertilizer, has a complex influence on plant growth (Box 4). The minimum amount required for sustaining optimum activity of soil life is in the range of 3-5 t/ha. In high yielding cereal fields the decomposition of very high amounts of straw must be aided by adding about 1 kg of N per 100 kg of straw.

Box 4. Effects of organic materials on plant growth (via the soil).
Improvement of physical soil properties, either directly or by activating living organisms in the soil;
- better soil structure as a result of soil loosening and crumb stabilization;
- better water-holding capacity and soil aeration;
- surface protection by mulch layer.

Influence on chemical properties:
- sorption of nutrients by humic acids;
- supply of nutrients from decomposition of humus and from dissolving action on soil minerals:
- fixation of nutrients in organic complexes (mainly a negative influence for a shorter or longer period);
- effects of growth regulators produced in soil (e.g. growth inhibitors accumulating in monocultures, and antibiotics protecting against some bacterial diseases).

Soil inoculants may, for example, contain Rhizobium spp. for inoculating seeds of leguminous crops where suitable strains are not present in sufficient quantity. The amount of nitrogen gained by legumes through an effective Rhizobium symbiosis is usually in the range 50-200 kg/ha N, whereas in paddy fields the Azolla/Anabaena symbiosis amounts up to 400 kg/ha N. Application of other bacteria to promote N fixation from the atmosphere, e.g. Azotobacter spp., or to enhance the release of P and K from soil minerals, has not yet proved successful.

Special problems concerning organic fertilizers, esp. manures, result from high rates of application of slurry (from cattle or from pigs) in certain areas, and from town compost and sewage sludge which may contain an excess of certain nutrients or be contaminated with heavy metals or toxic organic substances.

Mineral fertilizers

Numerous mineral fertilizers have been developed to supplement soil nutrients and to meet the high requirements of crops (Box 5). They are generally mineral salts, except for some organic chemicals such as urea which are easily converted into salts.

Box 5. Types of mineral fertilizers. (according to different criteria)

Method of production:

- "natural" (as found in nature or only slightly processed);
- synthetic (manufactured by industrial processes).

Number of nutrients:
- single-nutrient or straight fertilizers (whether for major, secondary or micro nutrients);
- multinutrient (multiple nutrient) or compound fertilizers, with 2, 3 or more nutrients:

Type of combination:
mixed fertilizers, i.e. either a physical mixture of two or more single-nutrient or multinutrient fertilizers (for granular products this may comprise a mixture of separate granules of the individual ingredients, or granules each containing these ingredients);
complex fertilizers, in which two or more of the nutrients are chemically combined (e.g. nitrophosphates, ammonium phosphates).

Physical condition:
- solid (crystalline, powdered, prilled or granular) of various size ranges;
- liquid (solutions and suspensions);
- gaseous (liquid under pressure, e.g. ammonia).

Mode of action:
- quick-acting (water-soluble and immediately available);
- slow-acting (transformation into soluble form required).

The customary classification into single- or multi-nutrient fertilizers usually refers only to the three major nutrients. Many so-called single-nutrient fertilizers actually supply more than one nutrient, e.g. ammonium sulphate which contains both N and S.

The nutrient content, or grade, may refer either to the total or to the available nutrient content, and may be expressed traditionally for some nutrients in oxide form (P2O5, K2O) or in elemental form (N, P, K).

In order to guarantee the farmer good quality and to exclude hazards, the composition of mineral fertilizers, and the trade in mineral fertilizers, are officially controlled in many countries (in Germany since 1918). Recently, with the object of obtaining better control of pollution, special regulations have been introduced for the application of certain organic fertilizers and manures (e.g. slurry). In the near future governmental regulations will be introduced for fertilizer application in general, in order to minimize any avoidable fertilizer-induced pollution or food contamination. Many fertilizer products, however, are not yet subjected to legal control, though there is an increasing voluntary standardization by producers in order to ensure uniformity of quality, especially for organic and organic-mineral fertilizers. Even so, there still remain a number of "unorthodox" fertilizers (mainly organic) on the market which, in spite of "expert" recommendations, have little real value.

Nitrogen fertilizers

They are valued according to their total N-content, the different N-forms (which determine the rate of action) and side-effects if any (Box 6).

Box 6. Types of N fertilizers (N content refers to total N)

Ammonium fertilizers:
- ammonia (80 % N), ammonium sulphate (21 % N), ammonium bicarbonate (17 % N), all moderately quick-acting. Uptake by plants can be retarded by addition of nitrification inhibitors, e.g. dicyandiamide (DCD).

Nitrate fertilizers:
- calcium nitrate (16 % N), sodium nitrate (16 % N), Chilean nitrate, all quick-acting and increasing soil pH.

Ammonium nitrate fertilizers:
- ammonium nitrate (about 34 % N), calcium ammonium nitrate which is a combination of ammonium nitrate and calcium carbonate (21-27 % N), ammonium sulphate nitrate (26-30 % N).

Amide fertilizers:
- urea (45-46 % N), calcium cyanamide (20 % N).

Solutions containing more than one form of N:
- urea ammonium nitrate solution (28-32 % N).

Slow release fertilizers:
- either derivatives of urea with N in large molecules, or granular water-soluble N fertilizers encased in thin plastic film, but slow or very slow-acting according to type of coating; partly including a quick-acting component.
- or other means of slow release, e.g. sulphur coated urea (SCU)

Multinutrient fertilizers containing N
- NP: nitrophosphate NP (20-23 % N, 20-23 % P2O5)
  monoammonium phosphate = MAP (11 % N, 52 % P2O5)
  diammonium phosphate = DAP (18 % N, 46 % P2O5)
  liquid ammonium polyphosphates (e.g. 12 % N, 40 % P2O5);
- NK;
- NPK.

Nitrate N in the soil solution is immediately available and thus acts quickly but is most liable to leaching. Plants take up N mainly in nitrate form. Ammonium N, although fully available, has a somewhat slower effect, because it is first adsorbed and then only gradually released and nitrified.

Amide N, e.g. as urea, must first be transformed to ammonium as a result of microbial action which is dependent upon temperature. However, urea applied as a foliar spray provides the quickest possible N supply, but for this purpose the biuret content must be below 0.3 %. Calcium cyanamide, in addition to its fertilizer value, has herbicidal and fungicidal properties due to intermediate decomposition products.

For most purposes the different forms of N give about the same yield responses (apart from any possible side-effects or gaseous losses which occur more with urea than with salts) because quick action, via the soil, is only needed in special cases, e.g. for application in spring at rather low temperatures. Considering the S-shaped curve of plants' nitrogen requirements during the vegetation period, most N fertilizers tend to act rather too quickly. This means that high rates given wholly at sowing time often provide too much for the young plant but not enough during the later stages. This can be compensated by dividing the total N application between a basal application at sowing and one or more topdressings. The N supply from slow release fertilizers is theoretically better adapted to the curve of N uptake but depends on temperature. Being more expensive, they are mostly confined to domestic garden use.

The most important side-effect of nitrogenous fertilizers to be considered is the acidifying effect of ammonium and amide fertilizers. Since this has to be counteracted by liming, it is convenient to express it in terms of the equivalent loss of CaO (Table 1).

Table 1. Soil acidification by nitrogen fertilizers.

Fertilizer	**Amount of CaO to compensate the soil acidification induced by 1 kg N***
Calcium ammonium nitrate (27 % N)	0.6

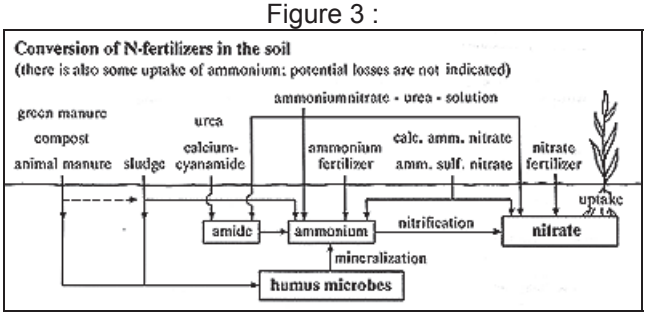
Ammonia, urea, ammonium nitrate	1
Diammonium phosphate, ammonium sulphate nitrate	2
Ammonium sulphate	3
* On the basis of 50 % utilization rate.	

N dynamics in soils are very complex (Fig. 3). The important process of nitrification (transformation of ammonium to nitrate by bacteria) proceeds rather quickly when temperatures are warm. At temperatures of 20-25 °C an application supplying 50-100 kg/ha N would nitrify in about one to two weeks.

Nitrification can be delayed for several weeks by adding special nitrification inhibitors to the fertilizer. This can be useful for preventing an undesirable accumulation of nitrate in vegetable crops or reducing loss by leaching.

The utilization rate of N in fertilizers is mostly about 50-70 % during the first year.

Figure 3 :



In terms of energy the production of 1 kg N in fertilizer requires the equivalent of about 1 litre of oil. The energy required for fertilizers is per area-wise about one-third of total energy requirement. However, the energy output of the principal crops is about two to three times the energy input. More effective fertilizer use, particularly of N fertilizers, therefore means a saving in energy.

Phosphate fertilizers

Box 7. Types of P fertilizers
(P2O5 content refers to 'available' portion, except for rock phosphate where it means total cotent)

Water-soluble types (quick-acting):

- single superphosphate (18-20 % P2O5);
- triple superphosphate (45 % P2O5).

Partly water-soluble types (quick- and slow-acting):
- partly acidulated phosphate (23-26 % P2O5, at least one-third water-soluble).

Slow-acting types:
- dicalcium phosphate (citrate-soluble);
- basic slag (citric acid-soluble).

Very slow-acting types:
- rock phosphate (finely-powdered soft type, e.g. 30 % P2O5), with reactivity indicated by formic acid-solubility; permitted minimum is about one-half of total P2O5 content).

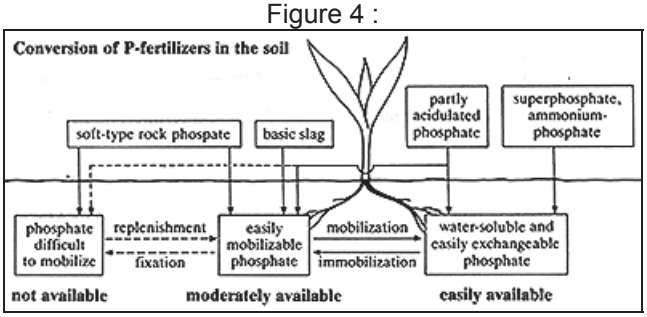
Multinutrient fertilizers containing P:
- NP (see N fertilizers, Box 6);
- PK (mixtures very commonly used);
- NPK (may contain about one-third or more water-soluble P for quick supply and two-thirds slow acting P for

continuous supply)

'Availability' is measured by solubility in specified extractants as an indication of the rate of transformation under various soil conditions. Water-soluble P (e.g. mono-calcium phosphate) is easily available to plants and remains available, though to a somewhat less extent after immobilization into other forms, this transformation being retarded by granulation and placement of the fertilizer. Citrate-/citric acid-soluble P is moderately available to plants and is suitable for many purposes over a wide range of acidic to neutral soil conditions except where quick action is required. Formic acid-soluble P in soft powdery rock phosphate is only very slowly available to plants; its reactivity (release of soluble P) is somewhat better where soils are warmer, moister and more acidic, but still above the acidity damage range.

Under conditions of intensive farming on well, or quite well supplied soils, for most purposes, the common phosphate fertilizers give about an equal yield response per unit of 'available' P2O5. Water-soluble P, however, is superior for crops with a short growing season and limited root system in deficient soils. The dynamics of phosphate fertilizers in soil is illustrated in Fig. 4.

Figure 4 :



NP fertilizers are generally superior in placement application; gaseous losses may occur from surface-applied diammonium phosphate on neutral soils. Phosphate fixation, i.e. the transformation of soluble fertilizer P into unavailable forms, is fortunately restricted to special soil conditions, e.g. high content of active Al and Fe, and is rather insignificant from a practical point of view in most soils. The utilization rate of P in fertilizers is usually about 15 % in the first year but only 1-2 % per year thereafter, with the result that only about two-thirds is taken up by the end of thirty years.

Potash fertilizers

These are mainly derived from geological saline deposits. Although low-grade, unrefined materials can be used directly, most fertilizer use is now in the form of higher-concentration products, all of which are water-soluble and quick-acting:

- potassium chloride, or muriate of potash (40-60 % K2O), the lower grades providing Na in addition to K2O, with or without Mg;
- potassium sulphate (50 % K2O), for Cl-sensitive crops (e.g.potatoes, tobacco);
- potassium magnesium sulphate, also known as sulphate of potash magnesia or Patentkali (e.g. 40 % K2O, 6 % Mg).

Recently, several industrial residues containing K, e.g. filter dust, have been developed for use as slower-acting forms, especially where it is desired to avoid loss by leaching.

Potash fertilizers should generally be applied at sowing time. The K+ ions are adsorbed in the soil and thus remain available, yet largely protected against leaching. However, split application is advisable (e.g. part autumn, part spring) in sandy soils where higher leaching losses may be expected. Some immobilization into clay lattice layers reduces availability but

strong fixation into completely unavailable forms is fortunately restricted to a few special soil types. The utilization rate of K in fertilizers is about 50-60 % during the first year.

Secondary nutrient fertilizers

Magnesium fertilizers are either quick-acting soluble salts (mainly sulphates) or in slow-acting form (their nutrient content may be expressed in terms of either Mg or MgO).

Quick-acting types:
- magnesium sulphate, in the forms of Epsom salts (10 % Mg) or kieserite (16 % Mg);
- potassium magnesium sulphate (see under potash fertilizers) or NMg fertilizers.

Slow-acting:
- magnesium carbonate (dolomitic lime).

The utilization rate and potential leaching loss of water-soluble Mg are similar to K.

For soils needing liming, the cheapest source of Mg is a lime containing Mg, of which a high rate of application can ensure a good supply of Mg for several years without detrimental effect.

Sulphur: The increasing incidence of sulphur deficiency, especially in crops with high requirements such as oilseed rape and legumes, has considerably increased the use of sulphur fertilizers. Water-soluble sulphates, e.g. potassium sulphate or magnesium sulphate (18 % S), are most effective for the treatment of growing crops, as also is ammonium sulphate (24 % S) where its acidifying effect can provide an added benefit. Slower-acting types such as pure gypsum (18 % S) or the by-product calcium sulphate of superphosphate (14 % S) should be used if leaching poses a problem. (Yellow) elemental sulphur (100 % S) may be applied either in powdered form or as the coating in sulphur-coated urea (10-20 % S). Both the water-soluble and slower-acting forms are suitable for application at sowing time.

Calcium is only applied other than for liming in cases of definite deficiency. Sources which do not raise the pH are:

Quick-acting:
- calcium chloride, solid or in solution;
- Ca components of foliar sprays.

Slow-acting: gypsum.

Soil application, though simple, is often disappointing because of restricted translocation within the plant; in such cases a foliar spray is preferable.

Micronutrient fertilizers

Apart from the widespread recognized areas of natural deficiencies of micronutrients, attention should be given to the possibility that they may become critical "minimum factors" as other critical factors are amended and yield levels rise. They may be applied either as single-nutrient fertilizers or as supplements in macronutrient fertilizers. The quickest and commonest method of correcting deficiencies is by foliar application.

Iron is usually applied as a foliar spray in the form of chelates such as Fe-EDTA (9 % Fe) or Fe-EDDHA (6 % Fe). For soil application the latter has the advantage that it is more stable in neutral soils.

Manganese deficiency occurs mainly in slightly acidic to neutral soils. Both Mn sulphate (24-32 % Mn) and Mn-EDTA (13 % Mn) are water-soluble and quick-acting, and are suitable for foliar or soil application; Mn oxides may be used as a means of increasing the soil's reserves. Indirect improvement of the soil supply may be achieved by using acidifying N fertilizers.

Zinc is usually applied to deficient crops as a foliar spray of Zn sulphate (e.g. 23 % Zn) or Zn chelate (e.g. Zn-EDTA), for soil application a rate of 5-10 kg/ha Zn is recommended.

Copper deficiency may most easily be corrected for a longer period by soil application of 5 kg/ha Cu as Cu sulphate or oxides, etc. Chelates or neutralized Cu sulphate (25 % Cu) are suitable for foliar spraying of deficient crops.

Boron needs vary widely. As a prophylactic treatment for crops with high demands, soil application of borax (11 % or 22 % B) is advisable, the rate depending on the crop (0.5-2.0 kg/ha B), but no more should be given than needed for that crop in order to avoid the risk of a damaging surplus affecting a succeeding crop with a low requirement. A better distribution can be obtained by incorporating the boron in e.g. phosphate or multinutrient fertilizers. Polyborates seem to be superior to borax for foliar application (at about 1 kg/ha).

Molybdenum is required in only very small amounts: 0.5-1.0 kg/ha Mo for soil application of water-soluble Na molybdate or ammonium molybdate (40-50 % Mo); less than 100 g/ha Mo for foliar application.

Multinutrient fertilizers

Since crops often require an additional supply of several nutrients, special combinations with different nutrient ratios offer a considerable simplification and facilitation of fertilizer application. Most combinations are of the complex NPK-type, which contains N in part as ammonium and part as nitrate, one-third of the phosphate in water-soluble form and K mostly as chloride.

Diagnosis of fertilizer requirements

The still widely used practice of deciding rates of fertilizer use on the basis of local experience or general data for crop requirements is certainly useful for obtaining at least medium yield levels, but neither very effective nor economic.

Soils differ widely in their capacity for providing nutrients, depending on the amount of total reserves, on mobilization or fixation dynamics, accessability of the chemically available nutrients to the roots, etc.

Therefore it remains necessary to assess empirically the nutrient status of soils and plants in order to provide guidelines for effective fertilizer use.

Box 8. Methods of diagnosis.

Optical (plant observation):
- extent of deviation from full green colour;
- identifiable deficiency symptoms;
- growth difference compared with plots without fertilizer.

Chemical:
- soil testing:
  - content of available nutrients as a basis for fertilizer requirements;
  - pH, salinity, etc.

- plant testing:
  - spot tests, e.g. on the leaf, or with extracts;
  - plant analysis, using nutrient contents as a basis for estimating additional requirements

Box 9. Checklist before applying fertilizers.

1. Are other agronomic factors (variety, plant protection, water, etc.) satisfactory?
2. Are basic requirements of soil fertility fulfilled? (pH, organic matter, stable porous soil structure, absence of compacted layer, good drainage, no salinity).
3. Which nutrients need not be considered in this particular soil?
(Many soils have adequate Ca, Fe, Mo, etc.)
4. Which nutrients need not be considered every year? (e.g. Mg may be supplied in liming material, Zn and Cu in long-lasting, slow-acting fertilizers, and S may not be needed for crops with low requirements)
5. What amounts of fertilizer P and K are needed at sowing time? (To be determined by soil testing or, in well supplied soils, estimated from nutrient removal by crop)
6. What kind and amount of N fertilizer is needed, and when ? (Either based on expected yield or soil testing)
7. Which nutrients may have special problems in this soil (e.g. fixation of Mn) or are needed in large amounts by particular plant species (e.g. S for oilseed rape, B for beet and legumes)?

Diagnosis by plant observation

A normal dark green colour characterizes a good nutrient supply, but any change to light green or yellowish tinges, suggests a deficiency, provided other factors such as extreme temperatures, diseases, spray damage, air pollution etc. are not responsible. The easiest means of diagnosis is by identification from colour photographs showing deficiency symptoms in the particular crop concerned (see also Box 10 below). The precise cause may, however, not be easy to establish from observation alone, particularly in cases of latent deficiency ("hidden hunger") for which chemical methods will usually be needed.

Box 10. Brief key to deficiency symptoms.

Symptoms	**Deficiency**
Symptoms appearing first on older leaves:	
Chlorosis starting from leaf tips	N
Necrosis on leaf margins	K
Chlorosis mainly between veins (which remain green)	Mg
Brownish, greyish, whitish spots (e.g. on cereals)	Mn
Reddish colour on green leaves or stem	P
Symptoms appearing first on younger leaves:	
Mottled yellow-green leaves with yellowish veins	S
Mottled yellow-green leaves with green veins	Fe
Brownish black spots (e.g. on legumes, potatoes)	Mn
Youngest leaf has white tip	Cu
Youngest leaf is brownish or dead (e.g. on beet)	B

A direct comparison in the field, sometimes called the "window" technique, may be obtained by using a small control plot without fertilizer (by stopping the distributor for some metres), which will often reveal an obvious colour difference. If it does not, then chemical methods are necessary.

Diagnosis by soil testing

Testing, which is based on the concept of "available" nutrients, in its usual form is rather simple and inexpensive, but the main problem lies in selecting and calibrating suitable extraction methods for particular soil types, sub-types or units. The value of these methods must be established empirically in the soil area concerned.

Box 11. Soil testing procedure and interpretation.

1. Representative soil sampling
About 20 auger cores are taken and mixed per hectare, or from test plot, from a depth, for most purposes, of 0-20 cm in arable land or 0-10 cm in grassland (sometimes also from the subsoil).

2. Preparation of sample.
Air-dried and sieved (<2 mm).

3. Extraction of available nutrients by appropriate method, e.g.:
for N: water (for nitrate) or calcium chloride (for ammonium);
for P: NaHCO3 (Olsen) or diluted HCl (Bray); or organic acids like lactic acid;
for K, Mg: ammonium acetate (for exchangeable K) or diluted and buffered mineral or organic acids (e.g. citric or lactic);
for Fe, Mn, Zn, Cu: chelating agents (e.g. EDTA, DTPA);
for B: hot water (Berger and Truog).

4. Chemical determination of extracted nutrients (expressed as %, mg/1oo g or mg/kg[ppm] of air-dried soil or per litre of soil).

5. Interpretation:
for P, K, Mg: in relation to expected relative yield without fertilizer
(see Table 2)

for Mn, Zn, Cu, B (see Table 3)

Table 2

Interpretation of soil test data for macro nurients				
Supply class	**Expected relative yield without fertilizer**	**Available (extractable) nutrients (mg/kg soil)**		
	%	**P**	**K**	**Mg**
very low	50	< 5	< 50	< 20
low	50-80	5- 9	50-100	20- 40
medium	80-100	10-17	100-175	40- 80
high	100	18-25	175-300	80-180
very high	100	>25	> 300	> 180
Methods: P: OLSEN (0.5 N NaHCO3; pH 8.5) K, Mg: MEHLICH (0.2 N NH4Cl + acetic acid, 0.015 N NH4F, 0.012 N HCl; pH 2.5); data for soils with medium CEC (10 - 20 mg/100 g)				

Table 3

Interpretation of soil data for micronutrients				
Element	Extraction method	**Available nutrients (mg/litre of soil = 1.5 kg min. soil) Ranges of supply**		
		Deficient	**Medium**	**Excess**
Mn	AAAc + pH correction	< 10 - 25	(100 - 500)	> 1 300
Zn	DTPA	< 0.4-0.6	(1 - 5)	> 10 - 20
Cu	AAAc-EDTA	< 0.8-1.0	(2 - 10)	> 17 - 25
B	hot water	< 0.3-0.5	(0.5- 2)	> 3 - 5

Methods: AAAc: Acid amm.-acetat. (0.5 M amm.-acetate + acetic acid; pH 4.7; with 0.02 M Na-EDTA respectively (LAKANEN and ERVIOE) (pH-correction for Mn: at pH 6.5 - factor 0.5; at pH 6 - factor 1; at pH 5.5 - factor 1.5) DTPA: 0.005 M; pH 7.3 (LINDSAY and NORWELL) Hot water: BERGER and TRUOG

The choice of a suitable method requires calibration either:

by field experiments, in which the relative crop yield with and without fertilizer is correlated with the relative extraction data from soil with and without fertilizer (r2 should be at least 60 % and preferably > 70 %);

or by plant nutrient content, i.e. by correlating the nutrient content in the plant at a particular stage with the nutrient extraction data from the soil (less expensive but still rather useful);

or by the appearance of deficiency symptoms, comparing the nutrient extraction data from the soil with the appearance or non-appearance of symptoms (useful for approximate classification in the case of micronutrients).

Diagnosis by plant analysis

The nutrient content of plants provides reliable information on their nutritional status at the date of sampling, thus giving a guide not only to any supplementary fertilizer needs of the current crop but also to the probable requirements of future crops. Although it is more costly than soil testing and needs more care in the handling of samples, this method is growing in importance(Box 12). Interpretation is usually based on the total contents of nutrients in leaves, or other suitable plant parts, in comparison with critical nutrient concentrations or "critical values"; but there are more sophisticated methods which either consider the "active" mobile contents (this can be somewhat better for e.g. Ca and Fe), or simple or complex nutrient ratios (e.g. Beaufil's Diagnostic and Recommendation Integrated System = D.R.I.S., which seems to have advantages for more detailed studies). It is essential to ensure that the "critical values" used relate to the expected level of yield (e.g. 80, 90 or 100 %); once properly established, they are applicable to the same crop worldwide.

Box 12. Plant analysis procedure and interpretation.

1. Representative sampling of specified plant part at stage corresponding with table of critical values.
2. Preparation of sample: same-day washing in distilled water, followed by oven-drying and careful grinding;
3. Chemical analysis: standard methods after dry or wet ashing (results expressed as %, %o or ppm = mg/kg).
4. Interpretation, e.g. by comparison with critical values or categories, which will determine whether immediate action is needed to correct a deficiency or to adjust a nutrient ratio; furthermore conclusions can be drawn whether the amount of fertilizer applied at sowing time was sufficient or should be increased for the next crop.

Diagnosis by fertilizer experiments

On account of the cost and labour involved, this method is much less suitable for individual farmers than for Advisory Services and Research Institutes. The experiments may be conducted either in the field or in pots (costing less but giving restricted information). Advice on field experimental design is given in Box 13 below. Although the statistical analysis of the yield results follows standard procedures, great care is needed to ensure that the correct conclusions are drawn concerning cause and effect. A statistically significant yield response to a particular fertilizer treatment might, for example, be due either to an increased supply of a subsidiary nutrient contained in that fertilizer or to an indirect effect on the availability of micronutrients in the soil rather than to the nutrient(s) which the fertilizer was intended to supply. Great care must also be taken in extrapolating the results, which strictly apply only to the soil area of the experimental plots under the conditions then prevailing, not to other areas and other crops in different seasons.

Box 13. Advice on design of fertilizer field experiments.

1. Provided the field has a reasonably uniform soil, a suitable choice of layout will probably compensate for small irregularities.
2. Good growth conditions must be ensured by careful management, including effective measures for weed control and crop protection.
3. The number of replications required (normally 4) depends on the purpose of the experiment.
4. Rates of fertilizer nutrient application should preferably be graded in equal steps from zero to that which possibly will give the maximum yield; a larger number of smaller steps gives more information but costs more.
5. The number of plots is determined by the number of different treatments and the number of replications, e.g. 5 rates of application (say O, 40, 80, 120, 160 kg/ha) with 4 replications requires 20 plots, or for example 3 rates of application of each of 3 nutrients (say N, P, K) with 4 replications would require 108 plots but this number may be reduced by including only the most important combinations (for example 4 replications of the 8 treatments NOPOKO, N0P1K1, N1POK1, N1P1KO, N1P1K1, N2P1K1, N1P2K1, N1P1K2 or N2P2K2 needs only 32 plots). The way in which the plots are laid out can range from simple blocks in a single row to more complicated multivariate designs such as the Latin square.
6. The total area of an experiment is usually limited by practical management considerations.
7. The results should be calculated by appropriate statistical methods, e.g. analysis of variance, and expressed in a manner indicating the statistical significance of differences in yield.

Significance, however, does not automatically mean "truth" of the results obtained.

Practical recommendations for fertilizer use

Recommendations based on any of the above methods of diagnosis must take into account the farmers' objectives and the economics of fertilizer use. Most farmers are interested in a system of fertilizer practice which will raise their level of crop yields and maintain that higher level, possibly over a long period. Those who are already achieving a medium yield level may wish to move into high-yield farming in which adequate supplies of all nutrients will need to be taken into account, while those who are now at a low yield level will need to concentrate first on the major "limiting factors".

The total amount of fertilizer N required is mainly established from response curves based on field experiments which are then transformed into local guidelines and modified in the light of farmers' long experience. However, correct timing, and the split between early application and later topdressings, are even more important than the total amount of N applied. In the example of a first (major) application of N in spring at the beginning of growth of autumn-sown crops or at the sowing of spring-sown crops, diagnosis may be based on the amount of mineral N (Nmin), mainly nitrate with some ammonium, in the 0-1 m soil layer; this amount expressed as kg/ha Nmin is then considered as if it were fertilizer which had already been applied. The desirable amount of Nmin for a particular crop is derived from field experiments. Taking the example of winter wheat for which the desired Nmin for high yield (8-10 t/ha) is about 120 kg/ha and the actual Nmin from a soil test in spring is 50 kg/ha, the difference to be applied in fertilizer would be 70 kg/ha N. Theoretically the amount to be applied at a later stage could be based on the "easily mineralizable organic N" in the soil at that later stage but the increase in accuracy compared with using the local guidelines is seldom enough to justify the extra cost. Even the Nmin, which is known to work well but is rather expensive, is more used by Advisory Services than by individual farmers.

The amounts of other macronutrients to be applied should be based on the concept of supplying and maintaining permanently an adequate nutrition level for the (high-yielding) varieties of crops to be grown. This implies the replacement of nutrients removed from the field at harvest or lost by leaching, erosion etc. Theoretically all additions and removals should be taken into account but this would complicate the calculation unnecessarily. Instead, it is usually sufficient to multiply the average nutrient content per unit yield at harvest by the expected yield per hectare. In estimating the amounts to be applied in mineral

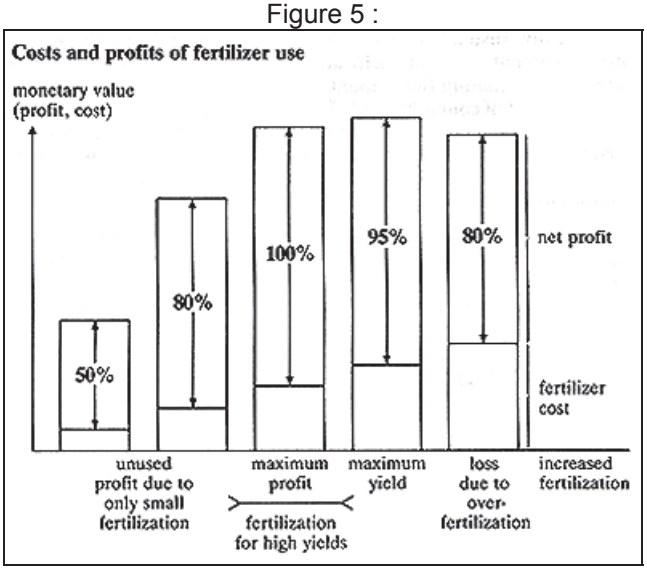
fertilizers, allowance must of course be made for any nutrients to be applied in other forms (organic manures, etc.). Preferably the calculations should be based on crop rotations as a whole, with due allowance for any expected "carry-over" from one crop to the next (and not lost by leaching), rather than on individual crops.

Foliar application provides a quick means of remedying nutrient deficiencies, particularly of micronutrients, but can seldom provide large enough quantities to meet the whole of the requirements for macronutrients except where repeated spraying, e.g. with N, is practicable.

Economics of fertilizer use

While in periods of food shortage the main aim may be a high level of food production, in market economies any input must produce a reasonable financial return. Because of the Law of Diminishing Returns the amount of fertilizer applied will be limited by the maximum profit obtainable, which will be at a slightly lower level of fertilizer use than that required for maximum yield.

Figure 5 :



Theoretically, from a comprehensive series of field experiments, it should be a simple matter to obtain yield response curves from which the economic optimum levels of fertilizer use can be calculated. The main problem is that such ideal response curves for a single nutrient often assume that there is no deficiency in any other nutrient, which is rarely the case in farmers' fields. Economic interpretation of response curves therefore needs to be undertaken with considerable caution. A further difficulty is that, in soils well supplied with nutrients such as P and K, unless the results of long-term field experiments extending over several years are available, it may not be possible to produce any yield response curve at all to these nutrients. In this case the practical solution is simply to apply fertilizer on the basis of nutrient removal, which is generally economic in the high yield range.

In those developing countries, where yields are lower, a simple rule of thumb is provided by the Value:Cost Ratio ("VCR"), i.e. the ratio between the value of the extra yield obtained by using fertilizer and the cost of the fertilizer applied. This ratio has been found to range from 1 to 10, but it is recommended that it should be at least 2, i.e. the value of the extra output obtained should be at least double the cost of the input. Here too, a measure of caution is needed. Unless the field experiments from which the ratio is derived were carried out on typically managed farmers' fields, there is always a danger that the economic value of the response may be slightly exaggerated, so that a purported ratio of 2 may perhaps in reality be only 1.5. This, of course, is one of the reasons why it is recommended that the VCR should be at least 2.

Fertilizer application

Fertilizers may be applied on or into the soil or directly to the plants. The aim is to apply them cheaply, uniformly and effectively. The method will depend on the type of material.

Box 14. Application methods according to type of material used.

Mineral fertilizers applied as nutrient sources:

- Solid water-soluble fertilizers are evenly distributed on to the soil surface (penetration into the root zone then takes place during leaching after dissolution by water) or are placed directly into the root zone, e.g. beneath the seed at sowing time.

- Solid water-insoluble fertilizers are distributed on to the soil surface and mechanically mixed into the arable layer (where this is impracticable, e.g. on grassland, there is a natural but slow penetration by soil organisms).

- Liquid fertilizers are:

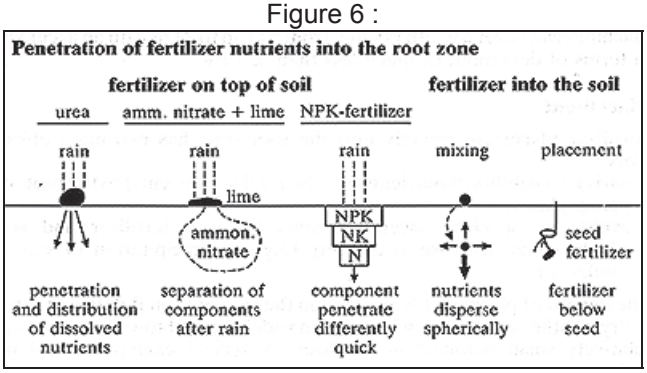
either sprayed on to the soil surface in their original concentration and left to penetrate (only suitable if no gaseous losses occur);
or mixed into the soil immediately after application (to prevent gaseous losses);
or injected directly into the soil;
or sprayed in diluted form directly on to the plants.

- Gaseous fertilizers are injected into the top layer of the soil (e.g. gaseous ammonia at a depth of at least 10 cm).

Bulky organic manures and amendments:

They should be evenly distributed on the soil surface to the extent possible and mixed into the arable layer, e.g. by harrow, disc harrow or plough, but some amendments need to be applied directly into the subsoil. Organic manures with a low nutrient content may be used as a mulch (surface protection layer).

Figure 6 :



Theoretically, finely powdered material mixed thoroughly into the arable layer would give the most uniform distribution within the root zone but this is often not necessary and too costly. The use of granular fertilizer represents a compromise between uniformity of distribution and ease of application. The granule size of water-soluble fertilizers is standardized , for example 90 % of the granules at 2-5 mm. Since water-insoluble fertilizer in similar granules of this size would generally act too slowly, they are usually granulated in such a way that powdery material is only loosely connected and the granules therefore disintegrate rapidly, though in all cases granules must be sufficiently stable to withstand the mechanical pressures encountered during transport and spreading.

Large super-granules are useful for some types of manual application.

In very large fields with exceptional heterogeneity in soil nutrient supply there may be an advantage in varying fertilizer use in conformity with the results of a close network of soil tests; in this case the process can be facilitated by a computer-assisted fertilizer application programme on the distributing machine.

The cost of distribution depends on the accuracy desired. Since medium accuracy suffices for most purposes, broadcasting by simple types of spinning disc distributors (with a spread of around 10-24 m) is very common. The amount of fertilizer to be distributed ranges from about 100 kg to more than 1 500 kg per hectare and the accuracy of distribution, in terms of deviaton from the desired rate, is usually about ± 10 %, up to a maximum of ± 20 %.

For more precise distribution, and especially for the application of fertilizers with varying physical characteristics, pneumatic or similar types of distributors are preferred, which either blow or throw the fertilizer through a number of distribution tubes on to the soil. Such machines can cover a width ranging from 5 m up to 24 m with an accuracy, in terms of deviation, of much less than ± 10 %.

Placement

Fertilizer placement directly into the root zone has two main objectives:

- early accessibility of nutrients (e.g. N, P, micronutrients) to the roots of young plants;
- avoidance of close, overall contact between fertilizer and soil which would decrease availability (e.g. by precipitation of water-soluble P).

The benefit of placement is greatest, in the case of P, on P-deficient soils, in dry weather conditions, with crops in widely spaced rows and receiving relatively small amounts of fertilizer. A typical example would be maize.

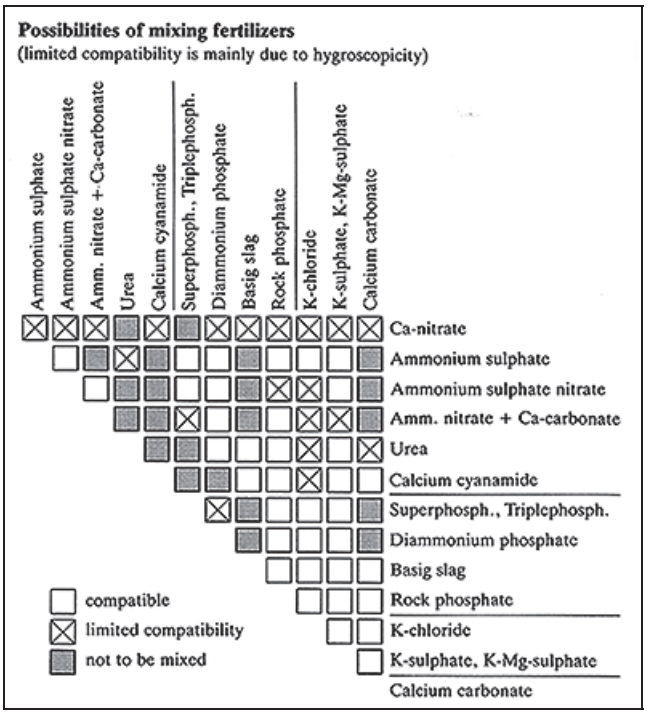
In contrast, wheat in humid areas with its narrower row-spacing would hardly justify the extra cost. For micronutrients, placement can take the form of actual attachment to the seed.

Bulk blending

Instead of distributing single-nutrient fertilizers separately or making use of factory-manufactured multinutrient fertilizers (complex fertilizers), the farmer may wish either to mix the fertilizers himself or to avail himself of the services of one of the retailers with mixing units who, in many countries, offer cheap mixing facilities. This provides the opportunity to prepare special blends with nutrient ratios to suit the particular farmer's needs appropriate to his own soils and crops. However, fertilizers which are to be mixed must be compatible both chemically and physically:

- chemically, so that no gaseous loss will occur nor any decrease in nutrient availability or caking due to chemical reactions
- physically, i.e. of similar granule size (e.g. in Europe 2-4 mm, or in USA 1-3.5 mm), to prevent segregation during transport and spreading and possibly also of similar density (urea, for example, can create problems through its lightness compared with other fertilizers), and to avoid probable yield reduction.

Figure 7 :



Liquid fertilizer application

Special spraying machinery is needed, especially for suspensions.

Advantages are the easy transfer from one container to another by pumping, the accuracy of distribution, the speed of application (covering 5-10 ha/hr), the possibility of incorporating fungicides , etc., and, in irrigated agriculture, the ease of inclusion of fertilizer in irrigation water.

Foliar application

Although possessing the advantage of quick action, the amount of fertilizer that can be distributed in a foliar dressing is limited by the sensitivity of the leaves to osmotic agents such as dissolved salts or (somewhat less) to organic chemicals such as urea. With the exception of some N fertilizers (due to the rather high tolerance of urea), foliar application can supply only very limited amounts of the primary nutrients compared with requirements. The situation is somewhat better for the secondary nutrients, but the best result are obtained with micronutrients because a relatively large proportion of the total requirement can be supplied in a single spraying. In cases of marked deficiency, repeated spraying is essential.

Spraying is most effective, and the risk of scorch is minimized, if the spray droplets do not dry too rapidly, i.e. on cloudy days and in the early morning or late afternoon. The amount of water used is often around 400 l/ha but progress is being made with lower volumes. Amounts of up to 30 kg N, 1 kg Mg and 0.2-0.5 kg micronutrients can be applied per hectare in a single foliar spray.

Fertilization, food quality and environmental pollution

Under conditions of food shortage, the major goal of fertilizer use is a high crop yield giving a lower priority to food quality and possible negative influences on the environment. However, when production efforts have resulted in meeting the food demand or even in a surplus, the

quality aspect and the potential pollution effects on soil, water and air receive the same importance than the crop yield itself or even more.

Fertilizers and food quality

This topic should be considered in a broad sense. Fertilizers can influence quality:

- either indirectly, by improving plant health, especially resistance to adverse climatic factors, diseases and pests;

- or directly, by increasing the content of essential and beneficial organic and mineral nutrients in human food and animal feed;

- or negatively, through their incorrect or imbalanced use or by the involontary addition of toxic substances.

Examples of the improved resistance of well nourished plants to adverse climatic factors:

- resistance to drought: a better supply of K improves their waterholding capacity, and P encourages early root growth and so ensures better survival in dry spells;

- resistance to frost and cold: increased by a better supply of K, P and some micronutrients (e.g. Mn, Cu);

- resistance to ultra-violet radiation: a good supply of Zn counteracts radiation-induced destruction of growth regulators.

Clearly, the damaging effects of plant diseases and pests cannot be completely eliminated simply by supplying abundant and balanced plant nutrition, but in many cases they can be contained and reduced to a lower and sometimes negligible level. Examples include:

- better resistance to some insect pests resulting from a good supply of K, as a result of better mechanical protection and a decrease in cell constituents attractive to insects;

- better resistance to fungal attack resulting from a good supply of boron.

- improved soil fertility also seems to result in soil fungi producing a better supply of antibiotics which protect plants therapeutically against some bacterial diseases. Further research is certainly needed in this border area between plant nutrition and plant protection with the aim of minimizing the need for protective sprays.

There are two separate aspects of food and fodder quality:

- market value, depending on easily recognizable external characteristics such as cleanness and absence of decay; furthermore, on the content of protein, suugar, etc. for the processing industry.

- nutritional value, comprising palatability (taste and smell, difficult to categorize), content of the many important organic and mineral nutritional constituents, and absence of undesirable or even dangerous toxic substances.

Box 15. Influence of nutrient supply on food quality.

N: better supply increases amounts of total and pure protein, protein quality (more of essential amino-acids), and some vitamins, especially B1; excessive supply tends to increase amide content, resulting in bad flavour after cooking, or to raise nitrate content unacceptably.

P: better supply improves protein quality and increases the content of some vitamins and of mineral phosphate, which is an important mineral nutrient; slightly increased radioactivity due to uranium present as natural impurity in P fertilizer seems to be of no importance whatsoever.

K: better supply increases carbohydrates, and especially vitamin C; as with P, slightly increased radioactivity coming from the naturally occurring K40 isotope is of no importance.

Other nutrients: the obvious advantage of having an optimum supply of all nutrients hardly needs mention. Individual nutrients which may adversely affect quality when supplied in excess include the heavy metals such as Zn and Cu. Unwanted contamination with the toxic heavy metal Cd may arise from the use of town wastes or, less importantly, from P fertilizers.

Fertilizers and health

Although the fertilizer-induced increase in the content of essential food constituents does not necessarily signify that fertilizers improve "health", it seems nevertheless to be so. Before the advent of fertilizer use, deficiency diseases in farm animals and humans were widespread : bone weaknesses due to lack of P, vitamin deficiencies due to inadequate plant nutrition, diseases in grazing livestock due to deficiencies of Cu and Co, for example. Furthermore, some virus and bacterial diseases seem to have diminished in their infective capacity as a result of improved nutrition. The considerable increase in human life expectancy must also be attributed in part at least to having more and better food, stemming in turn from fertilizer etc.

Even so, it has to be admitted that a significant proportion of the benefits to food quality are lost in processing, e.g. in the production of white bread, and may even be lost during cooking, as with some heat-sensitive vitamins.

In view of the established generally positive effect of fertilizer use on food quality, it is surprising that certain groups of consumers in the developed countries are requesting so-called "natural" food in the sense of food produced not only without chemical plant protection but also without the use of synthetic mineral fertilizers (quite apart from the entirely separate question of food additives such as preservatives and colourants).

A special market has been developed for such products of "organic farming" using either organic manure alone or together with "natural" mineral fertilizer such as rock phosphate. This is fully acceptable so long as scientific principles are observed and no unfounded claims are made for superior quality.

Mineral fertilizers, manures and environmental pollution

Whereas positive effects of fertilizer use on the environment are often overlooked, attention nowadays is focussed on negative aspects. Mineral and organic fertilizers are accused of:

- accumulation of dangerous or even toxic substances in soil from fertilizer constituents, e.g. Cd from mineral phosphate fertilizers or from town or industrial waste products;

- eutrophication of surface water, with its negative effect on oxygen supply (damaging to fish and other forms of animal life);

- nitrate accumulation in ground water, thus diminishing the quality of drinking water;

- unwanted enrichment of the atmosphere with ammonia from organic manures and mineral fertilizers, and with N2O from denitrification of excessive or wrongly placed N fertilizer.

As to contamination of soils with toxic heavy metals, it can easily be shown that mineral fertilizers make only a rather small contribution in comparison with, for example, town wastes. However, as soil fertility must be considered in the very long term and not only in decades or centuries, the annual addition should be kept at such a low level that the enrichment is negligible. Industrial waste products should always be carefully checked to determine whether they contain potentially toxic substances, and appropriate critical limits should be established.

Nutrient losses from the soil into surface and ground water (mainly nitrate by leaching and phosphate by erosion) occur even when fertilizers are not used, but they are increased slightly but unavoidably even by correct fertilizer use and are increased substantially by excessive or unbalanced use, which can be avoided.

Considerable leaching of nitrate is caused, for example, by:

- excessive application of organic liquid manure;
- intensively fertilized speciality crops;
- ploughing of grassland
- fertilizer application for over-optimistic yield expectations which fail to materialize (thus resulting in a surplus at the end of the vegetation period);
- part of the correctly estimated N requirement remaining unused because of other limiting factors not being taken into account, e.g. deficiencies of secondary or micronutrients;

In other words, N losses are mainly due to mistakes in fertilizer use or crop management, not to fertilizer use itself. Moreover, counter-measures can be taken to prevent loss of nitrate residues after harvesting (soil must not be left bare over winter) and to prevent soil erosion.

N loss by leaching seems to range from 10 to more than 100, in extreme cases more than 150 kg/ha N depending on the accuracy of fertilizer use and the extent of the preventive methods used. In Germany, at present the average appears to be far below the officially (but wrongly) discussed figure of 100 kg/ha N, but for most soils rather in the range of 30-60 kg/ha N. In any case, exaggerated overall averages do great injustice to farmers who apply fertilizers accurately and spend much effort in preventing excess leaching. From a scientific point of view, much more attention needs to be given to the enigma of N balance sheets before drawing premature conclusions on N losses.

Loss of phosphate by leaching (< 1 kg/ha P) is negligible, while loss by erosion is due to bad soil management rather than fertilizer use.

Atmospheric pollution by ammonia is mainly due to primitive methods of storing and spreading organic manure. N immission (involuntary intake from the air) ranges, in Central Europe, from 10 to 15 kg/ha N, with over 40 kg/ha N recorded in the vicinity of intensive animal husbandry.

Of the mineral fertilizers, only urea and ammonium sulphate might cause significant NH3-volatilization losses, especially if not incorporated (e.g. grassland, topdressing of cereals). To minimize these losses, incorporation into the soil or application before rain or irrigation is recommended.

The contention that agriculture contributes considerably to N2O production via denitrification, as a result of excessive or wrongly applied fertilizer N, is a serious problem, because this gas contributes to the destruction of the ozone layer in the stratosphere which protects against ultra-violet radiation. Official estimates, derived mainly under artificial conditions or by the difference method, showing losses of approximately 15 % or more of the applied N, are not

really substantiated; total denitrification losses in the range of 5-10 % of the applied N, of which only about 10 % is as N2O, seem to be more realistic, especially for soils under normal moisture conditions.

Since pollution of the environment should be minimized, governments are trying to control the avoidable negative influences by special laws.

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Barley *(Hordeum vulgare L.)*

French: Orge, escourgeon (winter barley); Spanish: Cebada; Italian: Orzo; German: Gerste

Crop data

Annual, winter- and spring-sown types; ears 2- or multiple-rowed; grains generally with glumes.

Harvested products: grain, straw, (occasionally) whole green plant.

Desired characteristics affecting fertilizer requirement:

In grain for livestock feed: high crude protein, especially lysine. In grain for processing for use in human foodstuffs: high-protein endosperm, lack of excrescences, low husk content.
In grain for malting: high starch, low crude protein, lack of excrescences.

Straw for bedding: should be dry, absorbent material.

Whole green plant for forage: high crude protein and energy, smooth glumes.

Sowing times: winter varieties should have completed tillering before the vegetative rest period, i.e. normally within 45 days (of real growth) from emergence. On the other hand, excessive early development of biomass is undesirable as it reduces winter hardiness.
Spring varieties should be sown as early as practicable, when temperature, moisture and other soil conditions permit.

Plant density: sowing rates for 2-rowed types are within the range of 320 - 365 grains/m^2 (at a desired optimum ear density of 700 - 800 ears/m^2). With multiple-rowed winter barley the following model calculation may serve as a guide:

Expected yield = 9 t/ha; required ear density = 600 ears/m^2. With an estimated germination rate of 95 %, an overwintering rate of 85 % and 2.7 ears per plant, the seeding rate should be 280 grains/m^2 (see also 2.3 Wheat).

Temperature limitations and the duration of the various growth phases are illustrated in the following table

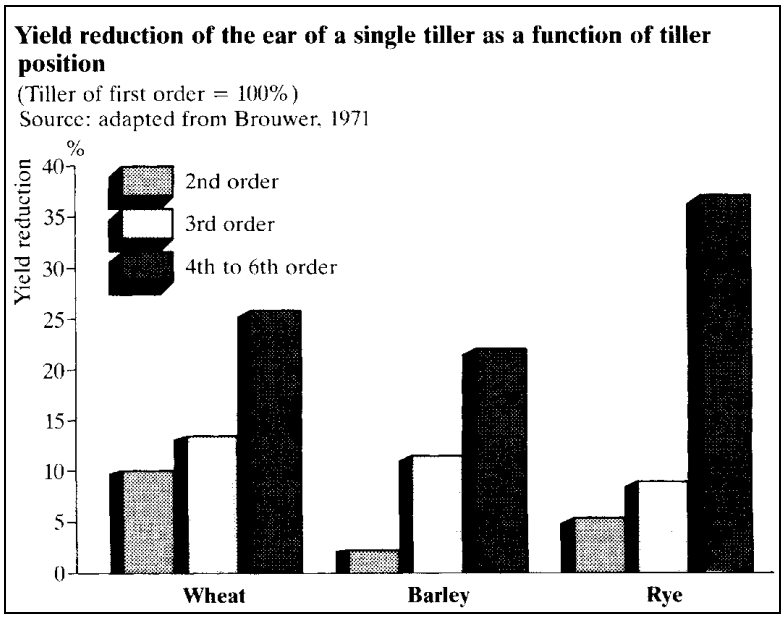
Growth stages and climatic limitations in the development of barley							
Development stage	**EC[1)]**	**Duration (days)**		**Temperature (°C)**			**Minimum water demand (mm)**
		Winter barley	**Spring barley**	**min.**	**Opt.**	**max.**	
Sowing and germination	0.1-0.9	7	7	2-4	20-25	27	
Seedling emergence and early growth	1.0-1.9	5	11				
Tillering, initiation of ear primordia	2.0-2.9	60[2)]	10		< 8		32
Beginning of stem elon gation and formation of ear primordia	3.0-3.9	17	19		< 9		52
Flag leaf, floret reduction, booting	4.0-4.9	14	14		< 14		
Ear emergence	5.0-5.9	16	24				70
Flowering and grain initiation	6.0-6.9	12	5		< 17		22
Grain formation	7.0-7.9	15	25		< 19		74
Maturing of the grain	8.0-9.2	19	21		19		
Total		165[2)]	136	1 700 - 2 100[3)]			
1) EC = Eucarpia Scale; 2) To be added: duration of vegetative rest (depending on location, e.g.: 120 days); 3) Total daily temperatures above 0° C (temperatures below 0° C deducted).							
Source: Aigner et al., 1988; modified							

Yield structure: The next table shows the (relative) changes of yield components in correlation to varying amounts of plant available water; assuming that water supply is the primary yield-determining factor in cereals:

Yield structure of barley as function of plant available precipitation*							
Winter barley	**Plant available precipitation**						
	150 mm	**250 mm**	**350 mm**	**450 mm**	**550 mm**	**650 mm**	**750 mm**
Grain yield	15	32	62	85	100 = 6.8 t/ha	113	119
Ear density	33	57	70	83	100 = 600/m2	117	125
Single ear weight	44	57	88	103	100 = 1.13 g	97	96
Optimal number of ears per plant = 2.5 - 3.5							
Spring barley	**Plant available precipitation**						
	100 mm	**150 mm**	**250 mm**	**350 mm**	**450 mm**	**550 mm**	**650 mm**
Grain yield	23	33	51	86	100 = 4.8 t/ha	112	116
Ear density	37	50	77	88	100 = 670/m2	112	125
Single ear weight	63	65	66	97	100 = 0.72 g	100	93
Optimal number of ears per plant = 2 - 3							
Relative to 550 mm plant available precipitation for winter barley, or to 450 mm for spring barley Plant available precipitation = amount solely available for crop growth, i.e. excluding evaporation, runoff, drainage and other losses.							
Source: Heyland, 1961							

The grain yield of barley is related to the amount of water consumption, which increases over-proportionally with increasing yield; the same is true of N uptake. If maximum utilization of water and applied nutrients is required for optimum grain yield, then the ratio of the number of plants per unit area to the number of ears per plant must be optimized; thus the crop should tiller heavily. This can be influenced, depending on water and N supply, by application of N. Depending on the quantity and timing of N application, around 250 l water per kg grain yield may be needed, the coefficient of productive tillering (ear-bearing tillers / total tillers) ranging between 0.39 and 0.60.

As shown in the figure it is not so important in barley as in wheat to control the uniformity of different orders of tillers. Unproductive tillering (caused for example by a too high or too late N fertilization in spring) should, however, be avoided.



Two-rowed barley varieties have lower ear weights, consequently a higher ear density than with multiple-rowed types is necessary to reach the same yield. Higher seeding rates and productive tillering promoted by N-fertilizer use are therefore necessary with two-rowed types (e.g. 700 - 800 ears/m^2 as compared to 550 - 600 ears/m^2 on fertile soils with ample water supply). With multiple-row types the target should be about three ears per plant, and with spring-sown two-row types two ears per plant. Because of the importance of floret reduction and grain formation in two-row types, an adequate nutrient supply must be ensured during shooting and after flowering.

When the crop is grown for malting, a variety should be selected which is appropriate to the expected wheather conditions.

If the crop is undersown, the undersown crop should not be grown so early that it might outgrow the barley and reduce resistance to lodging.

Nutrient demand/uptake/removal

Relative nutrient uptake of barley in relation to plant development									
(max. = 100)		Winter barley				Spring barley			
Stage	EC*	N	P2O5	K2O	Dry matter	N	P2O5	K2O	Dry matter
		per cent of maximum							
Early growth	1.0-1.9	0	0	0	1	0	0	0	1
Tillering	2.0-2.9	27	20	24	10**	25	18	24	6
Jointing	3.0-3.9	42	29	33	9**	37	27	36	11
Booting	4.0-4.9	59	45	51	13	53	42	62	19
Ear emergence	5.0-5.9	82	71	88	53	77	70	97	47
Flowering	6.0-6.9	100	88	100	79	100	91	100	71
Grain formation	7.0-7.9	97	100	79	100	100	100	97	100
Physiological maturity	8.0-9.2								
- total plant		97	100	76	91	96	97	88	87
- grain only		63	86	28	62	71	79	29	53
Maximum		kg/ha							
- in total dry matter		119	51	217	11 000	102	33	139	9 000
- in grain only		75	44	60	6 800	72	26	40	4 800
* EC = Eucarpia Scale ** Decrease due to loss of biomass during winter									
Source: various sources and own experimental results									

Nutrient uptake/removal - Macronutrients				
Type of barley	Yield base t/ha	kg/ha		
		N	P2O5	K2O
Winter barley	Total dry matter: 11	119	51	217
	Grain only: 6.8	75	44	60
Spring barley	Total dry matter: 9	102	33	139
	Grain only: 4.8	72	26	40
Source: adapted from Heyland, 1961				

Fertilizer recommendations

The same principles apply as for wheat, but the exact timing of split applications of N is more critical, especially for winter barley.

Since the root system of barley is less readily established than that of wheat, attempts have been made (in the UK) to promote longitudinal root growth by autumn application of clormequat chloride to young plants; but the efficacy of this method has not yet been fully confirmed.

Owing to the greater tendency of barley to lodge, as compared with wheat, stem stabilizers are being used in intensive growing systems. As chlormequat by itself does not give sufficient reduction in stem length, a combination of chlormequat chloride and etephon is favoured, with etephon alone being used for late applications.

Preferred forms of fertilizer nutrients

N - Quick acting forms are preferred for malting barley; forms releasing N too late in the growing period should be avoided in order to minimize the risk of too high a crude protein content in the grain. Slurry in particular can cause serious problems; if used at all for malting barley, only small amounts should be applied. Too late application of N fertilizer should be avoided.

P - a good supply is especially important for malting barley.

K - potassium chloride. As for P, adequate K should always be available. In general, K increases lodging-resistance and frost-hardiness.

Mg - barley reacts intensively to deficiency, producing leaf chlorosis; even though this may not always significantly reduce yield, it should be corrected by application of adequate Mg-containing fertilizers.

For further details refer to 'Fertilizer recommendations' and 'Calculation of nutrient rates' for 2.3 Wheat.

Generally, for the application of P and K three methods are practised:
- application in autumn on the stubble of the preceding crop, or with autumn ploughing;
- application in spring (in form of a NPK complex fertilizer) with the first N;
- for barley following sugarbeet or maize in the crop rotation, P and K are given to the preceding crop, the barley receiving only N.

On soils with satisfactory reserves of P and K, the applied nutrient rates depend on the expected grain yield.

	Winter barley (yield expectation 7 t/ha grain)	**Spring barley (yield expectation 6 t/ha grain)**
N	100 kg/ha N* + 40 kg/ha N as late topdressing	80 kg/ha N** 40 kg/ha N as late topdressing
P	120 kg/ha P2O5	100 kg/ha P2O5
K	120 kg/ha K2O	100 kg/ha K2O
* Timing and splitting as for winter wheat ** For malting barley: 40 - 50 kg/ha N, no N topdressing.		

Fertilizer practice in other countries

Canada

Most barley produced in Canada is spring seeded. It is grown both for feed and malting purposes, with small amounts used for human consumption.

Fertilizer recommendations vary from region to region, depending on environmental conditions. Soil testing is recommended for specific fertilizer recommendations.

In drier areas, band application of N is generally recommended. Limited amounts of N may be placed with the seeds. In the moister areas, spring broadcast and incorporated fertilizer applications are quite efficient. In all areas P should be seed-placed or banded with N. K is also more efficiently used if banded than if broadcast. Limited amounts (less than approx. 35 kg/ha K2O) can be placed with the seeds.

Alberta

Crop - Soil zone	kg/ha					
	N		**P2O5**		**K2O**	**S**
	Stubble	Fallow	Stubble	Fallow		
Feed barley						
- Brown	22- 62	5-22	0-28	17-40	-	-
- Dark Brown	40- 73	5-17	0-28	17-40	-	
- Thin Black	40- 90	5-34	17-40	17-40	-	
- Black & Grey Wodded	45-112	5-45	17-50	17-45	(*)	0-30**
Malting barley						
- Brown	-	-	-	-	-	-
- Dark Brown	28- 56	5-17	0-28	17-40	-	-
- Thin Black	40- 73	5-28	17-40	17-40	-	-
- Black & Grey Wooded	34- 90	5-40	17-45	17-45	(*)	0-30**

* Potassium or chloride may be required on some sandy, calcareous soils and soils with poor subsurface drainage adjacent to and on organic soils.
** Sulphur deficiency may occur on some wooded soils.

Manitoba

Barley following	kg/ha			
	N	**P2O5**	**K2O**	**S**
Fallow or legume	0- 30	30-45	(15-35)*	(15)**
Grass or grass-legume	30- 60	30-45	(15-35)*	(15)
Cereal	60-100	30-45	(15-35)*	(15)

* On sands, sandy loams and organic soils only
** When required, as sulphate

Sasketchewan

Soil zone	kg/ha				
	N		**P2O5**	**K2O**	**S**
	Stubble	**Fallow**			
Brown	15- 45	0-10	20-30	-	-
Dark Brown	20- 65	0-15	20-40	-	-
Black	50- 95	15-60	20-40	-	-
Dark Grey	55-100	20-65	20-40	0-40	0-20
Grey	55-105	20-65	20-40	0-40	10-20
Irrigated	75-120	50-80	20-40	-	-

India

- irrigated:
  60 kg/ha N, 30 kg/ha P2O5
  Half of the N and all P before or at sowing, the remaining N topdressed at the first irrigation.

- rainfed: 30 kg/ha N, 20 kg/ha P2O5
  All N and P before or at sowing.

P should be placed 5 cm below the seed; application rates are adjusted according to soil test results.

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Maize *(Zea mays L.)*

USA: Corn; French: Mais; Spanish: Maiz; Italian: Mais; German: Mais

Crop data

Annual. Harvested part: grain, used for human and livestock consumption. Lesser amounts are grown for harvest of the entire above-ground plants at physiological maturity to be made into silage for animal feed. In some areas, after the grain has been harvested the remainder is cut and used for animal feed. Yellow dent is primarily used for livestock feed, white dent is used primarily for production of meal and cereals for human consumption. Other dent lines have been bred for special purposes: e.g. waxy maize for production of amylopectin starch, high lysine maize for use in pig feed, high oil maize for production of vegetable oil for human consumption. Flint corns are grown in Central and South America, Asia and Southern Europe. Sweet corn was developed to be harvested in an immature stage for human consumption. Popcorn is used primarily for human consumption as freshly popped or other snack food items.

Adapted to a wide range of climates, the crop is mostly grown between latitudes 30° and 55°, principally in latitudes below 47°. In most areas it is sown in early spring but, due to its wide adaptation, it flowers at different times depending on the cultivar selected. Developed to take advantage of the length of growing season available, some will mature as soon as 60 days after emergence while others require over 40 weeks.

Plant densities vary considerably around the world, depending on cultivar and climate. In the more arid areas, densities as low as 15 000/ha can be found and 25 000/ha are common, but in humid or irrigated areas populations in excess of 75 000/ha give optimum production.

Spacing between rows ranges from 50 cm to 100 cm.

The crop will do well on any soil with adequate drainage to allow for the maintenance of sufficient oxygen for good root growth and activity, and enough water-holding capacity to provide adequate moisture throughout the growing season. Preferred pH 6.0-7.2. The amount of associated evapotranspiration varies with plant density, crop age, available soil water, atmospheric conditions, etc., from an estimated 0.20-0.25 cm/day for young plants to 0.48 cm/day for plants in the reproductive phase (Shaw 1977).

It is a warm weather crop, doing best when temperatures in the warm months range from 21° to 27° C. It does not do well when the mean summer temperature is below 19° C.

Nutrient removal - Macronutrients and Cl									
Yield t/ha	Source	Part	kg/ha						
			N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**	**Cl**
9.5	Barber & Olson, 1968	Grain	129	71	47	18	2.1	12	4.5
		Stover	62	18	188	55	55	9	76.0
6.3	Aldrich et al, 1986	Grain	100	40	29	9.3	1.5	7.8	-
		Stover	63	23	92	28	15	9	-

Nutrient removal - Micronutrients								
Yield t/ha	Source	Part	g/ha					
			Fe	**Mn**	**Cu**	**Zn**	**B**	**Mo**
9.5	Barber & Olson, 1968	Grain	110	60	20	190	50	6
		Stover	2020	280	90	190	140	3
6.3	Aldrich et al, 1986	Grain	-	70	40	110	-	-
		Stover	-	940	30	200	-	-

Plant nutrient requirements of maize (yielding 11.8 t/ha) during the growing season			
Plant age (days)	Nutrients absorbed (kg/ha/day)		
	N	**P2O5**	**K2O**
20-30	1.7	0.39	1.7
30-40	6.7	1.55	9.95
40-50	8.3	2.32	11.56
50-60	5.3	2.06	4.42
Source: Spies, 1973			

Plant analytical data

Critical plant nutrient levels in leaf opposite and below the ear at tasseling										
% of dry matter						ppm dry matter				
N	**P**	**K**	**Mg**	**Ca**	**S**	**Zn**	**Fe**	**Mn**	**Cu**	**B**
2.9	0.25	1.9	0.15	0.4	0.15	15	25	15	5	10
Source: Hoeft & Peck, 1991										

Published critical values found in various maize plant tissues - Macronutrients							
Plant part	Stage of growth	Source	% of dry matter				
			N	**P**	**K**	**Mg**	**Ca**
Ear leaf	Tasseling	Melsted et al, 1969	3.0	0.25	1.9	0.25	0.4
Ear leaf	Silking	Hanway et al, 1962			2.0		
6th leaf fom base	Silking	Tyner, 1946	2.9	0.3	1.3		
Leaves	Tasseling	Gallo et al, 1968	2.9	0.23	1.7-2.7		
Lower stems	Late summer	Goodall & Gregory, 1947		0.004	0.08		
Source: Jones & Eck, 1973							

Published critical values found in various maize plant tissues- Micronutrients								
Plant part	Stage of growth	Source	ppm dry matter					
			Fe	**Mn**	**Zn**	**Cu**	**B**	**Mo**
Ear leaf	Tasseling	Melsted et al, 1969	15	15	15	5	10	
Ear leaf	Tasseling	de L. Beyers & Coetzer, 1969			15			
Upper leaves	Not given	Berger et al, 1969					12	
Stems	Not given	Dios & Broyer, 1965						0.11
Source: Jones & Eck, 1973								

Fertilizer recommendations

Each of the following factors must be carefully integrated when determining the optimum rate for a particular field or part of a field:

- Inherent soil fertility.
Best determined by carefully calibrated and correlated soil test procedures for the particular soil type.

For N, soil sampling and analysis must be done annually, but even then the soil tests may not be reliable under high rainfall conditions, especially with sandy or very poorly drained soils. For relatively immobile nutrients like P, K and Mg, most research has shown that soil analysis conducted on samples collected every 3-4 years will be adequate. For micronutrients, the available soil test procedures have not been shown to be highly reliable and it is suggested that plant analysis should be used in conjunction with soil tests; deficiencies of many of the micronutrients are limited to soils with pH > 7.3.

- Yield potential.
Previous 5-year average yield plus 5 % is suggested as a basis for estimation, but, if fertilizer is applied post-emergence, adjustment may be made for delay in planting or inadequate stand.

- Previous crop.
Research has shown that maize will produce better when grown in rotation with another crop, especially a legume, probably as a result of diminished incidence of pests and diseases, reduction of the negative effect of continuous maize cropping, and a contribution of N from the legume. While much of the disadvantage of growing maize after maize can be overcome by applying N fertilizer, it is not possible to apply enough N to eliminate completely the yield difference between rotational and continuous maize.

- Soil reaction.
Maximum microbial activity for the release of essential nutrients from organic materials occurs at pH 6.0-7.0 and maximum availability of fertilizer nutrients is maintained. In most areas of the world, agricultural limestone is available to correct acid soils but economic means are often lacking for reducing the pH of alkaline soils.

- Timing.
Uptake of over half the N and P and 80 % of the K is accomplished before the crop reaches the reproductive stage. It is therefore imperative that an adequate supply of these major nutrients be available to the plants early and remain available throughout the growing season. Even though only small amounts are taken up early in the season, high

concentrations should be available in the root zone as the root system is small at that time and the soil is often cold.

N must be applied annually. Since it is subject to loss by leaching or denitrification, it is best applied near the time of crop need. On finer textured soils, silt loam or heavier, it is best applied as a pre-plant or side dressing; on coarse textured soils where leaching can be a problem, N is best applied as a side dressing or split application. If the crop is irrigated, 50-60 % pre-plant plus the rest through the irrigation system is an effective technique.

There is much greater flexibility in the time of application of P and K since they are relatively immobile. On many soils they may be broadcast either in the fall or spring with similar results, except on sandy soils where there is a possibility that the K might be leached out of the rooting zone (K should then be applied just before planting).

- Method of application.
On fields where the soil fertility status is at or above the desired level, there is little evidence to show any significant difference in yield associated with different methods of application. In contrast, on soils with a low nutrient status or a high P-fixing capacity, placement of the fertilizer within a concentrated band has been shown to result in higher yields, particularly at low rates of application. On higher-testing soils, although yield differences are unlikely, plant recovery of fertilizer nutrients in the year of application will usually be greater from a band placed 5 cm to the side and 5 cm below the seed than when broadcast.

Placement of fertilizer directly with the seed is sometimes referred to as "pop-up", but this is a misnomer as the crop does not emerge any sooner and may indeed emerge 1-2 days later than without such application. If used, pop-up fertilizer should contain all three major nutrients in the proportions N:P2O5:K2O=1:4:2. Under normal moisture conditions, the maximum safe amount of N + K2O for placement directly with the seed is 12-15 kg/ha in 100 cm rows and correspondingly more in closer rows. In excessively dry springs, even these low rates may result in reduced germination and/or damage to seedlings.

Forms of fertilizer

N: a mixture of ammonium and nitrate forms seems to be preferred, though all commonly available forms will perform well if properly applied. Within the USA Corn Belt, anhydrous ammonia is the most frequently used material; this must be injected to a depth of 15-20 cm to provide an adequate seal to prevent volatilization. Both urea and urea-AN solutions are excellent but, since the urea contained in these materials will quickly convert to ammonia, they should be incorporated in the soil within 3 days of application to prevent loss by volatilization. Although ammonium nitrate in solid form is also a satisfactory N source for maize, there is a potential for loss by leaching or denitrification if too much rain falls soon after application.

P: soluble sources including triple superphosphate or ammoniated phosphates perform well. Rock phosphate is not particularly effective since the crop grows best at pH > 5.7.

K: potassium chloride is the most economic source unless the soil is deficient in sulphur, in which case potassium sulphate can be used to provide both K and S.

S: precipitation carries with it not only the water essential for plant growth but also in many areas of the world a significant concentration of sulphur; values reported in the literature vary from 2 to 220 kg/ha S annually (the larger values in urban or industrial areas). Since a high-yielding maize crop removes less than 15 kg/ha, S in precipitation is often enough to meet crop need.

Micronutrients: while problems are not widespread, the most common are with Zn, Fe and Mn. These are most likely to be deficient where the pH is high; in that case, use of a chelate in a starter band or as foliar spray is the most effective treatment. Where Zn deficiency occurs on neutral to acid soils, an inorganic source such as zinc sulphate is an effective corrective.

Agricultural and industrial wastes: the value of those which are effective sources of nutrients for maize will depend on time and method of application. By-products containing free ammonia or compounds that will convert to ammonia must be incorporated into the soil soon after application to obtain maximum benefit. Since most of the nutrients will be in organic form, the amounts of the individual nutrients that will be available for the particular crop should be estimated from available data on their relative release rates. With industrial by-products, care must be taken to avoid applying excessive amounts of toxic metals.

Present fertilizer recommendations/practices

USA

N: recommendations throughout the eastern part of the Corn Belt and NE and SE USA are based on yield potential (t/ha) times 18 to 21 minus corrections for previous crop or application of organic manure. For maize after soybeans the correction is 45 kg/ha N and for maize after lucerne (alfalfa) approx. 100 kg/ha N. Procedures are being developed in the NE to modify the N recommendation according to the nitrate N concentration in the top 30 cm of soil collected at the 6th leaf stage of maize. Recommendations in the western part of the Corn Belt are based on soil nitrate levels to a depth of 60-120 cm.

P and K: recommendations are based on soil tests. If the test level is below that desired, enough is recommended to bring it up to the desired level and replace expected removal. If it is at the desired level, replacement is suggested at 7.7 kg P2O5 and 5.0 kg K2O per ton of expected yield. If the test level is above a level at which response is no longer expected, then no P or K is recommended until the field is retested 4 years later. In the eastern part of the region many producers have increased P levels in their soils to the point where additional P is not routinely needed. Soils in the western part of the region are inherently high in K, so fertilizer K is seldom recommended.

- Average fertilizer consumption on maize in the Corn Belt is, in the eastern part, 170 kg/ha N, 84 kg/ha P2O5, 78 kg/ha K2O, and in the western part 145 kg/ha N, 54 kg/ha P2O5, 56 kg/ha K2O. Farmers in NE USA use about 5-10 % less commercial fertilizer than those in the eastern part of the Corn Belt, because they have a lower yield potential and, being in a grain-deficient area which imports grain for livestock, they have more livestock waste available per unit of land area. Producers in SE USA, despite their somewhat lower yield potential, use about the same or slightly higher rates of commercial fertilizer than in the Corn Belt because their soils are more highly weathered and thus have less inherent fertility.

- About 47 % of total N supplied is in the form of anhydrous ammonia, 27 % as urea-AN solutions and urea 20 %. Well over 80 % of the total P is applied as an ammoniated material and virtually all the K is in the form of muriate.

- Where fall tillage is practised, P and K are broadcast before tillage; where tillage is in spring, fertilizer application is usually in late winter or early spring. In many areas, particularly where maize is sown in cool soil or where inherent fertility is low, some of the P and K is applied at sowing as a band application below and to the side of the seed. N is applied pre-plant in spring to a very high proportion of the area but is applied post-emergence where loss potential is high. Under irrigation, producers often apply 60 % of N pre-plant and the

remainder in successive applications of 20-25 kg/ha N through the irrigation system, so that all the N will have been applied by 2 weeks after pollination.

France

N: Rate of application (X) is determined as a function of predicted yield (Y) and the availability of N from the soil (FS), all as kg/ha, according to the equation:

X = 0.23 Y - FS.

In southern France the rate of application is set at 0.18 times the expected yield; in other regions the rate, when FS is not known, should not exceed 0.2 Y. The rate recommended for fields which regularly receive organic manure, as in the west and northeast parts of the country, is 50 kg/ha N. In Lorraine and Poitou-Charentes the recommended rate depends on soil type, previous crop and previous N content of the soil.

In southern France, one third of the N is applied at sowing and the rest at the 6-8 leaves stage. In the north, N is usually applied pre-plant.

P and K: Recommendations are based on soil test results. When these are not available, 100 kg/ha of each is usually adequate for high yield.

Both P and K are usually applied before sowing. In some areas, 100 kg/ha 18-46-0 starter is applied at sowing.

Zn and Cu: Applied before sowing to deficient soils at 10 kg/ha Zn and/or 6 kg/ha Cu.

Brazil

N: 50-90 kg/ha, mostly post-emergence.

P: 50-80 kg/ha P2O5, all at sowing.

K: 30-60 kg/ha K2O, all at sowing.

Preferred forms of primary nutrients: urea, ammoniated phosphates, potassium chloride.

S: 20 kg/ha S.

Zn: 5 kg/ha Zn.

S.E.Asia

Nutrient rates vary substantially between and within countries (see table). The higher rates are for hybrids and the lower for local varieties. Urea, ammoniated phosphates and potassium chloride are the preferred forms.

Country	kg/ha					
	Hybrids			**Local varieties**		
	N	P2O5	K2O	N	P2O5	K
Indonesia	120-180	45-60	30-60	45-90	30-45	0-30
Philippines	90-140	45-60	0-60	45-90	30	0
Thailand	45-120	45-60	0-60	30-45	30	0-30

India

N: 100-125 kg/ha N as urea, split into three applications.

P: 60 kg/ha P2O5, as processed phosphates applied mostly at sowing.

K: 30 kg/ha K2O, as potassium chloride applied mostly at sowing.

South Africa

N: Rates are calculated according to yield potential. 170 kg/ha, 45 kg/ha and 20 kg/ha N are recommended for yield levels of 8 t/ha, 3 t/ha and 2 t/ha respectively. N is applied within 8 weeks after planting, depending on rainfall distribution.

P: Rates are based on soil analysis (Bray 1, Bray 2, or Ambic 1) and yield potential (up to 300 kg/ha, 100 kg/ha or 45 kg/ha P2O5 for yields of 8 t/ha, 3t/ha or 2 t/ha respectively). At least a minimum of fresh P is applied each year (cold springs, P-fixing soils).

K: Rates are based on soil analysis (NH4OAc, Ambic 1), sub-soil K and clay content of the soil (up to 120 kg/ha, 45 kg/ha or 30 kg/ha for yields of 8 t/ha, 3 t/ha and 2 t/ha respectively).

Liming: Exchangeable acidity is reduced by liming to approx. 10 %.

Micronutrients: To prevent zinc deficiency, Zn-containing fertilizer mixtures are used on soils with < 1.5-2 ppm Zn. Because of acidic soils boron and molybdenum deficiency are common. Seed is usually treated with Mo; B-containing fertilizers or B-foliar sprays are used.

Further reading

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Oats *(Avena sativa L.)*

French: Avoine; Spanish: Avena; Italian: Avena; German: Hafer

Crop data

Annual, with autumn-sown (winter oats, in areas with mild winters) and spring-sown types.

Harvestable products: grain, straw and whole green plant.

Desired characteristics affecting fertilizer requirement:

In grain for flour or coarse meal for human consumption: high starch, protein and fat, absence of sprouting, no glumes. In grain for animal feed: high crude protein, low proportion of glumes.

Straw for animal feed: high cellulose, low lignin and ash. Straw for bedding: dry absorbent material.

Green plant for forage: high crude protein and energy contents. Green plant for silage: high content of easily soluble carbohydrates.

Oats for animal feed are preferably grown as mixed or dredge corn.

Varieties differ in water requirement, liability to micronutrient deficiency (particularly of Mn), disease resistance and proportion of glumes in the grain. The risk of lodging is greater than with wheat, especially because the use of plant growth regulators is not as safe as with wheat or barley.

Sowing time: Spring oats should be sown as early as possible at the beginning of the growing period. Winter oats should be sown sufficiently early to have at least three weeks for tillering before the onset of the vegetative rest period.

Seeding rate: for an expected yield of 5 t/ha grain, good conditions for emergence and varieties with high single panicle yield, 400 grains/m^2 are recommended.

Temperature limitations and the duration of the various growth phases are illustrated in the following table:

Growth stages and climatic limitations in the development of spring oats					
Stage	**EC[1)]**	**Duration (days)**	**Temperature (°C)**		
			min.	**opt.**	**max.**
Sowing and germination	0.1-0.9	13	3 - 5	25-30	30-37
Emergence and early growth	1.0-1.9	12			
Tillering, initiation of ear primordia	2.0-2.9	19	> -10[2)]	5	
Beginning of stem elongation, forming of ear primordia	3.0-3.9	19			
Flag leaf, floret reduction, booting	4.0-4.9	15			
Ear emergence	5.0-5.9	36			
Flowering and grain initiation	6.0-6.9	8	> 12	15-16	
Grain formation	7.0-7.9	40			
Maturing of the grain	8.0-9.2	5			
Total		167		2 100[3)]	
1) EC = Eucarpia Scale 2) Winter oats 3) Total daily temperatures above 0° C (temperatures below 0° C deducted)					
Source: Aigner et al., 1988; modified					

The next table shows the change in yield components relative to the amount of plant available precipitation. It should be noted that varieties differ in their water requirements and in harvest indices (i.e. proportions of grain in the total biomass at harvest, in terms of dry weight).

Yield structure of oats as function of plant available precipitation *					
	Plant available precipitation				
	350 mm	**450 mm**	**550 mm**	**650 mm**	**750 mm**
Grain yield	30	60	100 = 5.0 t/ha	120	138
Ear density	74	86	100 = 350/m2	114	120
Single ear weight	40	70	100 = 1.43 g	105	115
Optimal number of ears per plant = 1.5-2					
* Relative to 550 mm available precipitation. Plant available precipitation = amount solely available for crop growth, i.e. excluding evaporation, runoff, drainage and other losses.					
Source: various experimental results					

For spring oats the management of plant density should be similar to that of spring wheat (see “ wheat ”). For winter oats the management of plant density is similar to that of winter rye (see “ rye ”).

Nutrient demand/uptake/removal

Nutrient uptake/removal - Macronutrients				
Yield base	t/ha	kg/ha		
		N	**P2O5**	**K2O**
Biomass*: (DM)	9.8	134	50	235
Grain:	5.0	99	32	43
* Whole above-ground portion of plant				
Source: various experimental results				

Relative nutrient uptake of oats in relation to plant development				
Stage	**N**	**P2O5**	**K2O**	**Dry matter**
	per cent of maximum			
Early growth	0	0	0	1
Tillering	16	10	11	4
Jointing	31	14	19	13
Booting	34	20	31	33
Ear emergence	68	60	88	55
Flowering	85	74	100	77
Grain formation	97	100	98	100
Physiological maturity				
- Biomass(dry)	100	100	94	95
- Grain	74	64	18	51
Maximum (100 %)	**kg/ha**			
- Whole plant	134	50	235	9 800
- Grain only	99	32	43	5 000
Source: Aigner et al., 1988				

Fertilizer recommendations

The amount of nutrients to be applied depends on the expected yield. For a target of 3.5 t/ha an amount of 70 kg/ha N, 35 kg/ha P2O5 and 105 kg/ha K2O is needed. These figures depend on the location.

On sandy soils the extensive root system largely prevents loss of N by leaching and the available nutrients are much better utilized than, for example, by wheat. In areas where high yields (5 - 6t/ha) are possible, it is common to use a total of 11O kg/ha N minus N_{min}, including at least 40 kg/ha N in spring plus a late N application at booting of the panicles.

Since the crop needs 200 g Mn/t grain, adequate Mn is of special importance (apply 0.6 kg $MnSO_4$ per ton of grain = 1.9 kg/ha for the target yield of 3.5 t/ha, where required. For Cu deficiency, 50 - 100 g/ha copper sulphate should be applied; and for Mo deficiency, 0.8 kg/ha sodium molybdate. However, excess of Ni, Cu, Co, Mo or Zn may cause leaf chlorosis.

Further reading

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Rice *(Oryza sativa L.)*

French: Riz; Spanish: Arroz, Italian: Riso; German: Reis

On a global basis, rice ranks second only to wheat in terms of area harvested, but in terms of its importance as a food crop, rice provides more calories per ha than any other cereal food grain.

Crop data

Annual grass with round culms, flat leaves and terminal panicles. Varieties of growth duration ranging from 70 to 160 days exist in diverse environments. The grain is a caryopsis in which the single seed is fused with the wall which is the pericarp of the ripened ovary forming the grain which is the seed. Each rice panicle (which is a determinate inflorescence on the terminal shoot), when ripened, contains on average 80-120 grains, depending on varietal characteristics, environmental conditions and the level of crop management.

The bulk of the rice in Asia is grown during the wet season starting in June-July and dependence on rainfall is the most limiting production constraint for rainfed culture. Rice areas in South and Southeast Asia may, in general, be classified into irrigated, rainfed upland, rainfed shallow water lowland and rainfed deep water lowland areas.

Irrigated conditions being the most assured, the productivity of irrigated rice is highest, being in the range of 5-8 t/ha during the wet season and 7-10 t/ha during the dry season when very well managed. Though the average is often only in the range of 3-5 t/ha. The productivity of rainfed upland and deep water lowland rice, however, continues to be low and is static around 1.0 t/ha.

Seedlings 25-30 days old, grown in a nursery are usually transplanted at 20 x 15 or 20 x 10 or 15 x 15 cm spacing in a well prepared main field and normally this will have a population of 335 000 to 500 000 hills/ha (33 to 50 hills/m2), each hill containing 2-3 plants. Direct seeding is also practised. Being a crop that tillers, the primary tillers (branches) grow from the lowermost nodes of the transplanted seedlings and this will further give rise to secondary and tertiary tillers.

The floral organs are modified shoots consisting of a panicle on which are arranged a number of spikelets. Each spikelet bears a floret which, when fertilized, develops into a grain. A crop producing on average 300 panicles per m2 and 100 spikelets per panicle, with an average spikelet sterility of 15 % at maturity and a 1 000-grain weight of 20 g will have an expected yield of 5.1 t/ha.

Nutrient removal

Modern high-yielding varieties producing around 5 t/ha of grain, in general, can remove from the soil about 110 kg N, 34 kg P2O5, 156 kg K2O, 23 kg MgO, 20 kg CaO, 5 kg S, 2 kg Fe, 2 kg Mn, 200 g Zn, 150 g Cu, 150 g B, 250 kg Si and 25 kg Cl per ha. Removals of Si and K2O are particularly large if the panicles and straw are taken away from the field at harvest. However, if only the grains are removed and the straw is returned and incorporated back into the soil, the removal of Si and K2O is greatly reduced, although significant amounts of N and P2O5 are still removed.

Nutrient uptake/removal - Macronutrients (high yielding variety)						
Plant part	kg/t grain					
	N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
Straw	7.6	1.1	28.4	2.3	3.80	0.34
Grain	14.6	6.0	3.2	1.7	0.14	0.60
Total	22.2	7.1	31.6	4.0	3.94	0.94
Source: Calculated from S.K. De Datta, 1989 Data computed from the nutrient uptake data of rice var. IR 36 at a yield level of 9.8 t/ha of grain and 8.3 t/ha of straw (in the Philippines)						

Nutrient uptake/removal - Micronutrients (high yielding variety)							
Plant part	g/t grain					kg/t grain	
	Fe	**Mn**	**Zn**	**Cu**	**B**	**Si**	**Cl**
Straw	150	310	20	2	16	41.9	5.5
Grain	200	60	20	25	16	9.8	4.2
Total	350	370	40	27	32	51.7	9.7
Source: Calculated from S.K. De Datta, 1989 Data computed from the nutrient uptake data of rice var. IR 36 at an yield level of 9.8 t/ha of grain and 8.3 t/ha of straw (in the Philippines)							

Nutrient removal* - with and without N-fertilizer						
N-fertilizer applied	Yield t/ha	Plant part	kg/ha			
			N	**P2O5**	**K2O**	**S**
No N	Grain: 3.4	Straw	18	4.6	71	0.8
	Straw: 2.8	Grain	34	22.9	12	1.0
		Total	52	27.5	83	1.8
174 kg/ha N	Grain: 9.8	Straw	75	11.5	278	3.3
	Straw 8.2	Grain	143	59.5	31	4.9
		Total	218	71.0	309	8.2
* Rice variety IR 36 in farmer's field, Laguna, Philippines, 1983, dry season. Adapted from S.K. De Datta, 1987						

Nutrient uptake by rice-based cropping systems (India)				
Cropping system	Grain yield t/ha	kg/ha/year		
		N	**P2O5**	**K2O**
Rice-Rice	6.3	139	88	211
Rice-Wheat	8.8	235	92	336
Rice-Wheat Greengram	11.2	308	89	336

Modern high-yielding varieties, in general, remove nutrients to a greater extent than did their traditional counterparts in the past and such a rate of soil exhaustion can limit the long-term sustainability of rice production, unless the removals are replenished by supplementary application of fertilizers.

Soil analysis and critical nutrient level concept

Among the soil analysis techniques, determination of soil pH is the simplest and most informative analytical technique for diagnosing a nutrient deficiency or toxicity problem. The various soil and plant analysis methods for evaluating the N, P, K, S, Zn and Si availability to lowland rice on submerged soils have been extensively reviewed by Chang (1978).

The determination of available N by the waterlogged incubation and alkaline permanganate method, available P by the Olsen and Bray P1 methods, available K by exchangeable

potassium, available S with Ca $(H_2PO_4)_2.H_2O$, available Zn by extraction with buffered chelating agents or weak acids, and available Si by extraction with sodium acetate have been shown to record the best correlation with the response of rice to these elements.

Guidelines for N, P and K requirements based on soil analysis are:

Total N (%)	**N requirement**
< 0.1	high
0.1-0.2	moderate
> 0.2	low

Available N (ppm)	**N requirement**
50-100	high
100-200	moderate
> 200	low

Available P (Olsen, ppm)	**P requirement**
< 5	high
5-10	moderate
>10	low

Exchangeable K (me/100 g)	**K requirement**
> 0.2	low

The critical limits for micronutrients using soil analysis are presented below:

Critical deficiency levels in rice soils - Micronutrients			
Element	Method		Critical level (ppm)
B	Hot water		0.1 to 0.7
Cu	DTPA + $CaCl_2$	(pH 7.3)	0.2
Fe	DTPA + $CaCl_2$ $NH_4C_2H_3O_2$	(pH 7.3) (pH 4.8)	 2.5 to 4.5
Mn	DTPA + $CaCl_2$ 0.1 N H_2PO_4 and 3 N $NH_4H_2PO_4$	(pH 7.3)	1.0 15 to 20
Mo	$(NH_4)_2(C_2O_4)$	(pH 3.3)	0.04 to 0.2
Zn	0.5 N HCl Dithizone + $NH_4C_2H_3O_2$ EDTA + $(NH_4)_2CO_3$ DTPA + $CaCl_2$	 (pH 7.3)	1.5 0.3 to 2.2 1.5 0.5 to 0.8
Source: Adapted from S.K. De Datta, 1989			

- If the soil pH is more than 6.8, Zn-deficiency is most likely to occur, particularly so if the variety grown is not tolerant and efficient to utilize the available Zn.

Plant analysis data

Deficiency or toxicity symptoms usually occur when plants are young and so the whole plant samples are drawn at that stage for chemical analysis. However, identification of the exact stage of growth is very important in determining the critical limits. It is also to be remembered that the critical concentration of nutrients determined from greenhouse experiments often tends to be high and, as such, these values may not really be extrapolated and used for field crops. Tanaka and Yoshida (1970) recorded a list of critical concentrations of various

elements in the rice plant, which may be used as a rough guide for diagnostic purposes. However, whatever method is used, the correct correlation of the analytical data with yield data from reliable field experiments is decisive.

Plant analysis data (critical concentrations) - Macronutrients							
Plant part used for analysis	Growth stage	% of dry matter					
		N	**P**	**K**	**Mg**	**Ca**	**Si**
Leaf blade	Tillering	2.5 (D)*	0.1 (D)				
Straw	Maturity			1.0 (D)	0.10 (D)	0.15 (D)	5.0 (D)
* (D) = Deficiency							

Plant analysis data (critical concentration) - Micronutrients							
Plant part used for analysis	Growth stage	ppm dry matter					
		Fe	**Zn**	**Mn**	**B**	**Cu**	**Al**
Leaf blade	Tillering	(D)* 300(T)**					
Shoot	Tillering		10(D)	20(D) 2500(T)			300(T)
Straw	Maturity		1500(T)		3.4(D) 100(T)	6(D) 30(T)	
* (D) = Deficiency, ** (T) = Toxicity							

Nutrient absorption and translocation

A clear understanding of the different stages of growth and development of the crop and its nutritional requirements at these important stages is a pre-requisite for nutrient management.

In the case of N, the accumulation of N in the vegetative body is high during the initial growth stages and declines with age towards the later growth stages. Translocation of N from the vegetative organs to the grains becomes significant only after flowering. There is some translocation of carbohydrates from the vegetative plant parts to the grains after flowering and a large amount of carbohydrates acumulates in the grains. Protein synthesis is active during the vegetative stages and, during the reproductive stage, synthesis of cell wall substances (cellulose, lignin, etc.) becomes active, although the pace of protein synthesis also continues. It is only at the ripening stage that starch synthesis becomes active.

Nutrient mobility in the rice plant is in the sequence P > N > S > Mg > K > Ca. The elements that form immediate components of proteins have a high rate of mobility, while those that are continuously absorbed until senescence have a relatively low mobility. Thus, N, P and S, which are essential constituents of proteins, are absorbed rapidly during the active vegetative growth stage and are subsequently translocated to the grain after flowering. Other nutrients like Ca and K on the other hand, are absorbed at a rate matching the rate of dry matter production over the growth period.

Nutrient uptake at different growth stages

Based on temperate climate experience, Ishizuka (1965) has summarised nutrient uptake at different growth stages as follows:

- The percentage contents of N, P and K at the seedling stage increase progressively with growth and then decrease after reaching a maximum.
- The percentage of N in the plant decreases marginally after transplanting and then increases until the initiation of flowering. Subsequently the N content decreases continuously until the dough stage and then remains constant until ripening.
- The percentage of P declines rapidly after transplanting, then increases slowly and reaches a peak at flowering and then decreases until the dough stage.
- The percentage of K decreases gradually during the earlier growth of the plant but increases from flowering until ripening.
- Ca has a similar trend to K.
- The percentage of Mg is high from transplanting to the mid-tillering stage and then decreases gradually.
- The percentage of S decreases with growth.

Fertilizer management in rice

Rice farmers in India and most parts of South and Southeast Asia most commonly use N, P, K, S and Zn in the fertilizer schedule depending on soil types and seasonal conditions. Farmers in Japan, Korea and Taiwan also use silicon with advantage, although Si is not considered to be an essential element.

Since rice is pre-dominantly grown under wetland conditions, it is important to understand the unique properties of flooded soils for better management of fertilizers for this crop. When a soil is flooded, the following major chemical and electrochemical changes take place:

i) depletion of molecular oxygen,
ii) chemical reduction of soil,
iii) increase in pH of acid soils and decrease in pH of calcareous and sodic soils,
iv) increase in specific conductance,
v) reduction of Fe^{3+} to Fe^{2+} and Mn^{4+} to Mn^{2+},
vi) reduction of NO_3^- to NO_2^-, N_2 and N_2O,
vii) reduction of SO_4^{2-} to S^{2-},
viii) increase in supply and availability of N, P, Si and Mo,
ix) decrease in concentrations of water-soluble Zn and Cu, and
x) generation of CO_2, methane and toxic reduction products such as organic acids and hydrogen sulphide. These will have a profound influence on soil nutrient transformations and availability to rice plants.

The table below gives a broad indication of NPK recommendations for lowland rice in different countries:

Recommended/optimum levels of NPK for lowland rice (selected countries)					
Country	Region	Référence	Recommended/optimum levels (kg/ha)		
			N	**P2O5**	**K2O**
Bangladesh	Hathazari	Amin & Amin, 1990	80	28	17
Bhutan	Wangdipho-drang	Chettri et al, 1988	75	50	0
Egypt		Elgabaly, 1978	100	37	0
India	Haryana	Sharma et al, 1988	125	26	50
	Pattambi, Keraba	Alexander et al,1988	90	45	45
Indonesia	Lampang- Dry season Wet season West Java	Palmer et al, 1990	 140 80 115	 35 18 25	 30 30 40
Japan	Hyogo Prefecture	Sudo et al, 1984	170	122	170
Malaysia	MUDA	Jegatheesan, 1987	80	30	30
Pakistan	Muridke D.I. Khan Dist.	Zia, 1987 Gurmani et al, 1984	120 135	26 40	0 37
Philippines	Nueva Ecija Guadalupe, Laguna Tarlac	Aganon, 1987 UPCA, 1970	90 100 80	28 30 50	28 0 30
Sri Lanka		Balasuriya, 1987	73	58	58

Nitrogen

In lowland rice losses of applied N take place through: a) ammonia volatilization, b) denitrification, c) leaching, and d) runoff. The recovery of fertilizer N applied to rice seldom exceeds 30-40 %. Fertilizer N use efficiency in lowland rice may be maximized through a better timing of application to coincide with the stages of peak requirement of the crop, and placement of N fertilizer in the soil. Other possibilities, though much more costly, are the use of controlled-release N fertilizer or of urease and nitrification inhibitors, and finally, the exploitation of varietal differences in efficiency of N utilization.

Some general guidelines for efficient N management in rice	
Situation	Strategy
Upland (dryland)	Broadcast and mix basal dressing in top 5 cm of surface soil Incorporate topdressed fertilizer by hoeing-in between plant rows and then apply light irrigation, if available
Rainfed deep water	Apply full amount as basal dressing.
Lowland (submerged)	Use non-nitrate sources for basal dressing.
Soil very poor in N	Give relatively more N at planting.
Assured water supply	Can topdress every 3 weeks up to panicle initiation. Drain field before topdressing and reflood two days later.
Permeable soils	Emphasis on increasing number of split applications.
Short duration varieties	More basal N and early topdressing preferred.
Long duration varieties	Increased number of topdressing.
Colder growing season	Less basal N and more as topdressing.
Over aged seedlings used	More N at planting

N recommendations for dryland (rainfed upland) rice (India)		
Region	kg/ha	N application
Ranchi	60	in 3-4 split dressings
Bhubaneswar	75-90	in 3 split dressings
Dehradun	80	in 3 split dressings
Varanasi	80	50 % basal, 50 % 30-40 days after sowing
Source: AICRPDA, 1983		

General recommendations for N application to high yielding and improved varieties of lowland rice (India)		
State	kg/ha N	Application information
Andhra Pradesh	60-100	Rate depends on region, application in 2-3 split dressings.
Assam	60	For dry season rice: 1/3 basal, 1/3 at tillering, 1/3 at panicle initiation.
Haryana	120-150	1/3 basal, 1/3 3 WAP, 1/3 6 WAP.
Kerala	70- 90	For wetlands: Number of applications depends on varietal duration and soil type.
Karnataka	100	50 % basal, 25 % 25-30 DAP, 25 % at PI.
Meghalaya	60	Transplanted: 25 % basal, 50 % at tillering, 25 % at PI. Direct seeded: 1/3 20 DAS, 1/3 40 DAS, 1/3 at PI.
Orissa	50- 75	Medium and lowland; in 3 splits.
Punjab	125-150	1/3 basal, 1/3 3 WAP, 1/3 6 WAP.
Tamil Nadu	75-100	Rate depends on varietal duration. 50 % basal, 50 % topdressed.
Uttar Pradesh	100	50 % basal, 25 % tillering, 25 % at PI.
West Bengal	40-120	Rate depends on variety and soil test, in 2-3 applications.
Source: Compiled by Tandon, 1989		

Recommended rates of N for wet rice production (Taiwan)			
Variety	Region*	N rate** (kg/ha)	
		1st crop	**2nd crop**
Japonica (medium height)	CSE N	120-150 110-130	100-130 100-120
Japonica (in well drained soil)	CSE N	180-210 150	170-200 140
Indica (dwarf)	CSE	150-180	125-160
* C, S, E and N denote central, southern, eastern and northern Taiwan respectively. ** N rate should be raised by 10-20 % in direct-seeded culture, and by 20-40 kg/ha in calcareous soils. Conversely, it should¦ be decreased by 20-40 kg/ha in strongly acidic soils.			
Source: Lian, 1989			

During the recent past, a number of recommendations have evolved for increasing the efficiency of N use through non-conventional means, e.g. use of neem cake coated urea, urea supergranules (USG), soil-cured urea, etc. USG is recommended in Karnataka, Meghalaya and Orissa for root zone placement and for improving N-use efficiency in rice, although the material is yet to be commercially available. For neem-cake coated urea, 20-25 kg neem oil cake is needed per 100 kg urea.

Phosphorus

Although P availability generally increases in submerged soils, a significant response of lowland rice to P application is common in ultisols, oxisols, sulfaquept, andosol and some vertisols with high P fixation capacities. The response is higher during the cooler months of the dry season crop. Application of 60 kg P2O5 increases grain yield of rice in India, on average, by 0.50-0.75 t/ha (Pillai et al., 1985). In dryland rice, however, P use is particularly important and remunerative.

Specific P recommendations are usually based on soil-test values of available P. The soil-test based P recommendations may further be specified for different target yield levels. One such example is set out below.

P recommendations based on soil tests and target yields			
Target yield (t/ha)	kg/ha P2O5 recommended at different levels of Olsen P (kg/ha)		
	10	20	30
4.5	52	32	16
5.0	62	42	26
Source: Perumal Rani et al., 1985			

In wetland rice, the full rate of P fertilizer is generally recommended to be surface broadcast and lightly incorporated into the soil before planting, but some recommendations for split application are now emerging. In Haryana, India, it is recommended that P may also be profitably applied in two equal dressings - at puddling and 3 weeks after transplanting. In Bangalore, Kolar, Tumkur, Mandya and Mysore districts of Karnataka, India, application of 50 % at planting and 50 % at tillering is being recommended.

For upland (dryland) rice, the standard recommendation is to drill the full rate of P at or before sowing.

Factors like soil texture, P fertility status, seasonal conditions and duration of the variety are often taken into consideration. Examples are given below:

P recommendations for irrigated rice in Warangal District, India			
Variety	Season	P2O5 recommended kg/ha	
		Light soil	**Heavy soil**
HYV	Wet	50	<60
	Dry	45	60
Local	Wet	30	40
	Dry	40	50
Dept. of Agriculture, Hyderabad, 1985			

Recommended rates of P for wet rice production (Taiwan)			
Available P (Bray I)		P2O5 rate, kg/ha	
ppm	**Rating**	**1st crop**	**2nd crop**
0- 4	Very low	70-80	50-60
5-10	Low	60-70	40-50
11-20	Medium	40-60	20-40
21-50	High	20-40	0-30
Over 50	Very high	0-30	0-20
Source: Lian, 1989			

Potassium

Grain yield response to K is higher in dry season than in wet season rice. In India, average responses of 10 kg grain/kg K2O dry season and 8 kg grain/kg K2O wet season have been recorded in cultivators' fields (Pillai et al., 1985).

Although broadcasting and incorporating the whole K application at the time of puddling (before planting) is generally recommended, split application is also common in some areas.

General K recommendations (India)	
State	**Recommendation**
Andhra Pradesh	30-45 kg K2O/ha in K deficient and light soils.
Kerala	30-45 kg K2O/ha depending on variety and water regime.
Orissa	20-40 kg K2O/ha depending on variety and soil fertility.
Punjab	30 kg K2O/ha.
Tamil Nadu	38 kg K2O/ha for short duration varieties, 50 kg K2O/ha for medium-long duration varieties.
West Bengal	0-60 kg K2O/ha depending on variety, season and soil test.
Source: FAI, 1981	

Recommended rates of K for wet rice production (Taiwan)			
Available K (Mehlich's)		K2O rate*, kg/ha	
ppm	**Rating**	**1st crop**	**2nd crop**
0-15	Very low	60-70	80-90
16-30	Low	50-60	60-80
31-50	Medium	30-50	40-60
Over 50	High	0-30	0-40
* Plus 30 kg/ha K2O in the case of poorly drained soils.			
Source: Lian, 1989			

K recommendations for rice (Tamil Nadu, India), using the "per cent yield concept"			
District	Recommendations, kg/ha K2O for 87.5 % of the max. yield		
	Soil test	**Soil test**	**Soil test**
	160 kg/ha	**258 kg/ha**	**381 kg/ha**
	K2O	**K2O**	**K2O**
North Arcot	69	49	25
Ramnad	88	54	12
Tirunelveli	145	111	66
Source: Mosi & Lakshminarayana, 1987			

K application schedule (Kerala state, India) - example of split application of K		
Region	**Situation**	**Recommendation**
Kuttanad and Onattukara	Transplanted medium-late duration	50 % K before planting, 50 % 5-7 days before panicle initiation.
Wynad and hilly region	Transplanted long duration	50 % K one month after planting, 50 % before flowering.
Wynad and hilly region	Direct seeded	50 % 1 1/2 months after planting plus 50 % at panicle initiation.
Source: Kerala Agric. University, Package of Practices, 1986.		

Preferred nutrient forms

In the anaerobic environment of lowland rice soils, the only stable mineral form of N is NH4. Nitrate (NO3) forms of N, if applied, will enter the anaerobic zone and be subjected to heavy denitrification losses. At planting time, the basal dressing of N should never be supplied as nitrate. For topdressing the growing plants, however, NH4 and NO3 forms may be used with almost equal efficiency. Fully established rice can rapidly take up applied NO3 before it is leached down to the anerobic soil layer and can become denitrified.

Water soluble P sources such as single superphosphate (SSP) and diammonium phosphate (DAP) are generally recommended in neutral to alkaline soils. Water insoluble sources like rock phosphates are recommended for acid soils where they are effective at low pH values.

Muriate of potash is the most commonly used source of K.

Secondary and micronutrients

S deficiency has been reported from Bangladesh, Burma, Brazil, Indonesia, India, Nigeria, Philippines and Thailand (Jones et al., 1982). In Bangladesh, 20 kg/ha S is generally recommended in the form of gypsum for dry season rice, the residual effect of which can often meet the S requirement of the succeeding wet season rice crop. In Bangladesh, application of S along with NPK increases the grain yield by 30-79 % above that obtained by using NPK fertilizers alone (Bhuiyan & Islam, 1989). In India, although S is yet to be

introduced to the regular fertilizer schedule for rice, researchers have suggested application of 30 kg/ha S per crop at Delhi and 44 kg/ha S per two crops at Bhubaneswar, Orissa. In general, application of S containing fertilizers is advocated during the final land preparation.

Yield response of lowland rice to sulphur application (India)		
Location/soil type	**Grain yield response to sulphur (averaged over three years)**	
	kg/ha	% increase
Barrackpore (alluvial soils)	415	12
Hyderabad (red soils)	213	9
Bhubaneswar (lateritic soils)	372	12
Pantnagar (terai soils)	308	6
Source: Compilation of Tandon 1989		

Zn deficiency is the most widespread micronutrient disorder in lowland rice and application of Zn along with NPK fertilizer increases the grain yield dramatically in most cases:

Response of lowland rice to Zn application			
Country	**Soil characteristics**	**Zn level (kg/ha)**	**Optimum Response (t/ha)**
India	Calcareous red, pH 7.5 Saline-alkali, pH 10.6 Aquic Camborthid	10 10 11.2	1.8 1.0 1.4
Pakistan	Calcareous	100	2.6
Philippines	Calcareous Hydrosol	10 Root dipping in 2 % ZnO	4.8 4.4
Thailand		15	0.4
USA	Norman clay Crowley silt loam Crowley silt loam	9 27 8	7.0 0.7 2.4
Source: Jones et al., 1982			

Zn recommendations for lowland rice (India)	
State	**Recommendations* (kg/ha zinc sulphate)**
Tripura	15
Maharashtra	17.75
Kerala, Jammu-Kashmir and Karnataka	20
Meghalaya	20-25
Bihar, Delhi, Haryana, Himachal Pradesh, Rajasthan and West Bengal	25
Tamil Nadu	30
Madhya Pradesh	25 (light textured soil) 50 (heavy textured soil)
Andhra Pradesh	50
Punjab	62.5 + topdress 25, if necessary
* For soil application once in 2-3 seasons, during final land preparation.	
Source: Tandon, 1989.	

Alternative recommendations for all the provinces:

1. Root dipping of seedlings in 1-4 % ZnO suspension before transplanting.

2. Foliar spraying of 0.5 % zinc sulphate (+ 0.25 % slaked lime) solution at 30, 45 and 60 days after planting or more frequently.

Complementary use of organic manures

Wherever feasible organic manures like compost, farmyard manure, green manure or azolla should complement mineral fertilizer. Organic manures should not be considered merely as suppliers of nutrient elements; they play an important role in maintaining the long term fertility of rice fields through improvement of the physical and biological properties of the soil. About 5-10 t/ha of organic manures may be regularly applied to rice fields, preferably during the wet season, for realizing the maximum benefit from mineral fertilizers.

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Rye *(Secale cereale L.)*

French: Seigle; Spanish: Centeno; Italian: Segale; German: Roggen

Crop data

Annual, with autumn-sown (winter rye) and, rarely, due to low yields, spring-sown types.

Harvestable products: grain, straw and whole green plant

Desired characteristics affecting fertilizer requirements:

In grain for flour - high swelling ability of starch, high amylopectin, absence of sprouting. In grain for malting (for beer-making) - high starch, low crude protein, absence of sprouting. In grain for alcohol - high starch, low crude protein. In grain for animal feed (as coarse meal) - high crude protein, especially lysine.

Straw for bedding - dry absorbent material. Straw for cellulose and pasteboard - high cellulose, low lignin and ash. Straw for constructional insulation - dry mass, low bulk density.

Green plant for forage - high cellulose, low lignin and ash. Green plant for silage - high content of easily soluble carbohydrates.

Rye used for animal feed is best grown as a companion crop to a legume.

Varieties - those with open pollination are not very uniform; they can adapt themselves to poor soils, even to quite acid soils. Their resistance to lodging is too low for high yields; some improvement is achieved with tetraploid forms. So called 'hybrid varieties' (the correct breeding term should be 'synbrid') brought great progress in yield and lodging resistance and they can be as demanding as wheat. Triticale (hybrids of wheat and rye) tend to be a good compromise between wheat and rye. Winter rye has the advantage of producing a good growth in the autumn and so, with a better soil cover, reduces the risk of erosion and unproductive evaporation.

Sowing time - Winter rye should be sown sufficiently early to have at least three weeks for tillering before the winter rest period starts. Spring rye should be sown as soon as temperature and water supply permit.

Temperature limitations and the duration of the various growth periods are illustrated in the following table:

Growth stages and climatic limitations in the development of rye							
Stage	**EC[1)]**	**Duration (days)**		**Temperature (°C)**			**Daylength (h/d)**
		Winter rye	**Spring rye**	**min**	**opt.**	**max**	
Sowing and germination	0.1-0.9	7	7	1-2	25-30	30-37	
Seedling emergence and early growth	1.0-1.9	11	11	3-5			
Tillering, initiation of ear primordia	2.0-2.9	38[2)]	38		5		
Beginning of stem elongation and formation of ear primordia	3.0-3.9	18	19				> 12-14
Flag leaf, floret reduction, booting	4.0-4.9	12	12				
Ear emergence	5.0-5.9	9	7				
Flowering and grain initiation	6.0-6.9	21	20				
Grain formation	7.0-7.9	27	25		> 15		
Maturing of the grain	8.0-9.2	20	21				
Total		163[2)]	160		1 700-2 400[3)]		
1) EC = Eucarpia Scale; 2) To be added: duration of vegetative rest (depending on location; e.g.: 134 days); 3) Total daily temperatures above 0° C (temperatures below 0° C deducted).							
Source: Aigner et al., 1988; modified							

The following table shows the changes in yield components relative to the amount of plant available precipitation. It should be noted that varieties differ in their water requirements and in harvest indices (i.e. proportions of grain in the total biomass at harvest, in terms of dry weight), so some adaptation of the table may be needed. Also, yields of ears from third-order tillers are only a little lower than those of ears from the main stem of first-order tillers, with the result that more intensive tillering is desirable than, for example, in wheat.

Yield structure of rye as function of plant available precipitation*							
	Plant available precipitation						
	150 mm	**250 mm**	**350 mm**	**450 mm**	**550 mm**	**650 mm**	**750 mm**
Grain yield	20	32	68	100 = 5.9 t/ha	119	127	131
Ear density	44	67	89	100 = 450/m2	111	120	129
Single ear weight	46	48	76	100 = 1.31 g	107	106	101
Optimal number of ears per plant = 2-3							
* Relative to 450 mm available precipitation; Plant available precipitation = amount solely available for crop growth, i.e. excluding evaporation, runoff, drainage and other losses.							
Source: various experimental results							

Nutrient demand/uptake/removal

Nutrient uptake/removal of winter rye- Macronutrients				
Yield base	t/ha	kg/ha		
		N	**P2O5**	**K2O**
Biomass: (DM)	11.9	133	46	180
Grain:	5.9	87	35	36
Source: various experimental results				

Relative nutrient uptake of winter rye in relation to plant development				
Stage	**N**	**P2O5**	**K2O**	**Dry matter**
	per cent of maximum			
Early growth	0	0	0	1
Tillering	26	20	15	6
Jointing	35	30	24	10
Booting	49	46	42	26
Ear emergence	71	74	87	44
Flowering	81	96	100	78
Grain formation	94	98	97	100
Physiological maturity				
- Biomass	100	100	92	84
- Grain	65	76	20	49
Maximum (100 %)	**kg/ha**			
- Biomass	133	46	180	11 900
- Grain only	87	35	36	5 900
Source: Aigner et al., 1988				

Fertilizer recommendations

In general the same principles apply to rye and its crosses as to wheat but the level of nutrients needed varies considerably.

A recommendation for winter rye with an expected yield of 6 t/ha grain would be 120 kg/ha N, 120 kg/ha P2O5, 100 kg/ha K2O, with an additional 5 - 20 kg/ha N in minimum tillage systems. Attention needs to be given to the risk of lodging, which is greater than with wheat and is more difficult to prevent, due to the lack of appropriate breeding and because growth regulators do not work so well on rye. On poor soils and in succession to a crop which has received little fertilizer, some application of fertilizer in autumn is recommended. A smaller late application is needed than for other temperate cereals. On the other hand, if all the N is given in a single dressing, or if slurry is applied, then some loss of N must be expected.

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Sorghum and Millet

Sorghum is in general the most productive millet variety and is mostly cultivated in those areas with a sufficient water supply to achieve good yields. On drier locations sorghum is frequently replaced by Pennisetum americanum.

Sorghum *(Sorghum bicolor (L) Moench)*

French: Sorgho; Spanish: Sorgo; Italian: Sorgo; German: Mohrenhirse, Sorghum; Indian: Jowar; Arabic: Durra

Crop data

Annual.

Can be used for many purposes: as grain, animal feed, silage, for broom-making, as a sweetener (sugar sorghum), for fuel or as a building material.

Temperature for germination >15 °C, for optimal growth 25-30 °C.

Can be grown where rainfall over 250 mm, best with 800-1 200 mm during the vegetation period. Seldom irrigated.

Often found on light, sandy soils but yields are greater on good, rich soils.

Plant density:	under very dry conditions,	0.5 to 1-3 plants/m2
	under dry conditions,	5 to 10 plants/m2
	irrigated or for forage,	10 plants/m2

Early maturing varieties can be harvested 3-4 months after sowing; late varieties mature at 8-10 months. The crop can be cut for forage at 2 months or for silage at 3-4 months. Ratooning for forage or grain production is possible.

Nutrient demand/uptake/removal

Depends on soil conditions, stand, yield expectation and growth rate:

Nutrient uptake - Macronutrients			
	kg/ha		
Conditions	**N**	**P2O5**	**K2O**
Dryland, low yield	40	20	40
Wet, medium yield	60	35	75
Irrigated, high yield	100	50	125
Green forage	80	30	70
Silage	150	40	150
Top yield, grain or forage	250	60	250

Fertilizer requirements

Where farmers consider sorghum to be an important crop, organic manuring with f.y.m., compost or crop and food wastes is widely practised. This may be enough for low-yielding sorghum, but where such manures are scarce, inadequate or unavailable, mineral fertilizers must be used instead or as a supplement. The rates required depend on the ecological conditions and yield potential;

Fertilizer recommendations - Macronutrients			
	kg/ha		
Conditions	**N**	**P2O5**	**K2O**
Dryland, low yield	0- 20	0- 20	0- 20
Wet, medium yield	20- 60	20- 40	20- 50
Irrigated, high yield	60-100	40- 60	50-100
Green forage	40- 60	20- 40	40- 60
Silage	40- 80	20- 60	40- 80
Top yield, grain or forage	100-250	60-100	100-300

Under very dry conditions and with a yield potential below 1 t/ha, low rates of N and P2O5 may increase yield but fertilizer application is often inefficient and uneconomic. With better rainfall and potential yields of 1-2 t/ha, fertilizer N and P2O5 rates need to be doubled and in most cases should be accompanied by up to 60 kg/ha K2O. With irrigation and potential yields of 6 t/ha or more, the rates should be substantially increased.

With the higher rates of application, split dressings are generally recommended (60 % at sowing and 20-40 % at shooting or at the beginning of flowering) except for P2O5 which should all be included in the basal dressing at sowing. On P-fixing soils, band or lay-by application is more effective than broadcasting. Fertilizer use is most effective when the stand is not too thick and the soil pH is below 5.

For forage or silage, the crop must grow quickly and produce a dense vegetative mass. Therefore, the N-application has to be relatively high (40-80 kg/ha, or even above 100 kg/ha N). Especially under irrigated conditions and on light soils, the doses at sowing time may be as low as 20-40 kg/ha, with further split applications (50-80 kg/ha N) when the crop has reached a height of 50-80 cm.

After harvesting the first crop, for a ratoon crop a dressing of 20-60 kg/ha N should be given. A further 20-40 kg/ha K2O may also be advisable. The extra P needed (15-30 kg/ha P2O5) ought to be included in the basal dressing when sowing the first crop but, if omitted at that time, around 20 kg/ha P2O5 may be given as a lay-by application.

Preferred nutrient forms

N: ammonium sulphate or urea before sowing; urea or a nitrate fertilizer for topdressing.

P: in easily plant-available form before sowing; in granular form, placed (band or lay-by), for P-fixing soils.

K: before sowing, as sulphate or chloride.

Pearl or Bulrush Millet *(Pennisetum americanum (L) Leeke, Syn: P. glaucum, P. spicatum, P. typhoides)*

French: Millet à chandelle, millet perle ; Spanish: Mijo perla, lagartera; Italian: Miglio perlato, miglio africano; German: Perlhirse, Rohrkolbenhirse; Indian: Bajra

Crop data

Annual. Preferably grown under dryland conditions, as replacement for sorghum.

Main use is for grain production. Green stuff, stalks and stubble used as by-product for animal feed, building material and fuel.

Temperature for germination >15 °C, for optimal growth 25-30 °C.

Will grow where rainfall at least 200 mm, but better growth and higher yields with 500-750 mm/season (where rainfall exceeds this, sorghum would be the crop of choice). Seldom irrigated.

Often grown on light and sandy soils (more important crops are given preference on more fertile soils).

Plant density:	under very dry conditions,	0.5 to 1 plants/m2
	under dry conditions,	3 to 5 plants/m2
	under slightly moist conditions,	5 to 7 plants/m2

Early maturing varieties can be harvested after 2-3 months. Late maturing varieties of Pennisetum grown under cooler conditions need up to 10 months until harvest.

Nutrient demand/uptake/removal

Depends on soil fertility, yield potential, rainfall and plant density:

Nutrient uptake-Macronutrients			
Conditions	kg/ha		
	N	**P2O5**	**K2O**
Very dry, very low yield	30	15-20	30
Dry, low yield	40	20	40
Rainfed, average yield	60	40	70
Irrigated, intensive crop with high yield	100	60	100

Fertilizer requirements

If available, organic manure should be applied as much as possible. This may suffice if the yield potential is below 1 t/ha. Where higher yields are possible, the additional use of mineral fertilizer is recommended.

The N requirement ranges between 20 and 60 kg/ha N under rainfed conditions and may increase to 100 kg/ha if yields of 3-5 t/ha are aimed at. P2O5 is badly needed on soils low in P or with a high P-fixing capacity (pH very low or >7); depending on nutrient uptake and yield potential, rates of 15-40 kg/ha P2O5 for rainfed or around 60 kg/ha for irrigated conditions should be worked into the soil before sowing. K2O should be applied where the soil is deficient in this nutrient or yields are high as a result of applying N and P2O5, in which case 20-70 kg/ha K2O may become necessary under rainfed conditions; with irrigation as much as 100 kg/ha may be needed for the best results.

Fertilizer recommendations			
Conditions	kg/ha		
	N	**P2O5**	**K2O**
Very dry, low yield	0- 20	0- 20	0- 20
Wet, medium yield	20- 60	20- 40	20- 50
Irrigated, high yield	50-100	40- 60	40-100

Split dressings of N are advised under moister conditions, whether from higher rainfall or from irrigation. P2O5 should be worked into the soil before sowing, as indicated above; on soils with a high P-fixing capacity placement is very important and the fertilizer should be applied in granular form. K2O too should be applied before sowing.

Preferred nutrient forms

N: at sowing, as ammonium sulphate or urea; for topdressing, as ammonium nitrate or as nitrate.

P: at sowing, mainly as triple superphosphate but also as basic slag or rock phosphate. Late spray applications have often been promoted but have rarely been put into practice.

K: in sulphate or chloride form before sowing.

Further reading

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Wheat *(Triticum aestivum L., T. durum Desf.)*

French: Blé; Spanish: Trigo; Italian: Frumento; German: Weizen

Wheat is the world's most important cereal crop in terms of both area cultivated (232 million ha) and amount of grain produced (595 million t). It is widely grown throughout the temperate zones (in Northern Europe up to 60 °N) and in some tropical/sub-tropical areas at higher elevations. The major centres are: Europe (131 milion t grain, 27 million ha), the former USSR (108 million t grain, 48 million ha), North America (106 million t grain, 42 million ha), China (96 milion t grain, 30 million ha) and India (50 million t grain, 23 million ha). All these figures relate to 1990.

The following information is mainly for T. aestivum, though also generally valid for T. durum. Special data for durum wheat are given as an appendix to this end of chapter .

Crop data

Annual, autumn-sown (winter wheat) and spring sown types.

Harvested products: grain, straw, (occasionally) whole green plant.

Desired characteristics affecting fertilizer requirement:

In grain for milling for bread and pasta: high endosperm-protein: In grain for malting and brewing: high starch, low crude protein, absence of sprouting. For animal feed: high protein, especially lysine. For industrial starch: high starch concentration with slow starch ripening, high endosperm-protein. For alcohol production: high starch content, low crude protein.

Straw for litter and bedding: should be dry and absorbent. Straw for cellulose and for pasteboard: high starch, low lignin, low ash. Straw for constructional insulation: should be dry and of low bulk density.

Whole plant for green fodder: high protein, high energy. Whole plant for silage: high concentration of easily soluble carbohydrates

The quality of the protein in the grain depends largely on the variety. The protein/starch ratio in the grain depends both on the variety and on the way in which the crop has matured.

Preferred soils and soil conditions: Wheat (like barley) generally prefers the more fertile soils, but it can be grown on practically all types except very light sandy soils or peat soils, so long as the water requirement can be met and the nutrient demand is met by appropriate use of fertilizers.

Sowing times: winter wheat should be sown soon enough for there to be at least two leaves before the onset of the vegetative rest period. The available growth time for tillering should be ast least 21 days. Provided the ground is passable, spring wheat should be sown as soon as temperature and soil moisture permit.

Planting density: sowing rates are usually within the range of 140 - 380 grains/m2 for winter wheat (see the sample calculation page #) and 200 - 350 grains/m2 for spring wheat sown at respective row spacings of 13 - 20 cm for winter wheat or 12 -18 cm for spring wheat. But ideally the planned plant density should be based on the expected plant available precipitation, optimum ear density and number of ears per plant for the variety in question

(see table below) and the intended grain yield per hectare, adjusted for the expected percentage emergence and, for winter wheat, the expected percentage winter survival.

Limits of climatic values within the development of wheat								
Development stage	**Eucarpia Scale**	**Duration dayts**		**Temperature (°C)**			**Day-length (h/d)**	**Minimum water demand (mm)**
		Winter	**Spring**	**min.**	**opt.**	**max.**		
Sowing and germination	0.1-0.9	7	7	2-4	15-25	30-32		35-40
Emergence and early growth	1.0-1.9	28	12					
Tillering, initiation of ear primordia	2.0-2.9	35*	35		6-9			
Beginning of stem elongation, forming of ear primordia	3.0-3.9	25	19		<17		>13**	72
Flag leaf, floret reduction, booting	4.0-3.9	16	10					
Ear emergence	5.0-5.9	6	5		<19.5			14
Flowering and grain initiation	6.0-6.9	11	14		20-25			22
Formation of grains	7.0-7.9	20	20		20			
Maturing	8.0-9.2	21	21					
Total growth period (sum)		169*	143*	1 900-2 500***				
*) To be added: duration of vegetative rest period, depending on location, e.g. for Northern Europe 100 days for winter wheat or 29 days for spring wheat sown late January/early February **) Some varieties < 13 h/d ***) Sum of daily temperatures above 0 °C (temperatures below 0 °C deducted)								
Source: Aigner et al., 1988, changed and supplemented								

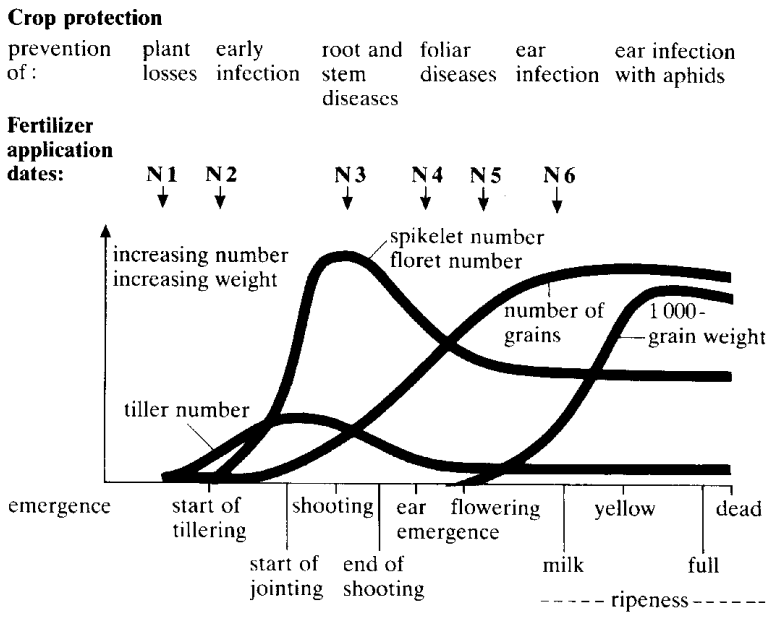
Yield structure of wheat as function of plant available precipitation*							
	Plant available precipitation (mm)						
	150	**250**	**350**	**450**	**550**	**650**	**750**
Grain yield	15	25	52	82	100 = 6.7 t/ha	116	127
Ear density	37	56	74	89	100 = 540/m2	111	120
Single ear weight	40	46	71	92	100 = 1.24 g	105	105
Relative to 550 mm available precipitation; Plant available precipitation = amount solely available for crop growth, i.e. excluding evaporation, runoff, drainage and other losses.							

For example, in a location with a plant-available precipitation of 550 mm and a target yield of 6.7 t/ha, the preferred plant density for a variety with an optimum ear density of 540 ears/m2 and an optimum of around 2.0 - 2.25 ears per plant would theoretically be in the range 215 to 270 plants/m2, say 240 plants/m2 (to reach the desired 540 ears/m2); but, to allow for an emergence rate of 95 % and an expected winter survival rate of 90 %, the actual sowing rate would need to be increased to 280 grains capable of germinating/m2 (i.e. 240 / 0.95 / 0.9). Similarly, with a plant-available precipitation of only 250 mm and an expected yield of 1.68 t/ha(= 25 % of 6.7), the preferred ear density would be 302 (= 56 % of 540) ears/m2, equivalent (with 2.1 ears/plant) to 144 plants/m2, and,with the same adjustments for percentage emergence and winter survival, to an actual sowing rate of 164/m2.

Where a crop is to be undersown, the intended plant density should be reduced by 10 - 15 % to protect the undersown plants.

The pattern of growth is illustrated in the next figure:

Pattern of yield formation of cereals



This model is valid only with the full plant protection programme and the indicated split dressings of N-fertilizer.

Source: Adapted from Aufhammer, 1976

Nutrient demand/uptake/removal

The amount and precise pattern of nutrient uptake will vary considerably with weather conditions. The figures given in the following table are indicative of uptake by a crop grown with 550 mm plant-available precipitation and yielding 6.7 t/ha of grain, as in the example quoted earlier.

Relative nutrient uptake in relation to plant development								
Growth stage	Winter wheat			Biomass * (dry)	Spring wheat			Biomass * (dry)
	N	**P2O5**	**K2O**		**N**	**P2O5**	**K2O**	
	Relative nutrient uptake as percentage of maximum							
Early growth	13	0	0	1	8	3	6	1
Tillering	24	9	10	4	25	17	36	2
Jointing	38	27	22	16	49	47	72	9
Booting	68	45	38	31	71	64	95	20
Ear emergence	84	75	90	58	97	100	100	36
Flowering	100	91	100	80	100	93	82	51
Grain formation	100	100	89	100	100	90	72	78
Physiological maturity								
-biomass*	95	90	77	100	100	86	68	100
-grain	70	71	20	49	78	86	20	50
	Maximum uptake (kg/ha) of each nutrient and maximum amounts (t/ha) of biomass* and grain							
Whole plant	187	55	252	13.7	129	58	125	9.0
Grain only	130	39	51	6.7	100	50	25	4.5
Whole above-ground portion of plant								
Source: Adapted from Aigner et al., 1988								

In general, for both winter and spring types, 60 % of the expected maximum total nutrient uptake (N + P2O5 + K2O) occurs by the middle or end of shooting.

Nutrient uptake/removal - Macronutrients				
Variety	Yield t/ha	kg/ha		
		N	**P2O5**	**K2O**
Winter wheat	Biomass			
	(DM): 13.7	187	55	252
	Grains: 6.7	130	39	51
Spring wheat	Biomass			
	(DM): 9.0	129	58	125
	Grains: 4.5	100	50	25
* Whole above-ground part of plant				
Source: Adapted from Aigner et al., 1988				

Plant analysis data

Tentative, preliminary values:

Plant analysis data - Macronutrients (beginning of stem elongation: Eucarpia Scale 3.0; whole above-ground plant)					
Soil classification (P2O5 and K2O)	% of dry matter				
	N	**P**	**K**	**Mg**	**Ca**
Class A - very low	2.0-2.9	0.20-0.29	2.00-2.99	0.07-0.09	0.20-0.29
Class B - low	3.0-3.4	0.30-0.39	3.00-3.99	0.10-0.13	0.30-0.39
Class C - medium or normal	3.5	0.40	4.00	0.14	0.40
Source: Aigner et al., 1988					

Plant analysis data - Micronutrients (beginning of stem elongation: Eucarpia Scale 3.0; whole above-ground plant)						
Soil classification (P2O5 and K2O)	ppm dry matter					
	Fe	**Mn**	**Zn**	**Cu**	**B**	**Mo**
Class A - very low	20-29	20-29	15-22	3.0-3.9	3.0-4.4	0.15-0.22
Class B - low	40-59	40-59	30-49	6.0-7.9	6.0-9.9	0.30-0.49
Class C - medium or normal	40-59	40-59	30-49	6.0-7.9	6.0-9.9	0.30-0.49
Source: Aigner et al., 1988						

Tissue analysis has been repeatedly tested as a means of evaluating nutrient demand by cereal crops but generally without success, on account of large differences in the development of individual plants and tillers. For this reason it has not been widely introduced in practice. Although tissue analysis for micronutrient deficiencies has received greater attention than that for the macronutrients, reliable results can only be obtained with extreme care during sample preparation.

Soil analysis

It is possible to calculate fertilizer rates from soil analysis data, but the available informaton is usually too inexact for intensive wheat growing. Moreover, the availability of soil nutrients changes with weather dynamics, and cereals themselves vary in their ability to mobilize and take up nutrients from a soil. Data from generalized soil extraction models should therefore be used with caution. In many countries where fertilizer recommendations are based on the results of soil tests, they are generally grouped according to ranges of nutrient levels in the soil (mainly for P, K, Ca and micronutrients) rather than specific values. ***For further information*** *on soil testing procedure and interpretation, please consult on the IFA Web site in the " Publications " section, " Fertilizer and Their Efficient Use, the chapter : Diagnosis by soil testing ".*

Fertilizer recommendations

General

Cereal growth is mainly influenced by temperature and precipitation. If these fall outside the desirable limits, the resultant adverse effect cannot in general be compensated by use of fertilizers (particularly if spikelet primordia are reduced as a result of excessive temperatures at the shooting stage). However, if only isolated plants are affected - as with moderate winter kill - the adjacent plants can still be stimulated by the supply of nutrients to form more tillers, thus minimizing the overall effect, except that maturing of the plants may be uneven.

Since, in intensive cereal production systems, the plant population is managed near to the risk of lodging, any mineral fertilizer applied must be distributed as evenly as possible, at the correct date, and the amount (particularly of N) must be carefully adjusted to take account of soil reserves and crop demand. Excessive N application may also increase liability to plant disease through causing too dense a plant population.

Unproductive tillering and secondary shoot formation should be avoided by restricting nutrient supply (especially of N) during early growth (stages N1 and N3 in the figure " Pattern of yield formation of cerelas ". Similarly, increasing the protein content of the grain can be achieved by increasing N supply at the appropriate period, in this instance at grain initiation and grain formation (stages N5 and especially N6). There is, however, a danger that too much N at this time can disturb the uniformity of the maturing process by delaying ripening, with the result that only the precursors of the proteins associated with the variety concerned are formed, thus lowering the quality of the protein in the grain. It is important to choose a variety which is appropriate to the intended yield and the expected weather conditions.

If there is a danger of applied nutrients becoming immobilized, especially P in alkaline or acid soils, corrective measures should be taken before application, e.g. by selection of acidifying fertilizers, or by liming, or, in arid regions, by flooding.

Because frost hardiness and resistance to drought are determined by the hydration of plant tissues, sufficient K should always be available.

Preferred nutrient forms

N - the efficiency of different types of N fertilizers in cereals is nearly identical. Differences are mainly the result of environmental influences (precipitation, temperature, denitrification). Normally they will contain nitrate and ammonium N, as ammonium nitrate, calcium ammonium nitrate, ammonium sulphate nitrate or urea-ammonium nitrate solution (UAN). Ammonia generally improves baking quality. If N fertilizers containing a nitrification inhibitor are used, they have the same effect as a depot application or concentrated band placement (see also 'Fertilizer application techniques').

P - the supply depends mainly on the availability of P in the soil. Therefore, the soil solution should be supplied with sufficient P. Especially in spring at the start of the growing period fertilizers containing water-soluble P2O5 are preferred, which can be applied advantageously together with K2O and the first N-application in form of an NPK compound fertilizer.

N-P - fertilizers containing N and P especially accelerate early growth of young plants, particularly when applied as band treatment below the seed ("starter application"). They are normally more effective than the corresponding single nutrient fertilizers. However, in contrast to nitrophosphates, diammonium phosphate or urea phosphate may injure the seedlings, especially on calcareous, dry soils, if applied at rates greater than 11 kg/ha N.

K - potassium chloride

S - on sulphur deficient crops S containing fertilizers may be used.

Mg - potassic fertilizers containing Mg. Critical Mg deficiency may also be treated by foliar application of magnesium sulphate (if necessary in combination with micronutrients, e.g. manganese).

Micronutrients - deficiencies may occur on hot, dry sites from the late shooting stage to flowering and may be treated by foliar sprays (see table at the end of chapter) (table 9).

Rates of fertilizer application

In intensive agriculture, the fertilizer application rates should primarily be based on the nutrient balance for the whole crop rotation, supported by soil analysis data and including a forecast of nutrient removal by the crop for which the fertilizer need is to be calculated.

Balance sheet for nutrients of a crop rotation: Maize - Winter Wheat - Winter Barley (example taken from practical farming)			
	Removal or supply of nutrients (kg/ha)		
	N	**P2O5**	**K2O**
1989 Crop: Wheat			
Crop residues from last year (none)	-	-	-
Legumes as previous main crop	-	-	-
Legumes as green manure	-	-	-
Farm yard manure, liquid manure	-	-	-
Fertilizer applied to crop	+180.0	+100.0	-
Yield (grains) = 5.8 t/ha*	-128.0	- 53.8	- 35.8
Straw (straw:grain = 1:0.9)	- 28.2	- 14.1	- 70.4
Losses	-	-	-
Balance 1	+ 23.8	+ 32.1	-106.2
1990 Crop: Barley			
Crop residues from last year (wheat straw 6.4 t/ha)	+ 28.2	+ 14.1	+ 70.4
Legumes as previous main crop	-	-	-
Legumes as green manure	-	-	-
Farm yard manure, liquid manure	+ 60.0	+ 30.0	+ 90.0
Fertilizer applied to crop	+180.0	-	-
Yield (grains) = 6.0 t/ha**	-120.0	- 48.0	- 24.0
Straw (straw:grain = 1:0.9)	- 33.0	- 13.9	- 79.2
Losses	-	-	-
Balance 2	+115.2***	-17.8	+ 57.2
1991 Crop: Maize			
Crop residues from last year (barley straw 6.6 t/ha)	+ 33.0	+ 13.9	+ 79.2
Legumes as previous main crop	-	-	-
Legumes as green manure	-	-	-
45 t/ha slurry from dairy cattle	+120.0	+ 90.0	+270.0
Fertilizer applied to crop	-	-	-
Yield (grains) = 8.0 t/ha	-144.0	- 51.2	- 27.2
Straw (straw:grain = 1:0.63)	- 96.0	- 70.4	-307.2
Losses	-	-	-
Balance 3	- 87.0	- 17.7	+ 14.8
Nutrient demand wheat 1992			
Grains (8.0 t/ha)	-160.0	- 67.2	- 44.8
Straw (7.2 t/ha)	- 29.0	- 14.0	- 72.0
Total nutrient demand in 1992	-189.0	- 81.2	-116.8
Difference between demand and supply (balance) from the rotation:	-137.0	- 84.6	-151.0
Nutrient amount needed for wheat in 1992 (approx.):	135 kg/ha N	85 kg/ha P2O5	150 kg/ha K2O
Unknown and unevitable losses or immissions of nutrients are not accounted for, assuming that they more or less balance each other. * When calculating the fertilizer rate, yield expectation was 9.0 t/ha ** When calculating the fertilizer rate, yield expectation was 10 t/ha *** Large N surplus due to the large difference between expected and obtained yield; a catch-crop should be introduced to prevent N losses			
Source: Aigner et al., 1988 and diverse others			

Should this nutrient balance sheet indicate a continuing substantial deficit of any particular nutrient, it is necessary to check whether the total nutrient reserves present in the soil are sufficient to sustain such a deficit and, if not, to investigate whether the production technique being used may result in too great a depletion of humus.

A large positive surplus in the balance sheet may be the result of overgenerous forecasting of crop yields. If not, then the calculation should be re-checked. A moderate surplus, on the other hand, may be regarded as a nutrient reserve (depending on soil conditions) for the next

2 - 4 years. In the example of the maize - winter wheat - winter barley rotation considered in the foregoing table, for example, a system of P "storage application" to build up a surplus of P in the balance sheet might be used on wheat to improve the grain quality. "Storage application" of K could similarly be used on winter barley in order to improve winter hardiness and lodging resistance. Sound agricultural practices have to be used within the crop rotation to avoid built-up or depletion of soil nutrients.

Fertilizer application rates for nutrients other than nitrogen

The precise amounts and timing of P and K fertilizers are not particularly important for wheat, as the crop yield does not respond significantly to PK dresings if the soil is already moderately or well supplied with these nutrients. The respective criteria to be followed are, firstly, a balanced nutrient cycle and, secondly, the date at which application is possible.

Thus, in practice, P and K are not applied to wheat so much for the wheat itself as for the requirements of the rotation as a whole (see also: 'Timing of fertilizer application').

If growth at any stage is limited by a specific factor such as nutrient supply, water, light or temperature, then the grain yield is limited irrecoverably unless it can be compensated by modifying a yield component occuring at a later stage of development. For instance, if the number of tillers per unit area is reduced by a deficiency in nutrient supply in the early stages, then the resultant reduction in ear density can, in theory, be compensated by improving the nutrient supply in time to promote spikelet initiation and to diminish the reduction in florets. The difficulty in practice lies in estimating the true supply of nutrient from the soil and in making due allowance for any delay in availability of fertilizer nutrient in unfavourable weather conditions, as well as for location-specific variations from one part of a field to another.

Nitrogen rates

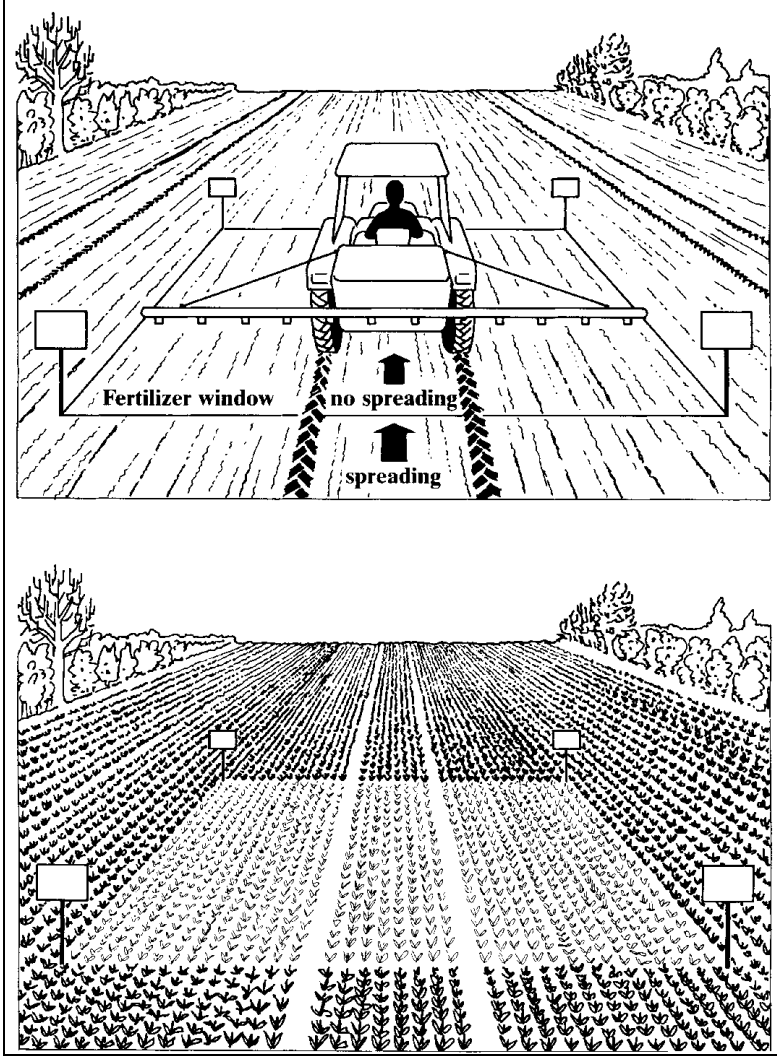
The exact calculation of the N demand for a specified yield is particularly important in intensive cereal production. A general guideline for the calculation of the N demand is given in the figure “ Model for calculating first N application (cereals) ”, but rather easier-to-use calulation 'models' are also employed. Such calculations - where undertaken - should begin, shortly before the early spring growing season, with analysis of the (probable) soil available N, based on the amount of mineralogically bound N known as N_{min}. This may relate either to nitrate plus ammonium N or, if necessary, to nitrate N alone. (For further information on use of the N_{min} technique as an indicator of the amount of mineralized and available N in the soil at the date of sampling, *please consult on the IFA Web site in the “ Publications ” section, “ Fertilizer and Their Efficient Use, the chapter : Practical Recommendations for Fertilizer Use .*

In Northern Europe, the results of experiments and observations indicate a release of about an additional 40 kg/ha N from the soil reserve of a normal fertile soil between the beginning of the early spring growing season, i.e. the date of soil sampling for N_{min} analysis, and the end of shooting. By subtracting this amount and the amount of mineral N available in the soil according to N_{min} analysis from the crop's expected requirement up to the end of shooting, one can obtain a reasonable estimate of the amount of N needed to come from fertilizers during this period. Thus, using the same example as in the table “ Relative nutrient uptake in relation to plant development ” for which the crop's total requirement up to the end of shooting was 157 kg/ha N (= 84 % of 187 kg/ha), and assuming that N_{min} analysis indicated 50 to 60 kg/ha N available in the soil at the time of sampling, then the amount to be supplied in fertilizers to provide enough N for the crop's needs up to end of shooting would be 57 to 67 kg/ha N (= 157 minus 40 minus 50 to 60 kg/ha).

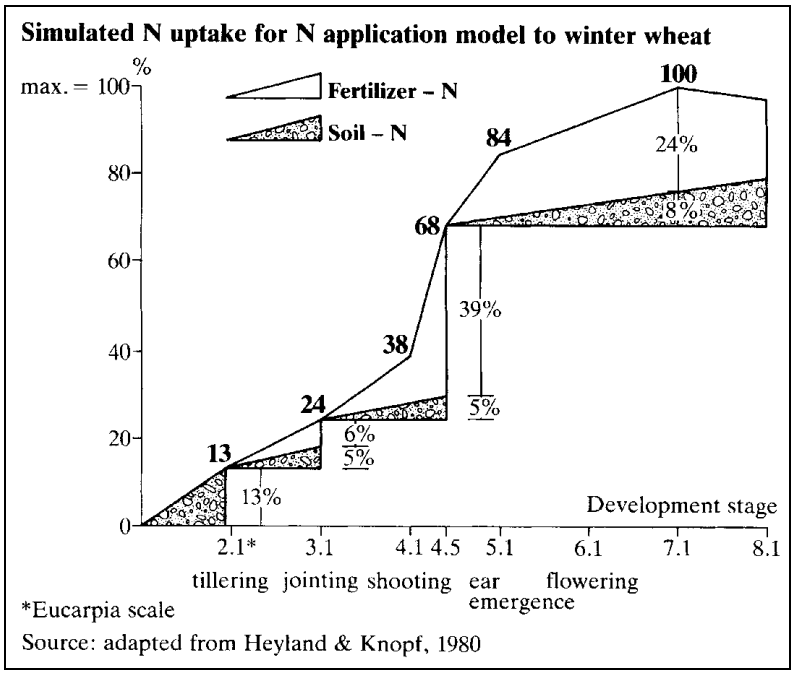
For the final "late N application", given at a stage when the yield can be estimated with a fair degree of accuracy, the amount needed can be obtained simply by subtracting the amount already supplied from the estimated maximum requirement. As a general rule of thumb, the "late N application" should be around 1 kg/ha N per 100 kg/ha expected yield of grain.

A practical system for the identification of N deficiency during growth is to leave a so-called "fertilizer window" in the field during N application. If the cereal plants within this window start to lighten in colour, then the plant-available N released by the soil has been taken up and the plants surrounding the "window" are benefiting only from the applied fertilizer N.

Method of producing a fertilizer window
(when passing the marked field area, the spreader is shut off)
Source: adapted from Rimpau, 1984



Most other methods for the calculation of N requirements make use of a model similar to that shown in the following figure, which aims to fit the N supply to the dynamics of N uptake consistent with maximum potential yield.



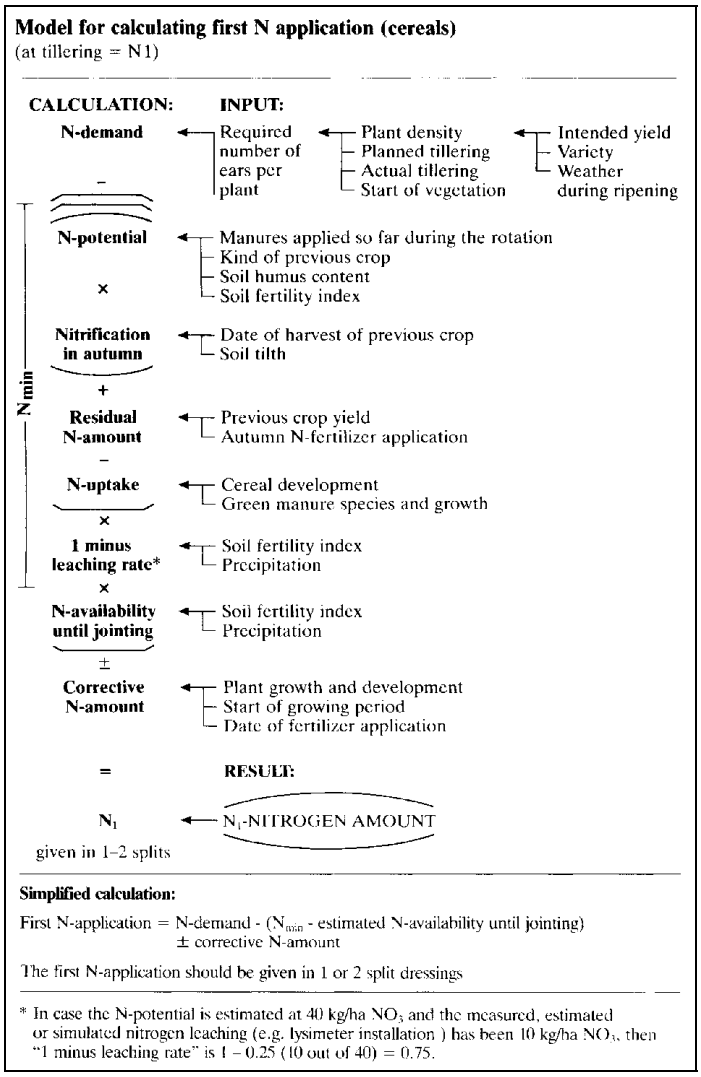
From the theoretical rate of N for each split application - to cover the crop's need only up to the next N application - there must be subtracted the estimated amount of N released from the soil during that period. If the resultant N rate is applied and the plants in an untreated "window" start to lighten, this shows, as indicated above, that N is avaible only from the applied fertilizer. For each subsequent application, the yield expectation and the amount of N available from the soil can be re-considered and the amount of fertilizer N to be applied is re-calculated. With this method it is possible to apply in each dressing as little as 30 kg/ha N; with the result that, if the calculated/estimated amounts prove to be wrong, the likely estimation error is then no more than 30 kg/ha N.

In deriving a figure for the total N requirement during growth, the crop variety must also be taken into account. Depending on a particular variety's straw/grain ratio and protein content, total N demand can vary by about 20 kg/ha.

Timely N application is especially important for spring varieties because of their shorter growth stages. More N is needed if plant residues are incorporated into the soil, because substantial amounts will be bound up during the process of decomposition. If straw is worked in, an additional 1 kg of N per 100 kg straw needs to be applied. Where such residues are regularly incorporated on a technically sound basis, a new nutrient cycle will be established by the soil micro-organisms in which additional N will no longer be necessary. The same applies to regular use of farmyard manure, in that an equilibrium will be established after two or three crop rotations which will allow the nutrients contained in the manure to be included in the total nutrient balance. However, on poor soils where soil fertility needs to be built up, 30 - 70 % immobilization of nutrients may be expected, so the calculated nutrient application rates must be increased for about the first 10 years.

The model shown in the figure above aims to achieve optimum ear weight (for the plant-available precipitation) combined with satisfactory yield stability, which in general is only possible with reasonably low ear densities. Fertilizer application at stage N1 of the figure " Pattern of yield formation of cereals " increases the numbers of tillers, at stage N2 the spikelet primordia, at stage N3 spikelet formation, at stage N4 reduction of florets, at stage N5 grain formation and at stage N6 grain development. The yield-diminishing effect of

unfavourable weather conditions at any of these stages may be compensated by stimulating the development of the appropriate later determined yield factors.



A computer-based model, "Anbauverfahren Bonn (Cropping System Bonn)" has been developed by Heyland and Kochs (BONAGRAR, 1984) designed to obtain site-specific plant populations related as closely as possible to optimum ear density and growth. With this system the necessary N-rates can be calculated for the application dates N1, N2 and N3 as shown in the figure above.

Timing of fertilizer application

Generally, for the application of P and K three methods are practized: (a) Application in autumn on the stubble of the preceding crop, or with autumn ploughing; (b) application in spring (as NPK fertilizer) with the first N. (c) For wheat following sugarbeet or maize, P and K are given to the preceding crop, the wheat receiving only N;

A number of practical points also have to be taken into account in deciding when to apply fertilizer, e.g. the availability of the necessary machinery and manpower, whether the soil is in a satisfactory condition to carry this machinery, and the actual weather conditions at the time. Particularly with N this can present a quite complex optimization problem in farm management, which can be solved more readily the higher the technical efficiency of the farm and the greater the extent to which use of the soil as a medium can be circumvented,

e.g. by more foliar fertilizer applications or by reducing the intervals between applications to the soil in such a way as to reduce the risk of nutrient fixation or of loss by leaching or by emission of N2 or NOx.

The crop takes up N from the soil solution whenever it is available, so promoting initiation and formation of those plant organs which are appropriate to the plants' stage of development at that time. If the objective is to maximize production of total biomass, this is of little importance; but, if the aim is to increase only grain production, then any surplus leaves and tillers that are formed will become undesirable competitors with the grain for available supplies of nutrients and water. For this reason, accurate timing is of vital importance in a system based on split dressings of fertilizer. If, however, accurate timing is impracticable due to weather conditions or poor organization, it may be advisable to restrict the number of dressings. with a possible reduction in the targeted yield.

While a single application may, of course, be given at the beginning of the growing period or at sowing, the unknown weather factor renders an accurate assessment of the final demand much less certain and there is little possibility of taking corrective measures.

It is important that the roots reach the full rooting depth as quickly as possible. If fertilizer is not incorporated into the soil, little or none should therefore be broadcast on the soil surface, especially when urea is used, because of its liability to volatilization losses. For application before sowing, incorporation or band application is necessary, particularly in areas with a dry spring.

The richer a soil biologically, the more are nutrients biologically fixed in the soil micro-organisms, a fact which is particularly important at the start of the growing period and at the beginning of tillering. N from the previous year's crop residues is, in general, mineralized in the autumn and is liable to be leached into the deeper parts of the topsoil with the winter precipitation. In spring the surface layer warms up while the deeper parts still remain frozen; the wheat plants and the soil micro-organisms then have to feed from the effectively nitrate-free surface layer. Under such conditions there may be insufficient N left for the wheat crop, so a small amount of N should even be applied shortly before the start of the growing period. A similar situation may arise later at the beginning of shooting, when the whole depth of the topsoil has been warmed and active soil organisms are again competing with the crop for nutrients; as this is a most important stage of crop growth, fertilizers are then again needed to supply the system with plant-available nutrients. This applies particularly to wheat grown on dairy farms whose soils are very rich biologically, but these farms should at least be able to apply slurry at that time.

Fertilizer application techniques

Broadcasting

- This calls for a sufficiently dry soil to permit the use of tractors and spreaders but nevertheless enough soil moisture for adaquate nutrient uptake by the roots from the soil solution. These conflicting requirements could be overcome by the use of aircraft, but still a rather expensive method. Mostly, however, the actual date of fertilizer application fails to coincide with the optimum for nutrient uptake.

Another difficulty is that P and K - and N when applied in form of urea - should always be incorporated into the soil, normally before sowing; but such incorporation, by aerating the top soil, results in a loss of moisture and of humus.

Particular attention must be given to absolute uniformity of N fertilizer distribution, both in and across the direction of travel. Evenness of distribution depends mainly on the physical quality

of the fertilizer (granule or prill size distribution, specific weight and surface characteristics), on the spreading system used (a single disc system is less accurate than a dual disc system, which in turn is less accurate than pneumatic and forced-feed precision spreaders) and on the rate and concentration of the fertilizer applied. With 120 kg/ha N applied as urea and an accuracy of spreading of ± 10 %, the overall range of error will be 24 kg/ha N. If the same total application is divided into six equal split dressings, the total overall range of error (assuming a random error distribution) is reduced theoretically to only 4 kg/ha N. However, multiple dressings not only involve more labour but also, more importantly, require spreaders capable of accurately distributing low rates of fertilizer.

Again, because of soil heterogeneity in large fields, the rate of application ought theoretically to be continously adjusted to take account of the varying nutrient requirements, particularly in intensive agriculture where the cropping system is operated near to the risk of lodging; but there are considerable practical difficulties in changing the rate during spreading. Some flexibility could, however, be achieved by using fertilizer solutions or foliar sprays. Where the crop is irrigated, crystalline fertilizers may be dissolved in the irrigation water or liquid fertilizer added, thus permitting accurate distribution at the optimum time.

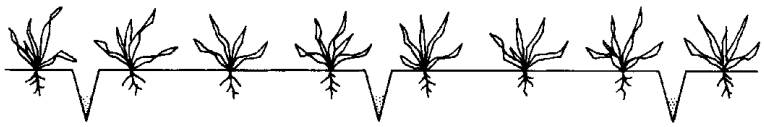
Band or storage application

- If it is desired to make fertilizer nutrients available only in those parts of the field which are actually covered by the wheat plants at tillering, band application is recomended. High nutrient concentration will then prevail in those particular areas, which may be used as long-lasting "fertilizer depots" because neither the roots nor the soil organisms will penetrate to the centre of the high concentrated zones and only their outer surfaces are exploited. This method is especially suited to nutrient-deficient and strongly nutrient fixing soils.

Possible technical solution for placement of deep band-application of fertilizer for cereals

Source: adapted from Halvorson, 1987

13 cm row spacing



Fertilizer placement as deep band

The greater efficiency of deep placement pre-plant urea ammonium nitrate solution compared with surface application is shown in the following table:

Effect of deep placed pre-plant UAN solution (Stafford County, KS; USA) Grain yield (t/ha)				
Application date	**8 July**		**15 August**	
	deep placed	broadcast	deep placed	broadcast
33 kg/ha N	1.99	1.89	2.24	2.3
67 kg/ha N	2.8	2.24	2.96	2.18
100 kg/ha N	3.3	2.99	3.33	2.66
Source: Gallagher et al., 1976				

However, injecting or "knifing-in" the fertilizer with coulters causes rupture of the soil surface parallel to the rows, so increasing the risk of greater germination of weed seeds. In already established cereal crops, the method is only applicable if the "depots" can be spaced widely enough and the plant density is low, both factors restricting the potential grain yield. If

injection nozzles are used, then it is necessary to have expensive technical systems and a stone-free soil.

The following conditions must be fulfilled in all cases where the "fertilizer depot" method is to be used:

- the amount of nutrient stored must supply the crop's total demand;

- the outer surface areas of the depots must be of an appropriate size to minimize both immobilization and leaching loss;

- placement should be at least 4 cm beside the seed or, even better, about 2.5 cm beside and 2.5 cm below the seed;

- in dry areas, placement can be up to 15 cm deep;

- the depots shold not be spaced more than 39 cm apart;

- the amount of nutrient stored should not exceed 100 kg/ha.

The fertilizers generally used are anhydrous ammonia, aquaeous ammonia and easily soluble P- or NP/NPK-fertilizers. Anhydrous ammonia should only be used where nitrification is low and where the winter precipitation is almost equal to the waterholding-capacity of the soil; a nitrification inhibitor should be added.

When non-pressure N solutions are used (e.g. urea ammonium nitrate), no more than 30 kg/ha N should be given at a time, preferably in conjunction with a pesticide spray (in which case the pesticide rate should be reduced - depending on droplet size - by up to 20 %, due to the reduced rate of evaporation of the spray solution and increased uptake through the leaves). N solutions are best applied in spring, to healthy, timely developed and dry leaf canopies. They should not be applied during flowering, nor if frost is expected. Minor leaf damage from leaf burn will have no effect on the final grain yield; any leaf burn of the flag leaf (which is important for grain filling) must, however, be avoided by using drag hoses attached to the nozzles of the spraying boom when giving a late N-application.

Foliar application

- The advantage of foliar application is the direct uptake of nutrients into the metabolism of the plant tissues. Thus, with a very low consumption of energy for transportation within the plant, the uptake is virtually independent of environmental factors such as soil moisture. The disadvantage is the limited amount that can be applied at one time, due to the risk of leaf burn; but this is of less concern provided proper attention is paid to the relevant limits of concentration, especially when there are a number of split dressings. Another advantage is the opportunity to combine fertilizer application with that of pesticides and growth regulators, in many cases with beneficial synergistic effects. In arid conditions the high concentration of combined sprays reduces evaporation, and so reduces vaporization of the active ingredients, thus permitting a reduction in pesticide usage with resultant benefit to the ecosystem.

Foliar application of dissolved granular fertilizer to cereals (assuming an application rate of 400 litres/ha)				
Fertilizer type	**Chemical formula**	**Concentration of fertilizer in the spray solution kg/100 l**	**Amount of fertilizer kg/ha**	**Amount of nutrients kg/ha**
Urea	CO(NH2)2	8-16	32-65	15-30 N
"Liquid nitrogen" (Non-pressure nitrogen solutions)	NH4NO3 + CO(NH2)2	0.5-1	2-4	1-1.8 N
Triple superphosphate	Ca(H2PO4)2	2	8	4.8 P2O5 2.0 CaO
Potassium sulphate	K2SO4	2	4	2.2 K2O 0.7 S
Magnesium sulphate	MgSO4.7H2O	2	8	1.3 MgO 1.5 S
Manganese sulphate	MnSO4.4H2O	1-2	4-8	1-2 Mn 0.6-1.25 S
Zinc sulphate	ZnSO4.7H2O	0.5	2	0.5 Zn 0.2 S
Copper sulphate	CuSO4.5H2O	0.5	2	0.5 Cu 0.3 S
Commercial foliar fertilizer		0.5-1	1-4	
Source: adapted from Finck, 1979				

Slurry

- Application of slurry to cereals causes great problems. Autumn application without added nitrification inhibitors may result in a large loss of nitrate by leaching, due to the limited uptake by the young crop. Application on frozen ground, while not harming the crop, may cause environmental problems. Spreaders with oversized tyres (to reduce pressure on the soil), applying the slurry through drag hoses, may be used for spring top dressing, provided "tramlines" have been laid down at sowing.

Present fertilizer practices

Germany

Fertilizer application rates Germany, Hesse (P, K, Mg) and Weser-Ems (micronutrients) based on an expected yield of 5 t/ha grain*								
Supply class	kg/ha							
	N**	**P2O5**	**K2O**	**MgO**	**Cu**	**Mn*****	**Zn**	**Mo**
A-very low	120	120	240	80	1.5	2 x 2	10	2
B-low	90	90	180					
C-medium or normal	60	60	120	40	0.5-1	2	5	1
D-high	30	30	60		1			
E-very high	-	-	-	-	-	-	-	-
* With other yield expectations the rates have to be adapted accordingly. ** N rates see: Calculation of fertilizer rates. *** As foliar spray; 2 x 2 = two foliar sprays with 2 kg/ha Mn each.								
Source: Aigner et al., 1988								

The table gives the recommended rates of nutrients for a wheat crop producing a yield of 5 t/ha grain. The ratios do not change much even with different water supply levels. This is also true of the maximum uptake and the demand at each developmental stage. On the other hand, if any factor such as nutrient supply, moisture, light or temperature is inadequate at

any stage, the kernel yield could be limited irreparably. The only chance of compensating for the resultant yield loss is by modifying a yield component associated with a later stage of development. For example, a poor supply of nutrients early in growth reduces the number of tillers per unit area, but the consequent reduction in ear density could in theory be compensated without difficulty by promoting the initiation of spikelets at EC stage 3.1 - 4.0 and diminishing the floret reduction at EC stage 4.1 - 5.1. In practice, however, it is not so easy to estimate accurately the available nutrient supply coming from the soil, or the magnitude of losses, e.g. by denitrification. The time interval elapsing between the application of fertilizer and nutrient uptake by the crop presents an additional and greater problem, particularly under unfavourable weather conditions, because the plant organs which are formed at the later stages of development are those which have less influence on yield production. In order to maximize the efficiency of fertilizer use, techniques will therefore need to be used which can quantify the various factors involved, by means of intensive site-specific crop observations, analysis, fertilizer management methods and/or computer-based simulation models.

China

(North China, winter wheat)

- Irrigated, expected grain yield 6 t/ha
  Basal application
  113 kg/ha N
  113 kg/ha P_2O_5
  broadcast and incorporated (15-20 cm)

  Topdressing (15-20 days after emergence)
  113 kg/ha N
  followed immediately by irrigation

- Rainfed, expected grain yield 4.5 t/ha
  Basal application
  150 kg/ha N
  90 kg/ha P_2O_5
  broadcast and incorporated (15-20 cm)

  No topdressing

Available organic manures are to be applied and incorporated well before sowing.

India

Irrigated timely-sown crop

- under assured irrigation
  N: 80-120 kg/ha N, depending on previous crop
  P: 40-60 kg/ha P_2O_5
  K: Based on soil test result

- under limited irrigation
  N: 60 kg/ha N
  P: 30 kg/ha P_2O_5
  K: Based on soil test result

Half the N and all the P and K are applied at or before sowing; P should be placed 5 cm below the seed. The remaining half of the N is topdressed at the first irrigation. N and P rates are adjusted according to soil test results.

Irrigated late-sown crop

N:	60-80 kg/ha N
P:	40-50 kg/ha P2O5
K:	Based on soil test results

All N, P and K are applied at sowing; rates are adjusted according to soil test results.

Durum wheat *(Triticum durum Desf.)*

French: Blé dur; Spanish: Trigo duro; Italian: Frumento duro; German: Durumweizen, Hartweizen

Durum wheat has been grown traditionally in semi-arid regions (e.g. in the Mediterranean, Turkey, North Africa) but is now found increasingly in other countries too (France, Germany, Austria, UK, Russia, Argentina and North America).

Crop data

Annual, autumn-sown types (winter durum), spring-sown types and types with dual sowing time.

Harvested product: grain - with special properties for the production of semolina for pasta (noodles) and couscous; in the UK also for puffed wheat and sugar puffs.

Quality: grains of durum wheat are richer in gluten, glassier and harder than grains of ordinary wheat (T. aestivum); their concentration of enzymes is also higher. Spring-sown durum types often give the best quality.

Yield level: about 10 % lower than that of wheat (T. aestivum), although the individual grains are about 7 - 10 % larger and 10 - 12 % heavier. Yields of autumn-sown durum wheat are normally greater than those of spring-sown types.

Preferred soils and climate: the crop prefers a fertile wheat soil with sufficient humidity up to the grain filling phase, mild winters (autumn-sown), cool springs and warm summers dry at harvest time. Vernalization is not necessary.

Seed density: 350 - 450 grains/m^2 depending on variety, sowing time and region; the seedbed should be dry and of very fine structure, and the depth of sowing not more than 3 cm because of the low germination power of durum wheat. The sowing time in Europe for autumn-sown types is October and for spring-sown types mid-February to mid-March.

Harvest: mid-July to early August

Fertilizer use

Rates and timing of fertilizer application are similar to those for T. aestivum. The recommendations given on the previous pages therefore also apply to this crop, including

those for the calculation of N rates. To obtain the desired glassy grains, late N topdressings of about 60 kg/ha N before ear emergence are necessary.

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Contributors: Liu Zongheng, Yu Zengshou, Soil and Fertilizer Institute, Shijiajuang, Hebei, P.R. China; R. Prasad, Professor of Agronomy, Indian Agricultural Research Institute, New Delhi, India; A.D. Halvorson, Research Leader, USDA Agricultural Research Service, Akron, CO, USA;
(Durum wheat) G. Jürgens, BASF Agricultural Research Station, Limburgerhof, Germany

Cassava *(Manihot esculenta Crantz)*

French: Manioc; Spanish: Yuca; Italian: Iucca, mandioca; German: Tapioka, Maniok; Portugese: Mandioca

Crop data

Annual (1-2 years). Harvested part: roots; in some cases young leaves. Grown in tropical and subtropical climates, generally at beginning or end of rainy season; cuttings 20 cm in length from mature stems are planted vertical, inclined or horizontal, 5-10 cm below soil surface.

Roots harvested 10-12 months after planting (but may range from 6 to 18 months; there is no fixed time of maturity).

Plant density: 8 000-16 000 per ha; often intercropped with rice, maize and grain legumes.

Optimum soil requirements for cassava

Preferably grown on light to medium soils, well-drained, pH 4.5-7.5. The crop is adapted to semi-arid conditions; it needs adequate soil moisture mainly during planting; once sprouted, it can withstand several months of drought; generally, it is not irrigated, but in some areas responds markedly to irrigation.

Cassava is well adapted to very acid soils with high levels of exchangeable Al. For that reason, is it seldom necessary to apply lime. The plant is also well adapted to low levels of available P, but requires fairly high levels of K, especially when grown for many years on the same plot. The crop is susceptible to Zn deficiency and often shows Zn deficiency symptoms at early stages of growth.

Optimum requirements:

pH 4.5-7.5; exchangeable Al < 80 % saturation; available P > 5 ppm; exchangeable Ca > 0.25 me/100 g dry soil; exchangeable K > 0.17 me/100 g dry soil; conductivity < 0.5 mmhos/cm; exchangeable Na < 2.5 % saturation; available Zn > 1 ppm; available Mn > 5 ppm; sulphate-S > 8 ppm.

Nutrient demand/uptake/removal

Nutrient uptake and removal - Macronutrients								
Yield t/ha	**Plant**	**Source**	**kg/ha**					
			N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
45	Fresh roots	Amarasiri,	62	23	197	36	17	3
	Whole plant	1975	202	73	343	179	183	15
37	Fresh roots	Howeler, 1985	67	38	122	14	22	7
	Whole plant		198	70	220	47	143	19
18	Fresh roots	Sittibusaya*	32	8	41	6	7	1
	Whole plant		95	23	77	52	23	7
9	Fresh roots	Sittibusaya*	13	2	5	3	4	0.2
	Whole plant		39	7	12	14	29	2
* personal communication								

Nutrient uptake and removal - Micronutrients							
Yield t/ha	Plant	Source	g/ha				
			Fe	Mn	Zn	Cu	B
37	Fresh roots	Howeler, 1985	900	60	170	30	70
	Whole plant		n.a.	1090	660	80	200

Plant analysis data

The youngest fully expanded leafblades without petioles (4th-5th leaf from top) at 3-4 months after planting provide the best indicator tissue. Petioles, stems and roots have much lower concentrations of N, P, and K. Adequate K is very important for starch synthesis and translocation and increases the plant's resistance to anthracnose.

Analysis of 4th - 5th leaf, 4 months after planting									
Supply	% of dry matter						ppm		
	N	P	K	Mg	Ca	S	Fe	Mn	Zn
Optimum	5.4	0.4	1.7	0.30	0.80	0.28	130	80	45
Deficient	4.7	0.3	1.0	0.27	0.65	0.24	100	45	25
Source: Howeler, 1983									

Fertilizer recommendations

Low yield level	
at 1 month after planting	half N, all P2O5 and half K20
at 3-4 months after planting:	half N and half K2O
High yield level	
at planting:	one-third N, all P2O5, one-third K2O
at 2 months after planting:	one-third N, one-third K2O
at 5 months after planting:	one-third N, one-third K2O

Nutrients may be applied either as organic manure, wood ash or mineral fertilizers. Organic manures and rock phosphate are generally incorporated in the whole soil during land preparations; wood ash is incorporated with the soil in the planting hole; while mineral fertilizers are applied in short bands near the planting stake and covered with soil, generally as a basal NPK dressing at planting or at 30 days, with 1 or 2 top dressings of N and/or K2O at 2-4 months after planting, when enough soil moisture is available. Rates depend on soil fertility status and yield levels:

Preferred nutrient forms:

There appears little difference between nitrate or ammonium forms of N. Being adapted to acid soil, cassava can make good use of rock phosphates, but generally responds better to soluble sources of P2O5 such as single or triple superphosphates. K can be applied as chloride or sulphate, but the former is generally cheaper. Wood ash is also a good source of K. Compound fertilizers are most convenient, if available, but they should either be high in N and K2O or should be supplemented by top dressings of urea and KCl. Zn can be applied as Zn sulphate to the soil if pH < 6.0. Foliar sprays of 1 % Zn sulphate can also be applied, or stakes can be dipped for 15 min in 2 % Zn sulphate before planting.

Present fertilizer recommendations:

Thailand - (north east and south-eastern region). 95 kg N, 45 kg P2O5, 95 kg K2O per ha. This can be applied as 300 kg/ha 15-15-15 at planting, followed at 3 months by a side dressing of a further 50 kg N and 50 kg K2O per ha.

India - (Kerala state): 12.5 t f.y.m., 100 kg N, 25 kg P2O5, 100 kg K2O per ha. The f.y.m. is incorporated before planting; half the N and K2O and all the P2O5 are banded near the stake at planting, and the remaining half of the N and K2O is side dressed near the plant at 45-60 days after planting.

Indonesia - (Java and southern Sumatra): 100 kg N, 50 kg P2O5, 100 kg K2O per ha. All the P2O5 and one-third of the N and K2O are applied at planting and two-third of the N and K2O at 3-4 months (after harvesting the intercrop).

Colombia - (Eastern Plains): 500 kg dolomitic lime, 80 kg N, 100 kg P2O5, 100 kg K2O, 20 kg S and 10 kg Zn per ha. The dolomitic lime is applied once every 4-5 years and is broadcast and incorporated before planting. P2O5 can be applied as rock phosphate, also broadcast and incorporated, or as triple superphosphate (TSP), diammonium phosphate (DAP) nitrophosphate or compound fertilizer. An alternative is to apply 200 kg/ha of DAP and 50 kg K2O/ha as KCl at planting followed at 3 months by 45 kg/ha N as urea and 50 kg/ha K2O as KCl, side dressed.

Brazil - (East-central states and Campo Cerrado): 30 kg N, 85 kg P2O5, 60 kg K2O per ha. The fertilizer is applied in the planting furrow at time of planting, but care should be taken that it is not in direct contact with the planting stake.

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Potato *(Solanum tuberosum L.)*

French: Pommes de terre; Spanish: Patatas; Italian: Patatas; German: Kartoffeln

Crop data

Annual, but some crops in some areas overwintered in the ground prior to harvesting.

Planted (tubers) spring to early summer; in some areas doublecropped in one season by planting very early in spring, harvesting early - before full maturity - then again planting in summer.

Planting from true potato seeds (TPS) occurs in parts of Africa, Asia and South America. TPS can be sown directly into the field (although this has not developed beyond the experimental stage) or sown to produce plants for transplanting into the field (a technique which appears to be less favoured currently) or grown under controlled conditions to produce high numbers of small tubers per unit area (minitubers) which are then grown on in the field in a relatively traditional manner.

Flowers: some cultivars rarely, if ever, produce full flowers; others produce flowers but sterile seed; whereas yet others produce flowers, fruits (commonly called berries) and viable seed.

Initiation of "daughter" tubers occurs 5-12 weeks after planting of "mother" tubers, presuming these latter to have reached the end of their normal dormancy at planting and depending upon the extent to which the mother tuber has been physiologically aged.

Plant density: 30 000-100 000 tubers/ha dependant on cultivar ("normal" range in temperate conditions 30 000 - 60 000 tubers/ha). Within this range, planting density is varied according to the final market requirement in tuber size, the availability of moisture (especially irrigation) and, in some countries, the relative cost of seed tubers and ware (consumption) tubers.

Potatoes will grow on most soils but, where mechanical harvesting is operated, then lighter and medium bodied soils cause less difficulty in harvesting when weather conditions are adverse. Potatoes are grown on organic as well as mineral soils. Soil pH requirement: minimum 5.5 but below about pH 4.8 growth is impaired. Alkaline conditions can adversely affect skin quality and highly alkaline conditions can induce micronutrient deficiencies.

Continuous cropping with potatoes only occurs in a few special situations and therefore it is normally necessary to strike a compromise in pH requirement between the needs of potatoes and other rotational crops. In most mineral soil situations a compromise recommendation for pH is 6.0-7.0. It is imprudent to apply lime immediately before a potato crop except in cases of extreme acicidity.

The crop responds well to moisture; irrigation is commonplace where it is grown on any scale. Irrigation at tuber initiation can affect the skin quality of daughter tubers by influencing phytopathogens, either favourably or adversely according to conditions, and amount of moisture present.

Nutrient uptake/removal

Nutrient demand/uptake/removal - Macronutrients								
Tuber yield t/ha	**Source**	kg/ha						
		N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**	**Cl**
100	Burton, 1989	250-450	35-65	350-550	10-30	10-20	20-40	20-100
57.9	Anderson and Hewgill, 1978	105 (1.8)*	60 (1.0)	315 (5.4)	12 (0.2)	5 (0.08)	- -	- -
77.7	Evans, 1977	250 (3.2)	80 (1.0)	325 (4.1)	-	-	-	-
90.0	ADAS (unpub-lished), 1976	306 (3.4)	93 (1.0)	487 (5.4)	19 (0.2)	8 (0.08)	-	-
50.0	Cooke, 1972	80 (1.6)	57 (1.1)	240 (4.8)	15 (0.3)	10 (0.19)	-	-
46.1	Widowson and Penny, 1975	154 (3.3)	47 (1.0)	290 (6.2)	-	-	-	-
50.0	Cooke, 1974	180 (3.6)	25 (0.5)	200 (4.0)	15 (0.3)	10 (0.20)	20 (0.4)	-
30.0 (whole plant)	Wirsing, 1990	150 (5.0)	25 (0.8)	200 (6.7)	20 (0.6)	60 (2.0)	-	-
36.0 (tops) (tubers)	Smith, 1990	58 153 211 (7.0)	8 32 40 (1.3)	112 209 321 (10.7)	22 16 38 (1.3)	40 5 45 (1.5)	6 10 16 (0.5)	- - - -
	Harris, 1978 quoting Kunkel et al., 1973	(2.68)	(0.62)	(3.93)	(0.27)	(0.07)	(0.21)	-
	Bedin; Malet, 1989	(3.2)	(1.6)	(6.0)	-	-	-	-
* in brackets kg/t tubers								

Nutrient demand/uptake/removal - Micronutrients							
Tuber yield t/ha	**Source**	kg/ha					
		Fe	**Mn**	**Zn**	**Cu**	**B**	**Mo**
100	Burton, 1989	0.5-2	0.1-2	0.4	0.1-0.2	0.1-0.2	0.1
	Harris, 1978 quoting Kunke et al., 1973	(0.042)*	(0.0134)	(0.018)	(0.0014)	(0.0062)	-
* in brackets kg/t tubers							

Plant analysis

Plant analysis data - Macronutrients									
Plant part	**Growth stage**	**Source**	**% of dry matter**						
			N	**P**	**K**	**Mg**	**Ca**	**S**	**Na**
Tuber	60 days 195 days	Kunkel et al., 1973	1.8 1.4	- -	- -	- -	- -	- -	- -
Foliage	-	Laughlin et al., 1974	0.19-3.8	0.16-0.25	-	-	-	-	-
Tuber	Mature	Painter and Augustin, 1976	1.16	-	1.47	0.084	0.019	-	0.022
Tuber	Mature	Rexen, 1976	1.23-1.50	-	-	-	-	-	-
Tuber	Mature	Kraus and Marschner, 1973	-	-	-	-	1.4-8.2	-	-
Aerial tissue	Actively growing	ADAS, 1990 (unpublished)	>3.5*s <1.5d	>0.2s -	>2s <1.5d	>0.1s	-	-	-
* s = satisfactory d = deficient									

Plant analysis data - NO3 and micronutrients						
Plant part	**Growth satge**	**Source**	**ppm dry matter**			
			NO3	**Fe**	**Cu**	**Mn**
Tuber	Mature	Painter and Augustin, 1976	-	15.7	1.3	-
Aerial tissue	Actively growing	ADAS, 1990 (unpublished)	- -	- -	- -	>30s* <20d
Petiole	Tuberisation	Kleinkopf und	15 000	-	-	-
(4th)	Tuber growth	Westerman,	15 000	-	-	-
	Maturation	1987	<10 000	-	-	-
* s = satisfactory d = deficient						

Fertilizer recommendations

Available organic manures are normally applied before seedbed preparation and ploughed or cultivated into the soil before planting. Frequently the nutritive value of such organic manures is unknown, and generally the amounts used are not recorded. It has been suggested that organic manures would enhance the water-holding capacity of the soil. However, pot experiments reported by Harris (1960) indicated such effects to be small, whilst Holliday et al.(1965) demonstrated yield benefits of up to 14 % from use of FYM in dry seasons. Experiments carried out by ADAS in England during the 1970s and 1980s have indicated that organic manures can have detrimental effects upon dry matter content (specific gravity) of potatoes for processing.

Timing of application of fertilizers

N: Potatoes use large amounts of N, frequently more than the total applied as fertilizer (Anderson & Hewgill, 1978). Normally all N is applied to the seedbed at planting but in cases

where leaching by rainfall is likely (e.g. on sandy soils) there are often benefits from applying half in the seedbed and half at tuber initiation. On heavier soils in temperate areas the benefits from splitting N application are less predictable.

P: generally best applied in spring.

K: Where large amounts (e.g. > 400 kg/ha K2O) are to be applied, they are probably better split. Theory would suggest several months between the two applications, the last at or about planting, but agricultural field practice suggests that 6-8 weeks between dressings in temperate conditions is adequate. On all except the lightest or most K-deficient soils, there appears to be no disadvantage in autumn application of K fertilizer followed by spring potato planting.

Micronutrients: deficiences are usually most efficiently corrected by foliar application.

Preferred nutrient forms

N: usually as readily soluble materials.

P: generally as materials containing a high proportion of water-soluble phosphate applied at or immediately prior to planting.

K: application as sulphate rather than chloride provides some small benefits in the form of relative increases in tuber dry matter (specific gravity) and is most appropriate to potatoes for processing. The chloride, however, through its effects on reducing tuber dry matter, has beneficial effects in reducing tuber bruising at harvest, other factors being unchanged. Large dressings of KCl at planting will have a deterious effect on crop emergence in dry soils.

Mg: can be applied as a foliar spray where deficiency symptoms occur in the absence of extenuating circumstances (e.g. drought); more normally it would be applied in a Mg-containing fertilizer or, as a routine on inherently Mg-deficient soils, in Mg-containing lime.

Fertilizer placement

Placement in close proximity but not in contact with seed tubers will produce more efficient fertilizer use. The risks of damage to tubers increase in dry conditions, on light soils, where pre-sprouted seed is planted, or where more than 250 kg/ha N plus K2O are applied together.

Present fertilizer practices

The Netherlands

Seed potatoes: fertilizer use depends upon cultivar and harvest date; the later a seed crop is harvested, the less N is applied.

Ware potatos:

N fertilizer requirement is calculated from the formula:
N requirement = 285 - (1.1 mineral N J/F*) (kg/ha)
* J/F - January/February

Mineral N is assessed in soil during January/February of the harvest year (= Mineral N J/F) and generally amounts to approximately 40 kg/ha N.

P and K requirements are calculated from soil status for these nutrients; whilst new soils high in Ca are given extra P.

Extra K is occasionally given to control bruising, presumably by its effect on dry matter.

Netherlands - Recommended nutrient rates			
Crop use	kg/ha		
	N	**P2O5**	**K2O**
Seed potatoes	90-130	120-180	ca. 200
Potatoes for human consumption	ca. 240	120-180	250-400
Industrial potatoes	ca. 240	120-180	100-200

France

France - Recommended nutrient rates			
Maturity group	kg/ha		
	N	**P2O5**	**K2O**
Early varieties	100-120	60-100	100-220
Second early varieties	80-100	60-100	100-200
Maincrop varieties	60- 80	60-100	100-200
New potatoes	120-150	100-150	200-250
Storage potatoes	150-200	80-110	300-400
Industrial potatoes	150-200	80-120	250-300

Boron is used in deficient soils at no more than 15-20 kg/ha because of its potential toxicity.

Germany: Former GDR

Nutrient supply to potatoes in organic manures and mineral fertilizers should match the crop requirement, which for P2O5 and K2O, is calculated from potential yield and soil nutrient status. A 30 t/ha crop would require 25 kg/ha P2O5 and 200 kg/ha K2O.

The N fertilizer recommendations which are set out below are for supplies from both mineral and organic sources for potatoes after a cereal crop (without overhead irrigation).

Germany (former GDR) - Recommended nitrogen rates				
Intended use	**Soil type**			
	Sand	**Loamy**	**Sandy**	**Loam/Clay**
	kg/ha N			
Ware potatoes for human consumption	130-170	130-170	130-170	120-160
Seed potatoes	80-120	80-120	80-120	70-100
Industrial potatoes for starch production	120-140	130-150	140-160	110-140

Where potatoes are intended for stock feed, an extra 20 kg/ha N is recommended. Where irrigation is used, a further 20 kg/ha N is advised.

N fertilizer dressings are sometimes split, especially on light sandy soils

Germany: Original FRG

Germany (original FRG) - Recommended nutrient rates*					
Type of potatoes	**Expected yield(t/ha)**	kg/ha			
		N	**P2O5**	**K2O**	**MgO**
Early potatoes	15-20	100-140	70- 80	180-240	50-60
	20-30	140-200	80-100	240-330	50-60
Maincrop potatoes	25-30	100-120	70- 80	250-275	50-60
	30-40	120-160	80-100	275-330	50-60
* For sites with good nutrient supply (Class C)					

Italy

Italy - Recommended nutrient rates			
Type of potatoes	kg/ha		
	N	**P2O5**	**K2O**
Early harvested potatoes	150-200	100-150	150-200
Human consumption potatoes	150-180	150-180	150-200
Industrial potatoes	120-150	150-180	100-150
Seed potatoes	100-120	120-160	100-150
Where FYM is used rates of mineral fertilizers are reduced proportionally.			

UK: England & Wales

Recommendations are given according to maturity on harvest date, soil nutrient status index for each nutrient, and soil type. N rates relate to previous cropping (index 0 is normal after cereals) and use of organic manures, whilst P2O5, K2O and Mg rates are based on soil analysis.

England & Wales - Recommended nutrient rates					
Variety	**Soil index**	**kg/ha**			
		N	**P2O5**	**K2O**	**MgO**
Early and canning potatoes	0	180a	350b	180	170
	1	130	300b	150	80
	2	80	250b	120	-
	3	-	250b	60	-
	over 3	-	250b	60	-
Seed potatoes (grown specifically for seed and burnt off early)	0	180	350	350	170
	1	130	300	300	80
	2	80	250	250	-
	3	-	150	150	-
	over 3	-	100	100	-
Second earlies and maincrop: Mineral soils	0	220	350	350	170
	1	160	300	300	80
	2	100	250	250	-
	3	-	200	150	-
	over 3	-	100	100	-
Second earlies and maincrop: Peaty soils (except moss soils)	0	130	350	350	170
	1	90	300	300	80
	2	50	250	250	-
	3	-	200	150	-
	over 3	-	200	100	-
Second earlies and maincrop: Organic, moss and warp soils	0	180	350	350	170
	1	130	300	300	80
	2	80	250	250	-
	3	-	200	150	-
	over 3	-	200	100	-
a For every early lifts, 150 kg/ha N is adequate					
b Early potatoes have shown much larger response on slightly acid soils in Dyfed and Cornwall than on calcareous soils in S.E. England. The latter have shown little response to P at soil P Index 4, and 125 kg/ha P2O5 is sufficient.					

Potatoes for processing: crops grown for 'French fry' production should receive the same fertilizer treatment as if grown for the ware market.
The variety Record grown for crisping should not receive more N than is recommended for ware.

Choice of source and rate of K will depend on whether higher dry matter or reduced internal bruising is the priority. High rates of animal manure can result in excessive nutrient application and reduced crisp quality.

UK: Scotland

Similar recommendations as for England & Wales, except as below:

Scotland - Recommended nutrient rates for specialist seed potato crops								
kg/ha								
N Soil N Status			**P2O5** Soil P2O5 status			**K2O** Soil K2O status		
Low	**Moderate**	**High**	**Low***	**Moderate**	**High**	**Low***	**Moderate**	**High**
100	70	40	250	200	150	160	110	60
* Soils of very low P or K status require additional 40 kg/ha P2O5 or K2O								

Adjustments for N			
	kg/ha soil N status		
	Low	Moderate	High
Average annual rainfall: > 850 mm	-25	-25	-25
Average annual rainfall: < 700 mm	+25	+25	+25
Sandy soils	+35	+35	+35
Very low organic matter (all arable)	+25	-	-

Sweden

Sweden - Recommended nutrient rates					
Use	Region	Variety	kg/ha		
			N	P2O5	K2O
Human consumption	South	Bintje	100-140	40-80	120-180
		King Edward	100-140	40-80	120-180
		Magnum Bonum	80-110	40-80	120-180
	Central	Bintje	100-120	40-80	120-160
		King Edward	80-100	40-80	120-160
		Magnum Bonum	60- 80	40-80	120-160
	North	Bintje	80-100	60-80	80-120
		King Edward	70- 90	60-80	80-120
		Magnum Bonum	50- 70	60-80	80-120
Starch/Alcohol	South	Dianella etc.	120-150	40-80	120-180
Chips	South	Bintje	100-120	60-80	120-200
(Pommes frites)	West	Saturna			
Seed	The whole of Sweden	Most	80-100	60-100	120-180
The recommended rates of P have generally a residual effect enough for a cereal crop.					

Finland

N requirements are taken from basic standards. Basic requirements for P2O5, K2O and Mg are calculated from soil analysis.

For all nutrients corrections are made for:

- effect of preceding crop
- organic manures
- irrigation/moisture stress
- method of fertilizer distribution, e.g. placement or broadcast

Computer programs are used to produce final individual recommendations.

USA: New York State

New York State - Recommended N rates		
Soil type	**Varieties**	**kg/ha N**
Mineral	Hudson, Kennebec	135-170
	Belrus, Monona, Norchip, Rosa, Superior, Wauseon*	170-200
Muck	Hudson, Rosa	85- 90
	Atlantic, Chieftain, Katahdin	110
	Belrus, Monono**, Norland**, Superior, Wauseon**	170
* Apply only 135-170 kg/ha N for early yields ** Apply only 110 kg/ha N for early yields		

New York State - Recommended P2O5 rates							
Soil type	**Soil test P**	**kg/ha P2O5 Soil iron + aluminium**			**kg/ha P2O5 pH**		
		<100	**100-200**	**>200**	**5.2**	**5.2-5.6**	**>5.6**
Mineral:							
- Upstate	<20	270-335	270-335	270-335	270-335	270-335	270-335
	≥20	135-170	170-270	270-335	270-335	170-270	135-170
- Long Island	<60	270-335	270-335	270-335	270-335	270-335	270-335
	≥60	135-170	170-270	270-335	270-335	170-270	135-170
Muck	<100	-	110	-	-	-	-
	≥100	-	55	-	-	-	-

New York State - Recommended K2O rates		
Soil type	**Soil Test K**	**kg/ha K2O**
Mineral	<175	270-335
	175-275	170-270
	275-375	85-170
	>375	55
Muck	<400	110
	≥400	55

New York State - Recommended MgO rates		
Soil type	**Soil test Mg**	**kg/ha MgO**
Mineral	<100 or pH <5.0	55
	≥100	none
Muck	<100	55
	≥100	none

For Long Island Mineral Soil apply Mg and at least 90 kg/ha each of N and P2O5 and all or part of K2O in bands at planting. If need is indicated by soil test, supplemental P2O5 and K2O can be applied between planting and emergence. The remainder of the N should be applied as a sidedressing when plants are 10-15 cm high.

For Upstate Mineral Soil, the entire application can be banded at planting, or at least 90 kg/ha N and all P2O5 and K2O can be applied at planting, with the remaining N applied as a sidedressing when the plants are 10-15 cm high.

In both locations, supplemental N applications may be needed if heavy rains occur early in the season, but they should not be made after plants are 20-25 cm high.

For Shallow Mucks, band all N at planting unless foliar applications of manganese sulphate are to be made; P2O5 and K2O can be broadcast or banded with N. For Deep Mucks, band or broadcast the required amount of fertilizer.

USA: Idaho

N recommendations vary from 22 to 337 kg/ha N based on the calculation:

$$\text{Requirement} = \frac{\text{Crop need - (mineralized N + soil test N)}}{\text{0.65 (\% recovery fertilizer N)}}$$

Idaho - P2O5 recommendations		
Soil test P 0 - 30 cm ppm	**kg/ha P2O5**	
	< 5 % free lime	**>15 % free lime**
5	180	450
10	90	360
20	0	180
30	0	0

Idaho - K2O recommendations	
Soil test K 0-30 cm ppm	**kg/ha K2O**
0	270
55	180
100	90
>158	0

Some N is applied in irrigation water. P2O5 and K2O can be applied in the autumn before planting of the spring potato crop.

Recommendations are not specifically given for sulphur but, where the level of S04-S in the 0-30 cm soil depth is less than 8 ppm, then treatment is advised in the form of water-soluble sulphate.

Canada

General recommendations for the Atlantic regions are:

kg/ha		
N	**P2O5**	**K2O**
110-150	120-180	120-180

Mg fertilization is rarely required.

India

Fertilizer recommendations based on data from the Departments of Horticulture of the various states				
State	**N**	**P2O5**	**K2O**	**farmyard manure t/ha**
	kg/ha			
Andhra Pradesh	100*	50	100	25-30
Assam				
- Irrigated	60	100	100	10
- Rainfed	60	50	50	10
Bihar	120*	80	60	20-30
Haryana	125-150*	50	100	50
Karnataka				
- Irrigated	125*	100	125	25
- Rainfed	75*	100	125	25
Madhya Pradesh	150*	100	100	25
Maharashtra	100	70	50	40-50
Punjab	125*	62	125	100
Orissa	100	62	100	-
Tamil Nadu	100	125	50	15
Uttar Pradesh	120*	80	100	20-30
West Bengal	150-200*	100-150	100-150	15
* Split application of N recommended				
Source : Tandaon, 1987				

South Africa

N recommendations		
Predicted yield t/ha	**kg/ha N**	
	Dryland	**Irrigation**
15	70	100
30	130	150
45	170	200
52	200	230

P2O5 recommendations		
Yield potential t/ha	**P2O5 status in soil**	**kg/ha P2O5**
10	a	30-100
20	b	40-115
30	c	50-130
40	d	60-145
50	e	70-160
a = ambic 0-4; Bray 1 0-5; Olsen 0-3 ppm P2O5		
e = ambic 20+; Bray 1 25+; Olsen 15 ppm P2O5		

Required K is calculated from cation exchange capacity and soil analysis. Mg and Ca are applied as dolomitic or calcitic limestone.

Other countries

Country	kg/ha		
	N	P2O5	K2O
Europe			
Austria	120	60	100
Hungary	138	125	236
Ireland	88	181	233
Poland	95	70	140
Asia			
Japan	120	152	136
South Korea	87	53	54
Sri Lanka	125	160	125
America			
Colombia	85	175	40
Dominican Republic	95	95	95
Venezuela	120-150	150-300	150-225
Africa			
Mauritius	78	78	120
Zimbabwe	0-160	0-310	0-130
Abstracted from International Potash Bureau Bulletin 8, 1983			

Further reading

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HARRIS, P.M. (ed.): The Potato Crop (A Scientific Basis for Improvement). Chapman and Hall, London, UK (1978)

Author: M.F. Askew, MAFF, Agricultural Development and Advisory Service, Wolverhampton, UK

Sweet Potato *(Ipomoea batatas Lam.)*

French: Patate douce; Spanish: Batata, Boniato, Camote; Italian: Patata dolce; German: Suesskartoffel

Crop data

Perennial root crop, belonging to the Convolvulaceae. Hexaploid (2n = 6x = 90) improved from Ipomoea trifida complex in Central America and northern S. America.

Used for food, animal feed and industrial materials.

Usually propagated from vine cuttings or sprouts from tubers, very rarely from seed.

Favourable growth temperature 20-30 °C (no growth below 15 °C or above 30 °C). Requires 4-6 months to produce economic yield. Growth period comprises two stages: the first, a period of extensive growth of fibrous roots, vines and leaves, and initiation of tuberous roots; the second, swelling of the tubers. In the former, a higher temperature favours vine and leaf growth; in the latter, a lower temperature favours swelling of the tubers.

The crop can be grown throughout the year in the tropics and sub-tropics but in warmer temperate regions it is grown as an annual.

Planting time in temperate regions: usually May/June for September/November harvest.

Plant density: 45 000-55 000/ha.

Relatively tolerant to drought but cannot withstand waterlogging. Preferred soil moisture content 60-75 % of maximum water-holding capacity.

Grows best on sandy loam soil and does poorly on clay soil.

Preferred soil pH 5.5-6.5. Sensitive to alkaline and saline soils.

Now grown in more than 100 countries worldwide but China accounts for 80 % of production.

Nutrient demand/uptake/removal

The following figures indicate the particular importance of both K2O (especially in the tubers) and N (in the leaves).

Nutrient demand/uptake - Macronutrients								
Part and yield	**Source**	**kg/ha**						
		N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**	**Na**
Leaf, 4 t/ha	Wenkam, 1983	25.6	8.7	25.4	4.2	2.1	-	-
Tuber, 10 t/ha		26.0	8.5	45.6	2.0	4.2	-	-
Leaf + tuber, 14 t/ha		51.6	17.2	71.0	6.1	6.3	-	-
Tuber, 10 t/ha	Bradburg, 1990	23.0	11.7	31.2	4.3	4.1	1.3	5.2
Leaf + tuber, 20 t/ha	Tsuno, 1972	83.6	17.4	130.8	-	-	-	-

Nutrient demand/uptake - Micronutrients							
Part and yield	**Source**	**kg/ha**					
		Fe	**Cu**	**Zn**	**Mn**	**Al**	**B**
Leaf, 4 t/ha	Wenkam, 1983	0.40	-	-	-	-	-
Tuber, 10 t/ha		0.40	-	-	-	-	-
Leaf + tuber, 14 t/ha		0.80	-	-	-	-	-
Tuber, 10 t/ha	Bradburg, 1990	0.49	0.17	0.59	0.11	0.82	0.11

Plant analysis data

Deficiency symptoms develop when the nutrient levels in the tissue (oven dry basis) fall respectively below 0.12 % P, 0.75 % K, 0.16 % Mg, 0.2 % Ca, 0.08 % S, 30 ppm Fe.

Fertilizer recommendations

Organic manure may be used to increase the soil fertility and the yield and quality of the tubers; this is a common practice in the traditional agriculture of many countries. In normal cultivation, however, mineral fertilizer is applied. Potash is especially important, and the ratio K2O:N should be kept fairly high (about 3:1) in order to promote swelling of the tubers.

A common recommendation in most countries is 35-65 kg N, 50-100 kg P2O5 and 85-170 kg K2O per hectare, which is usually applied in the form of NPK compound fertilizers with a high K content.

It has recently been suggested that part could be applied to the soil before planting and part as a topdressing, the relative proportions depending on the expected yield level as shown below:

Low yield level
Before planting - All N, all P2O5, one-half K2O.
40 days after planting - One-half K2O.

High yield-level
Before planting - Two-thirds N, all P2O5, one-half K2O.
40 days after planting - One-third N, one-half K2O.

Preferred nutrient forms

N: urea or ammonium nitrate/ammonium sulphate, preferable on alkaline soil.

P: water-soluble/citric acid-soluble form, such as superphosphate.

K: as sulphate.

Fertilizer practice in various countries - Nutrient rates							
Country/ region/ Season	**Source**	**Yield t/ha**	**Planting time (growth period in months)**	**a) Basal application b) Top dressing**			**Organic fertil.**
				kg/ha			
				N	**P2O5**	**K2O**	
China/ Jiangsu Japan Taiwan	Tsuno, 1972 Tsuno, 1972 Tsuno, 1972	20-30 20-30 16	May-June (4-5)* June (4-5) November (4-5)	a)100-110 b) 30- 50 a) 40- 80 b) - a) 70 b) -	100-120 30- 40 40-60 - 85 -	100-120 30- 40 50-100 50-100 100 50	Manure Manure -
Bangladesh	CIP 1989 Mackay, Palo-mar, Santico, 1989	10-15	October-November (4-5)	* a)**60 b) -	40 -	90 -	Manure
India/ Kharif	CIP, 1989 Mackay, Palo-mar, Santico, 1989	8,2	January -July (4)	a)** 35 b) 40	50 -	75 -	-
Malaysia	CIP, 1989	7-15	(4-5)	a)* 12 b) 15	12 15	17 15	-
Vietnam	CIP, 1989	5-10	(4-5)	* a)** 25 b) 15	20 -	40 20	Manure
USA/ Georgia	Granberry, Coldits, McClaurin, 1986	10-30	May-June (3-5)	a) 17- 42 b) 17- 42	38-84 38-84	50-126 50-126	-
Costa Rica/ Turrialba	Moreno, 1981	10.7	November	a) ** 65 b) -	10 -	30 -	-
South Africa	Jackson, 1974	20-30	Nov.-Dec. (4-5)	a) 40-50 b) 50	500 -	150 -	-
Papua New Guinea - Lowland - Highland	CIP, 1989 Villareal, 1982	 5-10 5-10	 (5-6) (8-12)	*			Compos t and ash
Micronesia	Mackay, Palomazr, Santico, 1989	16.6	-	a) ** 50 b) -	50 -	50 -	Manure
* Most farmers do not apply mineral fertilizers							
** Recommended rates of nutrients							

Further reading

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Yams *(Dioscorea spp)* **and Cocoyams etc.** *(Colocasia, Xanthosoma spp)*

French: Ignames & taro; Spanish: Names & taro; Italian: Ignames & taro; German: Yams & Taro

General Crop Data

Yams, one of the important staple food crops of the tropics, are particularly important in Africa which accounts for about 95 % of world production. Nutritionally they are rich in carbohydrate and are valuable sources of certain vitamins; they also contain 1.9-3.9 % protein.

In addition, precursors of Cortisone and related steroidal drugs are derived directly from some wild species (e.g. Dioscorea floribunda). Large scale cultivation for food is restricted largely to West Africa, Eastern Asia and the Caribbean region. In India they are grown only as a supplement to cereals.

Main edible species: D.esculenta, D.alata, D.rotundata.

Elephant yams (Amorphophallus campanulatus), not a species of Dioscorea, are usually consumed locally where they are grown, and are particularly important in India and Indonesia.

Amorphophallus rivieri is grown commercially in Japan for the production of mannose sugar.

Cocoyams are considered as subsistence and emergency food sources in many parts of the world, but in South Pacific islands and in some parts of Asia they are used as a major staple food crop.

Thanks to the high digestibility and very small size of starch granules from Colocasia (taro, old cocoyam), this crop can provide a good base for babyfoods, and the leaves are an excellent vegetable. It is a versatile crop which can be grown in both lowland and upland conditions.

Xanthosoma sp (tannia), unlike Colocasia, cannot stand waterlogging but it is a partial shade-loving crop. The cormels are rich in carbohydrates.

Individual Crop Data

Dioscorea esculenta (Lour) Burk.
English: Lesser yam, Lesser Asiatic yam, Chinese yam; French: Igname des blancs

Annual. Harvested part: tubers.
Planted: March/April (India), December/February (Africa).
Harvested 8 months after planting.
Plant density 12 500-17 000 plants/ha.
Prefers light friable sandy or gravelly soil, well drained and rich in organic matter. Generally rainfed.

Dioscorea alata L.
English: Greater yam, Water yam, Winged yam; French: Igname de Chine

Annual. Harvested part: tubers.
Planted: March/April (India), December/February (Africa).
Harvested 9-10 months after planting.

Plant density 10 000-12 500 plants/ha.
Can tolerate poorer soils than most other edible yams. Sandy loams suit very well. Responds well to manuring. Generally rainfed.

Dioscorea rotundata Poir
English: White yam, Guinea yam, Eight months yam; Spanish: Name blanco

Annual. Harvested part: tubers.
Planted: March/April (India), December/February (Africa).
Harvested 9-10 months after planting.
Plant density 10 000-12 500 plants/ha.
Prefers heavy soils and can tolerate high clay content.
Generally unirrigated.

Amorphophallus campanulatus (Roxb) Bl.
English: Elephant foot yam, Elephant yam, Sweet yam;

Annual. Harvested part: underground corms.
Planted: March/April (India):
Harvested 10 months after planting.
Plant density 10 000-12 500 plants/ha.
Prefers deep loamy soil worked to a fine tilth, not alkaline. Good drainage essential. Generally rainfed, but responds to irrigation.

Colocasia esculenta (L) Schott.
English: Taro, Old cocoyam; French: Taros; Spanish: Malanga (Colocasia)

Annual. Harvested part: corm and cormels.
Planting time not a major factor provided moisture and sunlight adequate, but, if rainfed, should be timed to precede or coincide with rainy season.
Harvest time depends on cultivar and method of cultivation, ranging from 6-8 months after planting (India, Nigeria) to 12-15 months after planting (Hawaii).
Plant density 27 000-49 000 plants/ha.
Grown in all kinds of soil types but prefers deep, well drained, friable loams, especially alluvial loams with high water-table. pH 5.5-6.5 preferred. Primarily adapted to moist environment but can be grown in wide range of conditions.

Xanthosoma sagittifolium (L.) Schott
English: Tannia, New cocoyam; French: Tannie, Chou caraibe; Spanish: Malanga amarilla, Mangarito

Annual. Harvested part: corm and cormels.
Planting time depends on adequate moisture; can be planted throughout the year where water supply is assured.
Harvested 9-12 months after planting.
Plant density 12 345-27 000 plants/ha.
Prefers deep well drained soils with pH 5.5-6.5. Can tolerate saline soils, but not water logging. Should be irrigated in the absence of adequate rainfall to produce high yields.

Nutrient demand/uptake/removal

Nutrient uptake/removal - Macronutrients								
Species	**Tuber yield t/ha**	**Source**	**kg/ha**					
			N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
Dioscorea alata (Greater yam)	35.95	Obigbesan and Agboola, 1978	128.6	38.9	195.0	18.7	5.6	-
	18.60	Kabeerathumma et al, 1987	75.0	21.3	108.8	6.5	16.3	-
Dioscorea rotundata (White yam)	26.90	Sobulo, 1972	205.0	29.8	135.0	-	-	-
	51.60	Irizarry and Rivera, 1985	189.0	57.3	259.0	61.0	127.4	-
	30.00	Kabeerathumma et al, 1987	91.2	34.4	124.9	9.8	25.8	-
	37.90	Obigbesan and Agboola, 1978	148.0	41.2	199.2	18.6	5.6	-
D. esculenta (Lesser yam)	17.87	Kabeerathumma et al, 1987	73.0	25.8	77.3	10.5	19.8	-
D. cayenensis (Yellow yam)	45.13	Obigbesan and Agboola, 1978	139.0	43.5	218.0	22.0	5.6	-
Amorphophallus campanulatus (Elephant foot yam)	33.0	Kabeerathumma et al, 1987	100.0	37.9	267.1	18.6	38.2	-
Colocasia esculenta (Taro)	17.0	Kabeerathumma et al, 1987	59.5 (124.8)	26.4 (44.0)	71.4 (162.0)	8.9 (26.8)	11.5 (45.6)	4.5 (11.8)*
	12.3	Mohankumar, 1986	29.0 (59.1)	10.3 (18.1)	37.2 (86.4)*	-	-	-
	34.9	Portieles et al, 1982	153.4	115.2	422.3	-	-	-
Xanthosoma sp. -Corm + cormel	21.1	Vicente-Chandler et al, 1982	100.0	28.6	145.2	40.0	18.2	-
-Edible cormel	11.5		(125.0)	(37.0)	(187.0)	(52.4)	(35.0)*	-

* Figures in parenthesis indicate total uptake inclusive of both tuberous and non-tuberous portion.

Nutrient uptake/removal - Micronutrients						
Species	**Tuber yield t/ha**	**Source**	**g/ha**			
			Fe	**Mn**	**Zn**	**Cu**
Dioscorea alata (Greater yam)	18.60	Kabeerathumma et al, 1987	1 400	180	134	157
Dioscorea rotundata (White yam)	30.00	Kabeerathumma et al, 1987	3 000	334	290	267
D. esculenta (Lesser yam)	17.87	Kabeerathumma et al, 1987	2 000	320	280	114
Amorphophallus ampanulatus (Elephant foot yam)	33.0	Kabeerathumma et al, 1987	3 920	530	490	90
Colocasia esculenta (Taro)	17.0	Kabeerathumma et al, 1987	1 186 (10250)	370 (1690)	327 (647)	37 (93)*
* Figures in parenthesis indicate total uptake inclusive of both tuberous and non-tuberous portion.						

Plant analysis data

Plant analysis data - Macronutrients									
Species	**Plant part**	**Stage of growth**	**Source**	**% of dry matter**					
				N	**P**	**K**	**Mg**	**Ca**	**S**
Dioscorea esculenta	Leaf with petiole	3rd month	Kabeerathumma et al, 1987	1.85	0.18	1.46	0.22	0.47	-
	First fully matured leaf with petiole	5th month	Vanderzaag et al, 1980	2.70	0.39	2.60	0.23	1.10	0.15
D. alata	Leaf with petiole	3rd month	Kabeerathumma et al, 1987	1.86	0.20	2.27	0.26	0.65	-
	Lamina/petiole	4th month	Obigbesan and Agboola, 1978	3.71	0.38	4.80*P	0.28	1.05	-
	First fully matured leaf with petiole	4th month	Vanderzaag et al, 1980	2.90	0.22	3.1	0.33	1.8	0.17
D. rotundata	Leaf with petiole	3rd month	Kabeerathumma et al, 1987	1.94	0.16	2.02	0.27	0.54	-
	First fully matured leaf with petiole	4th month	Kang and Wilson, 1981	2.94	0.19	0.96	-	2.82	-
	Lamina/ petiole	4th month	Obigbesan and Agboola, 1978	3.4	0.30	4.0*(P)	0.27	0.7	-
	First fully matured leaf	4th month	Abruna et al, 1982	3.1	0.10	2.11	0.42	1.68	-
Amorphophallus campanulatus (elephant foot yam)	Leaf blade	5th month	Kabeerathumma et al, 1987	4.05	0.55	3.82	0.33	0.65	-
Colocasia esculenta	Fully expanded leaf	4th month	Mohankumar, 1986	3.67 (3.0)	0.36 (0.34)	2.43 (2.0)	-	-	-
	Lamina	3rd month	Kabeerathumma et al, 1987	3.67	0.37	3.50	0.55	0.80	0.33
	Leaf blade of upland taro	3rd month	De la Pena & Plucknett, 1976	4.45	0.32	5.07	0.19	1.20	-
	Leaf blade of lowland taro	6th month	De la Pena & Plucknett, 1976	4.96	0.57	4.53	0.62	1.43	-
	Leaf blade of 3rd or 4th leaf	2nd month	Miyasaka and Bartholomew,1979	-	-	-	-	0.4 (CYV)	-
	Leaf of flooded taro	4th month	De la Pena et al, 1979	-	0.4 (OS)	-	-	-	-
	3rd leaf	3rd month	Leucas et al, 1977	4.5 (CYV) 1.8 (D)	0.56 (OS) 0.4 (CVY)	6.2 (CYV) 1.5 (D)	0.09 (D)	0.55 (D)	50 ppm as sulphate sulphur (D)
Xanthosoma sp.	Lamina	6th month	Vicente - Chandler et al, 1982	3.2	0.5	2.3	0.9 (0.3) (0.9) (OS)	1.3	-

	3rd & 4th leaf	-	Abruna et al, 1982	3.07	0.23	2.65	0.56	1.25	-
D = Deficiency OS = Optimum supply CVY = Critical value (yield) P = petiole analysis									

Fertilizer Recommendations

Response to fertilizers depends mainly on the native fertility of the soil as well as the variety in cultivation. In traditional bush fallow agriculture in West Africa, yams were usually grown during the first season after clearance from bush, thus obtaining reasonably good yields without using fertilizer. However, as land scarcity causes the fallow period to be shorter, so yields decrease progressively, and traditional farmers will need less persuasion to use fertilizer on their yams. Also, despite the widespread belief that fertilizer causes rapid deterioration of tubers in storage, Umanah (1973) could find no significant effect of applied fertilizer on weight loss or decay of tubers. On the contrary, application of P2O5 and K2O has been found to enhance storage life by decreasing sprouting in storage. In general, yams respond well to N, P2O5 and K2O. Response to P2O5 has been reported to be slight, partly on account of its low requirement and partly due to mycorrhizal fungi which help the roots to absorb P effectively from the soil. For this reason some of the earlier recommendations ignored or underestimated P2O5. Characteristic recommendations were: in Nigeria 125-250 kg/ha compound fertilizer (e.g. 20:0:20, 10:10:20, 12:12:18) or ammonium sulphate only: in Trinidad 400 kg/ha 11:11:13 or 200 kg/ha ammonium sulphate. Recommendations for individual species according to yield level and region are given below.

Owing to the shallow rooting system of yams, split application of N and K2O is advised: e.g. the first dressing one month after sprouting, and the second about 7-12 weeks after germination, depending on the availability of rain, or 3-5 months after planting. With mound planting, the fertilizer should be applied in a ring of about 15 cm radius around the plant. With ridge planting, it should be applied in a continuous band along the row about 10 cm away from the plants. The use of mulch is beneficial. Capping mounds with dry leaves or straw keeps the soil cool and minimizes moisture loss. In India addition of ash from burnt trash, with f.y.m. and dry leaf mulch is a common practice; 10-12 t/ha f.y.m. is beneficial. For elephant yams f.y.m. is applied at 25 t/ha, and a mulch of green leaves at 4-5 kg/plant.

For cocoyams, pig manure and cowdung were the organic manures commonly used. Fertilizer use was restricted by the belief that it results in the production of soft, watery corms with a lower market value; and palatability was found to be affected by high rates of fertilizer use. Evidence suggests, however, that these ill effects occur only when fertilizer is applied too late in the growth cycle. At least 3 months should be allowed between the application of the last dressing and harvest. Yield responses of cocoyams to fertilizer vary with soil conditions and method of cultivation: several reports of good responses have been published. Mulching is commonly practised; a mulch of 10-12 t/ha green manure has been reported to be particularly beneficial.

Preferred fertilizers used

N - urea or ammonium sulphate,
P - single and triple super phosphate,
K - mainly supplied as chloride; can also be supplied as sulphate.

Fertilizer recommendations for individual species:

Dioscorea esculenta

	Kg/ha			
	N	P2O5	K2O	MgO
(medium yield 15-20 t/ha, Kerala, India)				
On lateritic sandy clay loam (ultisol)				
Within 1 week after sprouting	40	60	40	-
One month later	40		40	-
On forest soil rich in organic matter:				
Within 1 week after sprouting	30	60	45	-
One month later	30		45	-

Dioscorea alata

	kg/ha			
	N	P2O5	K2O	MgO
(medium yield 12 1/2-25 t/ha, Kerala, India)				
On lateritic sandy clay loam (ultisol):				
Within 1 week after sprouting	40	60	40	-
One month later	40		40	-
(high yield > 25 t/ha, Puerto Rico)				
On acid corozal clay soil (aquic tropodults), limed to pH >5.5:				
At planting	500 kg/ha 10:10:10 mixture containing 30 kg/t of a mixture of 6 % Mg, 7.7 % Mn, 4.8 % Cu, 7 % Fe, 8 % Zn and 2.5 % B.			
After 5 months	500 kg/ha of same mixture			

Dioscorea rotundata

	kg/ha			
	N	P2O5	K2O	MgO
(low yield < 15 t/ha, Nigeria)				
On gravelly sandy loam:				
At planting	-	13	25	-
4 weeks after planting	30	-	-	-
10 weeks after planting	30	-	-	-
On light loamy ultisol:				
8 weeks after planting	35	25	40	-
(medium yield 25-30 t/ha, Kerala, India)				
On lateritic sandy clay loam (ultisol):				
Within 1 week after sprouting	40	60	40	-
One month later	40	-	40	-
(high yield > 30 t/ha, Puerto Rico)				
On Corozal clay soil (ultisol):				
2 months after planting	112	34	129	28
5 months after planting	112	34	129	28
Use of a 14:3:13:3 compound fertilizer is considered suitable.				
On Coto soil (oxisol):				
1 month after planting	102	34	85	-
3 months after planting	102	-	85	-

Amorphophallus campanulatus

	kg/ha			
	N	P2O5	K2O	MgO
(Kerala, India) **On lateritic sandy clay loam (ultisol):** 2 weeks after germination 1 month later	 50 50	 50 -	 75 75	 - -
Mulching with 10-12 t/ha green leaves is common practice.				

Colocasia esculenta

	kg/ha			
	N	P2O5	K2O	MgO
Upland (high yield, 6-7 month crop, Kerala, India) **On sandy clay loam soil with medium fertility:** 1 month after sprouting One month later	 40 40	 50 -	 50 50	 - -
Upland (high yield, 12 months crop, Hawaii) **On humic latosol:** At planting 2 months after planting 4 months after planting	 187 187 187	 280 - -	 93 93 93	 - - -
Lowland (high yield, 15 months crop, Hawaii) **On lowland paddy with yield potential > 50 t/ha:** At planting 2 months after planting 4 months after planting	 373 373 373	 280 - -	 93 93 93	 - - -
Lowland (high yield, Philippines) **On soil with yield potential 30-40 t/ha:** At planting 2 months after planting	 30 30	 15 15	 30 30	 - -

Colocasia, improved varieties

	kg/ha			
	N	P2O5	K2O	MgO
Johore Region, Malaysia **9-11 months crop on peat soil:** 1 month after planting 3 times at 5-weekly intervals 6 1/2 months after planting	 46 46 N = 85 (as ammonium sulphate)	 21.5 16	 46 23	 - 2.7
Perak Region, Malaysia; 1 month after planting 3 times at 5-weekly intervals 6 1/2 months after planting	 40 20 85 (as ammonium sulphate)	 40 20	 40 20	 - -
Fiji (Upland) **On humic latosol and alluvial soils with yields of 20-24 t/ha:** 6 weeks after planting 12 weeks after planting	 50 50	 25	 100	 -

Xanthosoma sp.

	kg/ha			
	N	**P2O5**	**K2O**	**MgO**
(medium yield, Kerala, India) **On lateritic sandy clay loam (ultisol):** 1 month after sprouting One month later	 40 40	 50 -	 50 50	 - -
(medium yield, Puerto Rico) **On Corozal clay soil, pH 5.8 :** At planting After 6 months **On clay soil (Vertic eutropepts):** 2 months after planting 6 months after planting	 500 kg/ha 10:10:10 mixture containing 30 kg/t of a mixture of 6 % Mg, 7.7 % Mn, 4.8 % Cu, 7 % Fe, 8 % Zn and 2.5 % B. 500 kg/ha of same mixture 500 kg/ha 10:5:15:3 mixture 500 kg/ha of same mixture			

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Author: S. Kabeerathumma, Central Tuber Crops Research Institute, Sreekariyam, Trivandrum, India

Chickpea *(Cicer arietinum L.)*

French: Pois chiche; Spanish: Garbanzo; Italian: Cece; German: Kichererbse
Local: Bengal gram, Chana, Homs

Crop data

Annual. Harvested part: seeds. Grown in dry climates, traditionally during the post-rainy season in South and Southeast Asia; during the winter rainfall season in West Asia, Middle East and South Mediterranean Region and in springtime in North America and North Mediterranean; benefits from deep sowing (10-12 cm) in unirrigated conditions.

Flowers 2-3 months after sowing.

Plants harvested 2.5-6 months after sowing (depending upon cultivars and latitude) at maturity for seed threshing.

Plant density: 150 000 (intended, but often much less due to poor seedling establishment) to 400 000 (irrigated) per ha; often intercropped with wheat, barley, sorghum, safflower or other pulses, but best results obtained with mustard or a chickpea-rice sequence.

Preferably grown on clay loam to sandy loam soils, well-drained, pH>7, but is often confined to poor lands and heavy black or red soils. Needs adequate soil moisture (15-40 %) during germination and seed development; excessive soil moisture at flowering reduces grain yield. It is generally unirrigated, but the area under irrigation is increasing due to responsiveness to appropriate water management.

Nutrient uptake/demand/removal

Nutrient uptake and removal - Macronutrients									
Yield kg/ha	**Plant part**	**Source**	**Location**	**kg/ha**					
				N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
1 373	Grain	Saxena,1984	Hyderabad	82	18	76	-	-	-
3 726	Grain	"	Hisar	200	32	201	-	-	-
1 500	Grain	Aulakh et al, 1985	India	91	14	60	18	39	9
1 150	Grain	Horton,1990	Australia	41	10	14	8	3	2

Nutrient uptake and removal - Micronutrients							
Yield kg/ha	Plant part	Source	Location	**g/ha**			
				Fe	**Mn**	**Zn**	**Cu**
1 500	Grain	Aulakh et al, 1985	India	1 302	105	57	17

Plant analysis

Plant analysis data (whole shoots) - Macro- and micronutrients								
Supply	Source	**% of dry matter**			**ppm dry matter**			
		N	**P**	**S**	**Mn**	**Cu**	**Zn**	**B**
Optimum	Robinson, 1986	-	0.26	-	120	4-35	12-500	-
Critical value (deficiency)	Reuter, 1986	2.3	0.24	0.15	-	-	-	40

Nutritional requirements:

Understanding of the mineral nutrition of chickpea is inadequate, but as marginal lands are often assigned to the crop or it has to rely on residual fertilizer from the previous crop, mineral deficiencies are commonly noticed.

It tolerates acid soils, but is very sensitive to salinity, alkalinity and high calcareousness. It requires quite high levels of soil N, especially under rainfed conditions if drought and deep sowing are reducing biological N fixation. It is well-adapted to high levels of available P, but requires fairly low levels of K. Deficiency symptoms of N, P, K and Mg appear initially in the older leaves. The crop is also susceptible to Fe, Zn, S, Cu and Mo deficiencies, symptoms generally appearing in young leaves.

Fertilizer recommendations

Fertilizers are generally applied at planting, broadcast and incorporated in the soil by a rough tillage as a basal NPKS dressing including some micronutrients, followed by a top dressing of N at flowering, when soil moisture is adequate. Rates depend on soil types and cropping system. Yield may be increased by seed inoculation if the crop is cultivated in new areas and/or after rice.

Preferred nutrient forms

N: preferably as urea. If inoculation is recommended, a suitable peat-based Rhizobium inoculant should be given.
P: better placed in the soil or as a foliar spray than broadcast. In a crop sequence, single or triple superphosphate, applied during the rainy season crop before chickpea, gives a better response than rock phosphate.
K: as chloride or sulphate, but preferably the latter if there is a risk of salinity.
S: should be applied separately if the regular macronutrient fertilizers are sulphur-free (e.g. urea, AN, TSP, DAP, concentrated NPK). Better broadcast and tilled in than placed. In irrigated soils, gypsum (18 % S) used to correct salinity problems is a good source of S.
Zn: as sulphate applied to the soil at planting.
Mo: as sodium or ammonium molybdate, preferably in seed coating.
Fe: as ferrous sulphate foliar spray if iron deficiency symptoms are identified.
Mn and/or Cu: in sulphate form as foliar spray or soil application.

Present fertilizer practices

- Australia
(NE Australia): 27-46 kg/ha P2O5 as double superphosphate, or 3-5 kg/ha N, 27-46 kg/ha P2O5 as monoammonium phosphate.

- India
(Tradional practices): Generally in rainfed areas no fertilizers or manures are applied to pulse crops; in some places farmers may apply a small amount of FYM (e.g. 8-15 t/ha) 3-4 weeks before sowing.
In irrigated areas, a small amount of mineral fertilizer (10-15 kg/ha N, 20-30 kg/ha P2O5) is either placed in the seed furrow or broadcast and mixed in.

(Improved practices): Irrigated areas, basal application of 18-20 kg/ha N, 40-50 kg/ha P2O5, plus K, Zn, S if required;
unirrigated areas, basal application of 10-15 kg/ha N, 20-30 kg/ha P2O5, followed by a foliar spray of 2 % urea at pod development.

- Nepal

20 kg/ha N as urea, 40 kg/ha P_2O_5 as single or triple superphosphate, 30 kg/ha K_2O as muriate of potash.

- Tunisia
45 kg/ha P_2O_5 as triple superphosphate broadcast and incorporated at planting. Occasionally 20 kg/ha N side-dressed, in the form of urea, near the plant at shooting.

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Contributor: Masood Ali, Directorate of Pulses Research, Kanpur, India

Field Bean [Broadbean, Fababean] *(Vicia faba L.)*

French: Feverole; Spanish: Haba comun; Italian: Fava; German: Ackerbohne

Crop data

Annual. Harvested part: grain (straw).
Sown autumn or early spring.
Flowers 7-8 months (winter variety) or 2-4 months (spring variety) after sowing.
Harvested 9-10 months (winter variety) or 5-7 months (spring variety) after sowing.
Target plant density 20-25 plants/m2 (winter variety) or 40-50 (traditional: 15) plants/m2 (spring variety).
Preferably grown on medium to heavy well drained, deep soil, pH 6-8.
Very sensitive to lack of water from flowering time.

Optimum soil requirements

Organic matter > 1.5 %; calcium carbonate < 4 % EC; conductivity < 4 mmohs/cm;
P (Olsen extractant) > about 20 ppm; K (ammonium acetate) > 66 ppm or (sodium acetate) > 75 ppm; available Fe and Mn > 5 ppm, available Zn > 1.5 ppm.
Susceptible to K deficiency (clayey soils), to P, Zn, Mn and/or Fe deficiency (intensive cropping in arid/semi-arid regions).

Nutrient demand/uptake

The crop requires about 300 kg/ha N (60 kg N/t beans), coming mainly from N fixed from the atmosphere.

Maximum uptake of P and K (around 70-80 kg/ha P2O5 and 250-300 kg/ha K2O) occurs about 100 days after sowing (spring variety), but a substantial proportion of the K taken up is returned to the soil later.

Nutrient removal in harvested beans and haulms - Macronutrients							
Yield/ha	**Source**	**kg/ha**					
		N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
6.5 t beans	El-Fouly et al,	240	49	94	-	-	-
21.0 t total	1989	405	102	258	-	-	-
6.0 t beans	Taureau, 1984	-	69	81	10	-	9
5.0 t beans	Poulain, 1989	-	65	91	11	7	-
3.7 t haulms	Poulain, 1989	-	5	64	6	40	-
4.0 t beans	Finck, 1982						
10.0 t total		240	73	192	55	120	28
3.1 t beans	NRC-GTZ, 1991						
9.5 t total		228	59	156	74	-	-

Nutrient removal in harvested beans and haulms - Micronutrients					
Yield/ha	**Source**	**g/ha**			
		Fe	**Mn**	**Zn**	**Cu**
6.0 t beans	Taureau, 1984	-	68	145	60
5.0 t beans	Poulain, 1989	-	75	255	75
3.7 t haulms	Poulain, 1989	-	81	66	10
3.1 t beans	NRC-GTZ, 1991				
9.5 t total		12 000	800	500	100

Plant analysis data

Nutritional disorders may be diagnosed by soil testing, visual deficiency symptoms (which, unfortunately, can often be confused with pathological or physiological disorders) or plant tissue analysis. Critical concentrations in the plant dry matter are summarized below. The youngest fully expanded trifoliate leaf blades without petiole at the 3rd node from stem apex, at 10 % flowering provide the best indicator tissue.

Plant analysis data (2nd/3rd leaf from apex at initial flowering) - Macronutrients					
Supply	**Source**	**% of dry matter**			
		N	**P**	**K**	**Mg**
Optimum	Neubert, 1976	3.0-4.5	0.22-0.3	1.0-1.75	-
	Reuter, 1988	4.8-5.3	0.3 -0.4	2.2-3.2	-
	NRC-GTZ, 1991	4.5-5.5	0.25-0.4	2.1-3.5	-
	Chapman, 1973	-	-	-	0.33-0.6
Deficient	NRC-GTZ, 1991	2.8	0.19	1.5	0.2

Plant analysis data (2nd/3rd leaf from apex at initial flowering) - Micronutrients					
Supply	**Source**	**ppm dry matter**			
		Fe	**Mn**	**Zn**	**Cu**
Optimum	NRC-GTZ, 1991	100-?*	58-200	40-75	13-20
	Chapman, 1973	-	68	-	-
Deficient	NRC-GTZ, 1991	?*	40	2	7
* Total Fe in leaves is not reliable for diagnosis.					

Deficiency symptoms

N: uniform yellowness and poor growth. May occur where root nodules have been destroyed by pea weevil larvae (Sitona lineatus) or where waterlogging and compacted soil conditions have damaged the root system.

P: dark green leaves; older leaves drop; thin stems; sparse flowering.

K: bluish green leaves, edges becoming black to brown, followed by marginal scorch; leaves form obtuse angle with stalk; short internodes.

Mg: leaves with central interveinal chlorosis and green margins, older leaves affected first.

Ca: blackening and death of plant.

B: terminal bud blackens and dies downward from tip; dark green, leathery leaves which shed early; few secondary branchings of roots.

Mn: leaves practically normal, or slight interveinal chlorosis and brown flecking; growing point dies; brown spots in centres of cotyledons (as with marsh spot of peas), affecting quality of seed crops.

Zn: shedding of leaves and flower buds; failure of seed pods to develop.

Mo: pale lower leaves; interveinal areas turn grey and wilt or collapse, followed by dark brown or black scorch; stems become defoliated.

Fertilizer recommendations

N: Many studies have shown that no fertilizer N is needed where Rhizobium leguminosarum is present in the soil. Fertilizer N reduces N fixation by the Rhizobium, and can promote excessive growth and stimulate weed competition. Although there appears to be no economic benefit from application of fertilizer N, it is used in some countries and applied to

the seed bed to aid establishment; a rate of 30-40 kg/ha N is enough but responses tend to be too erratic to make a recommendation. Where Rhizobium leguminosarum is absent, inoculation of the seed with bacterium is an alternative used in some countries.

P and K: Responses are dependent on residual levels in the soil, which depend in part on the previous cropping history. K is one of the most important major nutrients for field beans; although average responses are small, they are much larger where the soil is poor in readily available K. General recommendations are 80-120 kg/ha P2O5 and 100-200 kg/ha K2O, about one fifth of the P2O5 and one half of the K2O being considered as an advance for the next crop.

Secondary nutrients: S deficiency occurs widely throughout the world; for optimum production, soils should contain at least 10 ppm sulphate S and, below that level, 15-20 kg/ha S should be applied (generally, superphosphate with 18-25 % P2O5 brings enough S for crop maintenance). Mg deficiency is rare but can be corrected by application to soil, or to leaves as 5 % magnesium sulphate spray. Ca deficiency is very rare; uptake of 120 kg/ha by the plant is mostly returned to the soil at harvest time (but see above for the effect of acidity).

Micronutrients: only limited research has been conducted but where deficiencies are diagnosed they can mostly be readily corrected. Soils with < 0.5 water-sol. boron should be given 1-2 kg/ha B (McDole & Mahler 1984) but repeated foliar applications should never exceed 0.2 kg/ha B (Mahler et al. 1981). Co deficiency has been detected in some countries (Tisdale et al. 1985) and can be corrected by application of cobalt sulphate at 0.03-0.15 kg/ha Co. Cu deficiency is rare except perhaps on soils rich in organic matter; mineral soils should contain more than 4-6 ppm Cu (Lucas & Knezek 1972), Bull & Mahler 1983); in general, 3-8 kg/ha Cu is recommended for application to deficient soils, or foliar sprays of 15 % copper oxychloride, 1 % un-neutralized copper sulphate or 5 % neutralized copper sulphate. Fe deficiencies are temporary and occasionally encountered on alkaline soils, but are difficult to correct; foliar applications of chelate sulphate Fe at minimum 1 kg/ha Fe are usually given. Mn deficiency is mostly associated with alkaline soils or where lime has been applied, or sometimes with poor drainage; it can usually be corrected by several applications of manganese sulphate at 5 kg/ha in at least 300 l/ha water at the beginning of flowering. Mo deficiency is extremely difficult to distinguish from nutritional problems with N and S; it can be corrected by seed treament at 200-400 g/ha Mo or foliar application at 100-200 g/ha Mo. Zn deficiency is rare in acid soil but appears on alkaline soils (pH > 7), where the organic matter layer has been severely disturbed or levelled for irrigation; applications of 5 kg/ha Zn should be considered where deficiencies are known to occur.

Present fertilizer practices

Egypt

(Traditional varieties, 2.5 t/ha beans):
Basal fertilization, banded in the seedrows and incorporated in upper 10 cm of soil

36 kg/ha N
72 kg/ha P2O5

(Improved varieties, 5 t/ha beans, on medium and heavy soils):
Basal fertilization

36 kg/ha N
72 kg/ha P2O5

Two topdressings, 40 and 70 days after sowing, each of

57 kg/ha K2O
Micronutrient spray (60 g/ha Zn, 40 g/ha Mn, 20 g/ha Fe)

Seeds should be inoculated with Rhizobium before sowing.

United Kingdom.

The rates of application recommended by the Ministry of Agriculture, Fisheries and Food in relation to residual fertility (soil index) are given below. Experimental work indicates that these are good estimates which can be applied in most countries.

Soil index	Soil analysis values			Application rates		
	(mg/l)			(kg/ha)		
N, P, or K	Nitrate N	P	K	N	P2O5	K2O
0 very low	0-25	0-9	0-60	0	75	120
1 low	26-50	10-15	61-120	0	50	50
2 medium	51-100	16-25	121-240	0	0*	0*
3 high	100	25	240	0	0*	0*
* Maintenance treatment of 40 kg/ha may be applied to replace removal in crop.						

Survey work in the U.K. has shown that about 40 % of the field bean area receives no fertilizer because most U.K. soils on which the crop is grown have good residual fertility. A few research workers have been able to identify positive interactions between P and K at the levels applied, but this is likely to occur only where the base level of one of these nutrients is low.

China

Single superphosphate is used in most areas at about 375 kg/ha with 4.5 t/ha f.y.m. (equiv. 22 kg N, 12 kg P2O5, 19 kg K2O per ha). After seedling emergence, about 2.25 t/ha wood ash is applied (equiv. about 25 kg P2O5 and 150 kg K2O per ha). Finally, at 10 % flowering stage, an average of 7.5 t/ha f.y.m. is given (equiv. 36 kg N, 46 kg P2O5, 38 kg K2O per ha).

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Author: Ph. Plancquaert, Institut Technique des Céréales et des Fourrages (ITCF), Paris, France
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Field Pea *(Pisum sativum L.)*

French: Pois proteagineux; Spanish: Guisante; Italian: Pisello; German: Futtererbse

Crop data

Annual. Harvested part: grain.
Sown autumn or early spring.
Flowers 7-8 months (winter variety) or 3-4 months (spring variety) after sowing.
Harvested 8-9 months (winter variety) or 5-6 months (spring variety) after sowing.
Target plant density about 60-80 plants/m2.
Preferably grown on well drained, deep soil, without large stones, pH >6 (optimum pH 7).
Very sensitive to lack of water at flowering time.

Nutrient demand/uptake/removal

The crop needs about 50 kg N per metric ton of peas; generally, N in the soil or fixed by the plant is sufficient, and no fertilizer N is required. Field peas are, however, demanding in P and K; the total uptake of these two nutrients reaches a maximum at flowering (about 80 days after a spring sowing), after harvest a substantial amount returns to the soil. Maximum uptake for a crop yielding 5-6 t/ha is about 70-80 kg/ha P2O5 and 250-300 kg/ha K2O.

Nutrient demand/removal - Macronutrients								
Yield/ha	**Source**	**kg/ha**						
		N	**P2O5**	**K2O**	**MgO**	**CaO**	**Na2O**	**SO3**
6 t grain	Poulain et al,	-	63	115	13	7	0.4	-
3 t haulm	1989	-	4	80	9	90	4	-
6 t grain	TCF, 1984	-	60	90	15	-	-	30
3 t haulm		-	15	100	5	-	-	9
Maximum uptake								
Total plant	Poulain et al,	280	80	290	36	150	6	-
6 t Grain	1989	187	66	90	13	10	5	-
6 t Grain	Taureau, 1984	-	80	300	30	-	-	50

Nutrient demand/removal - Micronutrients				
Yield/ha	**Source**	**g/ha**		
		Cu	**Zn**	**Mn**
6 t grain	Poulain et al,	54	250	66
3 t haulm	1989	15	69	90
6 t grain	ITCF, 1984	84	-	90
3 t haulm		38	-	120
Maximum uptake				
Total plant	Poulain et al,	111	659	311
6 t Grain	1989	62	303	75
6 t Grain	Taureau, 1984	161	160	325

Plant analysis data

Nutritional disorders may be diagnosed by soil testing, by visual deficiency symptoms (which, unfortunately, can easily be confused with pathological and physiological disorders) or by plant tissue analysis. Critical concentrations in the plant dry matter are summarized below.

Peas mineral composition - Macronutrients								
Part	**Source**	**% of dry matter**						
		N	**P**	**K**	**Mg**	**Ca**	**Na**	**S**
Dry peas	Poulain et al, 1989	4.2	0.5	1.6	0.14	0.01	0.01	-
	Paul et al, 1978	3.5	0.3	1.0	0.12	0.06	0.04	0.13
Dry haulm	Poulain et al, 1989	1.1	0.12	2.3	0.22	1.7	0.11	-
	Coppenet, 1984	0.8	0.07	2.5	0.15	1.7	0.10	-
Mature leaves	Several authors (N)	0.2-0.3	0.02-0.05	0.06-0.14	0.01-0.03	0.08-0.22	- -	0.01-0.03
	ADAS (N)	-	>0.04	>0.2	>0.02	-	-	>0.02
Third leave at 4-8 node stage	Muehlbauer et al, 1989 (N)	0.5	0.04	0.25	0.04	0.1-0.3	-	0.06
	(D)	0.2	0.01	0.15	0.01	0.07	-	0.03
N = Normal; D = Deficiency								

Peas mineral composition - Micronutrients							
Part	**Source**	**ppm dry matter**					
		Cu	**Zn**	**Mn**	**Fe**	**B**	**Co**
Dry peas	Poulain et al, 1989	8.6	37	12	-	-	-
	Paul et al, 1978	5.0	35	-	-	-	-
Dry haulm	Poulain et al, 1989	7.0	50	42	-	-	-
	Coppenet, 1984	4.9	17	24	-	-	-
Mature leaves	Several authors						
	N	6-20	25-80	20-350	50 -	20-85	0.2-0.5
	D	-	<20	<20	-	<15	-
	E	-	>400	>500	-	>100	-
	ADAS N	>5	-	>20	>50	>20	-
Third leave at 4-8 node stage	Muehlbauer et al, 1989						
	N	33	75	110	0.05	-	-
	D	20	15	50	-	-	-
N = Normal; D = Deficiency; E = Excess							

Soil analysis

Critical soil levels of pH and mineral nutrients for acceptable yields are not absolute and can vary with soil type, moisture level and method of nutrient extraction. N-fixation in root nodules is sensitive to extremes of pH: nodulation and N-fixation are reduced where pH < 5.5 or with excessive salinity. As with field beans, soil P (Olsen extractant) should be at least about 20 ppm (Haddock & Luiton) and K (ammonium acetate) 66 ppm (Cutcliffe & Mantro) or 75 ppm (sodium acetate, McDole & Mahler).

Deficiency symptoms

See chapter 4.2 Field Bean.

Fertilizer recommendations

N: Many studies have shown that no fertilizer N is needed where Rhizobium leguminosarum is present in the soil. Where Rhizobium leguminosarum is absent, inoculation of the seed with bacterium is an alternative used in some countries.
P and K: Responses are dependent on residual levels in the soil. K is one of the most important nutrients; although average responses are small, they are much larger where the soil is poor in readily available K. General recommendations are 80-120 kg/ha P2O5 and 100-200 kg/ha K2O, broadcast before ploughing or cultivation before sowing, part being considered as an advance for the benefit of the next crop.

S deficiency seems to occur widely throughout the world in non-industrial areas; for optimum production, soils should contain at least 10 ppm sulphate S and, below that level, 15-20 kg/ha S should be applied (generally, superphosphate with 18-25 % P2O5 brings enough S for crop maintenance).
The other deficiencies of secondary and miconutrients occuring in Western Europe are those of: Mg, rare (application recommended when soils contain less than 0.5 meq Mg/100 g); Mn, frequent; Fe, temporary in some seasons. The occurrence and correction of micronutrient deficiencies are generally similar to those of field beans.

Present fertilizer practices

United Kingdom.

The rates of application recommended by the Ministry of Agriculture, Fisheries and Food in relation to residual fertility (soil index) are given below.

United Kingdom - Recommended rates of application								
Soil index N, P or K	Soil analysis values ppm			Application rates - kg/ha				
				Broadcast			Combine-drilled	
	NO3-N	P	K	N	P2O5	K2O	P2O5	K2O
0 very low	0- 25	0- 9	0- 60	0	50	50	50	150*
1 low	26- 50	10-15	61-120	0	25	40	25	50
2 medium	51-100	16-25	121-240	0	0	0	0	40
3 high	> 100	>25	>240	0	0	0	0	0
* Only 50 kg/ha K2O should be combine-drilled and the rest broadcast								

Most fertilizer is at present broadcast; with narrow row-spacing there is less advantage in combine-drilling. Fertilizers with enough P and K to correct deficiencies are worked into the soil in early spring.

France

France - Recommended rates of application on chalk soils (Crop rotation: Sugar beet-peas-wheat)						
	kg/ha					
	P2O5*			K2O**		
Soil/status/ppm	<120	180-200	>250	<100	180-200	>250
Sugar beet	200	180	100	400	200-250	180-200
Peas	100-150	0	0	0	0	0
Wheat	0	80	0	80	80	0***
* method Joret Liebat (oxalate) ** NH4-acetate method *** or 80 kg/ha K2O as maintenance application						

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Author: Ph. Plancquaert, Institut Technique des Céréales et des Fourrages (ITCF),Paris, France

Lentil *(Lens culinaris Medik.)*

French: Lentille; Spanish: Lenteja; Italian: Lenticchia; German: Linse
Local: Masur, Adas

Crop data

Annual. Harvested part: seeds.

As a cool season crop, lentils are grown in warm temperate, subtropical regions and in high altitudes of the tropics, traditionally during the winter in Asia, South and Central America, in spring in the Middle East and the South Mediterranean Region and in summer in the North Mediterranean Region and North America; seed is usually broadcast or sown by drilling in rows 25-30 cm apart.

Flowers 2-3 months after sowing.

Harvested: 3-51/2 months after sowing, at maturity with low moisture content (for drying for 1 week and then threshing).

Plant density: traditional, depending on cropping type, 200 000 (intended, but often much less due to poor seedling establishment resulting from disease or limited moisture) to 300 000 (irrigated)/ha.
South Mediterranean Region, 120 000 - 300 000/ha (often grown alone but may be intercropped with wheat, barley, rice, mustard.

Grows best on well-drained soils, but tolerates a wide range of soil conditions from black cotton and clay soils to sandy loams. Preferred pH 6.5-8.0; tolerates alkalinity. Usually rainfed, can resist drought but does not withstand poor drainage.

Nutrient demand/uptake/removal

Nutrient uptake/removal - Macronutrients					
Yield kg/ha	**Plant part**	**Source**	**kg/ha**		
			N	**P2O5**	**K2O**
2 000	Grain	Saxena, 1981	100	28	78
1 800	Grain	Prasad et al, 1990	108	19	55
1 000	Grain	Horton et al, 1990	44	11	-

Plant analysis

Plant analysis data - Macronutrients								
Stage	**Plant part**	**Source**	**% of dry matter**					
			N	**P**	**K**	**Mg**	**Ca**	**S**
45 DAS	Whole plant	Dwivedi, 1986	-	0.39	2.26	0.61	1.33	-

Plant analysis data - Micronutrients							
Stage	**Plant part**	**Source**	**ppm dry matter**				
			Fe	**Mn**	**Zn**	**Cu**	**B**
45 DAS	Whole plant	Dwivedi, 1986	134	45	25	-	-

Nutritional requirements

Many cultivars are salt-tolerant, especially at germination and sprouting. The plant is also adapted to low levels of available nutrients but thrives on fairly high available P.

Fertilizer recommendations

Fertilizers are usually applied at planting, generally as a basal P dressing. If the soils are poor in specific N-fixing bacteria, a basal NP dressing will be recommended, with a top dressing of N at flowering 2.5-3 months after sowing. Advice on K is very empirical; on-farm tests and recording are encouraged. If K is applied, it will be in the form of a basal NPK dressing. Rates depend on soil class and expected yield.

Preferred nutrient forms

N: as urea.
P: in soluble form as single or triple superphosphate. If both N and P are required, DAP is considered suitable.
K: as chloride or sulphate.

Present fertilizer practices

Australia

(Southern NSW, Victoria, South Australia): 27-46 kg/ha P2O5 as double superphosphate, or 3-5 kg/ha N, 27-46 kg/ha P2O5 as monoammonium phosphate.

India

(Tradional practices): Generally in rainfed areas no fertilizers or manures are applied to pulse crops; in some places farmers may apply a small amount of FYM (e.g. 8-15 t/ha) 3-4 weeks before sowing.
In irrigated areas, a small amount of mineral fertilizer (10-15 kg/ha N, 20-30 kg/ha P2O5) is either placed in the seed furrow or broadcast and mixed in.

(Improved practices): Irrigated areas, basal application of 18-20 kg/ha N, 40-50 kg/ha P2O5, plus K, Zn, S if required;
unirrigated areas, basal application of 10-15 kg/ha N, 20-30 kg/ha P2O5, followed by a foliar spray of 2 % urea at pod development.

Tunisia: 45 kg/ha P2O5 as TSP broadcast and incorporated at planting.

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Tropical Pulse Crops

Crop data

Pigeonpea *(Cajanus cajan (L.) Millsp.)*

French: Ambrevade, pois d'Angole; Spanish: Arveja de Angola, guandul; Italian: Pisello d'Angola; German: Taubenerbse

Perennial in habit but cultivated as an annual crop.
Harvested part: pods.
Generally sown at the beginning of rainy season (direct seeding). Post rainy season planting also successful in eastern India and Nepal.

Flowers 2 to 6 months after sowing. Harvested 4 to 10 months after sowing.

Plant density 100 000 plants (traditional) to 300 000 plants (intensive) for short-season cultivars and 50 000 plants (traditional) to 80 000 plants (intensive) for long-season cultivars as sole crop. In intercropping systems, 40 000 plants per ha for long-season cultivars.

Preferably grown on Entisols, Vertisols and Inceptisols, pH 5.5 to 8.0

Adapted to tropical and sub-tropical environments. Long-season cultivars generally grown as a rainfed crop but short-season cultivars are irrigated.

Urdbean *(Vigna mungo (L.) Hepper)*

French: Haricot velu; Italian: Fagiolo Urd; German: Linsenbohne

and

Mungbean *(Vigna radiata (L.) Wilczek)*

French: Amberique; Spanish: Judía de Mungo; Italian: Fagiolo Mungo; German: Mungobohne

Annual. Harvested part: pods.
Generally sown at the beginning of a rainy season (direct seeding); also at end of rainy season (South east Asia) and in spring/summer and late winter (India).

Flowers 1 to 2 months after sowing. Harvested 2 to 4 months after sowing.
Plant density 200 000 plants (traditional) to 300 000 plants (intensive) per ha for rainy season crop and 600 000 to 800 000 plants per ha for spring/summer crop.

Preferably grown on Vertisols, Entisols and Alfisols, pH 5.5 to 8.5.

Well adapted to tropical and sub-tropical environments. Rainy season crop generally unirrigated but spring/summer crop is essentially irrigated.

Cowpea *(Vigna unguiculata (L.) Walp.)*

French: Dolic asperge; Spanish: Judía espárago; Italian: Fagiolino asparago; German: Spargelbohne

Annual. Harvested part: pods.
Sown at the beginning of a rainy season, post-rainy season and spring/summer (direct seeding)

Flowers 1 to 2 months after sowing. Harvested 2 to 4 months after sowing
Plant density 200 000 plants/ha (traditional) to 300 000 plants (intensive) per hectare for rainy season crop and 400 000 plants per ha for spring/summer crop.

Preferably grown on Vertisols and Entisols, pH 5.0 to 8.0.

Adapted to arid to sub-humid tropics. Rainy season crop unirrigated but spring/summer crop is essentially irrigated one.

Nutrient uptake/demand/removal

Nutrient uptake/demand - Macronutrients								
Crop	**Yield t**	**Source**	**kg/ha**					
			N	**P2O5**	**K20**	**MgO**	**CaO**	**S**
Pigeonpea	2.0	Ahlawat, 1980	93.5	25.4	65.6	-	-	-
	1.2	Aulakh et al, 1985	85.0	18.3	75.0	24.9	32.2	9.0
	3.1	Dalal & Quilt, 1977	318.0	98.4	130.8	98.0	-	48.0
Mungbean	0.7	Kothari, 1984	91.9	12.8	83.6	-	-	-

Nutrient uptake/demand - Micronutrients						
Crop	**Yield t**	**Source**	g/ha			
			Fe	**Mn**	**Zn**	**Cu**
Pigeonpea	1.2	Aulakh et al, 1985	1 440	128	38	31

Plant analysis data - Macronutrients									
Crop	**Stage**	**Plant part**	**Source**	**% of dry matter**					
				N	**P**	**K**	**Mg**	**Ca**	**S**
Pigeonpea	Maturity	Leaf	Singh et al, 1978	1.96	0.25	-	-	-	-
		Shoot		1.02	0.13	-	-	-	-
		Grain		3.52	0.51	-	-	-	-
				Optimum concentration:					
	40-50 DAS	Whole shoot	Burton & Edward, 1981 Nicholas, 1965, Rosbrook & Edward, 1981 Sheldrak & Narayanan, 1979	4.0 (0.3)*	0.35-0.38 -	1.2-2.6 (0.78)	0.28 (0.17)	1.32 (0.84)	0.16-0.32 -
Urdbean	30 DAS	Whole plant	Pareek et al, 1978	3.67	0.32	1.74	-	2.4	0.33
				Optimum concentration:					
	45-50 DAS	Whole shoot	Dwivedi & Randhawa, 1974 Nowlan, 1980 Dange et al, 1964	3.6 (3.3)*	0.38 (0.28)*	- -	- -	- -	- -
Mungbean	Maturity	Grain	Dhama, 1982	3.75	0.46	-	-	-	-
		Straw		1.48	0.28	-	-	-	-
Cowpea	60 DAS	Shoot	Venkat Reddy & saxena, 1983	1.60	0.36	3.94	-	-	-
		Leaf		3.89	0.40	2.10	-	-	-
				Optimum concentration:					
	45 DAS	Whole shoot	Fox et al, 1977 Kashirad et al, 1978	2.76 (2.4)	0.27-0.35 (0.19-0.24)	- -	- -	- -	0.35-0.60 -
* Values in parantheses indicate the critical limit									

Plant analysis data - Micronutrients								
Crop	Stage	Plant part	Source	ppm dry matter				
				Cu	Zn	Mn	Fe	B
				Optimum concentration				
Pigeonpea	45-50 DAS	Whole shoot	Burton & Edward, 1981 Nicholas, 1965, Rosbrook & Edward, 1981 Sheldrak & Narayanan, 1979	12 (10)	48 (10-13)	19-95 (18)	151-191 (151)*	10-52 (10)
Urdbean	45-50 DAS	Whole shoot	Dwivedi & Randhawa, 1974 Nowlan, 1980 Dange, et al, 1984	- -	18 (14-15)	- -	- -	>20 -
				Optimum concentration:				
Cowpea	45 DAS	Whole shoot	Fox et al, 1977 Kashirad et al, 1978	- -	27-32 (18-24)	1000 (280)	(>100) (<70)	- -
* Values in parentheses indicate the critical limit.								

Fertilizer recommendations

Most tropical pulse crops are grown on marginal soils poor in fertility. The traditional practice is to apply small amount of organic manures with little or no mineral fertilizer. For sustained productivity, adequate and balanced fertilization is required.

Acidic soils of Indian Sub-continent and South-east Asia need liming at 2-4 t/ha to raise soil pH.

Phosphorus is the most important fertilizer nutrient for tropical pulses. As leguminous crops, they can obtain most of their nitrogen requirement by symbiotic N fixation. Response to potassium is generally small on account of the high available K status of the soils on which they are grown. On light sandy soils, chickpea and pigeonpea have responded to sulphur at 10-20 kg/ha S. On calcareous soils in northern India pulse crops showed a good response to a basal application of 20-25 kg/ha Zn SO4 or a foliar spray of 5 kg/ha Zn SO4. Iron chlorosis which can be corrected by a foliar spray of 0.5 % FeSO4, has also been observed in pulses growing on calcareous soils. Pigeonpea grown on Alfisols and Entisols has benefited from a basal application of 1 kg sodium molybdate and 10 kg borax per hectare.

General fertilizer recommendations			
Crop	**Time of aplication**	**Application rate kg/ha**	
		N	**P2O5**
Pigeonpea	All basal	20-30	40-80
Urdbean and Mungbean	All basal	10-20	30-60

Preferred nutrient forms

In the climatic and edaphic conditions in which tropical pulses are grown, the fertilizer nutrients should be applied in easily available form: N preferably as ammonium, P in water-soluble form and K as sulphate.

Current fertilizer practices

Indian Sub-continent (pigeonpea, urdbean and mungbean)

Generally in rainfed areas no fertilizers or manures are applied; in some places farmers may apply a small amount of f.y.m. (e.g. 8-15 t/ha) 3-4 weeks before sowing. In irrigated areas, a small amount of mineral fertilizer (10-15 kg/ha N, 20-30 kg/ha P2O5) is either placed in the seed furrow or broadcast and mixed in.

Under improved cultivation practices: irrigated, basal application of 18-20 kg/ha N, 40-50 kg/ha P2O5, plus K, Zn, S if required; unirrigated, basal application of 10-15 kg/ha N, 20-30 kg/ha P2O5, followed by a foliar spray of 2 % urea at pod development.

Diammonium phosphate, urea and single super phosphate are commonly used.

South East Asia (mungbean, urdbean)

Mostly grown without fertilizer. Organic manures, too, are seldom used. Acidic soils are limed to correct pH.

Under improved cultivation systems: basal application of 10-15 kg/ha N, 30-40 kg/ha P2O5, 20-30 kg/ha K2O; and for acidic soils lime at 2.5 t ha.

Australia (mungbean, cowpea, pigeonpea)

Phosphatic and potassium fertilizers at 40-60 kg/ha P2O5 and 30-40 kg/ha K2O as a basal dressing.

Africa (cowpea, pigeonpea).

Generally no fertilizer is applied. Under improved cultivation systems: basal application of 10-20 kg/ha N and 30-40 kg/ha P2O5.

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Author: Masood Ali, Directorate of Pulses Research, Kanpur, India

Coconut *(Cocos nucifera L.)*

French: Cocotier; Spanish: Cocotero; Italian: Cocco; German: Kokosnusspalme

Crop Data

Perennial. Harvested part: mature nuts (11 1/2-12 month-old), tender nuts (7-8 month-old).

Planted any time of the year, but preferably at start of rainy season.

Flowers: tall varieties 4 1/2-5 years from field-planting; dwarf varieties 2-3 years from field-planting; dwarf x tall hybrids 3-4 years from field-planting.

Differentiation of flower primordia: 22 months before flowering.

Nut maturity: 12 months after complete pollen fertilization of female flowers (buttons); 1 nut bunch matures each month; 10-14 bunches per palm at different maturity stages at any one time.

Planting density: 100-200 trees/ha on square system or 115-236 trees/ha on tri-angular system with 7 to 10 m spacing (dwarf varieties are planted at 7-8 m, hybrids at 8-8.5 m and tall varieties at 9-10 m spacing.

Preferably grown on deep, well-drained, light to medium textured soils, pH 6-7, rich in organic matter, high in fertility level (including soil chloride).

The crop is grown commercially between latitudes 20 °N and 20 °S; at altitudes of 600 m or less; with temperature range 24-29 °C, relative humidity 80-90 %, rainfall 1 500-2 300 mm evenly distributed throughout the year.

Generally, the crop is not irrigated except for seed production (seed gardens).

Average annual yields of 80-150 nuts per tree (2-4 t/ha of copra) are attained under favorable growing conditions, depending on the variety or hybrid type.

Nutrient demand/uptake/removal

One hectare of coconuts (average of 150 palms) producing 12-14 leaves and 100 nuts/tree/year contains in the harvest (matured bunches) the following amount of nutrients (per year): 49 kg N, 16 kg P2O5, 115 kg K2O, 5 kg Ca, 8 kg Mg, 11 kg Na, 64 kg Cl and 4 kg S. The husk contains 60 % of the K2O, 18 % of N and 26 % of Mg removed in the harvest. It is therefore recommended that wastes such as coconut husks and leaf fronds be left in the field to undergo decomposition and mineralisation so that nutrients eventually return to the crop.

Nutrient demand/uptake/removal - Macronutrients									
Yield	**Source**	**kg/ha**							
		N	**P2O5**	**K2O**	**Mg**	**Ca**	**S**	**Na**	**Cl**
1.5 t copra	Copeland, 1931	93	41	138	-	17	-	-	-
25 nuts/palm	Cooke, 1950	29	9	26	-	-	-	-	-
60 nuts/palm	Nathaniel, 1969	72	39	108	-	-	-	-	-
100 nuts/palm:	Khanna and Nair, 1977								
nuts only		120	18	85	-	-	-	-	-
whole palm		157	28	346	-	-	-	-	-
100 nuts/palm	Ouvier and Ochs, 1978	49	16	115	8	5	4	11	64
6.7t copra:	Ouvier and Ochs, 1978								
nuts only		108	39	232	15	9	9	20	125
whole palm		174	46	299	39	70	30	54	249
1 t copra	Ashgar, 1988	16.2	5	36	2	1.4	1.3	2.5	19.7

Plant analysis data

In foliar diagnosis, a composite sample of palms grown under similar conditions is collected at intervals. For a particular stage or age of coconuts, leaf sampling is done on the selected leaf rank (number) of the palm based on its phyllotaxy. Depending on the average count of living or functional leaves at sampling time, a guide to the proper leaf rank to be sampled was recommended by Magat and Prudente (1979) as follows:

Living leaves average count (5-10 leaves)	**Stage**	**Leaf rank to sample**
4-6	Nursery	1
7-12	Nursery/field	3 or 4
13-18	Pre-bearing	9
19 or more	Bearing	14

Critical levels for the 14th leaf at the bearing stage - Macronutrients									
Variety	**Source**	**% of dry matter**							
		N	**P**	**K**	**Mg**	**Ca**	**Na**	**Cl**	**S**
Local tall varieties	Fremond (1966)	1.8-2.0	0.12	0.8-1.0	0.20	0.50	0.4	0.5	0.15-
	Kanapathy (1971)	1.8	0.12	0.8-1.1	0.30	0.30	-	-	-
	Friend (1975)	2.0	0.14	1.0	0.26	0.55	0.2	-	-
	Magat (1979)	1.8 2.0 (OL)*	0.12 0.14 (OL)	0.8 1.0 (OL)	0.20 0.30 (OL)	0.30 0.50 (OL)	0.1 0.2 (OL)	0.3-0.4 0.5-0.6 (OL)	0.13 0.17 (OL)
	CRISL** '1988)	1.8-2.1	0.11-0.12	1.2-1.4	0.25-0.35	0.35-0.50	0.4	0.3-0.6	0.15-0.20
Dwarf varieties	Chew (1982)	1.8-2.0	0.12	0.6-0.8	0.25	0.15-0.20	-	-	-
Hybrids	Manciot et al (1979)	2.2	0.12	1.4	0.20	-	-	-	-
	Magat (1988)	1.8 2.0 (OL)*	0.12 0.13 (OL)	0.9 1.1 (OL)	0.30 0.33 (OL)	0.32 0.35 (OL)	0.15 0.17 (OL)	0.45 0.50 (OL)	0.15 0.16 (OL)
* Optimum Level ** Coconut Research Institute of Sri Lanka									

Coconut - Critical levels for the 14th leaf at the bearing stag - Micronutrients						
Variety	**Source**	**ppm dry matter**				
		B	**Fe**	**Mn**	**Zn**	**Cu**
Local tall varieties	Fremond (1966)	-	50	60	60	-
	Kanapathy (1971)	-	50	60	60	-
	Friend (1975)	14	115	185	15	12
	Magat (1979)	9-11	-	-	-	-
	CRISL** (1988)	8	40-115	60-120	60	12-13
Dwarf varieties	Chew (1982)	8	40-45	60	-	-

Fertilizer Recommendations:

Age	**Nutrient Rate (per tree)**					
	N	**P2O5**	**K2O**	**MgO**	**S**	**Cl**
Field-planting	30 g	30 g	90 g	50 g	18 g	66 g
6 months	40 g	50 g	0.15 kg	85 g	25 g	0.11 kg
1 year	0.10 kg	0.10kg	0.35 kg	125 g	60 g	0.26 kg
2 years	0.15 kg	0.15 kg	0.55 kg	0.25 kg	90 g	0.40 kg
3 years	0.20 kg	0.16 kg	0.70 kg	0.35 kg	0.12 kg	0.53 kg
4 years	0.30 kg	0.20 kg	1.00 kg	0.40 kg	0.18 kg	0.70 kg
5 years and older	0.40 kg	0.30 kg	1.20 kg	0.50 kg	0.24 kg	0.90 kg
Source: Magat (1988)						

The most common fertilizer combination is AS plus KCl thus supplying the four nutrients widely deficient in coconut soils: N, K2O, Cl and S.

Timing and frequency of application

Pre-bearing stage (vegetative), usually 1-3 year old coconuts: annual rate/palm in split application, half at the start of rainy season and half 6 months after or about a month before end of the rainy season. In areas with almost even distribution of rainfall throughout the year, apply first half of fertilizer anytime and the remaining half 6 months later.

Bearing stage: annual rate/palm in a single application, for areas with an even rainfall distribution (1.5-3 dry months) or in split applications for areas with distinct dry and rainy seasons.

Method of fertilizer application

First, circle-weed (remove all weeds) around the base of the palm, with a radius of 0.5-0.75 m for young palms and 1-1.5 m for bearing palms. Then, broadcast the fertilizer to each tree as uniformly as possible over the circle-weeded area. Finally, incorporate the broadcast fertilizer (to a depth of 5-8 cm) with the use of a suitable digging tool. This is necessary for N fertilizers (like ammonium sulfate) to minimize losses due to volatilization.

On steeply sloping and hilly areas, place the fertilizer, equally distributed, in 10-12 holes (10-15 cm deep and 7-10 cm wide) within 1-2 m radius around the base of palm.

Present fertilizer practices:

Philippines (Southeast Asian Region)

Most coconut areas in the Philippines are widely deficient in N, Cl, S and K2O and adquate in other nutrients. Generally, liming is not needed as coconut has a wide adaptability to soil acidity (pH 4.5-8).

Nursery - per seedling			
Age	**Fertilizer combination**		
	AS	**either + KCl**	**or common salt (if K is not needed)**
2	20 g	+ 25 g	+ 20 g
3	40 g	+ 45 g	+ 40 g
total	60 g	+ 70 g	+ 60 g

Field-planting to bearing stage - per tree			
Age	Fertilizer combination		
	AS	**either + KCl**	**or + common salt (if K is not needed)**
Coastal / within 2 km of coast)			
Field - planting	150 g	+ 100 g	+ 80 g
6 months	200 g	+ 150 g	+ 120 g
1 year	500 g	+ 500 g	+ 400 g
2 years	750 g	+ 750 g	+ 600 g
3 years	1.00 kg	+ 1.00 kg	+ 0.80 kg
4 years	1.25 kg	+ 1.25 kg	+ 1.00 kg
5 years or older	1.50 kg	+ 1.50 kg	+ 1.20 kg
Inland (over 2 km from coast)			
Field - planting	150 g	+ 200 g	+ 160 g
6 months	200 g	+ 250 g	+ 200 g
1 year	500 g	+ 600 g	+ 480 g
2 years	750 g	+ 900 g	+ 720 g
3 years	1.00 kg	+ 1.50 kg	+ 1.25 kg
4 years	1.25 kg	+ 1.70 kg	+ 1.35 kg
5 years or older	1.50 kg	+ 2.00 kg	+ 1.70 kg

Indonesia

The most common nutritional deficiency is N, followed by K and Mg. In certain areas, problems of Cl, P205, Ca and B exist (Mahmud and Allorerung, 1988). Current fertilizer recommendations widely used in the country are based on results of leaf analysis and soil analysis of field experiments and surveys.

Nursery - g per seedling				
Time	**Fertilizers**			
	Urea	**TSP**	**KCl**	**Kieserite**
Monthly	20	7.5	24	7.5
Source: Rompas, Lengkey and Mahmud, 1985				

Bearing Stage - g per palm and year			
Province	**Fertilizers**		
	Urea	**TSP**	**KCl**
West Java	90 - 1700	0 - 1050	0 - 1750
Central Java	600 - 1700	0 - 1050	0 - 200
East Java	250 - 1400	0 - 625	0 - 100
West Kalimantan	750 - 1600	0	0 - 100
East Kalimantan	450 - 1500	0 - 450	0 - 500
South Kalimantan	480 - 1250	0	0
North Sulawesi	600 - 2200	0 - 1100	0 - 1500
Central Sulawesi	300 - 1350	0	0 - 1350
Source: Mahmud and Allorerung, 1988			

Another fertilizer recommendation is based on altitude and soil type:

Soil type	Fertilizer in kg/palm and year		
	Urea	TSP	KCl
0 - 150 m above sea level			
Alluvial	1.10	0.20	1.30
Podzolic	1.20	0.30	1.60
Latosol	1.30	0.30	2.00
150 - 500 m above sea level			
Alluvial	0.90	0.60	1.30
Podzolic	1.00	0.90	1.60
Latosol	1.10	0.90	2.00
Source: Balittri Manado, 1984			

Malaysia

Nitrogen deficiency is common in all coconut soils, while P2O5, K2O and MgO levels are inadequate in some areas, particularly on acid, peaty and sandy soils. The fertilizers used for high yielding hybrid coconuts in Malaysia are as follows:

Age (years)	Fertilizer in kg/palm and year				
	17-8-17	Urea	Rock phosphate	KCl	Ground magnesium limestone
Coastal clay soils					
1	1.00	-	0.50	-	0.5
2	-	1.00	1.00	1.00	1.0
3	-	1.75	1.50	1.50	1.5
4	-	1.50	-	1.50	2.0
5	-	1.50	-	1.00	-
6	-	1.50	-	1.00	-
7	-	1.50	-	1.00	-
8	-	1.50	-	1.00	-
Acid sulphate soil					
1	1.00	-	0.75	-	-
2	-	0.75	1.00	1.25	3.0
3	-	1.50	1.50	2.00	-
4	-	1.50	1.50	1.50	3.0
5	-	1.25	-	1.25	-
6	-	1.25	-	1.25	-
7	-	1.25	-	1.25	-
8	-	1.25	-	1.25	-
Source: Chew, 1982					

Sri Lanka

The widely deficient nutrients in the country are N, P2O5, K2O and MgO. Countrywide, fertilizer application increases nut yield from 16 nuts to 72 nuts/tree per year (De Silva, 1981). The CRISL provides general fertilizer mixtures 13-12-17 and 12-6-32.

Young palms:

Age	kg per palm (13-12-17)
Field-planting	1.00*
6 months	0.50
1 year	0.60
1.5 years	0.60
2 years	0.80
2.5 years	0.80
3 years	1.00
3.5 years	1.00
4 years and up to bearing	1.20
* plus 1 kg dolomite + 10 kg dry cow dung per planting hole	

Adult palms:

Climatic Zone	Soil Type	kg per palm (12-6-32)
Wet (> 150 cm rainfall)	- gravel/cabook/sand - loam or clay	3* 2
Intermediate (100-150 cm rainfall)	- gravel/sand - loam or clay	3 2
Dry (< 100 cm rainfall)	- gravel - loam or clay - sand	3 2 2.5
* Plus dolomitic limestone at 2 kg/palm (wet zone); 1.5 kg/palm (in other areas) once in three years as a routine measure to meet Mg requirements.		

India

Based on fertilizer trials on West Coast Tall variety, Tall x Dwarf hybrids and Dwarf x Tall hybrids, general fertilizer recommendation (for palms yielding at least 50 nuts/palm per year) is as follows:

Condition	kg of nutrient per palm and year		
	N	P2O5	K2O
Rainfed/local talls	0.5 kg	0.3 kg	1.2 kg
Irrigated/hybrids	1 kg	0.6 kg	2.4 kg
Source: Nair, 1979			

Cote d'Ivoire (West Africa)

The main nutrients deficient are K2O, P2O5, MgO and N.

Stage	kg of fertilizer per palm and year			
	Urea	Single superphosphate	KCL	Kieserite
Year				
0	0.10	0.20	0.20	0.1
1	0.20	0.40	0.40	0.2
2	0.30	0.60	0.60	0.3
3	-	-	1.20	0.4
4	-	-	1.50	0.5
5	-	-	1.50	-
6	-	-	1.50	-
7 and over	-	-	1.00	-
Source: Zakra et al, 1986				

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Author: S.S. Magat, Agricultural Research and Development Branch, Philippine Coconut Authority

Groundnut *[Peanut] (Arachis hypogaea L.)*

French: Arachide; Spanish: Mani, cacahuete; Italian: Pistacchio di terra; German: Erdnuss; Portugese: Amendoim

Crop data

Grown as an annual but perennial growth is possible in climates which are warm until harvest. Types grown include: Runner, Spanish, Virginia, Valencia. It is a traditional smallholder low-input crop in W.Africa and a major cash crop on both small and large farms in India, China and USA. The fruits (nuts) are used for human food as peanut oil, peanut butter, candy and roasted, boiled and raw peanuts, etc. Shells (hulls) are a by-product residue sometimes used as fuel, e.g. for generation of electricity. Aboveground forage (hay or vines) often used for livestock feed.

Sown late spring, after soil temperatures warm in cool climates (15 April - 15 May in Southern USA), generally at 100-150 kg/ha in rows approx. 80-100 cm wide at depth of 3-8 cm dependent on moisture availability in the topsoil.

Flowering occurs 30-45 days after planting and continues for several weeks. Harvest of mature pods 120-145 days after planting.

Desirable plant densities: small-seeded types, approx. 160 000 plants/ha; large-seeded types, 125 000 plants/ha; late-maturing types in W.africa, 110 000 plants/ha.

Adapted to semi-arid and semi-humid conditions. The main growing areas receive 500-1 200 mm annual rainfall. Rain during the growing season is critical for good fruit development. The crop responds to irrigation when very dry conditions occur at planting or from flowering to maturity.

Prefers loose sandy soils, of types ranging from loamy sands to sandy clay loams, with good drainage. Soil pH can range from 5 to 7.8 but nutrient deficiencies and/or toxicities can be expected at the extremes of this wide range. The preferred pH range is 5.5-6.5. Deep tillage is favoured in order to bury residues from the previous crop, and for best development and easy recovery of the pods.

Methods of culture vary from planting on raised beds (in Southern USA), to flat planting, hole planting (in China) and ridge planting (in other parts of the world).

Nutrient demand/uptake/removal

Varies with yield, nutritional status and parts removed at harvest. Because of the high proportion of nutrients in the vines, the removal of nutrients is much less if the vines are returned to the soil than if they are removed. The relative proportions of macronutrients in various parts of mature plants are:

Partitioning of macro-nutrients at harvest					
Plant part	**Percentage of total**				
	N	**P**	**K**	**Mg**	**Ca**
Vines	46	41	65	70	88
Roots	9	11	8	14	7
Kernels	40	42	17	11	3
Hulls	5	6	10	5	2
Source: Longanathan and Krishnamoorthy, 1977					

Calculations of removal per hectare in USA, China and Senegal give rather similar figures when the differing yield levels are taken into account:

Nutrient removal - Macronutrients						
Country(pod yield)	**Plant part**	**kg/ha**				
		N	**P2O5**	**K2O**	**MgO**	**CaO**
USA (3 t/ha)						
	Pods	120	24	22	-	8
	Vines	72	24	58	-	71
	Total	192	48	80	-	79
China (1.023 t/ha)						
	Kernels	38	8	3	7	2
	Hulls	3	1	5	3	4
	Leaves	16	3	4	16	21
	Stems	7	2	5	15	9
	Total	64	14	17	41	36
(2.250 t/ha)	Total	150	30	82	-	91
(3.975 t/ha)	Total	195	38	90	-	-
(5.085 t/ha)	Total	259	49	124	-	256
(5.310 t/ha)	Total	270	55	165	-	261
Senegal - mean from 3 regions		48	7	15	8	10
Sources: Morrison, 1959; Univ. of Florida, 1967-76; Crop ecology group, Inst. of Botany, Acad. Sinica, 1977; Wang Zaixu, Sun Yanbao et al., 1982; Gillier & Silvestre, 1969.						

Only 10-20 % of the total uptake of nutrients occurs during the vegetative stage, the remainder being divided about equally between the reproductive and ripening stages:

Partitioning of total uptake of macronutrients by growth stage					
Growth stage	**Percentages of total uptake**				
	N	**P**	**K**	**Mg**	**Ca**
Vegetative	10	10	19	11	10
Reproductive	42	39	28	48	53
Ripening	48	51	53	41	37
Source: Adapted from Longanathan & Krishnamoorthy, 1977					

Plant analysis data

Leaf analysis is not well accepted as a good diagnostic tool for groundnuts; the final yield and quality of nuts do not generally relate well to leaf composition during growth. One factor that is probably involved is the restricted downward phloem movement of nutrients from the above-ground plant parts to the developing pods. However, leaf analysis is still important as a gauge of the nutritional health of the plant through much of its development.

Careful attention to plant age is necessary when interpreting foliar nutrient concentrations. Cox et al (1970) found that N and P concentrations decreased steadily from plant age 2 to 21 weeks; K increased nearly a full percent from week 2 to week 6 before decreasing slightly

with further age; Ca and S were erratic but did not decrease significantly over the whole time period; Mg tended to increase slowly to weeks 10-12 and then decrease slightly; Mn decreased slightly throughout; Zn increased to week 6, then decreased markedly to week 21; Cu and B changed only slightly.

Sufficiency ranges are:

Plant analysis data (sufficiency levels for leaves) - Macronutrients							
Plant par	**Time**	**% of dry matter**					
		N	**P**	**K**	**Mg**	**Ca**	**S**
7th leaf	40 DAP	3.3-3.9	0.15-0.25	1.0-1.5	0.30	2.0	0.19-0.25
Upper mature leaves	Bloom	3.0-4.5	0.20-0.50	1.7-3.0	0.30-0.80	1.25-2.0	0.20-0.35
Sources: Gillier & Silvestre, 1969; Plank, 1989							

Plant analysis data (sufficiency levels for leaves) - Micronutrients								
Plant part	**Time**	**ppm dry matter**						
		Ma	**Fe**	**B**	**Cu**	**Zn**	**Al**	**Mo**
Upper mature leaves	Bloom	20-350	50-300	20-60	5-20	20-60	<200	0.1-5.0
Sources: Gillier & Silvestre, 1969; Plank, 1989								

Data from China for various plant parts at different stages of development are shown below.

Plant analysis data (China) - Macronutrients						
Plant part	**Time**	**% of dry matter**				
		N	**P**	**K**	**Mg**	**Ca**
Leaf	Bloom	3.97	0.09	1.06	1.87	2.07
Leaf	Maturity	2.95	0.26	0.65	1.79	2.
Stem	Flowering	1.58	0.22	2.17	1.03	1.11
Stem	Maturity	1.15	0.12	0.79	1.63	1.12
Pod	Maturity	1.20	0.25	1.20	0.85	1.20
Seed	Maturity	4.76	0.46	0.24	0.58	0.17
Plant	Seedling	3.94	0.19	1.52	-	0.63 (root) 1.20 (leaf)
Plant	Flowering	3.86	0.23	1.31	-	0.76 (root) 1.64 (leaf)
Plant	Flowering	3.48	0.24	1.64	-	2.11 (leaf) 0.49 (young pod)
Plant	Maturity	3.70	0.28	1.59	-	2.70 (leaf) 0.89 (pod)
Plant	Maturity	2.52 2.61	0.17 0.22	0.86 0.79	0.43 0.41	0.53 unfertilized 0.48 fertilized
Source: Academia Sinica, 1977; Wang Zaixu et al., 1982; Cai Changbei, 1988.						

Plant analysis data (China) - Micronutrients				
Plant part	**Time**	**ppm dry matter**		
		Mn	**Fe**	**B**
Leaf	Bloom	-	200	-
Leaf	Maturity	70	63	-
Stem	Flowering	19	97	-
Stem	Maturity	17	198	-
Pod	Maturity	19	85	-
Plant	Seedling	-	-	25
Plant	Flowering	-	-	24
Plant	Flowering	-	-	23
Plant	Maturity	-	-	19
Source: Academia Sinica, 1977; Wang Zaixu et al., 1982; Cai Changbei, 1988.				

Some workers believe that nutrient ratios are more important than individual concentrations. Roche et al (1959) recommended ratios in the leaves of N: (N+P+K) = 0.5 to 0.65, P: (N+P+K) = 0.03 to 0.05 , and K: (N+P+K) = 0.32 to 0.40.

Since Ca is especially important for nut development, it has received considerable attention. Gaines et al (1991) found maximum yield and grade for the small-seeded runner type with 0.12 % Ca in the hulls and 0.04 % Ca in the nuts, and for the large-seeded Virginia type with 0.19 % and 0.058 % respectively.

Fertilizer recommendations

Preferably based on soil tests. If pH is low and liming is recommended, limestone should be broadcast and turned into the soil in early spring.

Most USA research indicates no response to N when the soil is limed to an adequate pH and inoculated with the correct Rhizobium sp. On the other hand, many responses to N have been obtained in other countries.

Few responses to P and K have been recorded in USA since the rotational crops usually have higher requirements. Responses have been reported, however, under very low fertility conditions in Africa and in particular in soils with high P fixation. Basal applications of P and K (and of N too in several countries) are usually broadcast and incorporated into the soil before planting. Some topdressings of small amounts of N are applied at the vegetative stage in China. Gypsum application (to supply Ca and S), by broadcasting or banding at flowering, is common. B and Mo are commonly applied in foliar sprays at the vegetative stage.

Preferred nutrient forms

Available organic manures are often applied and incorporated before planting. However, undecomposed organic matter may result in problems of germination and seedling disease if not buried. Generally most of the fertility is supplied by inorganic fertilizers including urea, concentrated superphosphate, and muriate of potash either as single nutrient fertilizers or in blends. Ca and Mg are supplied in limestone and/or Ca and S as gypsum. B and Mo, when applied, are commonly foliar sprays of soluble compounds.

Present fertilizer practices

USA

Limestone broadcast and turned into the soil before planting if pH < 5.5-6.0. P and K broadcast and incorporated before planting at rates of 45-90 kg/ha P2O5 and 50-95 kg/ha K2O depending on soil test. Also Mg at 50 kg/ha MgO if soil test is low and limestone is not applied.

Gypsum applied at bloom to small-seeded types, when Mehlich 1 extractable Ca < 150-250 mg/kg or Ca:Mg ratio < 3:1, broadcast at 250-315 kg/ha CaO or at proportionally lower rates for band application covering a smaller surface area. Double rates of gypsum applied to all large-seeded peanuts, regardless of Ca soil test.

Boron at 0.6 kg/ha B (unless soil level > 0.5 mg/kg B) usually in foliar sprays, split between the first two fungicide applications.

W. Africa

Recommendations from research are applied only where credit, subsidy and adequate supply are available.

For sowing on flat: fertilizer side-dressed or broadcast before 10th day of vegetation (drier areas).

For sowing on ridges: natural vegetation, and sometimes dry manure, buried at ridging and fertilizer side-dressed before 10th day of vegetation (frequent in wet areas with over 900 mm rainfall).

In Senegal: generally a basal dressing of 9-30-15 fertilizer.

In other countries: generally single superphosphate (60-100 kg/ha) or sulphate of ammonia and dicalcium phosphate; when available, cotton fertilizer is sometimes used.

Rock phosphate, available in many W. African countries, is being tested experimentally, either treated industrially to make it more soluble or applied as a basal fertilizer dressing.

The official quantities recommended are:

Senegal: 150 kg/ha 9-30-15, plus 400 kg/ha gypsum as a topdressing on seed multiplications of confectionery varieties 40 days after planting.

Burkina Faso: 75 kg/ha single superphosphate.

Tchad and Niger: 100 kg/ha single superphosphate.

Guinea-Bissau: 100 kg/ha 14-22-12 cotton fertilizer.

India

Fertilizer recommendations concerning the seven major oilseed growing states are given in the table below.

State	recommended rates (kg/ha)						farmyard manure t/ha	Gypsum kg/ha
	Rainfed			Irrigated				
	N	P2O5	K2O	N	P2O5	K2O		
Andhra Pradesh	20	40	20	30	40	50	10	500
Gujarat	12.5	25	0	25	50	0		
Karnataka	25	50	25	25	75	25-37		500
Madhya Pradesh	20	80	80	20	80	80	10	200
Maharashtra	12	25	0	25	50	0		
Rajasthan	15	60	0	15	60	0		
Tamil Nadu	10	10	45	17	34	54	12.5	250
Source: Tandon, 1990.								

China

For traditional varieties in North China with an expected pod yield of 3 000 kg/ha: 22.5 t/ha organic manure plus fertilizer containing 20 kg/ha P2O5, banded in the seed row and incorporated in the upper 10 cm of soil, followed by a topdressing of 30 kg/ha N.

For improved varieties on medium soils in the Province of Shan-Dong with an expected pod yield of 4 500 kg/ha: basal application at sowing time of 37.5 t/ha organic manure plus fertilizer containing 30 kg/ha P2O5 and 75 kg/ha K2O, and inoculation with groundnut Rhizobium, followed by first topdressing at seedling stage of 30 kg/ha N and second topdressing at first bloom of 10 kg/ha N.

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Oil Palm *(Elaeis guineensis Jacq.)*

French: Palmier à l'huile; Spanish: Palma de aceite; Italian: Palma da olio;
German: Oelpalme

Crop data

Perennial with a generic life of over 200 years, but the economic life is 20-25 years, (nursery 11-16 months, first harvest 32-38 months from planting, peak yield 5-10 years from planting).

Most widely used planting material consists of Dura x Tenera hybrids. Clonal palms are expected to come into commercial use in the late 1990s.

Harvested part: Fruit (Fruit bunch). Oil is obtained from fleshy mesocarp of fruit (45-55 % oil) and from kernels (50 % oil). Oil extraction rate (from mature fresh fruit bunches = FFB) 20-24 %.

Highest yield attained to-date (1990) 46 t/FFB = 10.6 t/ha crude palm oil/ha.

Planting density: 128-148 palms/ha, depending on planting material, soil and climate. Most common spacing 9m x 9 m triangular (= 143 palms/ha).

Grown in lowlands of the humid tropics, 15° N - 15° S with evenly distributed rainfall of 1 800-5 000 mm/year. Adapted to a wide range of soils and to low pH, but sensitive to high pH (above 7.5) and to stagnant water.

Nutrient demand/uptake/removal

Nutrient demand is highly variable, depending mainly on the yield potential determined by the genetic make-up of the planting material and on the yield limit set by climatic factors (water, effective sunshine, temperature). Yield and nutrient uptake are much higher, for example, in Malaysia than in Nigeria (see table). More important, however, is the fact that the efficiency of nutrient use (uptake/ton FFB) is much higher in Malaysia, the main reason being the heavy climatic stress in Nigeria (a long dry season, and limited hours of sunshine during the wet season). The response to most stress factors is a reduction in fruit output, while vegetative growth is little affected.

Nutrient uptake is low during the first year (transplanting shock!), but increases steeply between year one and year three (when harvesting commences) and stabilizes around years five to six.

Recognition of this fact has resulted in increased early applications of fertilizer, which combined with improved other agronomic practices (better planting material, higher nursery standards, more rigid culling etc.) has led to a dramatic increase in early yields (in third to sixth years from planting). In regions without any serious climatic constraint yields of over 25 t/ha FFB have been achieved in the second year of harvesting, largely as a result of higher early rates of fertilizer application.

Nutrient removal, immobilization and recycling in adult oil palms								
	Malaysia (24 t FFB/ha)				Nigeria** (9.7 t FFB/ha)			
	kg/palm/year				kg/palm/year			
	N	P2O5	K2O	MgO	N	P2O5	K2O	MgO
Removal in harvested fruit	0.49	0.18	0.76	0.23	0.20	0.09	0.28	0.05
Immobilized in palm tissue	0.27	0.05	0.56	0.12	0.18	0.06	0.13	0.17
Recycled	0.53	0.17	0.83	0.32	0.63	0.17	0.46	0.42
Total nutrient uptake	1.29	0.40	2.15	0.66	1.01	0.32	0.87	0.63
	%				%			
Removal as % of total uptake	38	44	35	35	20	29	31	8
	kg/ha				kg/ha			
Uptake per hectare (148 palms/ha)	191	62	318	98	149	48	236	93
	kg/t FFB				kg/t FFB			
Uptake per ton FF	8.0	2.5	13.2	4.2	15.5	5.0	13.3	9.6
Source: *Malaysia: Ng and Tamboo, 1967; ** Nigeria: Tinker & Smilde, 1963.								

Plant analysis data

Producing, as it does, leaves (and fruit) throughout the year at a relatively uniform rate, the oil palm is better suited to leaf analysis than most other crops. The reference tissue for adult palms is the centre part of the pinnae (without midrib) from the seventeenth frond.

Critical ranges of concentration of nutrient elements in oil palm leaves					
Age of trees	Range	% of dry matter			
		N	P	K	Mg
Young palms (under 6 years)	Deficiency	< 2.50	< 0.15	< 1.00	< 0.20
	Optimum	2.60-2.90	0.16-0.19	1.10-1.30	0.30-0.45
	Excess	> 3.10	> 0.25	> 1.80	> 0.70
Old palms (over 6 years)	Deficiency	< 2.30	< 0.14	< 0.75	< 0.20
	Optimum	2.40-2.80	0.15-0.18	0.90-1.20	0.25-0.40
	Excess	> 3.00	> 0.25	> 1.60	> 0.70

Critical ranges of concentration of nutrient elements in oil palm leaves(ctd.)						
Age of trees	Range	% of dry matter			ppm	
		Ca	S	Cl	B	Cu
Young palms (under 6 years)	Deficiency	< 0.30	< 0.20	< 0.25< 8< 3		
	Optimum	0.50-0.70	0.25-0.40	0.50-0.70	15-25	15-2
	Excess	> 1.00	> 0.60	> 1.00	> 35	> 15
Old palms (over 6 years)	Deficiency	< 0.25	< 0.20	< 0.25	< 8	< 3
	Optimum	0.50-0.75	0.25-0.35	0.50-0.70	15-25	5-8
	Excess	> 1.00	> 0.60	> 1.00	> 40	> 15

Optimal levels can vary widely with factors such as moisture regime, nutrient balance, type of planting material, spacing, etc., so it is advisable to think in terms of critical or optimal "ranges" rather than absolute "values".

Variations in K and Mg are generally greater than those of N and P, due to larger errors in field sampling as well as to greater sensitivity to climatic (moisture/sunlight) factors.

Nutrient deficiencies

N deficiency is usually associated with conditions of water-logging, heavy weed infestation and topsoil erosion. Symptoms are a general paling and stiffening of the pinnae which lose their glossy lustre. Extended deficiency will reduce the number of effective fruit bunches produced as well as the bunch size.

P deficient leaves do not show specific symptoms but frond length, bunch size and trunk diameter are all reduced. Most soils on which oil palms are grown are very low in P, but it appears to be very efficient in utilizing soil and fertilizer P.

K deficiency is very common and is the major yield constraint in sandy or peaty soils. The most frequent symptom is "confluent orange spotting". Pale green spots appear on the pinnae of older leaves; as the deficiency intensifies, the spots turn orange or reddish-orange and dessication sets in, starting from the tips and outer margins of the pinnae. Other symptoms are "orange blotch" and "mid-crown yellowing". In soils having a low water holding capacity (sands and peats) K deficiency can lead to a rapid, premature dessication of fronds.

Mg deficiency is a frequent problem in light textured soils and in acid soils where the topsoil has been eroded. Symptoms are a yellowing of leaflets of older fronds, with leaflets exposed to sunlight showing more severe yellowing than do shaded leaflets.

S deficiency has not (yet) reached levels of significance in any of the major growing areas. Current sources of N (Ammonium Sulphate) and Mg (Kieserite) appear to be sufficient to meet the S needs of the crop.

Outright Ca deficiency is rare and the largest benefit from applied Ca (lime or dolomite) is probably due to better growth of legume covers and improved P availability.

Although Cl is a micronutrient for most crops, oil palm (and coconut palms), for optimum growth and yield, require Cl in concentrations that bring it well into the range of a secondary nutrient. No clear deficiency symptoms have been observed so far, other than pronounced signs of water stress during dry spells.

B is so far the only micronutrient of general significance. Deficiency first appears as retarded growth of young tissues. Deficient leaves are misshapen, wrinkled, brittle, and dark green in color. Specific to oil palms are: "frond shortening", "blind leaf", "crinkle leaf", "hook leaf" and "fishbone leaf".

Cu deficiency is a common problem on peat soils and on highly leached sandy soils in high rainfall areas. Symptoms are severely stunted growth, chlorosis (starting from the tips or upper part of the youngest pinnae), shortening of internodes and dessication.

Only isolated cases of Fe, Zn and Mn deficiency have been detected.

Fertilizer recommendations

Fertilizer recommendations must take into account environmental (soil and climate) and economic (fertilizer price and produce price) constraints. Actual fertilizer use should be further modified by the level of agronomic management. Responses tend to be most vigorous from the first to the sixth year in the field.

Nursery programme

Fertilizer recommendations - Nursery programme			
Age of seedlings (weeks from planting)	**Quantity g/seedling**	**Frequency**	**Compound fertilizer (N - P2O5 - K2O - MgO)**
Weeks 8 and 10	3.5	bi-weekly	15 - 15 - 6 - 4
Weeks 12 - 24	7 - 10	bi-weekly	12 - 12 - 17 - 2 with miconutrients alternating with 15 - 15 - 6 - 4
Weeks 26 - 32	12 - 16	bi-weekly	12 - 12 - 17 - 2 with micronutrients alternating with 15 - 15 - 6 - 4
Weeks 34 - 48	25 - 36	bi-weekly	12 - 12 - 17 - 2 with micronutrients alternating with 15 - 15 - 6 - 4
Week 50 to field planting	30 - 40	every 3 weeks	12 - 12 - 17 - 2 with micronutrients alternating with 15 - 15 - 6 - 4

Healthy, well selected seedlings are a pre-condition for early and sustained high yield. In most cases granular multinutrient compound fertilizers (NPK Mg + micronutrients) are the preferred nutrient source for seedlings in the nursery. Where sub-soil is used to fill the polybags, extra dressings of Kieserite may be required (10-15 g every 6 to 8 weeks). Where compound fertilizers are not available, equivalent quantities of straight materials should be used.

Field Programme

N rates range from 1.5-8 kg/palm of ammonium sulphate (AS) or its equivalent, depending on the yield potential as affected primarily by the total rainfall and its distribution and the intensity of solar radiation.

P rates range from 0.5-2 kg/palm/year of triple superphosphate (TSP) or its equivalent.

K rates range from 1 to 5 kg/palm of KCl, depending on soil and yield potential.

Malaysia - Fertilizer input for maximum exploitation of the genetic Yield potential									
Soil series and taxonomy	**Annual rainfall (mm)**	**Annual water deficit (mm)**	**kg/ha/year**						
			N	**P2O5**	**K2O**	**MgO**	**B2O3**	**CuSO4**	**CaO**
Sagomana (Oxic Paleuqualt)	2582	97	Immature plants:*						
			60-120	75-110	65-105	12-20	4-8	-	20- 40
			Mature plants:*						
			170-230	70- 90	220-310	25-35	8-12	-	40- 60
Peat soil (Tropofibrist)	2638	83	Immature plants:*						
			50-100	65- 80	140-260	-	6-12	1-2	140-280
			Mature plants:*						
			120-160	50- 70	550-700	0-10	13-18	3-5	300-400
Holyrod (Oxic Distropept)	2921	99	Immature plants:*						
			50-100	80-120	120-200	20-40	6-12	-	40- 60
			Mature plants:*						
			170-200	100-130	400-500	45-60	12-16	-	60- 80
* Immature plants = up to 3 years; Mature plants = 4 to 7 years.									

To maintain good fertilizer response and high yields in older palms (selective) thinning is often necessary where interpalm competition has become a problem as a result of initial high density planting in combination with a high fertility regime and good rainfall.

Recent pioneering work on maximum exploitation of genetic yield potentials (MEGYP) in Malaysia (Ng et al. 1990) has demonstrated that through a combination of good overall agronomic husbandry and intensive and well balanced use of fertilizer, productivity of oil palms can be enhanced by as much as 30 to 40 %.

With good management and proper fertilizer use, differences between soils of widely different inherent fertility largely disappear, so long as moisture deficit does not become a limiting factor.

Preferred nutrient forms and methods of applications

N: the preferred source is ammonium sulphate (AS), but urea is expected to become a dominant source. To avoid N losses due to volatilization, urea must be applied at times of assured rainfall.
Other N sources include ammonium chloride (AC) and ammonium nitrate (AN).
With young palms N should be evenly distributed over the weeded circle; in older palms inter-row application is also possible if there is no competition from a ground cover. In no case should N be applied in a narrow band around the base of the palms, as this will increase losses and may cause severe root damage. High N rates should be split into several dressings. In areas with an extended dry period, the last N application should be at least 3-4 months before the beginning of the dry season (high N levels in leaves result in more water consumption).

P: water-soluble sources (TSP, DAP and NPK compounds) are prefered during the immature phase. Thereafter phosphate rocks (preferably of high reactivity) can be used, applied up to year 4 in the circle, and thereafter alternating between circle and inter-row or inter-row only (where surface erosion is a problem).

K: potassium chloride (KCl) is the sole K source of significance. Potassium chloride has the added benefit of supplying chloride - an element found to be often deficient in oil palms. With young palms (up to year 4) it should be spread evenly over the circle; with mature palms it should be broadcast in the inter-row.

Mg: Kieserite ($MgSO4. H2O$) is the main Mg source for oil palms. Other sources include sulphate of potash magnesia and dolomite (suitable for acid soils only).
Kieserite should be spread over the circle while dolomite should be applied to the inter-rows.

Cl: potassium chloride is the main Cl source. In high K soils ammonium chloride can be used as an efficient alternative Cl source.

B: Sources are mainly various forms of sodium borate ($Na2 B4 O7.10H2O$, $Na2 B4 O7 . 5H2O$).
Placement of B should be into the weeded circle. Axillary application, though effective, is not advisable as it may result in uneven B distribution in the palm and cause B toxicity.

Cu: a one time soil application of 50 g/palm of copper sulphate is usually sufficient to correct Cu deficiency. On some peat soils repeated applications may be required.

Present fertilizer use in major growing areas for young, mature palms (4-10 years old)						
Country	**Yield level (t/ha FFB)**	**Soil**	**kg/palm/year**			
			N	**P2O5**	**K2O**	**MgO**
Malaysia	26-32	Alluvial	0.90-1.35	0.33-0.50	1.80-3.00	0.135
	22-28	Inland (Sedentary)	0.84-1.25	0.50-0.65	2.40-3.30	0.27-0.40
	20-26	Peat*	0-0.45	0-0.35	2.40-3.00	-
Indonesia	22-30	Volcanic	0.90-1.35	0.46-0.70	1.70-1.80	0.35-0.40
	20-26	Podzolic (Sedentary)	0.90-1.35	0.70-0.90	1.50-3.00	0.35-0.40
Thailand	14-18	Sedentary	0.84-1.05	0.50-0.66	1.80-3.00	0-0.14
Papua New Guinea	20-28	Volcanic**	0.60-0.80	0.23-0.46	0-1.20	0-0.14
	18-26	Alluvial	0.40-0.60	0-0.23	1.80-2.40	-
West Africa	10-18	Sedentary sands	0.20-0.50	0.23	0.60-1.80	0-0.27
Latin America	16-24	Volcanic**	0.45-0.68	-	-	0.20-0.27
	17-25	Alluvial	0.45-0.68	0.45-0.68	0.60-1.20	0-0.27
	15-22	Sedentary	0.30-0.60	0.45-0.68	0.90-1.80	0.20-0.27
Borate is now commonly applied to young palms (up to year 6) at rates starting from 50 g/palm/year and increasing up to 150 g/palm/year. * During the immature phase finely ground limestone is applied twice at a rate of 6-8 kg/palm. ** When no potash (KCl) is used on K-rich, young volcanic soils, ammonium should be the preferred N source in order to provide adequate chloride.						

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Author: H.R. von Uexkull, Director (rtd), E. & S.E. Asia Program of the Potash & Phosphate Institute/International Potash Institute, Singapore

Oilseed Rape *(Brassica napus L. ssp. napus)*

French: Colza; Spanish: Colza; Italian: Ravizzone, colza; German: Raps

5.4.1 Winter oilseed rape - Example: Europe

Varieties in Europe have been changed completely to improve quality. All present cultivars are "00", which means without erucic acid in the oil and with low glucosinolate in the seeds.

Crop data

Annual. Harvested part, seeds (contained in pods).
Sowing date: winter type, late August- mid-October.

The cycle of winter oilseed rape lasts about 320 days, comprising:
- The autumnal stage from sowing to early winter. This stage conditions to a great extent the implantation of the crop and its root system; leaf and flower initiation, which are critical for leaf development and yield potential, also take place at this time.
- The vegetative rest period, lasting about 2-3 months according to region and ending when the daily average temperature is regularly >5 °C.
- Vegetation regrowth to flowering, lasting about 2 months and defined by a very active accumulation of dry matter. This is the essential period for the absorption of mineral elements. Leaf development and leaf area per unit of soil area are key factors in determining yield.
- Flowering (about 220 days after sowing) to knotting. This stage can last for about 3 weeks and is marked by high competition for C supply between different types of organs (flowers on the main stem, newly forming young pods and seeds in pods).
- The pod-filling period, when the 1 000-seed weight is determined. At maturity the total biomass produced can reach 12-14 t/ha, of which about 30 % is in the seeds, 40 % in pod walls and peduncles, 20 % in stems and 10 % in roots.

Plant density: 350 000 - 700 000/ha (the risk of lodging increases considerably with plant density > 800 000/ha).

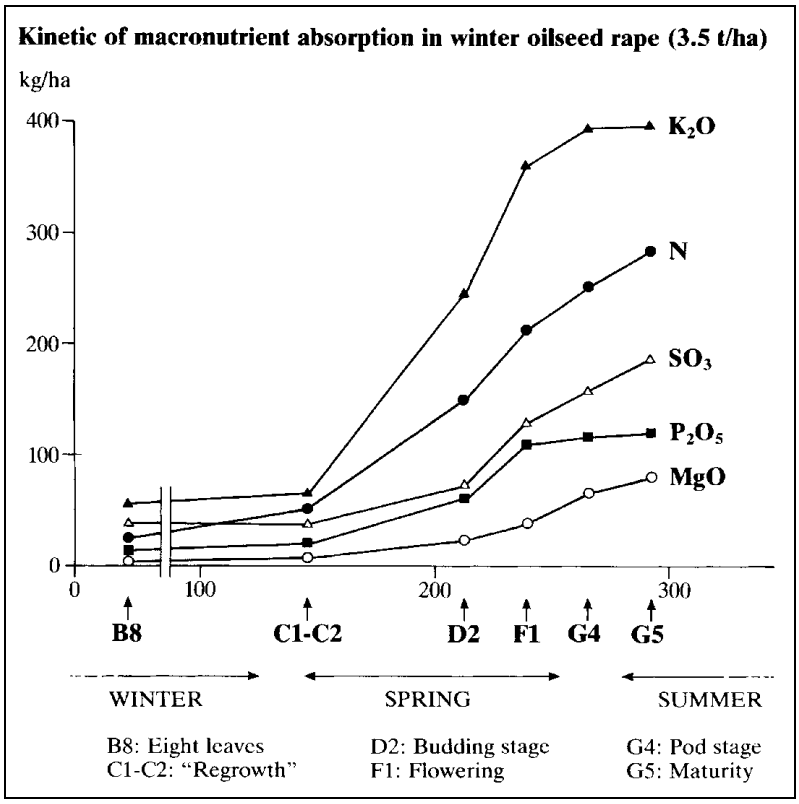
The yield is unaffected by row spacing within the range 17 cm (small grain drill) to 35 cm (high-precision drill).

The crop can be grown on a wide range of soils of varying pH.

Winter types require sufficiently low temperatures in early winter for flower initiation. Farmers in the South of France and in Spain, for example, need to pay particular attention to this point; where the winter is not cold enough, better results are generally obtained with alternative or spring cultivars, which unfortunately do not have such a high yield potential. In some areas in Southern Europe, or on light soils, farmers need to watch out for drought during the podfilling period.

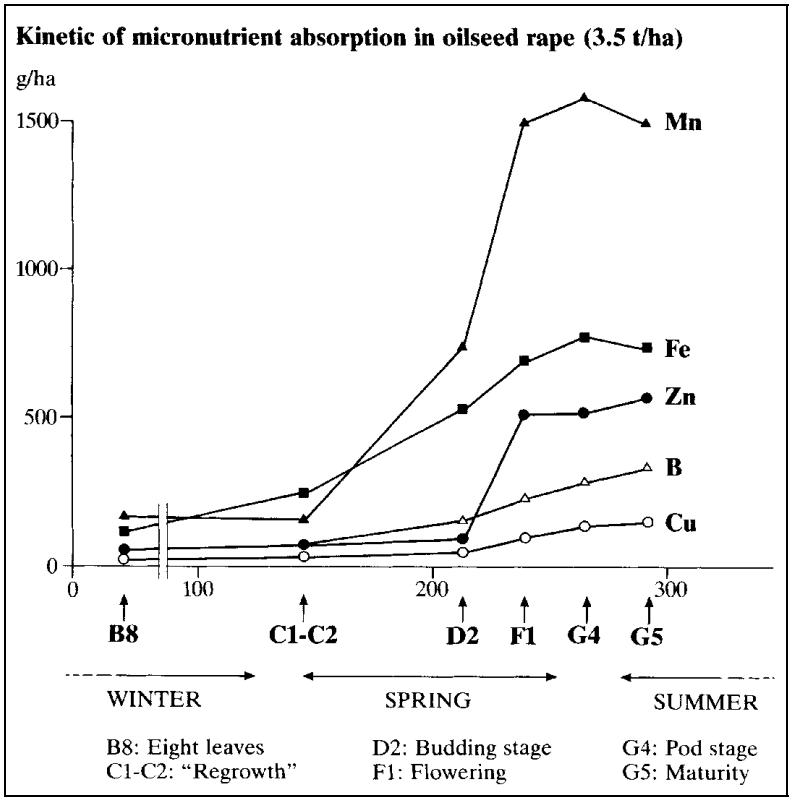
Nutrient demand/uptake/removal

Figure 1:



The two figures shom the important uptake of some 400 kg/ha K2O, particularly between vegetation regrowth and flowering, the steady uptake of N up to 280-300 kg/ha N at maturity and, among the micronutrients, the large uptake of Mn.

Figure: 2



Plant analysis data

Optimum concentrations (fully expanded lobate leaf on the main stem at the beginning of elongation) - Macronutrients						
Source	**% of dry matter**					
	N	**P**	**K**	**Mg**	**Ca**	**S**
CETIOM	5.5	0.58	2.5	0.12	1.1	0.53
Literature	4-4.7	0.35-0.5	3-4.4	0.12-0.1	1-2.2	0.68-0.7

Optimum concentrations (fully expanded lobate leaf on the main stem at the beginning of elongation) - Micronutrients						
Source	**ppm dry matter**					
	Fe	**Cu**	**Zn**	**Mn**	**B**	**Mo**
CETIOM	125	4.5	37.5	41	23	0.6
Literature	60-80	4-6.2	30-38	30-140	16-28	0.5-0.7

Fertilizer recommendations

N -

- Winter application can be advised only if: either there is a low N balance in the soil at sowing, in which case 30-50 kg/ha N may be applied, or sowing has been delayed and there is a need to stimulate growth in order to reach the optimum stage (8 leaves) for frost resistance, in which case 30-40 kg/ha N may be given.
- Spring application: the total requirement is about 200 kg/ha N, best divided into 2 applications, the first (50-70 % of the total) at the onset of regrowth (which, in France, could be at the end of January), and the second (30-50 % of the total) 2-3 weeks later. Where a high yield is expected, a third application, of 20-30 kg/ha N, may be worth giving about 10 days before the flowers are likely to be in full bloom.
- N is best applied as ammonium nitrate or in a solution containing 26-30 % N, but care must be taken when using fertilizer materials containing sulphur (see below).

P -

- Where soil pH is between 5.5 and 7.5, 60 kg/ha P_2O_5; or;
  on calcareous soils with pH >7.5, 70- 90 kg/ha P_2O_5; or,
  on acid soils with pH <5.5, 140-160 kg/ha P_2O_5.

K -

- On soils with high K availability, 60- 80 kg/ha K_2O; or,
  on soils with medium K availability, 120-150 kg/ha K_2O; or,
  on soils with poor K availability, 200-250 kg/ha K_2O.

S -

- This element is very important; on the one hand it increases yield (in France 75 kg/ha SO_3 in spring resulted in 8 cases out of 10 in an increase of 340 kg/ha in yield), but on the other hand it can increase the glucosinolate content of the harvested seeds. The best results, in increasing yield while maintaining quality, are to be expected from an application of 70-80 kg/ha SO_3 with the second application of N in spring.

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Author: A. Merrien, CETIOM, Département Etudes et Recherches - Section Physiologie-Nutrition, Paris, France

5.4.2 Spring oilseed rape *(Canola)* - Example: Canada

In Canada, oilseed rape cultivars are primarily Canola types, with low erucic acid and low glucosinolate content; there is a limited production of high erucic acid rapeseed cultivars but only on a contract basis.

Crop data

Annual. Harvested part: seeds.

Sown early spring: Argentine type (B. napus), 5-30 May; Polish type (B. campestris), 5 May-15 June.

Seeding rate: B. napus 6-8 kg/ha, B. campestris 5-7 kg/ha.

Flowers: B. napus approx. 48 days after sowing, B. campestris approx. 39 days after sowing.

Harvested: B. napus approx. 100 days after sowing, B. campestris approx. 88 days after sowing.

Preferably grown on loamy soils with good drainage and moisture-holding capacity, low salt concentration and pH 6.0-7.8.

The crop is adapted to cool, moist regions. It does not tolerate drought.

Nutrient demand/uptake/removal

Nutrient removal - Macronutrients					
Yield	**kg/ha**				
kg/ha		**N**	**P2O5**	**K2O**	**S**
1 960	Seed	74	36	18	13
	Straw	44	16	75	10
	Total	118	52	93	23

Plant analysis data

Whole above-ground part of plant of flowering stage:

Range	**% of dry matter**			
	N	**P**	**K**	**S**
Normal	3.2	0.40	2.5	0.5
Deficiency	2.0	<0.15	<1.2	<0.2

Recommendations for placement and timing of fertilizers

Good quality Canola oilseed contains 40-44 % oil and the meal contains 43-48 % protein, the lower figures generally relating to B. campestris. To produce seeds of good quality, special attention must be given to rate, placement and timing of fertilizer application. Excessive N application increases lodging, reduces oil content and results in green seeds that increase the chlorophyll content of the oil. Banding of P 2.5 cm below and 2.5 cm to the side of the seed is required for maximum yield, efficient use of P and increased oil content. The best responses in yield and seed quality are obtained when all the required nutrients are applied early in spring, at or immediately before sowing. The crop needs more S than either wheat or barley.

The generally recommended practice is to place N at rates greater than 7 kg/ha N, P at rates greater than 20 kg/ha P2O5, and all K and S, away from the seed to avoid seed injury. N should be applied in spring. P should be applied at sowing. K may be banded or broadcast and cultivated into the soil before sowing.

Fall fertilizer applications are generally less efficient than spring applications but are often preferred in order to spread the workload and to avoid drying and loosening the seedbed in spring. Band placement of fall applications will minimize loss. Elemental and other non-oxidized forms of S should be broadcast in the fall preceding sowing, to allow conversion to the plant-available sulphate form.

Preferred nutrient forms

For the soil and climatic conditions in which Canola is grown, all nutrients should be in easily plant-available form. If properly applied, N is equally effective as urea, anhydrous ammonia, nitrate and ammonium sources. P should be in water-soluble form, K as chloride or sulphate, and S as sulphate.

Present fertilizer practice in Canada

(for yields ranging from 1 000 to 2 000 kg/ha depending on available moisture)

Note that all Provinces recommend soil analysis and, when possible, plant analysis, for specific recommendations.

Manitoba (all soils):

Land	**kg/ha**			
	N	**P2O5**	**K2O**	**S**
Fallow or legume break	0- 35	30-45*	**	***
Grass and grass-legume break	35- 80	30-45*	**	***
Stubble	80-100	30-45*	**	***
* Not more than 20 kg/ha to be placed with the seed. ** Sands, sandy loams and organic soils use 35-70 kg/ha *** Grey luvisol and well drained soils deficient in S use 20 kg/ha.				

Saskatchewan:

Soil zone	kg/ha				
	N		P2O5	K2O	S
	Stubble	Fallow			
Brown	10- 40	0-10	15-40*	-	-
Dark brown	26- 65	0-15	15-40*	-	-
Black	55-100	20-65	15-40*	-	-
Dark grey	60-105	25-70	15-40*	0-40	0-30
Grey	60-110	25-70	15-40*	0-40	10-30
Irrigation	85-130	55-90	15-40*	-	-
* Up to 30 kg/ha may be placed with the seed.					

Alberta and British Columbia (Peace River Region):

Soil zone	kg/ha					
	N	P2O5	K2O	N	P2O5	K2O
	Fallow			Stubble		
Brown	5-15	10-20*	-	20- 55	10-15	-
Dark brown	5-30	15-20	-	35- 70	15-20	-
Thin black	5-35	15-25	-	30- 80	15-25	-
Black & Grey wooded	5-45	15-35	0-10	40-100	15-35	0-20
Irrigation				35-140	35-60	-
* Not more than 20 kg/ha to be placed with the seed.						

Further reading

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Authors: L.D. Bailey, C.A. Grant, Research Scientists, Agriculture Canada, Brandon Research Station, Brandon Manitoba, Canada

Olive *(Olea europaea L.)*

French: Olivier; Spanish: Olivo; Italian: Olivo; German: Olive; Arabic: Zeitoun

Crop data

Perennial evergreen tree. Harvested part: fruit, which is either consumed after processing ('table'varieties) or used to produce olive oil ('oil'varieties). Some varieties are used for both purposes.

It is cultivated all around the Mediterranean basin and in similar agro- ecological zones elsewhere.

In the Mediterranean area the flowering period is May and harvest starts in late October. The life of the leaves is about 3 years.

Planting density varies widely with soil, rainfall, variety and cultural practices, ranging from 17 trees/ha (Sfax area, Tunisia) to 500 trees/ha (in intensive and irrigated groves).

The crop is adapted to diverse soil conditions.

The yield varies from one site to another and from year to year. Biennial bearing is common, especially in semi-arid zones and in unirrigated groves.

Nutrient demand/uptake/removal

Nutrient uptake - Macronutrients				
Country	**Source**	**g/tree/year**		
		N	**P2O5**	**K2O**
Tunisia (Sfax)	Rey	578	67	502
France	Bouat, 1968	300	60	200
Spain (Jean)	Llamas, 1983	310	75	560
Italy	Pantanelli	276	142	488

Plant analysis data

The leaves used in foliar diagnosis should be the third or fourth pairs on the twig, taken during the winter; lower and apical leaves should be discarded.

Work at Montpellier, France, by Bouat (1968) established as optimum values:
N = 2.10 %; P2O5 = 0.35 %; K2O = 1.05 % of dry matter.

In Spain, Recalde (1975) observed generally lower levels of nutrients but in the same relative proportions: N = 0.68-2.20 %; P2O5 = 0.13-0.42 %; K2O = 0.21-1.80 % of dry matter.

Normal ranges found in Mediterranean counties, according to Bouat, are:

Plant Analysis Data - Macronutrients						
Range	**% of dry matter**					
	N	**P**	**K**	**Ca**	**Mg**	**S**
Minimum	1.01	0.05	0.22	0.56	0.08	0.02
Mean	1.77	0.12	0.80	1.43	0.16	0.12
Maximum	2.55	0.34	1.65	3.15	0.69	0.28

Plant Analysis Data - Micronutrients					
Range	ppm dry matter				
	Fe	Cu	Zn	Mn	B
Minimum	40.0	1.5	4.0	5.0	2.0
Mean	124.0	9.0	23.5	36.0	11.7
Maximum	460.0	78.0	84.0	164.0	24.5

Fertilizer recommendations

Some authors recommend an annual application of fertilizers equivalent to two to three times the amount removed in the harvested crop.

J.G. de Geus (1973) gave as a general recommendation 500-1 000 g N, 300-600 g P2O5 and 500-1 000 g K2O per tree, and suggested that these rates should be doubled for very high yielding olives under irrigation.

Spain

In Spain De La Vega de Luque (1969) recommended:

Spain - Fertilizer Recommendations				
Production level	g/tree			
	Winter application			Autumn application
kg/tree	N	P2O5	K2O	N
< 15	200	200	400	300
15-30	300	300	600	300
30-50	400	400	800	400
> 50	500	500	1000	500

Morocco

Morocco - Fertilizer Recommendations				
	g/tree			
	November			February
	N	P2O5	K2O	N
Rainfed	400	350	1000	700
Irrigated	550	460	1300	900

Peru

Peru - Fertilizer Recommendations			
	kg/ha		
	N	P2O5	K2O
Young trees	50	30	-
Bearing trees	80-130	60	60

Tunisia

50 kg N/tree for each year of age up to production, and 2 kg single superphosphate (16 % P2O5) and 1 kg KCl per bearing tree (Min.Ag.Tunisia 1976).

Fertilizers should be applied to the soil surface under the leaf canopy. Kechaou and Tnani (1978) using a fertilizer application of 1.5 kg N (surface appli- cation), 0.9 kg P2O5 and 1.4

kg K_2O (P_2O_5 and K_2O in-depth application) per tree in Sfax, Tunisia, have shown, that placement of phospho-potassic fertilizers at a depth of 20-30 cm by a localizer plough had the effect of depressing the yield.

Organic manures are also recommended where available, at a rate of 50 kg/tree every 3 years, especially in irrigated plantations (Loussert and Brousse 1978).

Further Reading

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MINISTRY OF AGRICULTURE, Tunisia: L'olivier. Brochure de vulgarisation (1976)

Authors: M. Lasram; M.T. Tnani, Institut National de la Recherche Agronomique de Tunisie, Ariana, Tunisia

Safflower *(Carthamus tinctorius L.)*

French: Carthame, safran batard; Spanish: Cartamo, alazor; Italian: Cartamo; German: Saflor

Crop data

Annual. Seeds are harvested for oil (30 to 45 %) used in human food (low cholesterol) or for industrial uses; meal for livestock protein supplement (24 % protein) which is high in fibre.

Planted in April/May (early spring) in areas which have at least 120 days of frost free periods, and hot summers. The crop is frost-tolerant in the seedling stage, withstanding temperatures of -7 °C. Typically grown at less than 1100 m altitude.

Sowing rates: 17 - 28 kg/ha for dryland and 28 - 39 kg/ha for irrigated conditions (pure live seed), at depths of 2.5-4 cm, after soil temperature has reached 4 °C. Row spacing: up to 35 cm. Soil crusting can be a problem in establishment.

Safflower is a deep-rooted, long-season crop that withstands periods of drought longer than other annuals. Optimum growing condition occur on deep, fertile, well-drained soils with a neutral pH. Plant height varies from 38 to 102 cm depending upon environmental conditions. It is particularly suited to dryland conditions where an annual crop is needed to use surplus water from recharge areas that otherwise would contribute to saline seeps*. The crop is very sensitive to excessive water. Safflower is a good rotation crop with small grains or on fallow in that it helps to break weed and disease cycles. It should not be grown in two consecutive seasons due to the lack of available water and the potential for serious damage from diseases.

** Saline seeps are areas with intermittent or continuous saline water discharge under dryland conditions, at or near the soil surface downslope from recharge areas (from which water percolating below the root zone flows laterally and so continues to the saline seepage).*

Nutrient demand/uptake/removal

Nutrient demand/uptake/removal - Macronutrients				
Yield kg/ha	**Source**	**kg/ha**		
		N	**P2O5**	**K2O**
1610 (grain + straw)	Halvorson & Black, 1985	68	-	-
1800 kg/ha	Thorup, 1984	91	23	68
2200 kg/ha	Singh & Singh, 1980	77	40	63

Plant analysis data

No information available.

Fertilizer recommendations

Fertilizer application should be based on residual nutrient levels as indicated by a preplant soil test, available soil water, soil type, level of management and availability of irrigation. On coarse-textured soils all fertilizer can be applied before sowing. On heavy-textured soils about half the N can be applied before sowing, the other half at early budding. Approximately 18 kg N/ha is needed for each kg/ha of seed. Excessive N applications can result in excessive foliage which depletes available water during seed development, thus decreasing seed yield and quality.

Nitrate N soil tests should be taken to a depth of 61 cm, while P and K tests should be taken to a depth of 15 cm. Deeper nitrate-N soil tests may give better guidance. Recently published fertilizer recommendations for North Dakota/USA based on target yield, residual nitrate-N, sodium bicarbonate extractable P and ammonium acetate extractable K, are shown below.

USA - North Dakota Recommendations for Broadcast Application - Macronutrients									
Target yield kg/ha	**Soil N plus fertilizer N needed* kg/ha N**	**P Soil Test Level(kg/ha)**				**K Soil Test Level(kg/ha)**			
		VL 0-10	**L 10-20**	**M 20-30**	**H over 30**	**VL 0-110**	**L 110-223**	**M 223-335**	**H over 335**
		kg P2O5/ha				**kg K2O/ha**			
672	34	17	11	0	0	22	17	0	0
896	45	22	11	0	0	34	22	0	0
1120	56	22	17	0	0	39	22	11	0
1344	67	28	17	0	0	50	28	11	0
1568	78	34	22	0	0	56	34	11	0
1792	90	39	22	11	0	67	39	11	0
2016	101	45	28	11	0	73	45	17	0
2240	112	50	28	11	0	84	50	17	0
2464	123	56	34	11	0	90	56	17	0
2688	134	62	34	11	0	101	62	22	0
* Subtract residual nitrate N (61 cm depth) from figures in this column.									

Preferred nutrient forms

Nutrients should be applied in readily available forms, consequently all standard N, P and K fertilizers are suitable sources. The best methods of application are dependent upon the fertilizer source, the equipment available and soil chemistry. For example, anhydrous ammonia should be applied 10-15 cm below the soil surface, while other ammonium N sources (urea, ammonium nitrate, solutions) should be broadcast and incorporated before sowing, particularly on high pH, calcareous soils.

Preliminary research indicates a distinct advantage to band placement of P over broadcasting, particularly on low testing soils.

Most soils of Western USA are high in K, but specific areas may be marginal or deficient and require fertilizer application for optimum production.

S deficiencies are not common, but may occur on sandy soils early in the growing season. Safflower has not responded to micronutrient applications in Montana, USA.

Present fertilizer practices

For Montana, USA, average dryland fertilizer recommendations would be 56 kg/ha 11-55-0 with the seed, followed by N topdressing of 34-68 kg/ha N; average irrigated fertilizer recommendations would be 78 kg/ha 11-55-0 with the seed and 68-90 kg/ha N topdressed.

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SINGH, U.B.; Singh, R.M.: Effect of graded levels of moisture regimes, N and P fertilization on seed yield, oil content and NPK uptake by safflower. Indian Journal Agronomy. 25 (1):9-17 (1980)

Author: J.S. Jacobsen, Department of Plant and Soil Science, Montana State University, Bozeman, USA

Soybean *(Glycine max [L.] Merr.)*

French: Soya; Spanish: Soja; Italian: Soja; German: Sojabohne

Crop data

Annual. Harvested part: seeds.

Sown February-August (Northern hemisphere).

Flowers 1-3 months after sowing.

Harvested 3-5 months after sowing.

Plant density:

Planting date	**Row width**		
	Wide > 75 cm	**Medium 30-75 cm**	**Narrow <30 cm**
	'000 plants/ha		
Early spring	150-300	200-450	250-700
Spring	150-300	200-450	250-600
Summer	150-200	150-300	200-450
Autumn	200-300	200-450	250-700

The crop is adapted to a wide range of climatic conditions. It is most susceptible to drought damage during flowering and grain filling. It is not generally irrigated.

Availability of essential nutrients is influenced by soil pH through its effects on Al saturation percentage and on nutrient fixation and release mechanisms. Highest soybean yields are usually produced when soil pH is between 6.2 and 7.0. In this range, adequate Ca and Mg are normally available. Soybeans grown on naturally acid oxisols and ultisols will generally produce to their potential at soil pH between 5.5 and 6.5. However, liming of these soils should also reflect the importance of exchangeable aluminum. Soils with low exchangeable Al with no soil solution Al generally will not benefit from lime application. This usually occurs at a soil pH of 5.5 or greater.

Nutrient demand/uptake/removal

Nutrient demand/uptake/removal - Macronutrients							
Part	**Source**	**kg/t grain**					
		N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
Grain only	(1)	65	14	23	5	4	2
Grain only	(2)	100	16	21	21	4	4
Grain only	(3)	65	11	20	4	4	2
Total		81	14	33	18	24	3
Grain only	(4)	64	11	20	4	4	2
Total		77	14	39	7	18	8
Grain only	(5)	38	8	16	19	9	n.a.

Nutrient demand/uptake/removal - Micronutrients							
Part	**Source**	**g/t grain**					
		Fe	**Mn**	**Zn**	**Cu**	**B**	**Mo**
Grain only	(1)	n.a.	20	17	16	n.a.	n.a.

Grain only	(3)	110	33	43	16	16	6
Total		366	90	61	25	39	7
Grain only	(5)	n.a.	n.a.	24	n.a.	n.a.	n.a.

(1) Adapted from "The Fertilizer Handbook", TFI, 1982
(2) Bataglia et al., 1977
(3) Bataglia & Mascarenhas, 1978
(4) Cordeio et al., 1979
(5) Guo Qingyuan, 1991 (personal communication)

Plant analysis data

Sufficiency range for upper fully developed leaf at initial flowering stage - Macronutrients				
% of dry matter				
N	**P**	**K**	**Mg**	**Ca**
4.62-5.50	0.26-0.50	1.71-2.50	0.26-1.00	0.36-2.00
Source: Small & Ohlrogge, 1978				

Sufficiency range for upper fully developed leaf at initial flowering stage - Micronutrients					
ppm dry matter					
Fe	**Mn**	**Cu**	**Zn**	**B**	**Mo**
51-350	21-100	10-30	21-50	21-55	1-5
Source: Small & Ohlrogge, 1978					

Fertilizer recommendations

Available organic manures should be applied and incorporated well before primary tillage. If necessary soil pH corrections should be made prior to doing primary tillage. Mineral fertilizers may be split into two or more applications on highighly leachable soils or soils which fix large amounts of nutrients. Most non-sandy soils are soils in which fixation is low; usually one application of fertilizer is satisfactory. On soils low in nutrients it is often more efficient to apply fertilizers in a band near the row at seeding.

N: Soybeans, being a legume fix adequate atmospheric N to produce yields of 3 000-4 000 kg/ha, if well nodulated. Johnson et al. (1975) found that adding N to well nodulated soybeans produced no yield increase. Fertilizer N added at planting will delay nodulation. Gascho et al. (1989) suggested that N applications during the vegetative stages result in decreases in nodulation in proportion to the rates applied. Adding N is recommended only when adequate nodulation is not expected.
N deficiency results in reduced chlorophyll development and a pale-green leaf colour. Growth, development and yield are reduced.

P: taken up througout the growing season. The period of greatest demand starts just before the pods begin to form and continues until about 10 days before the seeds are fully developed. Much P used in seed development is taken up early, stored temporarily in leaves, stems and petioles, and then translocated into the seed.
Stunted growth is usually the only symptom of P deficiency, though some leaf cupping and discolouration are possible.

K: Relatively large amounts of K are required. Hanway & Weber (1971) reported that the rate of uptake is highest during rapid vegetative growth and slows as seed formation begins. Uptake continues until two to three weeks before the seed is mature; it can be depressed by poor soil condition including compaction, excess moisture and poor aeration. Most K taken up moves to the roots by diffusion through moisture films around soil particles. As the water

content of a soil decreases, moisture films around the soil particles become thinner and the path length of ion movement increases and the movement of K to roots decreases.
Lawton (1945) reported that K uptake decreases if the oxygen content of the soil is low, therefore poor aeration would require higher available K.
Cold soils reduce the rate and extent of root growth and this can limit K uptake. When farmers plant earlier or adopt tillage practices that result in lower soil temperatures early in the growing season, such as no-till, higher levels of available K in the soil are likely to be needed for optimum growth.
K deficiency is easily recognized; chlorosis starts along the outside edges of leaves, especially the older leaves.

Mg and Ca: In most cases Mg and Ca needs are met when a suitable soil pH is maintained. Acid soils are often deficient in Mg; dolomite is therefore the preferred lime source.

S: Substantial amounts of S are taken up, but few research trials have shown responses to added sulphur. In the past, S was a component of several fertilizers applied as sources of other nutrients; and additional S from emissions from burning of fossil fuels may have been enough to make up the remaining requirements. Today, most fertilizers contain little sulphur, and emissions from fossil fuels are being reduced, so it may become more limiting in the future.
Deficiency is most likely to occur during cool, wet weather on highly leachable soils low in organic matter.

Micronutrients: Some of the first growers to experience micronutrient deficiencies are those who have supplied plenty of N, P and K and raised yields to the point where stress is put on the micronutrient-supplying capacity of soil.
Deficiencies of micronutrients are more common than on most other field crops. Shortages of Fe, Mn, Mo and Zn have been observed. Deficiencies are limited to rather specific soil situations, viz:

- Strongly weathered soils. These soils are old in terms of the extent of chemical and physical changes that have occurred. Many have lost nutrients within the soybean rooting zone through leaching.

- Coarse-textured soils. These soils were formed from rock materials low in micronutrients. Furthermore, rainfall penetrates much more rapidly and to a greater depth, thus causing more leaching of nutrients from sandy and gravelly soils than from loams, silt loams, and silty clay loams.

- Alkaline soils. The solubility, though not total supply, of several nutrients diminishes as pH rises; it is not feasible, however, to lower the pH to correct micronutrient deficiencies.

- Organic soils. These soils have a low mineral content.

Soil tests have not been nearly as satisfactory for determining micronutrient deficiencies as have plant analyses. Differential responses have been found between varieties to both Fe and Mn.

Where a micronutrient deficiency is acute, the visual response to small, corrective applications may be striking, but often the yield response, if any, is negligible and there is no visible change in plant growth. Carefully controlled research trials are usually needed for precise evaluation. The margin between too little, just enough, and too much of a micronutrient is very often narrow.

Fe and Mn: Deficiencies of both occur on soils with high pH. Symptoms are very similar: stunted plants with pale yellow to nearly white leaves, but with green veins, over the whole plant. Fe deficiency is accentuated by very dry soil conditions. Mn deficiency is most pronounced in cool wheather.
If soybeans are planted in rows wider than 50 cm and the planter can apply row fertilizer, then the use of an acid-forming row fertilizer will usually prevent Mn deficiency. The alternative method is foliar application of 5-10 kg/ha manganese sulphate when the plants have two or three trifoliate leaves.

B: Results with boron additions where soil tests indicate low supplies have been inconsistent. Conditions conducive to B shortage are: high pH (7.0-8.0); coarse-textured soil from which nutrients leach readily; low organic matter; and drought. Because of the scarcity and variability of results and the extreme susceptibility of soybeans to boron toxicity, growers should consult their agronomist for guidance.

Cu: seldom deficient, except on soils high in organic matter (mucks) with pH > 6.0.

Mo: Differs from other micronutrients in that availability increases with rising soil pH. It is seldom lacking for soybeans at pH 6.0 or above. Legumes need Mo for N fixation. The deficiency symptom is pale green or yellow plants indicating a nitrogen shortage in the leaf tissue. Not surprisingly, this usually does not occur on any soil that is high enough in N to make up for the lack of nodule fixation.
Sodium molybdate (30 % Mo) and ammonium molybdate (48 % Mo) are used as seed treatments to supply 30-40 g/ha Mo; soil treatment is 60-120 g/ha Mo, but the first step is always to establish the correct soil pH.

Zn: Deficiency is most likely on soils with a high pH, high P and low organic matter. Soybeans on sandy soils are more susceptible than those on finer-textured soils. The deficiency can be caused by heavy P fertilization, especially near the row, or by very high soil test levels for P. Deficient plants are stunted. The leaves are yellow or light green. The lower leaves may turn brown and drop. Flowers are scarce. The few pods that do set are abnormal and slow-maturing. In mild deficiencies, early growth is stunted and plants are very light green or chlorotic. Soil tests are available, but plant analysis is usually recommended for additional confirmation.
Zinc sulphate is the most widely applied inorganic source. A single broadcast application is normally adequate for 2-4 years, if soils do not have high pH or high P level. When chelated zinc is used, manufacturers' directions should be carefully followed. Application of organic manure often corrects Zn deficiency.

Preferred nutrient forms

All fertilizers applied should provide easily plant-available forms. On soils which are highly leachable and low in organic matter, K2O should be applied preferredly in the sulphate form.

Present fertilizer practices

USA:

Only P and K are routinely used. Primary and secondary nutrients are added when there is a predicted need from either plant or soil analysis. Common basal fertilizer rates are incorporated into the soil prior to seeding.

The three dominant soil tests used for P analysis are Bray P1, Mehlich and Olson; the numerical value for sufficiency differs for each extractant. Most soil tests for K use an extractant which indicates amounts of exchangeable K in the soil. The clay content is

sometimes used along with exchangeable K values to determine sufficiency levels for potassium. The Cation Exchange Capacity is used to indicate soil high in clay. Basal fertilizer rates are:

Soil test level	**Fertilizer nutrients recommended kg/ha**		
	P2O5	**K2O**	
		Low or normal clay content	**High clay content**
Low	40-60	100-150	120-180
Medium	30-40	50-100	70-100
Adequate	0-30	0- 50	0- 70
High	0	0	0
For expected yield of 2 500-2 700 kg/ha; for each additional 1 000 kg add an extra 10-15 kg/ha P2O5 and 20-30 kg/ha K2O.			

Brazil:

- Rio Grande do Sul and Santa Catarina

P: 10-140 kg/ha P2O5 (average about 50 kg/ha P2O5) on basis of soil analysis, % clay, number of successive soybean crops and expected yield.

K: <40-120 kg/ha K2O on basis of soil analysis, number of successive soybean crops and expected yield.

Liming: to pH 6.0

Micronutrients: When the pH is low, 8-10 g/ha Mo should be applied to the seeds.

- Paraná

P: 90-100 kg/ha P2O5 on soils used less than 3 years, or 20-50 kg/ha P2O5 on soils used 3 or more years, on basis of soil analysis and number of years of cropping.

K: up to 45 kg/ha K2O on soils used less than 3 years, or up to 60 kg/ha K2O on soils used 3 or more years, on basis of soil analysis and number of years of cropping.

Liming: on basis of formula:
NC = ((70 - V1) x T) / 100 x f
where :
NC - lime requirement
V1 - base saturation, before liming
T - CEC
f lime reactivity, usually 1.5

- Sao Paulo

P: 20-80 kg/ha P2O5 on basis of soil analysis.

K: up to 60 kg/ha K20 on basis of soil analysis.

S: 20 kg/ha S.

Liming: using same formula as in Paraná.

- Minas Gerais

P: 40-120 kg/ha P2O5 on basis of soil analysis.

K: 20-60 kg/ha K2O on basis of soil analysis.

Liming: using same formula as in Paraná.

- Cerrado Region

P: 60-100 kg/ha P2O5 on basis of soil analysis and % clay. Yields over 3 t/ha on soils with medium to good P supply require an additional 20 kg/ha P2O5 for each additional ton grain. A corrective application of partially acidulated phosphate or thermophosphate, broadcast and incorporated into the soil, is recommended at 50-120 kg/ha P2O5 for soils low in P (or 100-240 kg/ha P2O5 for soils very low in P).

K: 20 kg/ha K2O for each ton of grain expected. A corrective application of up to 100 kg/ha K20 is recommended for soils low in K. Where CEC is < 4 meq/100 cm3, half of the K may be applied at sowing and half 30 days after plant emergence.
Zn: 4 kg/ha Zn.

Liming: using same formula as in Paraná

China:

Sowing time	**Province**	**Yield level**	**Basal fertilization kg/ha**					**Remarks**
		t/ha	**N**	**P2O5**	**K2O**	**ZnSO4**	**H2BO 3**	
Spring	Heilong-jiang	2-2.25	45	45	-	-	-	30-45 kg/ha N topdressed 2 months after sowing
Spring	Hunan	3	-	45	120	-	6	60 kg/ha N topdressed 2 months after sowing
Summer	Henan	2-2.25	45	60	-	15	-	
		3	60	60	60	15	-	
Autumn	Hubei,Hunan	2.25	60	45	80	-	-	

The basal fertilization is banded in the seedrow and incorporated into the upper 15 cm of soil.

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Sunflower *(Helianthus annuus L.)*

French: Tournesol; Spanish: Girasol; Italian: Girasole; German: Sonnenblume

Crop data:

Annual species, member of the Compositae. Main stages of plant development:
- Emergence.
- Leaf development, from 1 to 25-30 leaves.
- Bud initiation progresses from the 8-10 leaves stage until the buds reach 15 mm diameter, when the potential member of seeds is determined.
- Flowering, 60-70 days after emergence. The period of seed setting is critical for the first yield component, the final member of seeds.
- Maturity, the period of oil accumulation, which is critical for the second yield component, the 1000-seed wheight, and when the leaves should be kept alive.

Sowing, March-April; harvest, September-October (Northern hemisphere). Commercial growing cycle 120-160 days. Quick-maturing cultivars are available for Northern France.

Sunflower seeds (kernels) have become a major source of oil (about 50 % of the dry weight), and of meal. The quality of the meal can be improved by dehulling before crushing, thus decreasing the crude fibre content.

Hitherto, sunflowers were grown on light and shallow soils but, with increasing prices, they can now be found on better soils. Yields are potentially up to about 5 1/2 to 6 t/ha but, due to diseases, water stress, etc., average around 3 t/ha. Under dry conditions a minimum of about 2 t/ha can be expected.

Nutrient demand/uptake/removal

Nutrient uptake/removal - Primary and secondary nutrients					
For yield of	**kg/ha**				
3.5 t/ha	**N**	**P2O5**	**K2O***	**MgO**	**CaO**
Uptake	131	87	385	70	210
Removal	66	54	82	14	10
Residues	65	33	303	56	200
* Note the high potassium uptake.					

Nutrient uptake/removal - Micronutrients					
For yield of	**g/ha**				
3.5 t/ha	**Fe**	**Cu**	**Zn**	**Mn**	**B***
Uptake	732	59	348	412	396
Removal	106	25	148	42	80
Residues	626	34	200	370	316
* Among the micronutrients, boron is one of the most critical.					

Plant analysis

Plant analysis data - Optimum concentration/Macronutrients								
Source	Sampled leaf	Sampling time	% of dry matter					
			N	P	K	Mg	Ca	S
CETIOM	5th and 6th leaves under the head	F1-Flowering	3.08	0.37	3.91	0.32	2.18	-
Literature	"	"	3.0-5.0	0.3-0.5	3.0-4.5	0.3-0.8	0.8-2.0	0.15-0.2

Plant analysis data - Optimum concentration/Micronutrients							
Source	**Sampled leaf**	**Sampling time**	**ppm dry matter**				
			Fe	**Cu**	**Zn**	**Mn**	**B**
CETIOM	5th and 6th leaves under the head	F1-Flowering	107	12.5	45.8	44.8	62.3
Literature	"	"	80-120	10-20	30-80	25-100	35-100

Fertilizer recommendations

50-80 kg/ha N applied before sowing (about 60-70 percent of N-uptake comes from the soil and subsoil; the rooting system can go down to 2 m depth). Too much N increases the risk of disease and lodging, with a consequent reduction in oil content.

Also at sowing, on rich soils, 60-80 kg/ha P2O5 and 100-120 kg/ha K2O. Or, on poor soils, 100-150 kg/ha P2O5 and 150-300 kg/ha K2O.

Special attention must be paid to boron nutrition: sunflower is very sensitive to boron deficiency on calcareous or sandy soils and in regions with demanding climate. Boron application may be made either at sowing time (1.2 kg/ha B), applied on the soil, or at the 10-leaves stage, as a foliar application of 500 g/ha B (0.1 % B solution).

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Author: A. Merrien, CETIOM, Département Etudes et Recherches - Section Physiologie, Paris, France

Tung *(Aleurites Fordii Hemsl.; Aleurites montana)*

French: Tung; Spanish: Tung; German: Tung

Crop data

Perennial.

Harvested part: fruits (which contain heavy-shelled seeds from which the kernels are extracted; it is from these kernels that the oil is obtained).

Sown late spring (direct seeding/transplanting).

Flowers early to mid-April 2-3 years after transplanting.

Harvested late October to early November.

Plant density: A. Fordii, traditional 1 125-1 200/ha, intensive 1 500-1 800/ha. A. montana, traditional 540-675/ha, intensive 750-900 ha.

Preferably grown at temperatures of 16-22 °C, falling in winter to 6-9 °C (minimum -8 °C), with an annual precipitation of 750-1 200 mm. Main commercial production in China is at 400-900 m (or even up to 1 200 m) above sea-level.

Prefers light to medium soils with pH 5.2-5.6 or up to 6.5. Liable to suffer from leaf roll and leaf scorch where soil pH is around 7.0-8.0.

Analytical data

Part	%					
	Moisture	Crude protein	Crude fat	Carbo-hydrate	Crude fibre	Ash
Shells	14.4	2.5	0.04	27.6	50.6	4.8
Kernels	4.0	16.6	57.5	12.6	2.7	3.6
Tung seed cake	9.8	46.2	6.5	29.5	3.6	4.4

Plant analysis data - Macronutrients			
Part	% of dry matter		
	N	P	K
Leaf	1.5	0.6	1.0
New branch	1.0	0.4	0.65
Stem	0.75	0.2	1.5
Shells	1.0	0.8	7.0
Tung seed cake	2.1	1.3	0.5

Fertilizer recommendations

Available organic materials such as pig manure, night soil, rapeseed cake and plant ash should be applied in February or March as a basal dressing before transplanting.

Mineral NPK fertilizers should be applied as a basal dressing, well mixed with the organic materials, followed by two fertilizer topdressings annually thereafter in April and September as indicated below:

For topdressing in next year after transplanting: 80 kg/ha N, 100 kg/ha K2O.

For topdressing 3-4 years after transplanting: 60 kg/ha N, 60 kg/ha P2O5, 60 kg/ha K2O.

For topdressing 5-6 years after transplanting: 100 kg/ha N, 60 kg/ha P2O5, 120 kg/ha K2O.

Present fertilizer practices

China, Hunan Province: For A. Fordii with an expected yield of 30-40 t/ha, basal application of 30 t/ha organic manure with mineral fertilizer containing 80 kg/ha N, 50 kg/ha P2O5, 80 kg/ha K2O. Topdressings of 50 kg/ha N in May and 50 kg/ha in September.

China, Szechuan Province: For traditional practices on medium red soil, basal application of 30 t/ha organic manure with mineral fertilizer containing 60 kg/ha N, 45 kg/ha P2O5, 90 kg/ha K2O. Topdressings in April and September each year of 7.5 t/ha green manure and plant ash and nitrogen fertilizer containing 60 kg/ha N.

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Sugarbeet *(Beta vulgaris L. ssp. vulgaris)*

French: Betterave à sucre; Spanish: Remolacha azucarera; Italian: Barbietola zuccherina; German: Zuckerruebe

Crop data

Sugarbeet grows best on slightly acid to neutral soils (pH = 5.8-7.0) of loose structure. Liming has to be carried out accordingly. The crop prefers a temperate climate, with sufficient precipitation (or irrigation) during the growing period, up to the ripening phase. The young seedlings are sensitive to frost. The number of plants per hectare varies from 35 000 - 90 000 in the various cropping areas, with beet yields of 17 - 80 t/ha (world average 1984: 33 t/ha) and saccharose contents of 13 - 20 %. A large number of plants per hectare, in addition to promoting beet yield and sugar content, also improves sugar extraction.

Nutrient demand/uptake/removal

As potential beet yields and thus nutrient removal figures for various sites may differ considerably, it is appropriate to express nutrient removal on a standard basis.

Nutrient removal by 10 t beets plus foliage								
Source	**Part**	**kg**					**g**	
		N	**P2O5**	**K2O**	**MgO**	**S**	**B**	**Mn**
Buchner & Sturm, 1985	B+F*	40-55	15-20	60-100	10-20	4	-	-
Finck, 1979	B+F	35-60	14-17	50- 70	14-19	-	-	-
Michigan State Univ., 1985	B	20	17	2	-	-	-	-
Faustzahlen, 1983	B+F	40-50	15-20	60-100	10-20	8	60-75	110-250
	B	15-22	8-10	20- 30	5-10	-	20-35	65-120
Hills et al, 1982	B+F	40-50	17	67	-	-	-	-
Beiss & Winner, 1975	B+F	~40	~14	~63	~14	-	-	-
* B = Beets, F = Foliage								
Amounts to be increased at low yield levels, to be decreased at high yield levels.								

Nutrient uptake/removal by 10 t beets, together with the foliage averages 40-50 kg N, 15-20 kg P2O5, 45-70 kg K2O, 12-15 kg MgO and 5 kg S, of which the beets themselves contain only about half. The calculation of nutrient demand must be based on the total removal, as the leaves in which the sugar synthesis primarily takes place must not suffer from nutrient deficiency. If the leaves are incorporated into the soil after the harvest, the nutrients they contain may be taken into account when estimating the fertilizer requirements of the next crop.

The approximate nutrient removal at an individual site can thus be easily calculated from the prospective yield and intensity of cultivation.

Variation of nutrient removal by sugarbeet depending on cultivation intensity							
Intensity	Yield t/ha	Beets/ha	Nutrient removal kg/ha				
			N	P2O5	K2O	MgO	S
Low	20	30 000	100	40	160	28	10
Medium	40	50 000	180	64	240	50	18
High	60	75 000	240	90	300	72	24

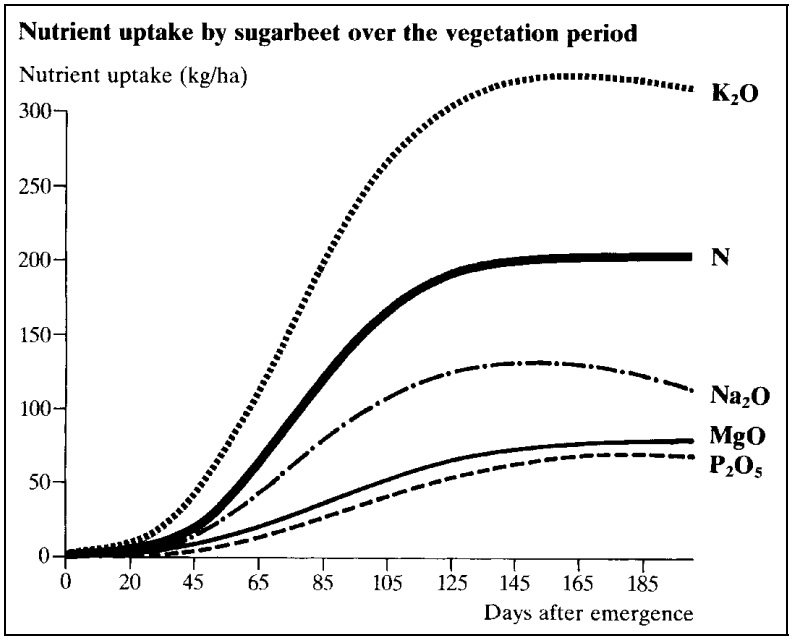
Plant analysis

Plant analysis data for fully developed leaf blades									
Source		% of dry matter						ppm DM	
		NO3-N*	P	K	Mg	Ca	SO4-S	B	Mn
Bergmann, 1976	Deficient	<0.45	<0.2	<0.6	<0.02	<0.1	<0.04	<20	<10
	Adequate	0.45-1.0	0.3-0.6	>1.0	>0.04	0.5-1.5	>0.05	>30	>30
Hills et al, 1982	Deficient	<0.5	<0.25	<0.5	<0.03	<0.1	-	<21	<10
	Adequate	>0.5	>0.3	>1.0	>0.5	>0.2	-	>35	>25
* Petioles									

Fertilizer recommendations

The aim of sugarbeet fertilization is to produce high yields with a high content of extractable sugar. However, fertilization is but one growth factor, yield and sugar content of the beets also depend significantly on climate (humidity, temperature), soil type and texture, cropping intensity (plants/ha) and crop management (plant protection) and on the soil's nutrient supply. It is therefore hardly possible to give a general recommendation for the fertilizer use in sugarbeet! It derives from the environmental conditions and the cultivation practices at the specific site.

Nutrient uptake precedes beet growth. An example showing the uptake of the macronutrients during the vegetation period is given in the figure below:



In consequence, the greater part of the nutrients should be applied in good time, i.e. before sowing. Maximum demand occurs approx. three to four months after sowing. The figure

shows that P uptake in particular develops very slowly, as the roots of the young beets have only a limited extraction capacity for this nutrient. Application and soil incorporation of easily soluble fertilizer P is therefore of special importance.

Since sugarbeet prefers nitrate, a proportion of the N application should be given as NO3. A subsequent dressing of N may be applied up to 90 days after sowing. The N supply should drop in particular significantly in the last third of the growing period, as excessive contents of N and protein adversely affect sugar extraction.

Sugarbeet can utilize either the natural soil supply or nutrients from organic manures and mineral fertilizers to meet its nutrient demand. Crucial for the fertilizer nutrient demand of sugarbeet is the content of "plant available" nutrients in the soil of the respective field. Under the aspect of economy and energy efficient crop production, agronomic soil analysis, considering as many site specific factors as possible, is very important for practical fertilizer usen. For the various beet growing regions different analysis methods have been developed for this purpose which can not be presented here in detail. However, it is important to use the method common in the specific area and well calibrated by field trials.

Fertilizer recommendation in accordance to soil nutrient status - Germany Expected yield: 50 t/ha beets					
Soil nutrient status	**Recommended rate kg/ha**				
	P2O5	**K2O**	**MgO**	**B***	**Mn***
A - very low	200	400	100	2	2x2
B - low	150	300			
C - adequate	100	200	50	1	2
D - high		50	100		
E - very high	0	0	0	0	0
* as foliar spray					
Source: Faustzahlen fuer Landwirtschaft, 1983					

As mentioned above, special attention has to be given to a targeted nitrogen fertilization of sugarbeet. A special method developed for this purpose in Germany is based on the amount of nitrate which is found at the beginning of the vegetation period in one hectar of soil to a depth of 90 cm. For a yield of approx. 50 to 60 t sugarbeets the optimum index value has been determined as 200 kg/ha N. I.e. with a nitrate content (NO3-N) of e.g. 70 kg/ha N found in the soil, the deficit of 130 kg/ha N has to be applied as mineral fertilizer.

Micronutrients

Sugarbeet has an increased demand for B and Mn, especially on soils with a high pH (> 6.8/7.0). 1-2 kg/ha B and 6-12 kg/ha Mn should be applied before sowing, or these nutrients may be included in a foliar spray.

Organic manures

Sugarbeet utilizes the nutrients from organic manures (farmyard manure, compost, organic wastes) quite well. They should be well incorporated into the soil before sowing. Their content of plant available nutrients - especially nitrogen - has to be taken into consideration when calculating the mineral fertilizer dose.

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Author: A. Buchner, Head (rtd.) Fertilizer Research and Development, BASF Aktiengesellschaft, Ludwigshafen/Rh., Germany

Sugarcane *(Saccharum officinarum L.)*

French: Canne à sucre; Spanish: Caña de azucar; Italian: Canna da zucchero; German: Zuckerrohr; Portuguese: Cane de acucar

Crop data

Perennial grass. Harvested part: stalks (sugar/sucrose extracted from juice after crushing).

True seed (fuzz) is used only for plant breeding; flowering is undesirable in the commercial crop. Commercial propagation is by cuttings or setts of 2-3 buds or eyes, usually 30 cm in length, planted in long furrows 15-30 cm deep, or in trenches, at 10 000-15 000 3-eye/bud setts/cuttings per hectare (one-third overlapping), with an intended stalk density at harvest of 90 000-150 000/ha.

Row spacing: high density, 75 cm; general, 90-100 cm, or for mechanized harvesting 125-150 cm.

Clump spacing within the row: 30-45 cm.

Planting time: practically throughout the year in equatorial belt; mostly in spring or late spring, or sometimes in autumn, in subtropical and subtemperate regions.

Maturity is reckoned by age and variety; hand refractometer Brix of at least 18° is usually a reliable index.

Harvesting: minimum 10 months, maximum 24 months after planting, mostly during the cooler and relatively dry part of the year. Usually the harvesting and planting operations overlap in order to avoid storing the planting material.

Crop cycle: One to four or five ratoons; followed by a green manure or arable crop before re-planting to sugarcane.

Soil: medium to heavy, pH 5.0-8.5. Liming required if pH < 5, or gypsum if pH > 9.5.

Climate: optimum temperature range 25-30 °C. High temperature, with high humidity and moist soil, is favourable for vegetative growth but it should be followed by cool dry wheather to promote ripening.

Irrigation: crop is adapted to semi-arid conditions with good availability of irrigation water. Requirement, 75-150 cm. Reasonably tolerant to waterlogging for up to 7-10 days except during tillering and sprouting.

Nutrient demand/uptake/removal

Nutrient uptake/removal - Macronutrients							
Country	**Source***	**kg/ton**					
		N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
India	Zende, 1983	1.2	0.46	1.44	-	-	-
Brasil	Malavolta, 1982	0.8	0.30	1.32	0.50	0.42	0.25
South Africa	Thompson, 1988	1.75	0.41	6.00	0.66	0.70	-
* Pooled data							

Nutrient uptake/removal - Micronutrients							
Country	**Source***	**g/ton**					
		Fe	**Mn**	**Zn**	**Cu**	**B**	**Mo**
Brazil	Malavolta, 1982	31	11	4.5	2.0	2.0	0.01
South Africa	Thompson, 1988	-	11	2.5	0.5	1.2	-
* Pooled data							

Plant analysis data

Crop logging and foliar diagnosis are widely used. Procedures have been outlined both for crops with an average of 12 months duration and for crops (in Hawaii) of about 2 years duration. Two systems have been evolved for the latter:

1. HSPA (Hawaiian Sugar Planters Association) 8-10 internode or stalk log procedure, supplemented by soil analysis;
2. Clement's crop log, using 3-, 4-, 5- and 6-leaf laminae for N, and sheath from the same leaves for P, K, Mg, Ca and micronutrients, and for the sugar index.

Optimum values for several varieties (HSPA - System)				
Crop age (months)	**%**			
	Moisture	**N**	**P**	**K**
6	86-88	>35	0.038-0.04	1.0
8	84-86	0.28-0.35	0.032-0.04	0.7-0.8
10	83-85	0.24-0.28	0.032-0.04	0.7-0.8

The first method is inappropriate for a one-year crop, and the second method is not directly applicable to a one-year crop without modification.

A third method, more widely practised, uses 3-leaf laminae at the boom stage of growth (samples being taken at 3, 4 1/2 and 5-6 months for plant cane and at 3-4 and 5-6 months for ratoons).

Plant analysis/crop log data - Macronutrients						
Country - Region	**Source**	**Age in months**		**% of dry matter**		
		Plant	**Ratoon**	**N**	**P**	**K**
British Guiana	Evans, 1965	3	2	2.4-2.5	0.21	1.25
Mauritius	Halais, 1962	6	4 ½	1.9	-	-
Puerto Rico	Samuels, 1965	-	-	1.5-2.0	0.18-0.25	1.62-2.0
South Africa	Meyer et.al, 1989	-	-	-	-	1.6-1.8
Pakistan	Wahab, 1961	-	-	1.6	0.2-0.22	2.0-2.2
Trinidad	Vlitos and Lawrie, 1963	-	-	1.7-2.0	0.2-0.29	1.1-1.9
USA Louisiana	Golden et al, 1965	-	-	1.5-1.7	0.18-0.22	1.2-1.8
Hawaii	(Clement's system)	3-6 elongating leaf sheath once in 35 days		1.85	0.08*	2.25*
	(HSPA system)	8-10 internodes once in 2 - 3 months		0.24-0.35	0.03-0.04	0.7-1.0
Brazil	Malavolta, 1982	-	-	1.9-2.0	0.2-0.24	1.1-1.3
India	Lakshmikantham, 1971	3-6 leaves within 4 months for N, sheath for P and K-		1.96	0.086*	1.99*
* Sugar free dry weight basis						

Plant analysis data - Micronutrients								
Country	**Source**	**Status**	**ppm dry matter**					
			Mn	**Fe**	**Cu**	**Zn**	**B**	**Mo**
British Guiana	Evans, 1956	Normal	40-250	40-100	5-15	20-100	10-40	0.08-0.8
		Deficient	25	5	4	10	4	0.04-0.05
		Toxic	400	200	-	-	-	-
Brazil	Malavolta, 1982	Adequate	100-500	200-500	8-10	25-50	15-50	0.15-0.30
		Deficient	40	100	1-5	15	5-10	0.05-0.10
South Africa	Meyer et al, 1971	-	-	11-270	49-915	-	10-55	1-10

Soil analysis

Critical limits of available nutrient status as a guide to fertilizer need:

Available nutrient status			
Country and region	**Source**	**P**	**K**
India	Jafri, 1983	20 kg P (Olsen)	100 kg exchangeable K
Queensland	Leverington,1962	66 ppm	0.2 meq./100 g
Barbados	Saint, 1962	-	125 ppm K
Fiji	Holford, 1966	5-20 ppm (modified Truog)	50-150 ppm (0.5 N acetic acid)
Philippines	Demetrio, 1960	32 ppm	150 ppm
South Africa	Wood, 1987	70 kg	112 ppm

Recommended rates of fertilizer P2O5 based on the results of soil analysis:

Method	**Level of available P**	**Likelihood of response**	**Recommended P2O5**

			rate (kg/ha)
Modified Truog	< 25 25-45 kg/ha >45	certain uncertain none	200-500 50-100 none
Modified Olsen	0- 6 7-14 ppm >15	certain uncertain none	100-170 60-100 none

Fertilizer recommendations

Fertilizer recommendations						
Country/region	**Source**	**No. of splits**	**Max. Duration of application**	**kg/ha**		
				N	**P2O5**	**K2O**
Argentina - Tucuman	Fogliata, 1970	-	-	100	adapted to requirement	
Australia - Queensland						
plant		-	-	56	25-80	75-110
ratoon		-	-	78	-	-
				(in addition to Bureau mixture)		
Bangla Desh		-	-	120	85	110
Brazil	Malavolta, 1987					
Sao Paulo						
plant		3 (N)	4-6 months	45-90	60-120	60-120
ratoon		1		45-90	30- 60	30- 60
British Guiana	Evans, 1963	-	-	65-90	65- 90	65- 90
India	Mathur, 1980					
Subtropic		3 (N)	4 months	100-250	60 as per requirement	80
Tropics		4 (N)	4-6 adsali crop in MR	200-300	120	120
		3 (N)	4 months for others	150-200	60	80
Indonesia		-	-	120	as per requirement	
Jamaica	Shaw de Innes, 1965	-	-	80-160	as per requirement	
Mauritius	Wong, 1968	-	-	100-125 (2-1-1 mixture)		
Mexico		-	-	120-200	40-80	40-120
Pakistan		-	-	90-160	-	-
Philippines	Lopez et. al,1965					
VMC		-	-	125	120	180
District - Luzon		-	-	120-140	-	-
Puerto Rico	Samuels, 1965	-	-	135-200	62	112
South Africa	Wood, 1989					
Coastal lowland						
plant		2(N;K)	10 weeks	100-120	40	100
ratoon				140	20	150
Natal Midland						
plant		-	-	80	60	125
ratoon		-	-	120	40	175
Lowveld						
plant		-	-	120	30	125
ratoon		-	-	100	10	175
USA- Hawaii	Hubert, 1963					
irrigated		2	10-12 months for 2-years crop	400	280	400-450
rainfed		2		300	280	400-450

N - The following formula for calculating the amount of N to be applied was evolved from extensive studies on cultivators fields in India (Lashmikantham, 1971):

Recommended rate (lb/acre N) = B/A x age correction x (100 - x)

Where:
A = Optimum N value minus N value when no N applied
B = Optimum N value minus test value from cultivator's field)
B/A = N index
X = N already given to the crop up to the time at which the fertilizer is to be applied.

N rates used in India range from 0-50 kg/ha N in U.P. and Bihar, compared with 250-300 kg/ha N in A. P., Karnataka and Maharashtra, to over 350 kg/ha N in the S.E. coastal area of Tamil Nadu. In general, the rate matches the intensity of irrigation which is higher in the tropics than in the subtropics (except Punjab).

As a simple rule of thumb, 1 kg N per ton of cane expected is given for plant cane and 1.25-1.50 kg N per ton of cane expected for ratoon crops. In other words the optimum for ratoons is at least 25 % greater than for plant cane.

The rate is adjusted to the extent of 10 % of the recommended rate to allow for leguminous green manure of compost and f.y.m. which has been used, but no correction is permissible for any residual N from one sugarcane crop to the next.

Enough N must be applied at an early stage of growth; in general, for a 12-months crop all should be applied within 3 months after planting. For a 2-year crop, or for areas with two rainy seasons and where tillering takes place in two flushes, a late supplementary N application is beneficial.

The effect of increasing N application is to increase the yields per hectare of both cane and sugar until the cane yield gradually levels off, while the sugar response falls off more sharply. Because reduction in quality may be associated both with too high a rate and with delayed application of N, it is essential to complete N application at least 5 months before the intended date of harvest.

In order to avoid loss by volatilization, all fertilizer N (including prilled urea) should be incorporated into the soil by deep placement.

P - For the plant crop and subsequent ratoon crop, all applied at planting in the furrow below the setts, normally as a basal dressing of a suitable compound fertilizer containing all the P and K and one-third to one-half of the total N required.

K - Large amount needed. In many countries the the amount applied, 100-200 kg/ha K2O, is similar to that of N. It is normally all included in the basal dressing, but there are areas (e.g. on sandy loams) where it is preferable to apply only one-half in the basal dressing and the remainder 6 months later.

K deficiency not only reduces the cane yield but also has an adverse effect on quality. K application often increases the percentage of sugar in the cane, and juice recovery, even where there is no increase in cane yield, particularly when harvest is delayed. The response to applied K has been observed to increase in successive ratoons.

Preferred nutrient forms

N - All sources equally effective.

P - Water-soluble forms preferred on soils rich in calcareous silt. More insoluble forms beneficial on acidic or lateritic soils (e.g. in Australia, large dressings of finely ground rock phosphate are used to build up soil reserves, followed by smaller annual applications of a soluble form of P).

K - Potassium chloride and potassium sulphate are equally effective.

Further reading

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Asparagus *(Asparagus officinalis L.)*

French: Asperge; Spanish: Espárrago; Italian: Asparago; German: Spargel

Under Temperate Conditions

Crop data

The crop, propagated by seed or crown, can be exploited for eight to ten years or more (up to 15 years). Tolerates extreme climatic conditions, but requires at least 12 °C in the soil with an optimum near 25 °C to produce shoots.

First picking should be delayed till the third year after planting. The yield reaches the maximum (6-8 t/ha) around the sixth year.

Plant density: 1 up to 4 plants/m2. Requires carefully prepared, sandy soil, rather neutral or slightly alkaline, deep and rich in organic matter.

Nutrient demand/uptake/removal

Nutrient uptake - Macronutrients (reported by various authors)					
Yield t/ha	**kg/ha**				
	N	**P2O5**	**K2O**	**CaO**	**MgO**
6	86	34	106	-	-
4.5	103	50	123	68	-
6.4	154	43	193	139	10
5	76	17	68	29	5
5.7	62	17	64	36	7
4	100	28	90	-	-
2.2	72	36	90	-	-
Average	93	32	105	68	7

Nutritional requirements increase after the turions picking; i.e. during maximum growth of the epigeous part of the plant.

Fertilizer recommendations

Before planting: 30-40 t/ha organic matter, 150-200 kg/ha P2O5 and 200-300 kg/ha K2O; applied in bands at bottom of planting furrow.

During first year: 50-100 kg/ha N split during spring and summer season.

Following years: (kg/ha) N: 100-150; P2O5: 100-150; K2O: 150-200.

If available, also manure. The P2O5 is to be applied before winter, N and K2O in two or three doses in summer.

N in nitrate form preferred; K2O as potassium chloride (sulphate form could increase the bitter taste of turions).

During the long growing cycle Ca availability in the soil should be increased; a high Ca:P-ratio is important for yielding purposes.

Asparagus might require applications of Mg and B.

Under Tropical/Subtropical Conditions

Crop data

Perennial. Harvested part: young shoots. Direct seeded and crown transplanting. Plant density: 15 000 to 25 500 plants per ha. Preferably grown in muck and light sandy soils well supplied with organic matter and pH range of 6.7 - 7.5. Does not tolerate acid soils but grows well in slightly alkaline and saline soils.

Target marketable yields in intensive commercial production = 1.5 - 4 t/ha.

Plant analysis data

Plant analysis data - Macronutrients (optimum fertility conditions)							
Plant part	**Growth stage**	**% of dry matter**					
		N	**P**	**K**	**Mg**	**Ca**	**S**
Fern	50 cm	2.4	0.2	1.9	0.4	0.4	0.1
Source: various							

Plant analysis data - Micronutrients (optimum fertility conditions)							
Plant part	**Growth stage**	**ppm dry matter**					
		Fe	**Mn**	**Zn**	**Cu**	**B**	**Mo**
Fern	50 cm	-	23	13	0	56	-
Source: various							

Fertilizer recommendations

P contributes to the improvement of spear quality, flavour, and texture, while K increases resistance to rust and helps maintain shoot quality. Mg is needed for carbohydrate storage. The crop also responds to B when soil supplies are deficient. Root inoculation with Glomus fasciculatum vesicular-arbuscular mycorrhizal fungus improves growth by improving P uptake and by decreasing the effects of Fusarium spp. diseases.

For the first year of establishment in the field apply all N, P and K before planting and field preparation. Annual treatments after field establishment should consist of split N applications, one-half before cutting begins and the remainder after harvesting has ended.

Present fertilizer practices

Senegal (Camberene)

During the first year of growth for field establishment in sandy soils of semi-arid areas, 15 t/ha organic manure and 40 kg/ha N, 180 kg/ha P2O5, and 90 kg/ha K2O are broadcast before planting. Annually sidedress all organic manure, P and K before earthing up. The annual N applications are split in two, one-half applied after earthing up and the rest after harvest.

USA (California)

Before field preparation on mineral soils apply 110 kg/ha N, 130 kg/ha P2O5, and 130 kg/h K2O; thereafter annually 110 kg/ha of N after the cutting season has ended.

Chile

In northern areas (for yields of 7- 10 t/ha): in winter, 50 kg/ha N, 50 kg/ha P2O5 and 50 kg/ha K2O and in summer a further 50 kg/ha N. In southern areas rates are doubled.

Further reading

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SIMS, W.L.; SOUTHER, F.D.; MULLEN, R.J.: Growing asparagus in California. Coop. Ext. Univ. Calif. Lf. 21447 (1988)

Beans (Phaseolus vulgaris L.)

Under Temperate Conditions

French Beans *(Phaseolus vulgaris L.)*

French: Haricots verts; Spanish: Judias; Italian: Fagiolini; German: Grüne Bohnen

Crop data

Harvested part: young immature pods, used fresh or for processing or freezing.

Plant density: 10-35/m2.

Yields 9-20 t/ha, depending mainly on variety and method of harvesting (manual or mechanized).

Temperature requirements rather high: optimum for growth, flowering and setting is near 20 °C (zero growth below about 10 °C); bush varieties tolerate low temperatures better than climbing varieties.

The crop is adaptable to different physical characteristics of the soil but is very intolerant of salinity in soil and water (salts exceeding 1.5 g/l in the water reduce the yield by about 50 %; a concentration of 0.25 g Cl/l reduces the yield by 25 %). The crop is also very sensitive to pH < 6.0; in acid soils deficiencies of Mg and Mn can often occur.

Fertilizer recommendations

In normal soil conditions, the crop benefits from a nitrogen supply resulting from a symbiotic association with Rhizobium. For out-of-season and greenhouse crops and generally when soil and wheather conditions are sub-optimal, direct applications of fertilizer N are needed; normally at rates of 40-50 kg/ha N in ammonium and/or nitrate form, before sowing and some weeks after emergence, thus meeting the requirements of the plant and permitting satisfactory symbiotic activity.

For P, the rates range from 50 to 100 kg/ha P2O5, according to soil status, distributed at sowing.

K may be applied at rates up to 120-150 kg/ha K2O; the K:N ratio should preferably be about 3:1.

Fertilizers should be well mixed into the soil, bearing in mind the plant's sensitivity to high salt concentration.

The crop is sensitive to excess of B and to deficiencies of Cu, Mo and particularly Zn. Problems with micronutrients are less liable to occur in garden crops than in large-scale field crops on account of the higher organic matter content of the soil.

Under Tropical/Subtropical Conditions

Beans *(Phaseolus vulgaris L.)*

Crop data

Directly seeded. Harvested: 40 - 60 (bush) and 50 - 70 (pole) days after seeding. Plant density: 100 000 to 330 000 plants/ha (bush beans); 20 000 - 35 000 plants/ha (pole beans). Preferably grown in sandy loam, friable soils free of nematodes and fungus diseases.

The crop is adapted to a wide range of environmental conditions from sea-level to highlands, preferably with temperatures in range 20 - 25 °C.

Target marketable yields in intensive commercial production = 3.5 - 8.5 t/ha.

Nutrient demand/uptake/removal

Nutrient uptake/removal - Macronutrients (optimum fertility conditions)					
Yield t/ha	**kg/ha**				
	N	**P2O5**	**K2O**	**MgO**	**CaO**
13	129	21	68	17	50
Source: various					

Plant analysis data

Plant analysis data - Macronutrients (optimum fertility conditions)							
Plant part	**Growth stage**	**% of dry matter**					
		N	**P**	**K**	**Mg**	**Ca**	**S**
Young mature leaf	Early flower	3.2	0.4	2.4	0.5	1.9	0.2
Source: various							

Plant analysis data - Micronutrients (optimum fertility conditions)							
Plant part	**Growth stage**	**ppm dry matter**					
		Fe	**Mn**	**Zn**	**Cu**	**B**	**Mo**
Young mature leaf	Early flower	137	92	23	11	26	1
Source: various							

Fertilizer recommendations

Both bush and pole types are very sensitive to salinity. Well decomposed organic manure should be used. Application of N early in growth is important in order to promote growth before effective atmospheric N fixation by nodule bacteria. The crop is sensitive to Mg deficiency (dolomitic lime to be applied).

Preferred nutrient forms

In loam soils the preferred N source is ammonium, to increase pod yields and N-fixing nodule formation.

Present fertilizer practices

Senegal (Camberene)

In light sandy soils in a semi-arid area apply 5 t/ha organic manure, 110 kg/ha N, 160 kg/ha P2O5, and 80 kg/ha K2O. Before planting, broadcast all the organic manure, 40 % of the P2O5 and 25 % of the K2O. At fifteen, thirty and forty days after planting, band 25 % of the N, 20 % of the P2O5 and 25 % of the K2O.

Brazil (Minas Gerais)

General recommendations are, firstly, 60 kg/ha N, 200 kg/ha P2O5 and 90 kg/ha K2O incorporated in the soil at planting and, secondly, 60 kg/ha N broadcast in two applications 15 and 30 days after planting.

Philippines (Los Banos)

Apply 100 kg/ha N, 200 kg/ha P2O5 and 100 kg/ha K2O. All of the fertilizer is applied at planting in bands 8 cm to the side and 3 cm below the seed.

Further reading

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Cabbage *(Brassica oleracea L., Capitata group)*

French: Chou; Italian: Cavolo; Spanish: Col; German: Kohl

Under Tropical/Subtropical Conditions

Crop data

Biennial. Harvested part: Leafy head. Both direct seeded and transplanted. Harvested 70 - 120 days after sowing. Plant density: 28 700 to 40 000 plants per ha. Preferably grown in loamy sand, pH 6 - 6.5. Not tolerant of acid soils.

Adapted to cool (16 - 20 °C), moist climates. Generally irrigated. Target marketable yields in intensive commercial production = 20 - 30 t/ha.

Nutrient demand/uptake/removal

Nutrient uptake/removal - Macronutrients (optimum fertility conditions)					
Yield t/ha	**kg/ha**				
	N	**P2O5**	**K2O**	**MgO**	**CaO**
29	121	32	106	5	21
Source: various					

Plant analysis data

Plant analysis data - Macronutrients (optimum fertility conditions)							
Plant part	**Growth stage**	**% of dry matter**					
		N	**P**	**K**	**Mg**	**Ca**	**S**
Wrapper leaf	Head	3.3	0.5	3.1	0.4	1.6	0.2
Source: various							

Plant analysis data - Micronutrients (optimum fertility conditions)							
Plant part	**Growth stage**	**ppm dry matter**					
		Fe	**Mn**	**Zn**	**Cu**	**B**	**Mo**
Wrapper leaf	Head	19	10	9	5	17	2
Source: various							

Fertilizer recommendations

Cabbage is a heavy feeder on fertilizer nutrients, except P. Heads will not form unless adequate N is given. Excessive N, on the other hand, may cause loose head formation and internal decay. The demand for P is greater during head formation. K deficiency can result in marginal necrosis and lower head quality, but an excess of K can cause the heads to open. The crop has a high S requirement and is sensitive to deficiencies of Mg and B.

Fertilizer applications are split, with part applied before planting. Ploughing-under of lime and compound fertilizer before planting is recommended. Use of a high analysis starter solution containing 0.75 kg nutrients per 100 litres is recommended when transplanting. The remainder of the crop's needs can then be met with in one or two applications during the growing season.

The N-fixing bacteria Azospirillum spp., which are present in many tropical soils, were recently found to promote cabbage foliage growth.

Present fertilizer practices

Brazil (Minas Gerais)

General recommendations are, firstly, 60 kg/ha N, 120 kg/ha P2O5, and 180 kg/ha K2O incorporated in the soil at planting and, secondly, 60 kg/ha N and 60 kg/ha K2O broadcast in 3 applications 15 and 30 days after planting and during heading closure. For improved yields also incorporate 30 t/ha of organic matter into the soil two weeks or more before planting.

Philippines (Los Banos)

240 kg/ha N, 60 kg/ha P2O5 and 60 kg/ha K2O, all the P2O5 and half the N and K2O applied in bands along the rows at planting and the remaining N and K2O sidedressed 8 - 10 cm deep one month after planting and watered immediately.

Senegal (Camberene)

On light sandy soil in a semi-arid area, 20 t/ha organic matter, 65 kg/ha N, 65 kg/ha P2O5, and 100 kg/ha K2O. All the organic manure and P2O5 and one-third of the N and K2O are broadcast before planting, one-third of the N and K2O is sidedressed 20 days after planting, and the remaining third of the N and K2O is sidedressed 40 days after planting.

India (Bangalore)

In sandy loams with pH 6.7: 150 kg/ha N, 80 kg/ha P2O5 and 40 kg/ha K2O. All the P2O5 and K2O and one-half of the N are applied before transplanting, and the remaining half of the N is applied 30 days after transplanting.

Further reading

GUPTA, A.: Effect of N and irrigation on cabbage production. Ind. J. Hort. Sci. 44, 241-244 (1987)

CSIZINSKY, A.A.: Nutrition of cole crops with the full-bed polyethylene mulch system in West-Central Florida. J. Plant Nutrition 10, 1489-1497 (1987)

Carrot *(Daucus carota L. var. sativus Hoff.)*

French: Carotte; Spanish: Zanahoria; Italian: Carota; German: Möhre

Under Temperate Conditions

Crop data

Biennial, cropped as annual. Harvested part: root.

Growth cycle for root production 100-180 days depending on cultivar.

Plant density: about 100/m2 (for half-long cultivars). Yields 25-50 t/ha (half-long cultivars). Requires light or well-textured soils, rich in decomposed organic matter; does not tolerate acid, alkaline or saline soils. Adapts well to climatic conditions.

Nutrient demand/uptake/removal

In kg for crop yielding 30 t/ha : N = 90-120, P2O5 = 30-45, K2O = 150-300.

Fertilizer recommendations

Adequate rate of fertilizer use in normal soils is estimated to be 120 kg N, 100 kg P2O5, 200-250 kg K2O per ha. It is recommended that one-quarter of the total N should be applied at sowing and the remainder in two topdressings. P2O5 and K2O should be incorporated in the soil before sowing. Direct use of organic manures should be avoided.

N deficiency impairs root colour by diminishing carotene synthesis. Excessive N may not only delay root growth and reduce keeping quality but may also increase the nitrate content, an important factor in roots to be processed for use in babyfoods. Excessive K can reduce uptake of Mg. Chloride salts should be avoided because excessive Cl reduces carotene content.

Boron is an important nutrient; foliar spray application is beneficial where calcareous soils contain <1 ppm B and where there is less than 18 ppm B in the leaves.

7.2.4 **Carrot** *(Daucus carota L.)*

Under Tropical/Subtropical Conditions

Crop data

Directly seeded. Harvested four months after sowing. Plant density: 435 600 plants/ha. Preferably grown in loose, deep, well-drained soil. The crop is adapted to highland conditions in the tropics. Target marketable yields in intensive commercial production = 25 - 37 t/ha.

Nutrient demand/uptake/removal

Nutrient uptake/removal - Macronutrients					
Yield t/ha	kg/ha				
	N	P2O5	K2O	MgO	CaO
43	126	71	175	20	224
Source: various					

Plant analysis data

Plant analysis data - Macronutrients						
Plant part	Growth stage	% of dry matter				
		N	P	K	Mg	Ca
Young mature leaf	Midseason	3.8	0.5	4.0	0.3	1.6
Source: various						

Plant analysis data - Micronutrients						
Plant part	Growth stage	ppm dry matter				
		Fe	Mn	Zn	Cu	B
Young mature leaf	Midseason	54	44	27	2	15
Source: various						

Fertilizer recommendations

Apply well decomposed organic matter to prevent forked root development. Excessive N may also be partly responsible for splitting or forked roots; heavy N treatments also promote foliage growth at the expense of root growth. Fertilizer should be applied at least 7 days before sowing, as the crop is susceptible to salt injury. Carrots respond well to B and to lime and Mg applications, as well as to N, P and K . N and K are applied before root enlargement, about 30 days after germination.

Present fertilizer practices

Senegal (Camberene)

On light sandy soils in semi-arid conditions apply 20t/ha of organic manure, 60 kg/ha N, 60 kg/ha P2O5 and 120 kg/ha K2O. All organic manure and P2O5 and 20 % of N and K2O broadcast before planting; 40 % of N and K2O sidedressed 30 and 60 days after planting.

Brazil (Minas Gerais)

General recommendations are, firstly, 40 kg/ha N, 320 kg/ha P2O5 and 240 kg/ha K2O incorporated in the soil at planting and, secondly, 80 kg/ha N and 40 kg/ha K2O broadcasted in 2 applications 15 and 30 days after planting.

Philippines (Los Banos)

A broadcast application of 500 - 600 kg/ha of a 10-25-25 mixture is recommended, or in K deficient soils 1 000 - 1 200 kg/ha 5-10-16 .

Further reading

BIENZ, D.R.: Carrot splitting and second growth in Central Washington as influenced by spacing, time of sidedressing and other cultural practices. Proc. Amer. Soc. Hort. Sci. 86, 406-410 (1965)

Cauliflower *(Brassica oleracea var. botrytis L.)*

French: Chou-fleur; Spanish: Coliflor; Italian: Cavolfiore; German: Blumenkohl

Under Temperate Conditions

Crop data

Most important of the numerous B. oleracea species, characterized by its distinctive large head.

Interval from sowing to harvest 100-180 days.

Plant density: 1.5 - > 5/m2. Yields 10-40 t/ha. Adaptable to wide range of pH (5.5-7.5) and tolerates brackish soils.

Considered a cool season crop; optimum temperature for growth is near 17 °C but also tolerates rather low temperatures.

Nutrient demand/uptake/removal

According to Anstett (1961), 198 kg N, 66 kg P2O5, 295 kg K2O, 21 kg MgO, 186 kg CaO, for crop yielding 37 t/ha (corymbs). Averaged from other authors, 175 kg N, 60 kg P2O5, 200 kg K2O, 25 kg MgO, 115 kg CaO, 45 kg S per ha.

Plant analysis data

Critical values for K in 60- to 75-day old plants are about 2 % in young leaves and 1 % in older leaves.

Fertilizer recommendations

Provided the soil organic matter content is satisfactory, suitable fertilizer application would comprise 150-250 kg/ha N, one-third before transplanting and the remainder as topdressings, 60-100 kg/ha P2O5, and normally 200-300 kg/ha K2O before planting. Mg, Ca and S may also merit consideration on the basis of soil analysis.

Special attention must be paid to B and Mo. The most common effect of B deficiency is hollowing of the stem due to cracking of the pith, and browning of the corymb (similar symptoms can be caused by excess of N). Mo deficiency can occur in acid soils; in these circumstances distribution of Mo salts to the soil or in foliar sprays may prove useful. K deficiency can also occur, causing shortening of internodes, thickening and curling of lamina, purple pigmentation along the leaf veins, inhibition of curd formation and floral bud necrosis.

Celery *(Apium graveolens L. var dulce (Mill.) Pers.)*

French: Céleri à côtes; Spanish: Apio blanco; Italian: Sedano da coste; German: Sellerie

Under Temperate Conditions

Crop data

Annual. Harvested part: petioles of the leaves. Variety "rapaceum" cropped for the roots.

Cultivation cycle three to five months in early and late varieties.

Plant density: Transplanting - top plants - in rows 50-80 cm apart and every 20-30 cm within the rows. Root spreading rather shallow within 25-30 cm of upper soil; does not tolerate drought conditions.
Yields 20-40 t/ha.

Prefers moderately acid soil, (pH value near 6.0, but can also be grown with pH 7-7.2) with high field capacity and rich in organic matter. Requires relatively cool wheather; for growth 7-24 °C, with an optimum near 15 °C.

Nutrient demand/uptake/removal

Heavy feeder on nutrients; on average in kg per t of yield: N = 2.5-6.5 (for 90 and 20 t/ha yielding level respectively); P2O5 = 2.0-2.5; K2O = 6.0-9.0; MgO = 0.2; CaO = 2.5-3.0.

For 81 t/ha of fresh epigeous parts Anstett indicates an uptake of (kg/ha) N = 206; P2O5 = 154; K2O = 240; MgO = 24; CaO = 316 and NaO = 124. Best celery yields are obtained with high doses of N during last phase of growth cycle (100-300 kg/ha).

Fertilizer recommendations

30-40 t/ha of manure, in addition to (in kg/ha) N = 200-300 (1/3 at planting; 2/3 in two or three shares as side dressing); P2O5 = 100-150 and K2O = 100-300 (according to soil analysis). For higher rates of K fertilization top dressing advisable.

Fertilizers containing Ca are to be avoided. KCl prefered in terms of product quality. Sensitive against Mg deficiency (leaf chlorosis, necrosis) and also B deficiency.

Cucumber *(Cucumis sativus L.)*

French: Concombre; Spanish: Pepino; Italian: Cetriolo; German: Gurke

Under Temperate Conditions

Crop data

Plant density: 1-2.5/m2 or more, depending on pruning and training system.0

Yields commonly 30-60 t/ha can reach 100 t/ha or more (up to 300 t/ha under glass).

Prefers sandy or well-structured soils rich in organic matter. Tolerates rather low pH (down to 5.5).

Requires a long warm season (optimum temperatures for growth: night, about 18 °C, and day, about 28 °C) and a high light intensity.

Nutrient demand/uptake/removal

Depends very much on yield. Typical values reported:

Yield 300 t/ha under glass: 450-500 kg N, 200-250 kg P2O5, 800-1 000 kg K2O,
130 kg MgO, 300 kg CaO per ha.

High-yielding outdoor crop: 170 kg N, 130 kg P2O5, 270 kg K2O per ha.

Yield 30 t/ha outdoors: 50 kg N, 40 kg P2O5, 80 kg K2O per ha.

Yield 15 t/ha outdoors: 47 kg N, 13 kg P2O5, 65 kg K2O per ha.

Fertilizer recommendations

Organic manures useful even for outdoor crops. In addition, on soils of normal nutrient content, fertilizer rates of 100 kg N, 100 kg P2O5, 200 kg K2O per ha are recommended for yield levels up to 30-40 t/ha. The N application should be split into several dressings according to the length of the harvesting cycle, preferably every 2 weeks if practicable.

Cucumber is very sensitive to N deficiency, which can alter the fruit shape, and is intolerant of salinity. Deficiencies of Mg and of B, Fe and Mn, can occur and demand direct application of these nutrients.

Under Tropical/Subtropical Conditions

Crop data

Annual. Harvested part: Fruit. Directly seeded. Flowers 35 - 45 days after planting. Harvested: 45 - 55 days after planting. Plant density: 33 000 to 54 450 plants/ha. Preferably grown in well drained, non saline soils. Adapted to a wide-range of soils, but will produce early in sandy soils. Generally irrigated.

Target marketable yields in intensive commercial production: 13 - 30 t/ha.

Nutrient demand/uptake/removal

Nutrient uptake/removal (outdoor) - Macronutrients					
Yield t/ha	**kg/ha**				
	N	**P2O5**	**K2O**	**MgO**	**CaO**
20	39	27	70	10	35
Source: various					

Plant analysis data

Plant analysis data - Macronutrients							
Plant part	**Growth stage**	**% of dry matter**					
		N	**P**	**K**	**Mg**	**Ca**	**S**
Young mature leaf	Fruit set	3.3	0.4	2.8	0.4	1.8	0.3
Source: various							

Plant analysis data - Micronutrients						
Plant part	**Growth stage**	**ppm dry matter**				
		Fe	**Mn**	**Zn**	**Cu**	**B**
Young mature leaf	Fruit set	108	60	23	8	25
Source: various						

Fertilizer recommendations

Cucumbers are sensitive to Mg deficiency and respond to Mn and Cu applications.

Present fertilizer practices

Senegal (Camberene)

On light sandy soils in a semi-arid area apply 20 t/ha of organic manure, 130 kg/ha N, 95 kg/ha P2O5, and 200 kg/ha K2O. Before planting broadcast all the organic manure and P2O5 and one-third of N and K2O. At 30, and again at 50 days after planting apply one-third of the N and K2O.

Brazil (Minas Gerais)

General recommendations are, firstly, 50 kg/ha N, 200 kg/ha P2O5 and 150 kg/ha K2O incorporated in the soil at planting and, secondly, 50 kg/ha N and 50 kg/ha K2O broadcast in two applications 15 and 30 days after transplanting. Greater yields are achieved by incorporating 20 t/ha organic matter two weeks or more before planting.

Philippines

In the dry season 120 kg/ha N, 120 kg/ha P2O5 and 120 kg/ha K2O. Band one-third at planting. When the vines have reached about 1 m in length, sidedress a second one-third. Sidedress the remaining one-third when the first fruit is about the size of an egg.

India (Assam)

In sandy loam soils with pH 6.5 and soil boron content of 0.58 ppm, apply 80 kg/ha N, 45 kg/ha P_2O_5, 85 kg/ha K_2O and a 0.25 % $Na_2B_4O_7 \cdot 10\ H_2O$ solution. Apply all the N, P_2O_5 and K_2O at planting. Spray the 0.25 % boron solution at the six leaf stage and at the flower bud initiation stage.

Further reading

HOCHMUTH, G. (ed.): Cucumber production guide for Florida. Florida Coop. Ext. Serv. Circ. 101E (1988)

MAURYA, K.R.: Effect of nitrogen and boron on sex ratio, yield, protein and ascorbic acid content of cucumber (Cucumis sativus L.). Indian J. Hort. 44, 239-240 (1987)

Eggplant [Aubergine]*(Solanum melongena L.)*

French: Aubergine; Spanish: Berenjena; Italian: Melanzana; German: Aubergine

Under Tropical/Subtropical Conditions

Crop data

The crop is well adapted to the tropics. Optimum temperature for seed germination is in the range of 21 - 24° C and for growth 21 - 29° C. The most satisfactory environments are in lowland coastal areas with relatively little temperature variation. Temperature extremes can diminish the activity of root systems and interfere with plant nutrition. The crop develops a strong taproot with a branched root system that does not spread widely but responds positively to well drained soils of medium texture that are moderately deep. Good aeration is essential.

World production averages 14 t/ha. Experimental plots have yielded more than 90 t/ha and some commercial growers produce 74 t/h but 30 t/ha should be considered an acceptable yield.

Nutrient demand/uptake/removal

The harvest extends over a long period of time in the tropics and the crop produces many fruits. As a result it requires fertilizers in large quantities especially nutrients that are easily released from the soil. In general its nutrition resembles that of tomato.

Nutrient uptake/removal - Macronutrients						
Plant part	**kg/40 t crop**					
	N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
Fruit	75	27	108	12	4	5
Total plant	207	46	340	-	-	15
Source: Paterson, 1989						

The figures given for S are not derived from actual data but are estimates based on typical N:S ratios in plants. The amount of S required in fertilizer is generally about 10 % of the N requirement minus the S supplied in rainfall and irrigation water and that taken up from the soil.

Plant analysis data

The table demonstrates the generally high concentration of nutrients in vegetative tissues especially N and K (for S, see remarks above).

Plant analysis data - Macronutrients (good plant nutrition and fertilizer practices)						
Plant part/	**% of dry matter**					
conditions of growth	**N**	**P**	**K**	**Mg**	**Ca**	**S**
Leaves*: well fertilized and limed; recently matured						
not mulched	4.9	0.45	4.6	0.65	1.6	0.4
mulched	5.2	0.6	5.0	0.55	1.3	-
5th leaf, 60 days**	2.73	0.42	3.9	-	-	-
Fruit***	2.1	0.31	3.0	0.23	0.12	-
* Paterson, 1989; ** Subbiah et al., 1987; *** Tindall,n.d. Paterson, 1989; Mortensen & Bullard; n.d.; Valenzuela, 1991						

Plant analysis data - Micronutrients (good plant nutrition and fertilizer practices)					
Plant part	**ppm dry matter**				
	Fe	**Mn**	**Zn**	**Cu**	**B**
Leaves: well fertilized and limed; recently matured					
not mulched	250	70	-	11	30
mulched	300	100	-	14	75
Source: Paterson, 1989					

Fertilizer recommendations

Fertilizer practices (recommended rates of fertilizer application for acceptable yields)				
Geographical area	**Nutrient use recommended kg/ha**			
	N	**P2O5**	**K2O**	**Other**
USA, East Coast 1)	200	185	190	Lime
Japan 2)	245	345	250	
Tropics/subtropics 3)	145	45	85	1 200 tree ash Lime
Brazil, Sao Paulo 4)	240	405±200	240±80	Lime (upto 70 % base saturation)
Brazil 5)	200	365±90	145±50	
Hawaii 6)	200	640	120	10 Zn; Lime
Puerto Rico 7)	400	205	240	

1) Sandy soils, temperate and subtropical climates.
2) General recommendation; soils generally influenced by volcanic ash.
3) For 39.5 t/ha crop. N and K mostly supplied in wood ash.
4) Recommendations are for soils with moderately high P sorption capacity, of medium P and K status. 20 % of N applied before planting, remainder in 4 equal applications 10, 25, 40 and 55 days after planting. P all applied before planting; K two-thirds before planting and remainder in 4 applications with the N. For low or high soil P, add or subtract 200 kg P2O5 before plantings. For low or high soil K, add or subtract 80 kg K2O before planting.
5) Recommendations based on general experience rather than substantial research.
6) The P requirement was such as would give 0.3 mg P/l soil solution. For soils in which iron and aluminium oxides dominate, typical P fertilizer requirements to attain 0.3 mg/l are in the range 690 - 2 750 kg/ha P2O5. Soil was high in non-exchangeable K.
7) The high application rate of fertilizer N seems to indicate a fertilizer efficiency of 50 % with a very small contribution from the soil. The rate of P fertilizer employed was not sufficient to match the high requirement for maximum production in this high P-sorbing Oxisol.

The table is a summary statement of fertilizer either used in experimental plantings or being recommended in several areas. They are not all in the tropics. The recommended rate for N (except in Puerto Rico) is around 200 kg/ha. The fertilizer requirement depends more on soil properties than on plant uptake. On sandy soils (USA, East Coast) the recommendation is

185 kg/ha P2O5; for moderately high P absorbing soils of Brazil the requirement is twice as great; for very high iron content (basalt) soils of Hawaii 3 to 4 times as great; and for soils developed on weathered volcanic ash it will be even greater. Virtually all recommendations emphasize the need for relatively high base saturation/low aluminum saturation, pH 6 - 7 (lime or wood ash applicaton).

Further reading

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Lettuce *(Lactuca sativa L.)*

French: Laitue; Spanish: Lechuga; Italian: Lattuga; German: Kopfsalat

Under Temperate Conditions

Crop data

Plant with a short cycle and very adaptable to different conditions. Grows at relatively low temperatures; high temperatures and long days promote the bolting.

Plant density: (heading cultivars) 7 and 12 plants/m2. Yield between 20 and 40 t/ha.

Prefers soils in good physical condition, rich in organic matter and rather neutral.

Nutrient demand/uptake/removal

Low in comparison to that of other vegetable crops; average nutrient uptake (different planting season, soil fertility, open air or protected cultivation) in kg/t of yield: N = 1.5-3; P2O5 = 1-1.5; K2O = 4-7; MgO = 0.2-0.4, CaO = 1-1.5; Total uptake for medium yield level (kg/ha): N = 50-100; P2O5 = 30-50; K2O = 120-200.

70 % of main nutrients are taken up during three to four weeks before harvesting.

Reliable index of satisfactory nutritional N-status could be given by the N-NO3 level in the veines of outer leaves: it should be about 8 000 ppm and never below 4 000 ppm.

Fertilizer recommendations

In soil containing more than 3 % of organic matter and in open-air applications of 100 kg/ha N and 50-100 kg/ha P2O5 are advisable. K2O, up to a maximum of 150-200 kg/ha, depending on soil status.

N doses to be increased for winter season crops. 1/3 of N is given before planting, 1/3 a month after transplanting and 1/3 a month before harvesting.

The plant accumulates nitrate N in its leaves up to concentrations near to safety limits set in some countries (3 000-4 000 ppm). NO3 accumulation is also enhanced by microelement deficiencies involved in N metabolism (mainly B, Mn, Mo, Cu).

Under Tropical/Subtropical Conditions

Crop data

Annual. Harvested part: leaf. Usually transplanted in cooler or highland areas. Harvested: 70 (early) to 85 (late) days after planting. Plant density: 33 000 to 60 000 plants/ha. Grows well in well-drained organic or mineral soils with pH 6 - 6.5. Not tolerant of acid conditions. The

crop is adapted to cool and dry environments. Target marketable yields in intensive commercial production: 30 - 34 t/ha.

Nutrient demand/uptake/removal

Nutrient uptake/removal - Macronutrients					
Yield t/ha	**kg/ha**				
	N	**P2O5**	**K2O**	**MgO**	**CaO**
18	68	21	130	8	20
Source: various					

Plant analysis data

Plant analysis data - Macronutrients (optimum fertility conditions)							
Plant part	**Growth stage**	**% of dry matter**					
		N	**P**	**K**	**Mg**	**Ca**	**S**
Young mature leaf	Head	2.8	0.48	4.2	0.32	0.92	0.1
Source: various							

Plant analysis data - Micronutrients (optimum fertility conditions)							
Plant part	**Growth stage**	**ppm of dry matter**					
		Fe	**Mn**	**Zn**	**Cu**	**B**	**Mo**
Young mature leaf	Head	128	51	38	8	19	0.03
Source: various							

Fertilizer recommendations

Adequate fertilizer levels are needed to hasten development, improve quality and reduce bolting in bolting-prone cultivars. The P requirement of lettuce is high in comparison with most vegetable crops. Best results are obtained if the cultivated volume of soil solution is brought to 0.2 - 0.4 mg/l P. If the P fertilizer rate is below optimum, efficiency is increased when banded in strips 8 cm wide and 5 cm below the seeds. Banding N together with P may also help to improve yields and the efficiency of fertilizer use.

Present fertilizer practices

USA (Florida)

Apply starter fertilizer to transplants. On mineral irrigated soils apply 120 kg/ha N, 160 kg/ha P2O5, and 280 kg/ha K2O. At planting broadcast in the bed or band all the P2O5 and one-quarter of the N and K2O. The remainder of the N and K2O is given in split dressings early in the growing season.

Senegal (Camberene)

In a light sandy soil in a semi-arid area apply 5 t/ha of organic manure, 125 kg/ha N, 180 kg/ha P2O5, and 280 kg/ha K2O. Before planting broadcast all the organic manure, 40 % of the P2O5 and 25 % of the N and K2O. At 30 days after planting apply 40 % of the P2O5 and 25 % of the N and K2O. At 50 days after planting band 25 % of N and K2O and the remaining 20 % of the P2O5. At 75 days after planting apply the remaining 25 % of the N and K2O.

Brazil (Minas Gerais)

General recommendations are, firstly, 30 kg/ha N, 120 kg/ha P2O5 and 90 kg/ha K2O incorporated in the soil at planting and, secondly, 60 kg/ha N broadcast in three applications 15 and 30 days after planting and during heading closure. An alternative is to apply 50 t/ha organic matter incorporated into the soil two or more weeks before planting.

Further reading

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SANCHEZ, C.A.; SWANSON, S.; PORTER, P.S.: Banding to improve fertilizer use efficiency of lettuce. J. Amer. Soc. Hort. Sci. 115, 581-584 (1990)

Melon *(Cucumis melo L.)*

French: Melon; Spanish: Melon; Italian: Melone; German: Melone

Under Temperate Conditions

Crop data

Annual, herbaceous, prostrate and branched. Many varieties are grown commercially. Normally monoecious, with male and female flowers within the same plant; fruits differ in shape, colour etc.

Growing season in Europe normally between April and September; outdoor ripening of first fruits about 90 days after transplanting or 120 days after sowing.

Yields 20-30 t/ha outdoors, 30-40 t/ha in protected cultivation.

Prefers deep soils, well supplied with organic matter, pH around neutral. Tolerates slightly alkaline or salty soils. Fairly resistant to moderate drought, but normally needs irrigation.

Requires warm temperatures and a high light intensity. Optimum temperatures: for germination, 24 °C-32 °C; for growth, night, 18 °-20 °C, and day, 24°-30 °C; and for ripening, 25°-30 °C. Minimum for germination, 13 °-15 °C. Zero growth, near 12 °C.

Nutrient demand/uptake/removal

Depends very much on yield level and growing conditions. Typical figures for outdoor crops: 50-120 kg N, 15-25 kg P2O5, 50-200 kg K2O, 70-100 kg CaO, and up to 20 kg MgO per ha. Highest rate of nutrient uptake occurs 4-6 weeks after planting, i.e. some weeks before the maximum rate of dry matter accumulation.

Plant analysis data

Satisfactory concentrations at the 6-8 leaf stage are, for nitrate N, not less than 7 500 ppm in the petioles and more than 5 000 ppm in the entire leaf, and for P corresponding values are about 2 000-2 500 ppm.

Fertilizer recommendations

In soils containing at least 3 % organic matter, an irrigated crop yielding 30-35 t/ha should receive a basal fertilizer application, placed along the rows before planting, of about 100 kg N, 100 kg P2O5, 150 kg K2O per ha, followed by a topdressing during fruit-swelling of 50-100 kg/ha each of N, P2O5 and K2O. Topdressing is generally not recommended for unirrigated crops. Protected crops usually require three topdressings unless slow-release fertilizers are used.

Melons are very sensitive to availability of Mg; deficiency (< 0.3 % in dry matter) results in shortage of chlorophyll, browning of tissues and chlorosis of leaves. In acid soils excess of Mn can occur; a concentration of this micronutrient of >1 000 ppm in the leaves can cause problems of phytotoxicity - yellowing of leaves, brown spots on stems, and necrosis. The crop is also sensitive to low availability in the soil of the micronutrients Fe, Mo and B.

Okra *(Abelmoschus esculentus [L.] Moench.)*

French: Gombo; Spanish: Ocra; Italian: Gombo; German: Okra

Under Tropical/Subtropical Conditions

Crop data

This is an important vegetable in tropical and subtropical areas. In the tender stage it is very nutritious and the season of availability for use is long. Young leaves and tender pods are eaten. The crop is cultivated either alone or in mixed culture with other crops. It is one of the most important vegetables in India, and in Nigeria alone it occupies 1.5 million hectares.

It requires a temperature of at least 16° C for germination. Temperatures of 20 to 30° C are appropriate for production. The crop will do well on soils ranging from sand to clay if both internal and surface drainage are good and soil fertility is maintained. Although okra is tolerant to some drought stress, supplementary irrigation is necessary for good production during extended drought.

Plant population densities from 16 000 to > 100 000 plants per hectare may give ever-increasing yields if all growth factors are abundantly supplied. However 40 000 plants/ha will produce near maximum yields if the plants are well spaced.

Although the yield potential is high (30 - 40 t/ha have been reported), actual yields are usually low because improved methods of production are exceptional rather than the rule and because many areas where the crop is grown are subject to extremes of weather which make yield prospects uncertain.

Nutrient demand/uptake/removal

Nutrient uptake/removal* - Macronutrients					
Plant part / yield	**kg/ha**				
	N	**P2O5**	**K2O**	**MgO**	**CaO**
Pods, 20 t/ha	79	32	89	15	29
Total plant**	200	73	156	45	38

* Based on data of Majanbu et al., 1986; Ahmad & Tullock-Reid, 1968 and FAO.
** Assumed that pods represented 45 % of total dry weight and that composition of vegetative parts was equal to leaves composition at 15 weeks after harvest.

Plant analysis data

The table gives the nutrient concentration in leaves and pods in approximate 20 t of harvested product.

Plant analysis data* - Macronutrients					
Plant part	**% of dry matter**				
	N	**P**	**K**	**Mg**	**Ca**
Leaves 15 weeks	4.1	0.35	1.9	0.62	3.2
Pods 8 weeks	3.3	0.60	3.1	0.38	0.7**

N-fertilizer application = 100 kg/ha N
* Based on data of Majanbu et al., 1986; Ahmad & Tullock-Reid, 1968; and FAO.
** FAO, 1968; assumed pods 12 % dry matter.

Fertilizer recommendations

Reliable general recommendations concerning practices cannot be made for two reasons; (1) environmental conditions, including soil factors, are so varied that generalizations will seldom if ever apply precisely, and (2) quality research upon which to base such generalizations is almost totally lacking. As a gross generalization it may be said that the requirement is similar to that of cotton.

The table below presents a few examples of fresh pod yields and leaf composition associated with various yield levels. Footnotes to the table suggests the conditions where these recommendations may apply. For example, recommendations in Brazil are based on soils of medium nutrient status. For soils low in P and K the recommendation is increased by 50 %, whereas for soils high in P and K recommendations are reduced by 50 %.

Generally, adjustments are necessary for fertilizer use efficiency and nutrient contributions from the soil, manure, residual fertilizers etc. As a first approximation, the efficiency of N fertilizer use is 50 % and many soils will deliver approximately 80 kg/ha N. Thus for a 20 t okra crop, the estimated N fertilizer requirement would be (200 - 80) / 0.5 = 240 kg/ha N, which was the N level employed for estimated maximum yield in South India (Bangalore). The P is applied at planting, together with 25 % of the N and 60 % of the K, followed by two applications of N and K2O approximately 20 and 40 days after seedling emergence.

Approximate okra yields (green pods), and N, P and K concentrations in leaves associated with various fertilizer rates						
Yield level (tons/ha)	**Leaf composition (% of dry matter)**			**Fertilizer rate (kg/ha)**		
	N	**P**	**K**	**N**	**P2O5**	**K2O**
	South India (Bangalore) 1)					
100 % predicted	4.0	0.31	2.6	240	172	288
80 % actual (6.5)	3.35	0.31	2.3	240	172	144
75 % (5.1)	3.3	0.29	2.25	120	0	144
50 % (4.1)	2.85	0.27	1.75	60	0	36
25 % (2.0)	<2.7	-	-	0	0	0
	Trinidad 2)					
100 % (16.5)	5.1	0.40	2.5	112	385	300
	Sao Paulo, Brazil 3)					
General recommendation for soils or medium P and K status	-	-	-	80	202	140
	Northern Nigeria (Samaru) 4)					
100 % (6.2)	4.13	0.35	1.88	100	30	40
	Southeast USA (Georgia and Florida) 5)					
General recommendations for soils very low in P and K	-	-	-	120	160-275	60-149

1) Data based on a graphic reexamination by one of the authors (RLF) of results presented by Kumar & Devarajura, 1988; using a boundary-line approach with some extrapolation.
2) Based on data by Ahmad and Tullock-Reid, 1968; with some obvious corrections. The soil is micaceous but with only 0.13 meq/100 g exchangeable K; thus K fixation accounts for high K fertilizer requirement.
3) Extracted from Van Raij et al., 1985; recommendation based on a soil of medium extractable P and K status but relatively high in P sorption capacity.
4) Extracted from data by Majanbu et al., 1985 & 1986. Soils averaged 0.2 meq exchangeable K and P sorption was low. Low P fertilizer rate was adequate but K fertilizer rate was insufficient for high yields, whence the low leaf K.
5) Composite fertilizer recommendations Hochmuth & Hanlon, 1989; Woodruff, 1927. The range represents greater P sorption by Georgia soils and greater K leaching by Florida sands.

Further reading

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Onion *(Allium cepa L.)*

French: Oignon; Spanish: Cebolla; Italian: Cipolla; German: Zwiebel

Under Temperate Conditions

Crop data

Biennial normally grown as an annual. Harvested part: bulb.

Usually direct sown (or transplanted) and harvested when bulbs are dry and reach dormancy phase.

Root system is shallow and fibrous.

Cycle length is rather long and depends on cultivar and environment; bulb formation generally requires 12-15 hour long days and high temperatures.

Plant density: 20-40 plants/m2

Yields: 30-40 t/ha (85 % moisture).

Friable seed bed required. Optimum pH >6.0; some cultivars perform well up to pH 7.5.

Onions grow well in a wide range of temperatures, but generally best when cool during early development and warmer near maturity.

Nutrient demand/uptake/removal

For 30 t/ha yield, 116 kg N, 44 kg P2O5, 144 kg K2O, 20 kg MgO, 131 kg CaO, per ha.

For yield of about 40 t/ha, 160 kg N, 76 kg P2O5, 115 kg K2O, 16.6 kg MgO, 128 kg CaO, per ha.

Although 50 % of uptake of the three major nutrients occurs during the last month before harvesting, the effect of N on growth and yield depends on its influence on the relative growth rate in the very early stages. Pungency of flavour is enhanced by fertilizers containing S but diminished by N in ammonium form.

Plant analysis data

Satisfactory N level in older leaves 2-3 % on dry matter basis; critical tissue level for N in younger, fully expanded leaves about 5 %.

Fertilizer recommendations

100-200 kg N, 100-200 kg P2O5, 200-300 kg K2O per ha, commonly recommended, with 20-40 kg/ha MgO for high-yielding crops (K2O dose may vary according to soil status). Trials in Europe have shown yield increases with up to 110-120 kg/ha N and 200 kg/ha P2O5. N application should be divided into two or three dressings (before sowing or planting, at the fully expanded leaf stage, and just before bulbification). Some experiments have shown the best results with slow-release N fertilizers.

Due to the shallow root system the fertilizers should be banded 8-10 cm below the seed row. P2O5 and K2O topdressings can prove useful on soils poorer in these nutrients. High nutrient availability is important during bulbification; in this phase a high K:N ratio is required.

Onions are very sensitive to Mg deficiency which causes chlorosis on alkaline soils. Zn, Mo and Cu may also present problems. Cu deficiency causes thin, poorly coloured bulb scales and impairs keeping quality.

Under Tropical/Subtropical Conditions

Crop data

Usually transplanted in the tropics. Harvested 90 - 150 days after planting. Plant density: 66 000 to 500 000 plants/ha. Grown in soils ranging from light sandy to organic with a pH range of 5.8 - 6.5 adapted to cool weather (15 - 20 °C) and low humidity.

Target marketable yields in intensive commercial production: 30 - 45 t/ha.

Nutrient demand/uptake/removal

Nutrient uptake/removal - Macronutrients					
Yield t/ha	**N**	**P2O5**	**K2O**	**MgO**	**CaO**
	kg/ha				
41	102	41	112	17	29
Source: various					

Plant analysis data

Plant analysis data - Macronutrients (optimum fertility conditions)						
Plant part	**Growth stage**	**% of dry matter**				
		N	**P**	**K**	**Mg**	**Ca**
Young mature leaf	Mid season	3.1	0.33	3.3	0.27	0.5
Source: various						

Plant analysis data - Micronutrients (optimum fertility conditions)						
Plant part	**Growth stage**	**ppm dry matter**				
		Fe	**Mn**	**Zn**	**Cu**	**B**
Young mature leaf	Mid season	10	27	7	0.73	9
Source: various						

Fertilizer recommendations

The unbranched root system is very inefficient in P uptake unless the root is mycorrhizal. Cu deficiency is common in peaty and acid soils; acid soils should be limed, since the crop does not tolerate acidity well. Mn deficiency may occur in alkaline soils.

All fertilizer should be applied close to the surface to be within reach of the shallow root system. A complex fertilizer, rich in P2O5 (e.g. 12-24-12) will improve bulb enlargement and yields. Even when mineral fertilizers are used, additional application of organic manure may enhance yields, quality and bulb size.

Present fertilizer practices

Senegal (Camberene)

In a light sandy soil in a semi-arid area apply 10 t/ha of organic matter, 55 kg/ha N, 125 kg/ha P_2O_5, and 140 kg/ha K_2O. Broadcast all the organic manure (well-decomposed) and three-quarters of the N, P_2O_5 and K_2O before planting, and band the remainder 35 days after planting.

Philippines (Los Banos)

Apply 120 kg/ha N, 240 kg/ha P_2O_5, and 120 kg/ha K_2O in the dry season. At planting apply all the P_2O_5 and K_2O and half the N as a basal dressing. The remaining half of N is sidedressed when bulbing is initiated. Sidedress about 10 cm deep and 8 cm away from the roots.

Pakistan (Baluchistan Province)

Apply 120 kg/ha N, 80 kg/ha P_2O_5 and 80 kg/ha K_2O. All of the P_2O_5 and K_2O and half the N are applied at transplanting, and the remaining N 4 weeks later.

Nigeria (Nsukka)

In acid ultisoils apply 2 t/ha CaO at least two weeks before transplanting. At twenty days after transplanting apply 75 kg/ha N, 70 kg/ha P_2O_5, and 180 kg/ha K_2O. At 35 days after transplanting apply a further 75 kg/ha of N.

Further reading

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Peas *(Pisum sativum L.)*

French: Pois; Spanish: Guisante; Italian: Piselli, German: Erbsen

Under Temperate Conditions

Harvested part: green seeds, used fresh or for processing, and mainly for freezing. Seed quality evaluation is based on resistance to pressure (tenderometer), water content (about 70 %) and sugar/starch ratio.

Plant density: 40-80/m2 depending on plant habit and sowing time. Yields 2.5-10.0 t/ha depending on variety and degree of seed maturity.

Interval from sowing to harvest: depends on growing season and variety (dwarf or climbing), with a minimum of about 70 days (corresponding to 600-700 degree-hours above 4.4 °C). Because the plant needs temperatures in the range 10-20 °C, it can be sown in different seasons; with higher temperatures the time from sowing to harvest is shorter but the product quality is poorer.

The crop requires well-drained soils which allow a good symbiotic activity on the roots.

Preferred pH 6.0-7.5. Calcareous soils may cause plant chlorosis.

Nutrient demand/uptake/removal

Yield t/ha	Source	(kg/ha)				
		N	P2O5	K2O	MgO	CaO
7	Anstett	146	44	125	12	62
10	Penningsfeld	125	43	88	10	152

About 60 % of this uptake is in the seeds.

Fertilizer recommendations

In normal soil conditions, N requirements are assured by symbiotic activity. Application of fertilizer N (at about 30 kg/ha N) is advisable only during the initial stages and/or in adverse climatic or soil conditions.

As regards other nutrients, application of 60-100 kg/ha P2O5 and about 110 kg/ha K2O is recommended at sowing. For topdressing, rates of about 50 kg/ha P2O5 and 100 kg/ha K2O are advisable. Direct contact of fertilizer with the roots should be avoided.

Adequate P benefits the Rhizobium symbiosis.

Low K may cause lack of colour in the pods.

The crop is sensitive to Cu deficiency, so application of this element before sowing may be useful, but problems regarding Cu availability and other micronutrients are rather rare in soils well supplied with organic matter.

Peppers *(Capsicum annuum L.)*

French: Piment; Spanish: Pimiento; Italian: Peperone; German: Gemüsepaprika

Under Temperate Conditions

Crop data

Can be considered as an annual when grown for sweet pepper production. Harvested part: fruits, which vary in shape, weight and colour. The root system consists of a deep tap root with branches spreading laterally about 50 cm, and adventitious roots.

Transplanted at 2-3/m2, spaced 40-50 cm apart in rows 70-100 cm apart. Can be direct sown but mainly in crops for processing. Yields 35-50 t/ha depending on cycle length and fruit characteristics. Protected cultivation under low or large tunnels, or in cold or heated greenhouses, very common.

Thrives well in deep, well-drained soils rich in organic matter. Tolerates a wide range of pH, with optimum 6.5-7.5. Sensitive to brackishness.

Temperatures required for germination, 18°-35 °C (optimum 29.5 °C); for growth, night 16°-18 °C, and day 20°-25 °C; and for fruit ripening, 18°-24 °C.

Nutrient demand/uptake/removal

180-400 kg N, 45-120 kg P2O5, 250-675 kg K2O, 32-50 kg MgO, 110-160 kg CaO per ha depending on yield, growing conditions and whether harvested green or final colour. Both uptake and plant growth proceed more rapidly from flowering to the gathering of the first fruits.

Fertilizer recommendations

For normal outdoor crops: 250-300 kg/ha N, of which 100 kg/ha before transplanting, followed by three to five topdressings (before flowering, at fruit set, and during harvest); 100-150 kg/ha P2O5 and 200-300 kg/ha K2O all before planting or, on sandy soils, in split dressings. The K2O rate may need adjusting according to soil analysis. Peppers respond well to N in the form of urea. K increases plantlet resistance to wilting.

The crop is rather sensitive to Mg deficiency.

7.2.10 Peppers *(Capsicum annuum L. var. grossum)*

Under Tropical/Subtropical Conditions

Crop data

Transplanted or direct seeded. Harvested: 65 - 80 days after transplanting. The roots thrive with good soil aeration. The crop is therefore preferably grown in sandy loams and loams

with good drainage, with pH between 6 - 6.8. The crop is adapted to warm conditions. Generally irrigated.

Target marketable yields in intensive commercial production: 11 - 25 t/ha.

Nutrient demand/uptake/removal

Nutrient uptake/removal - Macronutrients					
Yield t/ha	N	P2O5	K2O	MgO	CaO
	kg/ha				
21	70	16	92	18	67
Source: various					

Plant analysis data

Plant analysis data - Macronutrients (optimum fertility conditions)							
Plant part	Growth stage	% of dry matter					
		N	P	K	Mg	Ca	S
Young mature leaf	Early fruiting	3.7	0.3	3.4	0.4	1.0	0.2
Source: various							

Plant analysis data - Micronutrients (optimum fertility conditions)						
Plant part	Growth stage	ppm dry matter				
		Fe	Mn	Zn	Cu	B
Young mature leaf	Early fruiting	45	33	26	4	23
Source: various						

Fertilizer recommendations

The greatest absorption of nutrients occurs in the first 8 to 14 weeks of growth and again after the first fruit removal. Therefore, high N levels are required by the plant early in the growing season with supplemental applications after the fruit initiation stage. Improved N use efficiency and greater yields are achieved when N is applied under polyethylene mulches and with 12 weekly N applications in a drip irrigation system. At least 50 % of the total fertilizer N should be NO3-N.

For optimum P placement, band 5 - 8 cm deep in the rows. Blossom-end rot may result from Ca deficiency which may be corrected with foliar sprays of calcium chloride or calcium nitrate. The crop is also sensitive to Mg deficiency and has a low salt tolerance, but root inoculation with vesicular-arbuscular mycorrhizal fungi may improve growth under salt stress conditions.

Present fertilizer practices.

Senegal (Cambarene)

In a light soil in a semi-arid area apply 10 t/ha organic manure, 140 kg/ha N, 100 kg/ha P2O5, and 200 kg/ha K2O. At planting, basal applications of all the organic manure, dolomitic limestone if required, 60 % of the P2O5 and 15 % of the N and K2O. The balance of the N and K2O is applied in localized top dressings at 3-week intervals, beginning 15 days after planting. The balance of the P2O5 is either applied before flowering or it is split into three dressings at 6-week intervals.

Brazil (Minas Gerais)

General recommendations are, firstly, 60 kg/ha N, 240 kg/ha P2O5 and 180 kg/ha K2O incorporated in the soil at planting and, secondly, 240 kg/ha N and 50 kg/ha K2O broadcast in two applications 15 and 30 days after planting. Greater yields are obtained by incorporating into the soil 20 t/ha organic matter two or more weeks before planting.

India (Bangalore)

Applications on sandy loams in moderately fertilized soils are 150 kg/ha N, 80 kg/ha P2O5 and 40 kg/ha K2O. Broadcast all the P2O5 and K2O at planting time together with 50 kg/ha N. The remaining N is applied in equal split doses at 30 and 60 days after planting.

Further reading

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Plantain *(Musa accuminata x M. balbislana [AAB group])*

French: Plantain; Spanish: Plátano; German: Plantane, Kochbanane

Under Tropical/Subtropical Conditions

Crop data

Plantains (cooking banana) are closely related to the familiar dessert banana. In some respects they are more important than dessert banana because they are a staple diet in Southeast Asia, the Pacific Islands, Tropical America and Tropical Africa. The Cameroons, for example, produce taro, cassava and banana (including plantain) in approximately equal quantities, about 600 000 t each per year.

In some respects plantain seems to differ in nutrient requirements from dessert banana. The two crops also seem to differ in the way they respond to intensive management; and dessert bananas are often more productive through many successive crops, whereas successive yields of intensively managed plantains often decline rapidly.

Nutrient demand/uptake/removal

The following table is based on very limited data.

Nutrient uptake/removal - Macronutrients						
Production level	**kg/ha**					
	N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
Near maximum	250	46	702	100	252	24
Intermediate (75 %)	148	30	420	66	154	14
Low (50 %)	73	16	180	33	91	7
Source: Vicente-Chandler & Figurella, 1962; Del Valle et al., 1978; Fox et al., 1979						

Further data on contents of nutrients are given with "Fertilizer recommendations".

Plant analysis data

Plant analysis data - Macronutrients						
Production level	**% of dry matter (No. 3 leaf)**					
	N	**P**	**K**	**Mg**	**Ca**	**S**
Near maximum	3.4	0.19	3.5	0.25	0.7	0.26
Intermediate (75 %)	2.7	0.16	2.8	0.21	0.6	0.20
Low (50 %)	2.0	0.14	1.8	0.15	0.5	0.10
Source: Vicente-Chandler & Figurella, 1962; Del Valle et al., 1978; Fox et al., 1979						

Further "plant analysis data" is given with "Fertilizer recommendations".

Fertilizer recommendations

Plant nutrition problems may stem from a superficial and restricted root system - a problem of banana generally. However, reports frequently specify that the soil should be deep and fertile. Perhaps shallow rooting is as much a function of soil properties as it is of plant characteristics.

Fertilizer recommendations should be based on crop requirements for a particular expected yield, corrected for the ability of the environment to meet those requirements. But two

nutrients, N and K, are deficient for plantain growing in most soils. The quantity of N a soil can deliver to plantain can be estimated from the N taken up in a similar situation by a crop such as continuously-grown, minimum-till maize. N uptake by plantain is approx. 8 kg/t of fruit produced. Thus a yield of 30 t represents 250 kg of N removed (table nutrient uptake/removal) The deficit, uptake minus soil N delivered, should be supplied as fertilizer or manure, with allowance made for the inefficiency of the fertilizer and/or manure application.

The efficiency of N in the tropics is often approx. 50 %. In Hawaii (clay soil) estimates of N recovery in banana at harvest, in relation to N applied, were about 65 %. Probably much of the N was lost as volatilized ammonia resulting from spreading the fertilizer (urea) on dry ground and trash.

Dessert bananas, and probably plantain too, develop an effective mycorrhiza. The fungus-root association decreased the required concentration of P in soil solution from 0.1 mg/l to 0.05 mg/l. This is less than the required concentration for many vegetable crops and perhaps more than the requirement for maize. Such dilute solutions do not necessarily indicate low fertilizer requirements. Highly weathered, fine-textured soils of the tropics will typically require 100 to > 200 mg/kg P in soil to attain a sufficient P level for plantain. Such rates of phosphate are seldom used even in experiments. Thus reported fertilizer requirements are frequently low.

P percentage in plantain fruit is low and so also is the quantity of P contained in the fruit. P fertilizer required is more a matter of soil reactions with P than it is of crop need. Estimates of fertilizer requirements on a global scale demonstrate that Acrisols, Ferralsols and Andosols - important soils in the tropics - are high P-sorbing soils; the weighted mean P sorption is 900 kg/ha P_2O_5 to attain 0.02 mg P/l in solution. This is a little low for maximum plantain production.

Plantain is rich in K; a 30 t crop contains approx. 720 kg/ha K_2O of which 240 kg is removed in the fruit. Much of the remaining 480 kg should be available for a ratoon crop.

The requirement for S in plantain is approx. 7 % of the N requirement. However, S is taken up more sluggishly than N, and therefore should be supplied at approx. 10 % of the amount of N.

The following table is an example of estimated fertilizer requirements for plantain (plant crop) and the type of information needed to make reasonable estimates for specific locations. This example assumes yields of approximately 35 t/ha, typical fertilizer efficiencies for weathered soils, and low soil fertility. The table can be modified to suit local conditions and yield expectations by substituting more appropriate values for soil-supplied nutrients and fertilizer efficiencies. Such modifications depend on the amount of leaching reltive to expectation (based on rainfall-evaporation data, probability of N loss by NH_3 volatilization and denitrification, P and K sorption by the soil, S accretion via rainfall, etc.) The estimated N and K requirements are similar to the average fertilizer rates used in experiments that produced acceptable yields (average 34 t fruits/ha).

Estimated fertilizer requirements					
Plant crop	**N**	**P2O5**	**K2O**	**MgO**	**S**
Effective amounts from soil and rainfall (kg/ha)	60*	18**	216***	20	10
Total uptake by crop yielding 35 t/ha (kg/ha)	250	46	702	100	24
Balance to be obtained from fertilizer (kg/ha)	190	28	702	80	14
Assumed fertilizer use efficiency (%)	50	10	75	75	75
Amount to be provided in fertilizer (kg/ha)	380	280	648	107	19
Remark: Based on reasonable expectations of nutrient supplying power of wheathered soils, fertilizer efficiencies associated with moderate leaching, and plant uptake of nutrients associated with acceptable production of 35 t/ha of fruit.					
* Assume 0.15 % N, 4 % mineralization rate and 50 % efficiency.					
** Assume 10 t dry matter that is 0.1 % P and that this can be produced with no P fertilizer.					
*** Assume 0.15 meq exchangeable K/100 g soil which is 75 % available.					

Predicting fertilizer requirements for a ratoon crop is more uncertain than for a plant crop because of residual fertilizer effects and the efficiency of nutrient recovery from plant crop residues, which are difficult to evaluate. Soil contributions will usually decrease with time and residual fertilizer effects will increase if fertilizer is added in excess of crop removal. An evaluation of these residual effects requires estimates of nutrients removed in the fruit and long-term (residual) efficiencies of fertilizers. The first can be calculated from yield and composition data; the second is more in the realm of speculation, or at best, an educated guess. Such uncertainities notwithstanding, predicted requirements for a ratoon crop producing 35 t/ha of fruit are about 165 kg/ha N, 115 kg/ha P2O5, 288 kg/ha K2O, 40 kg/ha MgO and 25 kg/ha S.

These predictions are low in comparison with annual requirements (7-year average) for banana in Hawaii, but suitable data for ratoon plantain were not available for comparison. In the absence of specific data, local recommendations for dessert banana will be helpful.

Fertilizer requirements based on soil analysis have not been worked out for plantain as far, reference should be made to such worked out for banana.

Further reading

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Radish *(Raphanus sativus L.)*

French: Radis; Spanish: Rábano; Italian: Ravano; German: Rettich

Under Tropical/Subtropical Conditions

Crop data

Annual. Harvested part: root. Directly seeded. Harvested: 22 - 30 days after planting. Plant density: 363 000 to 522 000 plants (average) to 1 250 000 (very intensive) plants/ha.

Preferably grown in a rich fertile soil free of stones and clods, to allow for rapid growth and smooth root growth. Sandy soils are preferred for early yields, pH 5.5 - 6.8. The crop is adapted to cool growing conditions.

Target marketable yields in intensive commercial production: 11 - 25 t/ha.

Nutrient demand/uptake/removal

Nutrient uptake/removal - Macronutrients					
Yield t/ha	**kg/ha**				
	N	**P2O5**	**K2O**	**MgO**	**CaO**
19	276	89	389	76	147
Source: various					

Plant analysis data

Plant analysis data - Macronutrients							
Plant part	**Growth stage**	**% of dry matter**					
		N	**P**	**K**	**Mg**	**Ca**	**S**
Young mature leaf	20-30 days after planting	3.0	0.9	2.4	0.24	1.1	0.9
Source: various							

Plant analysis data - Micronutrients						
Plant part	**Growth stage**	**ppm dry matter**				
		Fe	**Mn**	**Zn**	**Cu**	**B**
Young mature leaf	20-30 days after planting	34	16	23	2	6
Source: various						

Fertilizer recommendations

Critical tissue P concentration in organic soils in Florida is 0.45 %. In B deficient soils 10 kg/ha of borax will increase ascorbic acid content and yields. B toxicity (B > 0.1 mg/l) should be avoided.

Present fertilizer practices

USA (Florida)

On irrigated mineral soils apply 100 kg/ha N, 134 kg/ha P2O5 and 134 kg/ha K2O. P and K should only be applied after soil analysis indicates deficiency. Broadcast all the P2O5 and

half the N and K2O at planting. Apply the remainder of the total N and K2O, 15 days after planting. These rates should support three radish crops grown in succession.

Philippines (Los Banos)

Broadcast 60 kg/ha N, 90 kg/ha P2O5 and 90 kg/ha K2O at planting. For radish, 300 to 400 kg/ha 10-25-25 mixture is recommended.

Brazil (Minas Gerais)

General recommendations are, firstly, 30 kg/ha N, 120 kg/ha P2O5 and 90 kg/ha K2O incorporated in the soil at planting and, secondly, 40 kg/ha N broadcast in two applications 10 and 20 days after planting. If available, incorporate 20 t/ha of organic matter into the soil two weeks or more before planting.

India (North Bihar)

In a sandy loam soil with pH 8.5 broadcast at planting 15 t/ha organic manure, 50 kg/ha N, 40 kg/ha P2O5, and 80 kg/ha K2O. If B deficiency occurs apply 10 kg/ha borax with the initial fertilizer package.

Further reading

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Spinach *(Spinacia oleracea L.)*

French: Epinard; Spanish: Espinaca; Italian: Spinacio; German: Spinat

Under Temperate Conditions

Crop data

Harvested part: Rosette of smooth or curly, fleshy leaves.

Growth cycle rather short (8 weeks for earlier cultivars). Requires rather low temperatures (15-18 °C optimum) for satisfactory leaf yield.

Root system: main, deep, tap root, numerous shallow rootlets.

Plant density: more than 100 plants/m2 with seed rate from 20-40 kg/ha. Yields 15-30 t/ha; hand-picked crops (leaves) can produce up to 50 t/ha.

Suffers from acid soils, prefers pH values between 6 and 7.5; tolerates brackish soils rich in organic matter and well-drained.

Nutrient demand/uptake/removal

In kg/ha; N = 60-100; P2O5 = 30-50; K2O = 100-230; MgO = 10-20; CaO = 20-30 for normally yielding crops; higher yielding crops twice these amounts. Maximum uptake (60 % of the total) only occurs during 2-3 weeks before harvesting.

Fertilizer recommendations

FYM not recommended for hygienic reasons. For a medium yielding crop 100-150 kg/ha N, 70-100 kg/ha P2O5 and 150-180 kg/ha K2O are recommended; doubled for higher yielding crops (hand-picked).

Two or more topdressings of N. Under field conditions highest yields were obtained with ammonium N and five applications up to a total of 200 kg/ha N.

All nutrients should be incorporated in the upper layer of soil (25-30 cm; shallow location of absorbing roots).

Sensitive to Mg deficiency (leaf chlorosis).

Especially during cloudy, winter seasons spinach can accumulate high quantities of nitrate N in its tissues (up to 2 500 ppm on fresh matter basis); N fertilization (doses, form, sharing) to be managed accordingly.

Under Tropical/Subtropical Conditions

Crop data

Direct seeded. Harvested: 37 - 45 days after planting. Plant density: 72 500 to 653 500 plants/ha. Preferably grown in sandy loams but can be grown in a variety of soils with pH between 6 and 7. The crop is adapted to cool growing conditions.

Target marketable yields in intensive commercial production: 9 - 17 t/ha.

Nutrient demand/uptake/removal

Nutrient uptake/removal - Macronutrients					
Yield t/ha	kg/ha				
	N	P2O5	K2O	MgO	CaO
21	131	34	226	42	41
Source: various					

Plant analysis data

Plant analysis data - Macronutrients (optimum fertility conditions)						
Plant part	Growth stage	% of dry matter				
		N	P	K	Mg	Ca
Young mature leaf	40 days after planting; harvesting	2.6	0.3	4.7	1.0	1.0
Source: various						

Plant analysis data - Micronutrients (optimum fertility conditions)							
Plant part	Growth stage	ppm dry matter					
		Fe	Mn	Zn	Cu	B	Mo
Young mature leaf	40 days after planting; harvesting	189	36	37	4	19	1
Source: various							

Fertilizer recommendations

Spinach has a moderate tolerance to salinity, but it is very sensitive to acidity. N demand is great just before harvest because of the flush in growth. When present in the rhizosphere, the N-fixing bacteria Azospirillum spp. promote the growth of this crop. It has a relatively high demand for B. In B-deficient soils broadcast 10 kg/ha borax with the initial fertilizer application.

Present fertilizer practices

USA (Florida)

For spinach grown on irrigated mineral soils apply: 90 kg/ha N, 120 kg/ha P2O5, and 120 kg/ha K2O. At planting broadcast all the P2O5 and one-quarter of the N and K2O. The remainder of the N and K2O is sidedressed in split dressings at 20 and 40 days after planting.

USA (North Carolina)

In sandy and sandy loam soils broadcast 130 kg/ha N, 95 kg/ha P2O5, and 160 kg/ha K2O before planting. Follow up with split applications of 50 kg/ha N, soon after planting.

Brazil (Minas Gerais)

Apply, firstly, 60 kg/ha N, 200 kg/ha P2O5 and 90 kg/ha K2O incorporated in the soil at planting and, secondly, 60 kg/ha N broadcast in two applications, 15 and 30 days after planting.

Further reading

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Sweet Corn *(Zea mays L. convar. saccharata Koern.)*

French: Mais sucré; Spanish: Maiz dulce; Italian: Mais dolce; German: Zuckermais

Under Tropical/Subtropical Conditions

Crop data

In the tropics field maize is more commonly grown as a vegetable than sweet corn, because most sweet corn varieties have been developed for long days in northern latitudes. However, sweet corn hybrids suitable for the tropics are now available (including several from Hawaii).

The crop can be grown in a wide variety of soils if they are naturally fertile or can be made fertile with appropriate fertilizers and/or organic manure. Optimum mean day temperatures are around 25° C. Temperatures in excess of 35° C can severely damage pollination, especially if accompanied with moisture stress from just before silk development extending through pollination. For crops grown in bright daylight conditions, an acceptable plant density is 60 000 per ha. Density should be less when extended periods of cloudy weather can be expected after full canopy development.

Nutrient demand/uptake/removal

The following table shows the generally high nutrient requirements for high yields of sweet corn; only Ca is low compared with many other crops. The requirements for nutrient quantity (and concentration) removed in a crop are similar to field maize. N and K contents are specially high, which explains the high fertilizer requirements.

Nutrient uptake/removal - Macro- and micronutrients (Zn)							
Yield	**kg/ha**						
	N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**	**Zn**
20 t fresh weight	208	60	228	25	42	14	0.21
Source: Fox, 1973; Daigger & Fox, 1971; and others							

Plant analysis data

Plant analysis data - Macro- and micronutrients (Zn) (good plant nutrition and fertilizer practices)							
Crop status	**% of dry matter**						
	N	**P**	**K**	**Mg**	**Ca**	**S**	**Zn**
Concentration* near maximum yield	2.7	0.26	2.25	0.15	0.56	0.24	0.0024
* Data are means of several sources including Chapman, 1966; Lorenz & Maynard, 3rd ed; Fox et al., 1964; Daigger & Fox, 1971; Fox, 1973. If data specifically for sweet corn were not available, data from field maize in Hawaii (Rashid & Fox, 1992) or from field maize grown on a highly weathered soil in Georgia, USA (Hargrove, 1985) were used to construct the data. In most cases extrapolation from lower to a higher yield was necessary.							

Fertilizer recommendations

Fertilizer used/recommended						
Area or soil conditions	Source	kg/ha				
		N	P2O5	K2O	S*	Zn**
Brazil (General)						
Sao Paulo	Van Raij et al., 1985	75	50	41	20	5
Minas Gerais	Lopes, 1989	60	80	50	-	-
Florida, leached mineral soils, irrigated	Hochmuth & Hanlon, 1989	90	121	120	-	-
Hawaii (General) low organic matter, otherwise fertile	Nakagawa, 1957	80	101	96	-	-
Maximum yield	Fox, 1973	220	101	96	10	10
Production Guidelines (General)	Valenzuela, 1991	103	96	110	-	-
* Apply with the N at about 10 % of the rate. ** Apply infrequently, only when needed, to Zn deficient soils.						

The N fertilizer requirement for maximum yields, as given for Hawaii (and it is the same in Nebraska, USA) is 220 kg/ha N. This is almost identical with N uptake given for an excellent (20 t/ha) crop. Any decrease in the rate of N fertilizer applied will diminish yields in a linear manner. This suggests that the usual fertilizer N recommendation is much below the optimum level. N should be given as a split application, one-quarter to one-half either before or at planting; and, depending upon the likelihood of N leaching and the visual appearance of the crop, the remainder in one or two applications up to approximately 40 days after germination.

Many soils will provide a substantial portion of the P required, but in the humid tropics most soils will not supply P in adequate concentration, especially at the seedling stage. Thus, an application of P near the seed is usually beneficial for early seedling vigour. The above recommendation for Sao Paulo, Brazil is for a soil of medium resin-extractable P status (7 to 15 mg/l P). The recommendation is doubled for soils with 0 - 6 mg/l and halved for soils with > 40 mg/l. The Florida recommendation is based on soils very low in P. Florida soils generally have low P absorbing capacities.

K is taken up in large quantities. Soils, however, even in the humid tropics, usually supply a substantial part of the K requirement. How much, can usually be estimated by soil analysis. The Brazil recommendation assumes a soil of medium exchangeable K status (1.6 meq/l). If exchangeable K exceeds 3 meq/l the recommendation is decreased by one half and if < 0.07, the recommendation is doubled. K is usually applied before or at planting except when leaching is severe as in the case of sandy soils or highly weathered soils which have little cation exchange capacitiy and little weatherable K minerals. In such cases the fertilizer K requirement approaches the amount of K removed in the crop; and the fertilizer is applied in split dressings like N.

The crop is among those most sensitive to Zn deficiency. Calcareous and strongly alkaline soils of the semi-arid tropics are suspect, as are also the highly weathered soils of the humid tropics. Eroded soils and low organic matter soils are at greatest risk. Zn uptake is very low; thus the effects of a substantial Zn application of 10 kg/ha Zn may persist for several years.

S deficiency is a greater problem than is generally recognized in the tropics. The crop is susceptible especially in the young growth stages. Many subsoils of the humid tropics contain much absorbed sulphate but it may be difficultly available or positionally unavailable. Rainwater contains sulphate. If the S concentration is <l mg/l, deficiency can be expected. If S deficiency is confirmed, apply S at 5 to 10 % of the N requirement.

Further reading

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Tomato *(Lycopersicon esculentum Mill.)*

French: Tomate; Spanish: Tomate; Italian: Pomodoro; German: Tomate

Under Temperate Conditions

Annual. Harvested part: fruits.

Direct sowing is practised, mainly in crops for processing, but transplanting is more usual.

Root system mainly (70 % or more) in upper 20 cm soil layer.

Cycle length varies with sowing time and growing methods. 3 000-4 000 degree(°C)-days are required between sowing and ripening of the first fruit.

Plant density: 3-4/m2.

Yields, in more typical areas: for processing, up to 40-50 t/ha; for table tomatoes, 20-60 t/ha outdoors, up to 100 t/ha or more under glass.

Can be grown on soils with a range of textures. Preferred pH 6.5-7.0. Tolerates brackishness up to conductivitys of about 2-3 mmho/cm. Fairly resistant to moderate drought but is usually irrigated.

Growth requires temperatures between 10 ° and 30 °C. In protected cultivation tomato is the most important crop in the Mediterranean area.

Nutrient demand/uptake/removal

100-150 kg N, 20-40 kg P2O5, 150-300 kg K2O, 20-30 kg MgO per ha for outdoor crop yielding 40-50 t/ha; or, for greenhouse crop on natural soil yielding over 100 t/ha, about 200-600 kg N, 100-200 kg P2O5, 600-1000 kg K2O per ha. The greatest rates of uptake of the three major nutrients occur during flowering and when the first fruits ripen.

Plant analysis data

Normal ranges in leaves of healthy plants: 2.8-4.9 % N, 0.4-0.7 % P2O5, 2.7-5.9 % K, 0.4-0.9 % Mg, 2.4-7.2 % Ca, 1.0-3.2 % S, 32-97 ppm B, 10-16 ppm Cu, 101-291 ppm Fe, 55-220 ppm Mn, 0.9-10.0 ppm Mo, 20-85 ppm Zn, all expressed on a dry matter basis.

Deficiency levels: <2 % N, <0.2 % P, <1.5 % K at vegetative stage or <2.5 % K at fruiting stage, <1 % Ca, <0.3 % Mg.

Fertilizer recommendations

FYM or other organic manures should first be applied at not less than 30 t/ha in order to increase the organic matter in the soil.

Outdoor crops yielding 40-50 t/ha should be given a basal fertilizer application, before planting, of about 50 kg N, 150-200 kg P2O5, 200-250 kg K2O per ha (less K2O if indicated by soil analysis), followed by 100-150 kg/ha N divided into two or three topdressings. Topdressing with K2O may also be useful for table tomatoes.

For protected crops, the rates and timing of fertilizer application need to be modified to take into account the higher yield, the longer growth cycle and the increased uptake per unit yield.

Nitrate N has proved more effective than ammonium N in the poorer light conditions in winter, and for use on plantlets low in carbohydrates.

Some attempts have been made to establish predictive models on which to base fertilizer application. A simple model for N fertilizer requirement has been proposed (Feigin et al., 1986), according to the formula:

$Nf = (Np / E) - Na$

where Nf = amount of N to be applied
Np = uptake by the crop estimated from data in the literature
E = nitrogen efficiency factor of 0.70 to 0.75
Na = soil N content (from analytical data) plus mineralizable N minus expected denitrification loss.

Tomatoes are rather sensitive to excess or deficiency of both macro- and micro- nutrients. Examples of likely deficiencies, particularly in soilless culture, other than those of N and P, are: K deficiency, affecting fruit quality; Ca deficiency, causing blossom end rot; Mg deficiency, in acid soils and in the presence of high levels of K; and deficiencies of B, Fe and Mn in calcareous soils.

Under Tropical/Subtropical Conditions

Crop data

Transplanted 35 to 50 days after seeding. Harvested 60 to 90 days after transplanting.
Plant density: 12 150 to 36 900 plants/ha. Preferably grown in sandy soils for an early harvest but adapted to sandy and clay soils, pH 6 - 6.5. Adapted to 20 - 24 °C temperatures and at least a 115-day growing season. In the tropics tomatoes are normally grown in the highlands or in the cooler season.

Target marketable yields in intensive commercial production: 27 - 37 t/ha.

Nutrient demand/uptake/removal

Nutrient uptake/removal - Macronutrients					
Yield t/ha	**kg/ha**				
	N	**P2O5**	**K2O**	**MgO**	**CaO**
24	177	46	319	43	129
Source: various					

Plant analysis data

Plant analysis data - Macronutrients (optimum fertility conditions)							
Plant part	**Growth stage**	**% of dry matter**					
		N	**P**	**K**	**Mg**	**Ca**	**S**
Young mature leaf	1/2 fruit	2.7	0.5	2.9	0.4	1.2	0.3
Source: various							

Plant analysis data - Micronutrients (optimum fertility conditions)							
Plant part	**Growth stage**	ppm dry matter					
		Fe	Mn	Zn	Cu	B	Mo
Young mature leaf	1/2 fruit	119	76	24	7	25	0.16
Source: various							

Fertilizer recommendations

N requirements are moderate during foliage growth until fruit set. P is very important for vigorous growth and fruit production. Near maximum yield is attained at approx. 0.2 mg P/l soil solution. Root inoculation with Glomus intraradices vesicular-arbuscular mycorrhizal fungi may improve growth through improved P uptake and decreased Fusarium oxysporum disease incidence.

K is required for fruit set and enlargement. Important growth stages for nutrient uptake include seedling establishment when maximum nutrient efficiency uptake occurs, and early fruiting when the maximum rates of K accumulation take place. Tomato should be supplied with Mg in soils deficient in this nutrient. Ca deficiencies may cause "blossom end rot" or apical necrosis.

Fertilizer applications through a drip irrigation system may improve the efficiency of N uptake.

Present fertilizer practices.

Senegal (Camberene)

In a light sandy soil in semi-arid conditions apply 20 t/ha organic matter, 70 kg/ha N, 200 kg/ha P2O5, and 240 kg/ha K2O distributed in the following manner: All organic matter and one-fifth of the N, P2O5 and K2O broadcast before planting, followed by four equal sidedressings of N, P2O5 and K2O at 20-day intervals beginning 15 days after planting.

For intensive tomato production under irrigation in sandy soil apply 190 kg/ha N, 225 kg/ha P2O5, and 300 kg/ha K2O. One-fifth of the N, P2O5 and K2O is broadcast before planting, and the remainder in four equal dressings at 15-day intervals beginning 15 days after planting.

Philippines

Fertilizer requirements are 96 kg/ha N, 192 kg/ha P2O5, and 96 kg/ha K2O. A basal dressing of all the P2O5 and one-half the N and K2O is placed 8 cm below and slightly to the side of the seedling root crown. The remaining N and K2O is sidedressed 1 month after transplanting. If planting in furrows, the fertilizer is applied in the furrows and covered with soil.

Pakistan (Baluchistan Province)

Fertilizer recommendations are 150 kg/ha N, 100 kg/ha P_2O_5, and 50 kg/ha K_2O. Apply all the P_2O_5 and one-half the N and K_2O before transplanting. The remainder of the N and K_2O is applied at 6 - 8 weeks after transplanting.

Venezuela (States of Lara, Aragua, Falcon, Nueva Esparta and Trujillo)

Sidedress 800 kg/ha of 12-24-12 fertilizer 15 - 20 days after planting.

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Watermelon *(Citrullus lanatus Thunb. Matsumara & Nakai)*

French: Pasteque; Spanish: Sandia; Italian: Cocomero; German: Wassermelone

Under Temperate Conditions

Crop data

Herbaceous annual of tropical African origin, trailing with branching stems up to 5 m long, and roots branched in upper 60 cm of soil. Harvested part: very heavy (2 kg to >20 kg), fleshy, multi-seeded fruits containing about 90 % water.

Plant density: 0.5-1.0/m2, sown with pre-germinated seeds or transplanted with pot plantlets in rows according to the cultivar.

Protected cultivation under low tunnels is increasingly common. Yields: 25-75 t/ha. Prefers light to medium soils, well-drained and slightly acid (up to pH 5.5).

Requires temperatures of 21°-35 °C for seed germination, 35 °C for quick emergence and 21°-30 °C for growth, with a long frost-free growing season. Although it can withstand dry conditions the crop should be irrigated, mainly during fruit growth.

Nutrient demand/uptake/removal

1.7 kg N, 1.3 kg P2O5, 2.7 kg K2O, 0.7 kg MgO per t of fruit. Peak uptake of nutrients and water occurs during fruit swelling.

Fertilizer recommendations

At least 30 t/ha of organic manure should be distributed along the rows or near plant holes. Experimental results suggest fertilizer application rates of 50-100 kg N, 100-150 kg P2O5, 140-200 kg K2O per ha, which could be applied along the rows but not too close to the seed in order not to damage germination. N should be applied in split dressings, at planting, at branching, just before flowering and after setting of the first fruit, preferably in nitrate form. K is considered important in preventing fruit cracking. Application of calcium sulphate may be needed to lower the pH of alkaline soils.

Watermelons are sensitive to deficiencies of Mg, B, Fe and Zn. Foliar sprays of these nutrients, in some cases, have proved useful in enhancing the sugar content of the fruit.

Under Tropical/Subtropical Conditions

Crop data

Annual. Harvested part: Fruit. Flowers about 80 days after planting. Harvested 75 - 95 days after planting.
Plant densitiy: 4 500 to 9 100 plants/ha. Preferably grown in sandy or sandy loam soils. Watermelons grown in heavy soils are subject to fruit cracking. Tolerant of acid soils. Adapted to a low ambient humidity.

Target marketable yields in intensive commercial production: 15 - 30 t/ha.

Nutrient demand/uptake/removal

Nutrient uptake/removal - Macronutrients					
Yield t/ha	**kg/ha**				
	N	**P2O5**	**K2O**	**MgO**	**CaO**
15	56	16	100	25	98
Source: various					

Plant analysis data

Plant analysis data - Macronutrients (optimum fertility conditions)							
Plant part	**Growth stage**	**% of dry matter**					
		N	**P**	**K**	**Mg**	**Ca**	**S**
Young mature leaf	Mid season	3.6	0.48	2.7	0.5	1.3	0.1
Source: various							

Plant analysis data - Micronutrients (optimum fertility conditions)						
Plant part	**Growth stage**	**ppm dry matter**				
		Fe	**Mn**	**Zn**	**Cu**	**B**
Young mature leaf	Mid season	33	30	15	4	15
Source: various						

Fertilizer recommendations

N or Mg deficiency will reduce fruit-set. Dolomitic limestone should be applied to offset Mg or Ca deficiencies. Pre-plant broadcast applications are recommended in preference to band-applied preplant fertilizer to prevent seedling salt injury. Cu deficiencies in organic soils may drastically reduce yields.

Present fertilizer practices

Senegal (Camberene)

In a light sandy soil in a semi-arid area apply 10 t/ha of organic manure, 60 kg/ha N, 60 kg/ha P2O5 and 120 kg/ha K2O in the following manner: All organic matter and one-third of the N, P2O5 and K20 broadcast before planting, another third sidedressed when the runners start to grow, and the remainder during the fruit enlargement stage.

Brazil (Minas Gerais)

Apply, firstly, 40 kg/ha N, 160 kg/ P2O5, and 90 kg/ha K2O incorporated in the soil at planting and, secondly, 100 kg/ha N broadcast in two equal dressings 15 and 30 days after transplanting.

Alternatively, incorporate 5 t/ha organic matter into the soil two or more weeks before transplanting.

Philippines

In the dry season apply 120 kg/ha N, 120 kg/h P2O5, and 120 kg/ha K2O. Band one third at planting, sidedress the second third when the vines have reached about 1 m in length, and the remaining third when the first fruit is about the size of an egg.

Hawaii

Apply 224 kg/ha N, 600 kg/ha P2O5, and 220 kg/ha K2O. Apply half the fertilizer at planting and the rest four weeks after planting. If the soils are low in Mg apply 170 kg/ha of magnesium sulphate.

Further reading

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Almond *(Prunus dulcis [Mill.] D.A. Webb var. dulcis)*

French: Amandier; Spanish: Almendro; Italian: Mandorlo; German: Mandelbaum

Crop data

Perennial. Harvested part: nuts
Transplanted in dormant season.
Flowers 3 - 4 years after transplanting.
Harvested 6 - 8 months after flowering.

Plant density (California, USA): 120-190 trees/ha.
Preferably grown on light to medium soils, pH 6-8.

Is adapted to a mediterranean climate; does best when irrigated, but is often grown on dry land.

USA (California) is the largest producer but it is also grown in Spain, France, Italy, Greece, Israel, and Bulgaria.

Plant analysis data

Plant analysis data - Macronutrients								
Plant part	**Stage of growth**	**Source**	**% of dry matter**					
			N	**P**	**K**	**Mg**	**Ca**	**S**
Leaf	Mid-season	Brown et al., 1953	2.8 (OS)	0.10 (OS)	3.0 (OS)	0.46 (OS)	1.61 (OS)	0.61 (OS)
Leaf	Mid-season	Proebsting & Serr, 1966		0.10 (OS)	1.0 (OS)			
Leaf	Mid-season	Weinbaum & Tate, 1964	2.2 (CVY)					
Leaf	Mid-season	Proebsting & tate, 1964						0.23 (OS)
Os = Optimum supply - CVY = Critical value (yield)								

Plant analysis data - Micronutrients						
plant part	**Stage of growth**	**Source**	**ppm dry matter**			
			Mn	**Zn**	**Cu**	**B**
Leaf	Mature basal leaves	Epsteim & Lillehand, 1942	96 (OS)			
Leaf	Mature	Hansen et al., 1962				23 (CVD) 38 (OS)
Leaf	Mature	Kester et al., 1961			2.6 (D) 3.9 (OS)	
Leaf	Mature	Procopiou, 1978		17 (OS)		
Hull	Mature	Kester et al., 1961			0.8 (D) 3.3 (OS)	
Kernel	Mature	Kester et al., 1961			1.6 (D) 12.1 (OS)	
Hull	Mature	Kester et al., 1961			1.4 (D)	

					2.7 (OS)	
Kernel	Mature	Kester et al., 1961			6.8 (D) 7.3 (OS)	
D = deficiency - OS = Optimum supply - CVD = Critical Value (deficiency)						

In California, Zn is a common problem.

Almonds are sensitive to excessive salts, especially of Na (leaf Na 0.30 % or more) and of Cl (leaf Cl 1.79 % or more).

In California, B toxicity (leaf B 87 ppm or more) and deficiency both present problems.

Present fertilizer practices

Almonds may be grown in cultivated or sod culture, so it may or may not be practicable to incorporate fertilizer into the soil. Fertilizer is usually given in a single dressing, sometimes in autumn, or, if in spring, then before bud-break. In general only N is applied annually, preferably as ammonium sulphate on alkaline soils or as ammonium nitrate on acid soils. P is preferably applied as triple superphosphate, and K as sulphate.

When supplying Zn as a foliar spray in mid-season, only ZnO should be used; ZnSO4 burns the foliage.

In USA the basal fertilizer is:

45-112 kg/ha N.
1.2 t K2O when problem occurs (California soils).
2 kg/ha B when problem occurs.

Zn deficiency is controlled by sprays of 19-29 g ZnSO4/l in the dormant period and by foliar sprays of 6 g ZnO/l applied in mid-season.

Further reading

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Author : D. Sparks, College of Agriculture, The University of Georgia, Athens, USA

Avocado *(Persea americana Mill.)*

French: Avocatier; Spanish: Aguacate; Italian: Avocado; German: Avocado

Crop data

Woody perennial.
Harvested part: fruits.
Flowers 2-3 months in spring.
Harvested 6-18 months after flowering (depending on cultivar).
Plant density: 200-400/hectare.
Preferably grown on light to medium soils; on heavy soils only when well-drained; pH 5-8.

The crop is adapted to tropical conditions (West Indian race) or to subtropical (Mexican or Guatemalan race).

Nutrient demand/uptake/removal

Nutrient removal (based on yield of 10 t/ha) - Macronutrients									
Cultivar	**Source**	**kg/ha**							
		N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**	**Cl**	**Na**
Fuerte	Lahav, 1980	11.3	3.9	23.5	8.3	2.9	8.0	1.5	0.8
Lula	Marchal, 1980	28.0	8.0	54.6	1.8	3.3	-	-	-

Nutrient removal (based on yield of 10 t/ha) - Micronutrients						
Cultivar	**Source**	**g/ha**				
		Fe	**Mn**	**Zn**	**Cu**	**B**
Fuerte	Lahav, 1980	90	20	40	10	40

Plant analysis

Leaf analysis data (optimum supply) - Macronutrients								
Country	**Cultivar**	**Source**	**% of dry matter**					
			N	**P**	**K**	**Mg**	**Ca**	**S**
California	Fuerte	Goodall et al., 1978	1.6-2.0	0.1-0.25	0.75-2.0	0.25-0.80	1.0-3.0	0.2-0.6
Martinique	Lula	Bertin, 1976	1.8-2.2	0.1-0.3	0.5-2.4	0.3-0.5	1.0-3.0	-

Leaf analysis data (optimum supply) - Micronutrients								
Country	**Cultivar**	**Source**	**ppm dry matter**					
			Fe	**Mn**	**Cu**	**Zn**	**B**	**Mo**
California	Fuerte	Goodall et al., 1978	50-200	30-500	5-15	30-150	50-100	0.05-1.0

Fertilizer recommendations:

Added N benefits the avocado almost universally. It is, therefore, regularly applied in four or more dressings annually, often broadcast or preferably via the irrigation system. P and K are applied only when leaf analysis indicates low levels.

Zn deficiency is second to N in its frequency of occurrence. It can be corrected by applying a Zn-containing fertilizer to certain types of soil or as a foliar spray. Foliar application is most effective when spring cycle leaves are expanded. It may be induced or aggravated by application of P-fertilizers or poultry manure.

Fe may be deficient when the crop is grown on calcareous soils or on soils with low oxygen content. The most effective application is chelated iron solution injected into the root zone.

Preferred nutrient forms:

N: as ammonium sulphate, or ammonium nitrate for alkaline soils; urea; calcium nitrate or calcium ammonium nitrate for acidic soils.

P: as superphosphate, before planting; or phosphoric acid, via the irrigation system.

K: as potassium chloride, when Cl in irrigation water is not high; or Potassium sulphate, in high Cl conditions especially when the plants are grafted on the chlorine-sensitive Mexican rootstocks.

Zn: as zinc sulphate, oxide or chelate.

Fe: as chelated compounds.

Present fertilizer practices

N application is based on foliar analysis and the cultivar:

Cultivar and race	**N in leaves (%of dry matter)**	**Recommended N application (kg/ha)**
Fuerte, Ettinger, Zutano	< 1.6	200
Other mostly Mexican	1.6-2.0 > 2.0	100 -
Hass, Nabal	< 1.8	250
Other mostly Guatemalan	1.8-2.2 > 2.2	150 -

The above amounts should be reduced on certain soils and when N is present in the irrigation water; they should be increased when trees have little new growth and pale leaf color, or heavy fruit set (over 10 t/ha for Fuerte or 15 t/ha for Hass).

USA, California:	375 kg/ha P_2O_5 every 3-5 years (in shallow hillside soils) 7.5-20 l/ha/year Zn (as ZnO or $ZnSO_4$)
USA, Florida:	200-250 kg/ha/year P_2O_5 240-380 kg/ha/year K_2O
Israel:	500 kg/ha K_2O
South Africa:	270 kg/ha/year P_2O_5 190 kg/ha/year K_2O Sometimes boron is also applied.

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Author: E. Lahav, Agricultural Research Organization The Volcani Center, Bet Dagan, Israel

Banana *(Musa spp)*

French: Banane; Spanish: Plátano, Banano; Italian: Banana; German: Banane

Crop data

Permanent crop with a succession of herbaceous generations by vegetative sprouting. Mostly triploid.

Harvested part: fruit (consumed either raw or cooked according to the cultivar; plantains form a specific group within the cooking cultivars). Leaf blades are frequently used as wrapping material, especially in Asia.

Plantations are renewed every 3-5, 10 or 30 years according to region and cropping system.

Planting material: corms, corm pieces or suckers (tissue-cultured plantlets are becoming increasingly used for intensive cropping).

Plant density:
- for the Cavendish group of sweet bananas (which account for nearly all international trade), 1 400 - 3 500/ha, according to sunshine, cultivar, cropping technique and marketing constraints such as the premium paid for larger-sized individual fruits. With intensive mechanization and/or very high plant densities, in widely spaced single or double rows closely planted within the row; otherwise evenly spaced square or rectangular planting.

- for all other types, 900 - 1 600/ha (exceptionally up to 2 500/ha).

(All figures for single-follower de-suckering technique).

Flowering (plant crop):
- Cavendish group, 5-10 months after planting, depending on planting material, climate, irrigation, cultivar and fertilizer use;
- other types, a shorter interval for a few small-sized diploids and longer for plantains etc;
- (ratoon crop): interval after harvest of previous crop is less than for plant crop but varies with de-suckering technique.

Time from flowering to harvest: 80-180 days for both plant and ratoon crops, depending mainly on climate ("heat sum").

Thus, for the Cavendish group the interval between successive harvests is 7-15 months, depending on climate, cultivar and cropping techniques.

Bananas are grown on many soil types; physical qualities are more important than chemical composition, because the roots are fragile, with a low penetrating power and a great need of oxygen. Preferred pH about 6.0, but successful crops may be obtained down to pH 4.0 without liming, where exchangeable Al is low (peat soils), - or up to pH 8.5 where potential metal deficiencies are well controlled or non-existent. Water requirements are high (150 mm/month) and water absorption capacity is low, so irrigation, of whatever kind (overhead, microjet, drip, etc), should maintain soil moisture within 60-100 % of the range between permanent wilting point and field capacity.

Nutrient demand/uptake/removal

Nutrient removal (harvested fruit, Cavendish group) - Macronutrients

kg/t whole bunches					
N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
1.7 ± 0.4	0.45 ± 0.15	6.0 ± 1.0	0.40 ± 0.15	0.30 ± 0.12	0.20 ± 0.05

Figures for other varieties are less well established but differ little from the above.

Nutrient uptake - Macronutrients									
Cultivar	**Source**	**Bunch yield t/ha/cycle**	**Sampling**	**kg/ha**					
				N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
Pisang Assam	Joseph, 1971	16	incomplete m.p.(-pulp)	40	18	343	80	45	-
Gros Michel	Martin-Prevel et al., 1968	26	whole matts	250	91	1 350	93	308	-
Dwarf Cavendish	Martin-Prevel et al., 1962	42	mother plants only	225	55	1 004	27-	122	-
Plantains (3cvs) min.	Marchal et al, 1979	32	whole matts	180	49	1 145	58	133	14
max.		48		226	69	1 625	66	196	20
Popoulou	Marchal et al, 1970	44	whole matts	370	108	2 440	114	252	31
Poyo, grand Nain	Martin-Prevel et al., 1965-66 Twyford et al., 1973-76	35-57	m.p., aver. Values	250	57	964	100	210	15
Robusta	Twyford et al, 1973-76	50	whole matts, extrapolated values	450	309+	2 109+	390+	420+	144
Poyo	Martin-Prevel et al., 1968	66	whole matts	450	70	1 500	80-	200	-
Grand Nain	Marchal et al, 1979	69	whole matts	293	69	1 325	108	224	29
Americani	Marchal et al, 1979	75	whole matts	294	92	1 565	124	266	36
Nanicao	Gallo et al, 1972	77	incompl. m.p (- corm, roots)	264	73	1 265	104	224	11

Poyo = Robusta, Valery (giant Cavendish) Grand Nain, Americani, Nanicao (semi-giant Cavendish)
Full références in : Martin-Plevel, 1980, 1987
m .p. := mother plants ; whole plants = mother plants + all existing followers
+ probably overestimated ; - Mg-deficient sites ; Also Cl = 300 kg/ha (Gallo et al., 1972)

Nutrient uptake - Micronutrients									
Cultivar	**Source**	**Bunch yield t/ha/cycle**	**Sampling**	**kg/ha**					
				Fe	**Mn**	**Zn**	**Cu**	**B**	**Mo**
Robusta	Twyford et	-	mother	2.7	4.3	0.47	0.18	0.84	-

	al, 1968		plants only						
Poyo	Marchal et al, 1971-72	32-57	mother plants only	4-15	4-9	(0.3)	(0.1)	-	-
Robusta	Twyford et al, 1973-76	50	whole matts, extrapolated values	4.5	26+	0.61	0.22	0.62	-
Nanicao	Gallo et al, 1972	77	incomplete mother plants (-corm, roots)	3.06	6.85	0.36	0.12	0.37	0.0013
Poyo = Robusta ; Nanicao giant (semi-Cavendish) Full references in : Martin-Prevel, 1980-1987 + probably overestimated ; Also Na = 4.2 kg/ha ; Al = 2.8 kg/ha (Gallo et al., 1972)									

Total uptake in normal conditions is, on the basis of available information:

Total nutrient uptake (normal growth conditions) - Macronutrients						
Variety	**kg/t whole bunch**					
	N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
Cavendish group	4-7	0.9-1.6	18-30	1.2-3.6	3-7.5	0.4-0.8
Other varieties	up to 10	up to 3.5	up to 60	1.2-3.6	up to 12	0.4-0.8

Variations are due to the harvest index (much higher for the Cavendish group than for many other varieties), nutrient status (except nitrogen, the nutrient content of fruit is less affected than that of other plant parts by deficiency or excess) and the de-suckering technique. The figures given are for total net uptake at the harvesting stage: they do not include amounts lost in pruning/de-suckering, or wilting of older leaves, or through leaching by rainfall. Note that roots may account for 5-10 % of the total uptake, and corm 10-12 % (5 % in the case of CaO).

K absorption is largest during bunch growth (hidden and visible); N and P continous uptake from planting (or sucker start) to bunch emergence.

Plant analysis data

The International Reference Sampling (IRS) method, which was agreed at a conference held in the Canary Islands in 1975, requires leaf samples to comprise inner (i.e. closest to the midrib) exact halves of strips taken from both sides of the lamina and at its exact mid-length, from the third youngest fully expanded leaf either at full bunch emission (all female hands and no more than three male hands being visible) or at approximate floral initiation. Because the choice of sample is highly critical, published data and the basis of other sampling methods or where the sampling method is not clearly stated should be treated with great caution and tested against IRS data by the user to obtain conversion tables.

Unfortunately, not enough experience has been accumulated with the IRS method to be able to assess definitely all its standards. The table below includes, in italics, some non-IRS data where sufficient information was available for conversion to be practicable, and, in parenthesis, values expressed as mere orders of magnitude where such conversion was not possible.

Leaf analysis standards (International Reference Sample - IRS) - Macronutrients								
Plant growth stage	**Nutritional status**	**% of dry matter**						
		N	**P**	**K***	**Mg***	**Ca***	**S**	**Cl**

Around flower initiation	Deficient (symptoms)	*<2.3*	*0.12*	*1.9*	*0.15-0.24*	*0.4(°)*	*0.21*	*-*
	Low	*2.3-3.3*	*0.13*	*<4.5*	*0.25-0.29*	*-*	*0.21-0.25*	*-*
	Optimum	*3.3-3.7*	*>0.14*	*4.5-5.0*	*0.30-0.40*	*0.8-1.3*	*>0.25*	*(1.0)*
	High (luxury)	*>3.7*	*-*	*>5.0*	*>0.40*	*>1.3*	*-*	*(2.0)*
	Excess (toxicity)	*-*	*0.3*	*5.5-6.5*	*-*	*-*	*-*	*(3.5)*
Just fully expanded bunch	Deficient (symptoms)	1.6-2.1	-	1.3-2.6	0.07-0.25	0.15	-	-
	Low	2.0-2.5	0.12-0.16	2.7-3.2	-	-	-	-
	Optimum	2.7-3.6	0.16-0.27	3.2-5.4	0.27-0.60	0.66-1.20	0.16-0.30	0.9-1.8
	High (luxury)	-	-	-	-	-	-	>2.0
	Excess (toxicity)	-	-	-	-	-	-	3.5

* K:Mg:Ca and K:N equilibria also to be considered.
Optimum for K:Mg:Ca in gramm-equivalents = 52-60:16-25:2-29.
(°) Up to 0.7 for first cycle issured from big corms

Italics: recalculated from non-IRS results (Martin-Prevel, 1987, and internal reports by IRFA/CIRAD)
Normal: figures used by J. Marchal for current practice by IRFA/CIRAD
Between brackets:order of magnitude (acc. Lahav and Turner, 1983, and Soto, 1985)

Leaf analysis standards (International Reference Sample - IRS) - Micronutrients							
Plant growth stage	**Nutritional status**	**ppm dry matter**					
		Fe	**Mn**	**Zn**	**Cu**	**B**	**Na**
Around flower initiation	Deficient (symptoms)	*77*	*25-100*	*14-(°)37*	*-*	*-*	*-*
	Low	*-*	*110-150*	*-*	*-*	*-*	*<100*
	Optimum	*≥100*	*160-2 500*	*>20*	*(9)*	*(11)*	*-*
	High (luxury)	*-*	*>2 500*	*-*	*-*	*-*	*>100*
	Excess (toxicity)	*300*	*≥4 800*	*-*	*-*	*-*	*>300*
Just fully expanded bunch	Deficient (symptoms)	-	40-150	6-17	<5?	<10?	-
	Low	-	-	-	-	-	< 60
	Optimum	80-360	200-1 800	20-50	6-30	10-25	-
	High (luxury)	-	2 000-3 000	-	-	-	>150
	Excess (toxicity)	-	>3 000	-	-	30-100	up to 3 500

P/Zn ratio (high in case of Zn deficiency) to be preferably considered
Italics: recalculated from non-IRS results (Martin-Prevel, 1987, and internal reports by IRFA/CIRAD)
Normal: figures used by J. Marchal for current practice by IRFA/CIRAD
Between brackets: order of magnitude (acc. Lahav and Turner, 1983, and Soto, 1985)

Fertilizer recommendations

Whilst N and K should be supplied according to the very high biomass requirements of the crop, attention must be given to maintaining an appropriate soil cationic balance. On most soil types this means (with pH around 6.0) about 80 % CEC saturation by K, Mg and Ca in the approximate proportions 1:3:6. Dressings of dolomite and/or limestone, for incorporation into the soil, should be calculated so as to achieve and maintain these proportions in the top 20 cm of soil. On highly unsaturated soils with a high cation exchange capacity, this may seldom be possible, in which case attention should be given principally to the K:Mg ratio, which should never exceed 1:2 in ferrallitic or sandy soils or 1:1 in volcanic or organic soils. The level of exchangeable K should preferably be raised to about 10 % of the total exchangeable cations by a basal application in the first year and subsequently maintained by dressings calculated to compensate for removals and leaching losses.

The amounts of N and K2O to be given to a plant crop should be calculated from the expected yield on a particular field and the total uptake per metric ton of whole bunches as quoted earlier. N application should be split into a number of dressings so as to provide a continuous supply from planting right through to harvest, with smaller and more frequent dressings where the risk of loss by leaching is higher (Godefroy et al, 1989), ranging from intervals of 1-3 months in relatively dry climates down to every 2-4 weeks in the humid tropics with suitable modification in seasons of high growth potential or in seasons affected by cold or drought. Subject to the demands of maintaining a correct cationic balance, the K application is generally divided in a rather similar manner to that of N except that dressings should be smaller at the beginning of the growth period and increased during the months immediately before and after flowering.

Similar calculations may be made for ratoon crops, making due allowance for the large losses resulting from chopping down the mother plants, and the more rapid growth during a shorter time period. In practice, the same average monthly rates as for the plant crop are generally adopted.

Preferred nutrient forms

Given adequate S, the cheapest forms of N and K fertilizers available may be used. Potassium nitrate, although acceptable in theory, is scarcely ever used except in irrigation water, due to its high cost and liability to loss by leaching. Where there is a need for added S, this can be given either in the form of a sulphate-based N fertilizer or, preferably on acid soils, potassium sulphate, so that S would account for 3-5 % of the total input unless abundant organic manure is used. Double K-Mg sulphates (Patentkali, etc) are useful where Mg deficiency is incipient. The preferred form of P depends on soil pH and P-fixation capacity. Where appropriate, rock phosphates can contribute, with lime and/or dolomite, towards CEC saturation; and low-P compound fertilizers are convenient except on soils where P-fixation capacity is high.

Present fertilizer practices

Departures from recommended usage often result in low yields, through under-use, or in poor quality and uneconomic production due to imbalanced or over-use or incorrect timing.

Organic manures are excellent for improving soil conditions and provide variable amounts of macronutrients, which must be taken into account if imbalances are to be avoided; they may also supply all the micronutrients needed. Cattle or chicken manures, at rates of 35-120 t/ha, are widely used in some countries, and in others, residues such as coffee pulp, cacao shells and composted town refuse, while copious mulching with grasses or branches has been common practice for decades in many regions. Where high yields are obtained by using only mineral fertilizers, the soil organic matter content can be improved by returning around 200

t/ha/year of plant residues, but care must be taken to ensure that adequate amounts of all macro- and micronutrients are provided.

N fertilizer is used almost everywhere unless abundant organic manure is applied; but even with abundant manure application K fertilizer must also be given except on volcanic soils containing very high reserves. Mg is considered the third most important nutrient, whether incorporated in a soil amendment or broadcast as a straight fertilizer (Epsom salts or kieserite) or in mixed or compound fertilizers.

Most fertilizers are hand-spread except when basal dressings are incorporated during land preparation. However, there is considerable controversy over the best method of placement. With good control of nematodes and soil aeration, an even broadcast would appear more logical, but applications in practice are often concentrated within a circle of 1.0-1.5 m diameter around the pseudo-stem, or (after flowering) in a crescent shape around the daughter plants. In mechanized fields the fertilizers are often spread along the rows.

Foliar feeding is efficient with the right nutrients and wetting agents. It is preferably used successfully for micronutrients, especially when they can be mixed with the aerial oil-fungicide sprays regularly applied against Sigatoka disease in tropical climates. Rates of 5-10 kg/ha Zn, B or Mn (in descending order of importance) applied in this way once to three times a year are sufficient, instead of soil applications of 20 kg/ha or more which are often ineffective because of blocking antagonisms.

Some individual growers apply amounts up to 1 200 kg/ha N, 800 kg/ha P2O5, 1 800 kg/ha K2O yearly, but the most common practices for Cavendish cultivars in various countries are summarised in the following table. The higher figures correspond broadly with the highest yields. The less productive stands, whether of Cavendish or other cultivars, receive less fertilizer but are less profitable.

Present fertilizer practices for "Cavendish" cultivars with good average yields unless differently stated				
Region/Country	**kg/ha/year**			**Other elements - Remarks (kg respect. t/ha)**
	N	**P2O5**	**K2O**	
Africa				
Cameroon	140-400	0	0-(800)	no K necessary on young volcanic soils, fertilizers must contain S
Cote d'Ivoire - peat soils - other soils	 100-330 300-500	 30-100 30-100	 700-1600 600-1200	 5-10 Cu, 500-2 000 dolomite 500- 2000 dolomite
South Africa	140-500	0-100	750-1600	organic manure 0-120 tons
Canary Is. - drip irrigat. - surf. irrigat.	 500-600 600-800	 200-300 300-450	 700-1000 900-1500	 organic or plastic mulch organic mulch
Morocco, greenhouses	440-750	0-285	800-1600	10-25 Mn, 0-200 MgO, 0-150 S manure
Egypt	380-2 500	55-300	0-950	org. manure - low yields without K
Middle East, Asia, Oceania				

Israel - coastal plain - Jordan valley	 400 400	 200 90	 1440 0	 organic manure (*) org. manure - no K necessary (*)
India	300-600	320-345	340-720	organic manure (*)
Taiwan	400	115	900	(*)
Australia - N Territories - Queensland - N.S. Wales	 110 280-370 180	 230 160-460 90-230	 760 480-1560 360- 720	 (*) (*) (*)
Latin America and Caribbean				
Brazil - Sao Paulo - other states	 250-500 0- 80	 125-240 0- 50	 500-950 0-120	 not "Cavendish", yield 5-20 t/ha
Costa Rica	300-450	0-160	600-750	50-200 MgO, 500-600 CaO
Honduras	290	0	0	no K necessary on most soils (*)
Martinique, Guadeloupe - andosoils andrecent pumices - other soils	 300-700 250-600	 0-180 0-150	 800-1350 600-1600	 80-270 MgO, 150-400 CaO CaO according to soil analysis
Jamaica	225	150	560	(*)
Sources: (*) Lahav and Turner, 1983 - Others: personal records and IRFA/CIRAD reports				

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Cashew *(Anacardium occidentale L.)*

French: Anacardier, acajou; Spanish: Anacardo, maranon; Italian: Anacardo; German: Kaschunuss

Crop data

Perennial.
Harvested parts: fruit (apple) and cashewnut. Processed produce: cashew kernels. Seedling progenies flower in 4 to 5 years but clonal progenies (softwood grafts) flower in third year.

Planting in pits of 60 cm x 60 cm x 60 cm during June-July. Seeds are either sown in situ or seedlings raised in polybags are transplanted at the onset of the monsoon. Vegetatively propagated plants (softwood grafts) are largely used.

Initial close planting at 3, 4, 5 m apart, results in early shading of the soil surface, suppresses weeds, conserves soil and moisture and provides high initial yield per unit area, but requires thinning to the final spacings of 8, 9, 10 m at 5 or 6 years when canopies and root systems are intermingled with those of neighbouring trees.

Often grown on marginal soils and also on waste land mostly unsuitable for other economic crops. Thrives well in sandy sea coast, fairly steep laterite slopes or rolling land with shallow top soils in India; alluvial soils in Sri Lanka; ferruginous soils in East and West Africa, Brazil and Madagascar; volcanic soils in the Philippines, Indonesia and Fiji Islands. The most fertile soils for cashew are virgin forest soils.

Grows best in warm moist, tropical climate with well-defined dry season of 4-5 months during the reproductive phase, followed by a wet season of 4-5 months (1 000-2 000 mm rainfall). Requires equable environment with maximum temperature 34 °C and minimum 20 °C. Optimum sunshine 1 285 hr (9 hrs/day) in flowering/fruit set period. Grown at altitudes from sea-level to 700 m and thrives well between 27 °N and 28 °S latitudes. Usually unirrigated but responds to summer irrigation.

Nutrient removal

Nutrient removal (30 year old tree) - Macronutrients			
Plant parts	**kg/tree**		
	N	**P2O5**	**K2O**
Leaf, stem & root	1.72	0.41	0.80
Fruit (155 kg)	0.37	0.12	0.28
Nuts (24 kg)	0.76	0.23	0.18
Total	2.85	0.76	1.26
Source: Mohapatra et al., 1973			

Leaf analysis

Leaf analysis data (Kenya) - Macronutrients						
Kind of tree	**% of dry matter**					
	N	**P**	**K**	**Mg**	**Ca**	**S**
Thrifty	1.98	0.21	1.69	0.20	0.09	0.15
Unthrifty	1.52	0.10	0.97	0.17	0.16	0.14
Source: Calton et al., 1961						

Leaf analysis data (Kenya) - Micronutrients	
Kind of tree	**ppm dry matter**

	Fe	Mn	Cu	B
Thrifty	45	95	16	9
Unthrifty	95	260	66	10
Source: Calton et al., 1961				

Leaf analysis data (India) - Macronutrients			
Season	**% of dry matter**		
	N	**P**	**K**
Before fruiting	1.41	0.09	0.63
After fruiting*	1.49	0.12	0.79
* fully matured leaves of previous season	1.19	0.06	0.46
* freshly matured leaves of current season	1.79	0.18	1.13
Source: Harishukumar et al., 1982			

Application of both lime and phosphate increased the NPK content of young leaves. In Malaysia, liming increased the pH of sandy soils and improved both P and K contents of young leaves.

Fertilizer recommendations

Fertilizer dose per year (India)			
Years after planting	**g/tree**		
	N	**P2O5**	**K2O**
1	170	40	40
2	350	80	80
from 3-4 onwards	500	125	125
15-20	750	250	250

The above annual rates are applied in two dressings, the first at the onset of the pre-monsoon period and the second during the post-monsoon period when the soil moisture condition is at its optimum; if only one application is given, it should be in the post-monsoon period when enough moisture is available. Organic manure must be applied at planting; a rate of 6 t fym per hectare provides for the better growth of young plants.

Foliar application

Foliar sprays of N as urea combined with an insecticide at the emergence of the flush and again at panicle initiation will ensure better fruit set and control the major seasonal pests. High-volume ground spray (2.0 %) or low-volume aerial spray have both been found to be economic.

Fertilizer and stage of growth

In India, young plants respond well to N and P.

In Madagascar too, N and P were found to be the most important nutrients during the pre-bearing stage, but at the bearing stage, K (together with N) is also important.

Flowering response of young cashew to N and P fertilizer			
	Percentage flowering		
	At 2 1/2 yrs	**At 3 1/2 yrs**	**At 4 1/2 yrs**
With added N and P	35 %	87 %	100 %
Without added N and P	0 %	8 %	31 %

Source: Lefebvre, 1970

Preferred nutrient forms

Urea is the most commonly used nitrogenous fertilizer.

Of phosphatic fertilizers for use on acid soils in India, the slow-release and more efficient ground Mussoorie (rock) phosphate is popular.

Fertilizer placement

On sandy soils, on laterite soils and on sloping land with heavy rainfall, application in a circular trench 25 cm wide and 15 cm deep, at a distance of 1.5 m from the trunk, is recommended. On red loamy soils with low rainfall, the fertilizer should be incorporated into the soil in a band 1.5 m wide, at a distance of 1.5 m (inner edge) to 3 m (outer edge) round each tree.

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Citrus *(Citrus spp., etc.)*

French: Agrumes; Spanish: Citricos; Italian: Agrumi; German: Zitrusfruechte

Citrus fruits belong to six genera (Fortunella, Eremocitrus, Clymendia, Poncirus, Microcitrus and Citrus) native to the tropical and subtropical regions of Asia and the Malay Archipelago, but the major commercial fruits are species of Citrus. The principal scions and rootstocks are:

Scions:
Orange (C. sinensis Osbeck)
Mandarin (C. reticulata Blanco)
Lemon (C. limon [L.] Burm.)
Lime (C. aurantifolia [Christ.] Swing.
Grapefruit (C. paradisi Macf.)
Pomelo (C. grandis [L.] Osbeck)

Rootstocks:
Rangpur lime (C. limonia Osbeck)
Rough lemon (C. jambhiri Lush.)
Sour orange (C. aurantium L.)
Cleopatra mandarin (C.reshni Hort.)
Trifoliata (P. trifoliata [L.] Raf.)

Crop data

Perennial, grown in a wide range of soil types and climatic conditions.

Soil: sand to loam to clay, acid to alkaline.

Climate: arid to humid where minimum temperature permits.

Plant density: 175-500 trees/ha, with newer and younger plantings having the higher density.

Useful life: 15-40 years depending on soil type. In general, trees on well drained sandy soils have a longer useful life than those on loam or clays where drainage is not as good.

Producing age:

- for intensive culture (with fertilizer use, and close monitoring and control of insect pests, diseases and weeds, and of irrigation and tree size): beginning second year, peak 10-30 years, yield 30-60 t/ha.

- for extensive culture (with fertilizer use, but only moderate monitoring and control of insect pests, diseases and weeds, and mostly unirrigated): beginning fourth year, peak 8-15 years, yield 15-25 t/ha.

The crop is grown in a belt between 40 °N and 40 °S, except at high elevation. Minimum temperature is the limiting factor, though the killing effect varies with variety, rootstock, dormancy of the trees, the absolute minimum temperature and its duration.

24 countries account for about 85 % of production. Brazil is the largest producer of oranges, and Japan of mandarins. Spain, Italy and Mexico lead in the production of lemon and limes, while USA produces the most grapefruit.

World production is expected to increase during the 1990s as new plantings in North and South America, the Mediterranean region and Asia come into full production. It is expected that the

volume of processed fruit will continue to increase while consumption of fresh fruit should remain relatively constant.

Nutrient demand/uptake/removal

Removal of mineral elements in the harvested fruit is one of the major considerations in formulating fertilizer recommendations. The table below shows the quantities of nutrient elements contained in one metric ton of fresh fruit. The large amounts of K reflect the high K content of citrus juice.

Nutrient removal - Macronutrients						
Variety	**grams per ton of fresh fruit**					
	N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
Orange	1 773	506	3 194	367	1 009	142
Mandarin	1 532	376	2 465	184	706	111
Lemon and lime	1 638	366	2 086	209	658	74
Grapefruit	1 058	298	2 422	183	573	90
Sources: Koo, 1958; Chapman, 1968; Malovolta, 1989.						

Nutrient removal - Micronutrients					
Variety	**grams per ton of fresh fruit**				
	Fe	**Mn**	**Zn**	**Cu**	**B**
Orange	3.0	0.8	1.4	0.6	2.8
Mandarin	2.6	0.4	0.8	0.6	1.3
Lemon and lime	2.1	0.4	0.7	0.3	0.5
Grapefruit	3.0	0.4	0.7	0.5	1.6
Sources: Koo, 1958; Chapman, 1968; Malovolta, 1989					

Soil analysis

This is useful for measuring pH, available P and certain exchangeable cations, notably Ca and Mg. However, because citrus trees are planted on a wide range of soil types, it would be difficult to establish standards for all soils. They are therefore usually developed for certain soil types in a given region.

It is usually more difficult to assess the N and K status in the soil because both these elements are subject to leaching, especially in humid regions.

Plant analysis data

Sufficient progress has been made to use leaf analysis as one of the guides in planning citrus fertilizer programmes. The table below relates to spring flush leaves 4-6 months old from non-fruiting terminals. Leaves from fruiting terminals (used in S. Africa and some S. American countries) have lower N, P and K and higher Ca and Mg contents than leaves of the same age from non-fruiting terminals, a fact which should be borne in mind when interpreting leaf analysis data.

Plant analysis data - Macronutrients **4 - 6 months old spring cycle leaves from nonfruiting terminals**	
Range	**% of dry matter**

	N	P	K	Mg	Ca	S
Deficient	<2.20	<0.09	<0.70	<0.20	<1.50	<0.14
Low	2.20-2.40	0.09-0.11	0.70-1.10	0.20-0.29	1.50-2.90	0.14-0.19
Optimum	2.50-2.70	0.12-0.16	1.20-1.70	0.30-0.49	3.00-4.90	0.20-0.39
High	2.80-3.00	0.17-0.29	1.80-2.30	0.50-0.70	5.00-7.00	0.40-0.60
Excess	>3.00	>0.30	>2.40	>0.80	>7.00	>0.60
Sources: Smith, 1966; Koo, 1984; Malavolta, 1989.						

Plant analysis data - Micronutrients 4 - 6 months old spring cycle leaves from nonfruiting terminals						
Range	**ppm dry matter**					
	Fe	**Mn**	**Zn**	**Cu**	**B**	**Mo**
Deficient	<35	<17	<17	<3	<20	<0.05
Low	36-59	18-24	18-24	3-4	21-35	0.06-0.09
Optimum	60-120	25-100	25-100	5-16	36-100	0.10-1.0
High	121-200	101-300	101-300	17-20	101-200	2.0-5.0
Excess	>200	>500	>500	>20	>250	>5.0
Sources: Smith, 1966; Koo, 1984; Malavolta, 1989.						

Perhaps the most distinctive feature of citrus nutrition is the large number of nutrient deficiencies that have been encountered in intensive cultivation. Foliar symptoms of deficiencies of N, P, K, Mg, Ca, Mn, Zn, Cu, Fe, B and Mo have been recognized in the field as well as in artificial culture. In order to obtain maximum yield of high quality fruit, it is essential to evaluate the crop's nutritional requirements locally and to provide a balanced fertilizer programme.

Fertilizer recommendations

Pre-plant

Available organic manures should be applied and well incorporated into the soil before planting. This practice is widely adopted in many developing countries. Typically, a planting hole about 1 metre in diameter and 1 metre deep is dug; compost, animal manure and green manure are mixed with the soil that has been dug out, and the mixture is then replaced in the hole before the tree is planted. Tree planting holes are not used in developed countries, because of scarcity of organic maure. On acid soils, limestone is usually added to the mixture for pH adjustment or, where organic manure is scarce, soil preparation consists simply of liming. On some alkaline soils in arid regions, pre-plant irrigation is often used to leach excess salts from the surface soil.

Young trees

N, P and K are the principal components of the recommended fertilizer treatment. The quantities applied vary with soils, fertility, local conditions and intensity of cultivation. Differences are illustrated in the table which shows recommendations from four citrus-producing countries in S. America, N. America, Africa and Asia respectively.

Fertilizer programmes for young trees							
Country	**Tree age in years**	**grams per tree**					
		N	**P2O5**	**K2O**	**MgO**	**Bone* meal**	**Wood** ash**
Brazil	1	100	0	20			

	2	160	160	80			
	3	240	240	160			
	4	360	320	320			
	5	480	400	400			
	6	600	480	480			
USA, Florida	1	200	200	200	65		
	2	330	330	330	110		
	3	440	440	440	150		
	4	500	500	500	165		
	5	580	580	580	190		
	6	640	640	640	220		
Côte d'Ivoire	1	65	0	20	200		
	2	85	0	30	400		
	3	115	55	120	400		
	4	280	80	170	600		
	5	380	110	230	800		
	6	540	160	350	600		
India	1	50	100			500	1 500
	2	100	200			1 000	3 000
	3	150	300			1 500	4 500
	4	200	400			2 000	6 000
	5	250	500			2 500	7 500
	6	300	600			3 000	9 000
* Bone meal: 3.0 % N; 25 % P2O5; 0.2 % K2O; 20 % Ca. ** Wood ash: 0.0 % N; 1.8 % P2O5; 5.5 % K2O; 2.2 % Mg; 23 % Ca.							

As can be seen from the table, the quantity of fertilizer applied increases with the age of the tree.

The area to which the fertilizer is applied also increases with the age of the tree. Generally where tree-planting holes are used, fertilizer is applied to the surface area of the hole because not many roots will grow beyond it. Where tree-planting holes are not used, the tree canopy serves as a guide; fertilizer is applied from about 30 cm beyond the drip line of the tree canopy inwards towards the centre of the tree.

The frequency of application of fertilizer decreases with the age of the tree, ranging from 5-7 applications in the first year to 3-4 applictions in the fifth year; but, where "fertigation" is practised, trees usually receive fertilizer 25-30 times a year during the first 5 years.

Country notes

Brazil
The crop is planted on loam soils which are relatively infertile but have enough organic matter to give a cation exchange capacity of about 7. Young trees appear healthy and well fertilized. Most plantings are unirrigated. Trees begin producing in the third year.

Florida, USA
The crop is grown on sandy soil with very low fertility and organic matter, and a cation exchange capacity of 2-3. Despite being in the humid region with adequate rainfall, irrigation is practised because of the sandy soils' low water-holding capacity. Growers follow an intensive programme;

large amounts of fertilizer are applied, mostly through "fertigation", and the young trees respond well; it is not uncommon for them to produce 20 kg of fruit in the second year.

Côte d'Ivoire
Cultivation is similar to that of many developing countries where much of the organic manure and mineral fertilizer used is incorporated in the planting hole. Only small quantities of mineral NPK fertilizer are applied during the first 3 years but appliction increases thereafter when the trees begin to produce fruit. Mg is a regular component of the young tree fertilizer in both Florida and the Ivory Coast but not in other regions.

India
Here, and in other Asian citrus-producing countries, K is not a regular component of the young tree fertilizer but comes from organic sources. It is questionable, however, whether the K requirement of citrus can be met from organic sources alone; symptoms of inadequate K are frequently observed in both India and China.

Mature trees

Fruit production and quality are the principal considerations. The macronutrients influence fruit quality both externally and internally. Nutrient removal in the expected yield, leaf analysis and soil analysis can all be of value in planning the fertilizer programme for mature trees.

N and K are the key elements; P is less critical because of the much smaller amount removed in the harvested fruit. Fertilizer for mature trees usually contains NPK in the proportions N:P2O5:K2O = about 1.0:0.2:1.0. Mg is included where it is needed. Around 3-6 kg N are required to produce one ton of fruit. N is often used as the basis from which to calculate the estimated requirements of other nutrients.

Examples of recommendations for mature trees				
Country	**kg/ha**			
	N	**P2O5**	**K2O**	**MgO**
Japan	150-350	115-205	115-235	-
Brazil	150-240	40- 80	90-320	-
Florida, USA	180-320	30- 60	180-360	75-210

Fertilizer for mature trees is usually given in two applications (fall-winter and spring) or in three applications (spring, summer, fall) yearly.

Micronutrients

In general, these do not affect fruit production or quality unless they are in severe deficiency, for which leaf analysis and visual observations are the best guides. Any deficiencies, or excess, should be corrected for the health of the tree.

Guidelines for the use of micronutrients, based on Florida experience, are summarized below. Some adjustment may be needed for other parts of the world, though it should be borne in mind that in Florida attention to micronutrients is a regular part of the citrus fertilizer programme and much research has entered into the development of these guidelines.

Guidelines for micronutrient application methods, timing and rate

Method and Rates	Mn	Zn	Cu	Fe	B	Mo
Foliar application (when trees have the most fully expanded new leaves)	Yes	Yes	Yes	No	Yes	Yes
Rate in grammes/litre	0.9	1.2	0.9	-	0.06	*
Soil application (any time)	Yes°	No	Yes	Yes	Yes	No
Rate in kg/ha	10.0	-	5.4	#	1.0	-

° Not recommended for alkaline soil.
* 34-36 g sodium molybdate per litre
# On acid soil, FeEDTA at 20 g Fe/tree; on alkaline soil, FeEDTAOH at 50 g Fe/tree

Further reading

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Deciduous Fruit

Apple *(Malus pumila Mill.)*
French: Pommier; Spanish: Manzano; Italian: Melo; German: Apfel

Pear *(Pyrus communis L.)*
French: Poirier; Spanish: Peral; Italian: Pero; German: Birne

Peach *(Prunus persica L.)*
French: Pecher; Spanish: Melocotonero; Italian: Pesco; German: Pfirsich

Apricot *(Prunus armeniaca L.)*
French: Apricotier; Spanish: Albaricoquero; Italian: Albicocco; German: Aprikose

Plum *(Prunus domestica L.; Prunus salicina Lindl.)*
French: Prunier; Spanish: Ciruelo; Italian: Susino; German: Pflaume

Sweet Cherry *(Prunus avium L.)*
French: Cerisier douce; Spanish: Cerezo des aves; Italian: Ciliego dolce; German: Suess-kirsche

Sour Cherry *(Prunus cerasus L.)*
French: Cerisier aigre; Spanish: Guindo griego; Italian: Ciliego amaro; German: Sauer-kirsche

Crop data

Perennial with winter rest. Harvested part: fruits.

Flowering late winter to middle spring; harvested 60 to 200 days after flowering, depending on species and cultivar.

Plant density: 250-700 (traditional) to 1 200-3 000 (intensive) plants/ha. Preferably grown on light to medium soils, pH 6.5-7.5.

Nutrient demand/uptake/removal

Nutrient demand/uptake/removal - Macronutrients									
Species	**Cultivar /Rootstock**	**Source**	**Yield t/ha**	**Kind of demand**	**kg/ha**				
					N	**P2O5**	**K2O**	**MgO**	**CaO**
Apple	Red	Greenham,	44.8	a	20.8	14.4	67.9	3.7	6.2
	Delicous	1980		b	18.4	9.6	17.2	3.8	64.1
				c	39.2	24.0	85.1	7.5	70.3
				d	47.6	7.6	62.9	30.1	120.1
				e	11.9	3.9	17.8	1.8	5.2
				f	11.8	5.3	4.3	2.8	39.2
				g	71.3	16.8	85.0	34.7	164.5
				h	110.5	40.8	170.1	42.2	234.8
	Cox's/M 2	Greenham,	23.3	a	16.5	7.3	40.4	2.2	1.7
		1980		b	9.5	3.7	8.8	2.3	15.0
				c	26.0	11.0	49.2	4.5	16.7
	Cox's/M 7	Greenham,	28.8	a	18.0	4.6	48.0	3.3	2.8
		1980		f	11.0	4.6	9.6	3.3	15.4
				b	4.0	<2.3	3.6	<1.7	16.8
				c	33.0	11.5	61.2	6.6	35.0
	-	Janik, 1986	34	a	38.0	7.8	43.1	9.3	4.8
Pear		Boulay,	25	a	14.0	3.9	39.8	5.0	8.0
		Decroux,		b	7.0	2.1	3.5	1.0	62.0
		Diris,		c	21.0	6.0	43.3	6.0	70.0
		1984		d	35.0	6.0	25.0	12.4	100.0
				f	10.0	2.1	4.4	1.0	34.0
				g	45.0	8.1	29.4	13.4	134.0
				h	66.0	14.1	72.7	19.4	204.0
Peach	-	Janik, 1986	28	a	33.6	7.8	98.2	9.3	4.8

a = removed in fruit
b = incorporated in framework
c = net uptake (a+b)
d = leaf-fall
e = dropping flowers and fruitlets
f = prunings
g = return to soil (d+e+f)
h = gross uptake (c+g)

Plant analysis

Apples - Optimum* leaf analysis values - Macronutrients

Country / Region	Source	Cultivar	Sampled leaf / Sampling time	% of dry matter					
				N	P	K	Mg	Ca	S
Australia	Reuter, Robinson, 1986	all	mid-3rd / late Jan. - mid. Feb.	2.00-2.40	0.15-0.20	1.20-1.50	0.21-0.25	1.10-2.00	0.20-0.40
Brazil Rio Grande Santa Catarina	Basso, 1990	all	mid-3rd / mid. +Jan. - mid. Feb.	2.00-2.50	0.15-0.30	1.20-1.50	0.25-0.45	1.10-1.70	- -
Canada Ontario	Cline, 1990	Red & Golden Delicious	mid-3rd / July 15-30	2.20-2.70	0.15-0.30	1.40-2.20	0.25-0.40	0.80-1.50	-
Denmark	Vang-Petersen,1990	all	mid-3rd / Aug. 15-30	2.00-2.50	0.18-0.26	1.30-1.70	0.20-0.40	0.70-1.20	-
France Loir et Cher	Gautier, 1975	Golden Delicious	mid-3rd / 70-90 days after bloom	2.30-2.50	0.16-0.18	1.80-2.00	0.22-0.26	-	-
		Red Delicious		2.40-2.60	0.17-0.19	1.60-1.80	0.25-0.30	-	-
Germany	Bergmann, 1988	al	mid-3rd / July - August	2.20-2.80	0.18-0.30	1.10-1.50	0.20-0.35	1.30-2.20	-
Hungary	Szucs, 1990	all	mid-3rd / July 15-30	2.00-2.70	0.12-0.20	1.00-1.60	0.27-0.40	1.20-1.80	-
Italy Trentino	Failla, 1991	Golden Delicious	lower 3rd / late July - early August	2.00-2.60	0.16-0.24	1.30-1.90	0.24-0.36	1.40-2.00	-
		Red Delicious		2.10-2.70	0.16-0.24	1.10-1.70	0.27-0.39	1.10-1.70	-
		Canada Reinette		1.80-2.40	0.26-0.38	1.00-1.60	0.24-0.36	1.40-2.00	-
Piedmont	Failla, 1990	Golden Delicious		2.10-2.70	0.12-0.20	1.20-1.80	0.20-0.32	1.00-1.60	-
		Red Delicious		2.30-2.90	0.14-0.22	1.10-1.70	0.22-0.34	0.90-1.50	-
Venetia/ Emilia	Lalatta, 1987	Rd & Gldn Delicious		2.10-2.50	0.16-0.21	1.50-1.90	0.30-0.40	1.30-1.80	-
Japan Nagano	Katou, 1990	Ralls Janet	mid-3rd / late July-early Aug.	3.40-3.60	0.17-0.19	1.30-1.50	0.27-0.40	0.80-1.30	-
Netherlands	Kodde, 1990	Golden Delicious	mid-3rd/ mid July-mid Aug	2.21-2.40	0.16-0.25	1.36-1.65	0.18-0.25	1.81-2.20	-
		Cox's		2.41-2.60	0.16-0.25	1.16-1.45	0.18-0.25	1.81-2.20	-
South Africa	Kotze, 1990	all	mid-3rd / Jan. 31	2.10-2.80	0.13-0.19	0.80-1.60	0.30-0.50	1.20-1.60	-
UK	Sharples, 1980	Cox's	mid-3rd / August	2.40-2.80	0.20-0.25	1.30-1.60	0.25-0.30	1.00-1.60	-
USA	Childers, 1973	all	mid-3rd / mid-summer	1.90-2.25	0.20-0.30	1.20-1.95	0.24-0.40	1.50-2.00	-

Oregon Washington	Righetti, 1989 Thoenis-sen, 1989	all Red Delicious	mid-3rd / August	2.00-2.30 1.80-3.00	0.14-0.55 0.15-0.40	1.20-2.00 1.30-2.50	0.25-0.60 0.25-1.00	1.10-2.50 1.20-2.60	0.10 0.20 0.12 0.20
U.S.S.R.	Kondakov, 1990	Antonovka & other	mid-3rd/second half of summer	1.80-2.50	0.13-0.22	1.00-1.50	0.23-0.35	1.00-1.40	-
* ranges in which yield and quality are satisfactory									

Apples - Optimum* leaf analysis values - Micronutrients								
Country/Region	**Source**	**Cultivar**	**Sampled leaf/Sampling time**	**% of dry matter**				
				Fe	**Mn**	**B**	**Zn**	**Cu**
Australia	Reuter, Robinson, 1986	all	mid-3rd / late Jan. - mid. Feb.	>100	50-100	21-40	20- 50	6-20
Brazil - Rio Grande & santa Catarina	Basso, 1990	all	mid-3rd / mid. Jan. - mid. Feb.	50-250	30-130	25-50	20-100	5-30
Canada - Ontario	Cline, 1990	Red & Golden Delicious	mid. - 3rd /July 15-30	25-200	20-200	20-60	15-100	-
France Loir et Cher	Gautier, 1975	Golden delicious	mid-3rd / 70-90 days after bloom	-	-	25-35	-	-
		Red Delicious		-	-	30-40	-	-
Germany	Bergmann, 1988	all	mid-3rd / July - Agust	-	35-100	25-50	15- 50	5-12
Hungary	Szucs, 1990	all	mid-3rd / July 15-30	100-300	50-200	25-50	25- 50	5-20
Italy Trentino	Failla, 1991	Golden delicious	lower 3rd / late July - eraly August	40-150	>8	20-40	>15	>1
		Red Delicious		40-150	>8	20-40	>15	>1
		Canada reinette		40-150	>8	20-40	>15	>1
Piedmont	Failla, 1990	Golden Delicious		60-200	>30	15-25	>10	>1
Japan - Nagano	Katou, 1990	Ralls Janet	mid-3rd / late July - early Aug.	-	50-200	30-50	30-50	10-30
South Africa	Kotze, 1990	all	mid-3rd / January 31	80-150	20- 90	25-40	30-50	5-10
UK	Sharples, 1980	Cox's	mid-3rd / August	-	31-100	25-30	15-25	5-10
USA	Childers, 1973	all	mid-3rd / mid-summer	100-300	-	30-40	25-50	5-20
Oregon	Righetti, 1989	all	mid-3rd / August	50-250	25-150	30-60	15-60	5-15
Washington	Thoenisse	Red		75-150	25-100	35-75	40-80	10-50

	n, 1989	Delicious							
* ranges in which yield and quality are satisfactory									

Pear - Optimum* leaf analysis values - Macronutrients									
Country/region	**Source**	**Cultivar**	**Sampled leaf/ Sampling Time**	**% of dry matter**					
				N	**P**	**K**	**Mg**	**Ca**	**S**
Australia	Reuter, & Robinson, 1986	all	mid-3rd / late Jan. - mid Feb.	2.30-2.70	0.14-0.20	1.20-2.00	0.30-0.50	1.50-2.20	0.17-0.26
Brazil - Rio Grande & Santa Catarina	Basso, 1990	all	mid-3rd / mid Jan. - mid Feb.	2.00-250	0.15-0.30	1.20-1.50	0.25-0.45	1.10-1.70	-
Canada -Ontario	MOAAF, 1989	all	mid-3rd / July 15-30	2.00-2.60	0.15-0.30	1.30-2.00	>0.20	-	-
Denmark	Vang-Petersen, 1990	all	mid-3rd / August 15-30	2.00-2.50	0.15-0.17	1.30-1.70	0.20-0.40	1.20-1.80	-
France -Loir -et-Cher	Gautier, 1975	Passe-Crassane Doyenne du Comice Conference	mid-3rd / 70-90 days after bloom	1.80-2.00 1.80-2.00 2.00-2.40	0.15-0.17 0.16-0.20 0.16-0.20	1.60-1.80 1.60-1.80 1.50-1.80	0.30-0.35 0.28-0.33 0.25-0.30	- - -	- - -
Germany	Bergmann, 1988	all	mid-3rd / July - Augut	2.20-2.80	0.15-0.30	1.20-2.00	0.20-0.35	1.20-1.80	-
Hungary	Szucs, 1990	all	mid-3rd / July 15-30	1.80-2.60	0.17-0.23	1.20-1.60	0.33-0.47	1.20-1.60	-
Japan -Nagano	Katou, 1990	Nijusseiki	mid-3rd / late July-early Aug.	2.50	0.12-0.14	0.80-1.40	0.27-0.40	2.30-3.00	-
Netherlands	Kodde, 1990	all	mid-3rd / mid July - mid Aug.	2.20-2.40	0.16-0.25	1.16-1.45	0.18-0.25	1.81-2.20	-
South Africa	Kotze, 1990	all	mid-3rd / Januaray 31	2.00-2.80	0.10-0.18	1.20-2.00	0.25-0.60	0.80-1.60	-
USA	Childers, 1973	all	mid-3rd / mid-summer	2.00-2.80	0.10-0.20	1.00-2.00	0.30-0.50	1.50-3.50	-
* ranges in which yield and quality are satisfactory									

Pear - Optimum* leaf analysis values - Micronutrients								
Country/Region	**Source**	**Cultivar**	**Sampled leaf/Sampling Time**	**ppm dry matter**				
				Fe	**Mn**	**B**	**Zn**	**Cu**
Australia	Reuter & Robinson, 1986	all	mid-3rd / late Jan.- mid. Feb.	60-200	60-120	20-40	20-50	9-20
Brazil -Rio Grande & Santa Catarina	Basso, 1990	all	mid-3rd / mid Jan. - mid Feb.	50-250	30-130	25-50	20-100	5-30
Denmark	Vang-Petersen,	all	mid-3rd / August 15-	-	30-150	20-50	-	-

	1990		30					
France -Loir-et-Cher	Gautier, 1975	Passe-Crassane Doyenne du Comice Conference	mid-3rd / 70-90 days after bloom -	- - -	- - -	20-25 20-25 25-30	- - 	- - -
Germany	Bergmann, 1988	all	mid-3rd /July-Aug.	-	30-100	20-50	15- 50	5-12
Hungary	Szucs, 1990	all	mid-3rd / July 15-30	>40	>30	20-50	15- 20	5-20
Japan -Nagano	Katou, 1990	Nijusseiki	mid-3rd / late July-early Aug.	-	60-200	-	50- 90	10-20
South Africa	Kotze, 1990	all	mid-3rd / January 31	60-240	25-100	22-29	10-35	3-20
USA	Childers, 1973	all	mid-3rd / mid-summer	100-250	20-75	20-50	15-40	4-10
* ranges in which yield and quality are satisfactory								

Peach - Optimum* leaf analysis values - Macronutrients									
Country/Region	**Source**	**Cultivar**	**Sampled leaf/Sampling Time**	**% dry matter**					
				N	**P**	**K**	**Mg**	**Ca**	**S**
Australia	Reuter & Robinson, 1986	all	mid-3rd / Jan.-Feb.	3.00-3.50	0.14-0.25	2.00-3.00	0.30-0.80	1.80-2.70	0.20-0.40
Brazil -Rio Grande & Santa Catarina	Basso, 1990	all	mid-3rd / 13-15 weeks after blomm	3.26-4.53	0.15-0.28	1.31-2.06	0.52-0.83	1.64-2.61	-
Germany	Bergmann, 1988	all	mid-3rd / July-Aug.	2.20-3.20	0.18-0.35	1.50-3.00	0.30-0.60	1.50-2.50	-
Hungary	Szucs, 1990	all	mid-3rd / first part Aug.	2.60-3.60	0.18-0.26	2.00-3.00	0.40-0.60	1.70-2.40	-
Italy -Piedmont -Tuscany -Latium Campania	Failla, 1991 Lalatta, 1987	all	lower 3rd / late July-early Aug.	 3.00-3.80 3.00-3.60 3.20-3.60	 0.19-0.27 0.16-0.22 0.16-0.21	 2.10-3.30 1.50-2.50 2.10-2.80	 0.35-0.55 0.40-0.60 0.65-1.00	 1.80-2.80 1.60-2.40 1.40-2.00	 - - -
Japan -Nagano	Katou, 1990	Ookubo	mid-3rd / mid June	3.40-3.50	0.20	1.60-2.00	0.27-0.40	-	-
South Africa	Kotze, 1990	all	mid-3rd / January 31	2.20-3.80	0.12-0.20	0.80-3.20	0.35-1.10	1.20-3.50	-
USA	Childers, 1973	all	mid-3rd / mid-summer	2.50-3.36	0.15-0.30	1.25-3.00	0.25-0.54	1.90-2.50	-
* ranges in which yield and quality are satisfactory									

Peach - Optimum* leaf analysis values - Micronutrients				
Country/Region	**Source**	**Cultivar**	**Sampled leaf/**	**ppm dry matter**

			Sampling Time					
				Fe	**Mn**	**B**	**Zn**	**Cu**
Australia	Reuter & Robinson, 1986	all	mid-3rd / late Jan.-Feb.	100-250	40-160	20-60	20-50	5-16
Brazil -Rio Grande & Santa Catarina	Basso, 1990	all	mid-3rd / 1 ”-14 weeks after bloom	100-230	31-160	34-63	24-37	6-30
Germany	Bergmann, 1988	all	mid-3rd / July-Aug.	-	35-100	20-60	15-50	7-15
Hungary	Szucs, 1990	all	mid-3rd / first part Aug.	120-150	20-140	20-80	15-30	4-12
Italy -Piedmont -Tuscany	Failla, 1991	all	lower 3rd / late July - early aug.	 50-250 80-300	 30-150 30-200	 18-30 22-32	 15-50 3-100	 3-30 1-16
Japan -Nagano	Katou, 1990	Ookubo	mid-3rd / mid-June	-	50-100	20-70	30-50	5-15
South Africa	Kotze, 1990	all	mid 3rd / January	60-240	30-140	24-45	18-50	3-20
USA	Childers, 1973	all	mid-3rd / mid-summer	124-152	20-142	20-80	15-30	4-12
* ranges in which yield and quality are satisfactory								

Plums - Optimum* leaf analysis values - Macronutrients								
Country/Region	**Source**	**Cultivar**	**Sampled leaf/Sampling Time**	**% of dry matter**				
				N	**P**	**K**	**Mg**	**Ca**
Australia	Reuter & Robinson, 1986	all	mid-3rd / Jan. - Feb.	2.40-3.00	0.14-0.25	1.60-3.00	0.30-0.80	1.50-3.00
Brazil -Rio Grande & Sata Catarina	Basso, 1990	all	mid-3rd / 13-15 weeks after bloom	3.26-4.53	0.15-0.28	1.31-2.06	0.52-0.83	1.64-2.61
Denmark	Vang-Petersen,1990	all	mid-3rd / Aug. 15-30	2.30-2.80	0.15-0.30	2.20-2.80	0.20-0.40	1.60-2.10
Germany	Bergmann, 1988	all	mid-3rd /July - Aug.	2.20-3.20	0.18-0.35	1.50-2.50	0.30-0.60	1.20-2.50
Hungary	Szucs, 1990	all	mid-3rd / July 15-Aug. 15	2.20-3.20	0.17-0.23	2.00-3.00	0.50-0.70	2.00-2.80
South Africa	Kotze, 1990	all	mid-3rd / january 31	2.30-3.20	0.16-0.25	2.20-3.50	0.36-0.87	1.20-2.30
USA	Childers, 1973	all	mid-3rd /mid-summer	1.80-2.10	0.14-0.25	1.50-250	0.18	2.00-4.00
* ranges in which yield and quality are satisfactory								

Plums - Optimum* leaf analysis values - Micronutrients				
Country/Region	**Source**	**Cultivar**	**Sampled leaf/sampling Time**	**ppm dry matter**

				Fe	Mn	B	Zn	Cu
Australia	Reuter & Robinson, 1986	all	mid-3rd /Jan.-Fen.	100-250	40-160	25-60	20-50	6-16
Brazil -Rio Grande & Santa Catarina	Basso, 1990	all	mid-3rd / 13-15 weeks after bloom	100-230	31-160	34-63	24-37	6-30
Germany	Bergmann, 1988	all	mid-3rd / July - Aug.	-	25-100	30-60	15-50	5-12
Hungary	Szucs, 1990	all	mid-3rd / July 15-Aug. 15	50-100	50- 90	25-50	30-50	7-10
South Africa	Kotze, 1990	all	mid-3rd / Jan. 31	80-135	22- 85	32-46	17-34	3-20
USA	Childers, 1973	all	mid-3rd / mid-summer	50-100	53- 93	33-50	25-50	7-10
* ranges in which yield and quality are satisfactory								

Apricot - Optimum* leaf analysis values - Macronutrients								
Country/Region	**Source**	**Cultivar**	**Sampled leaf/Sampling Time**	**% of dry matter**				
				N	**P**	**K**	**Mg**	**Ca**
Australia	Reuter & Robinson, 1986	all	mid-3rd / Jan. -Feb.	2.40-3.00	0.14-0.25	2.00-3.50	0.30-0.80	2.00-4.00
Germany	Bergmann, 1988	all	mid-3rd / July - Agu.	2.20-3.20	0.18-0.35	2.00-3.20	0.30-0.60	1.20-2.50
Hungary	Szucs, 1990	all	mid-3rd/2nd part of July	2.00-2.70	0.17-0.23	2.20-3.10	0.40-0.60	1.50-2.10
South Africa	Kotze, 1990	all	mid-3rd / Jan. 31	1.80-2.80	0.11-0.20	2.00-3.20	0.25-0.70	1.10-1.80
* ranges in which yield and quality are satisfactory								

Apricot - Optimum* Leaf Analysis Values - Micronutrients								
Country/Region	**Source**	**Cultivar**	**Sampled leaf/Sampling Time**	**ppm dry matter**				
				Fe	**Mn**	**B**	**Zn**	**Cu**
Australia	Reuter & Robinson, 1986	all	mid-3rd / Jan.-Feb.	100-250	40-160	20-60	20-60	5-16
Germany	Bergmann, 1988	all	mid-3rd / July - Aug.	-	30-100	20-60	15-50	5-12
Hungary	Szucs, 1990	all	mid-3rd / 2nd part of July	100-200	25-140	25-80	25-40	3-20
South Africa	Kotze, 1990	all	mid-3rd / Jan. 31	60-200	30-100	31-42	30-70	3-20
* ranges in which yield and quality are satisfactory								

Sweet Cherry - Optimum* leaf analysis values - Macronutrients								
Country/Region	**Source**	**Cultivar**	**Sampled leaf/Sampling Time**	**% of dry matter**				
				N	**P**	**K**	**Mg**	**Ca**
Australia	Reuter & Robinson,	all	mid-3rd / Jan.-Feb.	2.20-260	0.14-0.25	1.60-3.00	0.30-0.80	1.40-2.40

	1986							
Denmark	Vang-Petersen, 1990	all	mid-3rd / Aug. 15-30	2.60-3.20	0.15-0.30	1.40-1.90	0.20-0.40	1.60-2.10
Germany	Bergmann, 1988	all	mid-3rd /June-July	2.60-2.80	0.18-0.30	1.60-2.00	0.30-0.50	1.20-2.00
Hungary	Szucs, 1990	all	mid-3rd / July 1-15 after harvest	2.20-3.20	0.17-0.23	1.40-2.00	0.50-0.80	1.90-2.70
* ranges in which yield and quality are satisfactory								

Sweet Cherry - Optimum* Leaf Analysis Values - Micronutrients								
Country/Region	**Source**	**Cultivar**	**Sampled leaf/Sampling Time**	**ppm dry matter**				
				Fe	**Mn**	**B**	**Zn**	**Cu**
Australia	Reuter & Robinson, 1986	all	mid-3rd / Jan. - Feb.	100-250	40-160	20-60	20-50	5-16
Germany	Bergmann, 1988	all	mid-3rd / June - July	-	30-100	30-60	15-50	5-12
Hungary	Szucs, 1990	all	mid-3rd / July 1-15 after harvest	120-200	40-150	35-55	20-50	8-30
* ranges in which yield and quality are satisfactory								

Sour Cherry - Optimum* leaf analysis values - Macronutrients								
Country/Region	**Source**	**Cultivar**	**Sampled leaf/Sampling Time**	**% of dry matter**				
				N	**P**	**K**	**Mg**	**Ca**
Canada -Ontario	MOAAF, 1989	Montmorency	mid-3rd / July 15-30	2.20-3.00	-	1.50-2.10	>0.35	1.60-2.50
Germany	Bergmann, 1988	all	mid-3rd / June - July	2.80-3.20	0.20-0.35	1.60-2.00	0.30-0.50	1.30-2.00
USA	Childers, 1973	Montmorency	mid-3rd / mid-summer	2.33-3.27	0.23-0.32	1.25-1.92	0.49-0.74	1.62-2.60
USSR	Kondakov, 1990	Ljubskaja	mid-3rd / 2nd half of summer	1.80-2.50	0.13-0.22	1.30-2.00	0.23-0.35	-
* ranges in which yield and quality are satisfactory								

Sour Cherry - Optimum* leaf analysis values - Micronutrients								
Country/Region	**Source**	**Cultivar**	**Sampled leaf/Sampling Time**	**ppm dry matter**				
				Fe	**Mn**	**B**	**Zn**	**Cu**
Germany	Bergmann, 1988	all	mid-3rd / June - July	-	35-100	30-60	15-50	5-12
USA	Childers, 1973	Montmorency	mid-3rd /mid-summer	119-203	44-60	38-54	20-50	8-28
USSR	Kondakov, 1990	Ljubskaja	mid-3rd / 2nd half od summer	-	>10	>15	>6	>10
* ranges in which yield and quality are satisfactory								

Soil test data

Critical soil test values				
Below which, before planting fertilizers should be applied **Above which, before planting fertilizers need not be applied**				
Country/Region	**Source**	**Parameter, analytical methods, critical values**		
Brazil -Rio Grande & Santa Catarina	CDFDS, 1989	Organic matter (Walkey & Black): P (Mehlich) in ppm: K (Mehlich): Mg (KCl 1 N):	 clay >55 % clay 40-50 % clay 25-40 % clay 10-25 % clay <10 %	2 % 2.1 3.1 4.1 6.1 8.1 41 ppm 0.5 meq/ml
Canada -Ontario	Cline, 1990	P: 40 ppm K: 180 ppm		
Denmark	Vang-Petersen, 1990	P: 18-27 ppm K: 20-25 ppm Mg: 8-10 ppm		
France	Calvet & Villemin, 1987	Organic matter in %: P2O5 (Dyer) in ppm: P2O5 (Joret-Hebert) in ppm: K2O (exc.) in ppm: MgO (exc.) in ppm:	clay 1-10 % 2.7-2.2 clay 10-30 % 2 clay 30-45 % 2-3 CEC 50-150 meq/kg CEC 150-250 meq/kg CEC 250-350 meq/kg CEC 50-150 meq/kg CEC 150-250 meq/kg CEC 250-350 meq/kg CEC 50-150 meq/kg CEC 150-250 meq/kg CEC 250-350 meq/kg CEC 50-150 meq/kg CEC 150-250 meq/kg CEC 250-350 meq/kg	 160-250 250-325 325-375 95-150 150-200 200-245 75-225 225-325 325-425 75-100 100-150 150-200
Greece	Vasilakakis, 1990	P2O5 (Olsen) in ppm: K2O (NH4Ac) in ppm:	15-20 sandy soil 100 other 300	
Hungary	Szucs, 1990	P2O5 (Egner-Riehm) in ppm: K2O (Egner-Riehm) in ppm:	acid soil (pH<6.5) neutral soil alkaline soil clay <10 % clay 10-15 % clay 15-30 % clay 30-60 % clay >70 %	60- 80 80-100 100-120 80-100 120 160 200 250
Italy	Goldberg, 1986	P2O5 (Olsen) in ppm: K (NH4Ac): Mg (NH4Ac):	CEC <10 meq/100 g CEC 10-20 meq/100 g CEC >20 meq/100 g 5 % CEC 10 % CEC	40 60 80
Japan	Katou, 1990	P2O5 (Truog) in mg/100 g	10-25	

- Nagano		dry soil: K2O (NH4Ac): apple & pear: peach: MgO (NH4Ac): apple & pear: peach: CaO (NH4Ac): apple & pear: peach:	 CEC <10 meq/100 g CEC 10-20 meq/100 g CEC >20 meq/100 g CEC <10 meq/100 g CEC 10-20 meq/100 g CEC >20 meq/100 g CEC <10 meq/100 g CEC 10-20 meq/100 g CEC >20 meq/100 g CEC <10 meq/100 g CEC 10-20 meq/100 g CEC >20 meq/100 g CEC <10 meq/100 g CEC 10-20 meq/100 g CEC >20 meq/100 g CEC <10 meq/100 g CEC 10-20 meq/100 g CEC >20 meq/100 g	 5 % CEC 5 % CEC 3-5 % CEC 3-5 % CEC 3 % CEC 2-3 % CEC 20-23 % CEC 20 % CEC 15-20 % CEC 15-20 % CEC 15 % CEC 13-15 % CEC 55-72 % CEC 55 % CEC 42-55 % CEC 42-55 % CEC 42 % CEC 35-42 % CEC
South Africa	Kotze, 1990	P (Bray II ext.) in ppm: K (Bray II ext.) in ppm:	 sandy soil loamy soil clay soil	30 30- 50 50- 70 70-100
Switzerland	CRDF, 1985	Organic matter in %: P2O5 (Dirks-Scheffer mod.): K2O (Dirks-Scheffer mod.) in mg/100 g: Mg (Schachtschabel) in mg/100 g:	 clay <15 % clay 15-25 % clay >25 % clay <15 % clay 15-25 % clay >25 % clay <15 % clay 15-25 % clay >25 % clay <15 % clay 15-25 % clay >25 %	 1.2-1.5 1.8-2.5 2.5-3 16-24 x 0.0356 mg/100 g 8-16 x 0.0356 mg/100 g 6-9 x 0.0356 mg/100 g 3-6 2-4 1.2-2 5-7.5 7-10.5 12-18
USSR	Kondakov, 1990	Chernozem soil: P2O5 (Chiricov) in ppm: K2O (Kirsanov) in ppm: Chernozem soil: P2O5 (Machigin) in ppm: K2O (Machigin) in ppm: Red soil and subtropic podzolic soil: P2O5 (Oniani) in ppm: K2O (Oniani) in ppm:	 sandy-loam soil sandy-clay soil clay-loam soil sandy-loam soil sandy-clay soil clay-loam soil sandy-loam soil sandy-clay soil clay-loam soil	 150 80 120 180 40 200 300 400 400 150 200 300

Fertilizer recommendation

Available organic manures should be applied and well incorporated in soil (80-100 cm depth) before planting, especially if soil organic matter is low. If necessary, soil pH should be raised to 6.5 by liming.

P and K fertilizers should be applied before planting to reach the critical soil values, and to provide a nutrient reserve on which the trees can draw during their life. In young orchards, N, P and K fertilizers should be always applied, both to increase vegetative growth and to accelerate the onset of fruiting.

In mature orchards, applications of N, P, K, Mg and micronutrients should be based onyield, vegetative growth, leaf and soil analysis and visual symptoms of deficiency.

N should be applied just before bud-break in a single dressing or, on sandy soil, in two to three dressings; the later applications should be made about 30-60 days after the first. P and K fertilizers should be applied during the winter rest period. Micronutrients may also be applied as foliar sprays.

Preferred nutrient forms

The choice between nitrate, ammonium and urea forms of N should be made on the basis of meteorological and soil conditions. P must be water-soluble and K preferably in sulphate form.

Fertilizer recommendations: Before planting		
Country/Region	**Source**	**Recommendations**
All fruit trees		
France	Agenes & Terenne, 1988	P and K fertilizers are applied to reach the critical soil test values and to provide in the soil the whole amount that will be removed by the crop during its life.
Germany	Quast, 1990	Up to 200-300 kg/ha P2O5 and up to 400-600 kg/ha K2O based on soil analysis.
Greece	Vasilakakis, 1990	Organic matter: 30-40 t/ha when organic matter in soil is below 2 %. P2O5 based on soil analysis from 0 to 300 kg/ha. K2O based on soil analysis from 0 to 900 kg/ha
Hungary	Szucs, 1990	Organic matter based on soil analysis: from 40 to 100 t/ha. P2O5 based on soil analysis: from 9.3 (neutral or acid sandy soil) to 34.4 (alkaline clay soil) kg/ha x mg/kg soil to increase; up to 700 (sandy soil), 1400 (loamy soil) and 1800 (clay soil).* K2O based on soil analysis: from 9.3 (neutral or acid sandy soil) to 23.3 (alkaline clay soil) kg/ha x mg/kg soil to increase.*
Italy	Failla, 1990	Organic matter based on soil analysis 60-80 t/ha in relation to C/N ratio (max. 250 kg/ha N). P2O5: 0 - 300 kg/ha, K2O: 0-600 kg/ha, based on soil analysis.
South Africa	Kotze, 1990	P2O5, K2O and lime to reach the analytical soil standards.
USSR	Kondakov, 1990	20-60 kg/ha N, 60-180 kg/ha P2O5 and 60-180 kg/ha K2O.
Apple & Other		
Canada, Ontario	Cline, 1990	Liming to pH 6.5 up to 20 t/ha; 0-140 kg/ha P2O5 and 0-130 kg/ha K2O based on soil analysis.
Denmark	Vang-Petersen, 1990	P2O5 based on soil analysis: 170 kg/ha x mg/kg soil to increase.* K2O based on soil analysis: 30 kg/ha x mg/kg soil to increase.* Mg based on soil analysis: 42 kg/ha x mg/kg soil to increase.*
Apple & Pear		
Brazil, Rio Grande & Santa Catarina	CDFDS, 1989	Liming to pH 6 to 40 cm depth 0 - 320 kg/ha P2O5 and 0-200 kg/ha K2O based on soil analysis. 30 kg/ha of borax
Peach		
France	Agenes & Terenne, 1988	P and K fertilizers are applied to reach the critical soil test values and to provide in the soil the whole amount that will be removed ¦by the

		crop during its life, e.g.: P2O5: 400-600 kg/ha K2O: from 0 in sandy soil to 600-700 kg/ha in clay soil.
Peach & Plum		
Brazil, Rio Grande & Santa Catarina	CDFDS, 1989	Liming to pH 6 to 40 cm depth. P2O5: 0-120 kg/ha, K2O: 0-130 kg/ha, based on soil analysis.
Pear ---> Apple & Pear		
Plum ---> Peach & Plum		
* " P2O5, 34 kg/ha x mg/kg soil to increase" means: 34 kg/ha of P2O5 for each mg of available P2O5 per kg of soil intended to be increased; e.g.: soil analysis indicates 30 mg/kg of available P2O5, intended value is 35 mg/kg, consequently 34 x 5 = 170 kg of P2O5 have to be applied.		

Fertilizer recommendations: First years after planting		
Country/Region	**Source**	**Recommendations**
All fruit trees		
Italy	Failla, 1990	N: 100, 150-200, 200-250 g/tree in the 1st, 2nd and 3rd years respectively, up to 250 kg/ha in several dressings during the vegetative period.
South Africa	Kotze, 1990	N: 8.5, 17 and 25.5 g/tree/month in the 1st, 2nd and 3rd years respectively. K2O only in sandy soil and loam with over 50 % stones: 9, 18 and 27 g/tree/month in the 1st, 2nd and 3rd years respectively.
Apple		
Japan, Nagano	Katou, 1990	g/tree N P2O5 K2O 1 year old trees 60 24 48 5 year old trees 300 120 240 10 year old trees 600 240 180 15 year old trees 900 360 720 20 year old trees 1200 480 960 dwarfed trees: 1-2 year old trees 30 kg/ha N 3-4 year old trees 60 kg/ha N
Apple/9		
France	Decroux & Boulay, 1988	N: 30-40 and 30-70 g/tree in the 1st and 2nd year respectively.
Apple/MM106		
France	Decroux & Boulay, 1988	N: 0-50, 20-50 g/tree and 50-70 kg/ha in the 1st, 2nd and 3rd years respectively
Apple & Pear		
Brazil, Rio Grande & Santa Catarina	CDFDS, 1989	N: 15, 20 and 25 kg/ha in the 1st, 2nd and 3rd years respectively.
Peach		
France	Agenes & Terenne, 1988	N: 0-30 and 10-50 g/tree in the 1st and 2nd years respectively.
Peach & Plums		
Brazil, Rio Grande & Santa Catarina	CDFDS, 1989	N: 30, 60 and 90 g/tree in the 1st, 2nd and 3rd years respectively.
Pear ---> Apple & Pear		
Plums ---> Peach & Plums		

Fertilizer recommendations: Annually in mature orchards		
Country/Region	**Source**	**Recommendations**
All fruit trees		
USSR	Kondakov, 1990	Depending on leaf analysis, vegetative growth and yield. Standard rates in kg/ha: N = 0; P2O5 = 0-120; K2O = 0-120.
Apple		
Canada, Onatrio	Cline, 1990	Based on leaf analysis, growth, yield, fruit colour and storage life. Rates of N are based on tree age and tree population: Max. 35 g N/year of tree age when less than 500 trees/ha, e.g.: 8 year old trees, 800 trees/ha = 200 g/tree = 84 kg/ha; 8 year old trees, 1000 trees/ha = 125 g/tree = 125 kg/ha. Rates of P2O5: no P2O5 recommended for trees, 20 kg/ha P2O5 for

		cover crops. Rates of K2O: also based on tree population and tree size (cross section). Example: 5 cm cross section, 300 trees/ha = 40 kg/ha 5 cm cross section, 1000 trees/ha = 60 kg/ha 20 cm cross section, 300 trees/ha = 200 kg/ha 20 cm cross section, 1000 trees/ha = 300 kg/ha Lime: if pH below 5.6, 1-20 t/ha depending on pH and buffer pH.
France	Boulay & Decroux & Diris, 1984	Based on yield, growth and leaf analysis. Examples in kg/ha: Cultivar: "Golden Delicious": N = 80 - 120; P2O5 = 50-100; K2O = 100-250
	Decroux & Boulay, 1988	Cultivar: "Granny Smith" (50 t/ha yield): N = 60-100; P2O5 = 50-100; K2O = 100-250 Cultivar: "Red Delicious" (50 t/ha yield): N = 60-100; P2O5 = 50-100; K2O = 100-250.
Germany	Quast, 1990	Based on yield, growth and leaf analysis. Example in kg/ha: N = 0-90; K2O = 0-100; Mg and lime if necessary.
Greece	Vasilakakis, 1990	N = 1 kg/tree up to 350 kg/ha; P2O5 = 0-1 kg/tree up to 350 kg/ha in 2 years; K2O = 0-3 kg/tree up to 700 kg/ha in 2 years.
Apple		
Italy	Failla, 1990	Depending on leaf analysis, vegetative growth and yield. Example in kg/ha: N = 0-70; P2O5 = 0-25; K2O = 0-100.
Japan, Nagano	Katou, 1990	Depending on soil fertility rates in kg/ha: N = 120-200; P2O5 = 40-60; K2O = 100-140.
Apple & Pear		
Netherlands	Kodde, 1990	Depending on leaf analysis, vegetative growth and yield; adivsed rates in kg/ha: N = 0-100; P2O5 = 0-40; K2O = 0-300.
Apple, Pear & Peach		
South Africa	Kotze, 1990	Depending on yield, growth (N), soil (P2O5 and K2O) and leaf (P) analysis; minimum and maximum rates in kg/ha: N = 35-185; P2O5 = 0-115; K2O = 0-360.
Apple, Pear & Plum		
Denmark	Vang-Petersen, 1990	According to leaf analysis; standard rates in kg/ha when soil is kept clean: N = 0; P2O5 = 0; K2O = 120 and MgO = 12.
Apricot		
South Africa	Kotze, 1990	Depending on yield, growth (N), soil (P2O5 and K2O) and leaf (P) analysis; minimum and maximum rates in kg/ha: N = 15-140; P2O5 = 0-90; K2O = 0-290.
Apricott, Plum & Cherry		
Italy	Failla, 1990	Depending on leaf analysis, vegetative growth and yield; example in kg/ha: N = 50-150; P2O5 = 0-50; K2O = 0-150.
Peach		
France	Agenes & Terenne, 1988	Based on yield, growth and leaf analysis; example in kg/ha: N = 50-100; P2O5 = 50-100; K2O = 100-250.
Italy	Failla, 1990	Depending on leaf analysis, vegetative growth and yield; example in kg/ha: N = 50-150; P2O5 = 0-50; K2O = 0-200.
Japan, Nagano	Katou, 1990	Depending on soil fertility; rates in kg/ha: N = 140-180; P2O5 = 60-80; K2O = 100-120.
---> see also: Apple, Pear & Peach		
Pear		
France	Boulay, Decroux & Diris, 1984 Decroux & Boulay, 1988	Based on yield, growth and leaf analysis; example in kg/ha: Summer cultivars: N = 130-150; P2O5 = 50-100; K2O = 100-250
Pear		
Italy	Failla, 1990	Depending on leaf analysis, vegetative growth and yield; example in kg/ha: N = 0-100; P2O5 = 0-25; K2O = 0-100.

Japan, Nagano	Katou, 1990	Depending on soil fertility; rates in kg/ha: N = 160-200; P2O5 = 60-80; K2O = 120-160.
---> see also: Apple & Pear, Apple, Pear & Peach, Apple, Pear & Plum		
Plum		
France	Cordier, Huguet & Tabardon, 1988	Based on yield, growth and leaf analysis; example in kg/ha: N = 70-120; P2O5 = 80-120; K2O = 160-220.
South Africa	Kotze, 1990	Depending on leaf analysis, vegetative growth and yield; example in kg/ha: N = 25-137; P2O5 = 0-115; K2O = 0-360.
---> see also: Apple, Pear & Plum; Apricot, Plum & Cherry		
Sour Cherry		
Denmark	Vang-Petersen, 1990	According to leaf analysis; standard rates in kg/ha when soil is kept clean: Cultivar "Montmorency: N = 0; P2O5 = 0; K2O = 120; MgO = 20 Cultivar "Dard": N = 110; P2O5 = 0; K2O = 120; MgO = 20
Germany	Quast, 1990	Based on yield, growth and leaf analysis; example in kg/ha: N = 140-150; P2O5 = 80; K2O = 160-170.
Sweet Cherry		
Denmark	Vang-Petersen, 1990	According to leaf analysis; standard rates in kg/ha when soil is kept clean: N = 100; P2O5 = 0; K2O = 120; MgO = 20.
Germany	Quast, 1990	Based on yield, growth and leaf analysis; example in kg/ha: N = 100-120; P2O5 = 80; K2O = 160-200.
---> see also: Apricot, Plum & Cherry		

Further reading

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Grape or Grapevine *(Vitis vinifera L.)*

French: Vigne; Spanish: Vid, viña; Italian: Vite; German: Weinrebe

Most cultivated grapevines are of the species Vitis vinifera L.; a very small area is devoted to V.rotundifolia (Muscadines), V.labrusca (Concord) and interspecific hybrids. In most situations, V.vinifera varieties must be grafted onto Phylloxera-resistant rootstocks derived from species of American origin.

Crop data

Perennial. Harvested part: fruits.

Plant density: 1 000-10 000 vines per hectare.

Vineyards occupy 10 million hectares worldwide. They are located on various soils in zones with a Mediterranean, temperate or sub-tropical climate. The yield varies greatly (5-35 t/ha/year) depending on site, cultural conditions and end-use (fine wine, ordinary wine or table grapes). Preoccupation with quality often leads growers to limit vigour and yield.

Nutrient demand/uptake/removal

Mineral nutrient uptake depends on variety, rootstock, yield, soil, climate, etc. Total annual uptake in stems, leaves and fruit varies within the following ranges for a yield of 7-25 t/ha :

Nutrient demand/uptake/removal - Macronutrients					
Yield t/ha	kg/ha/year				
	N	P2O5	K2O	MgO	CaO
7 - 25	22-84	5-35	41-148	6-25	28-204

Nutrient demand/uptake/removal - Micronutrients					
Yield t/ha	**g/ha/year**				
	Fe	**B**	**Mn**	**Zn**	**Cu**
7 - 25	292-1 121	37-228	49-787	110-585	64-910
Source: Fregoni, 1984					

Stems and leaves, which are re-incorporated in the soil, account for about 70 % of the N uptake, and 60 % of the P2O5 and K2O, so the net removal in the harvest is clearly much smaller.

Plant analysis data

The method of foliar diagnosis for grapevines was developed at Montpellier, France, in about 1930. Since that date much information on the mineral composition of the leaf organs has been published throughout the world, but, because of the many factors involved, it is difficult to compare data from different sources.

Plant analysis data (normal growth) - Macronutrients								
Species or variety*	**Tissue samples**	**Sampling Time**	**Source**	**%o of dry matter**				
				N	**P**	**K**	**Mg**	**Ca**
L (Concord)	Petiole	July-August	Larson et al, 1956	8.2	2.0	20.1	4.4	17.5

L (Concord)	Petiole	July-August	Beattie Forhey, 1954	7.7	1.4	20.0	1.5	7.0
L (Concord)	Petiole	August-September	Shaulis & Kimball, 1956	17.0-34.0**	1.0-3.0	4.0-20.0	1.8-15.0	10.0-17.0
L-FH	Petiole	July	Bryant et al, 1956	8.9-13.8	1.2-4.1	20.5-53.1	1.7-5.6	11.8-20.6
L-FH	Petiole	August-September	Styles & Oberle, 1984	18.0-22.0**	0.8-3.0	11.0-20.0	2.5-5.0	7.5-17.0
L-FH	Petiole	Ripening	Parsons & Eaton, 1950	5.2-11.1	1.1-6.4	12.0-41.0	2.2-5.3	7.2-16.9
V	Blade	-	Shikhamany & Satyanarayana, 1971	-	1.9	6.5	3.3	8.0
L-FH	Petiole	July-August	Cahoon, 1980	9.0-13.0	1.6-3.0	15.0-25.0	2.6-4.5	10.0-18.0
V	Petiole	Bloom	Christensen et al., 1978	-	≥1.2	8.0-15.0	≥3.0	-
V	Leaves	-	Saini & Singh, 1975	25.0-27.0	2.0-2.3	13.0-15.0	-	23.0-25.0
V	Petiole	Ripening	Ryser, 1983	21.2-24.6	1.7-1.9	16.3-18.9	2.2-2.6	23.8-27.6
L-FH	Petiole	August-September	Cline, 1983	7.0-8.9	-	16.0-21.9	9.0-14.9	15.0-25.0

* L = V. labrusca, V = V. vinifera, FH = French hybrid. ** Bloom sample date
Source: Cahoon, 1985

Petiole analysis interpretation chart. Sampling time berry ripening (* dry matter basis)	
N > 6 %o*	normal N nutrition
P > 1.5 %o*	normal P nutrition
K/Mg < 1	K deficiency
K/Mg > 10	Mg deficiency
K/mg 2 to 8	normal K and Mg nutrition
B < 15 ppm*	boron deficiency
Source: Delas, 1990	

At present, foliar analysis is used to diagnose nutritional problems (deficiencies, imbalances and toxicities) and for adjusting fertilizer recommendations. Even if foliar analysis cannot be used directly to determine the amounts of fertilizer needed, it permits adaptation of fertilizer practices in relation to production objectives, provided results are interpreted in the light of regional norms for soil, climate, variety, rootstock and tillage practices.

The effects of plant nutrients on quality also need to be understood. Too much N, for example, can cause a reduction in the colour components of the fruit, and therefore of the wine, and can increase susceptibility to Botrytis cinerea, the causal agent of grey rot. Similarly, an excess of K can reduce the acidity of the berries and of the must, which can adversely affect the quality of the wine. Too much K can also increase susceptibility to "Stiellaehme" and induce Mg deficiency by antagonism between K and Mg.

Also, some nutrient elements may be provided in fungicides, e.g. S, used against Oidium; Cu in Bordeaux mixture : Mn and Zn as dithiocarbamates used against mildew.

Fertilizer recommendations

Before planting. The purpose of deep manuring is to provide minerals (P_2O_5, K_2O, MgO) which are not very mobile and to correct e.g. high acidity (which can induce Al or Cu toxicity) or low organic matter content. Depending on analysis of the soil and sub-soil, the amounts to be applied lie within the ranges: 0-600 kg/ha P_2O_5, 0-1 000 kg/ha K_2O, 0-300 kg/ha MgO, liming (where pH is below 6) at 2 000-10 000 kg/ha CaO, and 0-100 t/ha farmyard manure or equivalent.

Annual maintenance. For fine wine vineyards yielding less than 10 t/ha: 0-40 kg/ha N, 20-50 kg/ha P_2O_5, 60-100 kg/ha K_2O. For other vineyards : 60-120 kg/ha N (more if irrigated), 20-50 kg/ha P_2O_5, 100-150 kg/ha K_2O. In general, N is applied as an overall dressing in late winter or in spring. On clay soils with dry climates, P_2O_5 and K_2O are worked into the soil during the winter dormant period; on light soils with a wet climate, they may be topdressed together with the N.

Foliar application (usually of K, Mg, B or Fe) may be given in cases of deficiency.

Preferred nutrient forms

The grapevine is not very demanding in this respect.

As a source of K, the sulphate form is no more favourable to quality than the chloride form. However, the sulphate should preferably be used on saline soils or for heavy applications, e.g. 500-1 000 kg/ha K_2O.

In vineyards producing fine wine, organic amendments low in N (pine bark, peat, etc.) are preferred in order not to affect quality.

Fertilizer practices

There is very wide variation in current fertilizer practice. Within the same region, practice varies from vineyard to vineyard and from year to year, from no fertilizer at all to amounts well in excess of actual need. This situation is explained by the plant's low requirement, the slowness of diffusion of nutrient elements in the soil and fluctuations in economic conditions, but it represents a risk to the quality of production which may be threatened by nutrient deficiency, excess or imbalance. Rational fertilizer recommendations still need to be worked out, which must take into account basic knowledge of the soil and the needs of the plant together with the results of foliar analysis.

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Hazelnut or Filbert *(Corylus avelana L.)*

French: Noisetier; Spanish: Avellano; Italian: Nocciolo; German: Haselnuss

Crop data

Perennial. Harvested part: fruits.
Planted in late autumn (propagated by suckers).
Flowers in winter. Fruits 4 years after being planted.
Harvested in late July and early August.

Plant density: 400-600 trees ("Ocak" with 6-8 suckers) per ha.
Preferably grown on light to medium soils, pH 5.5-7.

The root system is shallow.

Filberts are generally grown in humid conditions. Irrigation is needed in dry summers. The crop is not grown economically where temperatures fall below - 7 °C; it is much affected by late frosts in spring.

Trees are generally not self-fertile and pollinating varieties are required.

Major commercial production is along the Black Sea coast of Turkey.

Nutrient demand/uptake/removal

Nutrient demand/uptake - Macronutrients					
Yield kg/ha	**Source**	**kg/ha**			
		N	**P2O5**	**K2O**	**CaO**
1 200	Carpentieri	19	9	12	16

Plant analysis data

Plant analysis data - Macronutrients								
Plant part	**Stage of growth**	**Source**	**% of dry matter**					
			N	**P**	**K**	**Mg**	**Ca**	**S**
Mid-terminal leaves	20-30 July	Genc, 1987	2.3 (OS) 1.9 (CVD)	0.15 (OS) 0.10 (CVD)	0.7 (OS) 0.3 (CVD)	- -	- -	- -
Mid-terminal leaves	20-30 July	Aydeniz et al, 1980	2.2 (OS)	0.19 (OS)	1.0 (OS)	0.37 (OS)	1.5 (OS)	0.22 (OS)
Mid-terminal leaves	14 July-14 August near harvest time	Painter, 1963	2.4 (OS)	0.15 (OS)	0.9 (OS)	0.24 (OS)	1.4 (OS)	-
OS = Optimum supply; CVD = Critical value (deficiency)								

Plant analysis data/Optimum supply - Micronutrients								
Plant part	**Stage of growth**	**Source**	**ppm dry matter**					
			Fe	**Mn**	**Cu**	**Zn**	**B**	**Mo**

Mid-terminal leaves	20-30 July	Aydeniz et al, 1980	550	330	13	52	33	0.09
Mid-terminal Leaves	14 July - 14 August	Painter, 1963	410	495	10	34	27	-

Special attention must be given to N and K application rates to obtain good vegetative growth and high yield with good quality. Annual shoot growth of 25 cm is enough for mature trees. High nitrogen applications increase shell thickness and reduce nut size. Correct potassium application diminishes the proportion of empty nuts.

Fertilizer recommendations

Prior to planting and young trees

Prior to planting and up to bearing age, organic and inorganic fertilizers should be applied according to soil analysis as follow:

Organic fertilizer like farmyard manure should be applied at around 30 ton/ha if the soil organic matter is below 2 per cent.

Where the soil pH is under 5.5, it should be raised to 6.5 by liming but not more than 5 t/ha should be given in a single dressing.

pH	**Application rate of lime or dolomite (t/ha)**	
	On clay loam	**On clay**
4.0	8	11
5.0	5	7
5.5	4	5

Mineral fertilizers and/or organic materials should be applied on the basis of soil analysis (as indicated below) broadcast and ploughed under before planting:

Phosphorus - Soil P - ppm	**Application rate - P2O5 kg/ha**
0- 7	40-50
7-14	30-40
> 14	Nil

Potassium - Soil K - ppm	**Application Rate - K2O - kg/ha**
0-100	150-200
100-200	100-150
> 200	Nil

Magnesium should be added in the form of dolomite at the rate of 3-4 t/ha where the soil Mg is less than 110 ppm.

Nitrogen should be applied annually at 100 g N per tree ("Ocak") to trees up to 5 years of age, in May in the first growing season and thereafter half in March and half in May.

Mature Trees

Fertilizer applications to mature trees should be based on leaf and soil analysis. Leaf samples should be taken at 3-year intervals and soil samples every 5 years.

Nitrogen fertilizer should be applied in March each year in a band 30-60 cm wide under the branches, an acidic or alkaline form being selected according to the soil pH.

Leaf N - % of dry matter	Application rate - g N per tree
< 1.5	500
1.5-2.1	400
2.2-2.4	300
> 2.4	Nil

Phosphate fertilizer may be applied at any time of the year, preferably in a band at the drip line of the tree (which is often better than broadcast application). A single application is usually effective for 2-3 years on soils slightly deficient in phosphorus.

Leaf P - % of dry matter	Application rate - g P2O5 per tree
< 0.14	500
0.14-0.16	300
> 0.16	Nil

Potassium fertilizer may be applied in a band together with nitrogen. A single application is effective for 2 years. Sulphate is preferable to muriate.

Leaf K - % of dry matter	Application Rate - g K2O per tree
< 0.5	1 000
0.5-0.8	750
> 0.8	Nil

Other important nutrients are boron and iron (deficiencies of these elements occur in the growing area in Turkey).

Where needed, a single application of 0.1 % boric acid or borax at 125 g/tree reduces the proportion of empty nuts. Iron deficiency may be remedied by applying Fe-EDDHA at 100 g/tree in March.

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Kiwifruit *(Actinidia deliciosa [Chev.] Liang et Ferg. var deliciosa)*

French: Kiwi; Spanish: Kiwi; Italian: Actinidia; German: Kiwi

Crop Data

Perennial vine. Harvested part: fruits.
Commercial production 3-5 years after planting.
Flowers approximately 60 days after bud burst in spring.
Fruit harvested in autumn about 160-180 days after flowering.

Plant density: 400 vines (traditional) to 800 vines (intensive) per ha.
Preferably grown in deep freely draining soils, pH 5.5-6.5.

Kiwifruit is a high water user and is generally irrigated.

Estimated annual uptake of nutrients

Nutrient uptake (estimated) - Macronutrients								
Vine age (years)	**Fruit Yield (t/ha fresh weight)**	**kg/ha/year**						
		N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**	**Cl**
3	10	74	23	116	22	97	13	29
4	20	126	37	193	35	162	21	48
>5	20	94	27	160	23	129	15	43
>5	30	129	39	219	35	176	22	59
>5	40	165	50	278	45	225	28	75
Source: Smith, Buwalda & Clark, 1989								

Note that the relatively high concentrations of N and K in the perennial tissues (structural roots, stem, and leaves) result in a disproportionate accumulation of these nutrients during vine development, such that developing vines may take up 50 % more N and 22 % more K than mature vines with similar fruit yields.

Plant Analysis Data

Plant analysis data (youngest fully expanded leaf) - Macronutrients							
Stage of Growth	**% of dry matter**						
	N	**P**	**K**	**Mg**	**Ca**	**S**	**Cl**
6 weeks	4.2	0.6	3.0	0.3	1.5	0.45	0.8
20 weeks	2.5 (OC)	0.25 (OC)	2.0 (OC)	0.45 (OC)	3.5 (OC)	0.40 (OC)	1.2 (OC)
6 weeks	1.5 (D)	0.12 (D)	1.5 (D)	0.10 (D)	0.2 (D)	0.18 (D)	0.2 (D)
6 weeks	5.5 (T)	1.2 (T)	-	-	-	-	2.5 (T)
OC = Optimum concentration - D = Deficient concentration associated with leaf symptoms - T = Toxic concentration associated with leaf symptoms							
Source: Smith, Asher & Clark, 1987							

Plant analysis data (youngest fully expanded leaf) - Micronutrients						
Stage of Growth	**ppm dry matter**					
	Fe	**Mn**	**Cu**	**Zn**	**B**	**Mo**

6 weeks	100	100	15	40	40	0.2
20 weeks	90 (OC)	150 (OC)	10 (OC)	20 (OC)	50 (OC)	0.2 (OC)
6 weeks	60 (D)	30 (D)	3 (D)	12 (D)	20 (D)	0.05 (D)
6 weeks	-	1200 (T)	-	1100 (T)	80 (T)	-
OC = Optimum concentration - D = Deficient concentration associated with leaf symptoms - T = Toxic concentration associated with leaf symptoms						
Source: Smith, Asher and Clark (1987)						

Special attention should be paid to B and Na as kiwifruit is extremely sensitive to excess of both of these elements in the soil. Kiwifruit is also exceptional in that it requires a comparatively high concentration of Cl in the tissues for growth. In general, production losses from nutrient disorders result mainly from a reduction in fruit number rather than a reduction in fruit size.

Fertilizer Recommendations

A mathematical model linked to a database has been developed for generating fertilizer recommendations for individual kiwifruit orchards. The model summarises nutrient fluxes within the orchard ecosystem from which a fertilizer requirement is derived. The fertilizer budget accounts for uptake, efficiency of recovery, cycling within the orchard, and any previous nutrient disorders. An example of information generated by the model is given below.

Nutrient inputs and recoveries for a typical mature kiwifruit orchard producing 30 t/ha of fruit situated 10 km from the coast							
Inputs/uptake	**kg/ha**						
	N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**	**Cl**
Inputs							
Fertilizer	170	128	240	60	168	65	200
Cycling (1)	87	21	64	30	162	17	45
Atmospheric (2)	31	0.5	8	7	10	12	116
Total	288	149	312	97	340	94	361
Uptake	129	39	219	35	176	22	59
% Recovery	45	26	70	36	52	23	16
(1) = includes nutrients in summer and winter prunings and root turnover. (2) = includes nutrients in rainfall and N fixed by legumes in the orchard sward.							
Source: Smith et al, 1988; Buwalda and Smith, 1988							

The benefits from using fertilizer recommendations generated from the kiwifruit fertilizer model as above can be gauged from a survey of kiwifruit orchards in the Bay of Plenty, New Zealand, during the 1989/90 season.

No. of orchards	**Basis of fertilizer recommendations**	**Average fertilizer use (kg/ha)**			**Average yield trays/ha***
		N	**P2O5**	**K2O**	
180	Kiwifruit Model	206	181	248	4533
78	Other	108	128	206	3842
* 1 tray = approx. 3.6 kg of fresh fruit					

Preferred nutrient forms

All nutrients should be applied in plant-available forms. K should be applied preferably as chloride (unless soil Cl is high, i.e. >150 ppm and/or soil N is low). K is much more readily absorbed by kiwifruit in chloride form than as sulphate. Yield differences of up to 30 % have been recorded in favour of the chloride.

Fertilizer practice

All fertilizers including N should be applied by bud-burst in spring to maximize flower numbers. For mature vines (>3 years) fertilizers should be broadcast over the entire orchard floor.

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Authors: G.S. Smith, C.J. Clark, J.G. Buwalda, MAF Ruakura Agricultural Centre, Hamilton, New Zealand

Macadamia *(Macadamia ternifolia F.V. Muell.)*

French: Macadamia; Spanish: Macadamia; Italian: Macadamia; German: Macadamia

Crop data

Perennial evergreen. Harvested part: fruits; nuts dehulled after harvest.
Transplanted in winter.
Flowers 4-5 years after transplanting.
Harvested about 7 1/2 months after flowering.

Plant density: Hawaii, 88-120 trees/ha; Australia, 182-212 trees/ha. In Hawaii, macadamia is largely grown in rockland of aa lava. Outside Hawaii, light soils are preferred as trees become chlorotic in heavier soils. Preferably pH is 5.5 - 6.5, but grows well up to pH 8.

Subtropical in its native range, but can be grown in tropical to cool subtropical climates. Irrigated in California (USA) and sometimes in Australia, but generally not in other tropical areas. Hawaii is the principal producer but production is increasing in South Africa, Guatemala, and Australia.

Nutrient demand/uptake/removal

Nutrient removal					
Source	**g/45 kg yield of fruit**				
	N	**P2O5**	**K2O**	**MgO**	**CaO**
Cooil et al, 1958	183	38	182	17	14

Plant analysis data

Plant analysis data - Macronutrients								
Plant part	**Stage of growth**	**Source**	**% of dry matter**					
			N	**P**	**K**	**Mg**	**Ca**	**S**
Leaf	current flush	Fox et al, 1976	0.9 (D) 1.5 (OS)	0.12 (OS)	0.84 (OS)	0.10 (OS)	0.40 (OS)	0.06 (D) 0.15 (OS)
Leaf	first flush	Fox et al, 1976	0.6 (D) 1.5 (OS)					0.08 (D) 0.20 (OS)
Leaf	oldest leaves	Fox et al, 1976	0.7 (D) 1.5 (OS)					0.10 (D) 0.25 (OS)
Leaf	current flush	Hue et al, 1988		0.12 (OS)				
Leaf	fully mature	Cooil et al, 1956	1.5 (OS)	0.06 (OS)	0.60 (OS)			
Leaf	fully mature	Cooil et al, 1966		0.10 (CVY)				
Leaf	fully mature	Hirae & Nagao, 1989		0.06 (D) 0.08 (OS)				
Leaf	fully mature	Stephenson & Cull, 1986	1.4 (OS)		0.60 (OS)			0.18 (OS)
Leaf	fully mature	Warner & Fox, 1972						0.02 (D) 0.24

								(OS)
D = Deficiency; OS = Optimum supply; CVY = Critical value (yield)								

Plant analysis data - Micronutrients							
Plant part	**Stage of growth**	**Source**	**ppm dry matter**				
			Fe	**Mn**	**Cu**	**Zn**	**B**
Leaf	4 months old	Labanauskas & Handy, 1970	10 (D)	3 (D)			
Leaf	current flush	Fox et al, 1976		141 (OS)		20 (OS)	
Leaf	fully mature	Stephenson & Cull, 1986					40 (OS)
Leaf	fully mature	Warner & Fox, 1972		100 (OS)	4.5 (OS)	15 (OS)	75 (OS)
D = Deficiency; OS = Optimum supply							

Macadamia growth is near maximum when leaf N:S ratio is 9:1 to 14:1. High P (about 0.10 % leaf P) may induce chlorosis and yield reduction. Low leaf Fe:Mn ratio (below 1,2:1), induced by the high P, causes the chlorosis.

Fertilizer recommendations

Most macadamia orchards are grown in sod culture and so fertilizer is not incorporated in the soil except before planting. When applied before planting, superphosphate, lime, Cu and Zn are incorporated according to the results of soil testing (Australia). After transplanting, fertilizer needs are determined by leaf (6-7 month-old leaves sampled at beginning of spring flush) and soil analysis.

If necessary, the soil pH needs to be raised to pH 5.5 to 6.5 by liming, calcitic or dolomitic depending on Mg need.

The mineral fertilizer for macadamia is a basal NPK with or without supplementary N. Applications are split.

Preferred nutrient forms

In Hawaii, N is preferably applied as ammonium sulphate, P as single or triple superphosphate and K as sulphate; in Australia, N is applied as urea or ammonium nitrate, P as superphosphate and K as muriate or sulphate.

Present fertilizer practices

Hawaii (“ aa lava ” soil)

For trees up to 7.5 cm trunk diamter: basal dressing of 8.4 g N/cm diameter, 16.0 g P2O5/cm diameter and 23.0 g K2O/cm diameter, applied January and July, and additionally 8.0 g N/cm diameter, applied April and October.

As trees increase in diameter, adjustments may have to be made according to leaf analysis.

Australia

Application is based on a standard quantity per tree to be multiplied by the tree age in years (up to 10). At year 10, the maximum application is reached and becomes the set application for older trees, except that, after thinning, the rate is increased proportionaly with each metre

increase in spacing between trees (assumed to be 5 m at 10 years). Further adjustments may be necessary according to leaf analysis.

The standard quantity is 69 g N, 10 g P2O5 and 48 g K2O per tree. In years 1 to 5, N and K are applied one-fifth each in February, April, July, October, and December. For trees 6 years and older, N is applied one-half in February, one-quarter each in April and October. P is supplied in a single application in March/April, and K one-third each in February, April and October.

When needed, foliar sprays of copper (6 g CuCl2. 3CuO/liter) and Zinc (1 g ZnSO/liter) are given once or twice per year.

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Pecan *[Carya illinoinensis (Wangenh.) K. Koch]*

French: Pecanier; Spanish: Pecano; Italian: Caria, noce di pecan; German: Pekannuss

Crop data

Perennial. Harvested part: nuts
Transplanted in dormant season.
Flowers 4-6 years after transplanting.
Harvested 6-6 1/2 months after flowering.

Plant density: 90 (traditional) to 240 (intensive) trees/ha. Trees are systematically removed as they increase in size and density is eventually reduced to 20 trees/ha.
Preferably grown on light to medium soils, pH 6-8.

Adapted to arid, semi-arid, and humid conditions. Irrigated by flood or sprinkler in arid and semi-arid climates; predominantly by drip, when irrigated in humid climates. Mostly grown in USA and Mexico.

Nutrient demand/uptake/removal

Nutrient removal - Macronutrients						
Yield t/ha	**Source**	**kg/ha**				
		N	**P2O5**	**K2O**	**MgO**	**CaO**
1.2 nuts	Sparks, 1975	9.7	5.3	5.4	1.0	5.2

Nutrient removal - Micronutrients							
Yield t/ha	**Source**	**g/ha**					
		Fe	**Mn**	**Zn**	**Cu**	**B**	**Mo**
1.2 nuts	Sparks, 1975	34	95	35	10	12	2.4

Plant analysis data

Plant analysis data - Macronutrients								
Plant part	**Stage of growth**	**Source**	**% of dry matter**					
			N	**P**	**K**	**Mg**	**Ca**	**S**
Mid-leaflet pair on mid-leaf	56-84 days after catkin fall	Sparks, 1968	2.2 (CVD)					
Leaf	6 weeks after budbreak	Alben, 1946	2.3 (CVD) 2.8 (OS)					
Leaf		Alben, 1947	1.2 (D)	0.10 (D)	0.12 (D)	0.08 (D)	0.43 (D)	
Leaf	19 weeks after nut germination	Sparks, 1978	1.7 (D)	0.08 (D)	0.44 (D)	0.08 (D)	0.12 (D)	0.08 (D)
			3.1 (OS)	0.23 (OS)	1.65 (OS)	0.64 (OS)	2.51 (OS)	0.33 (OS)
Mid-leaflet pair on mid-leaf	14 weeks	Sparks & baker, 1975	2.3 (CVD) 2.7 (OS)					
Leaf	11 weeks after nut germination	Sparks, 1986a				0.20 (D) 0.55 (OS)		
Leaf	14 weeks after after nut germination	Sparks, 1986b		0.14 (CVD) 0.22 (OS)				

Leaf	21 weeks after nut germination	Sparks, 1988		0.08 (D) 0.19 (OS)				
Leaf	15 weeks after budbreak	Upson & Sparks, 1969			0.27 (D) 1.51 (OS)			
Leaf	22 weeks after budbreak	Sharpe et al., 1951				0.14 (CVD) 0.31 (OS)		
Mid-leaflet pair on mid-leaf	56-84 days after catkin fall	Sparks, 1989	2.3 (CVD)	0.10 (CVD)	0.70 (CVD)	0.20 (D)	0.50 (CVD)	0.10 (D)
			2.7 (OS)	0.18 (OS)	1.3 (OS)	0.35 (OS)	1.3 (OS)	0.25 (OS)
D = Deficiency; OS = Optimum supply; CVD = Critical Value (deficiency)								

Plant analysis data - Micronutrients							
Plant part	**Stage of growth**	**Source**	**ppm dry matter**				
			Fe	**Mn**	**Cu**	**Zn**	**B**
Leaf	19 weeks after nut germintion	Sparks, 1978	102 (OS)	71 (OS)	18 (OS)	37 (OS)	41 (OS)
			-	-	-	31 (D)	6 (D)
Leaf, tree basis	56-84 days after catkin fall	Sparks & Payne, 1982	-	-	-	50 (OS)	-
			-	-	-	40 (CVD)	-
Mid-leaflet pair od mid-leaf, sholl bais	56-84 days after catkin fall	Hu & Sparks, 1991	-	-	-	14 (CVY)	-
			-	-	-	6 (CVD)	-
Leaf, tree basis	22 weeks after budbreak	Lane et al., 1965	-	-	-	44 (OS)	-
			-	-	-	16 (CVD)	-
Mid-leaflet pair on mid-leaf	56-84 days after catkin fall	Sparks, 1989	-	-	10 (OS)	50 (OS)	35 (OS)
			-	-	-	40 (CVD)	13 (CVD)
D = Deficiency; OS = Optimum supply; CVD = Critical value (deficiency); CVY = Critical value (yield)							

To produce nuts of good quality and obtain a good crop the following year, special attention has to be paid to leaf N, P, and K. Otherwise, fruiting suppresses these elements to the deficiency level. K is very important in minimizing winter injury. Young trees especially, are very sensitive to N:K imbalance. If the leaf N:K ratio is greater than about 0.5:1 the leaves are liable to scorch and defoliate in midseason.

Fertilizer recommendations

Most pecan orchards are grown in sod culture and so fertilizer usually is not mechanically incorporated into the soil. In flood and in sprinkler irrigated orchards, fertilizer is applied at about the time of budbreak. In unirrigated and in drip-irrigated orchards (humid regions), fertilizer is applied about 1 month before budbreak. In unirrigated orchards, the fertilizer is applied early to give time for spring rains to move the fertilizer into the root zone before budbreak. In most orchards, all the fertilizer is given in a single application, but there is a

developing trend to divide the total into two equal dressings about 60-70 days apart. The proportionate area receiving fertilizer varies with tree age:

Tree age (years)	Area fertilized per tree
1 - 3	1.8 m x 1.8 m
4 - 7	3.0 m x 3.0 m
8 - 9	3.7 m continuous strip
10 +	whole area

If necessary, the soil pH needs to be raised to pH 6 by liming, calcitic or dolomitic depending on Mg need.

Preferred nutrient forms

N is preferably applied as ammonium sulphate on alkaline soils and as ammonium nitrate on acid soils, P as single or triple superphosphate, K as muriate and Zn as sulphate.

Present fertilizer practices

USA

During the first year after transplanting a basal dressing of 56 kg/ha N, 112 kg/ha P2O5 and 448 kg/ha K2O followed by the same amounts about 2 1/2 months later. In subsequent years fertilization is based on leaf analysis. The sample, consisting of the middle leaflet pair from the mid-leaf, is collected 56-84 days after catkin fall. For orchards in production, the basal application is 112-280 kg/ha N, 50-112 kg/ha P2O5 and 112-280 kg/ha K2O.

In light soils, the applied N - K2O ratio should be 1:1; whereas, in heavy soils, the ratio should be 1.5:1 or greater. The higher N application should be adequately irrigated by flood or sprinkler. In acid soils and when needed, zinc is soil-applied (20 kg/ha Zn) but, in all regions, zinc is usually applied in foliar sprays [2.4 g ZnSO4 with UAN (1.3 ml) or KNO3 (3.7 g) per litre of water]. These sprays are applied at 15 day intervals from budbreak until after shoot elongation ceases (2 or 3 1/2 months for mature and young trees respectively).

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Pineapple *(Ananas comosus (L) Merr.)*

French: Ananas; Spanish: Ananas, pina; Italian: Ananasso; German: Ananas

Crop data

Planted late summer/early autumn.
Flowers 17-18 months after planting (plant crop) and again 12-14 months later (ratoon crop).
Harvested 21-23 months after planting (plant crop) and again 12-15 months later (ratoon crop).
Forced flowering 12 months after planting advances maturity by 2 to 3 months.

Plant density: 40 000 to 45 000 plants per ha; 2-row bed.
Preferably grown on light to medium-textured soils, pH 4.5 to 6.5.

This crop requires good surface and internal drainage and tolerates drought, but benefits very much from mulching and irrigation.

Nutrient demand/uptake/removal

Yield t/ha	Source	Uptake or removal	kg/ha				
			N	P2O5	K2O	MgO	CaO
100	Cowie, 1951	uptake	123	34	308	-	-
81	Stewart et al, in Py, 1956	uptake	574	126	1 631	-	-
		removal	67	19	238	-	-
55	Martin-Prével, 1961	uptake	205	58	393	42	121
		removal	43	17	131	10	17

Plant analysis data

Plant analysis data - Macronutrients							
Variety & plant part	Stage of growth	Source	% of dry matter				
			N*	P	K	Mg	Ca
Smooth Cayenne; yougest mature leaf, white basal part	-	Samuels & Gandia-Diaz, 1960	1.6-1.9 (OS)	0.16-0.20 (OS)	1.8-2.5 (OS)	-	-
ditto	9 months old	Hernandez-Medina, 1969	-	-	-	0.20 (CVY)	-
ditto	Grand growth (summer)	Su, 1969	1.4 (CVY)	0.18-0.20 (AD)	3.4 (CVY)	0.28 (CVY)	-
	15 months old	ditto	1.75 (CVY)	-	-	-	-
Red Spanish; yougest mature leaf, white basal part	22 months old	Cibes & Samuels, 1958	-	-	-	0.25 (OS)	0.27 (OS)

ditto	-	Samuels & Gandia-Diaz, 1960	1.7-2.2 (OS)	0.20-0.25 (OS)	3.5-4.0 (OS)	-	-
OS = Optimum supply; CVY = Critical value (yield); AD = Value observed where application was adequate. * Note that it is better to diagnose the nitrogen status on the basis of leaf colour; (optimum, intense green; excess, dark green; deficiency, pale or yellowish green) rather than leaf N analysis for which the critical value varies widely under different conditions and at different stages of growth.							

Plant analysis data - Micronutrients						
Variety & Plant part	**Stage of growth**	**Source**	**ppm dry matter**			
			B	**Mn**	**Zn**	**Cu**
Smooth Cayenne; yougest mature leaf, white basal part	Lyman & Dean, 1942	-	-	16-44 (OS)	-	
Red Spanish ; Youngest mature leaf, white basal part	22 months old	Cibes & Samuels, 1958	14 (OS)	14 (OS)	-	-
ditto	ditto	Steyn, 1961	-	-	-	9-12 (OS)
OS = Optimum supply						

Fertilizer recommendations

Organic manure or compost, where available, should be incorporated into the soil a week before planting. If the amount is small, it should be applied only around the plant row.

Liming is detrimental to pineapple except on extremely acid soils.

Mineral fertilizers:

Type of crop	**Recommended rate of nutrients, kg/ha**		
	N	**P2O5**	**K2O**
Plant crop (in 5-6 dressings)	500-650	0-180	300-600
Ratoon crop (in 3 dressings)	220-300	nil	100-200
Little potash is needed on soils with over 140 ppm exchangeable K.			

Timing of fertilizer dressings:

On medium-textured soils or where plastic mulch is used:			
Timing	**N**	**P2O5**	**K2O**
Plant crop			
Before planting (early autumn)	25 %	100 %	40 %
Early spring (at 6 months)	10 % (in solution with insecticide)		
Early summer (at 9 months)	20 %		30 %
Early autumn (at 12 months)	15 %		10 %
Late autumn (at 14 months)	10 %		
Early spring (at 18 months)	20 %		20 %

Ratoon crop			
Late summer (after hravest of plant crop)	40 %		60 %
Mid-autumn	20 %		
Early spring	40 %		40 %

On light soils without plastic mulching:			
Timing	**N**	**P2O5**	**K2O**
Plant crop			
Within 1 months of planting	15 %	33 %	20 %
Early spring (at 6 months)	20 %	33 %	20 %
Early summer (at 9 months)	20 %	33 %	30 %
Early autumn (at 12 months)	15 %		10 %
Late autumn (at 14 months)	10 %		
Early spring (at 18 months)	20 %		20 %
Ratoon crop			
Late summer (after harvest of plant crop)	40 %		60 %
Mid-autumn	20 %		
Early spring	40 %		40 %

From 12 months after planting, all fertilizers are applied to the axils of old leaves where the morning dew collects.

Where plastic mulch is used, the "third" dressing may be incorporated in basal dressing, which should then be broadcast and cultivated in.

Preferred nutrient forms

Pineapples respond better in both yield and quality to ammonium and urea N than to nitrate N, and to potassium sulphate than to the chloride. On sandy soils urea is inferior to ammonium sulphate.

Other nutrients

Mg-containing fertilizer is required where soil exchangeable Mg is below 70 ppm, magnesium-containing fertilizer should be applied to 50 to 120 kg/ha MgO. The optimum MgO : K2O ratio reported to be 1.0:2.5.

Where **micronutrient** deficiency occurs:

For inter-fruitlet corking, 0.3 % borax spray, 2 or 3 times at later stage of grand growth.

For leaf yellowing due to iron deficiency, 2-3 % ferrous sulphate spray monthly or as required (preferably mixed with urea).

For "crook-neck" and yellow leaf spots, one or two applications of 1 % zinc sulphate spray. For "green die-back" Bordeaux mixture at a rate containing 7 to 11 kg/ha copper sulphate.

Current fertilizer practices

Taiwan

For expected yield of 55-65 t/ha (plant crop) and 45-55 t/ha ratoon crop on medium-textured soil without plastic mulch or flower-forcing:

Timing	Application rates kg/ha		
	N	**P2O5**	**K2O**
Plant crop			
Basal	150	120	180
At 6 months	60*		
At 9 months	120		135
At 12 months (applied to lower leaf axils from this time on)	90		45
At 14 months	60		
At 18 months	120		90
Ratoon crop			
Immediately after harvest plant crop	120		90
2 months later	60		
At initial flowering	120		60
* Poured onto the plants in solution with insecticide			

With plastic mulch on bed, and flower-forcing treatment in early autumn (11-12 months after planting) :

Timing	Application rates kg/ha		
	N	**P2O5**	**K2O**
Plant crop			
Basal	180	100	200
At 1 month	60 (in solution)		
At 9 months	120		80
One week after flower-forcing treatment	120		40
After flowering	120		80

India

official recommendations in 5 states					
State	**Plants/ha (spacing)**	**N**	**P2O5**	K2O	FYM t/ha
		kg/ha (g/plant)			
Assam	44 000	530 (12)	90 (2)	530 (12)	15
Karnataka	(0.6 x 0.3 m)	350	130	440	30
Kerala	40 000	320	160	320	25
Tamil Nadu*	(0.6 x 0.3 m)	500	40	660	40-50
West bengal	40 000	400 (10)	200 (5)	400 (10)	20
* ZnSO4 and FeSO4 spary where required Source : Tandon, 1987					

In Assam FYM at last ploughing, all P and K and half of the N as band or line placement, the remaining half of the N as foliar spray (450 g urea in 10 l of water) in 6 applications. In other states N and K are applied in 2- equal splits.

Queensland Australia
For both plant and ratoon crops:

Timing	Application rates kg/ha		
	N	P2O5	K2O
Basal	-	50	450
At 2-monthly intervals therafter	15-25 (as 10% urea spary*)		
* 15 kg/ha N for young plants, rising to 25 kg/ha N for fully grown plants.			

Where leaf yellowing due to iron deficiency occurs, 3 % ferrous sulphate is included in the urea spray.

Where "crook-neck" occurs, 30 kg/ha each of zinc sulphate and copper sulphate are added to the basal fertilizer.

Further reading

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Author: N.R. Su, Council of Agriculture, Executive Yuan, Taipei, Taiwan

Pistachio *(Pistacia vera L.)*

French: Pistachier; Spanish: Alfoncigo, pistachero; Italian: Pistachio; German: Pistazie

Crop data

Perennial. Harvested part: fruits.
Scions are planted in autumn at 8-10 m spacing and are grafted about 3 years later.
Flowers end-April (in Turkey). Fruits 8-10 years after planting.
Male pollinators are required. Harvested end-August.
Plant density: 100-150 trees per hectare.
Pistachio has a deep and strong rooting system. It can be grown on all kinds of soils but prefers relatively deep, light soils with high lime content (pH 7.0-8.0). Sensitive to soil salinity. Strong and healthy growth is obtained in areas with cool winters and long, hot dry summers. In dry areas, additional irrigation is needed for good performance.
Major commercial production is in the southern part of Turkey.
Because of the wide spacing and the long period before coming into bearing, it is usual to intercrop with, e.g., grapevines (in arable conditions).

Nutrient demand/uptake/removal

Nutrient demand/uptake - Macronutrients					
Yield kg/ha	**Source**	**kg/ha**			
		N	**P2O5**	**K2O**	**CaO**
1 000	Woodroof, 1967	30	12	15	3

Plant analysis data

Leaf analysis data (optimum supply) - Macronutrients							
Plant part	**Stage of growth**	**Source**	**% of dry matter**				
			N	**P**	**K**	**Mg**	**Ca**
Mid-terminal compound leaves	At the end of July	Tekin et al, 1985	2.0	0.09	1.0	0.7	2.9

Leaf analysis data (optimum supply) - Micronutrients							
Plant part	**Stage of growth**	**Source**	**ppm dry matter**				
			Fe	**Mn**	**Zn**	**Cu**	**B**
Mid-terminal compound leaves	At the end of July	Tekin et al, 1985	105	35	18	48	140

Fertilizer recommendations

Organic materials such as farmyard manure should be applied at around 30 t/ha if the soil organic matter is below 2 per cent.

Mineral fertilizers and/or organic materials should be ploughed under and thoroughly cultivated in before planting, at rates based on topsoil (0-20 cm) and subsoil (20-40 cm) analysis (as indicated below):

Soil P (ppm)	Application rate (kg/ha P2O5)	Soil K (ppm)	Application rate (kg/ha K2O)	Soil Mg (ppm)	Application rate (kg/ha MgSO4)
0- 7	100-150	0-100	150-200	0-25	300-400
7-14	50-100	100-200	70-150	25-50	200-300
14-20	50	> 200	Nil	50-100	100-200
> 20	Nil				

Nitrogen fertilizer is not required before planting.

Established plantations up to 10 years old

Nitrogen fertilizer in March each year at 20 g/tree N (leaves are small and pale green when nitrogen is deficient). Other nutrients on the basis of soil analysis made at 3-4 year intervals.

In arable conditions the land should be cultivated in spring; cultivation and manuring help to retain moisture in the soil, maintain a good dust mulch and keep weeds under control.

Mature trees over 10 years old

Fertilizer applications should be based on leaf and soil analysis. Leaf samples should be taken at 3-year intervals and soil samples every 5 years. Where farmyard manure is used, the mineral fertilizer dressings may be reduced accordingly.

Nitrogen fertilizer should be applied in March each year, broadcast under the branches. Ammonium sulphate may be used where the soil is too alkaline.

Leaf N (% of dry matter)	Application rate (g N per tree)¦
< 1.8	500
1.8-2.2	400
2.2-2.5	300
> 2.5	Nil

Phosphate fertilizer should be applied at the beginning of March, every 2-3 years, placed in a band at least 30-40 cm deep at the drip line of the tree.

Leaf P (% of dry matter)	Application rate (g P2O5 per tree)¦
< 0.06	500
0.06-0.10	400
0.10-0.13	300
> 0.13	Nil

Potassium fertilizer may be applied in a band together with nitrogen, every 2-3 years. Sulphate is preferable to muriate.

Leaf K (% of dry matter)	Application rate (g K2O per tree)
< 0.4	750
0.4-0.8	500
> 0.8	Nil

Deficiencies of some micronutrients (Zn, Fe, Mn) may occur in future in S.E. Turkey. Iron deficiency may be remedied by applying Fe-EDDHA at 200-300 g/tree in March. For zinc and

manganese deficiencies, foliar spraying at petal fall with 0.4 per cent zinc and manganese sulphate is recommended.

Further reading

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Author: C. Genc, Atatuerk Horticultural Research Institute, Yalova-Istanbul, Turkey

Strawberry *(Fragaria x ananassa Duch.)*

French: Fraisier; Spanish: Fresón; Italian: Fragola; German: Erdbeere

Crop data

Fragaria x ananassa, with large fruit (10-40 g), is by far the most important cultivated species, but other species (e.g. *Fragaria vesca*, with small but better flavoured fruits), may also be grown.

Perennial but, as a crop, more frequently exploited for one year.

It is a typical rosette plant; the petiolate leaves and the inflorescence stalk originate from the short stem, rich in brownish scales (crown). Some axillary buds can develop into branches with long internodes (stolons) which produce a new leaf rosette and adventitious roots at the nodes. The stolons are used for propagation. The root system is mainly located in the upper 20 cm soil layer.

During the rest period strawberry can tolerate very cool conditions (down to -6°C); also during the vegetative and reproductive phases thermal requirements are rather low: zero growth point is near 5 °C; optimum conditions are 10-13 °C at night and 18-22 °C during daytime respectively.

In the Mediterranean area, short day cultivars are planted in summer or autumn; flowering occurs in winter and spring; harvest starts in spring.

Nutrient demand/uptake/removal

Nutrient removal per metric ton of fruit can amount to 6-10 kg N, 2.5-4.0 kg P2O5 and 10 or more kg K2O, with a nutrient ratio of about 2.5:1.0:3.5. Uptake amounts to about 200-250 kg/ha N, 100-150 kg/ha P2O5 and 400 kg/ha K2O.

Plant analysis data

A concentration in the petioles in the range 3 000-10 000 ppm nitrate N (on a dry matter basis) indicates an optimal nutritional status. At higher concentrations vegetative activity, mainly leaf growth, is stimulated to the detriment of fruit production.

Fertilizer practice

Factors to be taken into account are:

- the root system is rather weak and shallow (70 % in the upper 7 cm soil layer), just below the surface in mulched crops;
- the plant prefers light, rather sub-acid soil (pH 5.5-6.5) and suffers from brackish water and/or soil (from both Cl and Na);
- it is more or less susceptible, depending on the cultivar, to chlorosis due to iron deficiency;
- floral bud differentiation, on which the yield depends, occurs early and depends on the nutrient reserves of the plant;
- nutrient absorption is at a maximum during flowering.

Correct practice normally involves:

- use of manure or peat to improve soil fertility and to control pH;
- application of iron salts to calcareous soils to prevent plant chlorosis;

- application of 100 kg/ha N, 100 kg/ha P2O5 and 200 kg/ha K2O before planting;
- topdressing with N and K2O (50-100 kg/ha), throughout the cycle, normally by fertirrigation.

Slow release fertilizers have proved promising in meeting plant nutrient requirements and reducing the risk of leaching.

Fertilizers which increase the content of Na (e.g. sodium nitrate) or of Cl (e.g. potassium chloride), or of Ca (e.g. calcium nitrate), should be avoided due to the risk of chlorosis.

Application of chelates, or of foliar sprays with iron compounds, may be of use in checking negative effects caused by immobilization of Fe.

For further general information see introduction to chapter 'Temperate Vegetable Crops'.

Further reading

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See also references listed at the end of 'Temperate Vegetable Crops'.

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Tropical Fruit Trees

Mango *(Mangifera indica L.)*

French: Manguier; Spanish: Mango; Italian: Mango; German: Mango

Crop data

Perennial.
Planting at onset of rainy season.
Fruits 2 to 4 years after planting.
Fruits harvested 15 to 20 weeks after flowering.

Plant density: 123 plants/ha.
Preferably grown on well drained, deep, loamy soils, pH 5-6.

Well suited to regions with distinct wet and dry seasons, with at least 4-5 months dry period.

Plant analysis data

Plant analysis data (optimum supply) - Macronutrients							
Plant part	**Stage of growth**	**Source**	**% of dry matter**				
			N	**P**	**K**	**Mg**	**Ca**
4-7 month old leaves from midshoot of flush	n.a	Young & Koo, 1969	1.0-1.5	0.08-0.175	0.3-0.8	0.15-0.4	2.0-3.5
5th leaf from base of current flush	after harvest	Tengku Ab. Malek et. Al., 1989	1.0-1.5	0.08-0.18	0.3-0.8	0.15-0.4	2.0-3.5

To produce fruits of good quality and free from internal tissue breakdown ('softnose' disorders), the N/Ca and K/Ca ratios should be less than 0.5 and 0.2 respectively.

Fertilizer recommendations

10 kg organic manure and 50 g rock phosphate per plant should be put into 30 cm3 planting holes.

Plant age (years)	Rate per tree (kg per application)	Applications per year
1	0.15	4
2	0.25	4
3-5	1.0	2
6-8	1.5	2
> 8	2.0	2
In years 1 and 2, 15:15:15 or 14:13:9:2.5 (N:P2O5:K2O:MgO) fertilizers are used and from year 3 onwards either 12:12:17:2 or 12:6:22:2. Organic manures at 10-15 kg/plant/year are also generally given in split applications.		

Papaya *(Carica papaya L.)*

French: Papayer; Spanish: Papayero; Italian: Papajo; German: Papaya

Crop data

Perennial.
Planting generally at onset of rainy season.
Flowers 4-5 months after sowing.
Fruit matures about 5 months after flowering.

Plant density: 1 100 to 2 000 plants/ha.
Prefers a deep, friable and well drained soil, pH 6.0 - 6.5.

The crop is adapted to tropical and subtropical climates and is generally irrigated.

Plant analysis data

Plant analysis data (17th petiole) - Macro- and micronutrients					
Stage of growth	**Source**	**% of dry matter**			**ppm**
		N	**P**	**K**	**B**
Fruiting	Awada & Long, 1971	1.45*	-	3.61*	-
Fruiting	Awada, Suehisa & Kanehiro, 1975	-	0.18*	-	-
Fruiting	Chan & Raveendranathan, 1984	-	-	-	20-30*
* Critical value yield					

Fertilizer recommendations (Malaysia)

Soil pH should be raised to pH 6.0 - 6.5 by incorporating lime in upper 25 cm.

In planting hole: 200 g soluble phosphate/plant.

Vegetative phase:
1st month - 0 g/plant (15:15:15 fertilizer)
2nd month - 50 g/plant (15:15:15 fertilizer)
3rd month - 100 g/plant (15:15:15 fertilizer)

Fruiting phase:

From 5th month onwards at 2-month intervals, 350 g 12:12:17:2 fertilizer per tree.

To overcome boron deficiency (exemplified by "bumpy" fruits) apply evenly 5 g borax per plant or weekly sprays of 0.25 % borax solution (3-4 sprays).

Durian *(Durio zibethinus Murr.)*

French: Durion; Spanish: Durion; Italian: Durie; German: Durian

Crop data

Perennial.
Planting at onset of rainy season.
Fruits 5 to 6 years after planting.
Flowering to harvesting of fruit 110-120 days.

Plant density: 87 plants/ha.

Preferably grown in undulating areas on well-drained loamy soils.
Irrigation strongly recommended.

Nutrient removal in fruit production

Nutrient removal - Macronutrients						
Yield kg/ha	Source	kg/ha				
		N	P2O5	K2O	MgO	CaO
6720	Ng & Thamboo, 1987	16.1	6.6	33.5	5.4	2.8

Fertilizer recommendations

10 kg organic manure and 250 g rock phosphate per plant should be put into 60 cm3 planting holes.

Recommended fertilizer schedule (Malaysia)		
Plant age (years)	Rate per tree (kg/application)	Applicatons per year
1	0.15	4
2	0.3	4
3	1.0	3
4	2.0	3
5	2.5	3
6	4.0	2
7	5.0	2
8	5.0	2
> 8	6.0	2
In years 1 to 5 15:15:15 or 14:13:9:2.5 fertilizers are used, and from year 6 onwards 12:12:17:2 or 12:6:22:2.		

Passion fruit or Granadilla *(Passiflora spp)*

French: Passiflore comestible, granadille; Spanish: Granadilla, maracuya; Italian: Granadiglia; German: Passionsfrucht, Maracuja

Crop data

Woody perennial vines.
Planting at onset of rainy season.
Requires 8-12 months to produce commercial crops.
Flowering to fruit harvest: 60-90 days.

Plant density: 500- 1 500 vines/ha.
Grows well on deep (> 60 cm), fertile, well-drained loamy soils, pH 5.5-6.5.

Crop is adapted to tropical and subtropical climates with high rainfall (> 1 200 mm/year)

Nutrient uptake

Amounts of nutrients taken up by roots, stem and fruit of golden passion fruit vines after 370 days (1 500 vines/ha):

Nutrient uptake (1 500 vines/ha) - Macronutrients			
Plant part	Stage of growth	Source	kg/ha

			N	P2O5	K2O	MgO	S
Roots, stems and fruit	370 days	Menzel et al, 198	205	41	221	27	25

Nutrient uptake (1 500 vines /ha) - Micronutrients							
Plant part	**Stage of growth**	**Source**	**g/ha**				
			Fe	**Mn**	**Zn**	**Cu**	**B**
Roots, stems and fruit	370 days	Menzel et al, 1989	779	2 180	316	198	295

Plant analysis data

Plant analysis data (optimum supply) - Macronutrients							
Stage of growth	**Source**	**% of dry matter***					
		N	**P**	**K**	**Mg**	**Ca**	**S**
Before new growth or flush	Menzel et al, 1989	4.75-5.25	0.25-0.35	2.0-2.5	0.25-0.35	0.5-1.5	0.2-0.4
* Last fully expanded leaf from strong vines							

Plant analysis data (optimum supply) - Micronutrients						
Stage of growth	**Source**	**ppm dry matter***				
		Fe	**Mn**	**Zn**	**Cu**	**B**
Before new growth or flush	Menzel et al, 1989	100-200	50-200	45-80	5-20	25-100
* Last fully expanded leaf from strong vines						

Fertilizer recommendations

Soil pH should be raised to pH 5.5 or more by liming.

Recommended fertilizer schedule (Malaysia)		
Plant age (years)	**Rate per vine (g/application)**	**Applications per year**
1	300	3
2	450	2
3	450	2
> 3	450	2
15:15:15 fertilizer is used.		

Litchi *(Litchi chinensis Sonn.)*

French: Litchi; Spanish: Litchi; Italian: Litchi; German: Litchi

Crop data

Planted when wheather is neither too wet nor too dry.
Fruits 3 to 5 years after planting.
Harvested 5 to 6 months after flowering.

Plant density: 100 plants/ha
Grows well in a deep, well-drained loamy soil rich in organic matter, pH 5.5 to 6.0.

Crop thrives well under moist subtropical climate. Frost in winter and dry heat in summer are limiting factors for its successful cultivation. Can be grown without irrigation if rainfall is more than 1250 mm and is well distributed throughout the year.

Fertilizer recommendations (India)

25 kg farm yard manure, 2 kg bone meal and 0.3 kg muriate of potash per plant should be put into 1 m3 planting holes.

Recommended fertilizer schedule (India)				
Plant age (years)	**kg/plant/year**			
	FYM	**CAN**	**SSP**	**MOP**
1- 3	10-20	0.3-1.0	0.2 -0.6	0.05-0.15
4- 6	25-40	1.0-2.0	0.75-1.25	0.2 -0.3
7-10	40-50	2.0-3.0	1.5 -2.0	0.3 -0.5
> 10	60	3.5	2.25	0.6

The farm yard manure, superphosphate and muriate of potash are all applied in autumn while the calcium ammonium nitrate is given in two equal applications, i.e. in mid-February and after fruit harvest.

Rambutan *(Nephelium lappaceum L.)*

French: Litchi chevelu; Spanish: Rambutan; German: Rambutan

Crop data

Perennial.
Planting at beginning of rainy season.
Fruits about 3 years after planting.
Fruits mature about 15-18 weeks after flowering.

Plant density: 118 plants/ha.
Grows well on well drained, loamy soils, pH 5.0 to 6.0.
Generally not irrigated.

Nutrient removal in fruit production

Nutrient removal - Macronutrients						
Yield per ha	**Source**	**kg/ha**				
		N	**P2O5**	**K2O**	**MgO**	**CaO**
6720 kg fruits + 620 kg twigs	Ng & Thamboo, 1967	15.0	4.7	14.0	4.4	8.3

Fertilizer recommendations

10 kg organic manure and 250 g rock phosphate per plant should be put into 45 cm3 planting holes.

Recommended fertilizer schedule (Malaysia)		
Plant age (years)	**Rate per tree (kg/application)**	**Applications per year**
1	0.3	3

2	0.5	3
3	1.0	3
4	1.5	2
5	2.0	2
6	2.5	2
7-10	3.5	2
11-14	4.5	2
> 14	5.0	2
In years 1 to 3 15:15:15 or 14:13:9:2.5 fertilizers are used, and from year 4 onwards 12:12:17:2 or 12:6:22:2.		

Starfruit or Carambola *(Averrhoe carambola L.)*

French: Carambole; Spanish: Carambole; German: Sternfrucht

Crop data

Perennial.
Planting at beginning of rainy season.
Fruits 2 years after planting.
Fruits mature 60-70 days after flowering.

Plant density: 278 plants/ha.
Prefers well-drained, light to medium textured soils, pH 5.0 - 6.5.
Irrigation recommended.

Fertilizer recommendations

Soil pH should be raised to 5.0 - 6.5 by liming. 10 kg organic manure and 200 g rock phosphate per plant should be put into 60 cm3 planting holes.

Recommended fertilizer schedule (Malaysia)				
Plant age (years)	**Mineral fertilizer***		**Organic manure**	
	kg/tree at each application	**Applications per year**	**kg/tree at each application**	**Applications pêr year**
1	0.3	6	10	1
2	1.0	4	10	1
3	2.0	4	10	1
4	2.5	4	15	1
5	3.5	4	15	1
6	4.0	4	20	1
> 6	5.0	4	20	1
* In year 1, 15:15:15 fertilizer is used, and from year 2 onwards 12:12:17:2.				

Guava *(Psidium guajava L.)*

French: Goyavier; Spanish: Guyaba; Italian: Guaiava; German: Guave

Crop data

Perennial.

Planted at onset of rainy season.
Fruits within a year from planting.
Fruits harvested about 3 months after flowering.

Plant density: 1 000 plants/ha.
Generally adapted to wide ranges of soil types and pH but does well on deep and fertile loamy soil.
Performs well with irrigation.

Guava can be successfully grown in tropical and subtropical climates with good rainfall (2 000 - 3 000 mm/year).

Fertilizer recommendations

Soil pH should be raised to 5.0 - 6.5 by liming. 10 kg organic manure and 200 g rock phosphate per plant should be put into 60 cm3 planting holes.

Recommended fertilizer schedule (Malaysia)				
Plant age (years)	**Mineral fertilizer***		**Organic manure**	
	kg/tree at each application	**Applications per year**	**kg/tree at each application**	**Applications per year**
1	0.25	12	5	2
2	0.5	12	5	1
3	0.5	12	5	1
> 3	0.5	12	5	1
* In year 1, 15:15:15 fertilizer is used and from year 2 onwards 12:12:17:2.				

Sapota or Sapodilla [Ciku] *(Achras zapota L.)*

French: Sapotille; Spanish: Zapota, Sapotillo; Italian: Sapota; German: Sapotille

Crop data

Perennial.
Planting generally at beginning of rainy season.
Fruits 2-3 years after planting.
Fruits harvested 10-12 months after flowering.

Plant density: 123 plants/ha.
Prefers a well drained light to medium textured soil.
Generally not irrigated.

Adapted to warm, moist weather and grows in both dry and humid areas.

Fertilizer recommendations

5 kg organic manure and 200 g rock phosphate per plant should be put into 60 cm3 planting holes.

Recommended fertilizer schedule (Malaysia)				
Plant age (years)	**Mineral fertilizer***		**Organic manure**	
	kg/tree at each application	**Applications per year**	**kg/tree at each application**	**Applications per year**
1	0.15	6	4	2

2	0.3	6	8	2
3	0.75	4	8	2
4	1.5	4	-	-
5	2.5	4	-	-
6	3.0	4	-	-
> 6	3.5	4	-	-
*. In the first 2 years, 15:15:15 fertilizer is used and from year 3 onwards 12:12:17:2.				

Further reading

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Author: P. Raveendranathan, Fruit Research Division, Malaysian Agricultural Research and Development Institute (MARDI), Kuala Lumpur, Malaysia

Walnut *(Juglans regia L.)*

French: Noyer; Spanish: Nogal; Italian: Noce; German: Walnuss

Crop data

Perennial. Harvested part: nuts
Transplanted in dormant season.
Flowers 4-5 years after transplanting.
Harvested 4 1/2-5 months after flowering.

Plant density: 120-172 trees per ha. As trees increase in size, they are systematically removed and the density is eventually reduced to 30 trees per ha. Preferably grown on a deep well-drained silt loam soil, pH 6-8.

Adapted to wide climatic conditions, but the economically important Persian walnut is best adapted to long, warm, dry summers and mild winters. Irrigated in arid climates. Persian walnut is grown primarily in California (USA) and Europe.

Nutrient demand/uptake/removal

Nutrient removal - Macronutrients						
Yield	**Source**	**kg**				
		N	**P2O5**	**K2O**	**MgO**	**CaO**
0.9 t hull	0.9 Painter & Raese, 1965	7.6	1.6	10.4	1.5	6.2
0.9 t shell	Painter & Raese, 1965	2.0	0.2	4.8	0.5	2.8
0.9 t kernel	Painter & Raese, 1965	28.3	8.9	6.0	2.7	0.8

Plant analysis data

Plant analysis data - Macronutrients								
Plant	**Stage of growth**	**Source**	**% of dry matter**					
			N	**P**	**K**	**Mg**	**Ca**	**S**
Leaf	Mid-season	Serr, 1961b	>2.5 (OS)	>0.11 (OS)	>1.0 (OS)	0.30 (OS)	-	-
			≤2.5 (D)	≤0.10 (D)	≤1.0 (D)	0.20 (D)	-	-
Leaf	Mid-season	Serr, 1961a	-	0.12 (OS)	-	-	1.25 (OS)	-
			-	0.09 (D)	-	-	-	-
Leaf	Mid-season	Proebsting & Tate, 1964	-	-	-	-	-	0.21 (OS)
Leaf	Mid-season	Proebsting & Serr, 1966	2.5 (D)	-	1.3 (OS)	-	-	-
D = Deficiency; OS = Optimum supply								

Plant analysis data - Micronutrients						
Plant part	**Stage of growth**	**Source**	**ppm dry matter**			
			Mn	**Cu**	**Zn**	**B**
Leaf	Mid-season	Serr, 1961b	30 (OS)	>4 (OS)	20 (OS)	>35 (OS)
			20 (D)	3 (D)	15 (D)	≤35 (D)
Leaf	Mid-	Vanselow, 1945	35 (OS)	-	-	-

	season					
			7 (D)	-	-	-
Leaf	Mid-season	Proebsting & Serr, 1966	-	-	-	44 (OS)
D = Deficiency; OS = Optimum supply						

Fertilizer recommendations

Walnut plantations may be clean cultivated or gussed down, so it may or may not be practicable to incorporate the fertilizer into the soil. Fertilizer is usually applied in a single application. The time of application is often a matter of convenience, but it should be before budbreak.

Generally, only N is applied on an annual basis. In California, Zn is the second most common problem followed by lime-induced chlorosis: boron toxicity (400-1 000 ppm B in the leaf) is fairly common.

Preferred nutrient forms

N is preferably applied as ammonium sulphate (for alkaline soils) or ammonium nitrate (for acid soils), P as triple superphosphate and K as sulphate.

Present fertilizer practices

USA

Age (years)	**kg/tree**			
	N*	**P2O5**	**K2O**	**B**
Young trees				
1 - 2	0.03	5	6	0.05
3 - 4	0.45	5	6	0.05
5 - 7	0.68	5	6	0.05
Mature trees	**kg/ha resp. tree**			
	170	10/tree	280	6
* N is applied annually, other nutrients only when problems occur.				

Foliar sprays of Mg (4 g MgSO4/liter water) and Mn (4 g MnSO4/liter water) are applied late May/early June (California) where required

Further reading

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SERR, E.F.: Response of Persian walnut to superphosphate. Proc. Amer. Soc. Hort. Sci. 77, 301-307 (1961)

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Cotton *(Gossypium spp.)*

French: Coton; Italian: Cottone; Spanish: Algodon; German: Baumwolle; Portuguese: Algodao

Crop data

Annual. Harvested part: seed, for fibre.
Important uses of by-products: cotton seed oil; cake, meal and hulls (for feeding); linters.

Cotton is a major world fibre crop. It is grown under a very broad range of climates, soils and cultural practices. For the most part, it is grown in tropical latitudes. However, it is cultivated as far north as 43° N in the USSR and 45° N in China. There are four separate domesticated species of cotton grown in various parts of the world which include G. arboreum L. (diploid), G. herbaceum L.(diploid), G. hirsutum L. (tetraploid) and G. barbadense L. (tetraploid). G. arboreum and G. herbaceum are commonly referred to as Old World diploids due to their ploidy condition and origins being traced to locations in the Old World. Both G. hirsutum and G. barbadense evolved in the New World and are referred to as allelotetraploids. There are literally hundreds of varieties of these various domesticated species which have been developed for production around the world. G. hirsutum is the most extensively developed species and represents approx. 90 % of total world production.

Sown at an average soil temperature of at least 18 °C; between April 1 and May 25 (Georgia, USA), spring (China), early spring (March/Egypt), from May/June to August/September (India, North to South).
Sowing by dibbling or in rows, 18 - 24 kg/ha (Brazil), 25 - 40 kg/ha (USA), upto 50 - 150 kg/ha (Egypt).

Plant density (depending on spacing of rows and plants within rows) 55 000 plants/ha (India), 45 000 - 67 500 plants/ha (China/Yellow and Yangtze River) to 120 000 - 150 000 plants/ha (North China, short season), 60 000 - 90 000 plants/ha (USA), 55 000 - 125 000 plants/ha (Brazil) and from 100 000 (Egypt, traditional) to 168 000 plants/ha (intensive, recommended).
Temperature: Mainly tropical, grown at temperatures 25 - 35 °C.

Adapted to semi-arid conditions; in Brazil nearly all the cotton area is unirrigated; in India 70 % is rainfed; in China and Egypt it is irrigated (10 - 12 irrigations at 15 day intervals), also in USA (e.g. water use in Arizona, desert Southwest, equivalent to 1 040 mm).
Preferably grown on light to medium soils, pH 6 - 8; highly sensitive to soil acidity and water or soil salinity:

Influence of water (EC^*_w) or soil (EC_e) salinity on yield potential of cotton					
	Yield potential, %				
	100	90	75	50	0
	ohm^{-1}				
Ec_e	7.7	9.6	13	17	27
Ec_w	5.1	6.4	8.4	12	18
* EC = electrical conductivity					
Source: Adapted from Ayers & Westcot, 1985					

Flowers with direct seeding 75 - 80 days (China) to 80 - 120 days (Egypt) after sowing.

Harvested 140-160 (China) to 210 days (Egypt) after sowing.

Nutrient demand/uptake/removal

Depends on variety and location under both irrigated and rainfed conditions. G. arboreum requires less nutrients than G. hirsutum; hybrids require more than single varieties; short compact variety types are more efficient users of nutrients than tall late varieties.

Nutrient uptake/removal - Macronutrients									
Country	**Source**	**Variety**	**Yield of ginned cotton kg/ha**	**kg/ha**					
				N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
Brazil	Malavolta, 1987	G. hirsutum	(2 500)**	156	36	151	40	168	64
China*	¦An Yang, 1985	G. hirsutum	1 421	183	64	156	-	-	-
	An Yang, 1985	‘’	1 115	153	53	112	93	204	
	An Yang, 1985	‘’	941	128	45	90	-	-	-
USA	Mullins & Burmester, 1988	G. hirsutum	per 100 kg lint	19.9	5.2	18.4	-	-	-
* Nutrient uptake = All aerial parts plus part of roots - ** Seed cotton									

Nutrient uptake/removal - Micronutrients								
Country	**Source**	**Variety**	**Yield of ginned cotton kg/ha**	**g/ha**				
				Fe	**Mn**	**Zn**	**Cu**	**B**
Brazil	Malavolta, 1987	G. hirsutum	(2 500)*	2 960	250	116	120	320
China	An Yang, 1985	G. hirsutum	1 115	5 000	254	397	71	205
* Seed cotton								

Plant analysis data

Several analyses have been reported under different agroclimatic conditions and for different varieties. General sufficiency ranges for upland cotton (G. hirsutum) in the USA are given in the following tables:

Plant analysis data, sufficiency range - Macronutrients							
Country (Variety, time)	**Sampling**	**% of dry matter**					
		N	**P**	**K**	**Mg**	**Ca**	**S**
Brazil (Begin flowering)	Leaf next to fully develpd. flower	3.5-4.0	0.20-0.25	1.40-1.60	0.40-0.50	3.00-4.0	0.20-0.30
USA, Georgia (June or July)	Top mature leaves (20-30)	3.50-4.50	0.30-0.50	1.50-3.00	0.30-0.90	2.00-3.00	0.25-0.80

Plant analysis data, sufficiency range - Micronutrients						
Country (Variety, time)	Sampling	**ppm dry matter**				
		Fe	Mn	Zn	Cu	B
Brazil (Begin flowering)	Leaf next to fully develpd. Flower	60-80	20-40	10-15	30-40	20-30
USA, Georgia (June or July)	Top mature leaves (20-30)	50-250	25-300	20-200	5-25	20-80

Plant analysis data at different growth stages (India) - Macronutrients					
Variety	**Growth stage**	**Plant part**	**% of dry matter**		
			N	**P**	**K**
G. arboreum	Flower bud	Leaf	1.40	0.34	2.90
	stage (squaring)	Stem	0.42	0.18	1.70
	Flowering	Leaf	1.30	0.29	2.80
		Stem	0.26	0.13	1.30

	Harvesting	Leaf	1.08	0.56	2.70
		Stem	0.15	0.17	1.50
G. hirsutum	Flower bud	Leaf	2.67	0.41	2.10
	stage (squaring)	Stem	1.10	0.31	2.10
	Flowering	Leaf	2.30	0.40	1.98
		Stem	1.00	0.20	2.50
	Harvesting	Leaf	1.91	0.31	1.85
		Stem	0.77	0.17	1.90

Nitrate-N levels of the petioles at different growth stages in Western USA (Acala and other upland varieties) should be as follows:

Plant analysis data, petiole nitrate-N levels during growing season, USA (Variety: Acala, SJ-2)				
Range	**ppm NO_3^--N**			
	First squares	**First flowers**	**First bolls**	**First open bolls**
Sufficient for growing season	18 000	12 500	7 000	3 500
Deficient before end of growing season	12 000	7 500	3 000	1 500
Source: Bassett & George, 1983.				

Adequate concentration of NO_3^--N*				
Variety	**ppm NO_3^--N**			
	First quares	**First flowers**	**First bolls**	**First open bolls**
Pima	>10 000	>8 000	>4 000	>2 000
Delta Pine	>18 000	>14 000	>8 000	>4 000
* Contents of petioles from uppermost, fully developed leaves from upland and Pima cotton grown under irrigated conditions (Arizona, S.W. USA).				
Source: Pennington & Tucker, 1984				

Nutrient deficiency symptoms

N - lack of N reduces yield and quality; visible symptoms are light green plants, lower leaves yellow, drying to brownish colour.

P - deficient plants are short and small, growth is retarded, and maturity is postponed; typical symptoms include loss of chlorophyll and development of red discoloured leaves.

K - deficient plants are susceptible to stem blight leading to premature senescence; typical symptoms are bronzing and marginal necrosis of leaves adjacent to developing bolls; plants develop fully necrotic leaves and premature defoliation occurs.

Mg - typical symptoms are lower purplish red leaves with green veins.

S - deficient plants are dwarfed with green leaves.

Mn - deficient leaves are yellowish or reddish grey with green veins.

Zn - small leaves with interveinal chlorosis and shortened stems giving the plant a small bushy appearance; slow in development; leaves lose green colour and produce necrotic spots.

B - deficient plants will produce clittelums, and dieback of terminal buds which do not blossom and drop heavily in production; young leaves yellowish green, flower buds chlorotic.

Fertilizer recommendations

Due to the extensive variability in the conditions under which cotton is produced (soils, climate, varieties, species, cultural practices, etc.) it is very difficult to generalize recommendations for cotton fertility management.

N - increases height of plants, their boll-bearing capacity and seed weight, and improves seed cotton yield. Excess N - e.g. if cotton follows a good crop of soybeans - may result in rank growth, poor fruiting, delayed maturity, difficult defoliation and boll rot.
The N requirement of N is less for Asiatic cotton types than for American types, but it is greater for hybrids. The response to N is better in irrigated than in rainfed cotton. For better N response - as well as for other nutrients - proper weed control, insect and disease control is essential.
To obtain the highest efficiency from N, one-half (or one-third) should be given at the time of sowing, and the remainder split between (i) square formation and (ii) peak flowering and boll development. Split applications (side-dressing) will better meet the crop's requirements and reduce nitrate loss.

N given before or at planting can be broadcast together with P and K and incorporated into the soil; but placement at a depth of 5 cm below and 5 cm away from the plant has given a better yield (India) than broadcasting or simple drilling.

It is also common to apply part of the N as a foliar spray (1.5 - 2.0 % urea solution) in combination with insecticides at the time of peak flowering and the boll development stage. Foliar application has been found most beneficial on rainfed cotton.

P - improves root development, water use efficiency, the energy balance and the weight, oil and protein contents of the seed as well as fibre quality. The response to P (as well as to K) is normally better if it is given in form of NPK fertilizer (e.g. in India with N:P2O5:K2O ratio 2:1:1 or 3:1:1), before or at planting, incorporated into the soil or placed 7.5 - 10 cm below the moist soil surface.

When not applied pre-planting, P (and K) should be given at the latest at thinning (in the bottom of the furrow before watering).

K - improves fibre fineness and strength, and results in early maturity of the crop. The response to K will always be better if is it applied with N and P at preplanting and incorporated into the soil. Only on sandy, permeable soils it is appropriate to give a split application of K, e.g. with a second dressing at the time of thinning.

Mg - if soil status is low apply 45 - 50 kg/ha MgO; where liming is recommended, dolomitic limestone may be used.

S - if the applied NPK fertilizer does not contain S, at least 12 - 15 kg/ha S should be given in addition (e.g. in ammonium sulphate).

Mn - on soils with pH above 5.6, application of 2.8 kg/ha elemental Mn is recommended.

Zn - soils are deficient when Zn (DTPA extraction) drops below 0.55 ppm; in this case 4.5 kg/ha of e.g. 36 % zinc sulphate should be given (possibly as foliar spray with insecticides).

B - where needed, apply 0.6 kg/ha B (if possible in two foliar applications at 0.3 kg/ha each, tankmixed with insecticide spray).

Liming - the crop is very sensitive to soil acidity. It has repeatedly shown a marked response to lime when applied to an acid soil, particularly if pH has dropped below approx. 5.5. For

lasting results, lime must be applied periodically. Enough should preferably be applied some months before planting; in an emergency it can also be applied shortly before planting.
It should be thoroughly mixed with the soil. Split application may be necessary where the soil is very acid and correspondingly large quantities of lime are recommended; the first application should be disced into the soil, followed by mouldboard ploughing; this can be followed by a second application and another discing. Most fields should however be limed before the soil becomes so acid that yields are reduced.
(Attention: no reduced tillage practices should be used in cotton planting unless the soil has been well limed to a depth of 20 cm.)

Preferred nutrient forms

N - may be given in ammonium, nitrate or amide form (ammonium nitrate, ammonium sulphate, urea or NP or NPK fertilizer) containing N in ammonium or ammonium nitrate form.

P - water-soluble, or water- and citrate-soluble form (e.g. superphosphate, ammonium phosphate or ammonium nitrate phosphate, NP or NPK).

K - normally as potassium chloride in NPK fertilizers; also, where sulphur is lacking, as potassium sulphate.

Present fertilizer practices

Brazil (Sao Paulo)

At planting: 10 kg/ha N, 30 kg/ha S plus P2O5 and K2O (rates according to soil status, see below), applied in the furrows.

P and K rates at planting, according to soil analysis					
P(resin) µg/cm3	**P2O5 kg/ha**	**Exchangeable K meq/100 cm3**	**CEC meq/100 cm3**		
			0-4.0	4.1-8.0	> 8.0
			K2O kg/ha		
0- 6	100	≤ 0.007	60	60*	80*
7-15	80	0.008-0.15	60	60	60*
16-40	60	0.16-0.30	40	60	60
41-80	40	0.31-0.60	20	40	60
> 80	20	> 0.60	20	20	20
* 25 kg/ha K2O sidedressed with N					

When the crop is thinned, 15 - 50 kg/ha N (rate depending on soil organic matter) sidedressed, together with 25 kg/ha K2O where required.

B is either applied at planting (0.5 - 1.5 kg/ha B) or as a sidedressing (1.2 - 1.5 kg/ha B).

Lime (usually as dolomitic limestone) is topdressed and incorporated (0 - 20 to 0 - 30 cm) at a rate to raise base saturation to 70 %.

China

Before sowing or transplanting 30 - 45 t/ha organic manure is given.

N - part applied as basal dressing, the rest as a topdressing at flowering.

P, K, B and Zn - generally given as basal dressing: (B and Zn also as foliar spray at flowering).

On soils with adequate reserves of P and K according to soil analysis:
light soils, expected yield 900 kg/ha ginned cotton:

Preplanting:	40 kg/ha N
Topdressing at square stages:	
(45 - 55 days after sowing):	40 kg/ha N
(75 - 85 days after sowing):	65 kg/ha N

medium fertile soils, expected yield 1 067 kg/ha ginned cotton:

Preplanting:	53 kg/ha N
Topdressing at flowering	
(75 - 85 days after sowing):	65 kg/ha N

fertile soils, expected yield 1 467 kg/ha ginned cotton:

Preplanting:	40 kg/ha N
Topdressing at flowering	
(75 - 85 days after sowing):	50 kg/ha N

On soils with adequate reserves of K only, according to soil analysis; expected yield 1 164 kg/ha ginned cotton:

Preplanting:	50 kg/ha N, 87 kg/ha P2O5
Topdressing at flowering	
(75 - 85 days after sowing):	61 kg/ha N

On soils needing N, P and K according to soil analysis; expected yield 1 275 kg/ha ginned cotton:

Preplanting:	50 kg/ha N, 78 kg/ha P2O5, 70 kg/ha K2O
Topdressing at flowering	
(75 - 85 days after sowing):	62 kg/ha N

B - where needed, is generally applied at a rate of 12 kg/ha borax as a basal dressing; as foliar spray, 0.1 - 0.2 % borax solution is applied 2 - 3 times at 15 days intervals.

Zn - where needed, is generally applied as a basal dressing of 22 kg/ha zinc sulphate (Zn SO4.7 H2O = 23 % Zn); it is also sprayed - from the square stage - as 0.1 - 0.2 % zinc sulphate solution.

Egypt

Where available well decomposed organic manure is applied and incorporated in the soil before planting.

Fertilizer rates (based on long term field experiments in the cotton belt):

- for provinces in the Nile Delta
  N = 145 - 180 kg/ha (except for variety G77 receiving 160 - 200 kg/ha N)
  P2O5 = 35 - 70 kg/ha

- for provinces in Middle and Upper Egypt
  N = 160 - 200 kg/ha
  P2O5 = 35 - 70 kg/ha

K may be added, where needed, at a rate of 55 - 60 kg/ha K2O.

N is given in 2 split applications
(at thinning one month after planting, and one month later)

P and K - at preplanting or also at thinning.

India

Generally recommended N rate:

	kg/ha N	
	Irrigated	**Rainfed**
Traditional varieties	100	40 - 60
Hybrids	150 - 300	100 - 120

USA

N rate is used as a guide. Where cotton follows a good crop of soybeans N-rate is reduced by 20 - 30 kg/ha; where vegetative growth has been inadequate, N rate is increased by this amount.

Soil status	**K**				
P	**Very high**	**High**	**Medium**	**Low**	**Very Low**
	kg/ha N - P2O5 - K2O				
Very high	100-0-0	100-0-0	100-0-65	100-0-100	100-0-135
High	100-0-0	100-0-0	100-0-65	100-0-100	100-0-135
Medium	100-65-0	100-65-0	100-65-65	100-65-110	100-65-135
Low	100-110-0	100-110-0	100-110-55	100-110-110	100-135-135
Very low	100-135-0	100-135-0	100-135-65	100-135-135	100-155-155
Source: Coop. Ext. Service Clemson Univ., 1989					

Present practice in Southeastern USA

N - a basic rate of 65 kg/ha N is recommended for "average" loamy sand soils; this may be modified recording to residual N from previous crop, the possibility of irrigation, and the growth of cotton in the field in previous years.

Fertilizer recommendations according to soil P and K status - Southeastern USA				
	K status			
P status	**Low**	**Medium**	**High**	**Very high**
	kg/ha N - P2O5 - K2O			
Low	65-110-110	65-110-55	65-110-0	65-110-0
Medium	65-55-110	65-55-55	65-55-0	65-55-0
High	65-0-110	65-0-55	65-0-0	65-0-0
Very High	65-0-110	65-0-55	65-0-0	65-0-0

P2O5 and K2O soil analysis results				
Rating	**Coastal Plain**		**Piedmont, Mountain and Limestone Valley**	
	kg/ha			
	P2O5	**K2O**	**P2O5**	**K2O**
Low	0- 30	0- 80	0-20	0-135
Medium	30- 65	80-190	20-45	135-280
High	65-110	190-310	45-85	280-450
Very High	>110	>310	>85	>450

On fields which have produced rank growth, the total N rate is reduced to 45 - 55 kg/ha. The petiole monitoring programme can help lower early season rates without yield reduction as a result of N deficiency. With weekly monitoring, N can be applied well before actual deficiencies may appear.

Desired pH = 6.0

Remarks:
N - on fields having previously produced plants with excess vegetation, reduce the N rate; reduce by 20 - 30 kg/ha when the crop follows soybeans or other legumes in the rotation. On fields where vegetative growth has been unsatisfactory, increase N rate by 20 kg/ha. With N rates higher than 110 kg/ha, insect control is essential. Split application of N, with one-quarter to one-third before or at planting and the rest as side-dressing, is recommended.

P - on new ground low in P, double the P2O5 rate.

K - for first year cotton following hay crops, pasture or soybeans and with soil K status low or medium, increase the K2O rate by 45 - 55 kg/ha.

With irrigation, the P2O5 and K2O rates are increased by 25 %.

Middle South USA

On sandy loam and silt loam soils the recommended N rate is 60 - 90 kg/ha N; on silty clay loams, silty clays and clays, 100 - 135 kg/ha N.
The P and K rates are based on soil analysis, soil texture, (K) and irrigation potential:

Fertilizer recommendations - Middle South USA					
Soil type	**kg/ha**				
		Irrigated		**Dryland**	
	N	**P2O5***	**K2O***	**P2O5***	**K2O***
Silt loams and sandy loams	60- 90	0-110	0-135	0-90	0-110
Clays and clay loams	100-135	0-100	0-110	0-90	0- 90
* Depending on soil analysis.					

Southwest USA

Fertilizer recommendation - Southwest USA (Texas)				
Type of cotton	**kg/ha**			**Remarks**
	N	**P2O5**	**K2O**	
Upland	120	65	45	Apply one-half N at planting time and one-half at first square. Apply P2O5 and K2O at planting time.
Pima	100	65	45	

Preplant N fertilizer guidelines based on pre-season soil NO3-N levels (Arizona):

Preplant soil test level - ppm NO3-N	**Preplant N fertilizer rate - kg/ha**
0 - 5	30 - 55
10 - 15	20 - 30
>15	0

Split applications and/or water-run applications of N prior to the peak bloom period are recommended.

Western USA (California)

Recommended N-rate related to expected yield and soil NO3-N level									
Soil* nitrate level	**Expected yield - t/ha**								
	0.8	0.95	1.1	1.2	1.34	1.48	1.6	1.75	1.88
	kg/ha N								

50	55	85	115	140	170	200	200	200	200
70	25	55	85	120	150	180	200	200	200
90	0	20	55	90	130	165	200	200	200
110	0	0	20	60	100	140	180	200	200
130	0	0	0	20	65	100	150	195	200
150	0	0	0	0	25	75	120	165	200
170	0	0	0	0	0	35	85	130	180
190	0	0	0	0	0	0	45	95	145
210	0	0	0	0	0	0	10	60	110
230	0	0	0	0	0	0	0	25	75
250	0	0	0	0	0	0	0	0	50

* Soil nitrate levels (kg/ha NO_3^--N) are determined in the top 90 cm of soil after planting and converted into kg of soil-NO_3^- per hectare (multiply the NO_3^--N ppm concentration found in the top 90 cm of soil by 9.4 and then by the bulk density of the soil, e.g. 1.5 for clays to 1.7 for sands). This calculated amount of soil NO_3^--N is used along with the yield goal to estimate the fertilizer rate required to provide sufficient N for the projected yield.
Source: Zelinski, 1986

Western USA (Arizona)

Recommendations for P	
Soil test level* - ppm P	**Fertilizer P rates - kg/ha P2O5**
< 5	30 - 55**
> 5	0 - 0
* NaHCO3 extraction - ** Banded applications recommended.	

Further reading

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Flax *(Linum usitatissimum L.)*

French: Lin fibre; Spanish: Lino; Italian: Lino; German: Flachs, Lein

Crop data

Annual harvested part: whole plant with fibre production as the main target.

Sown early spring in maritime temperate areas, mid-spring in cool temperate or continental areas, autumn in warm temperate areas.

Flowers after 900 degree-days (sum of average daily temperatures in °C, nearly 70 days after a spring sowing).

Harvested (pulled) after 1 350 - 1 500 degree-days, or approx. 90-120 days after a spring sowing.

Plant density: 2 000 plants m2.

Mostly grown on deep loams, pH 6-7.
The crop is generally unirrigated except in warm temperate areas.

The water needs are important during the period of rapid growth (from 10 cm to flowering)

Nutrient demand/uptake/removal

Nutrient removal - Macronutrients							
Yield	**Source**	**kg/ha**					
		N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
6.5 t/ha (stems + leaves + fructifications)	Eyssautier (ITL), 1990	70	12	70	7	23	6
9.3 t/ha (stems + leaves + frutifications)	Eyssautier (ITL), 1988	105 120*	20	110 155**	10	40	-
16.5 t/ha (stems + leaves + fructifications)	Eyssautier (ITL), 1987	123 123*	30	230 230**	14	64	17
* maximum uptake at early flowering - ** maximum uptake at late flowering							

Nutrient removal - Micronutrients					
Yield	**Source**	**g/ha**			
		Mn	**Zn**	**Cu**	**B**
6.5 t/ha (stems + leaves + fructifications)	Eyssautier (ITL), 1990	130	120	32	100
9.3 t/ha (stems + leaves + fructifications)	Eyssautier (ITL, 1988	-	300	-	230
16.5 t/ha (stems + leaves + fructifications)	Eyssautier (ITL), 1987	113	264	73	290
* maximum uptake at early flowering - ** maximum uptake at late flowering					

Plant analysis data

Plant analysis data - Macronutrients								
Plant part	**Stage of growth**	**Source**	**% of dry matter**					
			N	**P**	**K**	**Mg**	**Ca**	**S**
Stems and leaves	2-3 cm	Eyssautier	2.9	0.33	2.60	0.25	1.14	0.33

		(ITL), 89/90						
Stems and leaves	15 cm	Eyssautier (ITL), 89/90	2.5	0.30	3.40	0.17	1.10	0.23
Stems and leaves	Begin of flowering	Eyssautier (ITL), 89/90	1.5	0.27	2.80	0.13	0.80	0.15
Stems and leaves	End of flowering	Eyssautier (ITL), 89/90	0.97	0.18	1.60	0.10	0.54	0.10
Stems and leaves	Pulling (harvest)	Eyssautier (ITL), 89/90	0.69	0.10	1.06	0.06	0.35	0.07
Fructifications	End of flowering	Eyssautier (ITL), 89/90	2.68	0.46	1.87	0.27	0.73	0.24
Fructifications	Pulling (harvest)	Eyssautier (ITL), 89/90	2.80	0.50	1.20	0.25	0.35	0.20

Plant analysis data - Micronutrients						
Plant part	**Stage of growth**	**Source**	**ppm dry matter**			
			Mn	**Cu**	**Zn**	**B**
Stems and leaves	2-3 cm	Eyssautier (ITL), 89/90	150	9	300	40
Stems and leaves	15 cm	Eyssautier (ITL), 89/90	160	7	45	28
Stems and leaves	Begin of flowering	Eyssautier (ITL), 89/90	54	7	20	25
Stems and leaves	End of flowering	Eyssautier (ITL), 89/90	30	5	16	18
Stems and leaves	Pulling (harvest)	Eyssautier (ITL), 89/90	20	3	16	13
Frucitfications	End of flowering	Eyssautier (ITL), 89/90	67	11	60	43
Fructifications	Pulling (harvest)	Eyssautier (ITL), 89/90	20	11	40	25

Flax is very sensitive to zinc deficiency. Half the Zn demand is taken up at the 2-3 cm stage. Boron is also important.

Fertilizer recommendations

It is inadvisable to apply lime or manure shortly before sowing. It is also inadvisable to bury the remains of the preceding crop, particularly if they cannot be quickly decomposed. Flax prefers well-rotted old organic matter.

Mineral fertilizer dressings should be based on the plant needs taking into account the expected yield.

P and K fertilizers can be incorporated in the soil in autumn for a spring sowing. However, in view of the limited needs of the crop, NPK fertilizer is usually supplied in a single application before sowing.

Preferred nutrient forms

Mineral fertilizers applied in spring must supply the plants' needs at the beginning of growth. N should preferably be in nitrate form or half in nitrate, half ammonia. K should preferably be in sulphate form ; Autumn K for a spring sowing can be in chloride form.

Present fertilizer practices

Western Europe

Rates are based on an expected yield of about 8-10 t/ha unretted straw (7 t/ha retted), taking into account the mineral N in the soil before sowing and the mineralization capacity of the soil:

- deep loams (kg/ha):
  N = 0-20; P2O5 = 70; K2O = 70.
- light soils, or soils with poor N mineralization capacity (kg/ha):
  N = 40-70; P2O5 = 70-100; K2O = 70-100.

Poland

For an expected yield of 5 t/ha unretted straw:

- dark soils (kg/ha):
  N = 30; P2O5 = 67; K2O = 90.
- podzols (kg/ha):
  N = 45; P2O5 = 67; K2O = 90.

Czechoslovakia

For an expected yield of 6 t/ha unretted straw (kg/ha):
N = 20-30; P2O5 = 20-40; K2O = 60-80.

USSR

For an expected yield of 5 t/ha unretted straw:

- podzols (kg/ha):
  N = 45; P2O5 = 90; K2O = 90.
- humiferous soils (kg/ha):
  N = 30-40; P2O5 = 60-90; K2O = 60-90.

China (Heilongjang)

For an expected yield level of 4 t/ha unretted straw:

- chernozems (kg/ha):
  N = 10-20; P2O5 = 50-70; K2O = 50-70

Further Reading

The reports of the following institutes are recommended:

IKWN (Instytut Krajowych Wlokien Naturalnych), 60-630 Poznan, Ul. Wojska Polskiego 71b, Poland

ITL (Institut Technique Agricole du Lin), 5 rue Cardinal Mercier, 75009 Paris, France

VSUTPL (Research and Breeding Institute of Technical Crops and Legumes, 78701 Sumperk-Temenice, Czechoslovakia

Author: C. Sultana, Institut Technique Agricole du Lin, Paris, France

Hemp *(Cannabis sativa L.)*

French: Chanvre; Spanish: Canamo; Italian: Canapa; German: Hanf

Crop data

Annual, herbaceous at start, becoming ligneous later.

Harvested part: stalk, for fibre (fibres develop between the primary bark and the cambium, while the inner part of the stalk is ligneous with a hollow centre).

Sown late March to early May, depending on region.

Sowing rate 250-450 viable seeds/m2 for fibre or 10-20 plants/m2 for seed production.

Vegetation period 115-125 days for fibre, 140-150 days for seed. Genetic yield potential of currently grown varieties 12-15 t/ha of dry above-ground portion, over 3t/ha fibre.

Prefers fertile soils, e.g. various kinds of chernozems in USSR and S. Europe.

Largest producers are USSR, India, China and some countries in central and southern Europe.

Nutrient demand/uptake/removal

Uptake by total above-ground portion at technological maturity

Nutrient uptake - Macronutrients						
Yield t/ha	**Source**	**kg/ha**				
		N	**P2O5**	**K20**	**MgO**	**CaO**
6.0	Bredemann, 1945	111	36	124	22	108
7.1	Jakobey, 1970	81	38	143	-	-
11.3	Ritz, 1972	109	64	118	-	-

Nutrient uptake is most intensive before and during flowering, when growth is complete.

Plant analysis data

Average composition of above-ground parts at technological maturity

Plant analysis data - Macronutrients						
Plant part	**% of dry matter**					
	N	**P**	**K**	**Mg**	**Ca**	**S**
Leaf	2.40	0.42	1.77	0.59	0.81	0.45
Bark	0.57	0.22	1.06	0.30	0.32	0.35
Stem	0.52	0.13	1.06	0.12	0.32	0.36
Total above-ground	1.13	0.22	1.17	0.27	0.47	0.39
Source: Starcevic, 1979						

Other authors have obtained comparable figures within the range 1.0-2.9 % N, 0.22 - 0.75 % P, 0.83-2.74 % K.

Fertilizer recommendations

The recommended rates of nutrients are 60-150 kg/ha N, 40-110 kg/ha P2O5, and 40-110 kg/ha K2O.

The plants take up nutrients more rapidly and grow better if a high concentration of nutrients is provided early in the season.

Timing of application

Level of application	Autumn - kg/ha			Spring - kg/ha		
	N	P2O5	K2O	N	P2O5	K2O
Low	-	45-60	45-60	45-60	-	-
Medium	40-60	60-90	60-90	40-60	-	-
High	50-60	60-90	60-90	70-90	20-30	20-30

At low rates, total N is applied before sowing; at medium rates, one half of N is applied in autumn and the other half before sowing; at high rates, 50-60 kg/ha N are applied in autumn, 50-60 kg/ha N before sowing and 20-30 kg/ha N are topdressed at the stage of three pairs of leaves.

N is applied in autumn in regions with moderate autumn and winter rainfall, on soils with a deep water-table, and when crop residues with an unsatisfactory C:N ratio have been ploughed under.

Preferred nutrient forms

Autumn N in ammonium and/or amide form, usually as complex NPK fertilizer or urea: spring N as calcium ammonium nitrate.

Rate and form of P depend on soil available P and pH. Watersoluble form is preferred.

K as KCl or K2SO4 , with a slight preference for the latter.

Present fertilizer use

Soil fertility (producing region)	Rates of application - kg/ha		
	N	P2O5	K2O
Fertile chernozem (USSR)	45- 60	45- 60	45- 60
Medium fertile chernozem (USSR, Southern Europe)	80-120	60- 90	60- 90
Less fertile soils (Central & SE Europe)	120-150	90-110	90-110

Further reading

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Author: Lj. Starcevic, Institute of Field and Vegetable Crops, Faculty of Agriculture Novi Sad, Yugoslavia

Jute *(Tossa jute = Corchorus olitorius L., White jute = Corchorus capsularis L.)*

French: Jute; Spanish: Yute; Italian: Corcoro; German: Jute

Crop data

Annual.
Harvested part: whole stem.
Sown early rainy season (February-April): direct seeding.
Harvested 3-5 months after sowing.
Spacing 25 cm x 7 cm = 570 000 plants/ha.
Preferred soil: river silt alluvium, pH 5.4 - 6.4.
Adapted to tropical and subtropical regions. Bangladesh and India have dominated world production but the crop is also grown in China, Thailand, Nepal, Burma etc.

Generally unirrigated.

Nutrient demand/uptake/removal

Nutrient uptake - Macronutrients						
Fibre yield - t/ha	**Source**	**kg/ha**				
		N	**P2O5**	**K2O**	**MgO**	**CaO**
C. olitorius (JRO-632) 3.0	Mandal et al, 1979	89	58	181	37	151
C. capsularis (JRC-212) 2.0	Mandal et al, 1979	63	31	159	43	99

Fertilizer recommendations

It is essential to neutralize soil acidity by liming.

Farmyard manure or compost at 5-10 t/ha should be applied and well incorporated during land preparation, together with all the mineral P and K needed. Mineral fertilizer is generally applied in the form of a basal NPK fertilizer which depends on the cropping pattern and on soil analysis. One third of the fertilizer N may be applied during land preparation or after the first weeding and thinning (30 days crop age) and the rest topdressed at 45 days crop age or after the final weeding.

Present fertilier practice

Bangladesh						
Major Zones	**Agroecological Region**	**Land & Soil Characteristics**	**Jute type/Varieties**	**Fertilizer rates - kg/ha**		
				N	**P2O5**	**K2O**
Jat	Young Brahma-	Highland	Tossa;	40	0	10
area	putra & Yamuna	Loam	0-9878			
	Floodplain	OM: low	0-4			
		pH: 5.5-6.8				
		K: medium				
	Madhupur tract	Highland,	Tossa;	40	0	20
		Loam	0-9897			
		OM: low	0-4			
		pH: 4.8-5.6				
		K: medium				
	Old Brahmaputra	Highland	White;	40	0	10

	Floodplain	Silt loam	D-154			
		OM: low	CVL-1			
		pH: 5.1-5.6	CVE-3			
		K: low	CC-45			
		Highland	White,	40	10	20
		Loam	as above			
		OM: medium				
		K: low				
	Lower Meghna	Highland	White,	30	0	0
	Floodplain	Silt loam	as above			
		OM: medium				
		pH: 5.0-6.0				
		K: low				
	Old Meghna	Medium	White,	30	0	0
	Floodplain	highland	as above			
		Silt loam				
		OM: medium				
		pH: 5.0-6.1				
		K:low				
		Medium	White,	30	0	10
		lowland	as above			
		Loam				
		pH: 5.5-6.5				
		K:low				
District	High Ganges River	Highland	Tossa;	40	0	10
	Floodplain	Silt loam	0-9897			
		OM: low	0-4			
		pH: 6.1-7.9				
		K: medium				
		Medium	Tossa,	45	0	10
		highland	as above			
		Loam				
	Low Ganges River	Highland	Tossa;	40	0	20
	Floodplain	Silt loam	as above			
		OM: low				
		pH: 6.2-7.7				
	Karatoya-Bengali	Highland	White;	40	0	10
	Floodplain	Silt loam	D - 154			
		OM: low	CVL - 1			
		pH: 5.4-5.7				
		Silty Clay-				
		loam	CVE - 3			
		K: low	CC - 45			
	Gopalgani-Khulna	Medium	White;	30	0	0
	Bils	highland,	as above			
		Clay				
		OM: medium				
		pH: 5.4				
		K: high				
Northern	Old Himalayan	Medium	Tossa;	40	10	20
	Piedmont plain	highland	9-9897			
		Loam	0-4			
		pH: 4.5-5.5				
		K: low				

	Tista Meander	Highland	Tossa;	40	0	20
	Floodplain	Loam	as above			
		OM: low				
		pH: 5.4-6.5				
		K: medium	White;	40	0	10
			as above			
		Medium				
		highland				
		OM: low	White; as above	40	10	20
Highland = land above normal flood-level.						
Medium highland = land normally flooded to about 90 cm in flood season.						
Medium lowland = land normally flooded to 90-180 cm in flood season.						
Fertilizer nutrient sources: N = urea, P = triple superphosphate, K = muriate of potash (60 % K2O).						

It is advisable to apply 5.0-7.7 t/ha well decomposed cow-dung 2-3 weeks before sowing, in which case the P and K fertilizers may be omitted except for the variety 0-9897 which would then need 10 kg P2O5 and 20 kg K2O/ha; for this variety the first dressing of urea should be reduced by 50 %; for other varieties no urea would be needed in the initial application and, in the second application (45 days after sowing) urea may be reduced by 25 %.

India							
Zone	**Agroecological Region**	**Land & Soil Characteristics**	**Jute type/Variety**	**Fertilizer rates - kg/ha**			**Remarks**
				N	**P2O5**	**K2O**	
1.	Gangetic	Gangetic alluvium	Tossa;	40	20	20	10 t/ha of
	West Bengal	sandy loam/loam	JRO-632				organic
		OM: medium/low	JRO-7835				manure
		K: medium	JRO-878				
			JRO-524				
2.	Teesta	grey alluvium	White;	45	20	20	Organic
	Mahananda	sandy loam/loam	JRC-321				manure and
	North	OM: low	JRC-212				liming of
	Bengal	pH: acidic					250-500 kg/ha is essential
3.	Upper Assam	North of bank of	Tossa;	40	20	20	Organic and
		Brahmaputra,	as above				liming as
		sand/loam	zone 1				above
		OM: low	White;	45	20	20	Organic &
		pH: acidic	as above				liming as
			zone 2				above
4	Lower Assam	Alluvium lateritic	Tossa;	30	20	20	Liming as
	and Meghalaya	area with red soil	as above				above
		and new clay loam	zone 1	40	20	20	
		OM: medium	White;				
		pH: mainly acidic	as				

			above				
			zone 2				
5.	Tripura	as above zone 4	Tossa;	30	20	20	Liming as
	Cachar		JRO-7835				above
	Surma		JRO-632				
	Valley		White;	40	20	20	
			JRC-321				
			UPC-94				
6.	Coshi	Grey alluvium,	White;	40	20	20	Liming as
	command	water stagnation	as above				above
		OM: medium to low	zone 2				
		pH: mainly acidic	Tossa;	30	20	20	
			JRO-7835				
			JRO-878				
7.	Mahanadi	Alluvium yellow	White;	45	20	20	Limins
	Delta	& red lateritic &	JRC-212				as above
		metamorphic rocks	JRC-7477				
		Clay and silt	Tossa;	40	20	20	Liming
		OM: medium/low	JRO-524				as above
		pH: mainly acidic	JRO-7835				
			TJ-40				
8.	Midnapore	Alluvium transpor-	White;	45	20	20	Liming
		ted buried late-	as above				as aboive
		rite areas	zone 7				
		Clay and silt					
		OM: medium					
		pH: generally acidic					
9.	North West	Grey alluvium with	Tossa;	40	20	20	Liming
	Bihar	poor and very	as above				
	U.P.	laterain	zone 7				
		OM: medium	White;	45	20	20	Liming
		pH: generally	as above				as above
		acidic	zone 7				
Liming at 250-500 kg/ha and application of 10 t/ha organic material with mineral fertilizers based on soil analysis ensure a profitable return.							

In addition to the 10 kg/ha of well decomposed cow-dung before sowing, fertilizer N should be applied in two topdressings, half after the second weeding (30 days after sowing) and the remainder 5-6 weeks after sowing. Alternatively, 25 kg/ha urea in solution may be sprayed on the foliage at a crop age of 35-60 days (but not when rain or high wind is likely or in scorching sun) according to the schedule below:

N foliar spray					
Urea kg/ha	**Water t/ha**	**Strength of spary solution**	**No.of sprays**	**Interval between sprys**	**Sprayer type**
12.50	90	14 %	2	15 days	Ultra-low volume
8.33	85	10 %	3	10 days	Ordinary knapsack
6.25	210	3 %	4	7 days	Hand

Further reading

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Author: N. Sermsri, International Jute Organisation, Dhaka, Bangladesh

Sisal *(Agave sisalana Perr.)*

French: Agave sisal; Spanish: Agave sisal; Italian: Canapa di sisal; German: Sisal-Agave

Crop data

A monocarpic perennial yielding 220-250 linear, lanceolate leaves with a terminal spine in the 6-9 years before it "poles" (flowers) and dies. The leaves are fleshy and contain parallel fibres. The tall inflorescence carries numerous bulbils after the flowers wither. Propagation either by bulbils or by suckers.

Hybrids, some outyielding and having a shorter production cycle than A. sisalana, were obtained in Tanzania as early as 1937 by crossing A. amaniensis ("blue sisal") with A.angustifolia and back-crossing the F1 with A. amaniensis.

Cutting usually starts after 2 1/2 years when the plant has about 100 leaves. At least 20-25 leaves should be left after each harvesting. Removal of suckers once a year is usually sufficient.

Plant density: up to 5 000 plants/ha (limit imposed by practical considerations although fibre yield positively correlated with plant population up to 10 000/ha). A double-row system is also used with bands of N-fixing cover crops (Pueraria phaseoloides, Dolichos argentus, Centrosema pubescens, Calopogonium mucunoides, Glycine javanica, Stizolobium aterrinium) providing a dense foliage which inhibits weed growth.

The crop is indigenous to Central America but is grown in several other regions, doing best between latitudes 10 ° N and S.

Average yields (kg/ha): Africa 697, Central America 516, S. America 673, Asia 1081 (FAO 1989). Field experiments in Kenya have shown, however, that well-managed sisal can yield 3.7 t/ha of fibre per year during a 74-month cycle, or for a hybrid 6.8 t/ha per year during a 110-month cycle (planting to poling).

Nutrient demand/uptake/removal

Nutrient removal - Macronutrients - Average content of 1 t sisal leaves (kg)							
Moisture	**Dry matter**	**N**	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
820	180	1.0	2.66	1.97	1.66	3.78	0.27
Note: The 180 kg dry matter comprises fibre 50 kg, pulp 130 kg.							

Nutrient removal - Micronutrient - Average content of 1 t sisal leaves (g)				
B	**Cu**	**Fe**	**Mn**	**Zn**
1.50	0.08	0.45	1.50	0.65
Note: The 180 kg dry matter comprises fibre 50 kg, pulp 130 kg.				

Nutrient removal can be substantially reduced by returning the sisal waste to the soil. Compost made from the bulky material obtained in the decorticating factory contains, per metric ton: 6 kg N, 1 kg P, 0.8 kg K, 1.6 kg Mg, 25 kg Ca, 2.5 kg S.

Foliar analysis

Leaf analysis data - Macronutrients - Satisfactory contents of freshly matured leaves					
% of dry matter					
N	**P**	**K**	**Mg**	**Ca**	**S**
2.4	0.18	2.5	0.40	0.8	0.10

Leaf analysis data - Micronutrients - Satisfactory contents of freshly matured leaves						
ppm dry matter						
B	**Cu**	**Fe**	**Mn**	**Mo**	**Zn**	**Co**
12	6	100	40	0.2	23	0.5

Fertilizer recommendations

Soil pH should be adjusted to 5.5-6.5 by liming (preferably using dolomitic limestone to supply Mg) before transplanting. Mottled-leaf symptoms and "bole-rot" stem disease occur more frequently in soils low in Ca.

Fertilizers are normally applied in bands alongside the rows; deep placement is unnecessary (except for transplants) as the root system of sisal is very shallow. The width of the bands is increased as the plants grow older.

Micronutrients

Boron deficiency may be avoided by including 0.5-1.0 kg/ha B in the fertilizer application.

Deficiencies of other micronutrients may be avoided or corrected by means of foliar sprays.

Present fertilizer practices

Recommendations for S. Brazil

after the first cutting 50 kg/ha N are sidedressed. P and K are applied according to soil P and K analysis:

Soil P - µg/cm3	**meq soil K/100 cm3**			
	< 0.07	0.08-0.15	0.16-0.30	> 0.30
	kg/ha N - P2O5 - K2O			
< 6	0-60-70	0-60-60	0-60-50	0-60-15
7 - 15	0-50-70	0-50-60	0-50-50	0-50-15
16-40	0-30-70	0-30-60	0-30-50	0-30-15
>40	0- 0-70	0- 0-60	0- 0-50	0- 0-15
Soil P and K; resin extraction - * N, see recommendation above (50kg/ha)				

Recommendations for N.E.Brazil

P and K according to soil analysis.

First year:

Nutrient	**Furrow for transplanting**	Sidedressing
	kg/ha	
Nitrogen (N)	20	40
Phosphorus (P2O5) - ppm soil P		
< 6	70	-

7-13	50	-
14-20	30	-
Potassium (K2O) - ppm soil K		
< 30	35	35
31-60	25	25
61-90	15	15
Soil P and K, Mehlich extraction		

Maintenance fertilization after first cutting (kg/ha/year):
N = 40;
P2O5 = 0 to 70 (according to soil analysis);
K2O = 30 to 70 (according to soil analysis);

Total rates split into 2 applications, one of them after cutting.

Recommendations for Africa

For maintenance, usually 50-100 kg/ha N, 0-120 kg/ha P2O5 and 30-60 kg/ha K2O per year in two equal dressings, the lower rates being applicable to soils high in organic matter, P and K.

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Author: E. Malavolta, Professor of Plant Nutrition, Universidade de Sao Paulo, Piracicaba, Brazil

Rubber *(Hevea brasiliensis Muell.Arg.)*

French: Hevea; Spanish: Hevea, seringa; Italian: Evea; German: Kautschuk

Crop data

Tree. Harvested part: latex (also timber at end of cycle of about 25-30 years).
Sown at onset of rainy season; bud-grafted, using clonal buds, 6-9 months later. Currently, seedling stocks are generally raised in bags, budded and the bud-grafted clonal plants are transplanted in the field. The plants attain a circumference of 45-50 cm at a height of 1.5 m after 5-6 years and are then ready for starting harvesting of latex.

Plant density: 450 trees/ha (commercial plantations) to 550 trees/ha (small farms). Often planted rectangularly, e.g. at 2.5 m x 9 m or at 2 m x 9 m. On sloping land, planted on contoured terraces.

Prefers deep, permeable, well-drained soil with pH 4.0-5.5.
Adapted to equatorial lowlands (below 200 m altitude). Traditionally grown between 10 °N and 10 °S but now extends to 20 °N (China) and 20 °S (Brazil). Mean annual rainfall 2 000 - 3 000 mm in 100-150 rainy days preferred.

Nutrient demand/uptake/removal

During the early unproductive phase (5 to 6 years), there is a substantial accumulation of vegetative matter and large amounts of nutrients are immobilized in the tree; from the fifth year, the tree recycles a considerable amount of vegetation and nutrients through leaf fall; and from about the sixth year nutrients are removed in the latex harvested.

Nutrients immobilized, returned and removed in a 30-year period - Macronutrients				
Kind of removal/return	**kg/ha**			
	N	**P2O5**	**K2O**	**MgO**
Total immobilized in 30-year period	1500-1800	458-573	1440-1680	300-365
Total returned in leaf litter (annual range in 5th to 30th years shown in brackets)	1400 (34-73)	82 (2.1-4.6)	426 (10.2-21.6)	275 (6.6-14.1)
Total removed in latex (annual range in 6th to 30th years shown in brackets)	485 (6.1-35.7)	94 (2.4-17.6)	418 (6.0-39.1)	120 (1.4-9.3)
Total litter fall in 5th-30th years 104 t/ha, ranging from 2.5 to 5.4 t/ha/yr,¦peaking in 9th-12th years. - Total yield of dry rubber in 6th-30th years 46.6 t/ha, ranging from 0.62 to 3.0 t/ha/yr, generally highest from 12th to 23rd years				

Total amounts of other nutrients immobilized in the trees:
1120-1 400 kg/ha CaO, 200-250 kg/ha S, 15 kg/ha Mn, 0.7-1.5 kg/ha B, 5 kg/ha Zn, 1 kg/ha Cu.

Analytical data

(Note that, though a particular nutrient may be present in the soil in adequate amounts, its uptake may be inhibited by the lack of another, e.g. lack of N can reduce uptake of K; also in acid koalinic soils low in C, the soil N level will not be built up even when adequate N is supplied. Soil analysis therefore needs to be complemented by leaf tissue diagnosis.)

Diagnostic criteria from soil analysis (top 30 cm) - Macronutrients

Status	C	Total N	P		K	Mg
			Sulphuric/perchloric extract	Bray II extract	6N HCl extract	6N HCl extract
	%	%	ppm	ppm	m.e. %	m.e. %
Low	<0.5	<0.10	<250	<11	≤0.50	≤0.75
Medium	0.5-1.5	0.10-0.25	250-350	11-20	0.50-2.0	0.76-3.0
High	1.5-2.5	0.25-0.40	350-600	20-30	2.0 -4.0	3.0 -8.0
Very high	>2.5	>0.40	>600	>30	>4.0	>8.0
Low/medium = deficiency to be expected. High/very high = soil reserves adequate but there is a need to monitor for imbalances with other nutrients, especially at the very high level.						
Source: Pushparajah, 1977						

Leaf analysis (for leaves at optimal age of about 90-150 days in shade of canopy) - Macro- and micronutrients								
Status	% of dry matter							ppm
	N			P	K		Mg	Mn
Low	(a) <3.2	(b) <3.3	(c) <2.9	<0.19	(d) <1.25	(e) <1.35	<0.2	<45
Medium	3.2-3.5	3.3-3.7	2.9-3.2	0.20-0.25	1.25-1.5	1.35-1.65	0.20-0.25	45-150
High	3.5-3.7	3.7-3.9	3.2-3.4	0.26-0.28	1.50-1.65	1.65-1.85	0.26-0.30	151-300
Very high	>3.7	>3.9	>3.4	>0.28	>1.65	>1.85	>0.30	>300
Toxic								>500
(a) = all clones except as specified in (b) and (c) (b) = RRIM 600, GTI and other clones with similar characteristics. (c) = clones susceptible to trunk snap. (d) = all clones except as specified in (e). (e) = RRIM 600, GTI, most PB clones and progenies. Low = well below optimal tending to visual deficiencies. Medium = sub-optimal. High = levels above which responses unlikely. Source: Pushparajah, 1977								

Indicative desired levels for other elements are: 0.5-0.7 % Ca, 0.20-0.25 % S, 60-80 ppm Fe, 15 ppm B, 1.5 ppm Mo, 20 ppm Zn, 10 ppm Cu.

Nutrient effects on latex quality

High N and Mg can adversely affect the technological properties of concentrate latex. Excessive Cu and Mn adversely affect the oxidative process of the rubber. Within the tree, excessive Mg and Ca can cause instability in the latex vessels resulting in early pre-coagulation on the excised bark, thus reducing the time of flow and yield.

Fertilizer recommendations

A typical fertilizer schedule for a rubber nursery (25 kg polybags) would be:

General fertilizer schedule for polybag nursery				
Time after planting (months)	**g/bag**			
	N	**P2O5**	**K2O**	**MgO**
At planting	-	20	-	-
1	0.6	1.0	0.5	0.15

2	1.3	2.0	1.0	0.3
3	1.3	2.0	1.0	0.3
4	2.0	3.3	1.5	0.4
5 - 5 ½ (budding) - After 1st whorl	1.3	2.0	1.0	0.3
Transplanting when second whorl has hardened				

During the unproductive phase, mineral fertilizers are applied five times in the first year, four in the second, three in each of the third, fourth and fifth years, and twice in the sixth year, according to the following schedule. An exception is that, where vigorous legume covers are maintained, fertilizer N is applied only in the first year.

General fertilizer schedule for immature rubber						
Region and country	**Soil and cover condition**	**Source**	**Total fertilizer nutrients in first 6 years from planting**			
			kg/ha			
			N	**P2O5**	**K2O**	**MgO**
AFRICA						
- Ghana	-	de Geus, 1973	225	250	200	70
- Liberia	-	de Geus, 1973	190	260	190	-
ASIA						
- India	Legume cover	RRI, India, 1977	200	200	116	21
	Non-legume cover		260	220	104	21
- Indonesia	East Java	Angkapradipta et al., 1970	290	310	130	50
	West Java and Sumatra		250	270	220	50
- Malaysia	Low K, no legume	Pushparajah &	640	250	170	50
	Low K, mixed legume	Mahmud, 1977	225	250	170	50
	Low K, pure legume		30	250	170	50
	High K, no legume		660	260	90	50
	High K, mixed legume		225	260	90	50
	High K, pure legume		30	260	90	50
- Sri Lanka	Low K soil	Yogoratnam, 1984	250	290	300	40
	High K soil		320	320	150	60
- Thailand	Low K, mixed legume	Wate Thainugul,	230	280	170	25
	High K, mixed legume	1987	260	290	90	10
LATIN AMERICA						
- Brazil	-	EMBRATER/EMBRAPA, 1980	205	200	125	45

Fertilizer recommendations for the productive phase of mature rubber according to soil classification and clone type are as follows:

Fertilizer recommendations for mature rubber (7th-25th years)								
Soil classification		**Annual fertilizer nutrients applied**						
Sub-Order	**Great Group**	**kg/ha/year**						
		N			**P2O5***	**K2O**		**MgO***
		(a)	(b)	(c)		(d)	(e)	
Udult	Paleudult	16	20	12	20	94	118	10
Orthox	Haplorthox	20	24	8	20	59	94	10
	Acrothox	16	20	8	20	59	94	10
Tropept	Dystropelpt, tropudult	20	24	12	20	59	94	10
Psamment	Quartzipsamment	16	28	12	20	140	180	10
Aquept	Tropaquept,	20	24	12	-	-	45	-

	Sulfaqueqt							
Notes: (a) For all clones (cultivars) except those mentioned below in (b) and (c). - (b) For clones RRIM 600, GTI and others with similar characteristics. - (c) For clones susceptible to trunk snap. - (d) For all clones except those mentioned in (e). - (e) For RRIM 600, GTI and most PB clones and progenies. - * The amounts of P2O5 and MgO indicated need not be given every year but may be omitted in 7-10 of the 19 years.								
Source: Chan et. al., 1972; Pushparajah et al., 1983								

Mn deficiency is widespread in some producing areas but it can be corrected by application of MnSO4 at 100 g/tree or 45 kg/ha.

Fertilizers for both immature and mature (i.e. productive) rubber are broadcast on the soil surface, the zone of application increasing with the canopy; but, if urea is used, it should be incorporated in the soil and preferably applied at the onset of annual refoliation.

The preferred sources of nutrients are: ammonium or nitrate N, relatively reactive phosphate P (except in first year, when soluble P is preferred), K as muriate and Mg as kieserite.

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Author: E. Pushparajah, International Board for Soil Research and Management (IBSRAM), Bangkok, Thailand

Cacao *(Theobroma cacao L.)*

French: Cacao; Spanish: Cacao; Italian: Cacao; German: Kakao

Crop data

Perennial tree. Harvested part: pods.

Sown directly in nursery in polybags (25 cm high, 12 cm diameter), usually arranged in double rows under shade. Planted out in the field after 7-8 months. First useful flowering occurs at 18-24 months after planting (the flowers appear on the trunk and older branches). Flowering is practically continuous, but favourable pollination periods occur about 2 months after the beginning of rainy seasons. Young fruits (called cherelles when young and pods when mature) grow slowly during first 40 days after pollination and are prone to wilting (average loss about 50 %); subsequent growth is more rapid up to 75 days; full maturity is reached in 150-180 days depending on climate. There are usually two harvest peaks (main and secondary) depending in equatorial regions on the rainfall pattern, and in sub-tropical regions (e.g. Bahia, Brazil) on temperature.

Plant density: around 950-1 330 trees/ha depending on soil fertility and climate.

Shade: temporary shade needed during first 3-4 years until closed canopy established (with fertilizers, early growth is enhanced by reducing to 75 % of full light intensity; without fertilizers, it is enhanced by reducing to light intensity of 50 %). With Upper Amazon hybrids, no permanent shade is advised except on poor soils. Shade reduces flowering and yield, and shaded cacao usually shows no response to N fertilizer. Windbreaks are preferable to shade in dry or cold areas.

Nutrient demand/uptake/removal

Nutrient demand/uptake/removal - Macro and micronutrients								
Growth stage	**Age (months)**	**Average nutrient requirements (whole plant*) - kg/ha**						
		N	P2O5	K2O	MgO	CaO	Mn	Zn
Nursery	5-12	2.5	1.4	3.0	1.9	3.3	0.04	0.01
Field:								
-immature	28	140	33	188	80	163	4.0	0.5
-first production	39	219	54	400	122	203	7.3	0.9
-mature	50-87	453	114	788	221	540	7.0	1.6
* Approx. 1 100 plants/ha Source: Thong and Ng (1978) in Wessel (1987)								

Nutrient removal in crop of 1 t/ha dry cacao beans (7 % humidity) with 1.4 t/ha husks:

Country	**Beans**					**Husks**					**Total**				
	N	**P2O5**	**K2O**	**MgO**	**CaO**	**N**	**P2O5**	**K2O**	**MgO**	**CaO**	**N**	**P2O5**	**K2O**	**MgO**	**CaO**
Cameroon	19.2	10.1	12.8	5.3	1.3	15.0	4.4	74.7	6.0	10.2	34.2	14.5	87.5	11.3	11.5
Côte d'Ivoire	22.1	6.8	9.0	2.3	0.6	13.2	4.2	51.9	9.3	4.2	35.3	11.0	60.9	11.6	4.8
Nigeria	22.7	9.2	10.1	-	-	17.0	5.3	93.0	-	-	39.7	14.5	103.1	-	-
West Malaysia	20.4	8.2	12.6	4.5	1.5	10.6	3.0	52.2	4.1	5.3	31.0	11.2	64.8	8.6	6.8
Sources: Boyer (1973), Lotode and Jadin (1981), Omotoso (1975), Thang and Ng (1978) in Wessel (1987)															

Plant analysis data

Nutrient content of near mature leaves (usually the second or third leaf of the last maturing flush is taken for analysis):

Plant analysis data - Macronutrients						
Country	**Nutritional state**	**% of dry matter**				
		N	**P**	**K**	**Mg**	**Ca**
Côte d'Ivoire	Normal	2.35-2.50	>0.18	>1.20	-	-
	Moderately deficient	1.80-2.00	0.10-0.13	1.00-1.20	0.30	0.30
	Severely deficient	<1.80	<0.08	<1.00	-	-
Trinidad	Normal	>2.00	>0.20	>2.00	>0.45	>0.40
	Moderately deficient	1.80-2.00	0.13-0.20	1.20-2.00	0.20-0.45	0.30-0.40
	Severely deficient	<1.80	<0.13	<1.20	<0.20	<0.30
Source: Loué, Murray, Spector in Snoeck,1984						

Plant analysis data - Micronutrients							
Country	**Nutritional state**	**ppm dry matter**					
		Fe	**Mn**	**B**	**Zn**	**Cu**	**Mo**
Cote d' Ivoire	Normal	100	200-500	25-40	20-60	-	-
	Moderately deficient	-	20	5	10-15	-	-
Trinidad	Normal	65-175	50-400	25-70	80-170	8-12	1.0-2.5
	Moderately deficient	-	-	-	20	8	-
	Severely deficient	50	-	10	16	4	0.5
Source: Loué, Murray, Spector in Snoeck (1984)							

In spite of much research, leaf analysis cannot be used as a basis for fertilizer recommendations but only to detect deficiencies or imbalances or trends in nutrient supply.

Soil analysis

A soil diagnostic method for fertilizer requirements has been devised by Jadin (1972, 1975, 1976, 1985). Composite soil samples are taken of upper horizons in the plantations (composite of 30 borings of 0-20 cm in each homogeneous type of soil, in the fertilized area around or between cacao trees) and their chemical analyses are compared with optimum levels as shown below, and with nutrient balances determined by observation and measurements of growth, flowering intensity and fruit set, cherelle wilt, pod development and production, number of beans per pod, pod values, and yield. Soil samples should be taken every 3-4 years in order to monitor changes.

Optimum values:

Available P (Olsen-Dabin or Olsen modified) 100 ppm

Total N in relation to sum of exchangeable bases
[(m.e. % K+Ca+Mg) + 6.15] / N %o = 8.9

Ratio between exchangeable bases as percentages of sum
K : Ca : Mg = 8 : 68 : 24

Base saturation at least 50-60 %

There is a very high correlation between Olsen-Dabin available P in the soil and flowering, as well as pod yield. Good correlations were also obtained between K application and flower set in Trinidad.

Fertilizer recommendations

As most feeder roots (80 %) are in the top 0-20 cm of soil and develop where the fertilizer is applied, fertilizer should always be given to the same area around each tree, except in the first years after planting:

- 1st year; at planting, in circle of radius 0.3 m around stem.

- 2nd year; in band 0.3-0.6 m around stem.

- subsequent years; either in band 0.6-1.0 m around stem or in 80 cm-wide strips between rows and between trees.

There is a computer programme at IRCC, Montpellier, with which to calculate recommended rates of application as below:

g P2O5/plant = 2.29 W (shortfall in soil below optimum level) / 1000

g K2O/plant = 47.1 W (shortfall in soil below optimum level) / 100

g MgO/plant = 20.2 W (shortfall in soil below optimum level) / 100

g CaO/plant = 28.1 W (shortfall in soil below optimum level) / 100

where W = corrected soil weight per plant (crown or strips) = apparent soil density x surface area x depth of 0.2 m

The optimum levels of K, Mg and Ca as m.e. % in the soil are computed from base saturation and a ratio of 8 % : 68 % : 24 % in the total of exchangeable bases (for worked example, see A in Present fertilizer practices in specific regions).

Nitrogen is required when (m.e. % E.B. + 6.15)/N %o > 8.9, but the amount needed is based on the results of field experiments and not on soil analysis. If the ratio is < 8.9, no nitrogen fertilizer is required.

Preferred nutrient forms

Nitrogen: urea usually recommended unless S seems to be a limiting factor, in which case sulphate of ammonia would be preferred.

Phosphorus: rock phosphate advised when soil pH < 5.5, provided liming not needed; and superphosphate (triple unless Ca is low in exchangeable bases, when single is preferred) if soil pH >5.5.

Potassium: muriate (KCl)

Magnesium: kieserite, unless Ca also needed.

Calcium: lime (or dolomitic limestone if Mg also needed).

Present fertilizer practices in specific regions

Cote d'Ivoire, Cameroon and other countries applying soil diagnostic method.

Examples of calculations:

Soil analysis chart				
	Soil n° 1		Soil n° 2	
Carbon %	1.6		1.2	
Nitrogen %o	1.5		1.0	
Avail. P ppm	48		12 (Olsen modified)	
Exchangeable bases:	% of total E.B.		% of total E.B.	
K m.e. %	0.50	2.7	0.12	4.8
Mg m.e. %	6.43	34.3	0.60	24.3
Ca m.e. %	11.83	63.0	1.75	70.9
Total E.B.	18.76		2.47	
			(Ammonium acetate extract)	
Exchange capacity	19.30 m.e. %		5.60 m.e. %	
Base saturation	97 %		44 %	
pH (water)	6.1		5.2	

Apparent soil density may vary according to soil texture from 1 100 to 1 400 kg/m3. Assume 1 400 in this case. Soil depth is 0.2 m.

So, in 1st year, W = (area of circle of radius 0.3 m) x 0.2 x 1 400
= pi x 0.3^2 x 280
= 79 kg

And in 2nd year, W = (area of band between circles of radius 0.6 m and 0.3 m) x 0.2 x 1400
= pi x (0.6^2 - 0.3^2) x 280
= 238 kg

And in 3rd year, W = (area of band between circles of radius 1.0 m and 0.6 m) x 0.2 x 1400
= pi x (1.0^2 - 0.6^2) x 280
= 563 kg

On soil no.1, available P should be raised to 100 ppm.

K:Ca:Mg ratio should be 8 : 64 : 24 but, as saturation is high, complete correction is impossible, so only K will be applied to raise K to 8 %, i.e. to 1.6 m.e. %; even this will oversaturate the exchange capacity by 2 %.

For N, (m.e. % E.B. + 6.15)/N %o = (18.76 - 6.15)/1.5 = 16.6, so nitrogen fertilizer is needed.

Rates of application (g/plant):	Year 1	Year 2	Year 3
Triple super (100-48)2.29 W/1000/0.44 =	21	64	152
KCl (1.6-0.5)47.1 W/100/0.60 =	68	206	486
Urea	50	75	100
Sulphate of ammonia could be applied instead of urea and ammonium phosphate instead of triple superphosphate.			

For soil no.2, available P should also be raised to 100 ppm.

Base saturation should be raised to 60 % and the K:Ca:Mg ratio should be 8:68:24, so:

Total E.B. = 5.6 x 0.6 = 3.36 m.e. %.
K = 3.36 x 0.08 = 0.27 m.e. %.

Ca = 3.36 x 0.68 = 2.28 m.e. %.
Mg = 3.36 x 0.24 = 0.81 m.e. %.

And (m.e. % E.B. + 6.15)/N %o = (2.47 + 6.15)/1.0 = 8.6

Rates of application, calculated in the same manner as for soil no. 1, are then:

Rates of application (g/plant)	Year 1	Year 2	Year 3
Triple super	36	107	258
KCl	9	28	66
Lime = (2.28-1.75)28.1 W/100/0.67 =	17	53	125
Kieserite = (0.81-0.60)20.2 W/100/0.27 =	12	37	88

Dolomitic limestone could be applied instead of lime and kieserite.

No nitrogen fertilizer is needed.

Brazil

The decision whether to apply fertilizers depends on the quality of plantation management, shade intensity, soil depth and drainage. The soil is analysed to monitor requirements.

In the Amazon region, natural high-fertility soils do not need fertilizers for the first 5 years after planting.

Requirements for low-fertility soils are based on critical levels of nutrients, particularly P and K, in the soil, as shown below. In new forest clearings, soil sampling to assess Ca and Mg levels should be done before and after burning. A composite sample of 12 borings per hectare is recommended. Sampling should be repeated every 3 years.

Nutrients	Critical levels	Criteria for fertilizing		Nutrients kg/ha					
		P	K	Amazonia			Bahia		
				N	P2O5	K2O	N	P2O5	K2O
Phosphorus	Low (L) < 6 ppm	L	L	30	90	60	60	90	90
	Average (A) 7-15 ppm	L	A	30	90	30	60	90	45
	High (H) >15 ppm	L	H	-	-	-	60	90	-
		A	L	30	60	60	60	45	90
Potassium	Low (L) ≤0.12 m.e.%(47 ppm)	A	A	30	60	30	60	45	45
	Average (A) 0.13-0.30 me.%	A	H	-	-	-	60	45	-
	(47-117 ppm)	H	H*	15	15	10			
	High (H) >0.30 me.% (117 ppm)	H	H**	-	-	-			
		Ca+Mg	Al	Liming:					
Calcium + Magnesium	Low (L) Amazonia = <2.0 m.e.%	L	L	No liming					
	Bahia = <3.0 me.%	L	A/H	t/ha CaO = 1.5 x Al					
Aluminium	(A/H) Amazonia > 3.0 m.e.%	A/H	L	No liming					
	Bahia > 0.4 m.e.%	A/H	A/H	(Al x (Al % - 15 %))/Al % = t/ha CaO					
* Maintenance fertilizing when cocoa prices are favourable; ** Profiting from residual effects of fertilizers when prices are low.									
Available P ppm according to Bray & Kurtz no.2 Al = Al m.e.%; Al % = Aluminium saturation minus 15 % (15 % is the acceptable level)									

Fertilizer N use should be moderate and based on light intensity and on field observation of possible deficiency symptoms. Urea may suffer some loss by volatilization of ammonia, particularly in alkaline soils or in soils rich in fresh organic matter. Soluble forms of P (mono-

or di-ammonium phosphate, single or triple superphosphate) are preferred, except on acid soils where rock phosphate may be used. Liming is needed only in exceptional circumstances where exchangeable Al > 3 m.e. %.

No fertilizers are usually given in the nursery but 0.5 % urea spray may be applied to the leaves if N deficiency symptoms appear. The first application in the field is usually given 2-4 months after planting, in a circle of 0.5 m radius around the stem, in 2 dressings per year. In subsequent years a basal dressing is given at the beginning of the rains, and supplemental N as indicated by field observations. In the second year the radius of the circle is increased to 1.0 m and from the third to the fifth years to 1.5 m. In later years, a blanket application between 4 trees on 9 m2.

Malaysia

An integrated approach is advocated, taking into consideration such factors as leaf nutrient content, age of trees, cropping level, soil type, amount of shade, leaching losses and other agronomic factors (tree vigour, harvesting system, etc).

Needs for N vary, depending on rainfall and shade. More N appears to be needed for high yield and refoliation of lightly shaded cacao. P, as basal dressing and planting-hole appliction of rock phosphate, benefits establishment and growth of young plants. For mature cacao, P is continuously required for sustained high yield. The need for K appears to decline as the trees mature, probably associated with the reduced growth rates and recycling of leaf litter and pod husks. Mg requirements have not been fully established, but there are indications that balanced nutrition is essential for high yield. Ca is needed for early establishment and growth but its role in mature trees needs further clarification.

Fertilizer recommendations

Nursery - compound fertilizer 12:12:17:2 with micronutrients, or 15:15:6:4 (slow-release 10:10:5:2 has also been used with success)

Age of seedling (months)	Rate (g)
2-3	1.0
3-4	1.5
4-5	2.0
> 6	3.0

Pre-planting and planting hole - on low-fertility acid inland soils (ultisol and oxisol)

- pre-planting:
400-600 kg/ha rock phosphate, 880 kg/ha ground limestone, ploughed into top 15 cm soil.

- planting hole:
150-230 g/plant rock phosphate.

Young cacao (Malaysian peninsular):				
Months after planting	**g/tree**			
	N	**P2O5**	**K2O**	**MgO**
1 ½	4.5	4.5	1.8	1.2
3, 5 and 8	7.5	7.5	3	2
11	12	12	17	2
14 and 17	17.5	8	21	2.2
20 and 23	23	11	28	3
27, 31, 35	35	16	42	4.5

Mature cacao - (kg/ha/year)			
Nutrient	**Soil analysis***	**Leaf (%)**	Fertilizer rate
N		< 2.0 2.0-2.6 > 2.6	100-150 60- 80 -
P2O5	< 15 ppm > 15 ppm	< 0.2 > 0.2 > 0.2	90-150 30- 60 -
K2O	< 0.3 m.e.% > 0.3 m.e.%	< 2.0 > 2.0 > 2.0	120-180 80-100 -
* Available P by Bray and Kurtz no.2; exchangeable K in m.e.%.			
Source: Ling, 1989			

Wyrley-Birch (1987), reviewing trials in Sabah 1964-1977, concluded that P was needed on the soils studied; K was also important, but not N.

Where soil nutrients are low, the accelerated removal of Ca by N fertilizers needs to be rectified by liming, which also helps to correct pH; losses of Mg are also significant. Consideration should also be given to possible effects of major nutrient application on micronutrients.

Tentative recommendations for mature cacao in Sabah:

	kg/ha	
Urea	90-110	(more may be needed if shade sparse)
Superphosphate (40 % P2O5)	110-170	
Sulphate of potash	110-280	(less K needed if pod
or muriate of potash	90-225	husks remain and are evenly distributed)
Kieserite	110-170	
Ground agricultural limestone	170-225	

Micronutrients

Boron and zinc are the more commonly reported deficiencies.

Remedy
- for B:
20-30 kg/ha borax (11.3 % sol.B) or other B-containing substances applied to the soil in a ring around the tree. Or foliar spray (200-300 g/l Solubor, Polybor or boric acid) repeated 3-4 times per year. Note that there may be a danger of toxicity after several years.

- for Zn:
10-20 kg/ha ZnSO4 applied to the soil. Or foliar spray of 1 % zinc sulphate or zinc oxide repeated 2-3 times per year.

Further reading

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Coffee *[Coffea arabica L. (Arabica coffee); Coffea canephora Pierre ex Froehner (Robusta coffee); Coffea liberica Bull ex Hiern. (Liberica coffee); Coffea excelsa Chev. (Excelsa coffee)]*

French: Café; Spanish: Café; Italian: Caffee; German: Kaffee

Arabica accounts for 80 per cent of the world coffee trade, and robusta most of the remaining 20 per cent. Liberica and excelsa together supply less than 1 per cent.

Crop data

Small perennial tree. Arabica typically up to 5 m, robusta up to 12 m, liberica up to 18 m.

Harvested part : fruits ("cherry" or "berry"). Processing removes the skin (epicarp), pulp (mesocarp), "parchment" (endocarp) and "silverskin" (integument) to leave the marketed product, "green bean" (endosperm).
Seed pre-germinated (optional) between wet sacks or in sand beds. Sown in nursery beds or pots. Vegetative propagation by grafting, budding and rooted cuttings is sometimes practised, particularly with robusta, and recently micropropagation techniques have been developed for arabica and robusta.

After 6-12 months in nursery, seedlings are transplanted to the field, usually at the beginning of the main rains.

Temporary shade and/or windbreaks are often provided for the first one or two years by shrubs such as Crotalaria spp and Tephrosia spp. Permanent shade may be provided by such trees as Albizia spp, Inga spp and Leucaena spp. Robusta is often shaded, dwarf arabica cultivars are usually not shaded, and tall arabica varieties may or may not be shaded.

Arabica is largely self pollinating, but robusta is cross-fertilized. First flowers 9-18 months after planting, and first crop matures during second or third year in the field. Following initiation, flower buds enter a dormancy phase which often coincides with a dry period. This dormancy is broken by the onset of rain. Fruits ripen 7-9 months after flowering in arabica, 9-11 months in robusta, about 12 months in excelsa, and up to 14 months in liberica.

Tree frame can be modified by various pruning systems. Individual trees having either a single bearing upright, or several, most commonly 2-5, bearing uprights. Low input coffee is often rarely, if ever, pruned, and individual trees may have 10 or more bearing uprights.

Plant densities vary from less than 1 000 to 10 000 trees/ha. Dwarf arabica cultivars are planted at the highest densities, of most commonly 3 000-7 000 trees/ha. Tall arabica cultivars are typically planted at 1 200 - 2 500 trees/ha, and robusta, liberica and excelsa from less than 1 000 to 2 000 trees/ha. Particularly in Brazil, it is common to plant two or more seedlings per station ("cova") at a relatively low density.

Mulch is sometimes applied, particularly to newly planted coffee, and in areas with a pronounced dry season. Leaf fall and prunings from coffee and shade trees can provide a significant proportion of mulching requirements.

Arabica grows best in a subtropical climate, which is frost-free and without strong winds; ideal temperature range 15°-24 °C; rainfall 1 500-2 500 mm per year, well distributed, but

with a drier period of 6-12 weeks. In some locations (e.g. Kenya) rainfall is supplemented by irrigation. Commonest altitudes 600-2 000 m in the tropics, although at higher latitudes (e.g. Brazil) it is grown below 600 m. Robusta, liberica and excelsa are more tolerant to heat, and flourish in the tropics from sea level to 1 100 m.

Coffee prefers deep (1-3 m, depending on climate) well drained, loamy soil which is slightly acid, and rich in humus and exchangeable bases, particularly potassium. Volcanic soils are often very suitable, although the crop is grown on a wide variety of soils.

Nutrient demand/uptake/removal

Nutrient removal in green bean - Macronutrients								
Coffee type	**Country**	**Source**	**kg/1000 kg green bean**					
			N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
Arabica	Brazil	Malavolta, 1990	17.0	2.5	18.7	2.6	3.9	1.3
Arabica	Papua New Guinea	Hart, 1969	22.0	4.6	20.4	3.3	2.1	-
Robusta	Papua New Guinea	Hart, 1969	25.4	4.6	24.0	4.0	3.6	-
Liberica	Equ.Afr.	Ledreux, 1928	28.0	6.4	45.0	-	-	-
Excelsa	Equ.Afr.	Ledreux, 1928	26.0	6.2	31.0	-	-	-

Nutrient removal in green bean - Micronutrients								
Coffee type	**Country**	**Source**	**g/1000 kg green bean**					
			Fe	**Mn**	**Zn**	**Cu**	**B**	**Mo**
Arabica	Brazil	Malavolta, 1990	61.2	20.4	12.2	13.6	16.3	0.05

If skins, pulp and parchment are not returned to the field, nutrients removed in whole fruit (equivalent to 1 000 kg green bean) are:

Nutrient removal in whole fruit - Macronutrients - (equivalent to 1000 kg green bean)								
Coffee type	**Country**	**Source**	**kg**					
			N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
Arabica	Brazil	Malavolta, 1990	34.8	5.7	64.5	4.6	9.9	2.9
Arabica	Kenya	Cannell and Kimeu, 1971	31.4	5.5	47.4	4.5	5.0	-
Arabica	India	Tandon, 1988	34.0	5.0	48.0	-	-	-
Robusta	India	Tandon, 1988	35.0	7.0	39.0	-	-	-
Robusta	Cote d'Ivoire	Snoeck and Duceau, 1978	33.4	6.2	43.8	4.1	5.5	-
Robusta	Indonesia	Roelofsen and Coolhaas, 1940	35.0	6.0	50.0	4.0	4.0	-

Nutrient removal in whole fruit - Micronutrients - (equivalent to 1000 kg green bean)								
Coffee type	**Country**	**Source**	**g**					
			Fe	**Mn**	**Zn**	**Cu**	**B**	**Mo**
Arabica	Brazil	Malavolta, 1990	112	50	84	32	51	0,12

Often, pruned leaves and branches are left in the field, but stems are removed for firewood or fences. Even the stems, however, contain significant quantities of nutrients:

Coffee type	**Country**	**Source**	**kg/ha**					
			N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
Arabica (1)	PNG	Willson, 1985	14.8	2.8	11.9	-	-	-
Arabica (2)	Brazil	Malavolta, 1990	105	14	119	13	84	2

Arabica (3)	Brazil	Malavolta, 1990	40	5	41	3	18	1
Robusta (4)	Cote d'Ivoire	Snoeck, 1983	34.8	9.1	17.4	13.9	66.1	-

Coffee type	Country	Source	g/ha					
			Fe	Mn	Zn	Cu	B	Mo
Arabica (2)	Brazil	Malavolta, 1990	1039	114	54	84	120	-
Arabica (3)	Brazil	Malavolta, 1990	471	45	11	28	33	-

Notes
(1) 1350 trees/ha, one stem per tree removed each year.
(2) 1904 covas/ha, 3-4 stems/cova, all cut 0.4 m above ground after 7 years.
(3) 1904 covas/ha 3-4 stems/cova, all cut 1.5 m above ground after 7 years.
(4) 1333 trees/ha, 4 stems/tree, all removed after 5 years.

Nutrient uptake

Nutrient uptake includes nutrients lost in crop and prunings, and nutrients used for the growth, development and maintenance of the tree:

Nutrient uptake - Macronutrients								
Coffee type	Country	Source	kg/ha/year					
			N	P2O5	K2O	MgO	CaO	S
Arabica (1)	Brazil	Malavolta, 1990	167.4	17.9	164.0	54.1	90.6	13.7
Arabica (2)	Costa Rica	Carvajal et al., 1969	172	36	119	20	53	-
Arabica (3)	Kenya	Cannell and Kimeu, 1971	150	23	180	25	77	-
Robusta (4)	Cote d'Ivoire	Snoeck, 1983	85.2	17.6	82.1	20.5	54.8	-
Robusta (5)	Indonesia	Saleh, 1983	53.2	10.5	80.7	16.5	28.0	-

Nutrient uptake - Micronutrients								
Coffee type	Country	Source	g/ha/year					
			Fe	Mn	Zn	Cu	B	Mo
Arabica (1)	Brazil	Malavolta, 1990	2202	746	489	360	199	-

Notes:
(1) Mundo Novo and Catuai varieties, 5.5-6.5 years old, 1250 covas/ha, 2 stems/cova, yield 3 000 kg/ha green bean.
(2) Bourbon variety, 3-4 years old growing in nutrient solution, equivalent uptake by 1 500 trees.
(3) Five year old trees, 1 500 trees/ha, yield 1 100 kg/ha gb.
(4) Five year old trees, 1 333 trees/ha, yield 1 000 kg/ha gb.
(5) 1 200 - 1 600 trees/ha, shaded, yield 1 500 - 2 000 kg/ha gb.

Nutrients recycled

Significant quantities of nutrients may be recycled in coffee leaf litter, prunings, pulp and parchment, and in shade tree leaf litter and prunings:

Nutrients recycled - Macronutrients								
Component	Country	Source	kg/ha/year					
			N	P2O5	K2O	MgO	CaO	S
Arabica (1) leaves	Brazil	Malavolta, 1990	37.5	6.0	46.4	29.0	43.7	3.1
Arabica (2) pulp + parchment	Brazil	Malavolta, 1990	17.8	3.2	45.8	2.0	6.0	1.6
Arabica (3) prunings	Brazil	Malavolta, 1990	322	41	341	50	207	11
Arabica (4)	Kenya	Cannell & Kimeu,	97.5	12.0	90.0	17.9	60.9	-

prunings + leaves		1971						
Robusta (5) leaves	Cote d'Ivoire	Snoeck, 1983	28.7	4.8	21.8	6.6	18.8	-
Erythrina (6) leaf litter + prunings	Costa Rica	Glover and Beer, 1986	182	48	187	45	183	-
Arabica + Erythrina leaf litter + prunings	Costa Rica	Glover and beer, 1986	366	69	317	80	340	-

Notes:
(1) Mundo Novo and Catuai varieties, 5.5-6.5 years old, 1 250 covas/ha, each with two trees.
(2) Pulp and parchment produced during processing of 1 000 kg green bean
(3) Mundo Novo variety, 1 904 covas/ha, with 3 or 4 stems per cova. Nutrients from leaves, branches and stems when pruned 0.4 m from the ground after 7 years growth.
(4) Ten year old, single stem coffee, 1 500 trees/ha. Nutrients from annual pruning and fallen leaves.
(5) Five year old trees with 4 stems/tree and 1 333 trees/ha.
(6) Erythrina poeppigiana pruned three times/year, 238 trees/ha.
(7) Erythrina poeppigiana, 238 trees/ha, pruned three times/year, and arabica coffee, 3 700 trees/ha, pruned once.

Nutrients recycled - Micronutrients								
Component	**Country**	**Source**	**g/ha/year**					
			Fe	**Mn**	**Zn**	**Cu**	**B**	**Mo**
Arabica (1) leaves	Brazil	Malavolta, 1990	751	78.7	160	200	97.5	-
Arabica (2) pulp + parchment	Brazil	Malavolta, 1990	51.0	29.5	71.4	18.4	34.7	-
Arabica (3) prunings	Brazil	Malavolta, 1990	2805	438	154	232	351	-

Notes:
(1) Mundo Novo and Catuai varieties, 5.5-6.5 years old, 1 250 covas/ha, each two trees.
(2) Pulp and parchment produced during processing of 1 000 kg green bean
(3) Mundo Novo variety, 1 904 covas/ha, with 3 or 4 stems per cova. Nutrients from leaves, branches and stems when pruned 0.4 m from the ground after 7 years growth.

Leaf analytical data

Sample 3rd or 4th pairs of leaves from tips of fruit-bearing branches, at mid-height of tree, on four quadrants; total of at least 40 leaves from 10 or more representative trees. Nutrient contents vary with season; it is often recommended that samples be taken at the end of the season when the crop matures, when nutrient levels are generally low. Differing critical levels from country to country are often attributable to differences in nutrient supply and/or retention capacities of soils:

Nutrient analysis data - Macronutrients							
Source	**Level**	**% of dry matter (60° C)**					
		N	**P**	**K**	**Mg**	**Ca**	**S**
Malavolta, 1990 Brazil (Arabica)	Deficient	<2.2	<0.10	<1.4	<0.26	<0.5	<0.10
	Marginal	2.2-2.6	0.10-0.14	1.4-1.8	0.26-0.30	0.5-0.9	0.10-0.14
	Adequate	2.7-3.2	0.15-0.20	1.9-2.4	0.31-0.36	1.0-1.4	0.15-0.20
	High	3.3-3.5	0.21-0.23	2.5-2.7	0.37-0.39	1.5-1.7	0.21-0.25
	Excess	>3.5	>0.23	>2.7	>0.39	>1.7	>0.25
Carvajal, 1984 Costa Rica (Arabica)	Deficient	<2.0	<0.09	<1.0	<0.10	<0.8	-

	Low	2.0-2.3	0.09-0.12	1.0-1.7	0.10-0.20	0.8-1.1	<0.2
	Adequate	2.3-2.8	0.12-0.20	1.7-2.7	0.20-0.35	1.1-1.7	0.2
	High	>2.8	>0.20	>2.7	>0.35	>1.7	-
CRF (undated) Kenya (Arabica)	Deficient	<2.5	<0.15	<2.2	<0.25	<1	<0.10
	Adequate	2.5-3.0	0.15-0.30	2.2-3.0	0.25-0.40	1.0-2.0	0.10-0.30
	Excessive	>3.0	>0.30	>3.0	>0.40	>2.0	>0.30
Krishnamurthy Rao, 1990 India (Arabica)	Low	<2.5	<0.10	<1.5	<0.30	<0.69	-
	Medium	2.5-3.5	0.10-0.15	1.5-3.5	0.30-0.40	0.69-1.11	0.10-0.18
	High	>3.5	>0.15	>3.5	>0.40	>2.0	>0.18
Willson, 1985 ,Compilation (Robusta)	Deficient	<1.8	<0.10	<1.2	<0.20	<0.4	<0.12
	Low	1.8-2.7	0.10-0.13	1.2-1.8	0.20-0.30	0.4-0.8	0.12-0.18
	Adequate	2.7-3.3	0.13-0.15	1.8-2.2	0.30-0.36	0.8-1.5	0.18-0.26
	High	>3.3	>0.15	>2.2	>0.36	>1.5	>0.26
Loué, 1957 Cote d'Ivoire (Robusta)	Very Low	<1.8	<0.11	<0.8	<0.30	<0.8	-
	Low	1.8-2.8	0.11-0.12	0.8-1.5	-	0.8-0.9	-
	Medium	2.8-3.3	0.12-0.15	1.5-2.5	0.30-1.10	0.9-1.2	-
	High	>3.3	>0.15	>2.5	>1.10	>1.2	-
Colonna, 1964 Central Africa Rep. (Excelsa)	Low	<1.59	<0.09	<0.73	<0.23	<1.92	-
	Medium	1.59-1.70	0.09	0.73-1.60	0.23-0.39	1.92-2.28	-
	High	>1.70	>0.09	>1.60	>0.39	>2.28	-

Nutrient analysis data - Micronutrients							
Source	**Level**	**ppm dry matter (60° C)**					
		Fe	**Mn**	**Cu**	**Zn**	**B**	**Mo**
Malavolta, 1990 Brazil (Arabica)	Deficient	<50	<40	<5	<4	<20	<0.10
	Marginal	50- 89	40-119	5- 7	4- 7	20-58	0.10-0.14
	Adequate	90-180	120-210	8-16	8-16	59-80	0.15-0.20
	High	181-220	211-300	17-25	17-30	81-90	0.21-0.30
	Excessive	>220	>300	>25	>30	>90	>0.30
Carvajal, 1984 Costa Rica (Arabica)	Deficient	-	<25	-	<10	<40	-
	Low	<75	25- 50	<6	10-15	40- 60	-
	Adequate	75-275	50-150	6-12	15-20	60-100	-
	High	-	>150	-	>20	>100	-
CRF (undated) Kenya (Arabica)	Deficient	<50	<50	<20	<10	<40	-
	Adequate	50-200	50-200	20-?	10-30	40-100	-
	Excessive	>200	>200	-	>30	>100	-
Krishnamurthy Rao, 1990, India (Arabica)	Low	-	-	-	<8	-	-
	Medium	100-300	80-200	10-15	10-15	40-45	0.10-0.15
	High	-	-	-	>15	-	-
Willson (1985) Compilation	Deficient	<40	<20	<13	<10	<20	<0.30

(Robusta)							
	Low	40- 70	20-35	13-20	10-15	20-35	0.30-0.50
	Adequate	70-200	35-70	20-40	15-30	35-90	>0.50
	High	>200	>70	>40	>30	>90	-
Loué, 1957 Côte D'Ivoire (Robusta)	Very Low	<75	<35	<3	<7	<25	-
	Low	75- 80	35-40	3-14	7-10	25- 40	-
	Medium	80-100	40-60	14-27	>10	40-150	-
	High	>100	>60	>27	-	>150	-
Note: Critical levels for molybdenum are tentative.							

Excessive levels of N, P or K, or low levels of Fe, have been shown to be detrimental to the cup quality of the beans. High levels of N also result in production of excessive foliage at the expense of flowers and hence cherries. Inadequate N or K leads to "dieback" and, in extreme cases, may even result in death of the tree. Inadequate potassium may also increase the proportion of "floaters", or empty cherries.

Several nutrient interactions are important. Excessive N can induce S deficiency. The leaf N : S ratio should be less than 35; if above 40, S is likely to be deficient. The N/P interaction is also important; generally the leaf N : P ratio should be between 7 and 35, or ideally between 18 and 22. High P can also induce Fe and Zn deficiencies. Deficiencies of Mn or S are often associated with high K. K/Mg antagonism is well established; generally, the leaf K : Mg ratio should be below 10. Other antagonisms include Fe/Mn and Ca/B.

Fertilizer recommendations

Nurseries: A well prepared bed or potting mixture, consisting of topsoil, sand, organic material and a P fertilizer, will provide adequate nutrition for the 6-12 months the seedlings spend in the nursery. Other nutrients may however be added to the potting mixture, and often small quantities of fertilizers are also applied as the seedlings develop. Foliar fertilizers are particularly effective.

Planting holes: should be filled with a mixture of topsoil, organic material and a P fertilizer. Other nutrients may be added in the planting hole, including limestone, dolomite or kieserite if appropriate. Some very acid soils are limed with a top dressing before planting.

Young coffee: for the first 2 or 3 years in the field it is usual to follow a standard fertilizer programme which provides relatively small, but frequent applications of N, P and K. Rates are calculated on a per tree basis, and increase as the trees develop. Particularly during the first year, foliar fertilizers provide a valuable supplement to ground-applied fertilizers, and can be used to supply the necessary micronutrients, especially Zn and B. Containing macro- and micronutrients, they are often beneficial during the dry season.
Ground-applied fertilizers are distributed in a gradually widening band around the tree, which extends from about 7 cm from the stem to slightly beyond the edge of the drip circle.

Mature coffee: When the trees are in bearing, fertilizer rates are calculated on the basis of soil and leaf analysis, expected yield, stage in the production cycle, and general condition of the crop: usually 150-300 kg/ha/year N and K2O and 0-150 kg/ha/year P2O5, split into three or four dressings. K is particularly necessary during the period of fruit expansion and ripening. Mg and S are also important and are usually provided as components of compound or other straight fertilizers.

Micronutrients are usually given in foliar fertilizers, which are often best applied as routine applications during the dry season. Serious deficiencies, however, are best treated with immediate foliar application in conjunction with a more concentrated ground-applied fertilizer.

Robusta coffee generally requires less nutrients than arabica.

Preferred nutrient forms

In most countries, at least some of the required nutrients are usually applied in the form of a compound fertilizer. Typically, one or two applications of an appropriate compound will provide adequate P, Mg and possibly S, and a proportion of the required N and K of which the balance is then provided by straight fertilizers.

Commonly used N fertilizers include urea, ammonium sulphate, ammonium sulphate nitrate, ammonium nitrate and calcium ammonium nitrate. In particularly acid soils urea and ammonium sulphate should be avoided if possible, and calcium ammonium nitrate be used. Urea should be mixed with the topsoil, particularly in drier weather, to reduce volatilization losses. Urea is commonly used in foliar sprays at concentrations of 1.0-1.5 % (over 2 % may produce leaf scorch).

K is provided as muriate or sulphate. If there is enough S in the compound and nitrogenous fertilizers, then muriate is preferred. If adequate S is not provided by the other fertilizers, then sulphate of potash should be considered. If muriate is used near the coast, it is important to monitor Cl leaf levels in order to avoid possible toxicity.

The most commonly used straight **P** fertilizers are single or triple superphosphate, diammonium phosphate and rock phosphate. Rock phosphate is best used in acid soils. Single superphosphate is also useful on acid soils, particularly those low in S and Ca. If diammonium phosphate is used, it also supplies some of the N requirement.

If additional **Mg** is needed, Kieserite, magnesium sulphate (as Epsom salts) or magnesium nitrate are usually recommended for soil application. Epsom salts or magnesium nitrate can also be used as 0.5-0.75 % foliar spray. Often, Mg and liming are both needed, in which case dolomite or dolomitic limestones are preferred.

B and **Zn** are the most commonly encountered micronutrient deficiencies. They may be treated respectively with Solubor, or Borax (foliar or soil-applied), and with zinc sulphate, zinc sulphate monohydrate, zinc oxide (foliar or soil-applied) or zinc chelates (foliar). **Fe** or **Mn** deficiencies may occur, particularly with robusta, and are usually treated with foliar sprays of Fe or Mn chelates or sulphates.

Present fertilizer practices

Brazil (Source: Malavolta, 1990)

Traditionally, coffee was planted at 1 400-1 900 covas/ha and 2-4 plants per cova. Recently higher densities (up to 5 000) have been used with one plant per hole. No shade is used. National average yields are around 600 kg/ha gb, but well managed low-density plantations achieve 1 200-1 500 kg/ha gb, and recently planted high-density plantations 1 800-2 400 kg/ha gb. The following practices refer to arabica.

Nurseries: Potting mixture = 1 000 litres topsoil plus 200 litres stable manure (or 50 litres chicken manure), 5 kg single superphosphate, 1 kg dolomitic limestone, 500 g muriate of potash, 5 g borax, 5 g zinc oxide (50 % Zn) and 5 g copper sulphate.

Planting holes: Before planting, acid soils are limed (topdressed and incorporated into top 30 cm at least) at rates necessary to increase base saturation to 70 %. Sometimes phosphogypsum is also topdressed to overcome subsoil acidity. Planting holes are prepared (40 x 40 x 40 cm), and refilled with soil plus 60-80 g P2O5 (as single superphosphate), 12-15

g K_2O (as muriate of potash), 200-500 g dolomitic limestone, 0.2 g boron, 0.2 g copper, 1 g zinc and organic fertilizers if available. If planting furrows are used, equivalent quantities of the above fertilizers are applied.

Young coffee: During the first year in the field each plant receives 4 dressings of 5 g N, and 2 dressings of 5-10 g K_2O. In the second year four times as much N and K_2O are applied, split into 3 dressings. In the third year, the application rates are double those of the second year, and 0-40 g P_2O_5 are added (depending on soil analysis).

Mature coffee: When the trees are in full bearing, fertilizer rates are based on soil and leaf analyses, and on expected yield: 200-300 kg/ha/year N and K_2O; 0-50 kg/ha/year P_2O_5 in 3-4 dressings (Sept. - Apr.). Micronutrients are controlled by 3-4 foliar sprays containing boric acid (0.3 %) and zinc sulphate (0.6-0.8 %). Copper is supplied as oxychloride used in coffee leaf rust control.

Kenya (Source: CRF, 1983 and CRF, 1990)

Traditionally most coffee was planted at 1 329 trees/ha, although more recent plantings, particularly if irrigated, have been at densities of 3 000-4 000 trees/ha. Traditional farmers average about 600 kg/ha green bean, while non-irrigated plantations average about 800 kg/ha green bean and irrigated plantations 1 300 kg/ha green bean. The better plantations exceed 2 000 kg/ha green bean. Most plantations are not shaded. The following practices are for arabica which represents the great majority of production.

Nurseries: Recommended potting mixture = 300 litres well sieved topsoil, 200 litres coarse sand, 100 litres farmyard manure, 250 g single superphosphate plus a soil insecticide. This is sufficient to fill about 220 polypots (30 x 17 cm when flat, volume 2.75 litres). Very little additional fertilizer is required although weekly applications of foliar feeds are considered beneficial during the final 6 months.

Planting holes: (60 x 60 cm dia) are dug during the dry season, and allowed to "weather" for 3 months before refilling with topsoil, 20 litres cattle manure (or well rotted coffee pulp), 100 g double superphosphate, 100 g dolomitic limestone (where necessary) and a soil insecticide. In acid soils 200 g single superphosphate are used in place of double superphosphate. If the Ca:Mg ratio is very low, ground limestone is used in place of dolomitic limestone. Holes are refilled one month before planting, in order to allow the soil to settle. When planting, a small hole just sufficient to accommodate the nursery seedling, is opened in the centre of the refilled hole.

Young coffee: During the first two years in the field 40 kg/ha/year N are applied, split into 3 dressings per year (March/April, May/June, November). Calcium ammonium nitrate is recommended if the soil pH is below 5.4; above pH 5.4, ammonium sulphate, ammonium sulphate nitrate, or urea are recommended. Fertilizer is applied in a broad band extending from about 30 cm from the stem to about 30 cm beyond the drip circle.

Mature coffee: 80, 100, 100-150 or up to 200 kg/ha/year N for expected yields of below 1 000, 1 000-1 500, 1 500-2 000, and over 2 000 kg/ha green bean respectively. More than 200 kg/ha/year N is not recommended. Annual amounts are split into 4 dressings in high rainfall or irrigated areas, or 3 dressings in moderate rainfall areas. If the soil pH (in water) is below 5.4, calcium ammonium nitrate (CAN) is recommended, twice for every one application of ammonium sulphate nitrate (ASN) or urea. If the pH is between 5.4 and 6.5, alternating applications of CAN and ASN are recommended, or urea; above pH 6.5, ammonium sulphate or ASN.

If the soil pH is below 5.4, P is given as 350 g single superphosphate/tree. Above pH 5.4, 150 g diammonium phosphate or 150 g double or triple superphosphate/tree are recommended before or during the main rains.

Soil exchangeable K level should be at least 0.5 me%, and K:Mg:Ca ratio 1:(2-3):(4-6). If additional K is required, 100-190 g muriate or sulphate/tree (135-250 kg/ha) are recommended once every 3-4 years at the beginning of the main rains. If Mg is needed 170 g calcined magnesite/tree are recommended every 3-4 years, supplemented by 0.5 % solution of Epsom salts as foliar feed. Ca is supplied as 340 g dolomitic limestone/tree (460 kg/ha) every 4-5 years.

Appropriate compound fertilizers (based on soil available P, and exchangeable cation levels) often replace one of the N applications 6 months before the main flowering. Foliar feeds are used to supplement soil applications during dry weather or times of stress.

Farmyard manure (20 litres/tree), cattle manure (40 litres/tree every 2-3 years), coffee pulp (25 t/ha dry material every 2-3 years) and Napier grass mulch (16-28 t/ha dry material for 2 years followed by 2-3 years without) are also recommended if available.

Lime is applied to acid soils at a rate of 500 g/tree for every 1 me% of exchangeable acidity.

Papua New Guinea (Source: CRI, 1990)

Coffee in Papua New Guinea is often shallow rooting (<50 cm) due to soil conditions, and annual rainfall totals range from 1 500-3 000 mm with no dry season, but a drier period July - Sept. Traditional low input, heavily shaded, coffee gardens produce up to 1 000 kg green bean/ha, although typically less than half of this is harvested. Rehabilitated, lightly shaded, coffee gardens can produce up to 1 500 kg/ha green bean, and unshaded commercial plantations average about 2 000 kg/ha green bean, with the better plantations exceeding 3 000 kg/ha green bean in good years. The following practices refer to arabica, which accounts for 95 % of production.

Nurseries: Recommended potting mixture = 300 litres sieved topsoil, 200 litres washed coarse river sand, 100 litres well-rotted coffee pulp, 250 g triple superphosphate and a granulated soil insecticide. This is sufficient for 125 polypots (30 x 15 cm , volume 4.8 litres). A complete foliar feed, containing macro- and micronutrients, is given monthly, and 5 g of high-P compound fertilizer is applied to each polypot after 6 months.

Planting holes (45 x 45 cm dia) are dug 1 - 2 weeks before planting, and immediately refilled (to avoid filling with rain) with a 4:1 mixture of topsoil and well rotted coffee pulp, plus 60 g diammonium phosphate or triple superphosphate (or 240 g rock phosphate). Seedlings are later planted in the centre of the refilled planting hole.

Young coffee: Tall varieties (e.g. Arusha, Mundo Novo, Blue Mountain) are typically planted at 2 500 trees/ha. During the first year in the field, they receive a complete foliar feed on the day of planting, and monthly thereafter. At 6 weeks, 70kg/ha ammonium sulphate and after 14 weeks, 100 kg/ha 15:15:6 (or similar) compound fertilizer are applied. Compact varieties (e.g. Caturra, Catimor) are typically planted at 3 000 - 5 000 trees/ha, and grow particularly rapidly in their early years. Application rates of soil-applied fertilizers are therefore increased to 140 kg/ha ammonium sulphate and 200 kg/ha 15:15:6.

During the second year in the field, tall varieties receive 100 kg/ha 15:15:6 in October and February, and 70 kg/ha ammonium sulphate in December and May. Complete foliar feeds are applied during the drier season in July, August and September. Compact varieties are fertilized according to expected yield.

Mature coffee in bearing (from year 3 for tall varieties, and from year 2 for compact varieties): Generally, recommended N:P2O5:K2O:MgO ratios are 4:1:4:1, and for expected yields of under 500, 1 000, 1 500, 2 000, 2 500 and over 2 500 kg/ha/year green bean respectively 60, 100, 140, 200, 300 and 400 kg/ha/year N are recommended. These rates are modified where necessary on the basis of the condition of the trees, and on soil and leaf analysis.

Fertilizer is not recommended for heavily shaded, unrehabilitated, traditional coffee gardens producing less than 500 kg/ha green bean. Rehabilitated coffee gardens are best fertilized with 2 dressings (October, April) of a compound fertilizer as close as possible to the recommended N: P2O5:K2O:MgO ratio of 4:1:4:1. Well managed rehabilitated coffee gardens and commercial plantations, producing more than 1 000 kg/ha green bean should be given a low-P/low-K compound in October and April/May, and straight N and K in December/January and February/March. Foliar sprays containing macro- and micronutrients should be given in July, August and September.

Urea and ammonium sulphate are the most commonly used N fertilizers, K is usually applied as muriate. If additional P is required, diammonium phosphate, triple superphosphate or rock phosphate is given. Supplementary Mg is provided in locally mined dolomite, kieserite or nitromagnesia at 50 kg/ha/MgO.

If, despite the routine foliar feeds, Zn deficiency is still a problem, one ground application of 30-60 kg/ha zinc sulphate or 15-30 kg/ha zinc sulphate monohydrate or 3 foliar sprays at four-week intervals of 1.2 kg zinc sulphate or 600 g zinc sulphate monohydrate in 300 litres water/ha, are recommended. Boron deficiencies are similarly treated with one ground application of 37.5 kg/ha Solubor (or 75 kg/ha borax, or 3 foliar sprays at four-week intervals of 900 g Solubor or 1.8 kg borax in 300 litres water/ha. Mo deficiency is treated by 3 applications of 150 g ammonium molybdate in 300 litres water/ha.

India (Source: Krishnamurthy Rao and Ramaiah, 1985)

India produces both arabica and robusta. The following practices refer to robusta, which is typically planted at 1 075 trees/ha, under a regulated mixed shade.

Nurseries: When seedlings have 4-5 pairs of leaves, each seedling receives 3-5 g diammonium phosphate and a pinch of urea.

Planting holes should receive triple superphosphate and coffee pulp.

Young coffee: During the first year in the field 45 kg/ha N, 30 kg/ha P2O5 and 45 kg/ha K2O are applied in three equal dressings. In the second and third years, these amounts are increased to 60 kg/ha/year N, 45 kg/ha/year P2O5 and 60 kg/ha/year K2O.

Mature coffee: For robusta producing up to 1 000 kg/ha/year green bean, 80 kg/ha/year N, 60 kg/ha/year P2O5 and 80 kg/ha/year K2O in two equal dressings in March (pre-blossom) and October (post-monsoon). For robusta producing over 1 000 kg/ha/year green bean, 120 kg/ha/year N, 90 kg/ha/year P2O5 and 120 kg/ha/year K2O in 3 equal dressings in March (pre-blossom), May (post-blossom/pre-monsoon) and October (post-monsoon).

Urea, ammonium sulphate and calcium ammonium nitrate are commonly used N fertilizers. P is given on acid soils as rock phosphate or basic slag, and on less acid soils as nitrophosphates. Both muriate and sulphate of potash are used as K fertilizers. Various compounds are also in use, and commercial plantations apply supplementary foliar sprays.

Côte d'Ivoire (Source: Snoeck, 1988)

The Ivory Coast is a major producer of robusta, typically grown without shade at about 1 333 trees/ha. Current recommendations are to plant 1 961 trees/ha. Yields range from less than 500 kg green bean/ha to about 2 000 kg/ha green bean, although well managed, elite clones can produce yields in excess of 3 000 kg/ha green bean.

Nurseries: Polypot nurseries are used occasionally for growing seedlings propagataed from seed but most commonly for rooted cuttings which are obtained from clonal cuttings gardens and initially raised in propagators until roots are 5-10 cm long. The potting mixture consists of sifted humus and rich topsoil, and the polypots are 25 cm x 12 cm diameter. Fertilizers applied depend on the topsoil used. Often no fertilizers are used, or only urea as a foliar feed (10 g urea/10 litres water) every 2 weeks. If necessary 3 g 15:15:15 compound fertilizer can be applied to each pot every 2 weeks from the 2nd to the 6th month.

Planting holes (40 x 40 x 40 cm) are dug at the beginning of the rainy season, and refilled the next day with topsoil, mixed if possible with compost or manure.

Young and mature coffee: Emphasis is placed on soil analysis, and to a much lesser extent, leaf analysis. On moderately leached soils (base saturation > 40 %), only urea is recommended during the first production cycle. During the first year in the field 43 kg/ha urea are recommended, split into 2 dressings in July and September. In the 2nd, 3rd and 4th years, rates are increased to 109, 174 and 196 kg/ha/year urea respectively, applied in March and September (i.e. at the onset of the two rainy seasons). In the 5th, 6th and 7th years the recommended rate is 217 kg/ha/year urea, applied in March and September. The coffee is stumped back during the 8th year and so the fertilizer rate is reduced to 109 kg/ha urea, applied in July and September. 500 kg 20:10:5 compound fertilizer ha/year are recommended throughout the second and subsequent production cycles.

On very leached soils (base saturation < 40 %), or soils with low cation exchange capacity (CEC < 5 me%), 12:6:20:4 compound is recommended: In the first year in the field, at 333 kg/ha split into 2 dressings in July and September, and in the 2nd and 3rd years at 666 and 1 000 kg/ha/year respectively, applied in March and September. This latter rate continues to the 7th year. The coffee is stumped-back in the 8th year and so the rate is reduced to 250 kg/ha/year, applied in July and September. 1 000 kg/ha/year 12:6:20:4 are recommended from the 9th year onwards.

Leaf analytical data are used to identify micronutrient deficiencies.

Further reading

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Hops *(Humulus lupulus L.)*

French: Houblon; Spanish: Lupolo; Italian: Luppolo; German: Hopfen

Crop data

Harvested part: umbels (cones) which develop from unpollinated flowers.

Rhizome cuttings, or young plants cultivated in the greenhouse, are used as planting material. Rootstocks develop during the first year, when little or no yield is produced.

Flowers July in northern hemisphere, December-January in southern hemisphere.

Harvested late August - early September (northern hemisphere), March - April (southern hemisphere). Whole plants are removed and then processed in a stationary picking machine; wastes (leaves, petioles, vines) are chopped and, in part, brought back to the field. Because the cones are immediately dried to 12 % moisture, yields are expressed in terms of dried hops.

Rootstocks sprout anew each year, life expectancy: 10-20 years.

Plant density: 1 800-2 200 plants/ha, with two vertical wires to each plant (variations depending on variety and site).

Preferably grown on medium soils (Sandy Brown Soil or Loess Brown Soil) with pH 5-7.

Without irrigation, in temperate climates with adequate precipitation during the vegetation period (Europe); or, with irrigation, in arid climates (USA, etc.).

Nutrient demand/uptake/removal

Nutrient removal (cones, plus leaves, petioles and vines) - Macro- and micronutrients								
Yield* t/ha	**Source**	**kg/ha**					**g/ha**	
		N	**P2O5**	**K2O**	**MgO**	**CaO**	**Zn**	**Fe**
1.86	Zattler, 1954/56	117	38	110	-	154	-	-
2.84	Marocke, 1957	204	60	167	-	296	-	-
2.42	Roberts, 1961	272	74	257	-	-	-	-
2.00	Rossbauer, 1978	150	45	160	40	190	600	1 630
* dried hops								

Plant analysis

Plant analysis data (average nutrient content of aerial parts) - Macronutrients				
% of dry matter				
N	**P**	**K**	**Mg**	**Ca**
2.58	0.34	2.21	0.39	2.35

Fertilizer use

Germany (example of temperate climate, without irrigation):

Planting year (baby hops):

- first N application when growth begins (late June), at 5-7 kg N/1 000 plants;
- second N application when plants are well developed (end of July), 5-7 kg N/1 000 plants.

- P and K fertilizer needed only on soils of low to medium content

Subsequent years - strung for harvest (for expected yield of 2 t/ha):

N at 180-200 kg/ha, which may be reduced if Nmin test in early March indicates more than 80 kg/ha (nitrate + ammonium) in top 90 cm of soil. For early-maturing varieties this is applied one-third in March, one-third in April, one-third late May; and for late-maturing varieties one-third in March, one-third late May, one-third early July. Usually given in ammonium nitrate form (e.g.CAN). Occasionally as 1-2 spray applications of urea ammonium nitrate solution (15-30 % concentration) from late May to mid-June to the soil and to the plants up to a height of 1 m, thereby simultaneously destroying excessive shoots.

P, K, Mg:

Nutrient levels in soil (mg/&àà g)			**Recommended fertilizer rates* kg nutrient per ha**		
P2O5	**K2O**	**Mg**	**P2O5**	**K2O**	**MgO**
15-25	15-35	7-25	45	160	40
*Rates may be increased by up to 100 % on soils lower in these nutrients or may be progressively reduced to zero at higher nutrient levels.					

Basal fertilization is generally applied in one operation (March).

- lime (where required, liming is done in the autumn):

Type of soil	**Desired pH**	**Maximum single application**	
		CaO (t/ha)	**CaCO3 (t/ha)**
Sand	5.3-5.7	-	2.0
Loamy sand	5.8-6.2	-	2.5
Sandy loam/silty loam	6.3-6.5	1.5	3.0
Clayey loam/clay	6.5-6.9	2.0	-

Micronutrients: hops are sensitive to Zn deficiency, which is generally associated with high soil pH and excessive soil P; it may be controlled by foliar sprays of 0.1-0.2 % zinc sulphate or chelate (3-5 sprays from tying to flowering), or Zn-containing fertilizers may be applied to the soil.

USA (example of arid climate, with irrigation):

N:
where soil N = 10 ppm, 160 kg/ha N
where soil N = 20 ppm, 110 kg/ha N

P2O5:
where soil P2O5 = 46 ppm, 126 kg/ha P2O5

K2O:

where soil K_2O = 72 ppm, 216 kg/ha K_2O

These rates may be increased by up to 50 % on soils where nutrient contents are lower, or reduced where they are higher than the amounts stated.

N is applied in three split dressings; P and K in a single application.

Zn: applied, where necessary, as foliar sprays of 0.15-0.18 % zinc sulphate or chelate.

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Author: G. Rossbauer, Bayerische Landesanstalt fuer Bodenkultur und Pflanzenbau, Hopfenforschung und Hopfenberatung, Wolnzach, Germany
Contributor: G. K. Lewis, Hopunion USA Inc., USA

Tea *(Camellia L. spp.):*

C. sinensis (L.) O. Kuntze - China type
C. assamica (Mast.) Wight ssp assamica - Assam or Indian type
C. assamica ssp lasiocalyx Planch ex Watt - Southern or Cambod type.

French: Thé; Spanish: Te; Italian: Te; German: Tee

Cultivated types are natural hybrids of the three major types or species and are referred to as China, Assam (Indian) or Cambod (Southern) hybrids according to their predominating character, particularly shoot size.

The China types are hardy and tolerant of extreme weather conditions; they have relatively small, erect, dark green leaves; they flower profusely and require more N than Assam types to sustain vegetative growth; their root cation exchange capacity is higher than that of Assam types and they require more K in the soil solution; they tolerate and respond to severe pruning and plucking. Assam types have supple, light green, long, flat or semi-erect, large leaves; they are high-yielding but require more care in handling; they are less tolerant of extreme weather conditions; they are more responsive to manuring and improved crop husbandry practices than other types. Cambod types are intermediate in character.

Crop data

Perennial. Harvested part: growing shoots (primarily 2 leaves and a bud; in some regions, up to 3 leaves and a bud).

Planting material: seedlings 10-18 months old raised from seed (hybrids, polyclonal and biclonal), single-leaf nodal cuttings from selected high-yielding bushes and grafted cuttings at the nursery stage; transplanted into the monsoon or rainy period.

Plants are trained into bushes by centring and plucking, and are harvested from the second year onwards at regular intervals of 7-20 days in the tropics and sub-tropics (all the year round between latitudes of up to 15-18 ° N and S, beyond which winter dormancy occurs at the rate of 30 days for every additional 3-5 ° from the equator) and at much longer intervals in temperate zones. The day length required for vegetative growth is 11 h 15 min. The economic life is believed to be around 40-60 years in many tea-growing regions; though, in South India, tea gardens over 100 years old are responding to manuring and good husbandry practices with productivity increasing every year.

Plant density: old plantings 6 400 plants/ha (1.2 m by 1.2 m, square), new plantings 10 000-20 000 plants/ha as single hedge (1.2 m by 0.6-0.8 m) or double hedge (1.20 - 1.35 m by 0.6-0.8 m by 0.6-0.8 m) along the contour.

Shade: tea is grown under shade (most commonly Grevillea - silver oak - and Albizzia) which allows 60-70 % penetration by sunlight. This reduces radiation damage at latitudes further from the equator, creates a favourable micro-climate during dry months and maintains a relatively much lower leaf temperature in summer, particularly in low latitudes.

Climate and soil

Tea is basically a rainfed crop cultivated commercially in humid zones of tropical, sub-tropical and temperate regions with alternating wet and dry periods, on well-drained soils preferably with pH 4.5-5.0 although in practice ranging from 3.0 to 6.5, and at altitudes from sea level to

2 300 metres. Its distribution extends from Georgia (USSR), 43 °N, to Corrientes in Argentina, 27 °S.

Major producers: India, China, Sri Lanka, USSR, Japan, Indonesia, Kenya.

Tea soils are managed by regular liming to pH 4.5 and 5.0; retaining prunings and litter from both tea and shade trees in the field itself; chemical methods of weed control, broadcasting of N and K fertilizers and placement of P at 15 - 25 cm depth to reduce soil erosion and leaving a buffer strip, where necessary, to avoid any direct loss of agricultural chemicals into waterways. Maintenance of soil organic matter is important for stabilizing soil structure. Soil conservation measures and "no-tillage" cultivation systems help to reduce erosion and to stabilize productivity in the long term.

Nutrient demand/uptake/removal

Only 8-13 % of the total dry matter produced, including roots, is harvested; this is about 13-17 % of the dry matter of the above-ground parts and 18-35 % of that of the parts formed after and above the level of pruning. The total amount of nutrients required to produce a given level of yield of marketable tea may be calculated from the chemical composition of the various plant parts and their relative proportions in the total dry matter (see Table).

Nutrient contents (to produce 100 kg marketable tea) - Macro- and secondary nutrients										
Plant part	**Proportiona dry matter**		**Nutrient contents in the various parts**							
			kg					**g**		
	kg	**%**	**N**	**P2O5**	**K2O**	**MgO**	**CaO**	**Al*** **	**Cl**	**Na**
1. Plucks (marketable tea)**	100	12.5	4.0	1.15	2.4	0.42	0.8	100	6	8
2. Foliage on the bush	120	15.0	3.9	0.98	1.1	0.60	2.1	120	15	9
3. Fallen leaves (litters)	80	10.0	2.6	0.66	0.7	0.40	1.4	80	10	6
4. Stems/branches	320	40.0	3.2	1.90	0.7	0.80	1.1	160	10	24
5. Roots	180	22.5	3.2	0.98	3.8	0.71	1.3	411	NA	27
Total	800	100.0	16.9	5.68	8.8	2.92	6.7	871	-	74
Removed by 1 & 4*	420	45.0	7.2	3.05	3.1	1.21	2.0	260	16	32
* removed as crop (1) and for fuel at the time of pruning (4). - ** 12.5 % in high yielding fields under intensive cultivation systems; in low/moderate yielding fields it varies between 8 to 10 %. - *** Al content in mature foliage can go several times (up to 20 times) the figures shown above. - NA Not available										
Sources : ¦Magambo and Othieno; 1977; Ranganathan and Natesan, 1988; Ling, Harding and Ranganathan, 1989; Natesan and Ranganathan, 1990; Annual Reports 1968 to 1989 UPASI Tea Research Institute, UPASI, Coonoor (India).										

Nutrient contents (to produce 100 kg marketable tea) - Micronutrients							
Plant part	**Proportional dry matter**		**Nutrient contents in the various parts (g)**				
	kg	**%**	**Zn**	**B**	**Cu**	**Fe**	**Mn**
1. Plucks (marketable tea)**	100	12.5	4	3	4	20	85
2. Foliage on the bush	120	15.0	4	9	12	40	180
3. Fallen leaves (litters)	80	10.0	3	6	8	27	120
4. Stems/branches	320	40.0	11	6	6	24	72
5. Roots	180	22.5	16	2	8	130	22
Total	800	100.0	38	26	38	241	479
Removed by 1 & 4*	420	45.0	15	9	10	64	157

* removed as crop (1) and for fuel at the time of pruning (4). - ** 12.5 % in high yielding fields under intensive cultivation systems; in low/moderate yielding fields it varies between 8-10 %.
Sources: Magambo and Othieno; 1977; Ranganathan and Natesan, 1988; Ling, Harding and Ranganathan, 1989; Natesan and Ranganathan, 1990; Annual Reports 1968-1989 UPASI Tea Research Institute, UPASI, Coonoor, India

Plant analysis data

Plant analysis is not very useful for predicting fertilizer needs, as the chemical composition of the tissues is little affected except in extreme cases of deficiency of long duration. However, comparison with the average figures shown in the table can prove useful in modifying current fertilizer schedules where the content of any particular nutrient(s) shows a decreasing trend over the years, and to correct imbalances.

For example, the ratio of K in growing shoots to that in older leaves gives an indication of withdrawal of that nutrient from mature foliage as a result of inadequate K supply; this was at one time used extensively in East Africa for correcting K imbalance.

The K content of mature leaves, and their ash, has been reported as a useful guide for identifying K deficiency in S. India, Indonesia, Taiwan and USSR. Healthy tissues contain > 0.8 % K and their ash > 8 % K.

The K index of living tea tissues expressed on a moisture basis, i.e. K % DM x (100 - % moisture)/ % moisture, varies little through the year. If it falls below 0.25 % at any time of the year, there is a need to increase the rates of K application in the fertilizer schedule. Succulent tissues are more sensitive in this respect than others.

The Ca content of the maintenance foliage should be kept between 0.6 % and 0.8 %. Crop depression occurs when it exceeds 0.8 %, and heavy metal toxicity (particularly of Mn) arises when it falls below 0.6 %. Ca, Zn and Fe can all accumulate in the roots, with restricted movement to the shoots. In soils high in Mn, availability in the soil should be kept low by heavy liming and by extra attention to the soil organic matter status.

Sensitive tissues for P are: flush in Indonesia and S. India, mature foliage in Sri Lanka, and second, third and fourth internodes in S. India. P in these tissues varies between 0.22 % and 0.50 % and declines with age.

Crop yield is reduced if the Na content of leaf tissues exceeds 150 ppm. In soils containing more than 60 ppm exchangeable Na, large quantities of K should be used to reduce the uptake of Na.

Al is accumulated passively by tea; its exact role is not yet understood. Aluminium - tea tannin complexes impart brightness and redness to black tea liquors, for which they are valued, and they affect the valuation of green tea liquors for the same reason.

Fertilizer recommendations

Fertilizer recommendations - Southern India			
Stage	**Nutrients**	**Nutrient kg/ha/year**	**Remarks**
1. Nursery	N:P2O5:K2O:MgO: Zn	30:30:30:10:3	Per 100 000 cuttings in 16 to 20 applications (once in 15 days)
2. Young tea			P in one blanket application
- 1st year	N:P2O5:K2O:MgO	90+180:270:30	placed at 15-25 cm depth.
- 2nd year	N:P2O5:K2O:MgO	90+240:360:40	N, K and Mg for soils with pH
- 3rd year	N:P2O5:K2O:MgO	90+300:450:50	below 4.5 and N

- 4th year	N:P2O5:K2O:MgO	90+300:300:50	and K for soils with pH above 4.5 in 4 dressings in the first year and 4-6 dressings (broadcast) thereafter; foliar application of Zinc sulphate at 6-10 kg/ha in 3-5 sprays as required.
3. Pruned fields			P in one blanket application placed at 15-25 cm depth.
- Rejuvenation pruning	P2O5 + N:K2O	90+180:360	N and K in 4-6 dressings in lighter forms of pruning and
- Hard pruning	P2O5 + N:K2O	90+210:315	2-3 in harder forms of pruning, 3-5 foliar sprays of in total 6-10 kg/ha Zinc sulphate.
- Medium, light, cut-across and skiffing	P2O5 + N:K2O	90+240:360	
4. Mature tea (low yielding less than 3000 kg/ha			P once in two years placed at 15-25 cm depth. N and K broadcast in 4-6 dressings; 6 kg/ha Zinc sulphate in 3 foliar sprays every year or 11.25 kg/ha in 5 foliar sprays in alternate years.
- 1 000 kg/ha	P2O5 + N:K2O	90+120:100	
- 2 000 kg/ha	P2O5 + N:K2O	90+120:170	
- 3 000 kg/ha	P2O5 + N:K2O	90+250:190	
5. Mature tea high yielding (above 3 000 kg/ha)			P once every year placed at15-25 cm depth. N and K broadcast in 4-6 dressings. 6-10 kg/ha Zinc sulphate every year in 3-5 foliar sprays.
- 3 000 kg/ha	P2O5 + N:K2O	70+250:250	For fields yielding below
- 4 000 kg/ha	P2O5 + N:K2O	70+350:350	4 000 kg/ha 20-25 kg/ha year
- 5 000 kg/ha	P2O5 + N:K2O	90+450:450	Mg as Magnesite and 75-125 kg/ha as magnesium silicate; for fields yielding above 4 000 kg/ha 40-50 kg/ha Mg as Magnesite and 125-150 kg/ha Mg as magnesium silicate.
- Liming to pH 4.5-5.0 recommended in year of pruning at rate based on soil pH, yield level, annual rainfall and length of pruning cycle.			
- N rates adjusted to yield of marketable tea. 10 kg N per 100 kg yield up to yield of 2 000 kg/ha (minimum 120 kg/ha N), plus 5 kg N per 100 kg additional yield over 2 000 kg/ha up to 3 000 kg/ha, and 10 kg N per 100 kg additional yield thereafter.			

Fertilizer recommendations - Northern India			
Stage	**Nutrient condition**	**Nutrient kg/ha/year**	**Remarks**
1. Nursery	N:P2O5:K2O	10:5:10, 2g/m^2, nursery bed	One part of fertilizer is mixed with 9 parts of sand and evenly broadcast (minimum of 8 fortnightly bed applications recommended).
2. Young tea			
- 1st year	N:P2O5:K2O	10:5:10 (8 :4 :8 - 12 :6 :12)	5 g in 3-4 applications per plant and application.
- 2nd year	N:P2O5:K2O	10:5:10 (16 :8 :16 - 24 :12 :24)	10 g in 3-4 applications per plant and application.
- 3rd year	N:P2O5:K2O	10:5:10 (32 :16 :32 - 48 :24 :24)	20 g in 3-4 applications per plant and application
- 4th year	N:P2O5:K2O	120:60:120 - 135 :67 :5 :135	as 10:5:10 in 3-4 split dressings.
3. Mature tea	N	90-135	A higher level of N up to 250 kg/ha and year is recently adopted in many

			commercial gardens (applied in several split dressings).
	P2O5	6.7	In 3 split dressings once in 3 years.
	K2O	100-120	A higher level of K2O up to 250 kg/ha and year is recently adopted in commercial gardens (applied in several split dressings).

Source: Tea Research Asociation - Tocklai Experimental Station, Jorkat, Assam (India) - Encyclopaedia on tea culture

Fertilizer recommendations - Sri Lanka			
Yield level kg/ha	**kg/ha per 100 kg marketable tea**	**kg/ha**	
	N*	**P2O5**	**K2O****
below 1 600	10	60	60- 80
above 1 600	10	60	90-120
* See footnote to table of fertilizer recommendations for S. India concerning N - ** N : K2O ratio of 2:1 recently introduced.			
Source: Kemmler, G., 1986			

Rates of N recommended for mature tea - Central Africa			
Age of bushes- years after planting	**kg/ha N**		
	High fertility	**Low fertility and replanted areas**	**All**
10	120	150	
15	155	160	
20	165	165	
21-30			170-185
31-40			200-215
41-50			230
Down pruned			90-110
N:P2O5:K2O ratio of 5:1:1 extensively used with supplementary P2O5 and K2O each at 150 kg/ha before pruning if required. Sources: Kemmler, G., 1986; Tolhurst, 1978.			

Fertilizer recommendations - Indonesia		
Age of bushes/yield level/type of soil	**Nutrients/Rate of application**	**Remarks**
Young tea		
- 1st year	8 g N + 2 g P2O5 + 2 g K2O/plant	MgO at 2 g/plant in podzols only
- 2nd year	12 g N + 3 g P2O5 + 3 g K2O/plant	
- 3rd year	12 g N + 2 g P2O5 + 4 g K2O/plant	
Mature tea in plucking		
- up to 1 200 kg/ha	N at 12 % pro rata on yield	N:P2O5:K2O ratio 4:1:2 in the year of pruning; from second year onwards 5:1:2 in wet months and 5:1:3 in dry months
- 1 200- 2 500 kg/ha	N at 10 % pro rata on yield	
- above 2 500 kg/ha	N at 8 % pro rata on yield	
= andosols	120 kg/ha N, 20 kg/ha P2O5, 40 kg/ha K2O	
= podzols	120 kg/ha N, 30 kg/ha P2O5, 30 kg/ha K2O, 30 kg/ha MgO	For every 1 000 kg of made tea
= latosols	120 kg/ha N, 30 kg/ha P2O5, 24 -72 kg/ha K2O	

- N:P2O5:K2O = 6:1:2 in andosols; 5:1:1 - 5:1:3 with a mean of 5:1:2 in latosols.
- N:P2O5:K2O:MgO ratio = 5:1:1:1 - 6:1.5:1.5:1.5 in podzols.
- Placement of fertilizers at 10 cm depth between tea rows.
- Leaf tissue analysis is used to monitor the health of bushes under the recommended manuring schedules.

Source: Isa Darmawijaya, 1985.

Fertilizer recommendations - Bangladesh		
Age of bushes	**Ratio of nutrients (N :P2O5 :K2O)**	**Rates kg/ha and year**
1. Young tea	1:1:1	9, 18, 36, 54 and 72 kg/ha N in the first five years from planting.
2. Mature tea		
- skiffing	1:0:0	90 kg/ha N plus foliar application of 1-2 % urea spray at 15 days interval
- light pruning	4:1:3	90 kg/ha N, 22 kg/ha P2O5, 68 kg/ha K2O
- medium pruning	1:1:1	90 kg/ha N, 90 kg/ha P2O5, 90 kg/ha K2O
Source: Rashid, 1985		

Present fertilizer practices

In South India and Sri Lanka, nursery plants receive a soluble 10:10:10:4 (N:P2O5:K2O:Mgo) nursery mixture which in South India also includes Zn. This is applied in 0.3 % solution at 10 litres per 4 m2 at fortnightly intervals after rooting commences, by means of a watering can fitted with a fine rose. In North India, NPK mixture diluted with sand is broadcast on the nursery beds. In some other regions, only straight N is applied in the nursery.

No fertilizer or organic manure is applied in South India at the time of planting; regular fertilizer applications begin 2 months later. In some other regions, organic manure (compost, bone meal, etc.) and rock phosphate or superphosphate are applied to planting holes at planting.

P is applied straight in South India, as rock phosphate placed at 15-25 cm depth once a year up to the first prune and once every 1-2 years thereafter, depending on yield level, at rates varying from 40 to 100 kg/ha P2O5. P is also applied as rock phosphate in most other tea-growing areas, except that superphosphate is used extensively in Africa, broadcast in mixtures or compounds.

N and K are broadcast in NK mixtures in South India, and, in other countries, in NPK mixtures supplemented by separate applications of straight N. Aerial application is favoured in Africa. Placement at 10 cm depth between the rows is recommended in Indonesia.

Urea is the major source of N, but ammonium sulphate is always used to provide part (15-20 %) of the total annual N requirement in South India. Ammonium sulphate is also used once every 3-4 years as a source of S as well as of N in North India.

In South India, calcium ammonium nitrate is used for pre-winter application.

The amount of N applied varies pro rate with yield in Sri Lanka, South India and Indonesia, the rates being slightly lower in South India due to the higher soil organic matter status. In Africa it is based on the age of the bushes and on the region and responses obtained in field trials.

The **N:K2O** ratio varies in South India with the stage in the pruning cycle, yield level and source of N : 1:2 or 2:3 in the pruned year and, in other years, 2:1 with ammonium sulphate, 1:1 with urea, 4:3 with calcium ammonium nitrate or 1:1 for yields over 3 000 kg/ha

irrespective of the source of N. K is applied as muriate. In Indonesia the N:K2O ratio varies with soil group and growth stage (young tea, pruned tea or mature tea). In other countries it is based mainly on regional trials. In Sri Lanka, K is used at rates of 60-80 kg/ha K2O for yields of up to 1 600 kg/ha or 90-180 kg/ha K2O for higher yields; a 2:1 N:K2O ratio has recently been introduced.

An **N:P2O5:K2O** ratio of 5:1:1 is widely used in Africa with a balancing application of P and K before prune if required.

Zn is widely applied, mainly as foliar application, in most tea-growing countries: as the sulphate in India and Sri Lanka, and as the oxide in African countries. **Mg** is given as magnesite or magnesium sulphate, or as magnesium silicate if Si is also needed.

Liming to maintain pH between 4.5 and 5.0 is done in South India once in 3-5 years in the year of prune, rates being based on factors determining loss of Ca from the soil (e.g. rainfall, yield and fertilizer use). Elsewhere liming is done on an ad hoc basis related either to soil testing or field trials, rates varying from 250 to 1 000 kg/ha/year through the pruning cycle.

Soil testing has been standardized in South India for pH, soil organic matter status, P, K, Mg, Ca, Na and Cu. Limiting nutrients are identified at critical growth stages and included in the fertilizer schedule as needed: Zn for yields over 1 000 kg/ha, Mg, Si and B for yields over 3 000 kg/ha, and Mo for yields over 4 500 kg/ha (Tandon & Ranganathan 1988).

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Tobacco *(Nicotiana tabacum L.)*

French: Tabac; Spanish: Tabaco; Italian: Tabacco; German: Tabak

Flue-cured tobacco
Note that Burley and Oriental Tobaccos are discussed separately later in this Section

Crop data (see under individual countries)

Annual. Harvested part: leaves

Nutrient demand/uptake/removal

Nutrient demand/uptake/removal - Macronutrients								
Yield (Cured leaf) kg/ha	Source	kg/ha						
		N	P2O5	K2O	MgO	CaO	S	Cl
2 000	Hawks & Collins, 1983	78	31	108	41	86	20	22
2 150	Ryding*	71	26	184	35	93	-	-
2 600	Ryding*	66	22	155	34	105	-	-
2 250	Mao Yen-Chen*	150	50	200	40	200	-	-
* Unpublished data								

Nutrient demand/uptake/removal - Micronutrients				
Yield (Cured leaf) kg/ha	Source	g/ha		
		Mn	Cu	B
2 000	Hawks & Collins,1983	800	40	80

Plant analysis data:

Plant analysis data (Cured leaf) - Macronutrients							
Source	% of dry matter						
	N	P	K	Mg	Ca	S	Cl
Collins*	2.03	0.23	2.13	0.48	2.02	0.69	0.83
Ryding*	1.74	0.32	4.32	0.64	2.32	-	-
Ryding*	1.72	0.24	3.17	0.61	2.51	-	-
Note: All nutrients in optimum supply - * Unpublished data							

Plant analysis data (Cured leaf) - Micronutrients				
Source	ppm dry matter			
	Mn	Cu	B	Na
Collins*	200	18	30	147
Note: All nutrients in optimum supply - * Unpublished data				

U.S.A.

Crop data

Crop transplanted March-June.

Flowers 60 days after transplanting when flowers are removed to 20 harvestable leaves per plant.
Harvesting begins 70 days after transplanting, removing about five lower leaves every 10-12 days.

Plant density: 13 000-15 000/ha.

Preferably grown on light sandy soils with about 1 % organic matter and pH of 5.7-6.0.

Requires about 25 mm rain or irrigation per week. Can withstand periods up to 1 month without rain or irrigation during the month after transplanting. Thereafter, especially as the plants reach the flowering stage, adequate water is essential for leaf development.

Fertilizer recommendations

Soil tests determine nutrient requirements and the need for lime. Dolomitic lime is used to adjust pH; it is important not to overlime.

N - 50-80 kg/ha N (total of basal and sidedressing applications), the amount of the basal dressing depending on depth and texture of the topsoil, the previous crop grown, the cultivar and the farmer's experience. Not more than 40 kg/ha N should be given during the transplant period. At least 50 % of the N in the transplanting fertilizer should be in nitrate form and the remainder in ammonium form; in some cases a favourable response can be expected up to 100 % of nitrate N. The N applied after the transplant period should be entirely in nitrate form.

P - as indicated by the soil test, all during the transplant period.

K - as indicated by the soil test. If the compound fertilizer used for the basal application does not provide enough K, more can be given in sidedressings of an NK fertilizer containing equal amounts of N and K2O.

S - on deep sandy soil, if needed, 22 kg/ha S.

The basal fertilizer is applied in two bands at transplanting or at 7-10 days after transplanting in order to minimize fertilizer salt injury and early season leaching losses

Zimbabwe

Crop data

Crop transplanted September-December (usually about 91-112 days after seeding).

Flowers 63-77 days after transplanting and flowers are removed leaving 18 reapable leaves on standard cultivars, or 22 on mammoth cultivars.
Harvesting (reaping) starts 70-77 days after transplanting; completed by 98-140 days; usually 2 leaves removed every 7-10 days.

Plant density: 14 500-18 500/ha.

Preferably grown on light sandy soils with pH about 5.5.

Usually requires about 25 mm water per week. About 20 % of the crop is irrigated. Can withstand long dry spells during early establishment.

Fertilizer recommendations

Lime often needed to increase soil pH to about 5.5.

Basal application of NPK mineral fertilizer at ridging to supply all N, P and K requirements. If leaching rains occur within 3-8 weeks after transplanting, N sidedressings (25 kg/ha N) is needed in order to obtain a high yield.

N is applied in ammonium nitrate form for the preplanting application and as ammonium nitrate and sodium nitrate for sidedressings. P is applied in water-soluble form, and K preferably as sulphate but on well-drained upland soils potassium chloride is beneficial as a supplier of 18-24 kg/ha Cl.

Present fertilizer practice
For Kutsaga Mammoth 10 and Kutsaga E1, two important cultivars which yield about 2 500 kg/ha:

On sands and loamy sands:

Basal application
N - early-ploughed 15-40 kg/ha N or late-ploughed 35-70 kg/ha N
P - 100-110 kg/ha P2O5
K - 90-110 kg/ha K2O

Sidedressing 3-8 weeks after planting (if leaching rain occurs)
N - 25 kg/ha N

On sandy loams and heavier soils:

Basal application
N - early-ploughed 10-30 kg/ha N or late-ploughed 20-55 kg/ha N
P - 140-160 kg/ha P2O5
K - 90-110 kg/ha K2O

Sidedressing 3-8 weeks after planting (if leaching rain occurs)

N - 15-25 kg/ha N

The basal fertilizer is applied in a double band 10 cm on either side of the seedling and about 15 cm from the top of the ridge, i.e. level with the root crown of the seedlings. The sidedressing is applied in a hole about 10 cm deep on the top of the ridge and 15 cm from the plant.

The reason for applying less N to an early-ploughed soil is because the organic matter has completely broken down by the time the tobacco is planted and mineralization of the organic N to ammonium- and nitrate-N occurs with the first rains. In contrast, on a late-ploughed soil the organic matter is still being broken down and the organic N is immobilized by micro-organisms; a much smaller amount is released. Generally only one flush of N occurs.

China

Crop data

Crop transplanted February-May (usually 55-70 days after seeding).

Flowers 55-60 days after transplanting.
Harvesting starts 70 days after transplanting, completed by 120 days.

Plant density: 16 000-22 500/ha

Grown on a wide range of soil types, including paddy soils, with pH 5.0-8.0.

Generally not irrigated except for the paddy soils.

Fertilizer recommendations

Organic manure is broadcast and incorporated before banding basal mineral NPK fertilizer in the plant row; sometimes more NPK or straight N fertilizer is sidedressed after transplanting.

N is mainly in the form of urea but sometimes ammonium nitrate, P in water-soluble form and K as potassium sulphate.

Present fertilizer practice

For the traditional cultivar, Honghuadajinyuan, yielding 2 250-2 650 kg/ha:

- Basal application, banded after organic manure incorporated into soil
N - 25-30 kg/ha N,
P - 33-60 kg/ha P_2O_5,
K - 50-60 kg/ha K_2O;

- Sidedressing 20 days after transplanting (dissolved in water and applied between plants)
N - 25-30 kg/ha N,
P - 33-60 kg/ha P_2O_5,
K - 50-60 kg/ha K_2O;

- Two further sidedressings 45 and 60 days after transplanting (dissolved in water and applied between plants), each consisting of
N - 12.5-15 kg/ha N,
P - 16.5-30 kg/ha P_2O_5,
K - 25-30 kg/ha K_2O.

For improved cultivars NC89, NC82, K326, G28, yielding 2 250-2 650 kg/ha in Northeast, Central and Western China:

- Basal application, all banded before transplanting
N - 60-90 kg/ha N,
P - 60-180 kg/ha P_2O_5,
K - 120-180 kg/ha K_2O

- Or basal application
N - 50-60 kg/ha N,
P - 50-60 kg/ha P_2O_5,
K - 100-120 kg/ha K_2O;
followed by
N - 25-30 kg/ha N,

P - 20-50 kg/ha P2O5,
K - 50-60 kg/ha K2O,
in planting holes at planting;

- Or basal application
N - 50-60 kg/ha N,
P - 75-90 kg/ha P2O5,
K - 150-180 kg/ha K2O;
followed by sidedressing 25 days after planting
N - 25-30 kg/ha N.

Effects on quality:

- K enhances leaf combustion.
- Cl concentration > 1.0 % progressively impairs leaf combustion.

Burley tobacco

Crop data (see under individual countries)

Annual. Harvested part: leaves

Nutrient demand/uptake/removal

Nutrient demand/uptake/removal - Macronutrients								
Yield (Cured leaf) kg/ha	Source	kg/ha						
		N	**P2O5**	**K2O**	**MgO**	**CaO**	**S**	**Cl**
2 913	Campbell*	202	22	161	11	146	28	22
* Unpublished data								

Nutrient demand/uptake/removal - Micronutrients								
Yield (Cured leaf) kg/ha	Source	g/ha						
		Mn	**Zn**	**Cu**	**B**	**Mo**	**Fe**	
2 913	Campbell*	200	200	70	90	0.4	900	
* Unpublished data								

Plant analysis data

Plant analysis data (cured leaf) - Macronutrients							
Type of reaping	Source	% of dry matter					
		N	**P**	**K**	**Mg**	**Ca**	**Cl**
Stalk cut	Nichols et al, 1962	4.17	0.28	3.38	-	5.57	0.56
Individual harvestings	Ryding, 1991	3.05	0.23	2.9	1.22	3.47	-
Stalk cut	Ryding, 1991	3.44	0.18	2.1	1.51	3.89	-
Note: All nutrients in optimum supply.							

Plant analysis data (cured leaf) - Micronutrients

Type of reaping	Source	ppm dry matter		
		Mn	B	Mo
Individual harvestings	Ryding, 1991	132	36	0.6
Stalk cut	Ryding, 1991	121	31	0.4
Note: All nutrients in optimum supply.				

U.S.A.

Crop transplanted May-early June (about 81 to 95 days after seeding).

Flowers 65-70 days after transplanting and flowers are removed to leave 20-24 harvestable leaves per plant.
Harvesting by stalk cutting in 95-110 days.

Plant density: 18 000 to 20 000/ha.

Preferably grown on silt loam soils but can be successfully produced on lighter, sandier or heavier soils. Desirable pH 5.7-6.3 (Mn toxicity likely if pH < 5.5).

Usually rainfed; less than 10 % of burley crop is irrigated. Can withstand extended periods of drought, especially when grown on silt loams or heavier soils.

Present fertilizer practices
For popular cultivars Kentucky 14 and Burley 21 x Kentucky 10 hybrid, yielding 2 400-3 000 kg/ha:

Basal fertilization				
Soil	kg/ha	Soil test level	kg/ha	
	N		P2O5	K2O
Silt loams	168-225	Low	168-224	336-448
sandier soils than silt loams	112-150	Medium	112-168	224-336
		High	0-112	0-224
		Very high	None	None

Sidedressing on sandier soils - (not later than 3 weeks after planting)
N = 56-75 kg/ha

N applied in nitrate form or as urea, P in water-soluble form, and K as sulphate. All the fertilizer is broadcast and incorporated with a disc before transplanting, except on sandy soils where leaching occurs. On sandy soils about one-half to one-third of the total nitrogen is banded as a sidedressing,. Heavy residues of non-leguminous materials, such as cornstalks or a mature rye cover crop incorporated before planting, may require an additional 55-110 kg/ha N.

Zimbabwe

Crop transplanted September-October (about 91-105 days after seeding).

Flowers 63-77 days after transplanting and flowers removed leaving 25-28 reapable leaves.

Harvesting starts about 63 days after transplanting and is completed in 91-112 days for stalk cut tobacco, and in 133-147 days for tobacco harvested throughout growth.

Plant density: 15 000/ha.

Preferably grown on sandy clay loams and heavier-textured soils with pH about 5.5. About 20 % of the burley crop is irrigated. It can withstand long dry spells during early establishment.

Present fertilizer practice
For standard Banket A1, Banket 102 and Burley 21 cultivars, yielding 2 500-3 000 kg/ha:

On sands and sandy loams:

- Basal application at ridging
N - 42 kg/ha N,
P - 102-120 kg/ha P_2O_5,
K - 90-105 kg/ha K_2O;

- Sidedressing 2 weeks after planting
N - 50 kg/ha N,
K - 25 kg/ha K_2O on sands;

- Sidedressing 4 and 6 weeks after planting
N - 40 kg/ha N.

On sandy clay loams and heavier soils:

- Basal application
N - 28-32 kg/ha N,
P - 120-140 kg/ha P_2O_5,
K - 90-105 kg/ha K_2O;

- Sidedressings as for sands/sandy loams.

N is applied as ammonium nitrate, P in water-soluble form and K as sulphate, although some in the chloride form is beneficial on sandy soil.

Oriental Tobacco

Annual. Harvested part: Leaves

Zimbabwe

Crop transplanted November-December (usually about 42-55 days after seeding).

Flowers are not removed.
Harvesting starts 21-28 days after transplanting, completed in 126-168 days.

Plant density: 150 000/ha.

Preferably grown on sandy soils with pH about 5.5, limed if necessary.

Grown under rainfed conditions at a lower altitude and in a drier and slightly warmer climate than flue-cured or burley.

Present fertilizer practice
For standard Samsun 090 cultivar, yielding 1 000-2 000 kg/ha:

Basal application in a single band about 10 cm from top of ridge and about 8 cm from plants in a "twin" row on the ridge at 90 cm centres
N - 0-10 kg/ha N,
P - 40-80 kg/ha P_2O_5,
K - 33-67 kg/ha K_2O.

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Fodderbeet *(Beta vulgaris ssp. vulgaris var. Alba)*

French: Betterave fouragère; Spanish: Remolacha forrajera; Italian: Barbabietola da foraggio; German: Futterruebe, Runkelruebe

Crop data

Biennial.
Harvested part: 1st year roots (= roots + stems) and normally also leaves.
Sown in spring (directly seeded in cultivated soil).
Does not flower in 1st year, but flowers in summer in 2nd year when grown for seed.

Fodder crop harvested 6-7 months after sowing.
Plant density: 60 000-90 000 plants/ha.
Can be grown on all soils with pH > 5.
In regions with summer drought, irrigation is needed on light sandy soils.

Nutrient demand/uptake/removal

Nutrient removal by crop yielding 10 t DM/ha roots and 5 t DM/ha leaves:

Nutrient uptake/removal - Macronutrients							
Yield	**Source**	**kg/ha**					
		N	**P2O5**	**K2O**	**MgO**	**CaO**	**Na2O**
Roots, 10 t DM	LIK, 1990	n.a.	37	180	22	32	20
Leaves, 5 t DM		n.a.	19	180	34	91	65
Total		n.a.	56	360	56	123	85
Roots, 10 t DM	LIK, 1974	102	37	182	20	27	20
Leaves, 5 t DM		129	31	181	34	91	65
Total		231	68	363	54	118	85

Nutrient removal - Micronutrients				
Yield	**Source**	**g/ha**		
		Mn	**Zn**	**Cu**
Roots, 10 t DM	LIK, 1990	270	250	70
Leaves, 5 t DM		450	225	65
Total		720	475	135
Roots, 10 t DM	LIK, 1974	300	270	42
Leaves, 5 t DM		450	235	64
Total		750	505	106

Plant analysis data

Plant analyses data - Macro- and micronutrients									
Plant part	**Stage of growth**	**Source**	**% of dry matter**					**ppm DM**	
			N	**P**	**K**	**Mg**	**S**	**Mn**	**B**
Normal levels									
Leaf	May	N & P:	4.8	0.73	6.0	0.78	-	-	-
Root	May	Knowles et al, 1934	2.7	0.21	4.8	-	-	-	-
Leaf	June	K:	4.6	0.53	5.0	0.75	-	-	-
Root	June	Draycott, 1972	2.4	0.41	2.5	-	-	-	-

Leaf	July	Mg:	3.0	0.40	4.0	0.48	-	50	50
Root	July	Durrant & Draycott, 1971	1.0	0.38	1.3	-	-	-	-
Leaf	Aug.	Mn & B:	2.0	0.34	3.5	0.23	-	-	-
Root	Aug.	Hengill, 1990	0.8	0.27	0.8	-	-	-	-
Leaf	Sep.		1.9	0.21	3.0	0.20	-	-	-
Root	Sep.		0.8	0.13	0.8	-	-	-	-
Deficiency levels									
Middle leaves	Half way through the life of the crop	Hengill (1990)	3.5	0.35	2.0	0.20	0.20	20	20

For good storage quality, special attention must be paid to B. Na can replace K up to approximately half the amount required.

Fertilizer recommendations

To reduce the risk of root diseases, a pH >6.0 is desirable.

Due to the long growing period, the crop can make effective use of organic manures. FYM can be ploughed into the soil before sowing. Urine/slurry can also be ploughed in before sowing or may be incorporated into the soil with a cultivator after ploughing but before sowing; in the latter case the amount must not exceed 40 t/ha to avoid germination problems. Urine/slurry may also be applied between the rows in early summer, preferably by injection.

The need for mineral fertilizers is calculated from the difference between the plants' total requirement and the amount of plant-available nutrients in the organic manures supplied. Because the crop is normally used on the farm on which it is grown, organic manure is usually available in quantity, so most of the plants' requirements are supplied in this form; often the only mineral fertilizer added is straight N.

The normal pattern of application is:

- two-thirds N, all P, all K before sowing (as organic manures)

- one-third N two months after sowing (as mineral fertilizer).

Preferred nutrient forms

N - ammonium or ammonium nitrate
P - water-soluble
K - as KCl.

Present fertilizer practices

Denmark

For expected yield of 10 t DM/ha roots and 3 t DM/ha leaves, on soil of good P and K status:

- 180 kg/ha N, 80 kg/ha P_2O_5, 240 kg/ha K_2O.

With lower or higher yield expectations the rates of P_2O_5 and K_2O are adjusted accordingly.

The N requirement may also be calculated (in kg/ha) as 245 minus Nmin in spring.

Germany

For soils of good P and K status (Stufe C):

- low yield (8 t DM/ha roots):120 kg/ha N, 60 kg/ha P2O5, 260 kg/ha K2O.
- normal yield (12 t DM/ha roots): 180 kg/ha N, 90 kg/ha P2O5, 310 kg/ha K2O.
- high yield (16 t DM/ha roots):220 kg/ha, 120 kg/ha P2O5, 350 kg/ha K2O.

Great Britain

For soils of good P and K status (index 3), after arable crops:

- 125 kg/ha N, 50 kg/ha P2O5, 175 kg/ha K2O (or 75 kg/ha K2O + 150 kg/ha Na).

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Fodder Legumes

French: Légumineuses fourragères; Spanish: Leguminosas forrajeras; Italian: Leguminosi da forragio; German: Futterleguminosen

Crop data

Fodder legumes are used extensively to conserve high quality feed as hay for animals. They are commonly perennial species with a capacity for rapid regrowth after cutting. Lucerne or alfalfa (Medicago sativa) is the most widely used fodder legume in temperate regions. However, other Medicago and Trifolium species such as red clover, crimson clover, berseem and white clover are also used on soils to which lucerne is poorly adapted. Fodder conservation is not widely practiced in tropical regions because seasonal conditions enable animals to access pasture throughout the year, and because of problems making hay or silage in the humid, high rainfall conditions that frequently occur during the growing season. Where fodder is conserved in the tropics, Dolichos lablab is the legume most frequently used.

Yields of fodder legumes vary considerably. Under dryland conditions, growth rates can limit the number of cuts per season, restricting yields to as low as 3 t/ha/year. Well fertilized, irrigated crops of modern cultivars in regions with long growing seasons may yield 15 to 20 t/ha each year.

Nutrient demand/uptake/removal

Nutrient demand is dependent upon yield. This crop puts a very heavy demand upon soil nutrient reserves because almost all of the plant top growth is removed from the field at each harvest. Frequent cutting of actively growing material produces the highest quality fodder. Since such material contains high concentrations of mineral nutrients this practice also increases the demand for nutrients. Listed below are some typical nutrient demands imposed by each tonne of high quality hay produced.

Nutrient uptake/removal - Macronutrients					
kg/ha per t hay					
N	P2O5	K2O	MgO	CaO	S
40	9	37	7	17	3

Nutrient uptake/removal - Micronutrients					
g/ha per t hay					
Fe	Mn	Zn	Cu	B	Mo
60	50	30	7	30	0.3

Nutrient removal

If all vegetative material is conserved as hay, the above amounts of nutrients are removed. Some may be returned later as dung from animal feeding areas. Some farmers may also permit animals to graze fodder legume areas, although this practice can result in grass and weed invasion.

Plant analysis data

Concentrations of mineral nutrients commonly found in healthy, adequately fertilized fodder legumes are shown below.

Plant analysis data - Macronutrients

Growth stage	Plant part	% of dry matter					
		N	P	K	Mg	Ca	S
Hay	Whole top	2.8-3.5	0.20-0.45	1.5-3.0	0.12-0.30	0.5-1.0	0.18-0.23
Vegetative	Top 15 cm	4.0-5.0	0.25-0.70	2.5-3.8	0.31-1.0	0.5-3.0	0.26-0.50
Early flower	Whole top	3.5-5.0	0.25-0.45	2.0-3.5	0.25-0.50	1.0-2.0	0.25-0.40

Plant analysis data - Micronutrients							
Growth Stage	Plant part	ppm dry matter					
		Fe	Mn	Zn	Cu	B	Mo
Hay	Whole top	45-70	25-100	12-40	5-10	25-40	0.15-0.30
Vegetative	Top 15 cm		30-120	20-80	8-15	30-60	
Early Flower	Whole top		25-100	15-60	5-15	25-60	0.20-0.40

Typical critical concentrations for yield are as follows:

Critical values for yield - Macronutrients							
Growth Stage	Plant part	% of dry matter					
		N	P	K	Mg	Ca	S
Hay	Whole top	2.8-3.1	0.20-0.25	1.4-1.8	0.10-0.12	0.4-0.6	0.18-0.20
Vegetative	Top 15 cm	4.0-4.5	0.25	2.2-2.6	0.25-0.30		0.23-0.26
Early flower	Whole top	3.0-3.5	0.22-0.25	1.7-2.0	0.20-0.25	0.7-1.0	0.20-0.25

Critical values for yield - Micronutrients							
Growth stage	Plant part	ppm dry matter					
		Fe	Mn	Zn	Cu	B	Mo
Hay	Whole top	40-45	22-27	10-12	4-7	22-27	0.12-0.16
Vegetative	Top 15 cm		30	20	10	30	0.5 -0.7
Early Flower	Whole top			15	5		0.2
Early Flower	Leaf						0.5

Present fertilizer practices

For establishment:

In view of the very high demand upon the soil for nutrients, the crop is commonly grown on fertile soils with a capacity to continue to supply nutrients.

It is important that the most effective strain of Rhizobium is used at planting and that plants are effectively nodulated on establishment.

Lucerne has a deep rooting system and so can tap nutrient reserves from deep in the soil profile. Species like lucerne are sensitive to soil acidity and liming may be necessary; sufficient lime to raise the soil pH to at least 6.5 should be incorporated into the surface soil 6-12 months before planting.

During the initial establishment of fodder legume stands all nutrient deiciencies must be corrected. A complete NPK fertilizer is most commonly used at planting. 15-20 kg/ha of N in the form of NP or DAP can aid establishment, particularly on soils low in organic matter.

For maintenance:

N fertilizers are not commonly used for the maintenance of fodder legumes as they tend to encourage grass growth and decrease the longevity of the stand.

P is applied as triple superphosphate broadcast on the surface annually. Rates of application depend upon the capacity of the soil to supply P and on the yield of legume; 45-90 kg/ha P2O5 is commonly given each year on responsive soils.

K : Regular topdressing with KCl is required to maintain productivity. The amount to be given is mostly based on soil test results. On low-K soils supporting high yields 240-480 kg/ha of K2O may be needed. Where high rates are necessary, split dressings reduce the risk of chloride injury from KCl.

Mg fertilizers are rarely required. When needed, dolomitic limestone is the principal source and is best applied before planting.

S deficiency is becoming more widespread and may be corrected by using single superphospate instead of triple, by applying gypsum, or by applying 10-20 kg/ha of elemental S per year.

Micronutrient deficiencies should be corrected at planting, or immediately when they are detected during the maintenance phase. Boron deficiency may be overcome by applying 4 kg/ha of borax every 2-3 years. 10 kg/ha of copper sulphate every 5 years is used to correct copper deficiency on most soils, but 20 kg/ha may be required on organic soils. Zinc deficiency may be corrected by applying zinc sulphate at the equivalent of 10-15 kg/ha Zn; particular attention should be given to the possibility of a deficiency of this nutrient on calcareous and on high-P soils. Molybdenum deficiency is common on slightly acid soils and may be corrected by applying molybdenized superphosphate (0.02 %) every third year.

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12.1 Temperate Grassland: Permanent Grass and Sown Grass or Leys

French: Herbage permanent et pâtures temporaires; Spanish: Pastizal permanente y prados temporarios; Italian: Erbaio permanente e pascolos temporaneos; German: Dauergruenland und Feldgras

Crop data

Generally permanent swards contain a wider range of grass species than sown grassland. However, in a particular environment, the production of permanent and sown perennial grassland is similar except that sown perennial grasses are likely to be 10-30 % more productive in the first harvest year. Short term leys of 1-2 years duration based on Lolium multiflorum and L. multiflorum x perenne are 20-30 % more productive than permanent or established sown swards. The main difference influencing fertilizer recommendations is in the amount of soil N likely to be available; the soil under a permanent sward releases more N than that under a sown sward.

Temperate climates are characterized by seasonal variation in temperature and radiation, and some precipitation throughout the year. The growing season is determined by temperature. In winter temperatures are too low to allow effective growth although in milder maritime areas some grass growth occurs. Temperature rises in spring, when growth starts, and continues rising until midsummer, then starts falling in late summer until growth ceases because of low temperature at some time in the autumn. Depending on latitude, elevation and in some areas proximity to the sea, temperatures are sufficiently high for active growth for 6-9 months in the year.

The other principal factor controlling growth is availability of water which depends seasonally on the precipitation : evapotranspiration ratio modified by the capacity of the soil (texture and rooting depth) to hold plant-available water. In general during the winter available water exceeds evapotranspiration, when surplus water drains through the soil, leading to the leaching of any surplus nitrate in the soil to groundwater and rivers. This causes concern in many places but can be minimized by good fertilizer practice. From spring into summer, evapotranspiration exceeds rainfall and a soil water deficit develops which restricts grass growth proportionally to the deficit. In areas of favoured grass growth with high summer precipitation, e.g. western U.K., Ireland and N.W. France, and upland and mountain areas, the deficit tends to be small. In continental areas on the other hand a deficit of some severity is to be expected every year. Hence the capacity to produce grass, and the optimum fertilizer rates needed, vary between places and between years.

Grass growth starts in spring, the expected date depending on latitude and elevation; it accelerates as temperature rises, reaching a peak at about flowering time in late spring/early summer, followed by a period of lesser growth and another lesser peak in late summer and a steady decline as temperature and radiation fall. Temperate pasture grasses are grown at high elevations (>2 000 m) in the tropics and sub-tropics where similar principles apply.

Production is principally influenced by environment and N supply, the potential being related to the length of the effective growing season and the amount of water available during that period, i.e. rainfall less evapotranspiration during the growing season plus available soil water. As a general guide, under good growing conditions (lowland areas, 7-8 months growing season, summer rainfall greater than about 400 mm) potential production is 10 000-14 000 kg/ha DM. Under poor growing conditions (upland areas with a short growing season, or shallow soils with less than 300 mm summer rainfall) it is 5 000-8 000 kg/ha DM.

Nutrient demand/uptake/removal

In most circumstances the amount of N mineralized in the soil is totally inadequate to support the potential production and additional N either in the form of fertilizer or from fixation associated with legumes, is necessary for a reasonable production level. Grass responds to fertilizer N in a characteristic way. The yield without fertilizer N is determined by the amount mineralized in the soil; there is a first, linear phase of response in which each increment of fertilizer N applied gives the same increment in yield, up to about 90 % of the potential yield for the particular conditions; and then a phase in which the incremental response decreases markedly until finally no further response is obtained. In general, the 90 % yield is obtained at a rate of fertilizer N (optimum rate) which is about 60 % of that required for maximum yield. Agriculturally and economically it is inefficient to apply fertilizer N above the optimum for particular conditions and management; otherwise substantial leaching may occur. The greater the potential production, the greater is the demand for N and the higher the optimum rate of application. Although the response in the linear phase tends to depend on growing conditions, for grazed grassland it is usually around 20-25 kg DM/kg fertilizer N applied and for grassland cut for conservation 25-30 kg DM/kg fertilizer N applied.

Nutrient uptake (approximate) - Macronutrients						
Yield level	**kg/ha**					
kg/ha DM	**N**	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
2 500	70	20	75	8	21	6
5 000	150	40	150	17	42	12
10 000	300	80	300	34	84	24
12 500	370	100	375	42	105	30

Nutrient uptake (approximate) - Micronutrients				
Yield level	**g/ha**			
kg/ha DM	**Fe**	**Mn**	**Zn**	**Cu**
2 500	250	400	100	20
5 000	500	800	200	40
10 000	1 000	1 600	400	80
12 500	1 250	2 000	500	100

Nutrient demand is a function of yield which is determined by growing conditions and N supply. Nutrient removal depends on management. Under grazing, the removal of nutrients is minimal. Where grassland is cut for conservation, considerable quantities are removed and need to be replaced, particularly of K and P, in the form of fertilizer or of animal manure balanced by fertilizer.

Plant analysis

Plant analysis data (ranges of analytical data) - Macronutrients				
% of dry matter				
P	K	Mg	Ca	S
0.18-0.50	1.38-3.43	0.10-0.39	0.4-1.0	0.13-0.75

Plant analysis data (ranges of analytial data) - Micronutrients			
ppm dry matter			
Fe	Mn	Zn	Cu
50-200	22-314	15-60	3.4-12.9

Critical values have been little used in determining fertilizer rates for temperate grassland, but approximate critical values for intensively managed grass would be 0.3-0.4 % P, 2.0 % K, 0.15-0.20 % S.

Fertilizer recommendations

The principles of fertilizer use on grassland are more complex than for annual crops because (i) nutrient demand is high throughout the growing season, (ii) production varies greatly with conditions of climate and soil, (iii) grassland encompasses a wide range of botanical composition and sward age, and (iv) management varies from differing intensities of grazing or cutting to combinations of these. All these factors influence the amount of fertilizer required and its effectiveness. This chapter is concerned with swards composed predominantly of grasses. For grass/legume swards, see chapter 12.2.

Nitrogen is the key and other nutrients are needed in proportion to the rate of N applied and nutrient cycling in the system.

Grassland systems with winter feeding and housing generate large amounts of stable manure and slurry. Although their composition varies widely, these are a useful source of K and P and can supply some N. During a 6-month winter feeding period a cow voids in dung and urine around 30 kg N, 12 kg P2O5, 40 kg K2O, 20 kg CaO and 10 kg MgO. This can be used most effectively if it is applied to the conservation area in spring in combination with fertilizer, and fertilizer rates should be reduced proportionately in order to take account of the amounts of nutrients returned. As a simple rule of thumb P and K in animal manures may be taken as roughly equal to the same amounts in fertilizers but N is only about half as effective as in fertilizer.

Recommendations for grazed grassland:

N: N fertilizer can be applied at any rate up to the optimum for particular conditions: any more will give a poor response and lead to leaching loss.

Growing conditions	Optimum rate kg/ha N	Likely yield kg/ha DM
Good	300-350	10 000-12 000
Moderate	250-300	8 000-10 000
Poor	200-250	5 000- 8 000
Establishment	60	

These rates take into account N returned by the grazing animal, which (at an optimum N rate under good growing conditions) is equivalent to about 100 kg fertilizer N per hectare.

N fertilizer must be applied regularly at 3- to 4-week intervals over the growing season, e.g. an annual rate of 300 kg/ha N would be applied in six dressings of 50 kg/ha N. Timing is critical (see below).

Allowance should be made for the likely availability of N through mineralization in the soil, which varies with soil and past management. Grazed permanent grassland or old sown grassland on clay and clay-loam soils will supply at least 100 kg/ha N and the optimum rate should be reduced by 50 kg/ha N for such conditions; on sandy soils with a low accumulation of organic matter the N supply will be low and no allowance need be made.

P and K: The rates below apply to established swards and at sowing:

Soil nutrient status	P2O5 (kg/ha)	K2O (kg/ha)
Deficient	60	60

Low	40	30
Moderate	20	0
Adequate	0	0
High	0	0

There is little loss of P or K under grazing. The principle is to apply enough to raise the soil indices to adequate levels and then to give annual applications to maintain this level. P and K are conveniently applied in a compound fertilizer also containing N at the beginning of the season. However on some calcareous and other soils the Mg content of the grass tends to be low where K availability is high during the initial period of growth, leading to hypomagnesaemia; where this risk is apparent, K application should be delayed until after the first flush of growth.

Recommendations for grassland cut for silage or hay:

N:

Growing conditions	**kg/ha N**			
	1st cut	2nd cut	Later cuts	Total
Good	120	100	60	280-340
Moderate	120	80	60	260-300
Poor	100	60	50	220-260

P and K:

Soil nutrient status	**kg/ha**			
	1st cut		**2nd cut**	
	P2O5	K2O	P2O5	K2O
Deficient	100	150	50	100
Low	60	120	30	90
Moderate	30	90	20	80
Adequate	30	30	20	30
High	0	0	0	0

For hay crops the N rate can be reduced if necessary to avoid too heavy a crop which is difficult to dry, and P and K reduced in proportion. The rates of P and K are based on the maintenance of adequate soil indices and replacement of nutrients removed in the cut crop.

Soil deficiencies other than of the primary nutrients are uncommon and, when severe, are locally recognized. S deficiency on intensively fertilized grassland is becoming recognized in temperate areas not subjected to atmospheric pollution, e.g. western Ireland, west Wales and western Scotland, and may be widespread in highland subtropical and tropical areas where temperate grasses are grown.

Timing of fertilizer N application

The first N dressing should not be given until the start of active growth and nutrient uptake by grass, which depends on locality. If given too soon, there is a likelihood of loss by leaching or demineralization.

Methods of assessing the right timing include:

"T200" = apply when the accumulated daily mean temperature over 0 °C from 1st January exceeds 200 °C.

Soil temperature = apply when the soil temperature at a depth of 10 cm reaches 5.5 °C.

Soil load-bearing capacity = apply in spring when the soil is sufficiently dry for a tractor but not later than 1st April (this method is for use on heavy soils).

In practice the start of active grass growth varies widely, e.g. in coastal areas with mild winters grass growth starts early and in northern upland areas growth starts late and there is variation between years: hence the date of the first application has to be adjusted according to circumstances. The same principles apply to temperate grassland in the southern hemisphere.

The last application should be given 6 weeks before the end of active growth, particularly on grazed grassland where there is an accumulation of mineral N in the topsoil. In western and northern Europe no further N should be applied after the end of August.

Avoiding pollution

Loss of nitrate from grassland is wasteful and results from poor practice. Nitrate loss from cut grassland is inherently lower than that from grazed grassland.

Sound practices to reduce the risk of loss include:

Rate of fertilizer use - do not exceed the optimum rate for the particular conditions involved.

Timing of fertilizer use - give the first dressing when there is active growth and uptake.

Use of manure and slurry - apply only when there is active growth and uptake.

In drought years - do not apply fertilizer N in dry conditions when there has not yet been a full response to earlier applications.

Preferred nutrient forms

N: The commonest forms are ammonium nitrate, CAN and urea. The N in compound fertilizers is usually in the form of ammonium nitrate but this should be checked.

P: Usually supplied as single or triple superphosphate in straight or compound form.

K: As chloride (muriate) or sulphate in either straight or compound form.

Present fertilizer practices

These recommendations are based on current practice in northern and western Europe, where intensive grassland production based on high inputs of nitrogen fertilizers has developed. In the temperate areas of Australasia, with comparatively low product prices/high fertilizer costs, intensive grassland production is based on legume/grass swards reliant on biological nitrogen fixation.

Further reading

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Temperate Grassland: Grass/Legume Swards

French: Pâturages à base de tréfle; Spanish: Pastoreos a base de trébol y hierba; Italian: Pratos mistos a trifoglio; German: Kleegras

Crop data

Temperate grass/legume associations for grazing and conservation are based on white clover (Trifolium repens) and perennial grasses; those for conservation are based on lucerne or alfalfa (Medicago sativa) and red clover (Trifolium pratense), most commonly in pure stands but sometimes with grass. In New Zealand, Maku lotus (Lotus uliginosus cv. Maku), which has been selected as a replacement for white clover on hill land, tolerates wet and acid soils and has a lower P requirement than white clover; this species is showing considerable promise in subtropical highland areas where temperate pasture species are grown.

The legume component of grass/legume swards provides the N source for the sward through the fixation of atmospheric N by Rhizobium bacteria in the root nodules. Where suitable Rhizobia are not present in the soil, legume seed must be treated with a specific inoculant to ensure effective nodulation. The legume component is independent of soil N; once the sward is established, the grass component obtains N from the legume through available N in the soil derived from the decay of legume root nodules and that recycled in urine from grazing livestock. Productivity depends on a substantial (i.e. 50 % legume at peak growth in mid-season) and vigorous legume component. Fertilizer practice and management are aimed at maintaining its vigour.

Application of fertilizer N reduces N fixation by the legume and increases the competitiveness of the grass, thus causing a reduction in the legume contribution. Single applications may, however be made in some circumstances to promote early growth or for a conservation crop.

A vigorous grass/white clover sward produces some 70-80 % of the yield of a pure grass sward receiving the optimum rate of fertilizer N but, because of the higher nutritive quality, animal output is about 90 % of that from a pure grass sward.

Soil pH should be maintained at >6.0 by regular liming.

Nutrient demand/uptake/removal

Nutrient uptake/removal (approximate) - Macronutrients Grass/white clover sward yielding 8 000 kg/ha DM					
kg/ha					
N	P2O5	K2O	MgO	CaO	S
320	69	240	33	189	25

Nutrient uptake/removal (approximate) - Micronutrients Grass/white clover sward yielding 8 000 kg/ha DM				
g/ha				
Fe	Mn	Zn	Cu	Mo
1 500	880	260	80	5

Uptake by grass/red clover or lucerne associations is of a similar order of magnitude but with a higher uptake of K2O (ca. 300 kg/ha K2O).

Plant analysis data

Ranges of analytical data and, in brackets, critical values for white clover (when grown in association with grass) are indicated below:

Plant analysis data - Macronutrients				
% of dry matter				
P	**K**	**Mg**	**Ca**	**S**
0.19-0.47	1.54-3.80	0.15-0.29	1.6-2.3	0.25-0.30
(0.30)	(2.00)	(0.15)	(1.0)	(0.25)

Plant analysis data - Micronutrients					
ppm dry matter					
Fe	**Mn**	**Zn**	**Cu**	**B**	**Mo**
102-448	40-63	22-31	5.4-9.7	25-35	0.05-0.64
(50)	(30)	(15)	(6.0)	(30)	(0.15)

Red clover and lucerne are of similar composition to white clover. The N content of legumes is high (2.4-4.5 %) and depends on the stage of growth.

Fertilizer recommendations

Nutrient demand is a function of the rate of growth. The grass and legume components compete actively for all nutrients, particularly K, and it is essential to maintain an adequate supply in the soil for both. This applies especially to P and K. Fertilizer recommendations are based on the maintenance of an adequate availability in the soil and the replacement of any nutrients removed. Under grazing, the removal is minimal but maintenance applications of fertilizer are necessary and should be based on regular soil analyses. When the sward is cut for conservation, considerable quantities of nutrients are removed and it is essential that P and K, and other nutrients as necessary, are replaced.

Legume growth and N fixation are influenced to a greater extent than grass growth by nutrient deficiency. Adequate Mo is essential for effective N fixation; where Mo deficiency is recognized locally (often in acid soils), then Mo must be applied, most conveniently in the form of molybdenized superphosphate. Other nutrients, such as S, Mg, Zn and B, should be applied where deficiencies are recognized.

Fertilizer requirements are the same whether the sward is permanent or temporary (sown as a ley):

P and K recommendations:

For grazed grass/legume swards		
¦Soil nutrient status	**kg/ha**	
	P2O5	**K2O**
Deficient	80	120
Low	60	80
Moderate	40	60
Adequate	20	0
High	0	0

For establishment	
Soil nutrient status	**kg/ha**

	P2O5	K2O
Deficient	80	120
Low	60	80
Moderate	40	60
Adequate	20	0
High	0	0

For silage and hay			
Soil nutrient status	kg/ha		
	First cut		Later cuts
	P2O5	K2O	K2O
Deficient	100	100	80
Low	80	80	60
Moderate	60	60	40
Adequate	30	40	0
High	0	0	0

N recommendations:

Grass/legume swards in general require no fertilizer N; any fertilizer N reduces the legume contribution and continued use reduces the legume component to an ineffective proportion. There are, however, three situations where fertilizer N might be applied:

- for establishment: 25-50 kg/ha N applied to the seedbed to encourage the growth of legume seedlings before the nodule system is fully established, and of the companion grass seedlings; this is only necessary on soils with a low N status;

- for early grazing: grass/white clover swards have a lower early production than grass swards receiving fertilizer N, but a spring application of 50 kg/ha N can improve early production;

- for the first cut for silage: grass/white clover swards have a more uniform seasonal pattern of production (when cut for conservation they yield 3-4 t DM/ha at the first cut followed by 2-3 t DM/ha at the second cut, compared with grass swards receiving intensive fertilizer N which give 5-7 t DM/ha at the first cut followed by 1-2 t DM/ha at the second cut), but a spring application of up to 100 kg/ha N will increase the first cut to about 5 t/ha DM, mainly grass, although there will be some reduction of the clover contribution and yield in mid-season.

Animal manures and slurry

Grass/legume swards respond well to stable manure or slurry which are useful sources of K and P and can replace some of the mineral fertilizer which would otherwise be required. Spring or early summer application to swards to be cut for conservation is the most effective method of use.

Preferred nutrient forms

P: - Usually applied as single or triple superphosphate, either straight or in compounds. Single superphosphate, which contains an appreciable amount of S, as well as Ca, is a useful fertilizer for S-deficient soil. On acid soils with a high Al content, on which some of the P applied in soluble forms becomes unavailable through fixation, part of the P requirement should be given in the form of ground rock phosphate. Basic slag, a by-product of the iron and steel industry, is a traditional source of P for grassland on acid soils but, with changes in the steelmaking process, it is no longer widely available.

K: - As chloride (muriate) or sulphate in straight or compound form.

Further reading

See chapter : Temperate grassland: Permanent grass and sown grass or leys

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Tropical Grasses

French: Graminées tropicales; Spanish: Gramineas tropicales; Italian: Erbe tropicali; German: Tropische Graeser

Crop data

Tropical pasture grasses cover a wide range of species. The environments in which they are grown include the wet tropics, the semi-arid tropics and the arid zone. Most of the species are perennials and are characterized by possessing the efficient C4 photosynthesis pathway. They are primarily used by grazing animals for meat, milk, hide or fibre production or for draft animals in cropping areas. They tend to occupy the lower fertility soils not used for cropping.

Semi-arid and arid tropical environments are characterized by strong seasonality. Grasses grow rapidly following the onset of rains in the wet season. They mature quickly and, during the dry season, the standing forage is high in fibre and low in protein and minerals. The low nutritive value of the forage during the dry period is the major limitation to animal production in such environments.

Nutrient demand/uptake/removal

Typical contents of nutrients in the whole tops of actively growing forage grasses as presented to animals are shown below. 8 000 kg/ha DM represents a high presentation yield for N-fertilized grasses in the better watered tropical environments. 4 000 kg dry matter per ha might be found in nitrogen fertilised grass pastures in areas subject to some drought, and 2 500 kg dry matter per ha in similar environments but without N-fertiliser.

Nutrient demand/uptake - Macronutrients						
Yield	**kg/ha**					
kg/ha DM	**N**	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
8 000	170	46	240	34	28	16
4 000	90	23	120	17	14	8
2 500	35	16	66	10	10	4

Nutrient demand/uptake - Micronutrients						
Yield	**g/ha**					
kg/ha DM	**Fe**	**Mn**	**Zn**	**Cu**	**B**	**Mo**
8 000	640	560	240	56	160	2.4
4 000	300	280	120	25	80	1.2
2 500	200	175	75	16	50	0.8

Nutrient removal

Where "cut-and-carry" feeding systems are used or where forage is harvested for hay or silage, the total amount of nutrient in the standing forage as shown above may be removed from the field. However, where animals are grazed on the fields, considerable quantities of mineral nutrients are returned to the field in dung and urine. The amount returned depends upon the number of hours per day that the animals graze on the pastures. In grazed situations there can be considerable redistribution of nutrients to rest, shade, watering and yarding areas.

Plant analysis data

Concentration ranges commonly found in the green leaf + stem (whole top) material of normal grass plants are shown below.

Plant analysis data - Macronutrients					
% of dry matter					
N	P	K	Mg	Ca	S
1.5-3.0	0.20-0.35	2.0-3.5	0.2-0.5	0.18-0.5	0.12-0.35

Plant analysis data - Micronutrients					
ppm dry matter					
Fe	Mn	Zn	Cu	B	Mo
50-200	30-500	15-50	5-12	10-30	0.2-0.5

Critical values for yield:

Critical nutrient concentrations for yield in the whole tops (leaves + stems) of actively growing green vegetative material have been determined for only a few nutrients. Typical values are:

Species	% of dry matter		
	P	K	S
Cenchrus ciliaris	0.25		0.07-0.12
Sorghum almum	0.20		-
Panicum maximum	0.20		0.07-0.12
Pennisetum clandestinum	0.22		0.07-0.12
Melinus multiflora	0.18		-
Digitaria decumbens	0.16		0.07-0.12
Paspalum dilatatum	0.25		0.07-0.12
Chloris gayana	0.22	0.50	0.07-0.12
Setaria sphacelata	0.21	1.6-1.8*	0.07-0.12
* 3.1 % for var. Kazungula			

Fertilizer recommendations

For establishment:
Any soil nutritional problems that will restrict the establishment and economic production of the grass pasture should be overcome. In acid soils in which grasses not adapted to acidic conditions are to be established, lime will be required. 35-115 kg/ha P2O5 is commonly recommended on P-deficient soils, depending on the degree of deficiency and the P-fixing characteristics of the soil. 35 kg/ha K2O is recommended when establishing grass pastures on sandy soils low in available K. 10-20 kg/ha S may be required on low-S soils, and any micronutrient deficiencies should be overcome when the pasture is being established.

For maintenance:
In high producing pastures, 35 kg/ha P2O5 each year or 70 kg/ha P2O5 every second year for most soils low in P. 100 to 300 kg/ha N in split applications during the year is recommended. The amount used depends upon the intensity of production from the pastures. On soils with less than 0.3 meq% K 60-120 kg/ha K2O should be applied in split applications during the year. 10 kg/ha S each year is recommended on S-deficient soils. The sulphur is usually supplied in superphosphate or ammonium sulphate. Where a more highly concentrated form of P is used, it may be necessary to apply S in elemental form.

Preferred nutrient forms

Urea, triple superphosphate or DAP, potassium chloride. If S is deficient ammonium sulphate, single superphosphate or flowers of sulphur may be used in the fertilizer blend. Micronutrients are mixed in fertilizer blends with major nutrients.

Present fertilizer practices

Wet Tropics: 125-250 kg/ha superphosphate or molybdenized superphosphate (0.02 % Mo) and 60-125 kg/ha muriate of potash applied annually; 300-400 kg/ha urea in split dressings after the wet season.

Humid subtropics: 125 kg/ha urea in spring and autumn; 60-125 kg/ha superphosphate or molybedenized superphosphate (0.02 % Mo) in spring.

Dung is often returned to pastures from yards associated with dairy farms and intensive feedlot feeding systems.

Further reading

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Author: F.W. Smith, CSIRO, Division of Tropical Crops and Pastures, St Lucia, Queensland, Australia

Tropical and Subtropical Grass/Legume Associations

French: Associations graminées-legumineuses tropicales et subtropicales; Spanish: Asociaciones graminéas-leguminosas tropicales y subtropicales; Italian: Assoziazoni erbe-leguminosi tropicali e sottotropicali; German: Tropische und subtropische Gras/Leguminosenbestaende

Crop data

Tropical grass pasture systems almost invariably suffer from N-deficiency. The low N levels in these grasses particularly during the dry season, restrict their utilization by grazing animals.

Use of legumes in association with grasses provides a means of improving the grazing animals' diet and of providing N to improve the growth and quality of the grass component. To achieve this a balanced mixture of grass and legume is sought. The optimum balance differs with species, seasons and the prevailing environmental conditions. A proportion of 20-30 % legume in the standing forage during the period of active growth is typical.

There are many species of tropical forage legumes. Their range of growth habit includes small erect plants, long twining plants, stoloniferous plants, small shrubs and small trees. Most are perennial. Some may contain toxic amino-acids or alkaloids and require special grazing management procedures. In order to be productive it is essential that an effective symbiosis is formed with the correct strain of Rhizobium; some tropical forage species have quite specific Rhizobium requirements.

Nutrient demand/uptake/removal

Typical nutrient contents in the whole tops of active forage grass/legume pastures, as presented to grazing animals, are shown below. The yield of 8 000 kg/ha DM represents high-producing pastures containing 20 % legume in a wet tropical environment; 5 000 kg/ha DM might be found in pastures containing 20 % legume in the humid subtropics or in the semi-arid tropics during the growing season.

Nutrient uptake/removal- Macronutrients						
Yield	**kg/ha**					
kg/ha DM	**N**	**P2O5**	**K2O**	**MgO**	**CaO**	**S**
8 000	202	48	228	37	46	18
5 000	125	30	144	23	27	11

Nutrient uptake/removal- Micronutrients						
Yield	**g/ha**					
kg/ha	**Fe**	**Mn**	**Zn**	**Cu**	**B**	**Mo**
8 000	680	560	260	56	160	3.2
5 000	420	350	160	35	100	1.9

Where "cut-and carry" feeding systems are used or where forage is harvested for hay or silage, the total amounts of nutrients shown in the tables above may be removed. However, in fields which are grazed, considerable amounts may be returned in dung and urine, the amount depending on the number of hours per day that the animals graze the pastures. This means that, with grazing systems there can be considerable redistribution of nutrients to rest, shade and yarding areas.

Plant analysis data

Concentration ranges commonly found in the green leaf and stem material (whole tops) of normal legume plants are shown below:

Plant analysis data - Macronutrients						
Plant part	**% of dry matter**					
	N	**P**	**K**	**Mg**	**Ca**	**S**
Whole top (leaf+stem)	2.5-5.0	0.20-0.40	1.0-3.0	0.30-0.50	0.5-1.2	0.20-0.50

Plant analysis data - Micronutrients						
Plant part	**ppm dry matter**					
	Fe	**Mn**	**Zn**	**Cu**	**B**	**Mo**
Whole top (leaf+stem)	50-200	30-100	15-50	5-12	10-40	0.5-1.0

The corresponding figures for the grass component are shown in chapter 12.3 on Tropical Grasses.

Critical concentrations for yield in the whole tops of actively growing green vegetative material from leguminous species are available for only a few nutrients. Typical values are:

Critical values for yield (legumes) - Macro-/Micronutrients					
Species	**% of dry matter**			**ppm DM**	
	P	**K**	**S**	**Cu**	**Zn**
Centrosema pubescens	0.16	0.75		4-5	20
Trifolium semipilosum			0.17		
Vigna unguiculata					17
Indigophera spicata				4-5	
Desmodium intortum	0.22	0.72	0.17		19
Desmodium uncinatum	0.23	0.80	0.17	4-5	
Neonotonia wightii	0.23	0.80	0.17		22
Lotononis bainesii	0.17	0.90	0.15		22
Macroptilium lathyroides	0.20	0.75	0.17	4-5	
Macroptilium atropurpureum	0.24	0.75	0.17		24
Stylosanthes guianensis	0.12	1.0		4-5	
Stylosanthes capitata	0.11-0.18	1.0-1.3			
Stylosanthes hamata	0.13		0.12-0.14		
Stylosanthes macrocephala	0.10	0.90			
Stylosanthes humilis	0.17	0.60	0.14		34
Vigna luteola	0.25				

Fertilizer recommendations

For establishment:

Any soil nutritional problems that will restrict the establishment and economic production of the pasture should be overcome. In acid soils in which species not adapted to acidic conditions are to be established, lime will be required. If the species are adapted to growth in acid soils but effective nodulation requires a higher soil pH, then lime pelleting of the seed can be useful. 35-115 kg/ha P2O5 is commonly recommended on P-deficient soils, depending on the degree of deficiency and the P-fixing characteristics of the soil. 35 kg/ha K2O is recommended when establishing pastures on sandy soils low in available K. 10-20 kg/ha S may be required on low S soils, and any microelement deficiencies should be overcome when the pasture is being established.

For maintenance:

In high producing pastures, 35 kg/ha P2O5 each year or 70 kg/ha K2O every second year for most soils low in P. On soils with less than 0.3 meq% K 60-120 kg/ha K2O should be applied in split dressings during the year. 10 kg/ha S each year is recommended on S-deficient soils. The sulphur is usually supplied through superphosphate but where high analysis forms of phosphate are used elemental S may be required. On acid soils growing species adapted to Al toxicity, Ca deficiency should be treated with lime or gypsum.

Where dung is being returned to pastures in dairying and intensive feedlot situations, due allowance should be made for the nutrients so returned.

Preferred nutrient forms

P: triple superphosphate, but, if S is deficient, single superphosphate or elemental sulphur may be used.

K: as chloride (muriate of potash).

Micronutrients: usually mixed in blends with major nutrients; alternatively, where pelleted seed is used, some micronutrients may be incorporated in the pellets provided due care is taken not to kill the associated Rhizobium inoculant.

Present fertilizer practices

In wet tropics: 125-250 kg/ha superphosphate applied annually; molybdenized superphosphate (0.02 % Mo) being used instead every third year.

In humid subtropics and semi-arid tropics: 60-125 kg/ha superphosphate applied in spring; molybdenized superphosphate (0.02 % Mo) instead every third year. 50 kg/ha muriate of potash annually on some light soils.

Dung is often returned to pastures from yards associated with dairy farms and/or intensive feedlot systems.

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Mulberry *Mulberry (Morus bombycis Koidz., Morus alba L., Morus latifolia Poiret, Morus nigra L., etc.)*

French: Mûrier; Spanish: Morera; Italian: Gelso; German: Maulbeerbaum

Crop data

Perennial, Harvested part: leaves
Transplanted early spring.
Harvested late summer/autumn in first year from transplanting, three or more times during the growing season from third year onwards.
Plant density: 10 000 (traditional low cut) to 20 000 (intensive) per hectare. Adapted to warm-temperate to tropical conditions. Generally unirrigated.

Effective soil depth 50 cm or more. Prefers light to medium textured soil without hard pan or high ground water-table or stagnant water within 80 cm of surface, and pH 5.5-6.5. Liming improves soil condition and ensures calcium nutrition of plant. Organic fertilizer is also required to maintain or increase soil fertility.

Nutrient uptake in soil-mulberry-silkworm relationship

The supply of mulberry leaves of high nutritive value is essential for the healthy growth of silkworm larvae and leads to a good production of high-quality cocoons.

Nutrient uptake and flow in the soil-mulberry-silkworm metabolic pathway for a leaf yield of 24 800 kg/ha/year and a cocoon yield of 1 240 kg/ha/year is given in the following table:

Japan/Nutrient uptake - Macronutrients*					
	kg/ha/year				
	N	**P2O5**	**K2O**	**MgO**	**CaO**
Total uptake	242	46	211	51	238
- of which retained in supporting parts of the tree	10	-5	-2	-3	10
- of which removed in harvested parts - leaves - shoots	 206 26	 41 9	 192 22	 50 5	 207 21
To cocoons	66	7	12	5	14
To faeces and other waste from silkworm rearing	140	34	180	45	193
* Leaf yield = 24 800 kg/ha/year; cocoon yield = 1 240 kg/ha/year					
Data compiled by H. Takagishi, 1990					

Plant analysis data

Japan/Plant analysis data - Macronutrients/August							
Sampled leaves*	**Sampling time**	**% of dry matter**					
		N	**P**	**K**	**Mg**	**Ca**	**Na**
Upper young leaves	August	3.90 ± 0.72	0.42 ± 0.05	1.94 ± 0.18	0.21 ± 0.18	0.83 ± 0.33	0.07 ± 0.01
Middle (medium-aged) leaves		3.36 ± 0.86	0.28 ± 0.08	1.68 ± 0.33	0.21 ± 0.11	1.19 ± 0.25	0.15 ± 0.15

* grown on farmers' fields
Source: Takagishi et al, 1988

Japan/Plant analysis data - Macronutrients/September						
Sampled leaves*	**Sampling time**	**% of dry matter**				
		N	**P**	**K**	**Mg**	**Ca**
Middle position on shoot	Late autumnal silkworm					
- leaf blade	rearing season	3.31	0.19	1.73	0.31	1.57
- petiole	September	1.13	0.11	3.75	0.77	1.37
Source: Takagishi et al., 1985						

Japan/Plant analysis data - Macronutrients/November						
Sampled leaves*	**Sampling time**	**% of dry matter**				
		N	**P**	**K**	**Mg**	**Ca**
Leaves	Overmature leaf stage/ November	2.44	0.26	2.34	0.54	2.98
Shoots		0.69	0.13	0.33	0.10	0.44
Aerial part of stump		0.39	0.05	0.41	0.07	0.62
Subterranean part of stump with roots, except fine roots		0.62	0.10	0.45	0.10	0.39
Fine roots		1.08	0.17	0.83	0.18	0.50
Mean		0.84	0.12	0.71	0.16	0.85
Source: Takagishi et al., 1987						

Japan/Plant analysis data - Macronutrients/excess-optimum-deficiency ranges					
	% of dry matter				
Range	**N**		**P**	**K**	
	Blade	Blade	Petiole	Blade	Petiole
Excess	>5.0	>0.39	>0.70	>2.1	>4.7
Optimal to luxury	3.0-5.0	0.13-0.22	0.09-0.22	0.5-2.1	0.5-4.7
Deficient	<2.5	<0.13	<0.09	<0.25	<0.25
Sources: N: Takagishi and Matsuda 1970 - P and K: Takagishi and Kawauchi 1985					

To produce leaves of high yield and good quality, special attention must be paid to balanced NPK fertilization. As mulberry is a nitrate-loving plant, favourable conditions are required to promote nitrification and long-term supply of nitrate-N.

Fertilizer recommendations and practice in Japan

In Japan (a warm-termperate country) it is recommended, firstly, that 15 tons/ha rice straw compost or other suitable organic manures are applied in late autumn or winter to maintain or enhance soil fertility. Alternatively, 100 parts (by weight) rice straw and 3 parts calcium cyanamide are applied together and well mixed into the soil between the rows in winter, so that the straw will decompose in the soil as the temperature rises in the spring and summer. Inorganic (mineral) fertilizers are generally applied, in the first year of establishment, at 50 % of the full recommended rates; in the second year at 70 % of the full rates; and, from the 3rd year onwards, at the full rates as set out below.

Japan/Recommended rates of nutrients - Macronutrients				
Soil types	**Soil conditions**	**kg/ha/year**		
		N	**P2O5**	**K2O**
Non-volcanic				
- Alluvial	Lowland medium-textured	300	140	120
- Colluvial	Sloping, medium-textured	300	140	150

- Red and yellow soil	Hill or plateau, medium to heavy textured	300	160	160
Volcanic				
Typic volcanic ash soil (Andosol)	Plateau, medium-textured	300	160	200
- Alluvial volcanic ash	Plateau or lowland, medium textured	300	150	150
- Colluvial volcanic ash	Sloping or plateau, medium textured	300	150	160
Sources: Ito & Mori, 1967 - Raw Silk Bureau, Min. Agr., Japan, 1968				

These rates are appropriate for the cultivation of mulberry for the grown silkworm (fourth and fifth instar) which consumes almost 95 % of the total amount of leaves required for the entire larval stage, and relate to an expected production of 1.2 tons cocoons/ha/year from 24 tons leaves/ha/year.
In northern Japan 60 % of the total fertilizer N, P2O5 and K2O are given in spring before sprouting and the remainder in June/July after the spring silkworm rearing season is over; and in southern Japan 50 % at each stage.

Preferred nutrient forms in Japan

N: ammonium or urea form.
P2O5: mainly in water-soluble form, with some part of citric soluble form.
K2O: usually applied in sulphate form.

The peat ball fertilizer, i.e., an organo-mineral ball fertilizer which contains woody peat as a matrix to hold inorganic fertilizers such as ammonium sulphate, calcium superphosphate, potassium sulphate and micronutrients, is widely used by sericultural farmers. Various types of multinutrient compound fertilizers are also used.

Fertilizer recommendations and practice in China

The Sericultural Research Institute, National Academy of Agricultural Sciences, indicates the following relationship between nutrient application rates and mulberry leaf yield.

PR China - Relationship between nutrient application rates and mulberry yield			
Mulberry leaf yield kg/mu*	**Nutrient application rates kg/mu***		
	N	**P2O5**	**K2O**
500	11.20	6.29	9.77
1000	20.50	7.18	15.70
1500	33.60	13.40	26.20
2500	50.60	22.60	36.50
* 1 mu = 0.0667 ha			

Another report suggests that the amounts of nutrients required to produce 2 500 kg mulberry leaf/mu (37 480 kg/ha):

Nutrient	**kg/mu**	**kg/ha**
N	44.3-47.0	664-705
P2O5	15.7-18.2	235-273
K2O	27.4-30.8	411-462

The N : P2O5 : K2O ratio in annual applications of fertilizer to mulberry is usually 6 : 3 : 5 for the grown silkworm, and 5 : 3 : 4 for young silkworms requiring nutritious young leaves. Fertilizers are applied in split dressings through the year in order to ensure a good yield of

high-quality leaves from spring through to the autumn as well as to maintain the vitality of the trees as a perennial crop.

An example of fertilizer practice in Zhejiang province is given below:

PR China/Zhejiang province - Nutrient rates/Macronutrients					
Season	**Product and rate**	**t/ha**	**kg/ha**		
			N	**P2O5**	**K2O**
Spring	Ammonium carbonate	0.39	65.2	-	-
	Night soil	15.15	72.7	40.9	65.1
Summer	Broad bean (as green manure)	26.52	155.0	39.4	130.0
	Ammonium carbonate	0.19	33.3	-	-
	Night soil	11.36	54.5	31.8	50.0
Autumn	Ammonium chloride (or ammonium sulphate)	0.13	31.8	-	-
	Night soil	15.15	72.7	40.9	65.2
Winter	Canal mud	189.39	227.0	123.0	105.0
Annual total			712.0	276.0	415.0

An example of fertilizer practice in Jiangsu province:

PR China/Jiansu province - Nutrient rates/Macronutrients					
Season	**Product and rate**	**t/ha**	**kg/ha**		
			N	**P2O5**	**K2O**
Spring	Night soil	15.15	72.7	40.9	65.2
	Cattle excreta*	11.36	54.5	31.8	48.5
	Ammonium sulphate	0.11	22.7	-	-
Summer(1)	Green manure	37.88	189.0	48.5	152.0
	Ammonium sulphate	0.23	45.4	-	-
	Cattle excreta*	15.15	72.7	40.9	65.2
Summer(2)	Clover	7.58	39.4	4.5	22.7
	Ammonium sulphate	0.15	30.3	-	-
	Cattle excreta*	15.15	72.7	40.9	65.2
Autumn	Cattle excreta*	11.36	54.5	30.3	48.5
	Ammonium sulphate	0.08	15.2	-	-
Annual total			670.0	238.0	467.0
* Mixed with night soil					

Fertilizer practices in S.E. Asia

Less inorganic fertilizers are applied to mulberry than in the countries mentioned above. Among organic manures recommended are: rice straw, cut grass, cassava stalks, sugar cane waste, kenaf stalk cores, sawdust, mulberry shoots, twigs and branches, fallen leaves, kitchen waste, silkworm faeces and litter, fowl droppings and cattle dung. On the other hand, temporarily recommended rates of inorganic fertilizers would be in kg/ha/year: N = 200, P2O5 = 100 and K2O = 130.

In Sulawesi, Indonesia, most sericultural farmers have been reluctant to use fertilizers in the mulberry fields, but N. Mori et al., 1982, have reported that dressings of urea (providing a total of 100 kg/ha/year N) gave a remarkable yield increase in the fertilizer demonstration and trial field.

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Bush Fruits

Blueberries and Cranberries *(Vaccinium spp.)*

French: Airelles & Canneberges; Spanish: Arandanos; Italian: Mirtillos & Mortellas; German: Blaubeeren & Preiselbeeren

Optimum soil pH for blueberries 4.5 - 5.0 and for cranberries even a little lower. Annual nutrient uptake is relatively small. Unselected wild varieties require lower fertility level than improved varieties. Indicative contents of nutrient elements in the leaf dry matter are shown in the table below:

Plant analysis data - Macronutrients				
Variety	**% of leaf dry matter**			
	N	**P**	**K**	**Mg**
High-bush (V. corymbosum)	2	0.18	0.5	0.2
Low-bush (V. angustifolium)	1.9	0.16	0.5	0.2
Rabbit-eye	1.5	0.1	0.15	0.1

The **leaf nutrient content** depends on the position on the plant, on dilution during rapid growth in spring, on the soil fertility level and on the available soil volume per plant. The N content declines markedly later in the season, e.g. in V. angustifolium from 2.01 % on 24 June to 1.26 % by 26 August or, with added N (75 kg/ha N on 5 May and 40 kg/ha N on 15 July), from 2.49 % on 24 June to 1.63 % on 26 August.

Fertilizer recommendations: Mulching is commonly practised in bush fruit cultivation to ensure nutrient recycling and outgrowth potential for superficial rootlets and to suppress weed competition, though in regions with severe winters intensive mulching may delay winter hardening of plant tissue. Because decaying organic materials incorporate free N and P, some addition of fertilizer is desirable. This input will return in available form in due course. The following rates of application are recommended for high-bush blueberries in Eastern Canada, based on a spacing of 3 m x 1.2 m or 2 690 bushes/ha:

Age in years since set in field	**Recommended rate of 10-10-10-2 fertilizer**	
	g/plant	**kg/ha**
Up to 3	25	75
4 - 5	40	115
6 - 7	85	230
8 - 9	125	345
10 and over	170	455
Source: Craig, 1980		

Raspberry *(Rubus spp.)* and related species and crosses *(loganberry, boysenberry, etc.)*

French: Framboise; Spanish: Frambuesa; Italian: Lampone; German: Himbeere

Optimum pH 6 -7. Nutrient uptake somewhat greater than that of blueberries.

Fertilizer recommendations (Netherlands): 50 t/ha farmyard manure at planting, followed by, for raspberries, in spring about 100 kg/ha N and 300 kg/ha K2O and from flowering to fruit-set about 60 - 70 kg/ha N in split dressings, and, for blackberries, in spring about 60 kg/ha N, 50 kg/ha P2O5 and 90 kg/ha K2O (best in form of a 12-10-18 fertilizer) and from flowering to fruit-set around 50 kg/ha N in split dressings.

Currants *(Ribes spp.)*

French: Groseille; Spanish: Grosella; Italian: Ribes; German: Johannisbeere

Optimum pH 6 - 7. Nutrient uptake greater than that of raspberries.

Indicative contents of nutrient elements in the leaf dry matter, varying with position on the plant and other factors (see Blueberries): 3 % N, 0.3 % P, 1.5 - 3.0 % K, 0.3 % Mg.

Fertilizer recommendations (Netherlands): 50 t/ha farmyard manure at planting, followed in spring by 300 kg/ha K2O and from flowering to fruit-set 60 - 70 kg/ha N in split dressings.

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Author: P.D.J. van der Vorm, Department of Soil Science and Plant Nutrition, Agricultural University Wageningen, The Netherlands

Dates and Figs

Date Palm *(Phoenix dactylifera L.)*

French: Dattier; Spanish: Palma datilera; Italian: Palma del dattero; German: Dattelpalme

Good quality staple food. Plant density under good growth conditions 100 - 140 bearing trees/ha. An adult palm contains 50 - 110 leaves with an average lifespan of 5 - 6 years. Potential fruit bunches develop in the internodes of new leaves. Transpiration by the leaves is about 320 litres water per kg leaves per year. The plant is relatively salt-tolerant, with almost no growth restriction up to 60 mmol salt per litre and about 60 % growth reduction at 1.2 % salt content. Yields 24 - 40 kg fruit per tree per year in hot desert areas with restricted water supply. In some areas in Egypt, yields up to about 12 t/ha fruit are obtained under intensive management.

N uptake when in bearing: approx. 650 g N/tree/year (assuming 50 kg dry matter/tree/year, containing 400 g N, in new vegetative tissue, and 70 kg edible portion of fruit/tree/year, containing a further 250 g N).

Plant analysis data:

Plant analysis data - Macronutrients								
Plant part	**N**	**P**	**K**	**Mg**	**Ca**	**S**	**Na**	**SiO2**
Leaves	(% in dry matter)							
(38-43 % DM)	0.68-1.04	0.08-0.12	2.1-3.9	0.2-0.5	0.13-0.5	0.09-0.13	0.02-0.11	0.2-0.3
Fruit	(% in edible portion)							
(22.5 % DM)	0.35	0.063	0.65	-	0.059	-	0.001	-

Fertilizer recommendations:

In India, 1.8 - 2.7 kg N/tree/year are recommended for maintenance of trees in bearing. This should be applied in conformity with the seasonal growth pattern, i.e. during periods of active growth and water availability (irrigation too should be adapted to the growth rhythm; in Iraq it is recommended that the majority of irrigation water should be supplied during the 5 - 7 months of active growth and bunch development).

In Egypt El-Hammady et al., 1987 recommend 0.5 kg N/tree/year in three split dressings, plus 165 g K2O/tree to increase fruit yield. Fertilizer may either be broadcast, coupled with appropriate irrigation, or may, if the equipment is available, be applied through drip irrigation (which improves the efficiency of use of both water and fertilizer).

Optimal use of locally available organic materials may be achieved by application of rapidly decomposing leguminous green manures (assisted by application of P fertilizer) and/or farmyard manure, by recycling of litter and by protecting the soil surface with mulch.

Fig *(Ficus carica L.)*

French: Figuier; Spanish: Higuera; Italian: Fico; German: Feige

A tree crop which is rather drought-tolerant and salt-tolerant but will not withstand high pH. Pruning to increase fruit size may start when the tree is large enough but during the first four years the prime interest is in vigorous plant growth. Removal of non-flower-bearing wood and

activation of dormant wood are commonly practised. Young plants with open tissue surfaces (resulting from propagation techniques) need protection from intensive insolation. Plant densities range from 40/ha under conditions of water stress to around 150/ha in ordinary plantations and to more than 1 000/ha where trees are headed back to a small size.

N uptake when in bearing: approx. 930 g N/100 kg fresh fruit (assuming a dry fruit/fresh fruit ratio of 0.35, with 0.47 % N in the dry fruit, and 1 kg dry leaves per 1 kg dry fruit, with 2.22 % N in the dry leaves).

Plant analysis:

Plant analysis data - Macronutrients						
Plant part	**% of dry matter**					
	N	**P**	**K**	**Mg**	**Ca**	**Na**
Leaves	2.0-2.5	0.1-0.3	1.0-1.6	0.8	3.0-3.6	-
Fruit	0.47	0.057-0.070	0.57-0.68	0.06	0.13-0.17	0.01

Fertilizer recommendations (India)

Young trees, 36 g/tree/year fertilizer N and 7 kg/tree/year farmyard manure.
Fully grown trees, 144 g/tree/year fertilizer N and 30 kg/tree/year farmyard manure. A typical nutrient content of fresh farmyard manure from cows in India may be taken as 15 - 55 kg N, 1.5 - 2.5 kg P2O5, 3.5 - 6 kg K2O and 1 - 1.5 kg MgO per 1 000 kg. Under conditions of regular annual application resulting in steady state decomposition (decay of manure = supply of manure), the recommended 30 kg farmyard manure correspond to 45 - 165 g N, so the total N recommendation for fully grown trees, including fertilizer N, amounts to about 200 - 300 g N/tree/year. Assuming 300 trees/ha, this corresponds to 60 - 90 kg/ha/year. The timing of N fertilizer application should be in accordance with seasonal growth rhythm and availability of soil water. Where fertilizer P is also needed, and 90 kg/ha fertilizer N is given annually, 40 kg/ha P2O5 may be applied in the form of superphosphate or available NP-fertilizers.

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Author: P.D.J. van der Vorm, Department of Soil Science and Plant Nutrition, Agricultural University Wageningen, The Netherlands

Drugs and Spices

Arecanut *[Betelnut] (Areca catechu L.)*

French: Noix d'arec; Spanish: Areca; German: Betelpalme

Often grown on soils of low fertility but, because selection has been based on tolerance to adverse growth conditions, many trees are relatively unresponsive to higher rates of fertilizer application.

Nutrient demand/uptake/removal: uptake around 80 kg/ha N, 30 kg/ha P2O5, 75 kg/ha K2O; potential return by leaf recycling 85 kg N, 25 kg P2O5 and 90 kg K2O per 10 t/year fresh leaves. Recycling may be assisted by addition of small amount of N and P to promote microbial breakdown. A regular minimal NP supply is also beneficial where mulch crops and green manure crops may compete for scarce nutrients.

General fertilizer recommendations: 160 kg/ha N, 60 kg/ha P2O5 and 180 kg/ha K2O, with 10 - 20 t/ha farmyard manure per year. N should be given in split dessings, P and K once a year.

Cinchona *(Cinchona spp.)*

French: Cinchone, quinquina; Spanish: Quinoa; Italian: Albero di China; German: Chinarindenbaum

Bark, which is source of drugs quinine and quinidine, is obtained first from young thinned trees and finally from larger trees at remaining plant density of 3 000 trees/ha. Indonesian practice is to start with 5 000 trees/ha and to coppice after 7 - 8 years. Fungal diseases (e.g. stripe cancer, caused by Phytophtora cinnamoni) and nematodes are troublesome in Zaire. Final production from a well managed plantation may reach 10 t/ha bark. Quinine content depends on plant type.

Plant analysis data: Typical leaf nutrient content is 2.0 % N, 0.04 % P, 0.60 % K, 0.57 % Ca, 0.12 % S.

General fertilizer recommendations: upwards from 70 kg/ha N per year. NPK compound with a relatively high P2O5 content (e.g. 15-15-15) may be given in P-retaining acid oxisols. Where there is a lack of available Ca, the crop responds to appication of a calcium salt.

Nutmeg *(Myristica fragrans Houtt.)*

French: Noix muscade; Spanish: Nuez moscada, miristica; Italian: Noce muscata; German: Muskatnuß

Dioecious perennial tree, 5 - 13 m high, normally planted 1 male to 10 females, spaced about 9 m x 9 m. Reproduction of high-yielding trees by grafting increases production. Yields range from less than 1 000 fruits/tree to 2 000 fruits/tree per year (fruit weight 8 g nut, 2 g mace). Young roots need protection against damage, waterlogging, drought and acidity. Young plants need an evenly distributed nutrient supply and should be protected against excessive sunlight.

General fertilizer recommendations:

General fertilizer recommendation, dependent on age			
Age (years)	**g/tree/year**		
	N	**P2O5**	**K2O**
5	100	90	250
10	200	180	500
15	300	270	750
≥20	400	360	1 000
On deep, fertile, volcanic soils, application may be limited to 200 g/tree N per year.			

Clove *(Syzigium aromaticum (L.) Merr. et Perr.)*

French: Clou de girofle; Spanish: Clavo de especia; Italian: Claventino, Garofano; German: Gewürznelke

Flower buds and fruit have a wide range of flavouring and medicinal uses. Plant spacing from wider than 8 m x 11 m in mixed cropping to 6 m x 6 m. Yields from less than 1 to several kg/tree/season; seasonal production may be rather irregular and both internal and external stresses cause wide variation in annual production. The root system is very sensitive and must be protected, particularly in the case of young plants.

General fertilizer recommendations: similar to, or slightly less than, those for nutmeg. Application should be regular, amounts depending on mulching and litter recycling practices and use of compost and/or farmyard manure.

Black Pepper *(Piper nigrum L.)*

French: Poivre; Spanish: Pimiento; Italian: Pepe; German: Pfeffer

Yields from a few hundred kg/ha berries at a common plant density of 1 370 vines/ha up to about 8 000 kg/ha from 5 000 intensively managed vines. Needs pruning, both to promote fruit-bearing laterals and to guide vigorous outgrowth. The crop is very responsive to intensity of management and nutrient supply. The root system is very sensitive to salt injury. Reports from S.E. Asia and Brazil indicate that fungal disease attack associated with excessive nutrient supply may shorten production potential. A reduction of disease incidence by mixed cropping has been reported from India.

Nutrient uptake by a full, vigorous crop of about 1 750 vines/ha: around 250 kg/ha N, 35 kg/ha P2O5, 205 kg/ha K2O, 20 kg/ha MgO and 45 kg/ha CaO.

Typical leaf nutrient content: 3.1 - 3.4 % N; 0.16 - 0.18 % P, 3.4-4.3 % K, 0.44 % Mg, 1.67 % Ca.

General fertilizer recommendations: 100 g N, 40 g P2O5 and 140 g K20 per vine per year. N should be given in split dressings, P and K once a year. These amounts may be doubled under intensive management where there is abundant water and sunshine. A small amount of Mg may be beneficial. Regular application of a compound NPKMg fertilizer, e.g. 12-12-17-2(MgO), is a good alternative. If soil pH is low, lime may be applied every 2 years at a rate of 200 - 500 g/vine.

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Oils and Fibres

Sesame (Sesamum indicum L.)

French: Sesame; Spanish: Sesamo; Italian: Sesamo; German: Sesam

Crop is relatively drought-tolerant and is often grown where water supply is limited. In these circumstances, plant density should not exceed 250 000/ha. Many varieties are available; local varieties are often relatively unresponsive to higher rates of fertilizer.

General fertilizer recommendations: 90 kg/ha N and, where P is needed, up to 90 kg/ha P2O5. Responses to N in the range of 7 - 10 kg seed per kg N applied have been reported, depending on inherent soil fertility and rate of N applied. Over 1 t/ha seed has been obtained in India with a fertilizer application of 60 kg/ha N, 40 kg/ha P2O5 and 24 kg/ha K2O.

Castor Bean (Ricinus communis L.)

French: Ricin; Spanish: Ricino; Italian: Ricino; German: Rizinus

Bean yields range from less than 0.5 t/ha per season, under conditions of severe growth constraint, to 3.5 t/ha per season, under intensive growth with high use of inputs. The oil content of the beans depends on variety, growth conditions and method of processing; 55 % is reported from USA.

General fertilizer recommendations: Up to 120 kg/ha N per crop, depending on plant type and availability of water; P2O5 at about 30 % of the N rate, or more on P-fixing soil; and possibly K2O at half the N rate, or more on K-fixing or on deficient weathered soil.

Jojoba *(Simmondsia chinensis Nutt.)*

Many-stemmed desert shrub originating from arid deserts of Mexico and USA, 0.6 - 4.6 m3 in size and aged up to 200 years. Seeds contain liquid wax with multiplicity of uses. Seeds range in size and density from 700/kg under good growth conditions to 5 300/kg under stress. Varieties exhibit a wide range of genetic variation in plant size, fruit density, delay in fruit-bearing, etc. Tolerates hot and dry conditions and is rather salt-tolerant.

General fertilizer recommendations: N needs to be applied before rainfall (or in "fertigation" management), also P2O5, preferably combined with mulch protection to withstand rapid litter-mineralization caused by extreme surface temperatures.

Abaca *[Manila Hemp] (Musa textilis Nee)*

French: Abaca, chanvre de Manille; Spanish: Abaca, canamo de Manila; German: Manilahanf

Closely related to the banana. Fibres are derived from pseudostem leaf sheaths. Fibre yields range from 0.3 - 1.7 t/ha/year in the Philippines to 1.5 - 2.5 t/ha/year in Ecuador. Grows best in rich volcanic soils. Special care is needed in heavy clays to ensure adequate root functioning.

Nutrient uptake by crop of 100 t/ha above-ground plant material yielding about 2 t/ha fibre: 280 kg/ha N, 30 kg/ha P2O5, 490 kg/ha K2O, 125 kg/ha CaO.

General fertilizer recommendations: N at 50 - 150 kg/ha/year, preferably in split dressings, together with sufficient K2O.

Ramie *[China Grass] (Boehmeria nivea (L.) Gaudich)*

French: Ramie blanche; Spanish: Ramio blanco; Italian: Ramia; German: Ramie, Chinanessel

Yields up to 140 t/ha/year fresh plant material from 2 to 6 harvests, giving about 1.4 t/ha clean fibre. Fibre has to be degummed from the stems, which requires expertise. Leaves are rich in protein, so forage use is possible. Required water supply is about 140 mm/month; plans do not withstand waterlogging.

General fertilizer recommendations: Split application of 200 kg/ha N, 60 kg/ha P2O5 and 90 kg/ha K2O per year. Increasing N rate to 300 kg/ha/year increases leaf protein and residual soil N. Composted decortication waste is returned to the soil in some places.

Roselle (Hibiscus sabdariffa L.)

French: Ketmie rose; Spanish: Canamo de Guinea, rosella; German: Rosellahanf

Crop data: Yields up to 70 - 90 t/ha/year fresh plant material, giving 3.5 - 4.5 t/ha clean fibre. Requires good rooting conditions. Time of planting is limited by photosensitivity.

General fertilizer recommendations: Commonly receives from 40 kg/ha N per crop in India where growth is limited by poor supply of available soil nutrients and water and other major constraints, to 250 kg/ha N in Egypt.
On Egyptian soil which yielded 2.84 t/ha fibre without fertilizer, the application of 117 kg/ha N, 77 kg/ha P2O5 and 140 kg/ha K2O inccreased the fibre yield to 4.65 t/ha.

Kenaf (Hibiscus cannabinus L.)

French: Kenaf; Spanish: Kenaf; German: Dekkanhanf, Hanf-Eibisch

Yields 0.5 - 2.5 t/ha fibre depending on variety and growth conditions.

General fertilizer recommendations: 40 - 80 kg/ha N, 20 - 50 kg/ha P2O5, 20 - 80 kg/ha K2O. Response to N rather variable but generally around 8 - 12 kg extra fibre per kg N applied.

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